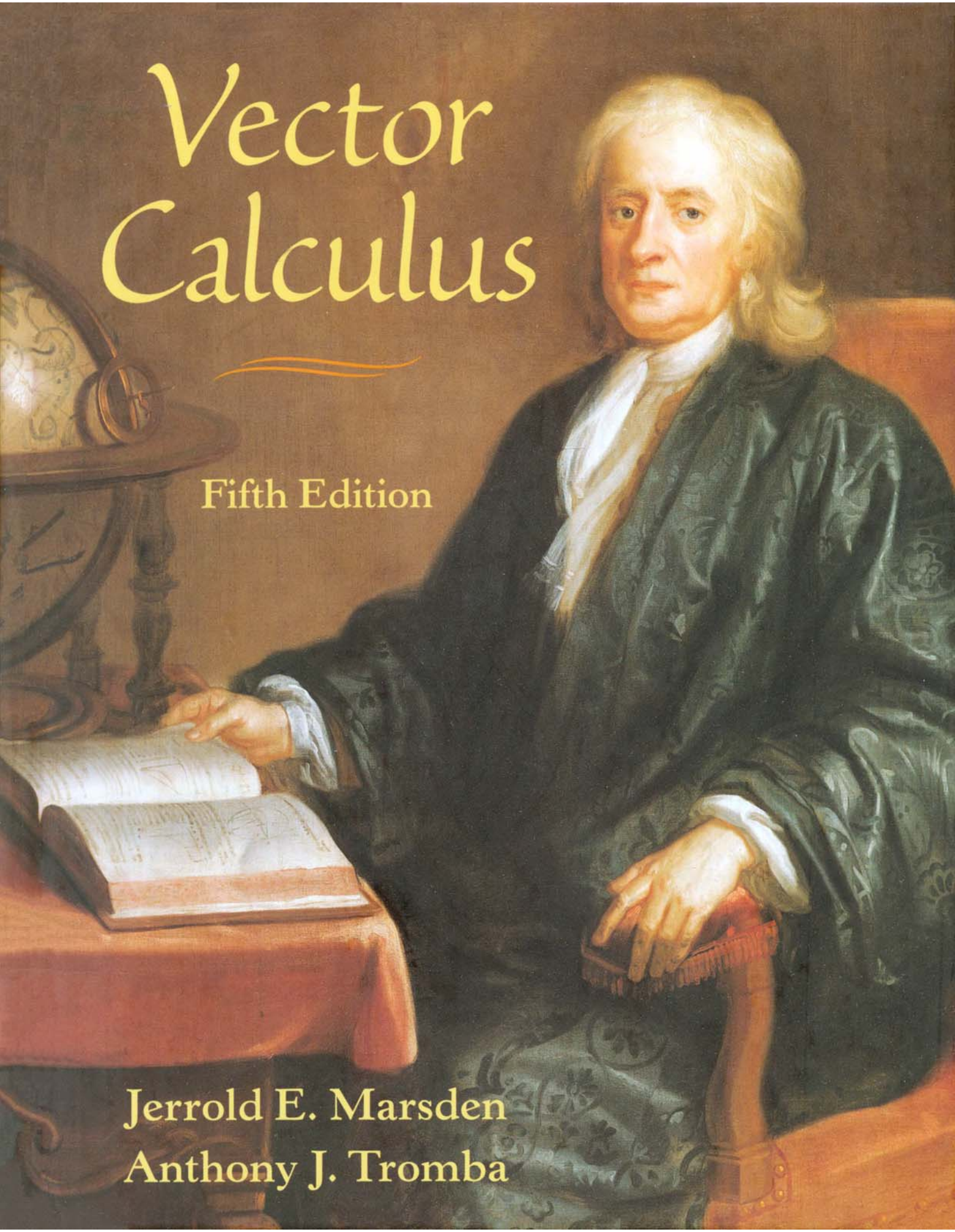
Vector Calculus

Fifth Edition



Jerrold E. Marsden
Anthony J. Tromba

DERIVATIVES

1. $\dfrac{dau}{dx} = a\dfrac{du}{dx}$
2. $\dfrac{d(u+v)}{dx} = \dfrac{du}{dx} + \dfrac{dv}{dx}$
3. $\dfrac{d(uv)}{dx} = u\dfrac{dv}{dx} + v\dfrac{du}{dx}$
4. $\dfrac{d(u/v)}{dx} = \dfrac{v(du/dx) - u(dv/dx)}{v^2}$
5. $\dfrac{d(u^n)}{dx} = nu^{n-1}\dfrac{du}{dx}$
6. $\dfrac{d(u^v)}{dx} = vu^{v-1}\dfrac{du}{dx} + u^v(\log u)\dfrac{dv}{dx}$
7. $\dfrac{d(e^u)}{dx} = e^u\dfrac{du}{dx}$
8. $\dfrac{d(e^{au})}{dx} = ae^{au}\dfrac{du}{dx}$
9. $\dfrac{da^u}{dx} = a^u(\log a)\dfrac{du}{dx}$
10. $\dfrac{d(\log u)}{dx} = \dfrac{1}{u}\dfrac{du}{dx}$
11. $\dfrac{d(\log_a u)}{dx} = \dfrac{1}{u(\log a)}\dfrac{du}{dx}$
12. $\dfrac{d\sin u}{dx} = \cos u\dfrac{du}{dx}$
13. $\dfrac{d\cos u}{dx} = -\sin u\dfrac{du}{dx}$
14. $\dfrac{d\tan u}{dx} = \sec^2 u\dfrac{du}{dx}$
15. $\dfrac{d\cot u}{dx} = -\csc^2 u\dfrac{du}{dx}$
16. $\dfrac{d\sec u}{dx} = \tan u\sec u\dfrac{du}{dx}$
17. $\dfrac{d\csc u}{dx} = -(\cot u)(\csc u)\dfrac{du}{dx}$
18. $\dfrac{d\arcsin u}{dx} = \dfrac{1}{\sqrt{1-u^2}}\dfrac{du}{dx}$
19. $\dfrac{d\arccos u}{dx} = \dfrac{-1}{\sqrt{1-u^2}}\dfrac{du}{dx}$
20. $\dfrac{d\arctan u}{dx} = \dfrac{1}{1+u^2}\dfrac{du}{dx}$
21. $\dfrac{d\operatorname{arccot} u}{dx} = \dfrac{-1}{1+u^2}\dfrac{du}{dx}$
22. $\dfrac{d\operatorname{arcsec} u}{dx} = \dfrac{1}{u\sqrt{u^2-1}}\dfrac{du}{dx}$
23. $\dfrac{d\operatorname{arccsc} u}{dx} = \dfrac{-1}{\sqrt{u^2-1}}\dfrac{du}{dx}$
24. $\dfrac{d\sinh u}{dx} = \cosh u\dfrac{du}{dx}$
25. $\dfrac{d\cosh u}{dx} = \sinh u\dfrac{du}{dx}$
26. $\dfrac{d\tanh u}{dx} = \operatorname{sech}^2 u\dfrac{du}{dx}$
27. $\dfrac{d\coth u}{dx} = -(\operatorname{csch}^2 u)\dfrac{du}{dx}$
28. $\dfrac{d\operatorname{sech} u}{dx} = -(\operatorname{sech} u)(\tanh u)\dfrac{du}{dx}$
29. $\dfrac{d\operatorname{csch} u}{dx} = -(\operatorname{csch} u)(\coth u)\dfrac{du}{dx}$
30. $\dfrac{d\sinh^{-1} u}{dx} = \dfrac{1}{\sqrt{1+u^2}}\dfrac{du}{dx}$
31. $\dfrac{d\cosh^{-1} u}{dx} = \dfrac{1}{\sqrt{u^2-1}}\dfrac{du}{dx}$
32. $\dfrac{d\tanh^{-1} u}{dx} = \dfrac{1}{1-u^2}\dfrac{du}{dx}$
33. $\dfrac{d\coth^{-1} u}{dx} = \dfrac{1}{u^2-1}\dfrac{du}{dx}$
34. $\dfrac{d\operatorname{sech}^{-1} u}{dx} = \dfrac{-1}{u\sqrt{1-u^2}}\dfrac{du}{dx}$
35. $\dfrac{d\operatorname{csch}^{-1} u}{dx} = \dfrac{-1}{|u|\sqrt{1+u^2}}\dfrac{du}{dx}$

INTEGRALS (An arbitrary constant may be added to each integral.)

1. $\int x^n\,dx = \frac{1}{n+1}x^{n+1} \quad (n \neq -1)$
2. $\int \frac{1}{x}\,dx = \log|x|$
3. $\int e^x\,dx = e^x$
4. $\int a^x\,dx = \frac{a^x}{\log a}$
5. $\int \sin x\,dx = -\cos x$
6. $\int \cos x\,dx = \sin x$
7. $\int \tan x\,dx = -\log|\cos x|$
8. $\int \cot x\,dx = \log|\sin x|$
9. $\int \sec x\,dx = \log|\sec x + \tan x| = \log\left|\tan\left(\frac{1}{2}x + \frac{1}{4}\pi\right)\right|$
10. $\int \csc x\,dx = \log|\csc x - \cot x| = \log\left|\tan\frac{1}{2}x\right|$
11. $\int \arcsin\frac{x}{a}\,dx = x\arcsin\frac{x}{a} + \sqrt{a^2 - x^2} \quad (a > 0)$
12. $\int \arccos\frac{x}{a}\,dx = x\arccos\frac{x}{a} - \sqrt{a^2 - x^2} \quad (a > 0)$
13. $\int \arctan\frac{x}{a}\,dx = x\arctan\frac{x}{a} - \frac{a}{2}\log(a^2 + x^2) \quad (a > 0)$
14. $\int \sin^2 mx\,dx = \frac{1}{2m}(mx - \sin mx\cos mx)$
15. $\int \cos^2 mx\,dx = \frac{1}{2m}(mx + \sin mx\cos mx)$
16. $\int \sec^2 x\,dx = \tan x$
17. $\int \csc^2 x\,dx = -\cot x$
18. $\int \sin^n x\,dx = -\frac{\sin^{n-1}x\cos x}{n} + \frac{n-1}{n}\int \sin^{n-2}x\,dx$
19. $\int \cos^n x\,dx = \frac{\cos^{n-1}x\sin x}{n} + \frac{n-1}{n}\int \cos^{n-2}x\,dx$
20. $\int \tan^n x\,dx = \frac{\tan^{n-1}x}{n-1} - \int \tan^{n-2}x\,dx \quad (n \neq 1)$
21. $\int \cot^n x\,dx = -\frac{\cot^{n-1}x}{n-1} - \int \cot^{n-2}x\,dx \quad (n \neq 1)$
22. $\int \sec^n x\,dx = \frac{\tan x\sec^{n-2}x}{n-1} + \frac{n-2}{n-1}\int \sec^{n-2}x\,dx \quad (n \neq 1)$
23. $\int \csc^n x\,dx = -\frac{\cot x\csc^{n-2}x}{n-1} + \frac{n-2}{n-1}\int \csc^{n-2}x\,dx \quad (n \neq 1)$

(*Continued on next page*)

24. $\displaystyle\int \sinh x\,dx = \cosh x$

25. $\displaystyle\int \cosh x\,dx = \sinh x$

26. $\displaystyle\int \tanh x\,dx = \log|\cosh x|$

27. $\displaystyle\int \coth x\,dx = \log|\sinh x|$

28. $\displaystyle\int \operatorname{sech} x\,dx = \arctan(\sinh x)$

29. $\displaystyle\int \operatorname{csch} x\,dx = \log\left|\tanh\frac{x}{2}\right| = -\frac{1}{2}\log\frac{\cosh x + 1}{\cosh x - 1}$

30. $\displaystyle\int \sinh^2 x\,dx = \tfrac{1}{4}\sinh 2x - \tfrac{1}{2}x$

31. $\displaystyle\int \cosh^2 x\,dx = \tfrac{1}{4}\sinh 2x + \tfrac{1}{2}x$

32. $\displaystyle\int \operatorname{sech}^2 x\,dx = \tanh x$

33. $\displaystyle\int \sinh^{-1}\frac{x}{a}\,dx = x\sinh^{-1}\frac{x}{a} - \sqrt{x^2 + a^2} \quad (a > 0)$

34. $\displaystyle\int \cosh^{-1}\frac{x}{a}\,dx = \begin{cases} x\cosh^{-1}\dfrac{x}{a} - \sqrt{x^2 - a^2} & \left[\cosh^{-1}\left(\dfrac{x}{a}\right) > 0, a > 0\right] \\ x\cosh^{-1}\dfrac{x}{a} + \sqrt{x^2 - a^2} & \left[\cosh^{-1}\left(\dfrac{x}{a}\right) < 0, a > 0\right] \end{cases}$

35. $\displaystyle\int \tanh^{-1}\frac{x}{a}\,dx = x\tanh^{-1}\frac{x}{a} + \frac{a}{2}\log|a^2 - x^2|$

36. $\displaystyle\int \frac{1}{\sqrt{a^2 + x^2}}\,dx = \log(x + \sqrt{a^2 + x^2}) = \sinh^{-1}\frac{x}{a} \quad (a > 0)$

37. $\displaystyle\int \frac{1}{a^2 + x^2}\,dx = \frac{1}{a}\arctan\frac{x}{a} \quad (a > 0)$

38. $\displaystyle\int \sqrt{a^2 - x^2}\,dx = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\arcsin\frac{x}{a} \quad (a > 0)$

39. $\displaystyle\int (a^2 - x^2)^{3/2}\,dx = \frac{x}{8}(5a^2 - 2x^2)\sqrt{a^2 - x^2} + \frac{3a^4}{8}\arcsin\frac{x}{a} \quad (a > 0)$

40. $\displaystyle\int \frac{1}{\sqrt{a^2 - x^2}}\,dx = \arcsin\frac{x}{a} \quad (a > 0)$

41. $\displaystyle\int \frac{1}{a^2 - x^2}\,dx = \frac{1}{2a}\log\left|\frac{a + x}{a - x}\right|$

42. $\displaystyle\int \frac{1}{(a^2 - x^2)^{3/2}}\,dx = \frac{x}{a^2\sqrt{a^2 - x^2}}$

43. $\displaystyle\int \sqrt{x^2 \pm a^2}\,dx = \frac{x}{2}\sqrt{x^2 \pm a^2} \pm \frac{a^2}{2}\log|x + \sqrt{x^2 \pm a^2}|$

44. $\displaystyle\int \frac{1}{\sqrt{x^2 - a^2}}\,dx = \log|x + \sqrt{x^2 - a^2}| = \cosh^{-1}\frac{x}{a} \quad (a > 0)$

45. $\displaystyle\int \frac{1}{x(a + bx)}\,dx = \frac{1}{a}\log\left|\frac{x}{a + bx}\right|$

46. $\displaystyle\int x\sqrt{a + bx}\,dx = \frac{2(3bx - 2a)(a + bx)^{3/2}}{15b^2}$

47. $\displaystyle\int \frac{\sqrt{a + bx}}{x}\,dx = 2\sqrt{a + bx} + a\int \frac{1}{x\sqrt{a + bx}}\,dx$

(*Continued at the back of the book*)

Vector Calculus

Fifth Edition

Jerrold E. Marsden
California Institute of Technology, Pasadena

Anthony J. Tromba
University of California, Santa Cruz

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Politics is for the moment.
An equation is for eternity.

A. EINSTEIN

Some calculus tricks are quite easy.
Some are enormously difficult. The fools
who write the textbooks of
advanced mathematics seldom take the trouble
to show you how easy the easy calculations are.

SILVANUS P. THOMPSON, *CALCULUS MADE EASY,* MACMILLAN (1910)

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Contents

Preface vii
Historical Introduction xiii

1 The Geometry of Euclidean Space 1

1.1 Vectors in Two- and Three-Dimensional Space *1*
1.2 The Inner Product, Length, and Distance *23*
1.3 Matrices, Determinants, and the Cross Product *38*
1.4 Cylindrical and Spherical Coordinates *65*
1.5 n-Dimensional Euclidean Space *74*
Review Exercises for Chapter 1 *88*

2 Differentiation 94

2.1 The Geometry of Real-Valued Functions *94*
2.2 Limits and Continuity *107*
2.3 Differentiation *127*
2.4 Introduction to Paths and Curves *141*
2.5 Properties of the Derivative *150*
2.6 Gradients and Directional Derivatives *163*
Review Exercises for Chapter 2 *173*

3 Higher-Order Derivatives; Maxima and Minima 181

3.1 Iterated Partial Derivatives *182*
3.2 Taylor's Theorem *193*
3.3 Extrema of Real-Valued Functions *203*
3.4 Constrained Extrema and Lagrange Multipliers *225*
3.5 The Implicit Function Theorem *246*
Review Exercises for Chapter 3 *255*

4 Vector-Valued Functions 261

4.1 Acceleration and Newton's Second Law *261*
4.2 Arc Length *274*
4.3 Vector Fields *285*
4.4 Divergence and Curl *294*
Review Exercises for Chapter 4 *313*

5 Double and Triple Integrals 317

5.1 Introduction *317*
5.2 The Double Integral Over a Rectangle *327*
5.3 The Double Integral Over More General Regions *341*
5.4 Changing the Order of Integration *349*
5.5 The Triple Integral *354*
Review Exercises for Chapter 5 *365*

6 The Change of Variables Formula and Applications of Integration 368

6.1 The Geometry of Maps from $\mathbb{R}^2$ to $\mathbb{R}^2$ *369*
6.2 The Change of Variables Theorem *376*
6.3 Applications *393*
6.4 Improper Integrals *406*
Review Exercises for Chapter 6 *417*

7 Integrals Over Paths and Surfaces 421

7.1 The Path Integral *421*
7.2 Line Integrals *429*
7.3 Parametrized Surfaces *451*
7.4 Area of a Surface *461*
7.5 Integrals of Scalar Functions Over Surfaces *474*
7.6 Surface Integrals of Vector Fields *483*
7.7 Applications to Differential Geometry, Physics, and Forms of Life *500*
Review Exercises for Chapter 7 *514*

8 The Integral Theorems of Vector Analysis 518

8.1 Green's Theorem *518*
8.2 Stokes' Theorem *532*
8.3 Conservative Fields *550*
8.4 Gauss' Theorem *561*
8.5 Some Differential Equations of Mechanics and Technology *576*
8.6 Differential Forms *588*
Review Exercises for Chapter 8 *605*

Answers to Odd-Numbered Exercises 609
Index 668
Illustration Credits 676

To Barbara and Inga for all their love and support

Preface

This text is intended for a one-semester course in the calculus of functions of several variables and vector analysis, which is normally taught at the sophomore level. In addition to making changes and improvements throughout the text, in this new edition we have added considerable material that presents the historical development of the subject and have also attempted to convey a sense of excitement, relevance, and importance of the subject matter.

Prerequisites

Sometimes courses in vector calculus are preceded by a first course in linear algebra, but this is not an essential prerequisite. We require only the bare rudiments of matrix algebra, and the necessary concepts are developed in the text. If this course is preceded by a course in linear algebra, the instructor will have no difficulty enhancing the material. However, we do assume a knowledge of the fundamentals of one-variable calculus—the process of differentiation and integration and their geometric and physical meaning as well as a knowledge of the standard functions, such as the trigonometric and exponential functions.

The Role of Theory

The text includes much of the basic theory as well as many concrete examples and problems. Some of the technical proofs for theorems in Chapters 2 and 5 are given in optional sections that are readily available on the book's Web site at www.whfreeman.com/MarsdenVC5e (see the description on the next page). Section 2.2, on limits and continuity, is designed to be treated lightly and is deliberately brief. More sophisticated theoretical topics, such as compactness and delicate proofs in integration theory, have been omitted, because they usually belong to a more advanced course in real analysis.

Concrete and Student-Oriented

Computational skills and intuitive understanding are important at this level, and we have tried to meet this need by making the book concrete and student-oriented. For example, although we formulate the definition of the derivative correctly, it is done by using matrices of partial derivatives rather than abstract linear transformations. We also include a number of physical illustrations such as fluid mechanics, gravitation,

and electromagnetic theory, and from economics as well, although knowledge of these subjects is not assumed.

Order of Topics

A special feature of the text is the early introduction of vector fields, divergence, and curl in Chapter 4, before integration. Vector analysis often suffers in a course of this type, and the present arrangement is designed to offset this tendency. To go even further, one might consider teaching Chapter 3 (Taylor's theorems, maxima and minima, Lagrange multipliers) after Chapter 8 (the integral theorems of vector analysis).

This fifth edition was completely reset, but retains and improves on the balance between theory, applications, optional material, and historical notes that was present in earlier editions.

Supplements

One of the main changes in this edition is in the supplement. They are as follows:

1. **Web Site.** The book's Web site contains the following materials:

   - **Internet Supplement,** a PDF file containing additional material suitable for projects as well as technical proofs and sample examinations with complete solutions.
   - **PowerPoint and KeyNote Slides** for instructors to use in presentations of the text's figures, as well as section-by-section summaries.
   - **LaTeX and PDF Files of Sample Exams** (on instructor's protected site)
   - **Updates**

   It is available to everyone and can be found at www.whfreeman.com/MarsdenVC5e.

2. **Student Study Guide with Solutions.** This student guide, written by Karen Pao and Fred Soon, contains helpful hints and summaries for the material in each section, contains the solutions to selected problems, and contains sample exams to help students in exam preparation. Problems whose solutions appear in the Student Study Guide have a colored number in the text, for easy reference. The guide has been revised and reset for the Fifth Edition of *Vector Calculus*. ISBN 0-7167-0528-1

3. **Instructor's Manual with Solutions.** This supplement contains material available only to instructors. This includes summaries of material and additional worked-out examples that are helpful in the preparation of lectures. It also contains additional solutions to problems and sample exams (some of them with complete solutions). ISBN 0-7167-0646-6

Final Exam Questions

There are practice exams available in the Student Study Guide, the Internet supplement, as well as in the Instructor's Manual. We also include some final exam questions (some of them challenging) for the reader's convenience on the book's Web site.

Of course, the level and choice of topics and the lengths of final exams will vary from instructor to instructor. Working these problems requires a knowledge of most of the main material of the book, and solving 10 of these problems should take the reader about 3 hours to complete. Some solutions are also given on the book's Web site.

We are excited about this new edition of *Vector Calculus*, especially the inclusion of the new historical material as well as the new discussions of interesting applications of vector analysis, both mathematical and physical. We hope that the reader will be equally pleased.

Jerry Marsden and Tony Tromba,
Caltech and UC Santa Cruz, Summer 2003.

Acknowledgements

Many colleagues and students in the mathematical community have made valuable contributions and suggestions since this book was begun. An early draft of the book was written in collaboration with Ralph Abraham. We thank him for allowing us to draw upon his work. It is impossible to list all those who assisted with this book, but we wish especially to thank Michael Hoffman and Joanne Seitz for their help on earlier editions. We also received valuable comments from Mary Anderson, John Ball, Patrick Brosnan, Andrea Brose, David Dresin, Gerald Edgar, Michael Fischer, Frank Gerrish, Mohammad Gohmi, Jenny Harrison, Jan Hogendijk, Jan-Jaap Oosterwijk, and Anne van Weerden (Uterecht), David Knudson, Richard Kock, Andrew Lenard, William McCain, Gordon McLean, David Merriell, Jeanette Nelson, Dan Norman, Keith Phillips, Anne Perleman, Oren Walter Rosen, Kenneth Ross, Ray Sachs, Diane Sauvageot, Joel Smoller, Francis Su, Melvyn Tews, Ralph and Bob Tromba, Steve Wan, Alan Weinstein, John Wilker, and Peter Zvengrowski. The students and faculty of Austin Community College deserve a special note of thanks, as do our students at both Caltech and UC Santa Cruz.

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A final word of thanks goes to those who helped in the preparation of the manuscript and the production of the book. For the earlier editions, we thank Connie Calica, Nora Lee, Marnie McElhiney, Ruth Suzuki, Ikuko Workman, and Esther Zack

for their excellent typing of various versions and revisions of the manuscript; Herb Holden of Gonzaga University and Jerry Kazdan of the University of Pennsylvania for suggesting and preparing early versions of the computer-generated figures; Jerry Lyons and Holly Hodder for their roles as our previous mathematics editors; Christine Hastings for editorial supervision; and Trumbull Rogers for his expert copyediting. For this fifth edition, we thank Matt Haigh and Wendy McKay for their help with TeX and Mathematica preparation of the material and also Terri Ward, the Mathematics Acquisitions Editor at W. H. Freeman for her excellent stewardship of the project, and Vivien Weiss for her excellent handling of production matters.

We will be maintaining an up-to-date web-based list of corrections and suggestions for the fifth edition and will be happy to receive any additional suggestions and corrections from our readers. Please send your request to either Jerrold Marsden (marsden@cds.caltech.edu) or Anthony Tromba (tromba@cats.ucsc.edu).

Historical Introduction:
A Brief Account

This, therefore, is Mathematics; *she reminds you of the invisible form of the soul; she gives life to her own discoveries; she awakens the mind and purifies the intellect; she brings light to our intrinsic ideas; she abolishes oblivion and ignorance which is ours by birth.*

Proclus, c. 450

Cum Deus Calculat Fit Mundus.
(As God calculates, so the world is created).

Leibniz, c. 1700

The word *mathematics* derives from the Greek word *mathema*, meaning knowledge, cognition, understanding, or perception, suggesting that the study of what we now call mathematics began by asking questions about the world. In fact, the historical evidence suggests that mathematics began about 2700 years ago as an attempt to comprehend nature. Unfortunately, in most mathematical expositions, historical motivations and contexts are often sacrificed. In this new edition, the authors continue to address this problem by increasing the discussion of historical and contextual material where appropriate. Therefore, before we dive into the mathematics of *Vector Calculus*, we briefly discuss the development of mathematics prior to and including the discovery of calculus.

Egyptian, Babylonian, and Greek Mathematics

It is generally acknowledged that mathematics developed in the seventh and sixth centuries B.C., somewhat after the Greeks had developed a uniform alphabet. This is not to say, however, that mathematical knowledge did not exist before the Greeks. In fact, the Egyptians and Babylonians knew many empirical facts centuries before the rise of the Greek civilization. For example, they could solve quadratic equations, compute the areas of certain geometric figures, such as squares, rectangles, and triangles, and they possessed a reasonably good formula for the area of a circle, using the value of 3.16 for π. They also knew how to compute certain volumes like the size of cubes, rectangles, rectangular solids, cones, cylinders, and (not surprisingly) pyramids. The ancients were also acquainted with the pythagorean theorem (at least empirically).

The Greeks, who settled throughout the Mediterranean, must have played an important role in preserving and spreading the mathematical knowledge of the Egyptians and the Babylonians. However, the Greeks were aware that there were different formulas for the same area or volumes. For example, the Babylonians had one formula for the volume of a frustum of a pyramid with a square base, and the Egyptians had another (see Figure 1).

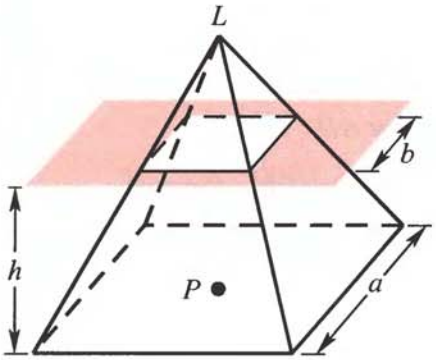


Figure 1 Volume of a frustum of a pyramid with a square base: $V = \frac{1}{3}h(a^2 + ab + b^2)$.

It is not surprising that the Egyptians (with the experience in pyramid construction) had the correct formula. Now, given two formulas, it was clear that only one could be correct. But how could one decide such an answer? Certainly it is not a question for debate, as would be the question of the quality of works of art. It is likely that the necessity to determine the answers to such questions is what led to the development of mathematical proof and to the method of deductive reasoning.

The person usually credited for the invention of rigorous mathematical proof was a merchant named Thales of Miletus (624–548 B.C.). It is Thales who is said to be the creator of Greek geometry, and it was this geometry (earth measure) as an abstract mathematical theory (rather than a collection of empirical facts) supported by rigorous deductive proofs that was one of the turning points of scientific thinking. It led to the creation of the first mathematical model for physical phenomena.

For example, one of the most beautiful geometric theories developed during antiquity was that of conic sections. See Figure 2.

Conics include the straight line, circle, ellipse, parabola, and hyperbola. Their discovery is attributed to Menaechmus, a member of the school of the great Greek philosopher Plato. Plato, a student of Socrates, founded his school *The Academy* (see Figure 3) in a sacred area of the ancient city of Athens, called Hekadameia (after the hero Hekademos). All later academies obtained their name from this institution, which existed without interruption for about 1000 years until it was dissolved by the Roman Emperor Justinian in A.D. 529.

Plato suggested the following problem to his students:

Explain the motion of the heavenly bodies by some geometrical theory.

Why was this a question of interest and puzzlement for the Greeks? Observed from the Earth, these motions appear to be quite complicated. The motions of the sun and the moon can be roughly described as circular with constant speed, but the deviations from the circular orbit were troublesome to the Greeks and they felt challenged to find an explanation for these irregularities. The observed orbits of the

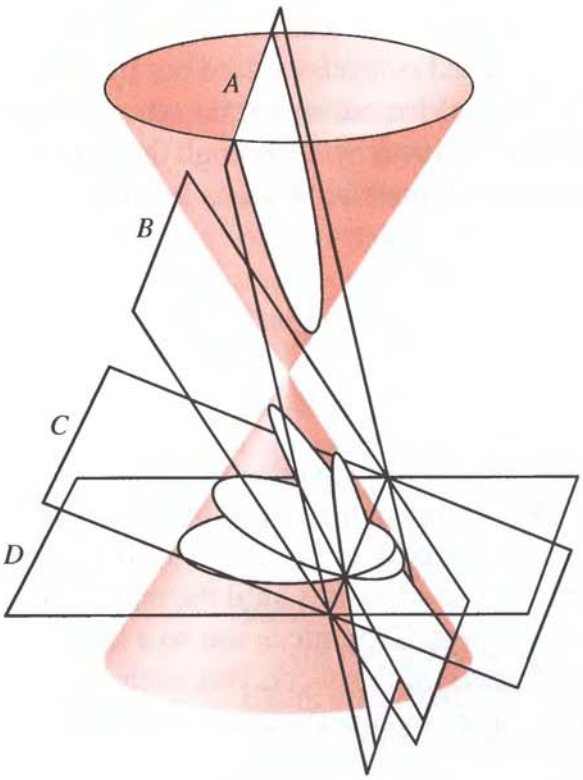


Figure 2 The conic sections: (A) hyperbola, (B) parabola, (C) ellipse, (D) circle.

planets are even more complicated, because as they go through a revolution, they appear to reverse direction several times.

The Greeks sought to understand this apparently wild motion by means of their geometry. Eudoxus, Hipparchus, and then Apollonius of Perga (262–190 B.C.)

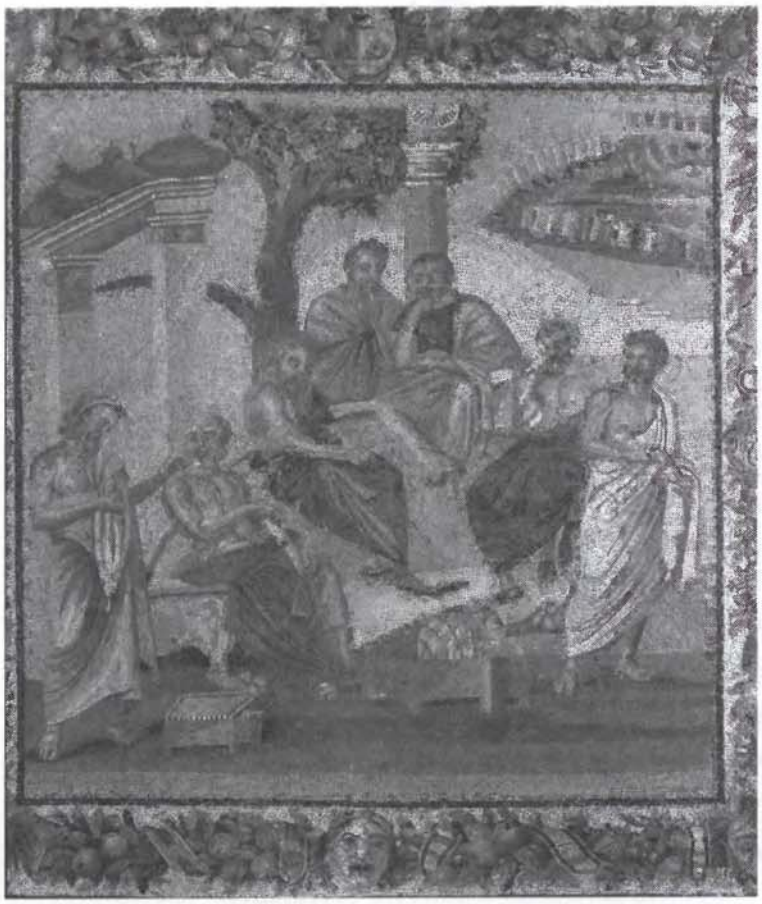


Figure 3 *Plato's Academy* (mosaic found in Pompeii, Villa of T. Siminius Stephanus, 86 × 85 cm, Naples, Archaeological Museum). With certainty the seven men have been identified as Plato (third from the left) and six other philosophers, who are talking about the universe, the celestial spheres, and the stars. The mosaic shows Plato's Academy, with the city of Athens in the background. It is probably a copy (from the first century B.C.) of a Hellenistic painting.

suggested that the celestial orbits could be explained by combinations of circular motion (that is, through the construction of curves called epicycles traced out by circles moving on other circles). This idea was to become the most important astronomical theory of the next two thousand years. This theory, known by us through the writings of the Greek astronomer Ptolemy of Alexandria, ultimately becomes known as the "Ptolemaic theory." See Figures 4 and 5.

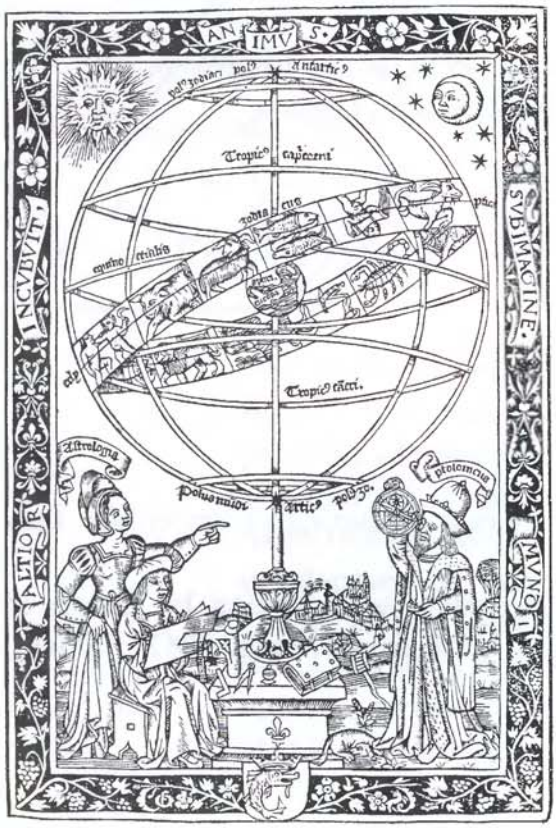


Figure 4 Woodcut from Georg von Peurbach's *Theoricae novae planetarum,* edited by Oronce Fine as a teaching text for the University of Paris (1515). It was the canonical description of the heavens until the end of the sixteenth century, and even Copernicus was to a large extent under the influence of this work. Peurbach described the solid sphere representations of Ptolemaic planetary models, which he probably based on Ibn al-Haytham's work "On the configuration of the world" (translated into Latin in the thirteenth century). The same frontispiece was used for the Sacrosbosco edition of the first four books of Euclid's *Elements* (in excerpts), which appeared under the title *Textus de Sphaera* in Paris (1521).

Figure 5 Ptolemy observing the stars with a quadrant, together with an allegoric Astronomia. (From Gregorius Reish, *Margarita Philosophica nova*, Strasbourg, 1512, an early compendium of philosophy and science.) In those days, Ptolemy was often depicted as a king, because he was erroneously thought to be descended from the Ptolemaic dynasty that ruled Egypt after Alexander.

Most of Greek geometry was codified by Euclid in his *Elements* (of Mathematics). Actually the *Elements* consist of thirteen books, in which Euclid collected most of the mathematical knowledge of his age (circa 300 B.C.), transforming it into a lucid, logically developed masterpiece. In addition to the *Elements*, some of Euclid's other writings were also handed down to us, including his *Optics* and the *Catoptrica* (theory of mirrors).

The success of Greek mathematics had a profound effect on views of nature. The Platonists, or followers of Plato, distinguished between the world of ideas and the world of physical objects. Plato was the first to propose that ultimate truth or understanding could not come from the material world, which is constantly subject to change, but only from mathematical models or constructs. Thus, infallible knowledge could be attained only through mathematics. Plato not only wished to use mathematics in the study of nature, but he actually went so far as to attempt to substitute mathematics *for* nature. For Plato, reality lies only within the realm of ideas, especially mathematical ideas.

Not everyone in antiquity agreed with this point of view. Aristotle, a student of Plato, criticized Plato's reduction of science to the study of mathematics. Aristotle thought that the study of the material world was one's primary source of reality. Despite Aristotle's critique, the view that mathematical laws governed the universe took a firm hold on classical thought. *The search for the mathematical laws of nature was underway.*

After the death of Archimedes in 212 B.C., Greek civilization went into a period of slow decline. The final blow to Greek civilization came in 640 A.D. with the Moslem conquest of Egypt. The remaining Greek texts housed in the great library in Alexandria were burned. Those scholars who survived migrated to Constantinople (now part of Turkey), which had become the capitol of the Eastern Roman Empire. It was in this great city that what survived of Greek civilization was preserved for its rediscovery by European civilization some five hundred years later.

Indian and Arabian Mathematics

Mathematical activity did not, however, cease with the decline of Greek civilization. In the middle of the sixth century, somewhere in the Ganges Valley in India, our modern system of numeration evolved. The Indians developed a number system based on ten, with ten rather abstract symbols from zero to nine looking "roughly" as they do today. They developed rules for addition, multiplication, and division (as we have today), a system infinitely superior to the Roman abacus, which was used (by a special class of servants called *arithmeticians*) throughout Europe until the fifteenth century. See Figure 6.

After the fall of Egypt, came the rise of Arab civilization centered in Baghdad. Scholars from Constantinople and India were invited to study and to share their knowledge. It was through these contacts that the Arabs came to acquire the learning of the ancients as well as the newly discovered Indian system of numeration. See Figure 7.

It was the Arabs who gave us the name *Algebra*, which comes from the book by the astronomer Mohammed ibn Musa al-Khuwarizmi titled "Al-Jabr w'al muqabola,"

Figure 6 Arithmetician performing a calculation on a counter-abacus.

which means "restoring" or "balancing" (equations). Al-Khuwarizmi is also responsible for a second profoundly influential book entitled "Kitab al jami' wa'l tafriq bi hisab al hind" (Indian Technique of Addition and Subtraction), which described and clarified the Indian decimal place value system.

Al-Khuwarizmi also gave us another name for a fundamental branch of science, the word *algorithm*. Latinized, his name became Algorism, then Algorismus, and finally Algorithm. The term initially represented the Indian system of numeration, but ultimately came to be used in its modern computational sense.

The decline of Arab civilization coincided with the rise of European civilization. The dawn of the modern age began when Richard the Lionhearted reached the walls of Jerusalem. From approximately 1192 through around 1270, the Christian knights brought the learning of the "infidels" back to Europe. Around 1200–1205, Leonardo of Pisa (also known as Fibonacci), who had traveled extensively in Africa and Asia Minor, wrote his interpretation (in Latin) of Arabic and Greek mathematics. His

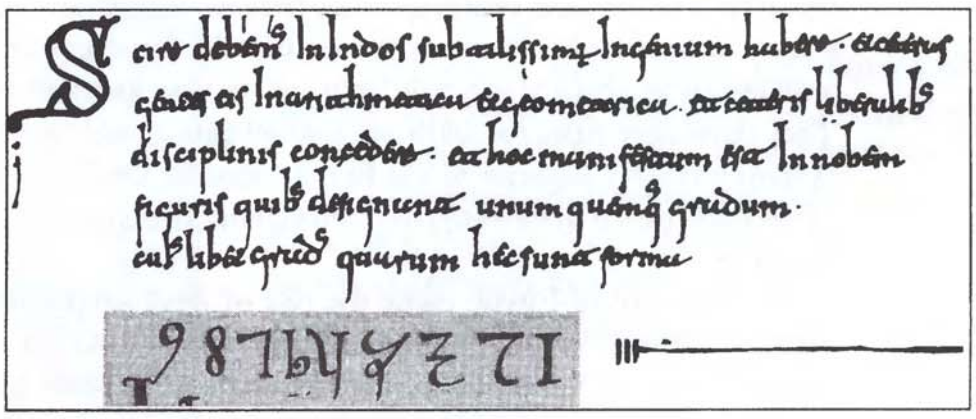


Figure 7 Detail from the *Codex Vigilanus* (976 A.D. northern Spain). The first known occurrence of the nine Indo-Arabic numerals in Western Europe. (Escurial Library, Madrid.)

historic texts brought the work of al-Khuwarizmi and Euclid to the attention of a large audience in Europe.

European Mathematics

Around 1450 Johann Gutenberg invented the printing press with movable type. This, combined with the advent of linen and cotton paper obtained from the Chinese, dramatically increased the rate of the dissemination of knowledge. The steep rise in trade and manufacturing fueled the growth of wealth and dramatic change in European societies from feudal to city-states. In Italy, the mother of the Renaissance, we see the rise of extraordinarily wealthy states such as Venice under the Doges and Florence under the Medicis.

The needs of the rising merchant class accelerated the adoption of the Indian system of numeration. The teachings of the Catholic Church, which rested on absolute authority and dogma, began to be challenged by the ideas of Plato. From Plato, scholars learned that *the world was rational and could be understood*, and that the means of understanding nature was through mathematics. But this sharply contradicted the teachings of the church, which taught that God designed the universe. The only possible resolution of this apparent contradiction was that "God designed the universe mathematically" or that "God is a mathematician."

It is perhaps surprising how much this point of view inspired the work of many sixteenth- to eighteenth-century mathematicians and scientists. For if this were indeed the case, then by understanding the mathematical laws of the universe, one could come closer to an understanding of the Creator himself. Believe it or not, this point of view survives to this day. The following is a quote from Paul Dirac, a Nobel Prize–winning physicist and a creator of modern quantum mechanics.

> *It seems to be one of the fundamental features of nature that fundamental physical laws are described in terms of a mathematical theory of great beauty and power, needing quite a high standard of mathematics for one to understand it. You may wonder: Why is nature constructed along these lines? One can only answer that our present knowledge seems to show that nature is so constructed. We simply have to accept it. One could perhaps describe the situation by saying that God is a mathematician of a very high order, and He used very advanced mathematics in constructing the universe. Our feeble attempts at mathematics enable us to understand a bit of the universe, and as we proceed to develop higher and higher mathematics we can hope to understand the universe better.*

Mathematics began to see further advances and applications. In the sixteenth and seventeenth centuries, al-Khuwarizmi's algebra was significantly advanced by Cardano, Vieta, and Descartes. The Babylonians had solved the quadratic equation, but now two thousand years later, del Ferro and Tartaglia solved the cubic equation, which in turn led to the discovery of imaginary numbers. These imaginary numbers were later to play a fundamental role, as we shall see, in the development of vector calculus. In the early seventeenth century, Descartes, perhaps motivated by the grid technique used by Italian fresco painters to locate points on a wall or canvas, created, in a moment of great mathematical inspiration, coordinate (or analytic) geometry.

This new mathematical model enables one to reduce Euclid's geometry to algebra and provides a precise and quantitative method to describe and calculate with space curves and surfaces.

Early on, Archimedes' great work in statics and equilibrium (centers of gravity, the principle of the lever—which we study in this book) was absorbed and improved upon, leading to dramatic engineering achievements. In a building spree that remains astonishing to this day, engineering advances made possible the rise of an incredible number of cathedrals throughout Europe, including the stunning Duomo in Florence, Notre Dame in Paris, and the Great Cathedral in Cologne, to mention a few. See Figure 8.

Figure 8 Duomo.

Filippo Brunelleschi (1377–1446) studied the works of Euclid and Hipparchus and was the first artist to employ mathematics extensively. The mathematical principles of perspective were eventually completed by Piero della Francesca (1410–1492). Mathematicians and engineers were recruited by warring princes to fuel the development of advanced weapons and ballistic science. The most famous among these was none other than Leonardo da Vinci, who in the last years of his life was employed by the Duke of Milan. It was in these final years that he painted the "Mona Lisa," now housed in the Louvre in Paris. See Figure 9.

However, as in Greek times, it was astronomy that was to give mathematics its greatest impetus. It is not surprising that the Greek astronomers placed the Earth and not the sun at the center of our universe, because on a daily basis we see the sun both rise and set. Still, it is interesting to ask if the Greeks, who were such marvelous thinkers, at least tested the heliocentric theory, which places the sun at the center of the

Figure 9 Leonardo, self portrait.

universe. In fact, they did. In the third century B.C., Aristarchus of Samus taught that the Earth and other planets move in circular orbits around a fixed sun. His hypotheses were, for several reasons, rejected. First, the opposing astronomers reasoned that if the Earth were indeed moving, one should be able to sense it. Second, how would objects, circulating with us, be able to stay on a moving Earth? Third, why are the clouds not lagging behind the moving Earth?

Such arguments were to be used again in the sixteenth century against the Polish astronomer Nicolas Copernicus (see Figure 10), who in 1543 introduced the heliocentric theory (the planets move in orbit around the sun). His book *Revolutionibus Orbium Coelestium* (*On the Revolution of the Heavenly Orbits*) was to initiate the "Copernican revolution" in science and to give the world a new word, *revolutionary*.

In 1619, the German astronomer Johannes Kepler (see Figure 11), using the astronomical calculations of the Danish astronomer Tycho Brahe, showed that the planetary

Figure 10 Nicolaus Copernicus (1473–1543).

orbits were in fact elliptical, the same ellipses that the Greeks had studied as abstract forms some 2000 years earlier (see Figure 12).

But Kepler's law of elliptical orbits was only one of three laws he discovered governing planetary motion. Kepler's second law states that if a planet moves from a point A to another point B in a certain amount of time T, and also moves from A' to B' in the same time, and if S is a focus of the orbital ellipse, then the sections SAB are $SA'B'$ have equal areas (see Figure 13). Kepler's third law was that the square of time T a planetary body requires to complete an orbit is proportional to a^3, where a is the great axis of the elliptical orbit. In equation form, $T^2 = Ka^3$, where K is some constant (we shall derive this law for circular orbits in Chapter 4).

Figure 11 Johannes Kepler (1571–1630).

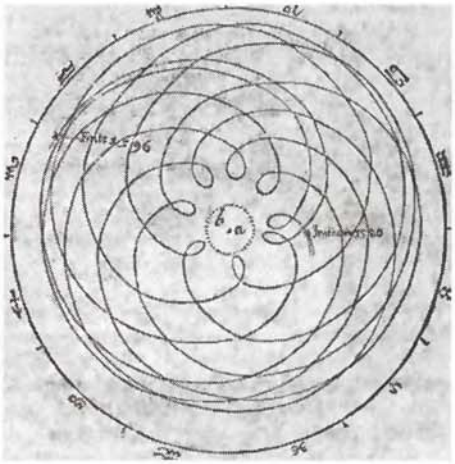


Figure 12 The motion of Mars. From Kepler's *Astronomia Nova* (1609).

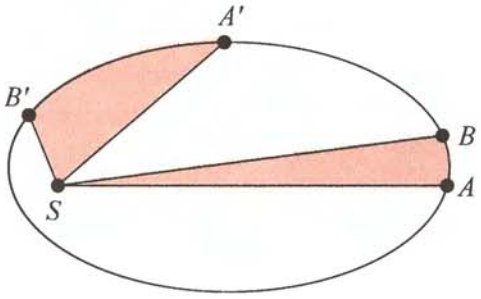


Figure 13 Kepler's second law.

Profound as these observations were, an explanation of why these laws held was lacking. However, by the middle of the seventeenth century, it was fully understood that a change of velocity requires the action of forces, but how these forces influenced motion was not at all clear. In 1674 Robert Hooke, in an attempt to explain Kepler's laws,

assumed the existence of an attractive force the sun must exert on the planets, a force that decreased with planetary distance. Hooke's theory, however, was only qualitative.

Newton

What was also seriously lacking was a quantitative, precise definition of both velocity and acceleration. This was ultimately solved by the invention of calculus by both Isaac Newton and Gottfried Wilhelm Leibniz (see Figure 14). Hooke was never able to achieve an understanding of the profound ideas behind the infinitesimal calculus. However, during the period of 1679–1680 Hooke discussed his ideas with Newton, including his conjecture that the force the sun exerts on the planets was actually inversely proportional to the square of the planetary distance.

Figure 14 Gottfried Wilhelm Leibniz (1646–1716).

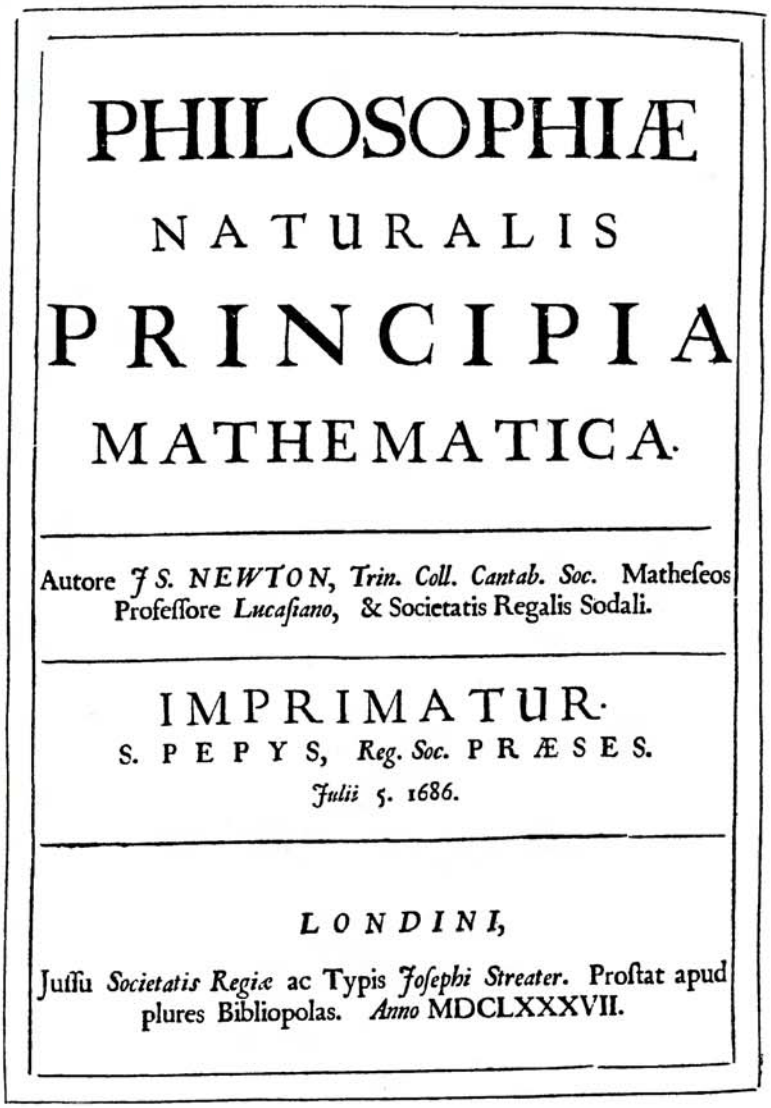
After Sir Christopher Wren, amateur astronomer, architect of the city of London and London's magnificent St. Paul's Cathedral, issued a public challenge to "theoretically determine" the orbits of the planets, Isaac Newton took a serious interest in the problem. Perhaps acting on rumors, the great British astronomer Edmund Halley (1656–1743) in August 1684 visited Newton in Cambridge and asked him directly what the orbit of a planet would be under an inverse square force. Newton answered that it had to be an ellipse. As the stunned Halley asked him how he knew this, Newton's famous reply was "*Why I have calculated it.*" Halley ultimately urged Newton to publish his results as a book, and these appeared in 1686 in Newton's now legendary *Principia*. See Figure 15.

This book, often and justly referred to as the foundation of modern science, had an immediate dramatic impact. Alexander Pope wrote:

Nature and nature's laws lay hid at night,
God said, "Let Newton be" and all was light.

On the front cover of this text, we see Newton holding open a copy of his *Principia*.

Although Newton did not use calculus in the *Principia*, convincing arguments have been put forward that Newton originally used his calculus to derive the



PHILOSOPHIÆ
NATURALIS
PRINCIPIA
MATHEMATICA.

Autore *J S. NEWTON*, *Trin. Coll. Cantab. Soc.* Matheſeos Profeſſore *Lucaſiano*, & Societatis Regalis Sodali.

IMPRIMATUR.
S. PEPYS, *Reg. Soc.* PRÆSES.
Julii 5. 1686.

LONDINI,

Juſſu *Societatis Regiæ* ac Typis *Joſephi Streater*. Proſtat apud plures Bibliopolas. *Anno* MDCLXXXVII.

Figure 15 The frontispiece of the two-lines print of the *Principia*, carrying the imprint "Prostat apud plures Bibliopolas," which is sometimes called the "first issue" of the first edition. The "export copy" (with the three lines "Prostant Venales apud Sam Smith... aliosq; nonnullos Bibliopolas") is called the second issue of the first edition. This distinction between the first and second issues seems to be quite unfounded. It has been suggested that Halley made an agreement with Smith concerning foreign sales; in fact, most of Smith's fifty copies were apparently sold on the continent.

trajectories of the planetary orbits from the inverse square law.* The *Principia* provided profound evidence that the universe, as the early Greeks had understood, was indeed designed mathematically. Incidentally, it was Newton who first conceptualized force as a *vector*, although he provided no formal definition of what a vector was. Such a formal definition had to wait for William Rowan Hamilton, a century and a half after the *Principia*. It was for this achievement and his creation of calculus itself that we chose Newton for our cover.

The invention of the calculus and the subsequent development of vector calculus was the true beginning of modern science and technology, which has changed our world so dramatically. From the mathematics of Newton's mechanics to the profound intellectual constructs of Maxwell's electrodynamics, Einstein's relativity, and Heisenberg's and Schrödinger's quantum mechanics, we have seen the discoveries of radio, television, wireless communications, flight, computers, space travel, and countless engineering marvels.

Underlying all these developments was mathematics, an exciting adventure of the mind and a celebration of the human spirit. It is in this context that we begin our account of Vector calculus.

*We shall study the problem of planetary orbits in Section 4.1 and further in the Internet supplement.

Prerequisites and Notation

We assume that students have studied the calculus of functions of a real variable, including analytic geometry in the plane. Some students may have had some exposure to matrices as well, although what we shall need is given in Sections 1.3 and 1.5.

We also assume that students are familiar with functions of elementary calculus, such as $\sin x$, $\cos x$, e^x, and $\log x$ (we write $\log x$ or $\ln x$ for the natural logarithm, which is sometimes denoted $\log_e x$). Students are expected to know, or to review as the course proceeds, the basic rules of differentiation and integration for functions of one variable, such as the chain rule, the quotient rule, integration by parts, and so forth.

We now summarize the notations to be used later. Students can read through these quickly now, then refer to them later if the need arises.

The collection of all real numbers is denoted $\mathbb{R}$. Thus $\mathbb{R}$ includes the ***integers***, $\ldots$, $-3, -2, -1, 0, 1, 2, 3, \ldots$; the ***rational numbers***, p/q, where p and q are integers ($q \neq 0$); and the ***irrational numbers***, such as $\sqrt{2}$, π, and e. Members of $\mathbb{R}$ may be visualized as points on the real-number line, as shown in Figure P.1.

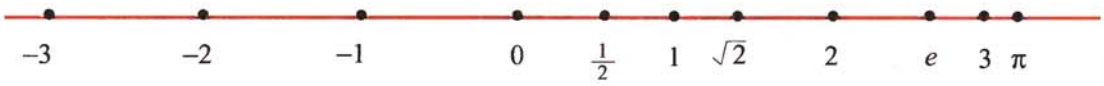


Figure P.1 The geometric representation of points on the real-number line.

When we write $a \in \mathbb{R}$ we mean that a is a member of the set $\mathbb{R}$, in other words, that a is a real number. Given two real numbers a and b with $a < b$ (that is, with a less than b), we can form the ***closed interval*** $[a, b]$, consisting of all x such that $a \leq x \leq b$, and the ***open interval*** (a, b), consisting of all x such that $a < x < b$. Similarly, we can form half-open intervals $(a, b]$ and $[a, b)$ (Figure P.2).

Figure P.2 The geometric representation of the intervals $[a, b]$, (c, d), and $[e, f)$.

The ***absolute value*** of a number $a \in \mathbb{R}$ is written $|a|$ and is defined as

$$|a| = \begin{cases} a & \text{if } a \geq 0 \\ -a & \text{if } a < 0. \end{cases}$$

For example, $|3| = 3$, $|-3| = 3$, $|0| = 0$, and $|-6| = 6$. The inequality $|a + b| \leq |a| + |b|$ always holds. The ***distance from*** a ***to*** b is given by $|a - b|$. Thus, the distance from 6 to 10 is 4 and from -6 to 3 is 9.

If we write $A \subset \mathbb{R}$, we mean A is a ***subset*** of $\mathbb{R}$. For example, A could equal the set of integers $\{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\}$. Another example of a subset of $\mathbb{R}$ is the set $\mathbb{Q}$ of rational numbers. Generally, for two collections of objects (that is, sets) A and B, $A \subset B$ means A is a subset of B; that is, every member of A is also a member of B.

The symbol $A \cup B$ means the ***union*** of A and B, the collection whose members are members of either A or B (or both). Thus,

$$\{\ldots, -3, -2, -1, 0\} \cup \{-1, 0, 1, 2, \ldots\} = \{\ldots, -3, -2, -1, 0, 1, 2, \ldots\}.$$

Similarly, $A \cap B$ means the ***intersection*** of A and B; that is, this set consists of those members of A and B that are in both A and B. Thus, the intersection of the two sets above is $\{-1, 0\}$.

We shall write $A \backslash B$ for those members of A that are not in B. Thus,

$$\{\ldots, -3, -2, -1, 0\} \backslash \{-1, 0, 1, 2, \ldots\} = \{\ldots, -3, -2\}.$$

We can also specify sets as in the following examples:

$$\{a \in \mathbb{R} \mid a \text{ is an integer}\} = \{\ldots, -3, -2, -1, 0, 1, 2, \ldots\}$$
$$\{a \in \mathbb{R} \mid a \text{ is an even integer}\} = \{\ldots, -2, 0, 2, 4, \ldots\}$$
$$\{x \in \mathbb{R} \mid a \leq x \leq b\} = [a, b].$$

A ***function*** $f\colon A \to B$ is a rule that assigns to each $a \in A$ one specific member $f(a)$ of B. We call A the ***domain*** of f and B the ***target*** of f. The set $\{f(x) \mid x \in A\}$ consisting of all the values of $f(x)$ is called the ***range*** of f. Denoted by $f(A)$, the range is a subset of the target B. It may be all of B, in which case f is said to be ***onto*** B. The fact that the function f sends a to $f(a)$ is denoted by $a \mapsto f(a)$. For example, the function $f(x) = x^3/(1 - x)$ that assigns the number $x^3/(1 - x)$ to each $x \neq 1$ in $\mathbb{R}$ can also be defined by the rule $x \mapsto x^3/(1 - x)$. Functions are also called ***mappings***, ***maps***, or ***transformations***. The notation $f\colon A \subset \mathbb{R} \to \mathbb{R}$ means that A is a subset of $\mathbb{R}$ and that f assigns a value $f(x)$ in $\mathbb{R}$ to each $x \in A$. The ***graph*** of f consists of all the points $(x, f(x))$ in the plane (Figure P.3).

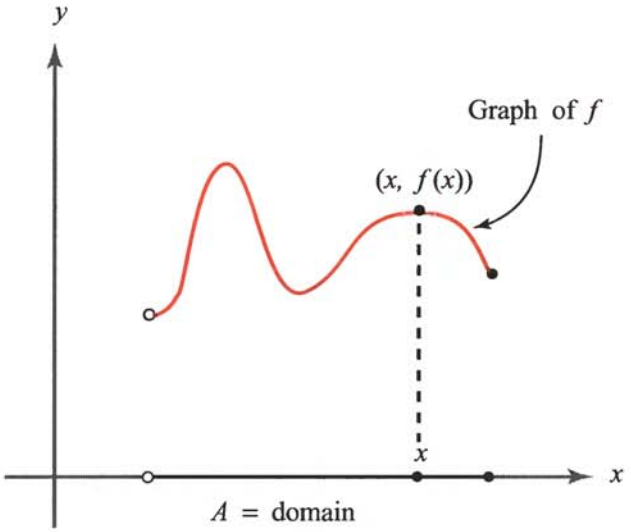


Figure P.3 The graph of a function with the half-open interval A as domain.

The notation $\sum_{i=1}^{n} a_i$ means $a_1 + \cdots + a_n$, where $a_1, \ldots, a_n$ are given numbers. The sum of the first n integers is

$$1 + 2 + \cdots + n = \sum_{i=1}^{n} i = \frac{n(n+1)}{2}.$$

The ***derivative*** of a function $f(x)$ is denoted $f'(x)$, or

$$\frac{df}{dx},$$

and the ***definite integral*** is written

$$\int_a^b f(x)\,dx.$$

If we set $y = f(x)$, the derivative is also denoted by

$$\frac{dy}{dx}.$$

Readers are assumed to be familiar with the chain rule, integration by parts, and other basic facts from the calculus of functions of one variable. In particular, they should know how to differentiate and integrate exponential, logarithmic, and trigonometric functions. Short tables of derivatives and integrals, which are adequate for the needs of this text, are printed at the front and back of the book.

The following notations are used synonymously: $e^x = \exp x$, $\ln x = \log x$, and $\sin^{-1} x = \arcsin x$.

The end of a proof is denoted by the symbol ■, while the end of an example or remark is denoted by the symbol ▲.

1

The Geometry of Euclidean Space

Quaternions came from Hamilton . . . and have been an unmixed evil to those who have touched them in any way. Vector is a useless survival . . . and has never been of the slightest use to any creature.

Lord Kelvin

In this chapter we consider the basic operations on vectors in two- and three-dimensional space: vector addition, scalar multiplication, and the dot and cross products. In Section 1.5 we generalize some of these notions to n-space and review properties of matrices that will be needed in Chapters 2 and 3.

1.1 Vectors in Two- and Three-Dimensional Space

Points P in the plane are represented by ordered pairs of real numbers (a_1, a_2); the numbers a_1 and a_2 are called the ***Cartesian coordinates of*** P. We draw two perpendicular lines, label them as the x and y axes, and then drop perpendiculars from P to these axes, as in Figure 1.1.1. After designating the intersection of the x and y axes as the origin and choosing units on these axes, we produce two signed distances a_1 and a_2 as shown in the figure; a_1 is called the x ***component*** of P, and a_2 is called the y ***component***.

Points in space may be similarly represented as ordered triples of real numbers. To construct such a representation, we choose three mutually perpendicular lines that meet at a point in space. These lines are called x ***axis***, y ***axis***, and z ***axis***, and the point at which they meet is called the ***origin*** (this is our reference point). We choose a scale on these axes, as shown in Figure 1.1.2.

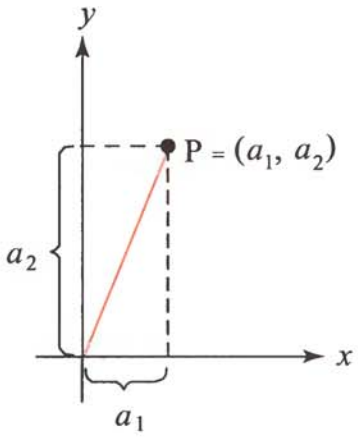


Figure 1.1.1 Cartesian coordinates in the plane.

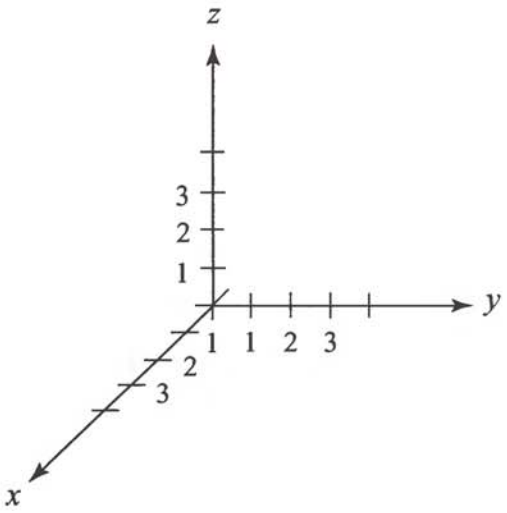


Figure 1.1.2 Cartesian coordinates in space.

The triple (0, 0, 0) corresponds to the origin of the coordinate system, and the arrows on the axes indicate the positive directions. For example, the triple (2, 4, 4) represents a point 2 units from the origin in the positive direction along the x axis, 4 units in the positive direction along the y axis, and 4 units in the positive direction along the z axis (Figure 1.1.3).

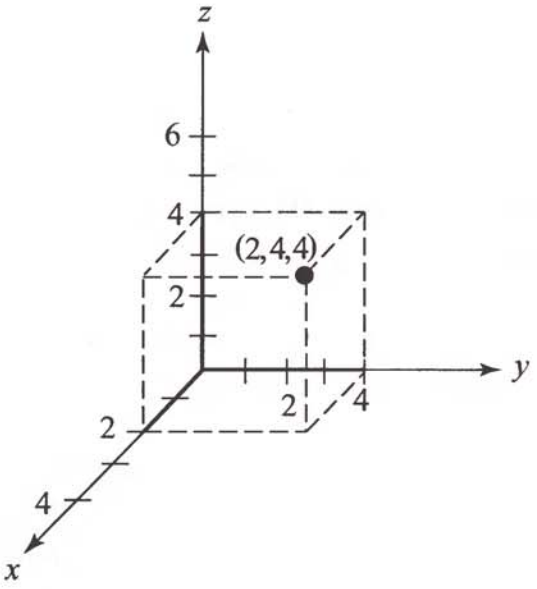


Figure 1.1.3 Geometric representation of the point (2, 4, 4) in Cartesian coordinates.

Because we can associate points in space with ordered triples in this way, we often use the expression "the point (a_1, a_2, a_3)" instead of the longer phrase "the point P

that corresponds to the triple (a_1, a_2, a_3)." We say that a_1 is the x ***coordinate*** (or first coordinate), a_2 is the y ***coordinate*** (or second coordinate), and a_3 is the z ***coordinate*** (or third coordinate) of P. It is also common to denote points in space with the letters x, y, and z in place of a_1, a_2, and a_3. Thus, the triple (x, y, z) represents a point whose first coordinate is x, second coordinate is y, and third coordinate is z.

We employ the following notation for the line, the plane, and three-dimensional space:

(i) The real number line is denoted $\mathbb{R}^1$ or simply $\mathbb{R}$.

(ii) The set of all ordered pairs (x, y) of real numbers is denoted $\mathbb{R}^2$.

(iii) The set of all ordered triples (x, y, z) of real numbers is denoted $\mathbb{R}^3$.

When speaking of $\mathbb{R}^1$, $\mathbb{R}^2$, and $\mathbb{R}^3$ simultaneously, we write $\mathbb{R}^n$, where $n = 1, 2,$ or 3; or $\mathbb{R}^m$, where $m = 1, 2, 3$. Starting in Section 1.5 we will also study $\mathbb{R}^n$ for $n = 4, 5, 6, \ldots$, but the cases $n = 1, 2, 3$ are closest to our geometric intuition and will be stressed throughout the book.

Vector Addition and Scalar Multiplication

The operation of addition can be extended from $\mathbb{R}$ to $\mathbb{R}^2$ and $\mathbb{R}^3$. For $\mathbb{R}^3$, this is done as follows. Given the two triples (a_1, a_2, a_3) and (b_1, b_2, b_3), we define their ***sum*** to be

$$(a_1, a_2, a_3) + (b_1, b_2, b_3) = (a_1 + b_1, a_2 + b_2, a_3 + b_3).$$

EXAMPLE 1

$$(1, 1, 1) + (2, -3, 4) = (3, -2, 5),$$
$$(x, y, z) + (0, 0, 0) = (x, y, z),$$
$$(1, 7, 3) + (a, b, c) = (1 + a, 7 + b, 3 + c). \quad \blacktriangle$$

The element $(0, 0, 0)$ is called the ***zero element*** (or just ***zero***) of $\mathbb{R}^3$. The element $(-a_1, -a_2, -a_3)$ is the ***additive inverse*** (or ***negative***) of (a_1, a_2, a_3), and we will write $(a_1, a_2, a_3) - (b_1, b_2, b_3)$ for $(a_1, a_2, a_3) + (-b_1, -b_2, -b_3)$.

The additive inverse, when added to the vector itself, of course produces zero:

$$(a_1, a_2, a_3) + (-a_1, -a_2, -a_3) = (0, 0, 0).$$

There are several important product operations that we will define on $\mathbb{R}^3$. One of these, called the *inner product*, assigns a real number to each pair of elements of $\mathbb{R}^3$. We shall discuss it in detail in Section 1.2. Another product operation for $\mathbb{R}^3$ is called *scalar multiplication* (the word "scalar" is a synonym for "real number"). This product combines scalars (real numbers) and elements of $\mathbb{R}^3$ (ordered triples) to yield elements of $\mathbb{R}^3$ as follows: Given a scalar α and a triple (a_1, a_2, a_3), we define the ***scalar multiple*** by

$$\alpha(a_1, a_2, a_3) = (\alpha a_1, \alpha a_2, \alpha a_3).$$

EXAMPLE 2

$$2(4, e, 1) = (2 \cdot 4, 2 \cdot e, 2 \cdot 1) = (8, 2e, 2),$$
$$6(1, 1, 1) = (6, 6, 6),$$
$$1(u, v, w) = (u, v, w),$$
$$0(p, q, r) = (0, 0, 0). \quad \blacktriangle$$

Addition and scalar multiplication of triples satisfy the following properties:

(i)	$(\alpha\beta)(a_1, a_2, a_3) = \alpha[\beta(a_1, a_2, a_3)]$	(associativity)
(ii)	$(\alpha + \beta)(a_1, a_2, a_3) = \alpha(a_1, a_2, a_3) + \beta(a_1, a_2, a_3)$	(distributivity)
(iii)	$\alpha[(a_1, a_2, a_3) + (b_1, b_2, b_3)] = \alpha(a_1, a_2, a_3) + \alpha(b_1, b_2, b_3)$	(distributivity)
(iv)	$\alpha(0, 0, 0) = (0, 0, 0)$	(property of zero)
(v)	$0(a_1, a_2, a_3) = (0, 0, 0)$	(property of zero)
(vi)	$1(a_1, a_2, a_3) = (a_1, a_2, a_3)$	(property of the unit element)

The identities are proven directly from the definitions of addition and scalar multiplication. For instance,

$$\begin{aligned}(\alpha + \beta)(a_1, a_2, a_3) &= ((\alpha + \beta)a_1, (\alpha + \beta)a_2, (\alpha + \beta)a_3) \\ &= (\alpha a_1 + \beta a_1, \alpha a_2 + \beta a_2, \alpha a_3 + \beta a_3) \\ &= \alpha(a_1, a_2, a_3) + \beta(a_1, a_2, a_3).\end{aligned}$$

For $\mathbb{R}^2$, addition and scalar multiplication are defined just as in $\mathbb{R}^3$, with the third component of each vector dropped off. All the properties (i) to (vi) still hold.

EXAMPLE 3 Interpret the chemical equation $2NH_2 + H_2 = 2NH_3$ as a relation in the algebra of ordered pairs.

SOLUTION We think of the molecule N_xH_y (x atoms of nitrogen, y atoms of hydrogen) as represented by the ordered pair (x, y). Then the chemical equation given is equivalent to $2(1, 2) + (0, 2) = 2(1, 3)$. Indeed, both sides are equal to $(2, 6)$. $\blacktriangle$

Geometry of Vector Operations

Let us turn to the geometry of these operations in $\mathbb{R}^2$ and $\mathbb{R}^3$. For the moment, we define a ***vector*** to be a directed line segment beginning at the origin, that is, a line segment with specified magnitude and direction, and initial point at the origin. Figure 1.1.4 shows several vectors, drawn as arrows beginning at the origin. In print, vectors are

usually denoted by boldface letters such as **a**. By hand, we usually write them as $\vec{a}$ or simply as a, possibly with a line or wavy line under it.

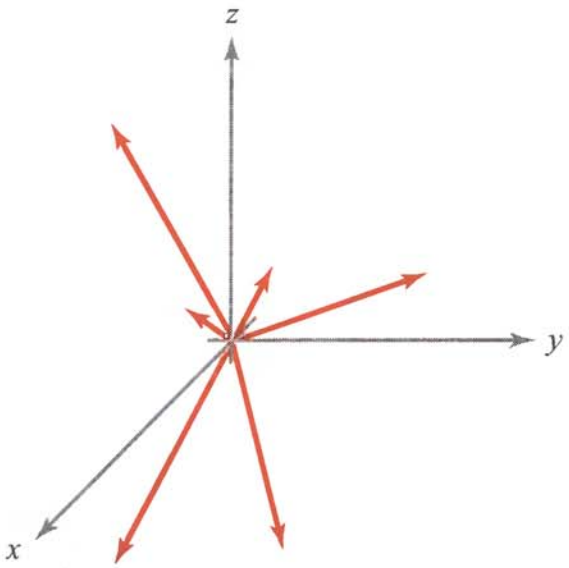


Figure 1.1.4 Geometrically, vectors are thought of as arrows emanating from the origin.

Using this definition of a vector, we associate with each vector **a** the point (a_1, a_2, a_3) where **a** terminates, and conversely, we can associate a vector **a** with each point (a_1, a_2, a_3) in space. Thus, we shall identify **a** with (a_1, a_2, a_3) and write $\mathbf{a} = (a_1, a_2, a_3)$. For this reason, the elements of $\mathbb{R}^3$ not only are ordered triples of real numbers, but are also regarded as vectors. The triple (0, 0, 0) is denoted **0**. We call a_1, a_2, and a_3 the ***components*** of **a**, or when we think of **a** as a point, its ***coordinates***.

Two vectors $\mathbf{a} = (a_1, a_2, a_3)$ and $\mathbf{b} = (b_1, b_2, b_3)$ are equal if and only if $a_1 = b_1$, $a_2 = b_2$, and $a_3 = b_3$. Geometrically this means that **a** and **b** have the same direction and the same length (or "magnitude").

Geometrically, we define vector addition as follows. In the plane containing the vectors $\mathbf{a} = (a_1, a_2, a_3)$ and $\mathbf{b} = (b_1, b_2, b_3)$ (see Figure 1.1.5), form the parallelogram having **a** as one side and **b** as its adjacent side. The sum $\mathbf{a} + \mathbf{b}$ is the directed line segment along the diagonal of the parallelogram.

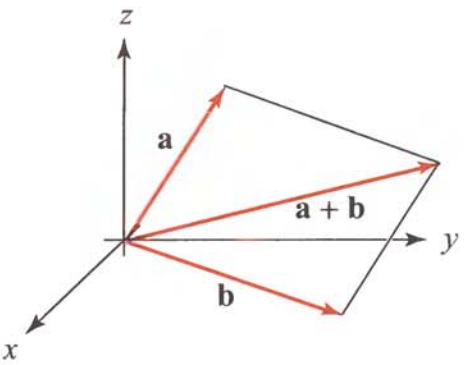


Figure 1.1.5 The geometry of vector addition.

This geometric view of vector addition is useful in many physical situations, as we shall see in the next section. For an easily visualized example, consider a bird or an airplane flying through the air with velocity $\mathbf{v}_1$, but in the presence of a wind with velocity $\mathbf{v}_2$. The resultant velocity, $\mathbf{v}_1 + \mathbf{v}_2$, is what one sees; see Figure 1.1.6.

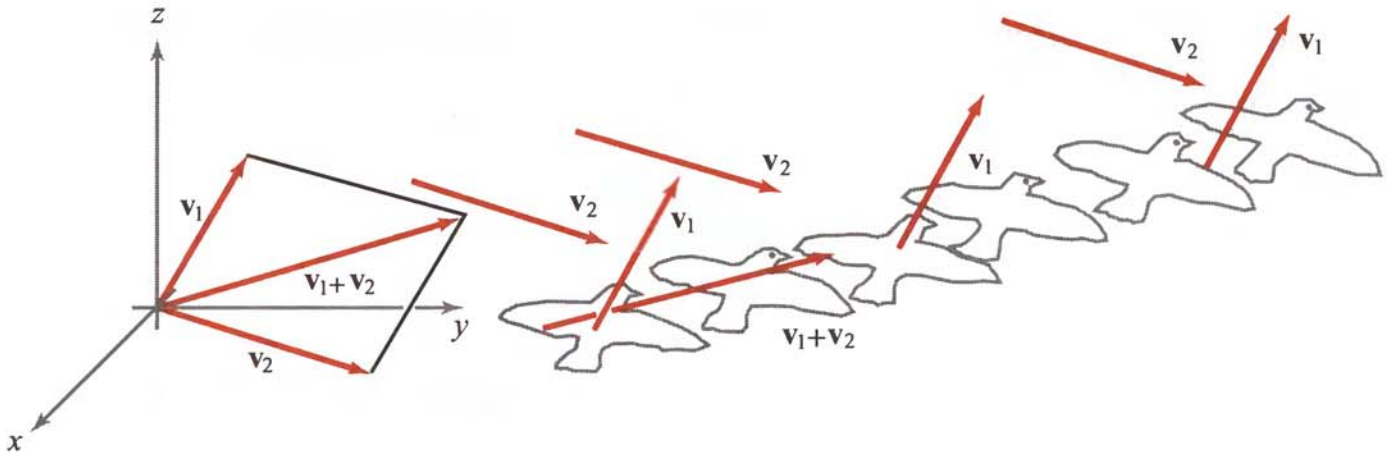


Figure 1.1.6 A physical interpretation of vector addition.

To show that our geometric definition of addition is consistent with our algebraic definition, we demonstrate that $\mathbf{a} + \mathbf{b} = (a_1 + b_1, a_2 + b_2, a_3 + b_3)$. We shall prove this result in the plane and leave the proof in three-dimensional space to the reader. Thus, we wish to show that if $\mathbf{a} = (a_1, a_2)$ and $\mathbf{b} = (b_1, b_2)$, then $\mathbf{a} + \mathbf{b} = (a_1 + b_1, a_2 + b_2)$.

In Figure 1.1.7 let $\mathbf{a} = (a_1, a_2)$ be the vector ending at the point A, and let $\mathbf{b} = (b_1, b_2)$ be the vector ending at point B. By definition, the vector $\mathbf{a} + \mathbf{b}$ ends at the vertex C of parallelogram OBCA. To verify that $\mathbf{a} + \mathbf{b} = (a_1 + b_1, a_2 + b_2)$, it suffices to show that the coordinates of C are $(a_1 + b_1, a_2 + b_2)$. The sides of the triangles OAD and BCG are parallel, and the sides OA and BC have equal lengths, which we write as OA = BC. These triangles are congruent, so BG = OD; since BGFE is a rectangle, EF = BG. Furthermore, $\mathrm{OD} = a_1$ and $\mathrm{OE} = b_1$. Hence, $\mathrm{EF} = \mathrm{BG} = \mathrm{OD} = a_1$. Since OF = EF + OE, it follows that $\mathrm{OF} = a_1 + b_1$. This shows that the x coordinate of $\mathbf{a} + \mathbf{b}$ is $a_1 + b_1$. The proof that the y coordinate is $a_2 + b_2$ is analogous. This argument assumes A and B to be in the first quadrant, but similar arguments hold for the other quadrants.

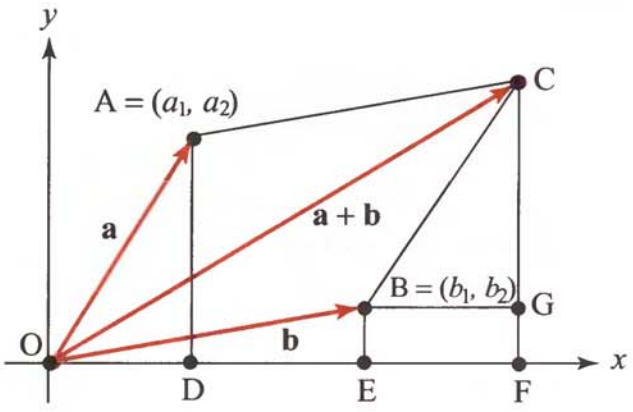


Figure 1.1.7 The construction used to prove that $(a_1, a_2) + (b_1, b_2) = (a_1 + b_1, a_2 + b_2)$.

Figure 1.1.8(a) illustrates another way of looking at vector addition: in terms of triangles rather than parallelograms. That is, we translate (without rotation) the directed line segment representing the vector **b** so that it begins at the end of the vector **a**. The endpoint of the resulting directed segment is the endpoint of the vector

$\mathbf{a} + \mathbf{b}$. We note that when $\mathbf{a}$ and $\mathbf{b}$ are collinear, the triangle collapses to a line segment, as in Figure 1.1.8(b).

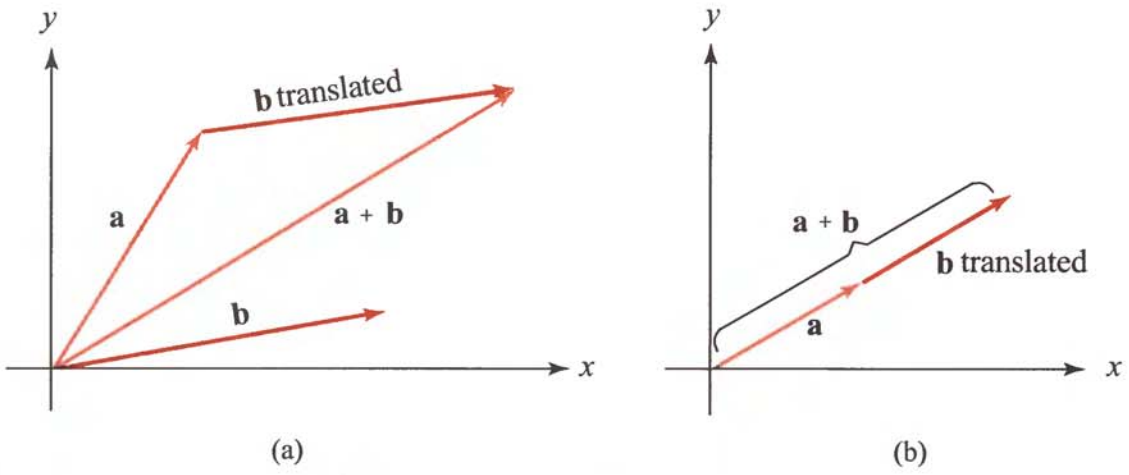


Figure 1.1.8 (a) Vector addition may be visualized in terms of triangles as well as parallelograms. (b) The triangle collapses to a line segment when $\mathbf{a}$ and $\mathbf{b}$ are collinear.

In Figure 1.1.8 we have placed $\mathbf{a}$ and $\mathbf{b}$ *head to tail*. That is, the tail of $\mathbf{b}$ is placed at the head of $\mathbf{a}$, and the vector $\mathbf{a} + \mathbf{b}$ goes from the tail of $\mathbf{a}$ to the head of $\mathbf{b}$. If we do it in the other order, $\mathbf{b} + \mathbf{a}$, we get the same vector by going around the parallelogram the other way. Consistent with this figure, it is useful to let vectors "glide" or "slide," keeping the same magnitude and direction. We want, in fact, to regard two vectors as the *same* if they have the same magnitude and direction. When we insist on vectors beginning at the origin, we will say that we have ***bound vectors***. If we allow vectors to begin at other points, we will speak of ***free vectors*** or just ***vectors***.

Vectors Vectors (also called *free vectors*) are directed line segments in [the plane or] space represented by directed line segments with a beginning (tail) and an end (head). Directed line segments obtained from each other by parallel translation (but not rotation) represent the same vector.

The components (a_1, a_2, a_3) of $\mathbf{a}$ are the (signed) lengths of the projections of $\mathbf{a}$ along the three coordinate axes; equivalently, they are defined by placing the tail of $\mathbf{a}$ at the origin and letting the head be the point (a_1, a_2, a_3). We write $\mathbf{a} = (a_1, a_2, a_3)$.

Two vectors are added by placing them head to tail and drawing the vectors from the tail of the first to the head of the second, as in Figure 1.1.8.

Scalar multiplication of vectors also has a geometric interpretation. If α is a scalar and $\mathbf{a}$ a vector, we define $\alpha\mathbf{a}$ to be the vector that is $|\alpha|$ times as long as $\mathbf{a}$, with the same direction as $\mathbf{a}$ if $\alpha > 0$, but with the opposite direction if $\alpha < 0$. Figure 1.1.9 illustrates several examples.

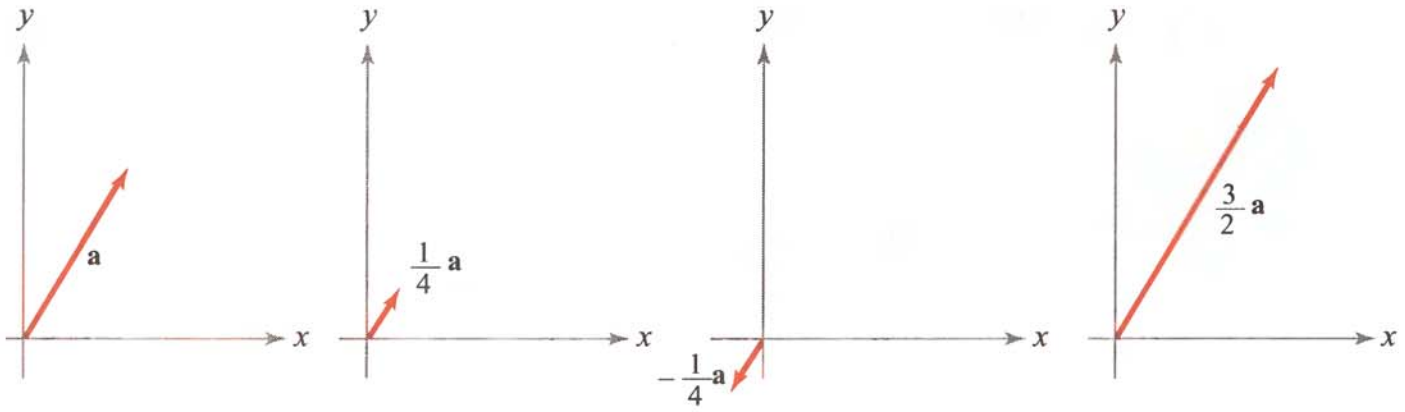


Figure 1.1.9 Some scalar multiples of a vector **a**.

Using an argument based on similar triangles, one finds that if $\mathbf{a} = (a_1, a_2, a_3)$, and α is a scalar, then

$$\alpha\mathbf{a} = (\alpha a_1, \alpha a_2, \alpha a_3).$$

That is, the geometric definition coincides with the algebraic one.

Given two vectors **a** and **b**, how do we represent the vector $\mathbf{b} - \mathbf{a}$ geometrically, that is, what is the geometry of vector subtraction? Because $\mathbf{a} + (\mathbf{b} - \mathbf{a}) = \mathbf{b}$, we see that $\mathbf{b} - \mathbf{a}$ is the vector that one adds to **a** to get **b**. In view of this, we may conclude that $\mathbf{b} - \mathbf{a}$ is the vector parallel to, and with the same magnitude as, the directed line segment beginning at the endpoint of **a** and terminating at the endpoint of **b** when **a** and **b** begin at the same point (see Figure 1.1.10).

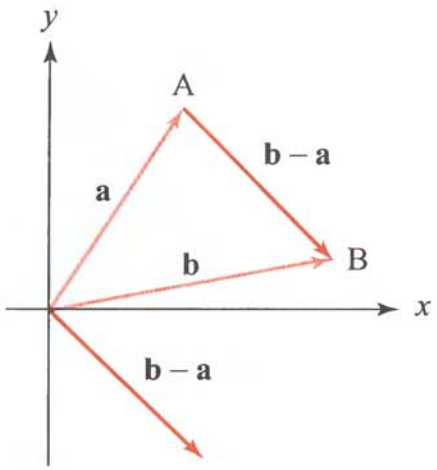


Figure 1.1.10 The geometry of vector subtraction.

EXAMPLE 4 Let **u** and **v** be the vectors shown in Figure 1.1.11. Draw the two vectors $\mathbf{u} + \mathbf{v}$ and $-2\mathbf{u}$. What are their components?

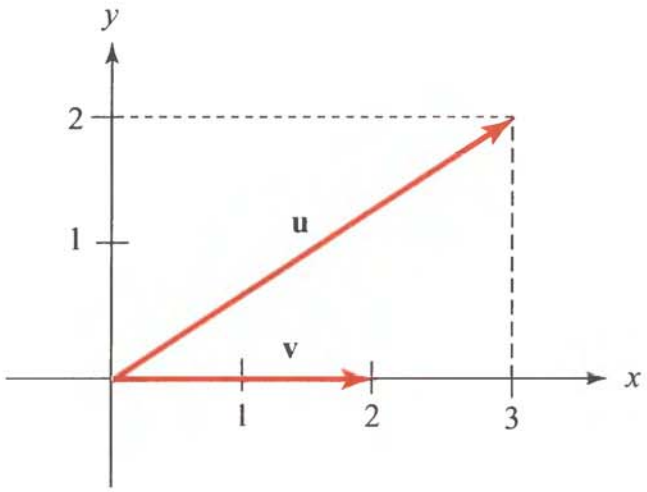


Figure 1.1.11 Find $\mathbf{u} + \mathbf{v}$ and $-2\mathbf{u}$.

SOLUTION Place the tail of $\mathbf{v}$ at the tip of $\mathbf{u}$ to obtain the vector shown in Figure 1.1.12.

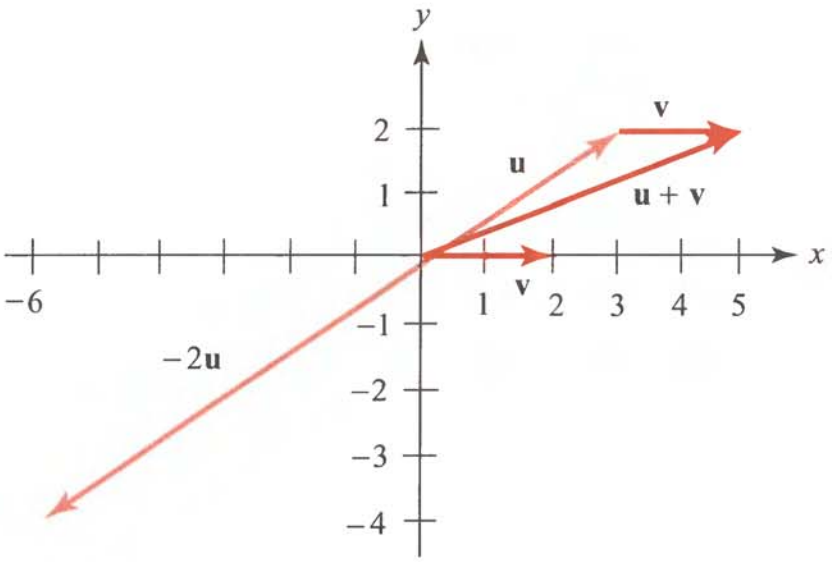


Figure 1.1.12 Computing $\mathbf{u} + \mathbf{v}$ and $-2\mathbf{u}$.

The vector $-2\mathbf{u}$, also shown, has length twice that of $\mathbf{u}$ and points in the opposite direction. From the figure, we see that the vector $\mathbf{u} + \mathbf{v}$ has components (5, 2) and $-2\mathbf{u}$ has components $(-6, -4)$. ▲

EXAMPLE 5

(a) Sketch $-2\mathbf{v}$, where $\mathbf{v}$ has components $(-1, 1, 2)$.

(b) If $\mathbf{v}$ and $\mathbf{w}$ are any two vectors, show that $\mathbf{v} - \frac{1}{3}\mathbf{w}$ and $3\mathbf{v} - \mathbf{w}$ are parallel.

SOLUTION

(a) The vector $-2\mathbf{v}$ is twice as long as $\mathbf{v}$, but points in the opposite direction (see Figure 1.1.13).

(b) $\mathbf{v} - \frac{1}{3}\mathbf{w} = \frac{1}{3}(3\mathbf{v} - \mathbf{w})$; vectors that are multiples of one another are parallel. ▲

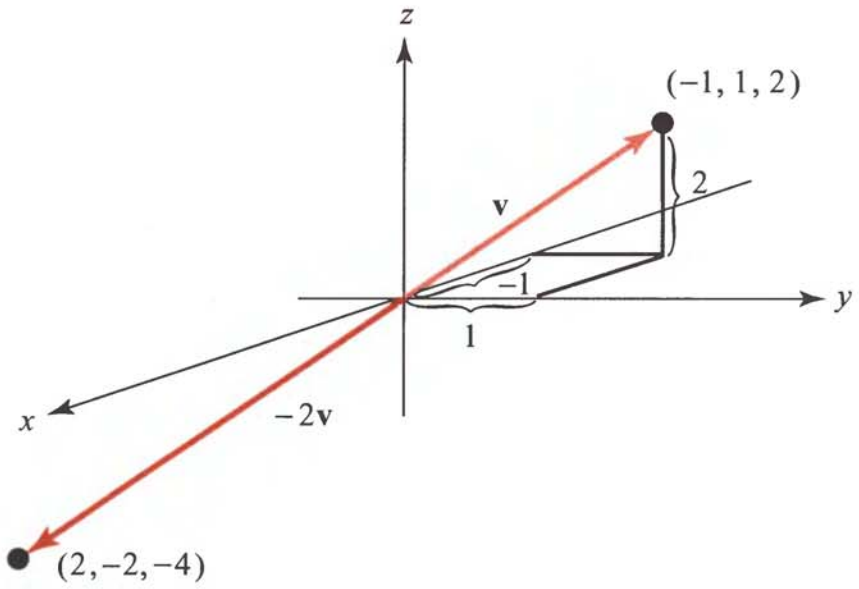


Figure 1.1.13 Multiplying $(-1, 1, 2)$ by -2.

The Standard Basis Vectors

To describe vectors in space, it is convenient to introduce three special vectors along the x, y, and z axes:

i: the vector with components $(1, 0, 0)$

j: the vector with components $(0, 1, 0)$

k: the vector with components $(0, 0, 1)$.

These ***standard basis vectors*** are illustrated in Figure 1.1.14. In the plane one has the standard basis **i** and **j** with components $(1, 0)$ and $(0, 1)$.

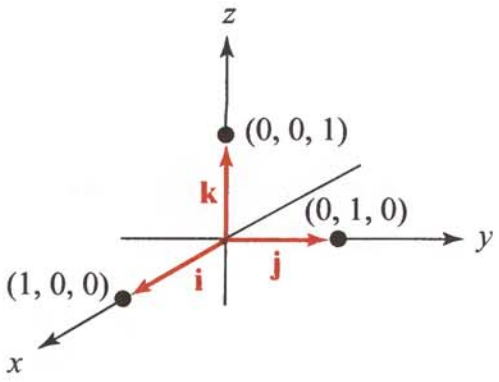


Figure 1.1.14 The standard basis vectors.

Let **a** be any vector, and let (a_1, a_2, a_3) be its components. Then

$$\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k},$$

because the right-hand side is given in components by

$$\begin{aligned} a_1(1, 0, 0) + a_2(0, 1, 0) + a_3(0, 0, 1) &= (a_1, 0, 0) + (0, a_2, 0) + (0, 0, a_3) \\ &= (a_1, a_2, a_3). \end{aligned}$$

Thus, we can express every vector as a sum of scalar multiples of **i**, **j**, and **k**.

The Standard Basis Vectors

1. The vectors **i**, **j**, and **k** are unit vectors along the three coordinate axes, as shown in Figure 1.1.14.
2. If **a** has components (a_1, a_2, a_3), then

$$\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}.$$

EXAMPLE 6 Express the vector whose components are $(e, \pi, -\sqrt{3})$ in the standard basis.

SOLUTION Substituting $a_1 = e, a_2 = \pi$, and $a_3 = -\sqrt{3}$ into $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$ gives

$$\mathbf{v} = e\mathbf{i} + \pi\mathbf{j} - \sqrt{3}\mathbf{k}. \quad \blacktriangle$$

EXAMPLE 7 The vector $(2, 3, 2)$ equals $2\mathbf{i} + 3\mathbf{j} + 2\mathbf{k}$, and the vector $(0, -1, 4)$ is $-\mathbf{j} + 4\mathbf{k}$. Figure 1.1.15 shows $2\mathbf{i} + 3\mathbf{j} + 2\mathbf{k}$; the student should draw in the vector $-\mathbf{j} + 4\mathbf{k}$. $\blacktriangle$

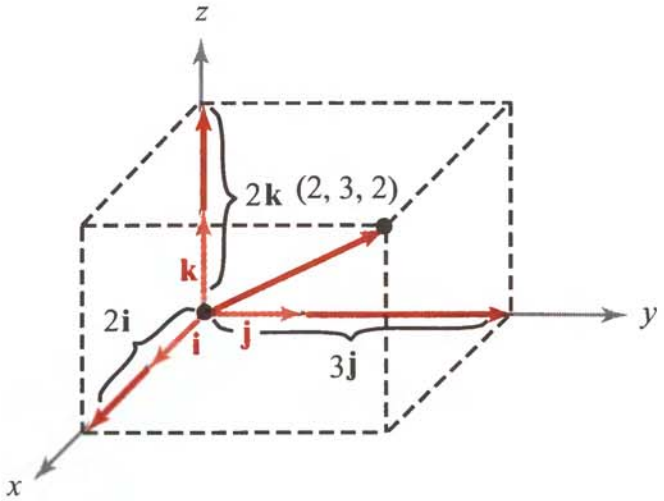


Figure 1.1.15 Representation of (2, 3, 2) in terms of the standard basis vectors **i**, **j**, and **k**.

Addition and scalar multiplication may be written in terms of the standard basis vectors as follows:

$$(a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}) + (b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}) = (a_1 + b_1)\mathbf{i} + (a_2 + b_2)\mathbf{j} + (a_3 + b_3)\mathbf{k}$$

and

$$\alpha(a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}) = (\alpha a_1)\mathbf{i} + (\alpha a_2)\mathbf{j} + \alpha(a_3)\mathbf{k}.$$

The Vector Joining Two Points

To apply vectors to geometric problems, it is useful to assign a vector to a *pair* of points in the plane or in space, as follows. Given two points P and P′, we can draw the vector **v** with tail P and head P′, as in Figure 1.1.16, where we write $\overrightarrow{PP'}$ for **v**.

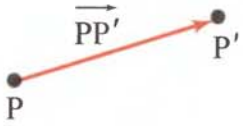


Figure 1.1.16 The vector from P to P′ is denoted $\overrightarrow{PP'}$.

If $P = (x, y, z)$ and $P' = (x', y', z')$, then the vectors from the origin to P and P′ are $\mathbf{a} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ and $\mathbf{a}' = x'\mathbf{i} + y'\mathbf{j} + z'\mathbf{k}$, respectively, so the vector $\overrightarrow{PP'}$ is the difference $\mathbf{a}' - \mathbf{a} = (x' - x)\mathbf{i} + (y' - y)\mathbf{j} + (z' - z)\mathbf{k}$. (See Figure 1.1.17.)

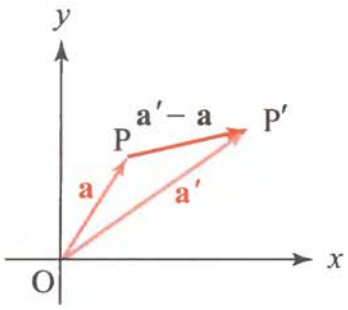


Figure 1.1.17 $\overrightarrow{PP'} = \overrightarrow{OP'} - \overrightarrow{OP}$.

The Vector Joining Two Points If the point P has coordinates (x, y, z) and P′ has coordinates (x', y', z'), then the vector $\overrightarrow{PP'}$ from the tip of P to the tip of P′ has components $(x' - x, y' - y, z' - z)$.

EXAMPLE 8

(a) Find the components of the vector from (3, 5) to (4, 7).

(b) Add the vector **v** from (−1, 0) to (2, −3) and the vector **w** from (2, 0) to (1, 1).

(c) Multiply the vector **v** in (b) by 8. If the resulting vector is represented by the directed line segment from (5, 6) to Q, what is Q?

SOLUTION

(a) As in the preceding box, we subtract the ordered pairs: (4, 7) − (3, 5) = (1, 2). Thus the required components are (1, 2).

(b) The vector **v** has components (2, −3) − (−1, 0) = (3, −3), and **w** has components (1, 1) − (2, 0) = (−1, 1). Therefore, the vector **v** + **w** has components (3, −3) + (−1, 1) = (2, −2).

(c) The vector $8\mathbf{v}$ has components $8(3, -3) = (24, -24)$. If this vector is represented by the directed line segment from (5, 6) to Q, and Q has coordinates (x, y), then $(x, y) - (5, 6) = (24, -24)$, so $(x, y) = (5, 6) + (24, -24) = (29, -18)$. ▲

EXAMPLE 9 Let $P = (-2, -1)$, $Q = (-3, -3)$, and $R = (-1, -4)$ in the xy plane.

(a) Draw these vectors: $\mathbf{v}$ joining P to Q; $\mathbf{w}$ joining Q to R; $\mathbf{u}$ joining R to P.

(b) What are the components of $\mathbf{v}$, $\mathbf{w}$, and $\mathbf{u}$?

(c) What is $\mathbf{v} + \mathbf{w} + \mathbf{u}$?

SOLUTION

(a) See Figure 1.1.18.

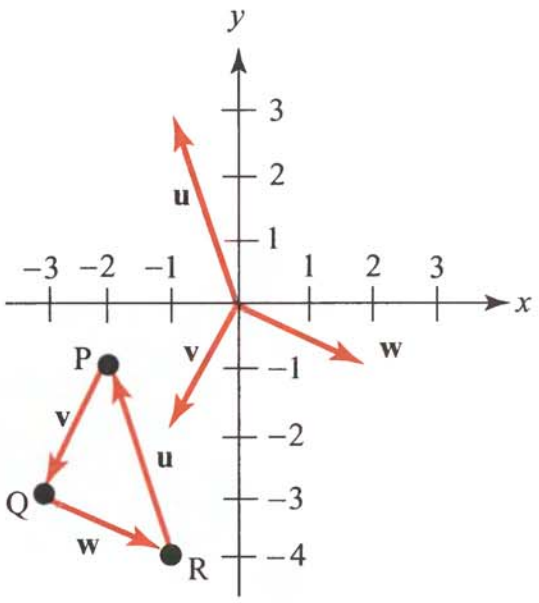


Figure 1.1.18 The vector $\mathbf{v}$ joins P to Q; $\mathbf{w}$ joins Q to R; and $\mathbf{u}$ joins R to P.

(b) Because $\mathbf{v} = \overrightarrow{PQ}$, $\mathbf{w} = \overrightarrow{QR}$, and $\mathbf{u} = \overrightarrow{RP}$, we get

$$\mathbf{v} = (-3, -3) - (-2, -1) = (-1, -2),$$
$$\mathbf{w} = (-1, -4) - (-3, -3) = (2, -1),$$
$$\mathbf{u} = -(-1, -4) + (-2, -1) = (-1, 3).$$

(c) $\mathbf{v} + \mathbf{w} + \mathbf{u} = (-1, -2) + (2, -1) + (-1, 3) = (0, 0)$. ▲

Geometry Theorems by Vector Methods

Many of the theorems of plane geometry can be proved by vector methods. Here is one example.

EXAMPLE 10 Use vectors to prove that the diagonals of a parallelogram bisect each other.

SOLUTION Let OPRQ be the parallelogram, with two adjacent sides represented by the vectors $\mathbf{a} = \overrightarrow{OP}$ and $\mathbf{b} = \overrightarrow{OQ}$. Let M be the midpoint of the diagonal OR, N the midpoint of the other diagonal, PQ. (See Figure 1.1.19.)

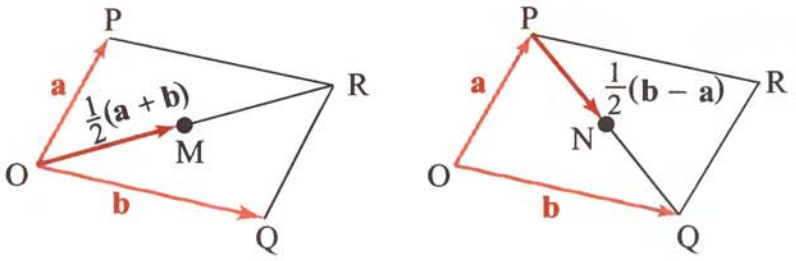


Figure 1.1.19 If the midpoints M and N coincide, then the diagonals OR and PQ bisect each other.

Observe that $\overrightarrow{OR} = \overrightarrow{OP} + \overrightarrow{OQ} = \mathbf{a} + \mathbf{b}$ by the parallelogram rule for vector addition, so $\overrightarrow{OM} = \frac{1}{2}\overrightarrow{OR} = \frac{1}{2}(\mathbf{a} + \mathbf{b})$. On the other hand,

$$\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP} = \mathbf{b} - \mathbf{a}, \qquad \text{so} \qquad \overrightarrow{PN} = \tfrac{1}{2}\overrightarrow{PQ} = \tfrac{1}{2}(\mathbf{b} - \mathbf{a}),$$

and hence

$$\overrightarrow{ON} = \overrightarrow{OP} + \overrightarrow{PN} = \mathbf{a} + \tfrac{1}{2}(\mathbf{b} - \mathbf{a}) = \tfrac{1}{2}(\mathbf{a} + \mathbf{b}).$$

Because $\overrightarrow{OM}$ and $\overrightarrow{ON}$ are equal vectors, the points M and N coincide, so the diagonals bisect each other. ▲

Equations of Lines

Planes and lines are geometric objects that can be represented by equations. We shall defer until Section 1.3 a study of equations representing planes. However, using the geometric interpretation of vector addition and scalar multiplication, we will now find the *equation of a line l that passes through the endpoint of the vector* **a**, *with the direction of a vector* **v** (see Figure 1.1.20).

As t varies through all real values, the points of the form $t\mathbf{v}$ are all scalar multiples of the vector **v**, and therefore exhaust the points of the line *passing through the origin* in the direction of **v**. Because every point on l is the endpoint of the diagonal of a parallelogram with sides **a** and $t\mathbf{v}$ for some real value of t, we see that all the points on l are of the form $\mathbf{a} + t\mathbf{v}$. Thus, the line l may be expressed by the equation $\mathbf{l}(t) = \mathbf{a} + t\mathbf{v}$. We say that l is expressed ***parametrically***, with t the parameter. At $t = 0$, $\mathbf{l}(t) = \mathbf{a}$. As t increases, the point $\mathbf{l}(t)$ moves away from **a** in the direction of **v**.

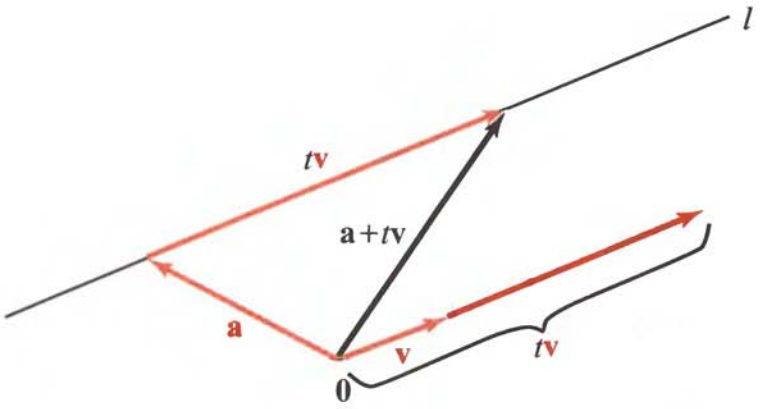


Figure 1.1.20 The line l, parametrically given by $\mathbf{l}(t) = \mathbf{a} + t\mathbf{v}$, lies in the direction $\mathbf{v}$ and passes through the tip of $\mathbf{a}$.

As t decreases from $t = 0$ through negative values, $\mathbf{l}(t)$ moves away from $\mathbf{a}$ in the direction of $-\mathbf{v}$.

Point-Direction Form of a Line The equation of the line l through the tip of $\mathbf{a}$ and pointing in the direction of the vector $\mathbf{v}$ is $\mathbf{l}(t) = \mathbf{a} + t\mathbf{v}$, where the parameter t takes on all real values. In coordinate form, the equations are

$$x = x_1 + at,$$
$$y = y_1 + bt,$$
$$z = z_1 + ct,$$

where $\mathbf{a} = (x_1, y_1, z_1)$ and $\mathbf{v} = (a, b, c)$. For lines in the xy plane, one simply drops the z component.

EXAMPLE 11 Determine the equation of the line l passing through (1, 0, 0) in the direction $\mathbf{j}$. See Figure 1.1.21.

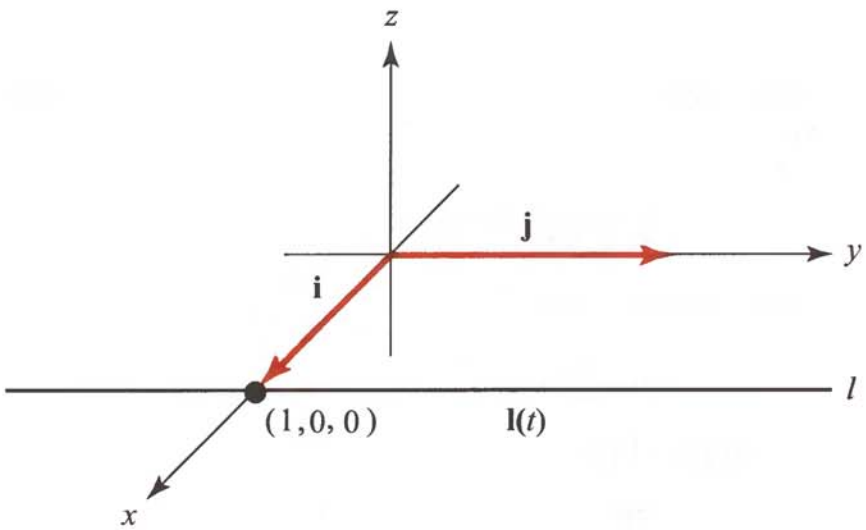


Figure 1.1.21 The line l passes through the tip of $\mathbf{i}$ in the direction $\mathbf{j}$.

SOLUTION The desired line can be expressed parametrically as $\mathbf{l}(t) = \mathbf{i} + t\mathbf{j}$. In terms of coordinates,

$$\mathbf{l}(t) = (1, 0, 0) + t(0, 1, 0) = (1, t, 0). \quad \blacktriangle$$

EXAMPLE 12

(a) Find the equations of the line in space through the point $(3, -1, 2)$ in the direction $2\mathbf{i} - 3\mathbf{j} + 4\mathbf{k}$.

(b) Find the equation of the line in the plane through the point $(1, -6)$ in the direction of $5\mathbf{i} - \pi\mathbf{j}$.

(c) In what direction does the line $x = -3t + 2$, $y = -2(t - 1)$, $z = 8t + 2$ point?

SOLUTION

(a) Here $\mathbf{a} = (3, -1, 2) = (x_1, y_1, z_1)$ and $\mathbf{v} = 2\mathbf{i} - 3\mathbf{j} + 4\mathbf{k}$, so $a = 2$, $b = -3$, and $c = 4$. From the box above, the equations are

$$x = 3 + 2t, \qquad y = -1 - 3t, \qquad z = 2 + 4t.$$

(b) Here $\mathbf{a} = (1, -6)$ and $\mathbf{v} = 5\mathbf{i} - \pi\mathbf{j}$, so the required line is

$$\mathbf{l}(t) = (1, -6) + (5t, -\pi t) = (1 + 5t, -6 - \pi t);$$

that is,

$$x = 1 + 5t, \qquad y = -6 - \pi t.$$

(c) Using the preceding box, we construct the direction $\mathbf{v} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$ from the coefficients of t: $a = -3$, $b = -2$, $c = 8$. Thus, the line points in the direction of $\mathbf{v} = -3\mathbf{i} - 2\mathbf{j} + 8\mathbf{k}$. $\blacktriangle$

EXAMPLE 13 Do the two lines $(x, y, z) = (t, -6t + 1, 2t - 8)$ and $(x, y, z) = (3t + 1, 2t, 0)$ intersect?

SOLUTION If the lines intersect, there must be numbers t_1 and t_2 such that the corresponding points are equal:

$$(t_1, -6t_1 + 1, 2t_1 - 8) = (3t_2 + 1, 2t_2, 0);$$

that is, all three of the following equations hold:

$$t_1 = 3t_2 + 1,$$
$$-6t_1 + 1 = 2t_2,$$
$$2t_1 - 8 = 0.$$

From the third equation, $t_1 = 4$. The first equation then becomes $4 = 3t_2 + 1$; that is, $t_2 = 1$. We must check whether these values satisfy the middle equation:

$$-6t_1 + 1 \stackrel{?}{=} 2t_2.$$

Since $t_1 = 4$ and $t_2 = 1$, this reads

$$-24 + 1 \stackrel{?}{=} 2,$$

which is false, so the lines do not intersect. ▲

Notice that there can be many equations of the same line. Some may be obtained by choosing instead of $\mathbf{a}$, a different point on the given line, and forming the parametric equation of the line beginning at that point and in the direction of $\mathbf{v}$. For example, the endpoint of $\mathbf{a} + \mathbf{v}$ is on the line $\mathbf{l}(t) = \mathbf{a} + t\mathbf{v}$, and thus, $\mathbf{l}_1(t) = (\mathbf{a} + \mathbf{v}) + t\mathbf{v}$ represents the same line. Still other equations may be obtained by observing that if $\alpha \neq 0$, the vector $\alpha\mathbf{v}$ has the same (or opposite) direction as $\mathbf{v}$. Thus, $\mathbf{l}_2(t) = \mathbf{a} + t\alpha\mathbf{v}$ is another equation of $\mathbf{l}(t) = \mathbf{a} + t\mathbf{v}$.

For example, both $\mathbf{l}(t) = (1, 0, 0) + (t, t, 0)$ and $\mathbf{l}_1(s) = (0, -1, 0) + (s, s, 0)$ represent the same line since both are in the direction $\mathbf{i} + \mathbf{j}$ and both pass through the point $(1, 0, 0)$; $\mathbf{l}$ passes through $(1, 0, 0)$ at $t = 0$ and $\mathbf{l}_1$ passes through $(1, 0, 0)$ at $s = 1$.

Therefore, the equation of a line is not uniquely determined. Nevertheless, it is customary to use the term "the" equation of a line. Keeping this in mind, let us derive *the equation of a line passing through the endpoints of two given vectors* $\mathbf{a}$ *and* $\mathbf{b}$. Because the vector $\mathbf{b} - \mathbf{a}$ is parallel to the directed line segment from $\mathbf{a}$ to $\mathbf{b}$, we calculate the parametric equation of the line passing through $\mathbf{a}$ in the direction of $\mathbf{b} - \mathbf{a}$ (Figure 1.1.22). Thus,

$$\mathbf{l}(t) = \mathbf{a} + t(\mathbf{b} - \mathbf{a}); \qquad \text{that is,} \qquad \mathbf{l}(t) = (1 - t)\mathbf{a} + t\mathbf{b}.$$

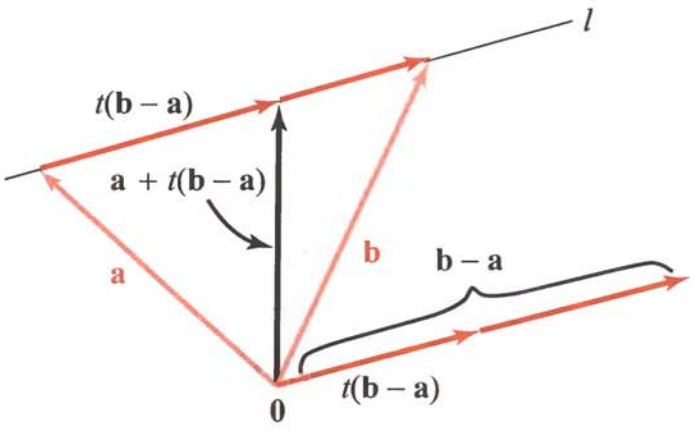


Figure 1.1.22 The line l, parametrically given by $\mathbf{l}(t) = \mathbf{a} + t(\mathbf{b} - \mathbf{a}) = (1 - t)\mathbf{a} + t\mathbf{b}$, passes through the tips of $\mathbf{a}$ and $\mathbf{b}$.

As t increases from 0 to 1, $t(\mathbf{b} - \mathbf{a})$ starts as the zero vector and increases in length (remaining in the direction of $\mathbf{b} - \mathbf{a}$) until at $t = 1$ it is the vector $\mathbf{b} - \mathbf{a}$. Thus, for

$\mathbf{l}(t) = \mathbf{a} + t(\mathbf{b} - \mathbf{a})$, as t increases from 0 to 1, the vector $\mathbf{l}(t)$ moves from the endpoint of $\mathbf{a}$ to the endpoint of $\mathbf{b}$ along the directed line segment from $\mathbf{a}$ to $\mathbf{b}$.

If $P = (x_1, y_1, z_1)$ is the tip of $\mathbf{a}$ and $Q = (x_2, y_2, z_2)$ is the tip of $\mathbf{b}$, then $\mathbf{v} = (x_2 - x_1)\mathbf{i} + (y_2 - y_1)\mathbf{j} + (z_2 - z_1)\mathbf{k}$, and so the equations of the line are

$$x = x_1 + (x_2 - x_1)t,$$
$$y = y_1 + (y_2 - y_1)t,$$
$$z = z_1 + (z_2 - z_1)t.$$

By eliminating t, these can be written as

$$\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1}.$$

Parametric Equation of a Line: Point–Point Form The parametric equations of the line l through the points $P = (x_1, y_1, z_1)$ and $Q = (x_2, y_2, z_2)$ are

$$x = x_1 + (x_2 - x_1)t,$$
$$y = y_1 + (y_2 - y_1)t,$$
$$z = z_1 + (z_2 - z_1)t,$$

where (x, y, z) is the general point of l, and the parameter t takes on all real values.

EXAMPLE 14 Find the equation of the line through $(2, 1, -3)$ and $(6, -1, -5)$.

SOLUTION Using the preceding box, we choose $(x_1, y_1, z_1) = (2, 1, -3)$ and $(x_2, y_2, z_2) = (6, -1, -5)$, so the equations are

$$x = 2 + (6 - 2)t = 2 + 4t,$$
$$y = 1 + (-1 - 1)t = 1 - 2t,$$
$$z = -3 + (-5 - (-3))t = -3 - 2t. \quad \blacktriangle$$

EXAMPLE 15 Find the equation of the line passing through $(-1, 1, 0)$ and $(0, 0, 1)$ (see Figure 1.1.23).

SOLUTION Letting $\mathbf{a} = -\mathbf{i} + \mathbf{j}$ and $\mathbf{b} = \mathbf{k}$ represent the given points, we have

$$\mathbf{l}(t) = (1 - t)(-\mathbf{i} + \mathbf{j}) + t\mathbf{k} = -(1 - t)\mathbf{i} + (1 - t)\mathbf{j} + t\mathbf{k}.$$

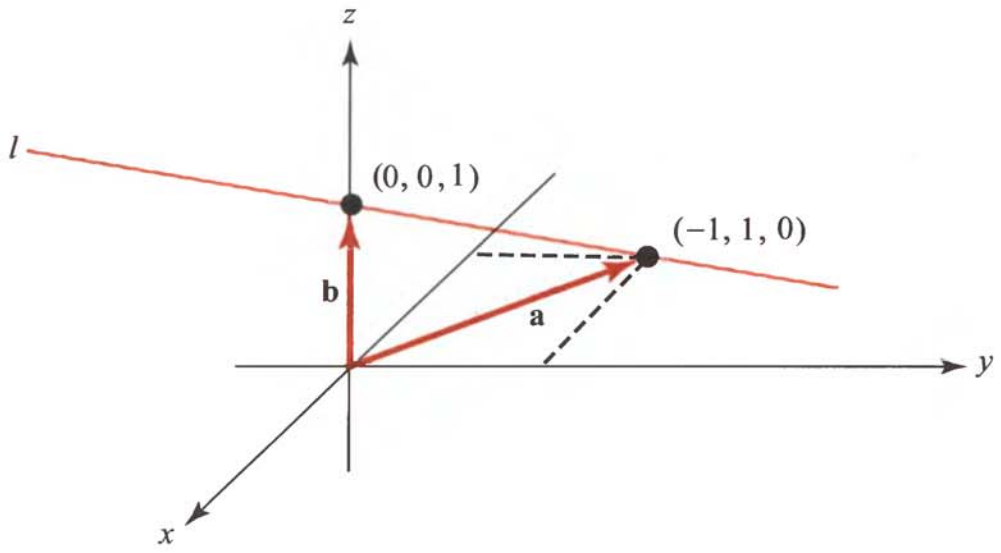


Figure 1.1.23 Finding the equation of the line through two points.

The equation of this line may thus be written as

$$\mathbf{l}(t) = (t - 1)\mathbf{i} + (1 - t)\mathbf{j} + t\mathbf{k},$$

or, equivalently, if $\mathbf{l}(t) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$,

$$x = t - 1, \qquad y = 1 - t, \qquad z = t. \quad \blacktriangle$$

The description of a line *segment* requires that the domain of the parameter t be restricted, as in the following example.

EXAMPLE 16 Find the equation of the line segment between (1, 1, 1) and (2, 1, 2).

SOLUTION The *line* through (1, 1, 1) and (2, 1, 2) is described in parametric form by $(x, y, z) = (1 + t, 1, 1 + t)$, as t takes on all real values. When $t = 0$, the point (x, y, z) is (1, 1, 1), and when $t = 1$, the point (x, y, z) is (2, 1, 2). Thus, the point (x, y, z) lies between (1, 1, 1) and (2, 1, 2) when $0 \le t \le 1$, so the line *segment* is described by the equations

$$\begin{aligned} x &= 1 + t, \\ y &= 1, \\ z &= 1 + t, \end{aligned}$$

together with the inequalities $0 \le t \le 1$. $\blacktriangle$

We can also give parametric descriptions of geometric objects other than lines.

EXAMPLE 17 Describe the points that lie within the parallelogram whose adjacent sides are the vectors **a** and **b** based at the origin ("within" includes points on the edges of the parallelogram).

SOLUTION Consider Figure 1.1.24. If P is any point within the given parallelogram and we construct lines l_1 and l_2 through P parallel to the vectors **a** and **b**, respectively, we see that l_1 intersects the side of the parallelogram determined by the vector **b** at some point $t\mathbf{b}$, where $0 \leq t \leq 1$. Likewise, l_2 intersects the side determined by the vector **a** at some point $s\mathbf{a}$, where $0 \leq s \leq 1$.

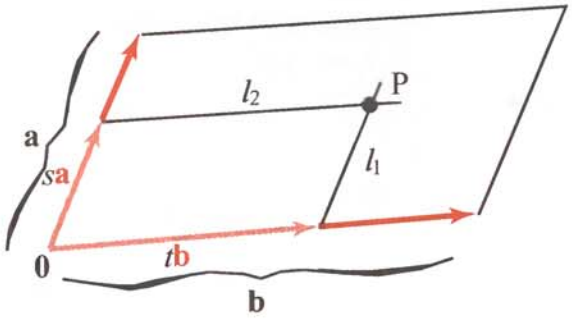


Figure 1.1.24 Describing points within the parallelogram formed by vectors **a** and **b**, with vertex **0**.

Note that P is the endpoint of the diagonal of a parallelogram having adjacent sides $s\mathbf{a}$ and $t\mathbf{b}$; hence, if **v** denotes the vector $\overrightarrow{OP}$, we see that $\mathbf{v} = s\mathbf{a} + t\mathbf{b}$. We conclude that all the points in the given parallelogram are endpoints of vectors of the form $s\mathbf{a} + t\mathbf{b}$ for $0 \leq s \leq 1$ and $0 \leq t \leq 1$. Reversing our steps, we see that all vectors of this form end within the parallelogram. ▲

As two different lines through the origin determine a plane through the origin, so do two nonparallel vectors. If we apply the same reasoning as in Example 17, we see that the entire plane formed by two nonparallel vectors **v** and **w** consists of all points of the form $s\mathbf{v} + t\mathbf{w}$ where s and t can be any real numbers, as in Figure 1.1.25.

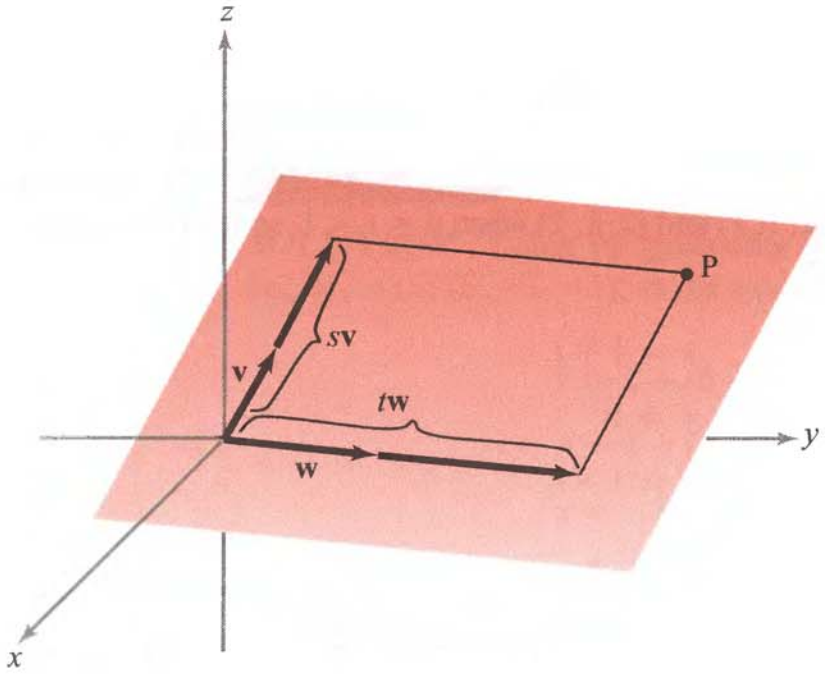


Figure 1.1.25 Describing points P in the plane formed from vectors **v** and **w**.

We have thus described the points in the plane by two parameters. For this reason, we say the plane is ***two-dimensional***. Similarly, a line is called ***one-dimensional*** whether it lies in the plane or in space or is the real number line itself.

The plane determined by **v** and **w** is called the plane ***spanned by*** **v** and **w**. When **v** is a scalar multiple of **w** and $\mathbf{w} \neq \mathbf{0}$, then **v** and **w** are parallel and the plane degenerates to a straight line. When $\mathbf{v} = \mathbf{w} = \mathbf{0}$ (that is, both are zero vectors), we obtain a single point.

There are three particular planes that arise naturally in a coordinate system and that will be useful to us later. We call the plane spanned by vectors **i** and **j** the xy plane, the plane spanned by **j** and **k** the yz plane, and the plane spanned by **i** and **k** the xz plane. These planes are illustrated in Figure 1.1.26.

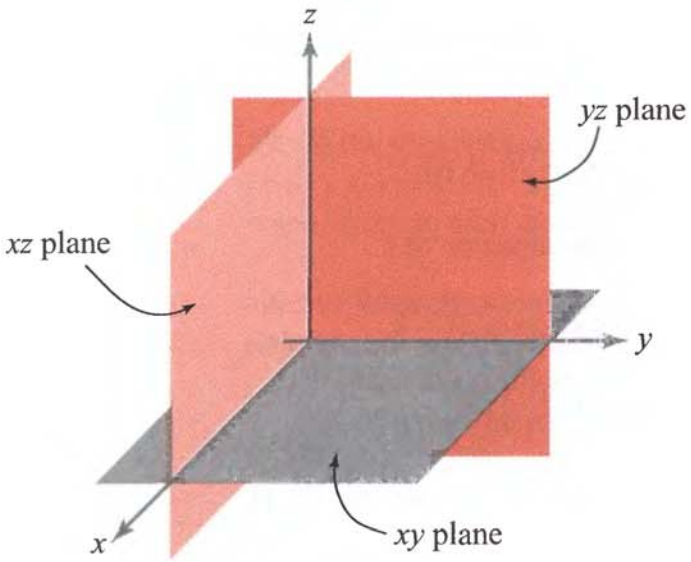


Figure 1.1.26 The three coordinate planes.

EXERCISES

(*Exercises with colored numbers are solved in the Study Guide.*)

Complete the computations in Exercises 1 to 4.

1. $(-21, 23) - (?, 6) = (-25, ?)$

2. $3(133, -0.33, 0) + (-399, 0.99, 0) = (?, ?, ?)$

3. $(8a, -2b, 13c) = (52, 12, 11) + \frac{1}{2}(?, ?, ?)$

4. $(2, 3, 5) - 4\mathbf{i} + 3\mathbf{j} = (?, ?, ?)$

In Exercises 5 to 8, sketch the given vectors **v** *and* **w**. *On your sketch, draw in* $-\mathbf{v}$, $\mathbf{v} + \mathbf{w}$, *and* $\mathbf{v} - \mathbf{w}$.

5. $\mathbf{v} = (2, 1)$ and $\mathbf{w} = (1, 2)$

6. $\mathbf{v} = (0, 4)$ and $\mathbf{w} = (2, -1)$

7. $\mathbf{v} = (2, 3, -6)$ and $\mathbf{w} = (-1, 1, 1)$

8. $\mathbf{v} = (2, 1, 3)$ and $\mathbf{w} = (-2, 0, -1)$

9. What restrictions must be made on x, y, and z so that the triple (x, y, z) will represent a point on the y axis? On the z axis? In the xz plane? In the yz plane?

10. (a) Generalize the geometric construction in Figure 1.1.7 to show that if $\mathbf{v}_1 = (x, y, z)$ and $\mathbf{v}_2 = (x', y', z')$, then $\mathbf{v}_1 + \mathbf{v}_2 = (x + x', y + y', z + z')$.

(b) Using an argument based on similar triangles, prove that $\alpha\mathbf{v} = (\alpha x, \alpha y, \alpha z)$ when $\mathbf{v} = (x, y, z)$.

In Exercises 11 to 17, use set theoretic or vector notation or both to describe the points that lie in the given configurations.

11. The plane spanned by $\mathbf{v}_1 = (2, 7, 0)$ and $\mathbf{v}_2 = (0, 2, 7)$

12. The plane spanned by $\mathbf{v}_1 = (3, -1, 1)$ and $\mathbf{v}_2 = (0, 3, 4)$

13. The line passing through $(-1, -1, -1)$ in the direction of $\mathbf{j}$

14. The line passing through $(0, 2, 1)$ in the direction of $2\mathbf{i} - \mathbf{k}$

15. The line passing through $(-1, -1, -1)$ and $(1, -1, 2)$

16. The line passing through $(-5, 0, 4)$ and $(6, -3, 2)$

17. The parallelogram whose adjacent sides are the vectors $\mathbf{i} + 3\mathbf{k}$ and $-2\mathbf{j}$

18. Find the points of intersection of the line $x = 3 + 2t$, $y = 7 + 8t$, $z = -2 + t$, that is, $\mathbf{l}(t) = (3 + 2t, 7 + 8t, -2 + t)$, with the coordinate planes.

19. Show that there are no points (x, y, z) satisfying $2x - 3y + z - 2 = 0$ and lying on the line $\mathbf{v} = (2, -2, -1) + t(1, 1, 1)$.

20. Show that every point on the line $\mathbf{v} = (1, -1, 2) + t(2, 3, 1)$ satisfies the equation $5x - 3y - z - 6 = 0$.

21. Determine whether the lines $x = 3t + 2$, $y = t - 1$, $z = 6t + 1$, and $x = 3s - 1$, $y = s - 2$, $z = s$ intersect.

22. Do the lines $(x, y, z) = (t + 4, 4t + 5, t - 2)$ and $(x, y, z) = (2s + 3, s + 1, 2s - 3)$ intersect?

In Exercises 23 to 25, use vector methods to describe the given configurations.

23. The parallelepiped with edges the vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ emanating from the origin.

24. The points within the parallelogram with one corner at (x_0, y_0, z_0) whose sides extending from that corner are equal in magnitude and direction to vectors $\mathbf{a}$ and $\mathbf{b}$.

25. The plane determined by the three points (x_0, y_0, z_0), (x_1, y_1, z_1), and (x_2, y_2, z_2).

Prove the statements in Exercises 26 to 28.

26. The line segment joining the midpoints of two sides of a triangle is parallel to and has half the length of the third side.

27. If PQR is a triangle in space and $b > 0$ is a number, then there is a triangle with sides parallel to those of PQR and side lengths b times those of PQR.

28. The medians of a triangle intersect at a point, and this point divides each median in a ratio of 2 : 1.

Problems 29 and 30 require some knowledge of chemical notation.

29. Write the chemical equation $CO + H_2O = H_2 + CO_2$ as an equation in ordered triples (x_1, x_2, x_3) where x_1, x_2, x_3 are the number of carbon, hydrogen, and oxygen atoms, respectively, in each molecule.

30. (a) Write the chemical equation $pC_3H_4O_3 + qO_2 = rCO_2 + sH_2O$ as an equation in ordered triples with unknown coefficients p, q, r, and s.
(b) Find the smallest positive integer solution for p, q, r, and s.
(c) Illustrate the solution by a vector diagram in space.

31. Find a line that lies entirely in the set defined by the equation $x^2 + y^2 - z^2 = 1$.

1.2 The Inner Product, Length, and Distance

In this section and the next we shall discuss two products of vectors: the inner product and the cross product. These are very useful in physical applications and have interesting geometric interpretations. The first product we shall consider is called the *inner product*. The name *dot product* is often used instead.

The Inner Product

Suppose we have two vectors **a** and **b** in $\mathbb{R}^3$ (Figure 1.2.1) and we wish to determine the angle between them, that is, the smaller angle subtended by **a** and **b** in the plane

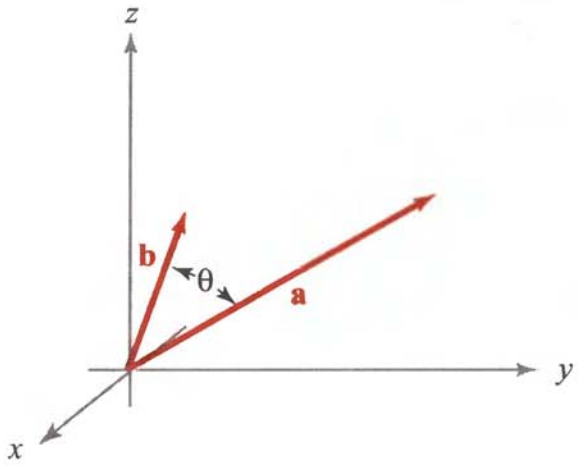


Figure 1.2.1 θ is the angle between the vectors **a** and **b**.

that they span. The inner product enables us to do this. Let us first develop the concept formally and then prove that this product does what we claim. Let $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$ and $\mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$. We define the ***inner product*** of $\mathbf{a}$ and $\mathbf{b}$, written $\mathbf{a} \cdot \mathbf{b}$, to be the real number

$$\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3.$$

Note that the inner product of two vectors is a scalar quantity. Sometimes the inner product is denoted $\langle \mathbf{a}, \mathbf{b} \rangle$; *thus*, $\langle \mathbf{a}, \mathbf{b} \rangle$ *and* $\mathbf{a} \cdot \mathbf{b}$ *mean exactly the same thing.*

EXAMPLE 1

(a) If $\mathbf{a} = 3\mathbf{i} + \mathbf{j} - 2\mathbf{k}$ and $\mathbf{b} = \mathbf{i} - \mathbf{j} + \mathbf{k}$, calculate $\mathbf{a} \cdot \mathbf{b}$.

(b) Calculate $(2\mathbf{i} + \mathbf{j} - \mathbf{k}) \cdot (3\mathbf{k} - 2\mathbf{j})$.

SOLUTION

(a) $\mathbf{a} \cdot \mathbf{b} = 3 \cdot 1 + 1 \cdot (-1) + (-2) \cdot 1 = 3 - 1 - 2 = 0.$

(b) $(2\mathbf{i} + \mathbf{j} - \mathbf{k}) \cdot (3\mathbf{k} - 2\mathbf{j}) = (2\mathbf{i} + \mathbf{j} - \mathbf{k}) \cdot (0\mathbf{i} - 2\mathbf{j} + 3\mathbf{k})$
$= 2 \cdot 0 - 1 \cdot 2 - 1 \cdot 3 = -5.$ ▲

Certain properties of the inner product follow from the definition. If $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ are vectors in $\mathbb{R}^3$ and α and β are real numbers, then

(i) $\mathbf{a} \cdot \mathbf{a} \geq 0$;

$\mathbf{a} \cdot \mathbf{a} = 0$ if and only if $\mathbf{a} = \mathbf{0}$.

(ii) $\alpha\mathbf{a} \cdot \mathbf{b} = \alpha(\mathbf{a} \cdot \mathbf{b})$ and $\mathbf{a} \cdot \beta\mathbf{b} = \beta(\mathbf{a} \cdot \mathbf{b})$.

(iii) $\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}$ and $(\mathbf{a} + \mathbf{b}) \cdot \mathbf{c} = \mathbf{a} \cdot \mathbf{c} + \mathbf{b} \cdot \mathbf{c}$.

(iv) $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$.

To prove the first of these properties, observe that if $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$, then $\mathbf{a} \cdot \mathbf{a} = a_1^2 + a_2^2 + a_3^2$. Because a_1, a_2, and a_3 are real numbers, we know $a_1^2 \geq 0$, $a_2^2 \geq 0$, $a_3^2 \geq 0$. Thus, $\mathbf{a} \cdot \mathbf{a} \geq 0$. Moreover, if $a_1^2 + a_2^2 + a_3^2 = 0$, then $a_1 = a_2 = a_3 = 0$; therefore, $\mathbf{a} = \mathbf{0}$ (zero vector). The proofs of the other properties of the inner product are also easily obtained.

It follows from the Pythagorean theorem that the ***length*** of the vector $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$ is $\sqrt{a_1^2 + a_2^2 + a_3^2}$ (see Figure 1.2.2). The length of the vector $\mathbf{a}$ is denoted by $\|\mathbf{a}\|$. This quantity is often called the ***norm*** of $\mathbf{a}$. Because $\mathbf{a} \cdot \mathbf{a} = a_1^2 + a_2^2 + a_3^2$, it follows that

$$\|\mathbf{a}\| = (\mathbf{a} \cdot \mathbf{a})^{1/2}.$$

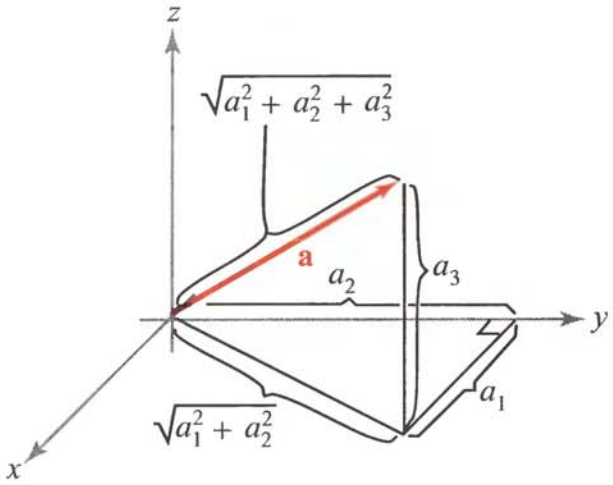


Figure 1.2.2 The length of the vector $\mathbf{a} = (a_1, a_2, a_3)$ is given by the Pythagorean formula: $\sqrt{a_1^2 + a_2^2 + a_3^2}$.

Unit Vectors

Vectors with norm 1 are called ***unit vectors***. For example, the vectors **i**, **j**, **k** are unit vectors. Observe that for any nonzero vector **a**, $\mathbf{a}/\|\mathbf{a}\|$ is a unit vector; when we divide **a** by $\|\mathbf{a}\|$, we say that we have ***normalized*** **a**.

EXAMPLE 2

(a) Normalize $\mathbf{v} = 2\mathbf{i} + 3\mathbf{j} - \frac{1}{2}\mathbf{k}$.

(b) Find unit vectors **a**, **b**, and **c** in the plane such that $\mathbf{b} + \mathbf{c} = \mathbf{a}$.

SOLUTION

(a) We have $\|\mathbf{v}\| = \sqrt{2^2 + 3^2 + (1/2)^2} = (1/2)\sqrt{53}$, so the normalization of **v** is

$$\mathbf{u} = \frac{1}{\|\mathbf{v}\|}\mathbf{v} = \frac{4}{\sqrt{53}}\mathbf{i} + \frac{6}{\sqrt{53}}\mathbf{j} - \frac{1}{\sqrt{53}}\mathbf{k}.$$

(b) Because all three vectors are to have length 1, a triangle with sides **a**, **b**, and **c** must be equilateral, as in Figure 1.2.3. Orienting the triangle as in the figure, we take $\mathbf{a} = \mathbf{i}$, then necessarily

$$\mathbf{b} = \frac{1}{2}\mathbf{i} + \frac{\sqrt{3}}{2}\mathbf{j}, \qquad \text{and} \qquad \mathbf{c} = \frac{1}{2}\mathbf{i} - \frac{\sqrt{3}}{2}\mathbf{j}.$$

Note that indeed $\|\mathbf{a}\| = \|\mathbf{b}\| = \|\mathbf{c}\| = 1$ and that $\mathbf{b} + \mathbf{c} = \mathbf{a}$. ▲

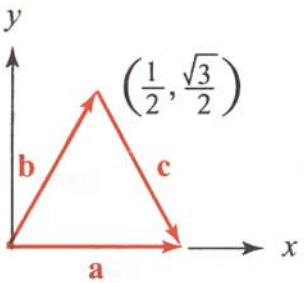


Figure 1.2.3 The vectors **a**, **b**, and **c** are represented by the sides of an equilateral triangle.

In the plane, define the vector $\mathbf{i}_\theta = (\cos\theta)\mathbf{i} + (\sin\theta)\mathbf{j}$, which is the unit vector making an angle θ with the x axis (see Figure 1.2.4).

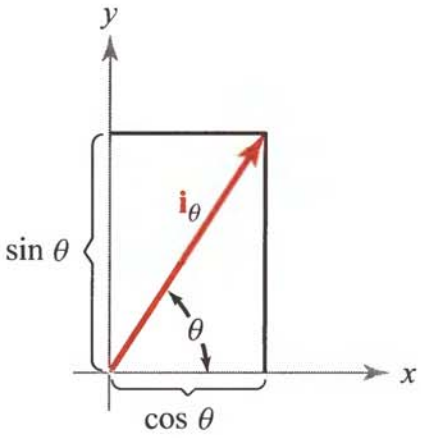


Figure 1.2.4 The coordinates of $\mathbf{i}_\theta$ are $\cos\theta$ and $\sin\theta$; it is a unit vector because $\cos^2\theta + \sin^2\theta = 1$.

Distance

If $\mathbf{a}$ and $\mathbf{b}$ are vectors, we have seen that the vector $\mathbf{b} - \mathbf{a}$ is parallel to and has the same magnitude as the directed line segment from the endpoint of $\mathbf{a}$ to the endpoint of $\mathbf{b}$. It follows that the distance from the endpoint of $\mathbf{a}$ to the endpoint of $\mathbf{b}$ is $\|\mathbf{b} - \mathbf{a}\|$ (see Figure 1.2.5).

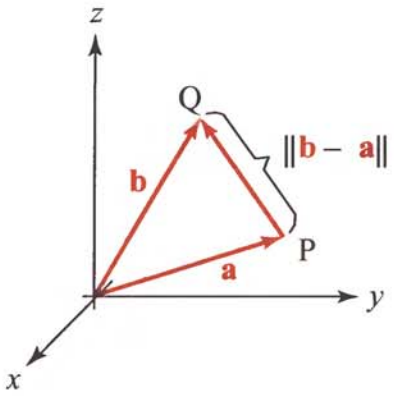


Figure 1.2.5 The distance between the tips of $\mathbf{a}$ and $\mathbf{b}$ is $\|\mathbf{b} - \mathbf{a}\|$.

Inner Product, Length, and Distance Letting $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$ and $\mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$, their ***inner product*** is

$$\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3,$$

while the ***length*** of $\mathbf{a}$ is

$$\|\mathbf{a}\| = \sqrt{\mathbf{a} \cdot \mathbf{a}} = \sqrt{a_1^2 + a_2^2 + a_3^2}.$$

To ***normalize*** a vector $\mathbf{a}$, form the vector

$$\frac{\mathbf{a}}{\|\mathbf{a}\|}.$$

The ***distance between*** the endpoints of $\mathbf{a}$ and $\mathbf{b}$ is $\|\mathbf{a} - \mathbf{b}\|$, and the ***distance between*** P and Q is $\|\overrightarrow{PQ}\|$.

EXAMPLE 3 Find the distance from the endpoint of the vector **i**, that is, the point (1, 0, 0), to the endpoint of the vector **j**, that is, the point (0, 1, 0).

SOLUTION $\|\mathbf{j} - \mathbf{i}\| = \sqrt{(0-1)^2 + (1-0)^2 + (0-0)^2} = \sqrt{2}$. ▲

The Angle Between Two Vectors

Let us now show that the inner product does indeed measure the angle between two vectors.

THEOREM 1 Let **a** and **b** be two vectors in $\mathbb{R}^3$ and let θ, where $0 \leq \theta \leq \pi$, be the angle between them (Figure 1.2.6). Then

$$\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos\theta.$$

It follows from the equation $\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos\theta$ that if **a** and **b** are nonzero, we may express the angle between them as

$$\theta = \cos^{-1}\left(\frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|}\right).$$

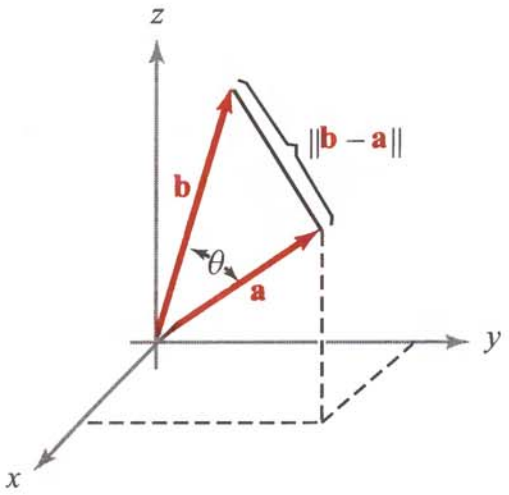


Figure 1.2.6 The vectors **a**, **b**, and the angle θ between them; the geometry for Theorem 1 and its proof.

PROOF If we apply the law of cosines from trigonometry to the triangle with one vertex at the origin and adjacent sides determined by the vectors **a** and **b** (as in the figure), it follows that

$$\|\mathbf{b} - \mathbf{a}\|^2 = \|\mathbf{a}\|^2 + \|\mathbf{b}\|^2 - 2\|\mathbf{a}\| \|\mathbf{b}\| \cos\theta.$$

Because $\|\mathbf{b}-\mathbf{a}\|^2 = (\mathbf{b}-\mathbf{a})\cdot(\mathbf{b}-\mathbf{a})$, $\|\mathbf{a}\|^2 = \mathbf{a}\cdot\mathbf{a}$, and $\|\mathbf{b}\|^2 = \mathbf{b}\cdot\mathbf{b}$, we can rewrite the above equation as

$$(\mathbf{b}-\mathbf{a})\cdot(\mathbf{b}-\mathbf{a}) = \mathbf{a}\cdot\mathbf{a} + \mathbf{b}\cdot\mathbf{b} - 2\|\mathbf{a}\|\|\mathbf{b}\|\cos\theta.$$

We can also expand $(\mathbf{b}-\mathbf{a})\cdot(\mathbf{b}-\mathbf{a})$ as follows:

$$\begin{aligned}(\mathbf{b}-\mathbf{a})\cdot(\mathbf{b}-\mathbf{a}) &= \mathbf{b}\cdot(\mathbf{b}-\mathbf{a}) - \mathbf{a}\cdot(\mathbf{b}-\mathbf{a}) \\ &= \mathbf{b}\cdot\mathbf{b} - \mathbf{b}\cdot\mathbf{a} - \mathbf{a}\cdot\mathbf{b} + \mathbf{a}\cdot\mathbf{a} \\ &= \mathbf{a}\cdot\mathbf{a} + \mathbf{b}\cdot\mathbf{b} - 2\mathbf{a}\cdot\mathbf{b}.\end{aligned}$$

Thus,

$$\mathbf{a}\cdot\mathbf{a} + \mathbf{b}\cdot\mathbf{b} - 2\mathbf{a}\cdot\mathbf{b} = \mathbf{a}\cdot\mathbf{a} + \mathbf{b}\cdot\mathbf{b} - 2\|\mathbf{a}\|\|\mathbf{b}\|\cos\theta.$$

That is,

$$\mathbf{a}\cdot\mathbf{b} = \|\mathbf{a}\|\|\mathbf{b}\|\cos\theta. \quad \blacksquare$$

EXAMPLE 4 Find the angle between the vectors $\mathbf{i}+\mathbf{j}+\mathbf{k}$ and $\mathbf{i}+\mathbf{j}-\mathbf{k}$ (see Figure 1.2.7).

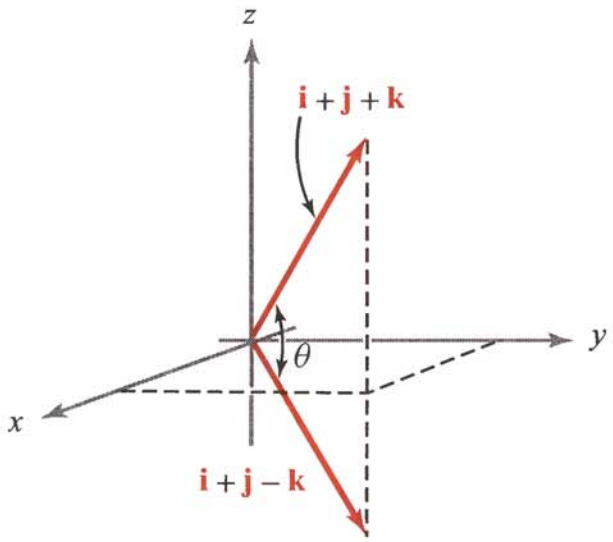


Figure 1.2.7 Finding the angle between $\mathbf{a} = \mathbf{i}+\mathbf{j}+\mathbf{k}$ and $\mathbf{b} = \mathbf{i}+\mathbf{j}-\mathbf{k}$.

SOLUTION Using Theorem 1, we have

$$(\mathbf{i}+\mathbf{j}+\mathbf{k})\cdot(\mathbf{i}+\mathbf{j}-\mathbf{k}) = \|\mathbf{i}+\mathbf{j}+\mathbf{k}\|\|\mathbf{i}+\mathbf{j}-\mathbf{k}\|\cos\theta,$$

and so

$$1+1-1 = (\sqrt{3})(\sqrt{3})\cos\theta.$$

Hence,

$$\cos\theta = \tfrac{1}{3}.$$

That is,

$$\theta = \cos^{-1}(\tfrac{1}{3}) \approx 1.23 \text{ radians } (71°). \quad \blacktriangle$$

The Cauchy–Schwarz Inequality

Theorem 1 shows that the inner product of two vectors is the product of their lengths times the cosine of the angle between them. This relationship is often of value in problems of a geometric nature. An important consequence of Theorem 1 is:

COROLLARY: Cauchy–Schwarz Inequality For any two vectors **a** and **b**, we have

$$|\mathbf{a} \cdot \mathbf{b}| \leq \|\mathbf{a}\| \|\mathbf{b}\|$$

with equality if and only if **a** is a scalar multiple of **b**, or one of them is **0**.

PROOF If **a** is not a scalar multiple of **b**, then θ, the angle between them, is not zero or π, and so $|\cos\theta| < 1$, and thus the inequality holds; in fact, if **a** and **b** are both nonzero, *strict* inequality holds in this case. When **a** is a scalar multiple of **b**, then $\theta = 0$ or π and $|\cos\theta| = 1$, so equality holds in this case. ■

EXAMPLE 5 Verify the Cauchy–Schwarz inequality for $\mathbf{a} = -\mathbf{i} + \mathbf{j} + \mathbf{k}$ and $\mathbf{b} = 3\mathbf{i} + \mathbf{k}$.

SOLUTION The dot product is $\mathbf{a} \cdot \mathbf{b} = -3 + 0 + 1 = -2$, so $|\mathbf{a} \cdot \mathbf{b}| = 2$. Also, $\|\mathbf{a}\| = \sqrt{1+1+1} = \sqrt{3}$ and $\|\mathbf{b}\| = \sqrt{9+1} = \sqrt{10}$, and it is true that $2 \leq \sqrt{3} \cdot \sqrt{10}$ because $\sqrt{3} \cdot \sqrt{10} > \sqrt{3} \cdot \sqrt{3} = 3 \geq 2$. ▲

If **a** and **b** are nonzero vectors in $\mathbb{R}^3$ and θ is the angle between them, we see that $\mathbf{a} \cdot \mathbf{b} = 0$ if and only if $\cos\theta = 0$. Thus, *the inner product of two nonzero vectors is zero if and only if the vectors are perpendicular*. Hence, the inner product provides us with a convenient method for determining whether two vectors are perpendicular. Often we say that perpendicular vectors are ***orthogonal***. The standard basis vectors **i**, **j**, and **k** are mutually orthogonal and of length 1; any such system is called ***orthonormal***. We shall adopt the convention that the zero vector is orthogonal to all vectors.

EXAMPLE 6 The vectors $\mathbf{i}_\theta = (\cos\theta)\mathbf{i} + (\sin\theta)\mathbf{j}$ and $\mathbf{j}_\theta = -(\sin\theta)\mathbf{i} + (\cos\theta)\mathbf{j}$ are orthogonal, because

$$\mathbf{i}_\theta \cdot \mathbf{j}_\theta = -\cos\theta \sin\theta + \sin\theta \cos\theta = 0$$

(see Figure 1.2.8). ▲

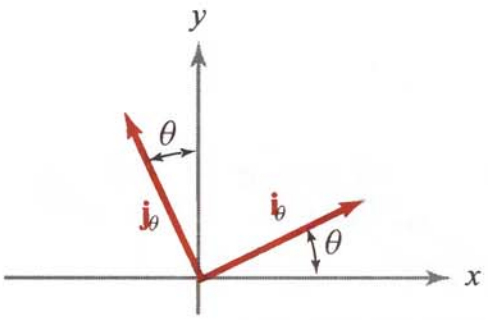


Figure 1.2.8 The vectors $\mathbf{i}_\theta$ and $\mathbf{j}_\theta$ are orthogonal and of unit length, that is, they are orthonormal.

EXAMPLE 7 Let $\mathbf{a}$ and $\mathbf{b}$ be two nonzero orthogonal vectors. If $\mathbf{c}$ is a vector in the plane spanned by $\mathbf{a}$ and $\mathbf{b}$, then there are scalars α and β such that $\mathbf{c} = \alpha\mathbf{a} + \beta\mathbf{b}$. Use the inner product to determine α and β (see Figure 1.2.9).

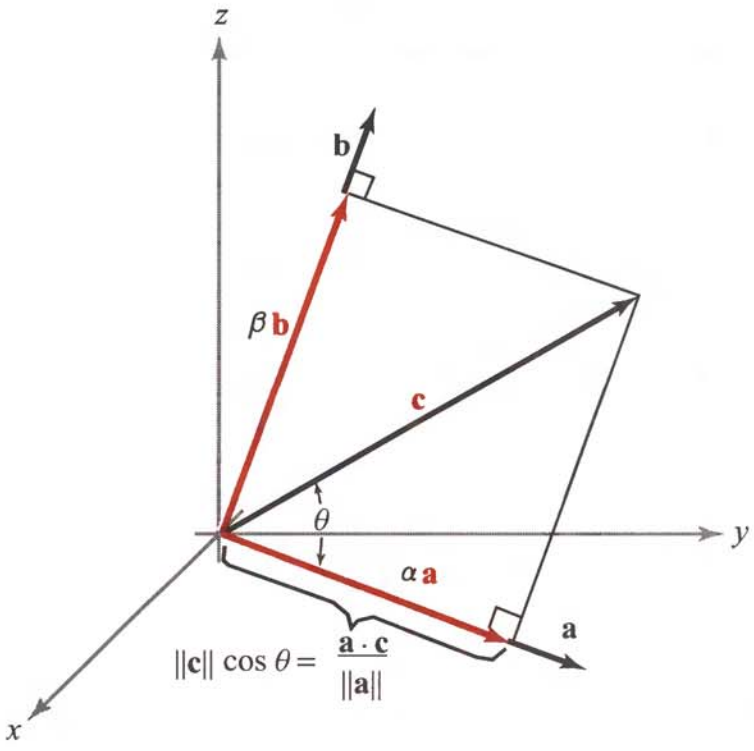


Figure 1.2.9 The geometry for finding α and β, where $\mathbf{c} = \alpha\mathbf{a} + \beta\mathbf{b}$.

SOLUTION Taking the inner product of $\mathbf{a}$ and $\mathbf{c}$, we have

$$\mathbf{a} \cdot \mathbf{c} = \mathbf{a} \cdot (\alpha\mathbf{a} + \beta\mathbf{b}) = \alpha\mathbf{a} \cdot \mathbf{a} + \beta\mathbf{a} \cdot \mathbf{b}.$$

Because $\mathbf{a}$ and $\mathbf{b}$ are orthogonal, $\mathbf{a} \cdot \mathbf{b} = 0$, and so

$$\alpha = \frac{\mathbf{a} \cdot \mathbf{c}}{\mathbf{a} \cdot \mathbf{a}} = \frac{\mathbf{a} \cdot \mathbf{c}}{\|\mathbf{a}\|^2}.$$

Similarly,

$$\beta = \frac{\mathbf{b} \cdot \mathbf{c}}{\mathbf{b} \cdot \mathbf{b}} = \frac{\mathbf{b} \cdot \mathbf{c}}{\|\mathbf{b}\|^2}. \quad \blacktriangle$$

Orthogonal Projection

In the preceding example, the vector $\alpha\mathbf{a}$ is called the ***projection of*** $\mathbf{c}$ ***along*** $\mathbf{a}$, and $\beta\mathbf{b}$ is its ***projection along*** $\mathbf{b}$. Let us formulate this idea more generally. If $\mathbf{v}$ is a vector, and l is the line through the origin in the direction of a vector $\mathbf{a}$, then the ***orthogonal projection of*** $\mathbf{v}$ ***on*** $\mathbf{a}$ is the vector $\mathbf{p}$ whose tip is obtained by dropping a perpendicular line to l from the tip of $\mathbf{v}$, as in Figure 1.2.10.

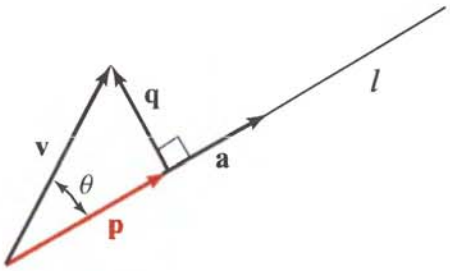


Figure 1.2.10 $\mathbf{p}$ is the orthogonal projection of $\mathbf{v}$ on $\mathbf{a}$.

Referring to the figure, we see that $\mathbf{p}$ is a multiple of $\mathbf{a}$ and that $\mathbf{v}$ is the sum of $\mathbf{p}$ and a vector $\mathbf{q}$ perpendicular to $\mathbf{a}$. Thus,

$$\mathbf{v} = c\mathbf{a} + \mathbf{q},$$

where $\mathbf{p} = c\mathbf{a}$ and $\mathbf{a} \cdot \mathbf{q} = 0$. Taking the dot product of $\mathbf{a}$ with both sides of $\mathbf{v} = c\mathbf{a} + \mathbf{q}$, we find $\mathbf{a} \cdot \mathbf{v} = c\mathbf{a} \cdot \mathbf{a}$, so $c = (\mathbf{a} \cdot \mathbf{v})/(\mathbf{a} \cdot \mathbf{a})$, and hence

$$\mathbf{p} = \frac{\mathbf{a} \cdot \mathbf{v}}{\|\mathbf{a}\|^2}\,\mathbf{a}.$$

The length of $\mathbf{p}$ is

$$\|\mathbf{p}\| = \frac{|\mathbf{a} \cdot \mathbf{v}|}{\|\mathbf{a}\|^2}\,\|\mathbf{a}\| = \frac{|\mathbf{a} \cdot \mathbf{v}|}{\|\mathbf{a}\|} = \|\mathbf{v}\| \cos\theta.$$

Orthogonal Projection The ***orthogonal projection*** of $\mathbf{v}$ on $\mathbf{a}$ is the vector

$$\mathbf{p} = \frac{\mathbf{a} \cdot \mathbf{v}}{\|\mathbf{a}\|^2}\,\mathbf{a}.$$

EXAMPLE 8 Find the orthogonal projection of $\mathbf{i} + \mathbf{j}$ on $\mathbf{i} - 2\mathbf{j}$.

SOLUTION With $\mathbf{a} = \mathbf{i} - 2\mathbf{j}$ and $\mathbf{v} = \mathbf{i} + \mathbf{j}$, the orthogonal projection of $\mathbf{v}$ on $\mathbf{a}$ is

$$\frac{\mathbf{a} \cdot \mathbf{v}}{\mathbf{a} \cdot \mathbf{a}} \mathbf{a} = \frac{1 - 2}{1 + 4} (\mathbf{i} - 2\mathbf{j}) = -\frac{1}{5} (\mathbf{i} - 2\mathbf{j})$$

(see Figure 1.2.11). ▲

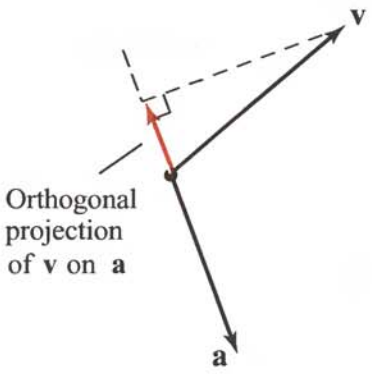


Figure 1.2.11 The orthogonal projection of $\mathbf{v}$ on $\mathbf{a}$ equals $-\frac{1}{5}\mathbf{a}$.

The Triangle Inequality

A useful consequence of the Cauchy–Schwarz inequality, which is called the ***triangle inequality***, relates the lengths of vectors $\mathbf{a}$ and $\mathbf{b}$ and of their sum $\mathbf{a} + \mathbf{b}$. Geometrically, the triangle inequality says that the length of any side of a triangle is no greater than the sum of the lengths of the other two sides (see Figure 1.2.12).

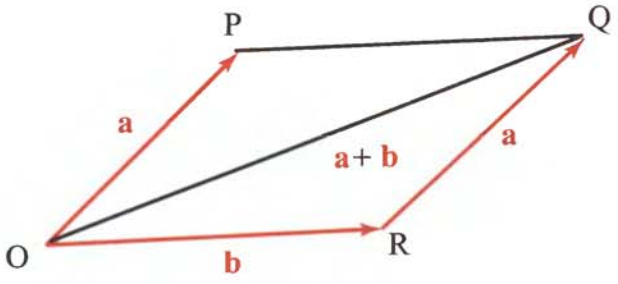


Figure 1.2.12 This geometry shows that $\|OQ\| \leq \|OR\| + \|RQ\|$ or, in vector notation, that $\|\mathbf{a} + \mathbf{b}\| \leq \|\mathbf{a}\| + \|\mathbf{b}\|$, which is the triangle inequality.

THEOREM 2: Triangle Inequality For vectors $\mathbf{a}$ and $\mathbf{b}$ in space,

$$\|\mathbf{a} + \mathbf{b}\| \leq \|\mathbf{a}\| + \|\mathbf{b}\|.$$

PROOF While this result may be clear geometrically, it is useful to give a proof using the Cauchy–Schwarz inequality, as it will generalize to n-dimensional vectors. We consider the square of the left-hand side:

$$\|\mathbf{a}+\mathbf{b}\|^2 = (\mathbf{a}+\mathbf{b})\cdot(\mathbf{a}+\mathbf{b}) = \|\mathbf{a}\|^2 + 2\mathbf{a}\cdot\mathbf{b} + \|\mathbf{b}\|^2.$$

By the Cauchy–Schwarz inequality, we have

$$\|\mathbf{a}\|^2 + 2\mathbf{a}\cdot\mathbf{b} + \|\mathbf{b}\|^2 \leq \|\mathbf{a}\|^2 + 2\|\mathbf{a}\|\|\mathbf{b}\| + \|\mathbf{b}\|^2 = (\|\mathbf{a}\| + \|\mathbf{b}\|)^2.$$

Thus,

$$\|\mathbf{a}+\mathbf{b}\|^2 \leq (\|\mathbf{a}\| + \|\mathbf{b}\|)^2;$$

taking square roots proves the result. ■

EXAMPLE 9

(a) Verify the triangle inequality for $\mathbf{a} = \mathbf{i} + \mathbf{j}$ and $\mathbf{b} = 2\mathbf{i} + \mathbf{j} + \mathbf{k}$.

(b) Prove that $\|\mathbf{u}-\mathbf{v}\| \leq \|\mathbf{u}-\mathbf{w}\| + \|\mathbf{w}-\mathbf{v}\|$ for any vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$. Illustrate with a figure in which $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ have the same base point.

SOLUTION

(a) We have $\mathbf{a}+\mathbf{b} = 3\mathbf{i}+2\mathbf{j}+\mathbf{k}$, so $\|\mathbf{a}+\mathbf{b}\| = \sqrt{9+4+1} = \sqrt{14}$. On the other hand, $\|\mathbf{a}\| = \sqrt{2}$ and $\|\mathbf{b}\| = \sqrt{6}$, so the triangle inequality asserts that $\sqrt{14} \leq \sqrt{2} + \sqrt{6}$. The numbers bear this out: $\sqrt{14} \approx 3.74$, while $\sqrt{2} + \sqrt{6} \approx 1.41 + 2.45 = 3.86$.

(b) We find that $\mathbf{u}-\mathbf{v} = (\mathbf{u}-\mathbf{w}) + (\mathbf{w}-\mathbf{v})$, so the result follows from the triangle inequality with $\mathbf{a}$ replaced by $\mathbf{u}-\mathbf{w}$ and $\mathbf{b}$ replaced by $\mathbf{w}-\mathbf{v}$. Geometrically, we are considering the shaded triangle in Figure 1.2.13. ▲

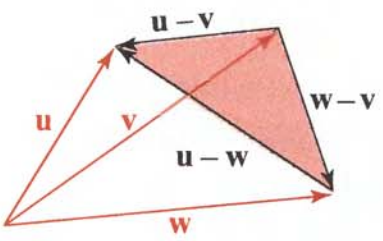


Figure 1.2.13 Illustrating the inequality $\|\mathbf{u}-\mathbf{v}\| \leq \|\mathbf{u}-\mathbf{w}\| + \|\mathbf{w}-\mathbf{v}\|$.

Physical Applications of Vectors

A simple example of a physical quantity represented by a vector is a displacement. Suppose that, on a part of the earth's surface small enough to be considered flat, we introduce coordinates so that the x axis points east, the y axis points north, and the unit of length is the kilometer. If we are at a point P and wish to get to a point Q,

the ***displacement vector*** $\mathbf{d} = \overrightarrow{PQ}$ joining P to Q tells us the direction and distance we have to travel. If x and y are the components of this vector, the displacement of P to Q is "x kilometers east, y kilometers north."

EXAMPLE 10 Suppose that two navigators who cannot see one another but can communicate by radio wish to determine the relative position of their ships. Explain how they can do this if they can each determine their displacement vector to the same lighthouse.

SOLUTION Let P_1 and P_2 be the positions of the ships, and let Q be the position of the lighthouse. The displacement of the lighthouse from the ith ship is the vector $\mathbf{d}_i$ joining P_i to Q. The displacement of the second ship from the first is the vector $\mathbf{d}$ joining P_1 to P_2. We have $\mathbf{d} + \mathbf{d}_2 = \mathbf{d}_1$ (Figure 1.2.14), and so $\mathbf{d} = \mathbf{d}_1 - \mathbf{d}_2$. That is, the displacement from one ship to the other is the difference between the displacements from the ships to the lighthouse. ▲

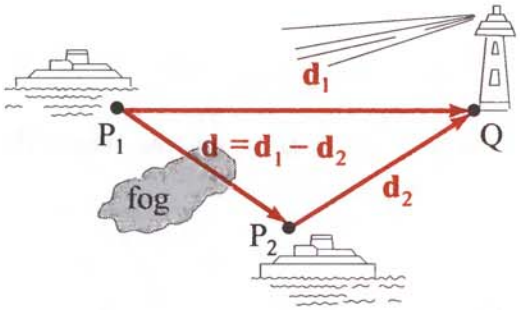


Figure 1.2.14 Vector methods can be used to locate objects.

We can also represent the velocity of a moving object as a vector. For the moment, we will consider only objects moving at uniform speed along straight lines. Suppose, for example, that a boat is steaming across a lake at 10 kilometers per hour (km/h) in the northeast direction. After 1 hour of travel, the displacement is $(10/\sqrt{2}, 10/\sqrt{2}) \approx (7.07, 7.07)$; see Figure 1.2.15.

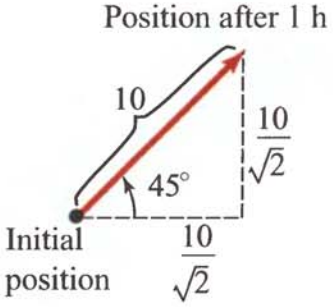


Figure 1.2.15 If an object moves northeast at 10 km/h, its velocity vector has components $(10/\sqrt{2}, 10/\sqrt{2}) = 10(1/\sqrt{2}, 1/\sqrt{2})$, where $(1/\sqrt{2}, 1/\sqrt{2})$ is the unit vector of the northeast direction.

The vector whose components are $(10/\sqrt{2}, 10/\sqrt{2})$ is called the *velocity vector* of the boat. In general, if an object is moving uniformly along a straight line, *its* ***velocity vector*** *is the displacement vector from the position at any moment to the position* 1 *unit of time later*. If a current appears on the lake, moving due eastward at 2 km/h, and the boat continues to point in the same direction with its

engine running at the same rate, its displacement after 1 hour will have components given by $(10/\sqrt{2}+2,\ 10/\sqrt{2})$; see Figure 1.2.16. The new velocity vector, therefore, has components $(10/\sqrt{2}+2,\ 10/\sqrt{2})$. We note that this is the sum of the original velocity vector $(10/\sqrt{2},\ 10/\sqrt{2})$ of the boat and the velocity vector (2, 0) of the current.

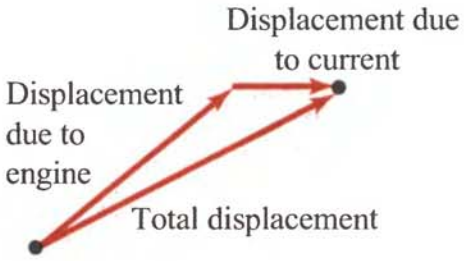


Figure 1.2.16 The total displacement is the sum of the displacements due to the engine and the current.

Displacement and Velocity If an object has a (constant) velocity vector $\mathbf{v}$, then in t units of time the resulting displacement vector of the object is $\mathbf{d} = t\mathbf{v}$; thus, after time $t = 1$, the displacement vector equals the velocity vector. See Figure 1.2.17.

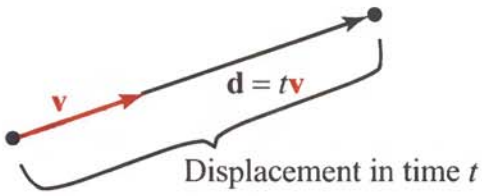


Figure 1.2.17 Displacement = time × velocity.

EXAMPLE 11 A bird is flying in a straight line with velocity vector $10\mathbf{i} + 6\mathbf{j} + \mathbf{k}$ (in kilometers per hour). Suppose that (x, y) are its coordinates on the ground and z is its height above the ground.

(a) If the bird is at position (1, 2, 3) at a certain moment, what is its location 1 hour later? 1 minute later?

(b) How many seconds does it take the bird to climb 10 meters?

SOLUTION (a) The displacement vector from (1, 2, 3) after 1 hour is given by $10\mathbf{i} + 6\mathbf{j} + \mathbf{k}$, so the new position is $(1, 2, 3) + (10, 6, 1) = (11, 8, 4)$. After 1 minute, the displacement vector from (1, 2, 3) is $\frac{1}{60}(10\mathbf{i} + 6\mathbf{j} + \mathbf{k}) = \frac{1}{6}\mathbf{i} + \frac{1}{10}\mathbf{j} + \frac{1}{60}\mathbf{k}$, and so the new position is $(1, 2, 3) + (\frac{1}{6}, \frac{1}{10}, \frac{1}{60}) = (\frac{7}{6}, \frac{21}{10}, \frac{181}{60})$.

(b) After t seconds $(= t/3600$ hours), the displacement vector from (1, 2, 3) is $(t/3600)(10\mathbf{i} + 6\mathbf{j} + \mathbf{k}) = (t/360)\mathbf{i} + (t/600)\mathbf{j} + (t/3600)\mathbf{k}$. The increase in altitude is the z component, namely, $t/3600$. This will equal 10 m $(= \frac{1}{100}$ km) when $t/3600 = \frac{1}{100}$, that is, when $t = 36$ s. ▲

EXAMPLE 12 Physical forces have magnitude and direction and may thus be represented by vectors. If several forces act at once on an object, the resultant force is represented by the sum of the individual force vectors. Suppose that forces $\mathbf{i} + \mathbf{k}$ and $\mathbf{j} + \mathbf{k}$ are acting on a body. What third force $\mathbf{F}$ must we impose to counteract the two—that is, to make the total force equal to zero?

SOLUTION The force $\mathbf{F}$ should be chosen so that $(\mathbf{i} + \mathbf{k}) + (\mathbf{j} + \mathbf{k}) + \mathbf{F} = \mathbf{0}$; that is, $\mathbf{F} = -(\mathbf{i} + \mathbf{k}) - (\mathbf{j} + \mathbf{k}) = -\mathbf{i} - \mathbf{j} - 2\mathbf{k}$. (Recall that $\mathbf{0}$ is the *zero vector*, the vector whose components are all zero.) ▲

EXERCISES

1. Calculate $(3\mathbf{i} + 2\mathbf{j} + \mathbf{k}) \cdot (\mathbf{i} + 2\mathbf{j} - \mathbf{k})$.

2. Calculate $\mathbf{a} \cdot \mathbf{b}$ where $\mathbf{a} = 2\mathbf{i} + 10\mathbf{j} - 12\mathbf{k}$ and $\mathbf{b} = -3\mathbf{i} + 4\mathbf{k}$.

3. Find the angle between $7\mathbf{j} + 19\mathbf{k}$ and $-2\mathbf{i} - \mathbf{j}$ (to the nearest degree).

4. Compute $\mathbf{u} \cdot \mathbf{v}$, where $\mathbf{u} = \sqrt{3}\mathbf{i} - 315\mathbf{j} + 22\mathbf{k}$ and $\mathbf{v} = \mathbf{u}/\|\mathbf{u}\|$.

5. Is $\|8\mathbf{i} - 12\mathbf{k}\| \cdot \|6\mathbf{j} + \mathbf{k}\| - |(8\mathbf{i} - 12\mathbf{k}) \cdot (6\mathbf{j} + \mathbf{k})|$ equal to zero? Explain.

In Exercises 6 to 11, compute $\|\mathbf{u}\|$, $\|\mathbf{v}\|$, *and* $\mathbf{u} \cdot \mathbf{v}$ *for the given vectors in* $\mathbb{R}^3$.

6. $\mathbf{u} = 15\mathbf{i} - 2\mathbf{j} + 4\mathbf{k}$, $\mathbf{v} = \pi\mathbf{i} + 3\mathbf{j} - \mathbf{k}$

7. $\mathbf{u} = 2\mathbf{j} - \mathbf{i}$, $\mathbf{v} = -\mathbf{j} + \mathbf{i}$

8. $\mathbf{u} = 5\mathbf{i} - \mathbf{j} + 2\mathbf{k}$, $\mathbf{v} = \mathbf{i} + \mathbf{j} - \mathbf{k}$

9. $\mathbf{u} = -\mathbf{i} + 3\mathbf{j} + \mathbf{k}$, $\mathbf{v} = -2\mathbf{i} - 3\mathbf{j} - 7\mathbf{k}$

10. $\mathbf{u} = -\mathbf{i} + 3\mathbf{k}$, $\mathbf{v} = 4\mathbf{j}$

11. $\mathbf{u} = -\mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$, $\mathbf{v} = -\mathbf{i} - 3\mathbf{j} + 4\mathbf{k}$

12. Normalize the vectors in Exercises 6 to 8. (Only the solution corresponding to Exercise 7 is in the Student Guide.)

13. Find the angle between the vectors in Exercises 9 to 11. If necessary, express your answer in terms of $\cos^{-1}$.

14. Find the projection of $\mathbf{u} = -\mathbf{i} + \mathbf{j} + \mathbf{k}$ onto $\mathbf{v} = 2\mathbf{i} + \mathbf{j} - 3\mathbf{k}$.

15. Find the projection of $\mathbf{v} = 2\mathbf{i} + \mathbf{j} - 3\mathbf{k}$ onto $\mathbf{u} = -\mathbf{i} + \mathbf{j} + \mathbf{k}$.

16. What restrictions must be made on the scalar b so that the vector $2\mathbf{i} + b\mathbf{j}$ is orthogonal to (a) $-3\mathbf{i} + 2\mathbf{j} + \mathbf{k}$ and (b) $\mathbf{k}$?

17. Find two nonparallel vectors both orthogonal to (1, 1, 1).

18. Find the line through $(3, 1, -2)$ that intersects and is perpendicular to the line $x = -1 + t, y = -2 + t, z = -1 + t$. [HINT: If (x_0, y_0, z_0) is the point of intersection, find its coordinates.]

19. A ship at position $(1, 0)$ on a nautical chart (with north in the positive y direction) sights a rock at position $(2, 4)$. What is the vector joining the ship to the rock? What angle θ does this vector make with due north? (This is called the ***bearing*** of the rock from the ship.)

20. Suppose that the ship in Exercise 19 is pointing due north and traveling at a speed of 4 knots relative to the water. There is a current flowing due east at 1 knot. The units on the chart are nautical miles; 1 knot = 1 nautical mile per hour.

(a) If there were no current, what vector $\mathbf{u}$ would represent the velocity of the ship relative to the sea bottom?

(b) If the ship were just drifting with the current, what vector $\mathbf{v}$ would represent its velocity relative to the sea bottom?

(c) What vector $\mathbf{w}$ represents the total velocity of the ship?

(d) Where would the ship be after 1 hour?

(e) Should the captain change course?

(f) What if the rock were an iceberg?

21. An airplane is located at position $(3, 4, 5)$ at noon and traveling with velocity $400\mathbf{i} + 500\mathbf{j} - \mathbf{k}$ kilometers per hour. The pilot spots an airport at position $(23, 29, 0)$.

(a) At what time will the plane pass directly over the airport? (Assume that the plane is flying over flat ground and that the vector $\mathbf{k}$ points straight up.)

(b) How high above the airport will the plane be when it passes?

22. The wind velocity $\mathbf{v}_1$ is 40 miles per hour (mi/h) from east to west while an airplane travels with air speed $\mathbf{v}_2$ of 100 mi/h due north. The speed of the airplane relative to the ground is the vector sum $\mathbf{v}_1 + \mathbf{v}_2$.

(a) Find $\mathbf{v}_1 + \mathbf{v}_2$.

(b) Draw a figure to scale.

23. A force of 50 lb is directed 50° above horizontal, pointing to the right. Determine its horizontal and vertical components. Display all results in a figure.

24. Two persons pull horizontally on ropes attached to a post, the angle between the ropes being 60°. Person A pulls with a force of 150 lb, while B pulls with a force of 110 lb.

(a) The resultant force is the vector sum of the two forces. Draw a figure to scale that graphically represents the three forces.

(b) Using trigonometry, determine formulas for the vector components of the two forces in a conveniently chosen coordinate system. Perform the algebraic addition, and find the angle the resultant force makes with A.

25. A 1-kilogram (1-kg) mass located at the origin is suspended by ropes attached to the two points $(1, 1, 1)$ and $(-1, -1, 1)$. If the force of gravity is pointing in the direction of the vector $-\mathbf{k}$, what is the vector describing the force along each rope? [HINT: Use the symmetry of the problem. A 1-kg mass weighs 9.8 newtons (N).]

26. Suppose that an object moving in direction $\mathbf{i} + \mathbf{j}$ is acted on by a force given by the vector $2\mathbf{i} + \mathbf{j}$. Express this force as a sum of a force in the direction of motion and a force perpendicular to the direction of motion.

27. A force of 6 N (newtons) makes an angle of $\pi/4$ radian with the y axis, pointing to the right. The force acts against the movement of an object along the straight line connecting $(1, 2)$ to $(5, 4)$.

(a) Find a formula for the force vector $\mathbf{F}$.

(b) Find the angle θ between the displacement direction $\mathbf{D} = (5 - 1)\mathbf{i} + (4 - 2)\mathbf{j}$ and the force direction $\mathbf{F}$.

(c) The ***work done*** is $\mathbf{F} \cdot \mathbf{D}$, or equivalently, $\|\mathbf{F}\|\|\mathbf{D}\| \cos\theta$. Compute the work from both formulas and compare.

1.3 Matrices, Determinants, and the Cross Product

In Section 1.2 we defined a product of vectors that was a scalar. In this section we shall define a product of vectors that is a vector; that is, we shall show how, given two vectors $\mathbf{a}$ and $\mathbf{b}$, we can produce a third vector $\mathbf{a} \times \mathbf{b}$, called the *cross product* of $\mathbf{a}$ and $\mathbf{b}$. This new vector will have the pleasing geometric property that it is perpendicular to the plane spanned (determined) by $\mathbf{a}$ and $\mathbf{b}$. The definition of the cross product is based on the notions of the matrix and the determinant, and so these are developed first. Once this has been accomplished, we can study the geometric implications of the mathematical structure we have built.

2×2 Matrices

We define a 2×2 ***matrix*** to be an array

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

where a_{11}, a_{12}, a_{21}, and a_{22} are four scalars. For example,

$$\begin{bmatrix} 2 & 1 \\ 0 & 4 \end{bmatrix}, \quad \begin{bmatrix} -1 & 0 \\ 1 & 1 \end{bmatrix}, \quad \text{and} \quad \begin{bmatrix} 13 & 7 \\ 6 & 11 \end{bmatrix}$$

are 2×2 matrices. The ***determinant***

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$$

of such a matrix is the real number defined by the equation

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}. \tag{1}$$

EXAMPLE 1

$$\begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} = 1 - 1 = 0; \qquad \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = 4 - 6 = -2; \qquad \begin{vmatrix} 5 & 6 \\ 7 & 8 \end{vmatrix} = 40 - 42 = -2. \quad \blacktriangle$$

3×3 Matrices

A 3×3 ***matrix*** is an array

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

where, again, each a_{ij} is a scalar; a_{ij} denotes the entry in the array that is in the ith row and the jth column. We define the ***determinant*** of a 3×3 matrix by the rule

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}. \tag{2}$$

Without some mnemonic device, formula (2) would be difficult to memorize. The rule to learn is that you move along the first row, multiplying a_{1j} by the determinant of the 2×2 matrix obtained by canceling out the first row and the jth column, and then you add these up, remembering to put a minus in front of the a_{12} term. For example, the determinant multiplied by the middle term of formula (2), namely,

$$\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix},$$

is obtained by crossing out the first row and the second column of the given 3×3 matrix:

$$\begin{bmatrix} \cancel{a_{11}} & \cancel{a_{12}} & \cancel{a_{13}} \\ a_{21} & \cancel{a_{22}} & a_{23} \\ a_{31} & \cancel{a_{32}} & a_{33} \end{bmatrix}.$$

EXAMPLE 2

$$\begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 1 \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} - 0 \begin{vmatrix} 0 & 0 \\ 0 & 1 \end{vmatrix} + 0 \begin{vmatrix} 0 & 1 \\ 0 & 0 \end{vmatrix} = 1.$$

$$\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix} = 1 \begin{vmatrix} 5 & 6 \\ 8 & 9 \end{vmatrix} - 2 \begin{vmatrix} 4 & 6 \\ 7 & 9 \end{vmatrix} + 3 \begin{vmatrix} 4 & 5 \\ 7 & 8 \end{vmatrix} = -3 + 12 - 9 = 0. \quad \blacktriangle$$

Properties of Determinants

An important property of determinants is that interchanging two rows or two columns results in a change of sign. For 2×2 determinants, this is a consequence of the definition as follows: For rows, we have

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{21}a_{12} = -(a_{21}a_{12} - a_{11}a_{22}) = -\begin{vmatrix} a_{21} & a_{22} \\ a_{11} & a_{12} \end{vmatrix}$$

and for columns,

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = -(a_{12}a_{21} - a_{11}a_{22}) = -\begin{vmatrix} a_{12} & a_{11} \\ a_{22} & a_{21} \end{vmatrix}.$$

We leave it to the reader to verify this property for the 3×3 case.

A second fundamental property of determinants is that *we can factor scalars out of any row or column*. For 2×2 determinants, this means

$$\begin{vmatrix} \alpha a_{11} & a_{12} \\ \alpha a_{21} & a_{22} \end{vmatrix} = \begin{vmatrix} a_{11} & \alpha a_{12} \\ a_{21} & \alpha a_{22} \end{vmatrix} = \alpha \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = \begin{vmatrix} \alpha a_{11} & \alpha a_{12} \\ a_{21} & a_{22} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{12} \\ \alpha a_{21} & \alpha a_{22} \end{vmatrix}.$$

Similarly, for 3×3 determinants we have

$$\begin{vmatrix} \alpha a_{11} & \alpha a_{12} & \alpha a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \alpha \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & \alpha a_{12} & a_{13} \\ a_{21} & \alpha a_{22} & a_{23} \\ a_{31} & \alpha a_{32} & a_{33} \end{vmatrix},$$

and so on. These results follow from the definitions. In particular, if any row or column consists of zeros, then the value of the determinant is zero.

A third fundamental fact about determinants is the following: *If we change a row (or column) by adding another row (or, respectively, column) to it, the value of the determinant remains the same.* For the 2×2 case, this means that

$$\begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} = \begin{vmatrix} a_1 + b_1 & a_2 + b_2 \\ b_1 & b_2 \end{vmatrix} = \begin{vmatrix} a_1 & a_2 \\ b_1 + a_1 & b_2 + a_2 \end{vmatrix}$$

$$= \begin{vmatrix} a_1 + a_2 & a_2 \\ b_1 + b_2 & b_2 \end{vmatrix} = \begin{vmatrix} a_1 & a_1 + a_2 \\ b_1 & b_1 + b_2 \end{vmatrix}.$$

For the 3×3 case, this means

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 + b_1 & a_2 + b_2 & a_3 + b_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 + a_2 & a_2 & a_3 \\ b_1 + b_2 & b_2 & b_3 \\ c_1 + c_2 & c_2 & c_3 \end{vmatrix},$$

and so on. Again, this property can be proved using the definition of the determinant.

EXAMPLE 3 Suppose

$$\mathbf{a} = \alpha\mathbf{b} + \beta\mathbf{c}; \qquad \text{that is, } \mathbf{a} = (a_1, a_2, a_3) = \alpha(b_1, b_2, b_3) + \beta(c_1, c_2, c_3).$$

Show that

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = 0.$$

SOLUTION We shall prove the case $\alpha \neq 0$, $\beta \neq 0$. The case $\alpha = 0 = \beta$ is trivial, and the case where exactly one of α, β is zero is a simple modification of the case we prove. Using the fundamental properties of determinants, the determinant in question is

$$\begin{vmatrix} \alpha b_1 + \beta c_1 & \alpha b_2 + \beta c_2 & \alpha b_3 + \beta c_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

$$= -\frac{1}{\alpha} \begin{vmatrix} \alpha b_1 + \beta c_1 & \alpha b_2 + \beta c_2 & \alpha b_3 + \beta c_3 \\ -\alpha b_1 & -\alpha b_2 & -\alpha b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

(factoring $-1/\alpha$ out of the second row)

$$= \left(-\frac{1}{\alpha}\right)\left(-\frac{1}{\beta}\right) \begin{vmatrix} \alpha b_1 + \beta c_1 & \alpha b_2 + \beta c_2 & \alpha b_3 + \beta c_3 \\ -\alpha b_1 & -\alpha b_2 & -\alpha b_3 \\ -\beta c_1 & -\beta c_2 & -\beta c_3 \end{vmatrix}$$

(factoring $-1/\beta$ out of the third row)

$$= \frac{1}{\alpha\beta} \begin{vmatrix} \beta c_1 & \beta c_2 & \beta c_3 \\ -\alpha b_1 & -\alpha b_2 & -\alpha b_3 \\ -\beta c_1 & -\beta c_2 & -\beta c_3 \end{vmatrix}$$

(adding the second row to the first row)

$$= \frac{1}{\alpha\beta} \begin{vmatrix} 0 & 0 & 0 \\ -\alpha b_1 & -\alpha b_2 & -\alpha b_3 \\ -\beta c_1 & -\beta c_2 & -\beta c_3 \end{vmatrix}$$

(adding the third row to the first row)

$$= 0. \quad \blacktriangle$$

Closely related to these properties is the fact that *we can expand a* 3×3 *determinant along any row or column* using the signs in the following checkerboard pattern:

$$\begin{vmatrix} + & - & + \\ - & + & - \\ + & - & + \end{vmatrix}$$

For instance, the reader can check that we can expand "by minors" along the middle row:

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = -a_{21}\begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} + a_{22}\begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} - a_{23}\begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix}.$$

Let us redo the second determinant in Example 2 using this formula:

$$\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix} = -4\begin{vmatrix} 2 & 3 \\ 8 & 9 \end{vmatrix} + 5\begin{vmatrix} 1 & 3 \\ 7 & 9 \end{vmatrix} - 6\begin{vmatrix} 1 & 2 \\ 7 & 8 \end{vmatrix} = (-4)(-6) + (5)(12) + (-6)(6) = 0.$$

Historical Note

Determinants appear to have been invented and first used by Leibniz in 1693, in connection with solutions of linear equations. Maclaurin and Cramer developed their properties between 1729 and 1750; in particular, they showed that the solution of the system of equations

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 &= b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 &= b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 &= b_3 \end{aligned}$$

is

$$x_1 = \frac{1}{\Delta}\begin{vmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{vmatrix}, \quad x_2 = \frac{1}{\Delta}\begin{vmatrix} a_{11} & b_1 & a_{13} \\ a_{21} & b_2 & a_{23} \\ a_{31} & b_3 & a_{33} \end{vmatrix}, \quad x_3 = \frac{1}{\Delta}\begin{vmatrix} a_{11} & a_{12} & b_1 \\ a_{21} & a_{22} & b_2 \\ a_{31} & a_{32} & b_3 \end{vmatrix}$$

where

$$\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix},$$

a fact now known as ***Cramer's rule***. While this method is rather inefficient from a numerical point of view, it is of theoretical importance in matrix theory. Later, Vandermonde (1772) and Cauchy (1812), treating

determinants as a separate topic worthy of special attention, developed the field more systematically, with contributions by Laplace, Jacobi, and others. Formulas for volumes of parallelepipeds in terms of determinants are due to Lagrange (1775). We shall study these later in this section. Although during the nineteenth century mathematicians studied matrices and determinants, the subjects were considered separate. For the full history up to 1900, see T. Muir, *The Theory of Determinants in the Historical Order of Development* (reprinted by Dover, New York, 1960).

Cross Products

Now that we have established the necessary properties of determinants and discussed their history, we are ready to proceed with the cross product of vectors.

DEFINITION: The Cross Product Suppose that $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$ and $\mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$ are vectors in $\mathbb{R}^3$. The ***cross product*** or ***vector product*** of $\mathbf{a}$ and $\mathbf{b}$, denoted $\mathbf{a} \times \mathbf{b}$, is defined to be the vector

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \mathbf{k},$$

or, symbolically,

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}.$$

Even though we only defined determinants for arrays of *real* numbers, this formal expression involving *vectors* is a useful memory aid for the cross product.

EXAMPLE 4 Find $(3\mathbf{i} - \mathbf{j} + \mathbf{k}) \times (\mathbf{i} + 2\mathbf{j} - \mathbf{k})$.

SOLUTION

$$(3\mathbf{i} - \mathbf{j} + \mathbf{k}) \times (\mathbf{i} + 2\mathbf{j} - \mathbf{k}) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & -1 & 1 \\ 1 & 2 & -1 \end{vmatrix} = -\mathbf{i} + 4\mathbf{j} + 7\mathbf{k}. \quad \blacktriangle$$

Certain algebraic properties of the cross product follow from the definition. If $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ are vectors and α, β, and γ are scalars, then

(i) $\mathbf{a} \times \mathbf{b} = -(\mathbf{b} \times \mathbf{a})$

(ii) $\mathbf{a} \times (\beta\mathbf{b} + \gamma\mathbf{c}) = \beta(\mathbf{a} \times \mathbf{b}) + \gamma(\mathbf{a} \times \mathbf{c})$ and $(\alpha\mathbf{a} + \beta\mathbf{b}) \times \mathbf{c} = \alpha(\mathbf{a} \times \mathbf{c}) + \beta(\mathbf{b} \times \mathbf{c})$.

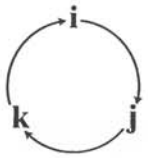
Note that $\mathbf{a} \times \mathbf{a} = -(\mathbf{a} \times \mathbf{a})$, by property (i). Thus, $\mathbf{a} \times \mathbf{a} = \mathbf{0}$. In particular,

$$\mathbf{i} \times \mathbf{i} = \mathbf{0}, \qquad \mathbf{j} \times \mathbf{j} = \mathbf{0}, \qquad \mathbf{k} \times \mathbf{k} = \mathbf{0}.$$

Also,

$$\mathbf{i} \times \mathbf{j} = \mathbf{k}, \qquad \mathbf{j} \times \mathbf{k} = \mathbf{i}, \qquad \mathbf{k} \times \mathbf{i} = \mathbf{j},$$

which can be remembered by cyclicly permuting $\mathbf{i}, \mathbf{j}, \mathbf{k}$ like this:



To give a geometric interpretation of the cross product, we first introduce the triple product. Given three vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$, the real number

$$(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}$$

is called the ***triple product*** of $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ (in that order). To obtain a formula for it, let $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$, $\mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$, and $\mathbf{c} = c_1\mathbf{i} + c_2\mathbf{j} + c_3\mathbf{k}$. Then

$$\begin{aligned}(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} &= \left(\begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \mathbf{k} \right) \cdot (c_1\mathbf{i} + c_2\mathbf{j} + c_3\mathbf{k}) \\ &= \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} c_1 - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} c_2 + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} c_3.\end{aligned}$$

This is the expansion by minors of the third row of the determinant, so

$$(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}.$$

If $\mathbf{c}$ is a vector in the plane spanned by the vectors $\mathbf{a}$ and $\mathbf{b}$, then the third row in the determinant expression for $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}$ is a linear combination of the first and second rows, and therefore $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = 0$. In other words, *the vector* $\mathbf{a} \times \mathbf{b}$ *is orthogonal to any vector in the plane spanned by* $\mathbf{a}$ *and* $\mathbf{b}$, *in particular to both* $\mathbf{a}$ *and* $\mathbf{b}$.

Next, we calculate the length of $\mathbf{a} \times \mathbf{b}$. Note that

$$\begin{aligned}\|\mathbf{a} \times \mathbf{b}\|^2 &= \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix}^2 + \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix}^2 + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix}^2 \\ &= (a_2b_3 - a_3b_2)^2 + (a_1b_3 - b_1a_3)^2 + (a_1b_2 - b_1a_2)^2.\end{aligned}$$

If we expand the terms in the last expression, we can recollect them to give

$$(a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2) - (a_1b_1 + a_2b_2 + a_3b_3)^2,$$

which equals

$$\|\mathbf{a}\|^2\|\mathbf{b}\|^2 - (\mathbf{a}\cdot\mathbf{b})^2 = \|\mathbf{a}\|^2\|\mathbf{b}\|^2 - \|\mathbf{a}\|^2\|\mathbf{b}\|^2\cos^2\theta = \|\mathbf{a}\|^2\|\mathbf{b}\|^2\sin^2\theta$$

where θ is the angle between $\mathbf{a}$ and $\mathbf{b}$, $0 \le \theta \le \pi$. Taking square roots and using $\sqrt{k^2} = |k|$, we find that $\|\mathbf{a}\times\mathbf{b}\| = \|\mathbf{a}\|\|\mathbf{b}\||\sin\theta|$.

Combining our results, we conclude that $\mathbf{a}\times\mathbf{b}$ is a *vector perpendicular to the plane $\mathcal{P}$ spanned by* $\mathbf{a}$ *and* $\mathbf{b}$ *with length* $\|\mathbf{a}\|\|\mathbf{b}\||\sin\theta|$. We see from Figure 1.3.1 that this length is also the area of the parallelogram (with base $\|\mathbf{a}\|$ and height $\|\mathbf{b}\sin\theta\|$) spanned by $\mathbf{a}$ and $\mathbf{b}$. There are still two possible vectors that satisfy these conditions, because there are two choices of direction that are perpendicular (or normal) to $\mathcal{P}$. This is clear from Figure 1.3.1, which shows the two choices $\mathbf{n}_1$ and $-\mathbf{n}_1$ perpendicular to $\mathcal{P}$, with $\|\mathbf{n}_1\| = \|-\mathbf{n}_1\| = \|\mathbf{a}\|\|\mathbf{b}\||\sin\theta|$.

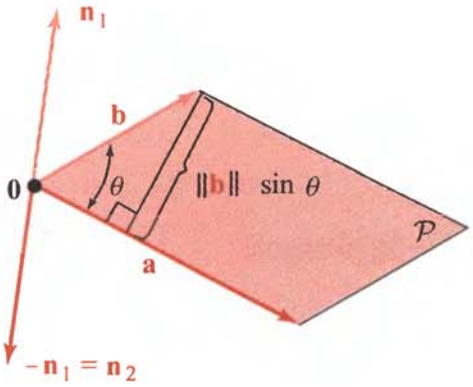


Figure 1.3.1 $\mathbf{n}_1$ and $\mathbf{n}_2$ are the two possible vectors orthogonal to both $\mathbf{a}$ and $\mathbf{b}$, and with norm $\|\mathbf{a}\|\ \|\mathbf{b}\|\ |\sin\theta|$.

Which vector represents $\mathbf{a}\times\mathbf{b}$, $\mathbf{n}_1$ or $-\mathbf{n}_1$? The answer is $\mathbf{n}_1$. Try a few cases such as $\mathbf{k} = \mathbf{i}\times\mathbf{j}$ to verify this. The following "right-hand rule" determines the direction of $\mathbf{a}\times\mathbf{b}$ in general. Take your *right hand* and place it so your fingers curl from $\mathbf{a}$ toward $\mathbf{b}$ through the *acute* angle θ, as in Figure 1.3.2. Then your thumb points in the direction of $\mathbf{a}\times\mathbf{b}$.

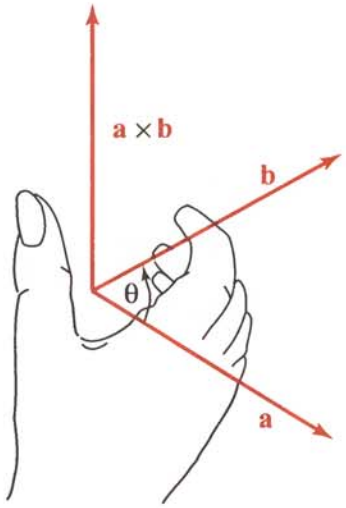


Figure 1.3.2 The right-hand rule for determining in which of the two possible directions $\mathbf{a}\times\mathbf{b}$ points.

The Cross Product

Geometric definition: $\mathbf{a} \times \mathbf{b}$ is the vector such that:

(1) $\|\mathbf{a} \times \mathbf{b}\| = \|\mathbf{a}\| \|\mathbf{b}\| \sin\theta$, the area of the parallelogram spanned by $\mathbf{a}$ and $\mathbf{b}$ (θ is the angle between $\mathbf{a}$ and $\mathbf{b}$; $0 \leq \theta \leq \pi$); see Figure 1.3.3.

(2) $\mathbf{a} \times \mathbf{b}$ is perpendicular to $\mathbf{a}$ and $\mathbf{b}$, and the triple $(\mathbf{a}, \mathbf{b}, \mathbf{a} \times \mathbf{b})$ obeys the right-hand rule.

Component formula:

$$(a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}) \times (b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$$= (a_2b_3 - a_3b_2)\mathbf{i} - (a_1b_3 - a_3b_1)\mathbf{j} + (a_1b_2 - a_2b_1)\mathbf{k}.$$

Algebraic rules:

1. $\mathbf{a} \times \mathbf{b} = \mathbf{0}$ if and only if $\mathbf{a}$ and $\mathbf{b}$ are parallel or $\mathbf{a}$ or $\mathbf{b}$ is zero.
2. $\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$.
3. $\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}$.
4. $(\mathbf{a} + \mathbf{b}) \times \mathbf{c} = \mathbf{a} \times \mathbf{c} + \mathbf{b} \times \mathbf{c}$.
5. $(\alpha\mathbf{a}) \times \mathbf{b} = \alpha(\mathbf{a} \times \mathbf{b})$.

Multiplication table:

		Second factor		
	$\times$	$\mathbf{i}$	$\mathbf{j}$	$\mathbf{k}$
	$\mathbf{i}$	$\mathbf{0}$	$\mathbf{k}$	$-\mathbf{j}$
First	$\mathbf{j}$	$-\mathbf{k}$	$\mathbf{0}$	$\mathbf{i}$
factor	$\mathbf{k}$	$\mathbf{j}$	$-\mathbf{i}$	$\mathbf{0}$

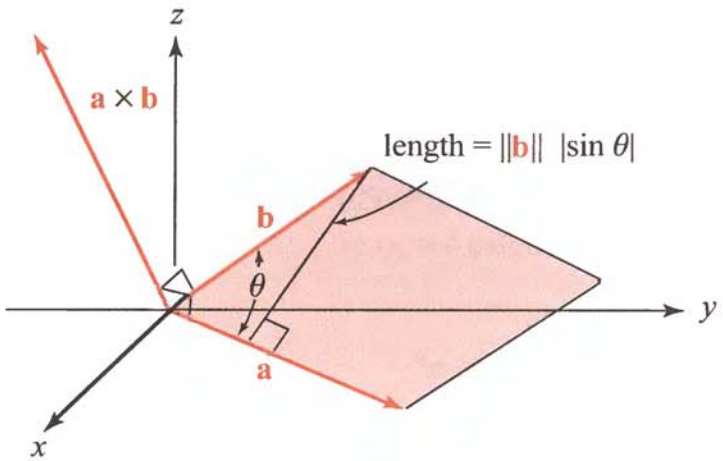


Figure 1.3.3 The length of $\mathbf{a} \times \mathbf{b}$ is the area of the parallelogram formed by $\mathbf{a}$ and $\mathbf{b}$.

EXAMPLE 5 Find the area of the parallelogram spanned by the two vectors $\mathbf{a} = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$ and $\mathbf{b} = -\mathbf{i} - \mathbf{k}$.

SOLUTION We calculate the cross product of $\mathbf{a}$ and $\mathbf{b}$ by applying the component or determinant formula, with $a_1 = 1$, $a_2 = 2$, $a_3 = 3$, $b_1 = -1$, $b_2 = 0$, $b_3 = -1$:

$$\begin{aligned}\mathbf{a} \times \mathbf{b} &= [(2)(-1) - (3)(0)]\mathbf{i} + [(3)(-1) - (1)(-1)]\mathbf{j} + [(1)(0) - (2)(-1)]\mathbf{k} \\ &= -2\mathbf{i} - 2\mathbf{j} + 2\mathbf{k}.\end{aligned}$$

Thus, the area is

$$\|\mathbf{a} \times \mathbf{b}\| = \sqrt{(-2)^2 + (-2)^2 + (2)^2} = 2\sqrt{3}. \quad \blacktriangle$$

EXAMPLE 6 Find a unit vector orthogonal to the vectors $\mathbf{i} + \mathbf{j}$ and $\mathbf{j} + \mathbf{k}$.

SOLUTION A vector perpendicular to both $\mathbf{i} + \mathbf{j}$ and $\mathbf{j} + \mathbf{k}$ is their cross product, namely, the vector

$$(\mathbf{i} + \mathbf{j}) \times (\mathbf{j} + \mathbf{k}) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix} = \mathbf{i} - \mathbf{j} + \mathbf{k}.$$

Because $\|\mathbf{i} - \mathbf{j} + \mathbf{k}\| = \sqrt{3}$, the vector

$$\frac{1}{\sqrt{3}}(\mathbf{i} - \mathbf{j} + \mathbf{k})$$

is a unit vector perpendicular to $\mathbf{i} + \mathbf{j}$ and $\mathbf{j} + \mathbf{k}$. $\blacktriangle$

EXAMPLE 7 Derive an identity relating the dot and cross products from the formulas

$$\|\mathbf{u} \times \mathbf{v}\| = \|\mathbf{u}\|\|\mathbf{v}\| \sin\theta \qquad \text{and} \qquad \mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\|\|\mathbf{v}\| \cos\theta$$

by eliminating θ.

SOLUTION Seeing $\sin\theta$ and $\cos\theta$ multiplied by the same expression suggests squaring the two formulas and adding the results. We get

$$\|\mathbf{u} \times \mathbf{v}\|^2 + (\mathbf{u} \cdot \mathbf{v})^2 = \|\mathbf{u}\|^2\|\mathbf{v}\|^2(\sin^2\theta + \cos^2\theta) = \|\mathbf{u}\|^2\|\mathbf{v}\|^2,$$

so

$$\|\mathbf{u} \times \mathbf{v}\|^2 = \|\mathbf{u}\|^2\|\mathbf{v}\|^2 - (\mathbf{u} \cdot \mathbf{v})^2.$$

This identity is interesting because it establishes a link between the dot and cross products. $\blacktriangle$

Geometry of Determinants

Using the cross product, we may obtain a basic geometric interpretation of 2×2 and 3×3 determinants. Let $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j}$ and $\mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j}$ be two vectors in the plane. If θ is the angle between $\mathbf{a}$ and $\mathbf{b}$, we have seen that $\|\mathbf{a} \times \mathbf{b}\| = \|\mathbf{a}\|\|\mathbf{b}\||\sin\theta|$ is the area of the parallelogram with adjacent sides $\mathbf{a}$ and $\mathbf{b}$. The cross product as a determinant is

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & 0 \\ b_1 & b_2 & 0 \end{vmatrix} = \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \mathbf{k}.$$

Thus, the area $\|\mathbf{a} \times \mathbf{b}\|$ is the absolute value of the determinant

$$\begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} = a_1b_2 - a_2b_1.$$

Geometry of 2×2 Determinants The absolute value of the determinant $\begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix}$ is the area of the parallelogram whose adjacent sides are the vectors $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j}$ and $\mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j}$. The sign of the determinant is + when, rotating in the counterclockwise direction, the angle from $\mathbf{a}$ to $\mathbf{b}$ is less than π.

EXAMPLE 8 Find the area of the triangle with vertices at the points (1, 1), (0, 2), and (3, 2) (Figure 1.3.4).

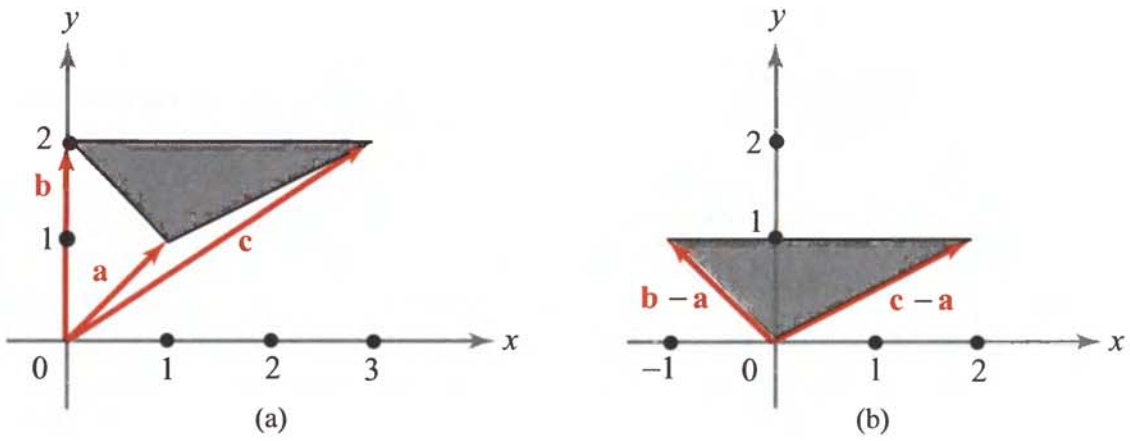


Figure 1.3.4 (a) Find the area A of the shaded triangle by expressing the sides as vector differences (b) to get $A = \|(\mathbf{b} - \mathbf{a}) \times (\mathbf{c} - \mathbf{a})\|/2$.

SOLUTION Let $\mathbf{a} = \mathbf{i} + \mathbf{j}$, $\mathbf{b} = 2\mathbf{j}$, and $\mathbf{c} = 3\mathbf{i} + 2\mathbf{j}$. It is clear that the triangle whose vertices are the endpoints of the vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ has the same area as the triangle with vertices at $\mathbf{0}$, $\mathbf{b} - \mathbf{a}$, and $\mathbf{c} - \mathbf{a}$ (Figure 1.3.4). Indeed, the latter is merely a

translation of the former triangle. Because the area of this translated triangle is one-half the area of the parallelogram with adjacent sides $\mathbf{b} - \mathbf{a} = -\mathbf{i} + \mathbf{j}$, and $\mathbf{c} - \mathbf{a} = 2\mathbf{i} + \mathbf{j}$, we find that the area of the triangle with vertices (1, 1), (0, 2), and (3, 2) is the absolute value of

$$\frac{1}{2}\begin{vmatrix} -1 & 1 \\ 2 & 1 \end{vmatrix} = -\frac{3}{2},$$

that is, 3/2. ▲

There is an interpretation of determinants of 3×3 matrices as volumes that is analogous to the interpretation of determinants of 2×2 matrices as areas.

Geometry of 3×3 Determinants The absolute value of the determinant

$$D = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

is the volume of the parallelepiped whose adjacent sides are the vectors

$$\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}, \qquad \mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}, \qquad \text{and} \qquad \mathbf{c} = c_1\mathbf{i} + c_2\mathbf{j} + c_3\mathbf{k}.$$

To prove the statement in the box above, we refer to Figure 1.3.5 and note that the length of the cross product, namely, $\|\mathbf{a} \times \mathbf{b}\|$, is the area of the parallelogram with adjacent sides $\mathbf{a}$ and $\mathbf{b}$. Moreover, $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = \|\mathbf{a} \times \mathbf{b}\| \|\mathbf{c}\| \cos \psi$, where ψ is the angle that $\mathbf{c}$ makes with the normal to the plane spanned by $\mathbf{a}$ and $\mathbf{b}$. Because the volume of the parallelepiped with adjacent sides $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ is the product of the area of the base $\|\mathbf{a} \times \mathbf{b}\|$ and the altitude $\|\mathbf{c}\| |\cos \psi|$, it follows that the volume is $|(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}|$. We saw earlier that $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = D$, so the volume equals the absolute value of D.

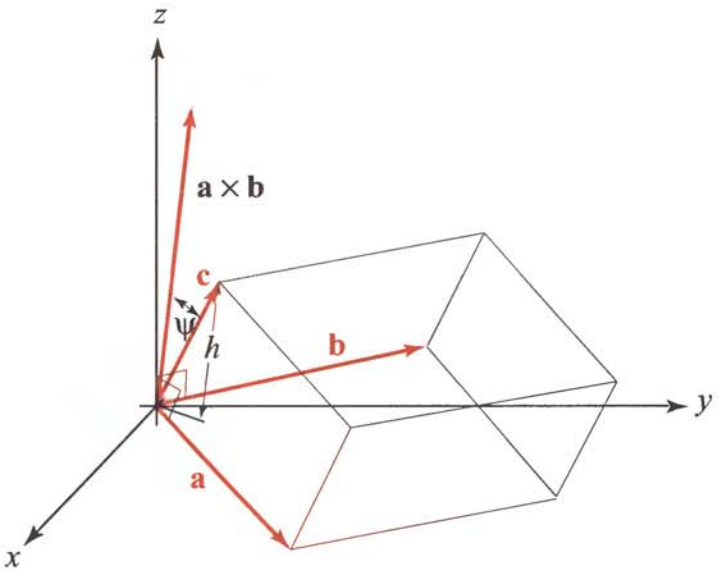


Figure 1.3.5 The volume of the parallelepiped spanned by $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ is the absolute value of the determinant of the 3×3 matrix having $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ as its rows.

EXAMPLE 9 Find the volume of the parallelepiped spanned by the three vectors $\mathbf{i} + 3\mathbf{k}$, $2\mathbf{i} + \mathbf{j} - 2\mathbf{k}$, and $5\mathbf{i} + 4\mathbf{k}$.

SOLUTION The volume is the absolute value of

$$\begin{vmatrix} 1 & 0 & 3 \\ 2 & 1 & -2 \\ 5 & 0 & 4 \end{vmatrix}.$$

If we expand this determinant by minors by going down the second column, the only nonzero term is

$$\begin{vmatrix} 1 & 3 \\ 5 & 4 \end{vmatrix}(1) = -11,$$

so the volume equals 11. ▲

Equations of Planes

Let $\mathcal{P}$ be a plane in space, $P_0 = (x_0, y_0, z_0)$ a point on that plane, and suppose that $\mathbf{n} = A\mathbf{i} + B\mathbf{j} + C\mathbf{k}$ is a vector normal to that plane (see Figure 1.3.6). Let $P = (x, y, z)$ be a point in $\mathbb{R}^3$. Then P lies on the plane $\mathcal{P}$ if and only if the vector $\overrightarrow{P_0P} = (x - x_0)\mathbf{i} + (y - y_0)\mathbf{j} + (z - z_0)\mathbf{k}$ is perpendicular to $\mathbf{n}$, that is, $\overrightarrow{P_0P} \cdot \mathbf{n} = 0$, or, equivalently,

$$(A\mathbf{i} + B\mathbf{j} + C\mathbf{k}) \cdot [(x - x_0)\mathbf{i} + (y - y_0)\mathbf{j} + (z - z_0)\mathbf{k}] = 0.$$

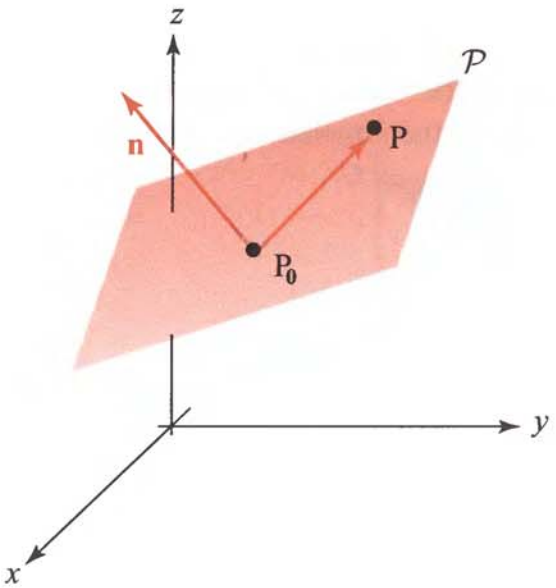


Figure 1.3.6 The points P of the plane through P_0 and perpendicular to $\mathbf{n}$ satisfy the equation $\overrightarrow{P_0P} \cdot \mathbf{n} = 0$.

Thus,

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0.$$

Equation of a Plane in Space The equation of the plane $\mathcal{P}$ through (x_0, y_0, z_0) that has a normal vector $\mathbf{n} = A\mathbf{i} + B\mathbf{j} + C\mathbf{k}$ is

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0;$$

that is, $(x, y, z) \in \mathcal{P}$ if and only if

$$Ax + By + Cz + D = 0$$

where $D = -Ax_0 - By_0 - Cz_0$.

The four numbers A, B, C, and D are not determined uniquely by the plane $\mathcal{P}$. To see this, note that (x, y, z) satisfies the equation $Ax + By + Cz + D = 0$ if and only if it also satisfies the relation

$$(\lambda A)x + (\lambda B)y + (\lambda C)z + (\lambda D) = 0$$

for any constant $\lambda \neq 0$. Furthermore, if A, B, C, D and A', B', C', D' determine the same plane $\mathcal{P}$, then $A = \lambda A'$, $B = \lambda B'$, $C = \lambda C'$, $D = \lambda D'$ for a scalar λ. Consequently, A, B, C, D are ***determined by $\mathcal{P}$ up to a scalar multiple***.

EXAMPLE 10 Determine an equation for the plane that is perpendicular to the vector $\mathbf{i} + \mathbf{j} + \mathbf{k}$ and contains the point (1, 0, 0).

SOLUTION Using the general form $A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$, the plane is $1(x - 1) + 1(y - 0) + 1(z - 0) = 0$; that is, $x + y + z = 1$. ▲

EXAMPLE 11 Find an equation for the plane containing the three points (1, 1, 1), (2, 0, 0), and (1, 1, 0).

SOLUTION *Method 1.* This is a "brute force" method that you can use if you have forgotten the vector methods. The equation for any plane is of the form $Ax + By + Cz + D = 0$. Because the points (1, 1, 1), (2, 0, 0), and (1, 1, 0) lie in the plane, we have

$$\begin{aligned} A + B + C + D &= 0, \\ 2A \qquad\quad + D &= 0, \\ A + B \qquad + D &= 0. \end{aligned}$$

Proceeding by elimination, we reduce this system of equations to the form

$$\begin{aligned} 2A + D &= 0 \quad \text{(second equation)} \\ 2B + D &= 0 \quad (2 \times \text{third} - \text{second}), \\ C &= 0 \quad \text{(first} - \text{third)}. \end{aligned}$$

Because the numbers A, B, C, and D are determined only up to a scalar multiple, we can fix the value of one of them, say $A = 1$, and then the others will be determined uniquely. We get $A = 1$, $D = -2$, $B = 1$, $C = 0$. Thus, an equation of the plane that contains the given points is $x + y - 2 = 0$.

Method 2. Let $\mathrm{P} = (1, 1, 1)$, $\mathrm{Q} = (2, 0, 0)$, $\mathrm{R} = (1, 1, 0)$. Any vector normal to the plane must be orthogonal to the vectors $\overrightarrow{\mathrm{QP}}$ and $\overrightarrow{\mathrm{RP}}$, which are parallel to the plane, because their endpoints lie on the plane. Thus, $\mathbf{n} = \overrightarrow{\mathrm{QP}} \times \overrightarrow{\mathrm{RP}}$ is normal to the plane. Computing the cross product, we have

$$\mathbf{n} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & 1 & 1 \\ 0 & 0 & 1 \end{vmatrix} = \mathbf{i} + \mathbf{j}.$$

Because the point $(2, 0, 0)$ lies on the plane, we conclude that the equation is given by $(x - 2) + (y - 0) + 0 \cdot (z - 0) = 0$; that is, $x + y - 2 = 0$. ▲

Two planes are called ***parallel*** when their normal vectors are parallel. Thus, the planes $A_1x + B_1y + C_1z + D_1 = 0$ and $A_2x + B_2y + C_2z + D_2 = 0$ are parallel when $\mathbf{n}_1 = A_1\mathbf{i} + B_1\mathbf{j} + C_1\mathbf{k}$ and $\mathbf{n}_2 = A_2\mathbf{i} + B_2\mathbf{j} + C_2\mathbf{k}$ are parallel; that is, $\mathbf{n}_1 = \sigma\mathbf{n}_2$ for a constant σ. For example, the planes

$$x - 2y + z = 0 \qquad \text{and} \qquad -2x + 4y - 2z = 10$$

are parallel, but the planes

$$x - 2y + z = 0 \qquad \text{and} \qquad 2x - 2y + z = 10$$

are not parallel.

Distance: Point to Plane

Let us now determine the distance from a point $\mathrm{E} = (x_1, y_1, z_1)$ to the plane $\mathcal{P}$ described by the equation $A(x - x_0) + B(y - y_0) + C(z - z_0) = Ax + By + Cz + D = 0$. To do so, consider the unit normal vector

$$\mathbf{n} = \frac{A\mathbf{i} + B\mathbf{j} + C\mathbf{k}}{\sqrt{A^2 + B^2 + C^2}},$$

which is a unit vector normal to the plane. Drop a perpendicular from E to the plane and construct the triangle REQ shown in Figure 1.3.7. The distance $d = \|\overrightarrow{\mathrm{EQ}}\|$ is the length of the projection of $\mathbf{v} = \overrightarrow{\mathrm{RE}}$ (the vector from R to E) onto $\mathbf{n}$;

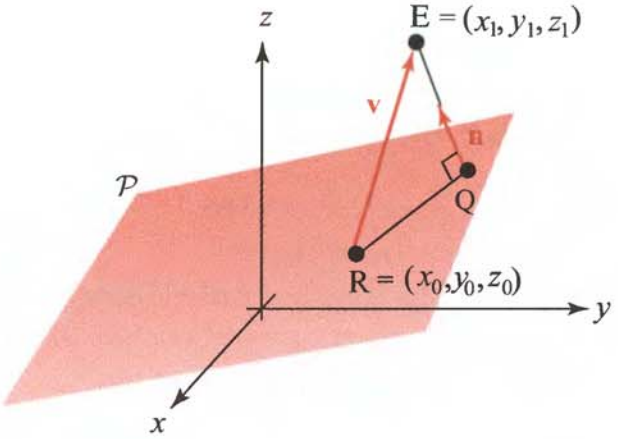


Figure 1.3.7 The geometry for determining the distance from the point E to plane $\mathcal{P}$.

thus,

$$\text{Distance} = |\mathbf{v} \cdot \mathbf{n}| = |[(x_1 - x_0)\mathbf{i} + (y_1 - y_0)\mathbf{j} + (z_1 - z_0)\mathbf{k}] \cdot \mathbf{n}|$$
$$= \frac{|A(x_1 - x_0) + B(y_1 - y_0) + C(z_1 - z_0)|}{\sqrt{A^2 + B^2 + C^2}}.$$

If the plane is given in the form $Ax + By + Cz + D = 0$, then for any point (x_0, y_0, z_0) on it, $D = -(Ax_0 + By_0 + Cz_0)$. Substitution into the previous formula gives the following:

Distance from a Point to a Plane The distance from (x_1, y_1, z_1) to the plane $Ax + By + Cz + D = 0$ is

$$\text{Distance} = \frac{|Ax_1 + By_1 + Cz_1 + D|}{\sqrt{A^2 + B^2 + C^2}}.$$

EXAMPLE 12 Find the distance from $Q = (2, 0, -1)$ to the plane $3x - 2y + 8z + 1 = 0$.

SOLUTION We substitute into the formula in the preceding box the values $x_1 = 2$ $y_1 = 0, z_1 = -1$ (the point) and $A = 3, B = -2, C = 8, D = 1$ (the plane) to give

$$\text{Distance} = \frac{|3 \cdot 2 + (-2) \cdot 0 + 8(-1) + 1|}{\sqrt{3^2 + (-2)^2 + 8^2}} = \frac{|-1|}{\sqrt{77}} = \frac{1}{\sqrt{77}}. \quad \blacktriangle$$

Historical Note

The Origins of the Vector, Scalar, Dot, and Cross Products

Quadratic Equations, Cubic Equations, and Imaginary Numbers. We know from Babylonian clay tablets that this great civilization possessed the quadratic formula, enabling them (in verbal form) to solve quadratic equations. Because the concept of negative numbers had to wait until the sixteenth century to see the light of day, the Babylonians did not consider either negative (or imaginary) solutions.

With the Renaissance and the rediscovery of ancient learning, Italian mathematicians began to wonder about the solutions of cubic equations, $x^3 + ax^2 + bx + c = 0$, where a, b, and c are positive numbers.

Around 1500, Scipione del Ferro, a professor in Bologna (the oldest European university) was able to solve cubics of the form $x^3 + bx = c$, but kept his discovery secret. Before his death, he passed his formula to his successor, Antonio Fior, who for a while also kept the formula to himself. It remained a secret until a brilliant, self-taught mathematician named Nicolo Fontana, also known as Tartaglia (the stammerer), appeared on the scene. Tartaglia claimed he could solve the cubic, and Fior felt he needed to protect the priority of del Ferro, and so in response challenged Tartaglia to a public competition.

We are told that Tartaglia was able to solve all of the thirty cubic equations posed by Fior. Amazingly, some scholars believe that Tartaglia discovered the formula for solutions to $x^3 + cx = d$ only days before the contest was to take place.

The greatest mathematician of the sixteenth century, Gerolamo Cardano (1501–1576)—a Renaissance scholar, mathematician, physician, and fortuneteller—gave the first published solution of the general cubic. Although born of modest means, he (like Tartaglia) rose, through effort and natural brilliance, to great fame. Cardano is the author of the first book on games of chance (marking the beginning of modern probability theory) and also of *Ars Magna* (the Great Art), which marks the beginning of modern algebra. It was in this book that Cardano published the solution to the *general* cubic $x^3 + ax^2 + bx + c = 0$. How did he get it?

While working on his algebra book, and aware that Tartaglia was able to solve forms of cubic, Cardano, in 1539, wrote to Tartaglia asking for a meeting. After some cajoling, Tartaglia agreed. It was at this meeting that, in exchange for a pledge of secrecy (and we know how these generally go), Tartaglia revealed his solution, from which Cardano was able to derive a solution to the general equation, which then appeared in *Ars Magna*. Feeling betrayed, Tartaglia led a scathing attack on Cardano, leading to a small soap opera.

What is important for us, at the moment, is that as a consequence of the method of solution, something very strange occurred. Consider the cubic $x^3 - 15x = 4$. Its only positive root is 4. However, the Tartaglia–Cardano solution formula yields

$$x = \sqrt[3]{2 + \sqrt{-121}} + \sqrt[3]{2 - \sqrt{-121}} \tag{1}$$

as the positive root. Thus, this number must be equal to 4. Yet this must be *nonsense*, because inside the cube root we are taking the square root of a negative number—at the time, an absolute impossibility. This was a real shock. Over 100 years later, in 1702, when Leibniz, codiscoverer of calculus, showed the great Dutch scientist Christian Huygens the formula

$$\sqrt{6} = \sqrt{1 + \sqrt{-3}} + \sqrt{1 - \sqrt{-3}} \tag{2}$$

Huygens was completely flabbergasted, and remarked that this equality "defies all human understanding." [Try, informally, to verify both formulas (1) and (2) for yourself.]

Whether nonsense or not, Tartaglia and Cardano's formula forced mathematicians to confront square roots of negative numbers (or *imaginary numbers*, as they are called today). This historical incident is another example that negates the (widespread) view that mathematics is "made up" by mathematicians. As is often the case, *it is the mathematics itself that speaks to us.*

The Maturing of Complex Numbers. For well over two centuries, numbers like $i = \sqrt{-1}$ were looked at with great suspicion. The square root of any negative number can be written in terms of i; for example, $\sqrt{-a} = \sqrt{(a)(-1)} = \sqrt{a}\sqrt{-1}$. In the middle of the eighteenth century, the Swiss mathematician Leonhard Euler connected the universal cosmic numbers e and π with the imaginary number i. Whatever i was or meant, it necessarily follows that

$$e^{\pi i} = -1,$$

that is, e "raised to the power πi equals -1. Thus, these cosmic numbers, reflecting perhaps some deeper mystery, are in fact connected to each other by a very simple formula.

At the beginning of the nineteenth century, the German mathematician Karl Friedrich Gauss was able to prove the *fundamental theorem of algebra*, which says that any nth-degree polynomial has n roots (some or all of which may be imaginary; that is, the roots have the form $a + bi$, where, as earlier, $i = \sqrt{-1}$ and where a and b are real numbers).

By the middle of the nineteenth century, the French mathematician Augusten-Louis Cauchy and the German mathematician Bernhard Riemann had developed the differential calculus for functions of one complex variable. An example of such a function is $F(z) = z^n$, where

$z = a + bi$. In this case, the usual formula for the derivative, $F'(z) = nz^{n-1}$, still holds. However, by introducing imaginary numbers, Cauchy was able to evaluate "real integrals" that heretofore could not be evaluated. For example, it is possible to show that

$$\int_0^\infty \frac{\sin x}{x}\, dx = \frac{\pi}{2}$$

and that

$$\int_0^\pi \log \sin x\, dx = -\pi \log 2,$$

These were stunning results.

In summary, the solution of the cubic equation, the fundamental theorem of algebra, and the evaluation of real integrals proved how valuable it was to consider imaginary numbers $a + bi$, even though they were not (at least not yet) on *terra firma*. Did they really exist or were they simply phantoms of our *imagination*, and thus truly *imaginary*?

Hamilton's Definition of Complex Numbers. Many mathematicians after Cardano made important contributions to imaginary (or complex) numbers, including Argand, Wessel, and Gauss—all of whom represented them geometrically. However, the modern, intellectually rigorous definition of a complex number is due to the great Irish mathematician William Rowan Hamilton (see Figure 1.3.8). After Newton, who created the vector concept through his invention of the notion of force, Hamilton was, beyond any doubt, the most important and singular figure in

Figure 1.3.8 Sir William Rowan Hamilton (1805–1865).

the development of vector calculus. It was Hamilton who gave us the terms *vector* and *scalar quantity*.

William Rowan Hamilton was born in Dublin, Ireland, at midnight on August 3, 1805. In 1823, he entered Trinity College, Dublin. His university career, by any standard, was phenomenal. By his third year, Trinity offered him a professorship, the Andrew's Chair of Astronomy, and the State named him Royal Astronomer of Ireland. These honors were based on his theoretical prediction (in 1824) of two entirely new and unexpected optical phenomena, namely, internal and external conical refraction.

By 1827 he had become interested in imaginary numbers. He wrote that "the symbol $\sqrt{-1}$ is absurd, and denotes an impossible extraction . . ." He set out to put the idea of a complex number on a firm logical foundation. His solution was to define a complex number $a + bi$ as a point (a, b) in the plane $\mathbb{R}^2$, much as we do today. Thus, the imaginary number bi for Hamilton was simply the point $(0, b)$ on the y axis. The difference between complex numbers and the Cartesian plane was that Hamilton followed the proforma multiplication of complex numbers:

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i,$$

and defined a new multiplication on the complex plane:

$$(a, b) \cdot (c, d) = (ac - bd, ad + bc).$$

Thus, $i = \sqrt{-1}$ just disappears into the point $(0, 1)$, and the mystery and confusion over complex numbers disappears along with it.

From Complex Numbers to Quaternions. From Hamilton's interpretation, *complex numbers are nothing more than the extension of real numbers into a new dimension, two dimensions*. Hamilton, however, also did fundamental work in mechanics, and he knew well that two dimensions were too limiting for the space analysis necessary for understanding the physics of the three-dimensional world. Therefore, Hamilton set out to find a triplet system; that is, an acceptable[1] multiplication scheme on points (a,b,c) in $\mathbb{R}^3$, or, as it were, on vectors $a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$.

By 1843, Hamilton realized that his quest was hopeless. But then, on October 16, 1843, Hamilton discovered that what he could not achieve for $\mathbb{R}^3$ he could achieve for $\mathbb{R}^4$; he discovered *quaternions*, an entirely new number system.

[1]For him, "acceptable" meant that the associative law of multiplication would hold.

Let us revisit the important historical moment in Hamilton's own words:

> But on the 16th day of the same month—which happened to be a Monday, and a Council Day of the Royal Irish Academy—I was walking in to attend and preside, and your mother was walking with me, along the Royal Canal, to which she had perhaps driven; and although she talked with me now and then, yet another *under-current* of thought was going on in my mind, which gave at last a *result*, whereof it is not too much to say that I felt at *once* the importance. An *electric* circuit seemed to *close*; and a spark flashed forth, the herald (as I *foresaw*, *immediately*) of many long years to come of definitely directed thought and work, by *myself* if spared, and at all events on the parts of *others*, if I should even be allowed to live long enough to distinctly communicate the discovery. Nor could I resist the impulse—unphilosophical as it may have been—to cut with a knife on a stone of Brougham Bridge, as we passed it, the fundamental formula with the symbols i, j, k; namely
>
> $$\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = \mathbf{ijk} = -1,$$
>
> which contains the *Solution of the Problem*, but of course, as an inscription, has long since moldered away.[2]

Hamilton had realized that the multiplication he had been searching for could be introduced on 4-tuples (a, b, c, d), which he had denoted by

$$a + b\mathbf{i} + c\mathbf{j} + d\mathbf{k}.$$

The a was called the *scalar part* and $b\mathbf{i} + c\mathbf{j} + d\mathbf{k}$ was called the *vector part*, which in reality, as with complex numbers, meant the point (a, b, c, d) in $\mathbb{R}^4$. The multiplication table he introduced was

$$\mathbf{ij} = \mathbf{k} = -\mathbf{ji}$$

$$\mathbf{ki} = \mathbf{j} = -\mathbf{ik}$$

$$\mathbf{jk} = \mathbf{i} = -\mathbf{kj}$$

$$\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = \mathbf{ijk} = -1.$$

Hamilton continued to *passionately* believe in his quaternions until the end of his life. Unfortunately, historical development went in another direction.

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[2] *North British Review* **14** (1858): 57.

The first step away from the quaternions was in fact taken by a firm believer in the importance of quaternions, namely, Peter Guthrie Tait, who was born in 1831 near Edinburgh, Scotland. In 1860, Tait was appointed to the Chair of Natural Philosophy at Edinburgh University, where he remained until his death in 1901. In 1867, he wrote his *Elementary Treatises on Quaternions*, a text stressing physical applications. His third chapter was most significant. It was here that Tait looked at the quaternionic product of two vectors:

$$\mathbf{v} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k} \qquad \text{and} \qquad \mathbf{w} = a'\mathbf{i} + b'\mathbf{j} + c'\mathbf{k}.$$

Then the product $\mathbf{vw}$, as defined by Hamilton, yields:

$$\begin{aligned}(a\mathbf{i} + b\mathbf{j} + c\mathbf{k})(a'\mathbf{i} + b'\mathbf{j} + c'\mathbf{k}) \\ = -(aa' + bb' + cc') + (bc' - cb')\mathbf{i} + (ac' - ca')\mathbf{j} + (ab' - ba')\mathbf{k}\end{aligned}$$

or, in modern form:

$$\mathbf{vw} = -(\mathbf{v} \cdot \mathbf{w}) + \mathbf{v} \times \mathbf{w},$$

where $\cdot$ is the modern dot or inner product of vectors and $\times$ is the cross product. Tait discovered the formulas

$$\mathbf{v} \cdot \mathbf{w} = \|\mathbf{v}\|\|\mathbf{w}\| \cos\theta \qquad \text{and} \qquad \|\mathbf{v} \times \mathbf{w}\| = \|\mathbf{v}\|\|\mathbf{w}\| \sin\theta,$$

where θ is the angle formed by $\mathbf{v}$ and $\mathbf{w}$. Moreover, he showed that $\mathbf{v} \times \mathbf{w}$ was orthogonal to $\mathbf{v}$ and $\mathbf{w}$, therefore giving a *geometric* interpretation of the quaternionic product of two vectors.

This began the move away from the study of quaternions and back to Newton's vectors, with the quaternionic product eventually being replaced by two separate products, the inner product and the cross product.

By the way, you might wonder why Hamilton did not at first discover the cross product, since it is a product on $\mathbb{R}^3$. The reason is that it did not have a fundamental property that he required—namely, it was not associative:[3]

$$0 = (\mathbf{i} \times \mathbf{i}) \times \mathbf{k} \neq \mathbf{i} \times (\mathbf{i} \times \mathbf{k}) = -\mathbf{k}$$

---

[3]Interestingly, if one is willing to continue to live with nonassociativity, there is also a vector product with most of the properties of the cross product in $\mathbb{R}^7$; this involves yet another number system called the *octonians*, which exists in $\mathbb{R}^8$. The nonexistence of a cross product in other dimensions is a result that goes beyond the scope of this text. For further information, see the *American Mathematical Monthly*, **74** (1967), pp. 188–194, and **90** (1983), p. 697, as well as J. Baez, "The Octonians," *Bulletin of the American Mathematical Society*, **39** (2002), pp. 145–206. One can show that systems like the quaternions and octonians occur only in dimension 1 (the reals $\mathbb{R}$), dimension 2 (the complex numbers), dimension 4 (the quaternions), and dimension 8 (the octonians). On the other hand, the "right" way to extend the cross product is to introduce the notion of *differential forms*, which exists in *any* dimension. We discuss their construction in Section 8.6.

THE MOVE AWAY FROM QUATERNIONS. The scientists ultimately responsible for the demise of quaternions were James Clerk Maxwell (see Figure 1.3.9), Oliver Heaviside, and Josiah Willard Gibbs, a founder of statistical mechanics. In the 1860s, Maxwell wrote down his monumental equations of electricity and magnetism. No vector notation was used (it did not exist). Instead, Maxwell wrote out his equations in what we would now call "component form." Around 1870, Tait began to correspond with Maxwell, piquing his interest in quaternions.

Figure 1.3.9 James Clerk Maxwell (1831–1879).

In 1873, Maxwell published his epic work, *Treatise on Electricity and Magnetism*. Here (as we shall do in Chapter 8), Maxwell wrote down the equations of the electromagnetic field using quaternions, thus motivating physicists and mathematicians alike to take a closer look at them. From this manuscript many have concluded that Maxwell was a supporter of the "quaternionic approach" to physics. The truth, however, is that Maxwell was reluctant to use quaternions. It was Maxwell, in fact, who began the process of separating the *vector* part of a product of two quaternions (the cross product) from its *scalar* part (the dot product).

It is known that Maxwell was troubled by the fact that the scalar part of the "square" of a vector ($\mathbf{vv}$) was always negative ($-\mathbf{v} \cdot \mathbf{v}$), which in the case of a velocity vector could be interpreted as negative kinetic energy—an unacceptable idea!

It was Heaviside and Gibbs who made the final push away from quaternions. Heaviside, an independent researcher interested in electricity and magnetism, and Gibbs, a professor of mathematical physics at Yale, almost simultaneously—and independently—created our modern system of vector analysis, which we have just started to study.

In 1879, Gibbs taught a course at Yale in vector analysis with applications to electricity and magnetism. This treatise was clearly motivated by the advent of Maxwell's equations, which we will be studying in Chapter 8. In 1884, he published his *Elements of Vector Analysis*, a book in which all the properties of the dot and cross products are fully developed. Knowing that much of what Gibbs wrote was in fact due to Tait, Gibbs's contemporaries did not view his book as highly original. However, it is one of the sources from which modern vector analysis has come into existence.

Heaviside was also largely motivated by Maxwell's brilliant work. His great *Electromagnetic Theory* was published in three volumes. Volume I (1893) contained the first extensive treatment of modern vector analysis.

We all owe a great debt to E. B. Wilson's 1901 book *Vector Analysis: A Textbook for the Use of Students of Mathematics and Physics Founded upon the Lectures of J. Willard Gibbs*. Wilson was reluctant to take Gibbs's course, because he had just completed a full-year course in quaternions at Harvard under J. M. Pierce, a champion of quaternionic methods; but he was forced by a dean to add the course to his program, and he did so in 1899. Wilson was later asked by the editor of the Yale Bicentennial Series to write a book based on Gibbs's lectures. For a picture of Gibbs and for additional historical comments on divergence and curl, see the Historical Note in Section 4.4.

EXERCISES

1. Verify that interchanging the first two rows of the 3×3 determinant

$$\begin{vmatrix} 1 & 2 & 1 \\ 3 & 0 & 1 \\ 2 & 0 & 2 \end{vmatrix}$$

changes the sign of the determinant.

2. Evaluate the determinants

(a) $\begin{vmatrix} 2 & -1 & 0 \\ 4 & 3 & 2 \\ 3 & 0 & 1 \end{vmatrix}$

(b) $\begin{vmatrix} 36 & 18 & 17 \\ 45 & 24 & 20 \\ 3 & 5 & -2 \end{vmatrix}$

(c) $\begin{vmatrix} 1 & 4 & 9 \\ 4 & 9 & 16 \\ 9 & 16 & 25 \end{vmatrix}$

(d) $\begin{vmatrix} 2 & 3 & 5 \\ 7 & 11 & 13 \\ 17 & 19 & 23 \end{vmatrix}$

3. Compute $\mathbf{a} \times \mathbf{b}$, where $\mathbf{a} = \mathbf{i} - 2\mathbf{j} + \mathbf{k}$, $\mathbf{b} = 2\mathbf{i} + \mathbf{j} + \mathbf{k}$.

4. Compute $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$, where $\mathbf{a}$ and $\mathbf{b}$ are as in Exercise 3 and $\mathbf{c} = 3\mathbf{i} - \mathbf{j} + 2\mathbf{k}$.

5. Find the area of the parallelogram with sides $\mathbf{a}$ and $\mathbf{b}$ given in Exercise 3.

6. A triangle has vertices $(0, 0, 0)$, $(1, 1, 1)$, and $(0, -2, 3)$. Find its area.

7. What is the volume of the parallelepiped with sides $2\mathbf{i} + \mathbf{j} - \mathbf{k}$, $5\mathbf{i} - 3\mathbf{k}$, and $\mathbf{i} - 2\mathbf{j} + \mathbf{k}$?

8. What is the volume of the parallelepiped with sides $\mathbf{i}$, $3\mathbf{j} - \mathbf{k}$, and $4\mathbf{i} + 2\mathbf{j} - \mathbf{k}$?

In Exercises 9 to 12, describe all unit vectors orthogonal to both of the given vectors.

9. $\mathbf{i}, \mathbf{j}$

10. $-5\mathbf{i} + 9\mathbf{j} - 4\mathbf{k}$, $7\mathbf{i} + 8\mathbf{j} + 9\mathbf{k}$

11. $-5\mathbf{i} + 9\mathbf{j} - 4\mathbf{k}$, $7\mathbf{i} + 8\mathbf{j} + 9\mathbf{k}$, $\mathbf{0}$

12. $2\mathbf{i} - 4\mathbf{j} + 3\mathbf{k}$, $-4\mathbf{i} + 8\mathbf{j} - 6\mathbf{k}$

13. Compute $\mathbf{u} + \mathbf{v}$, $\mathbf{u} \cdot \mathbf{v}$, $\|\mathbf{u}\|$, $\|\mathbf{v}\|$, and $\mathbf{u} \times \mathbf{v}$, where $\mathbf{u} = \mathbf{i} - 2\mathbf{j} + \mathbf{k}$, $\mathbf{v} = 2\mathbf{i} - \mathbf{j} + 2\mathbf{k}$.

14. Repeat Exercise 13 for $\mathbf{u} = 3\mathbf{i} + \mathbf{j} - \mathbf{k}$, $\mathbf{v} = -6\mathbf{i} - 2\mathbf{j} - 2\mathbf{k}$.

15. Find an equation for the plane that

(a) is perpendicular to $\mathbf{v} = (1, 1, 1)$ and passes through $(1, 0, 0)$.
(b) is perpendicular to $\mathbf{v} = (1, 2, 3)$ and passes through $(1, 1, 1)$.
(c) is perpendicular to the line $\mathbf{l}(t) = (5, 0, 2)t + (3, -1, 1)$ and passes through $(5, -1, 0)$.
(d) is perpendicular to the line $\mathbf{l}(t) = (-1, -2, 3)t + (0, 7, 1)$ and passes through $(2, 4, -1)$.

16. Find an equation for the plane that passes through

(a) $(0, 0, 0)$, $(2, 0, -1)$, and $(0, 4, -3)$.
(b) $(1, 2, 0)$, $(0, 1, -2)$, and $(4, 0, 1)$.
(c) $(2, -1, 3)$, $(0, 0, 5)$, and $(5, 7, -1)$.

17. (a) Show that two parallel planes are either identical or they never intersect.
(b) How do two nonparallel planes intersect?

18. Find the intersection of the planes $x + 2y + z = 0$ and $x - 3y - z = 0$.

19. Find the intersection of the planes $x + (y - 1) + z = 0$ and $-x + (y + 1) - z = 0$.

20. Find the intersection of the two planes with equations $3(x - 1) + 2y + (z + 1) = 0$ and $(x - 1) + 4y - (z + 1) = 0$.

21. (a) Prove the two triple-vector-product identities

$$(\mathbf{a} \times \mathbf{b}) \times \mathbf{c} = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{b} \cdot \mathbf{c})\mathbf{a} \quad \text{and} \quad \mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}.$$

(b) Prove $(\mathbf{u} \times \mathbf{v}) \times \mathbf{w} = \mathbf{u} \times (\mathbf{v} \times \mathbf{w})$ if and only if $(\mathbf{u} \times \mathbf{w}) \times \mathbf{v} = \mathbf{0}$.

(c) Also prove that $(\mathbf{u} \times \mathbf{v}) \times \mathbf{w} + (\mathbf{v} \times \mathbf{w}) \times \mathbf{u} + (\mathbf{w} \times \mathbf{u}) \times \mathbf{v} = \mathbf{0}$ (called the ***Jacobi identity***).

22. (a) Prove, without recourse to geometry, that

$$\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = \mathbf{v} \cdot (\mathbf{w} \times \mathbf{u}) = \mathbf{w} \cdot (\mathbf{u} \times \mathbf{v}) = -\mathbf{u} \cdot (\mathbf{w} \times \mathbf{v})$$
$$= -\mathbf{w} \cdot (\mathbf{v} \times \mathbf{u}) = -\mathbf{v} \cdot (\mathbf{u} \times \mathbf{w}).$$

(b) Use part (a) and Exercise 21(a) to prove that

$$(\mathbf{u} \times \mathbf{v}) \cdot (\mathbf{u}' \times \mathbf{v}') = (\mathbf{u} \cdot \mathbf{u}')(\mathbf{v} \cdot \mathbf{v}') - (\mathbf{u} \cdot \mathbf{v}')(\mathbf{u}' \cdot \mathbf{v}) = \begin{vmatrix} \mathbf{u} \cdot \mathbf{u}' & \mathbf{u} \cdot \mathbf{v}' \\ \mathbf{u}' \cdot \mathbf{v} & \mathbf{v} \cdot \mathbf{v}' \end{vmatrix}.$$

23. Verify Cramer's rule.

24. Find an equation for the plane that passes through the point $(2, -1, 3)$ and is perpendicular to the line $\mathbf{v} = (1, -2, 2) + t(3, -2, 4)$.

25. Find an equation for the plane that passes through the point $(1, 2, -3)$ and is perpendicular to the line $\mathbf{v} = (0, -2, 1) + t(1, -2, 3)$.

26. Find the equation of the line that passes through the point $(1, -2, -3)$ and is perpendicular to the plane $3x - y - 2z + 4 = 0$.

27. Find an equation for the plane containing the two (parallel) lines

$$\mathbf{v}_1 = (0, 1, -2) + t(2, 3, -1) \qquad \text{and} \qquad \mathbf{v}_2 = (2, -1, 0) + t(2, 3, -1).$$

28. Find the distance from the point $(2, 1, -1)$ to the plane $x - 2y + 2z + 5 = 0$.

29. Find an equation for the plane that contains the line $\mathbf{v} = (-1, 1, 2) + t(3, 2, 4)$ and is perpendicular to the plane $2x + y - 3z + 4 = 0$.

30. Find an equation for the plane that passes through $(3, 2, -1)$ and $(1, -1, 2)$ and that is parallel to the line $\mathbf{v} = (1, -1, 0) + t(3, 2, -2)$.

31. Redo Exercises 19 and 20 of Section 1.1 using the dot product and what you know about normals to planes.

32. Given vectors $\mathbf{a}$ and $\mathbf{b}$, do the equations $\mathbf{x} \times \mathbf{a} = \mathbf{b}$ and $\mathbf{x} \cdot \mathbf{a} = \|\mathbf{a}\|$ determine a unique vector $\mathbf{x}$? Argue both geometrically and analytically.

33. Determine the distance from the plane $12x + 13y + 5z + 2 = 0$ to the point $(1, 1, -5)$.

34. Find the distance to the point $(6, 1, 0)$ from the plane through the origin that is perpendicular to $\mathbf{i} - 2\mathbf{j} + \mathbf{k}$.

35. (a) In mechanics, the ***moment M*** *of a force* $\mathbf{F}$ *about a point* O is defined to be the magnitude of $\mathbf{F}$ times the perpendicular distance d from O to the line of action of $\mathbf{F}$. The ***vector moment*** $\mathbf{M}$ is the vector of magnitude M whose direction is perpendicular to the plane

of O and $\mathbf{F}$, determined by the right-hand rule. Show that $\mathbf{M} = \mathbf{R} \times \mathbf{F}$, where $\mathbf{R}$ is any vector from O to the line of action of $\mathbf{F}$. (See Figure 1.3.10.)

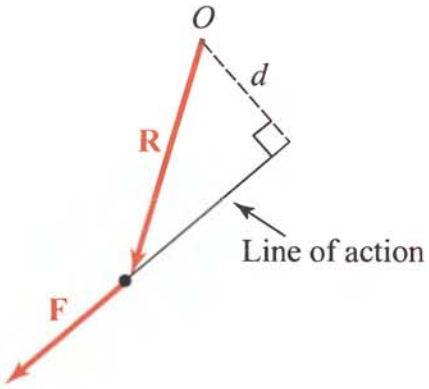


Figure 1.3.10 Moment of a force.

(b) Find the moment of the force vector $\mathbf{F} = \mathbf{i} - \mathbf{j} + 2\mathbf{k}$ newtons about the origin if the line of action is $x = 1 + t, y = 1 - t, z = 2t$.

36. Show that the plane that passes through the three points $A = (a_1, a_2, a_3)$, $B = (b_1, b_2, b_3)$, and $C = (c_1, c_2, c_3)$ consists of the points $\mathrm{P} = (x, y, z)$ given by

$$\begin{vmatrix} a_1 - x & a_2 - y & a_3 - z \\ b_1 - x & b_2 - y & b_3 - z \\ c_1 - x & c_2 - y & c_3 - z \end{vmatrix} = 0.$$

(HINT: Write the determinant as a triple product.)

37. Two media with indices of refraction n_1 and n_2 are separated by a plane surface perpendicular to the unit vector $\mathbf{N}$. Let $\mathbf{a}$ and $\mathbf{b}$ be unit vectors along the incident and refracted rays, respectively, their directions being those of the light rays. Show that $n_1(\mathbf{N} \times \mathbf{a}) = n_2(\mathbf{N} \times \mathbf{b})$ by using *Snell's law*, $\sin\theta_1 / \sin\theta_2 = n_2/n_1$, where θ_1 and θ_2 are the angles of incidence and refraction, respectively. (See Figure 1.3.11.)

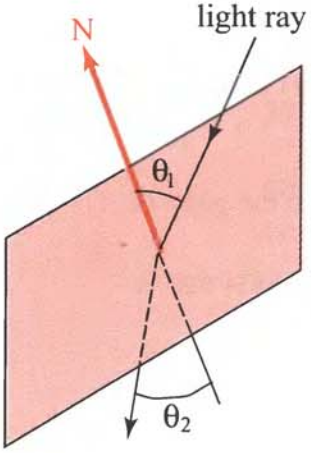


Figure 1.3.11 Snell's law.

38. Justify the steps in the following computation:

$$\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 10 \end{vmatrix} = \begin{vmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 7 & 8 & 10 \end{vmatrix} = \begin{vmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -6 & -11 \end{vmatrix} = \begin{vmatrix} -3 & -6 \\ -6 & -11 \end{vmatrix} = 33 - 36 = -3.$$

39. Show that adding a multiple of the first row of a matrix to the second row leaves the determinant unchanged; that is,

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 + \lambda a_1 & b_2 + \lambda b_1 & c_2 + \lambda c_1 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}.$$

[In fact, adding a multiple of any row (column) of a matrix to another row (column) leaves the determinant unchanged.]

1.4 Cylindrical and Spherical Coordinates

A standard way to represent a point in the plane $\mathbb{R}^2$ is by means of rectangular coordinates (x, y). However, as the reader has probably learned in elementary calculus, polar coordinates in the plane can be extremely useful. As portrayed in Figure 1.4.1, the coordinates (r, θ) are related to (x, y) by the formulas

$$x = r\cos\theta \qquad \text{and} \qquad y = r\sin\theta,$$

where we usually take $r \geq 0$ and $0 \leq \theta < 2\pi$.

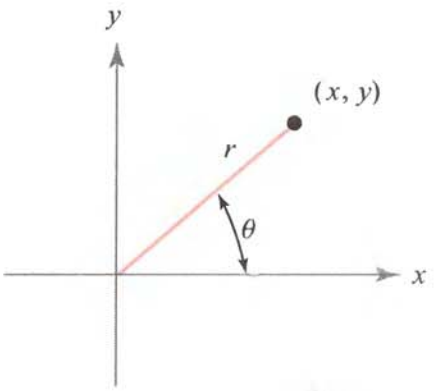


Figure 1.4.1 The polar coordinates of (x, y) are (r, θ).

Readers not familiar with polar coordinates are advised to study the relevant section of their calculus texts. We now set forth two ways of representing points in space other than by using rectangular Cartesian coordinates (x, y, z). These alternative coordinate systems are particularly well suited for certain types of problems, such as the evaluation of integrals using a change of variables.

Historical Note

In 1671, Isaac Newton wrote a manuscript entitled *The Method of Fuxions and Infinite Series*, which contains many uses of coordinate geometry to sketch the solutions of equations. In particular, he introduces the polar coordinate system, among various other coordinate systems.

In 1691, Jacob Bernoulli published a paper also containing polar coordinates. Because Newton's manuscript was not published until after his death in 1727, credit for the discovery of polar coordinates is usually attributed to Bernoulli.

Cylindrical Coordinates

DEFINITION The ***cylindrical coordinates*** (r, θ, z) of a point (x, y, z) are defined by (see Figure 1.4.2)

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad z = z. \tag{1}$$

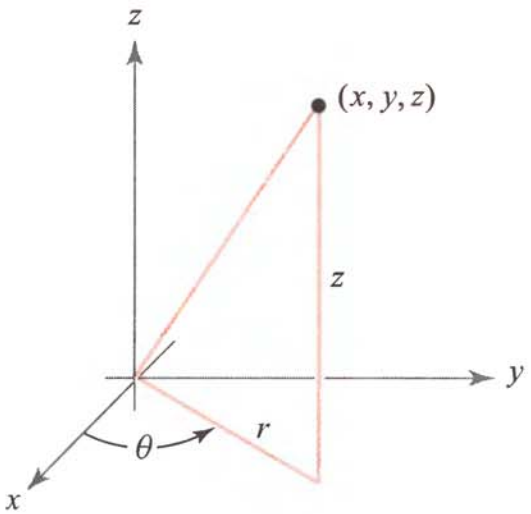


Figure 1.4.2 Representing a point (x, y, z) in terms of its cylindrical coordinates r, θ, and z.

To express r, θ, and z in terms of x, y, and z, and to ensure that θ lies between 0 and 2π, we can write

$$r = \sqrt{x^2 + y^2}, \qquad \theta = \begin{cases} \tan^{-1}(y/x) & \text{if } x > 0 \text{ and } y \geq 0 \\ \pi + \tan^{-1}(y/x) & \text{if } x < 0 \\ 2\pi + \tan^{-1}(y/x) & \text{if } x > 0 \text{ and } y < 0, \end{cases} \qquad z = z,$$

where $\tan^{-1}(y/x)$ is taken to lie between $-\pi/2$ and $\pi/2$. The requirement that $0 \leq \theta < 2\pi$ uniquely determines θ and $r \geq 0$ for a given x and y. If $x = 0$, then $\theta = \pi/2$ for $y > 0$ and $3\pi/2$ for $y < 0$. If $x = y = 0$, θ is undefined.

In other words, for any point (x, y, z), we represent the first and second coordinates in terms of polar coordinates and leave the third coordinate unchanged. Formula (1) shows that, given (r, θ, z), the triple (x, y, z) is completely determined, and vice versa, if we restrict θ to the interval $[0, 2\pi)$ (sometimes the range $(-\pi, \pi]$ is convenient) and require that $r > 0$.

To see why we use the term *cylindrical coordinates*, note that if the conditions $0 \leq \theta < 2\pi$, $-\infty < z < \infty$ hold and if $r = a$ is some positive constant, then the locus of these points is a cylinder of radius a (see Figure 1.4.3).

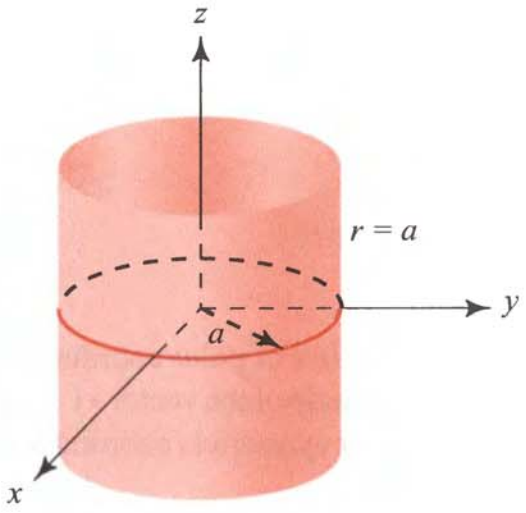


Figure 1.4.3 The graph of the points whose cylindrical coordinates satisfy $r = a$ is a cylinder.

EXAMPLE 1 (a) Find and plot the cylindrical coordinates of (6, 6, 8). (b) If a point has cylindrical coordinates $(8, 2\pi/3, -3)$, what are its Cartesian coordinates? Plot.

SOLUTION For part (a), we have $r = \sqrt{6^2 + 6^2} = 6\sqrt{2}$ and $\theta = \tan^{-1}(6/6) = \tan^{-1}(1) = \pi/4$. Thus, the cylindrical coordinates are $(6\sqrt{2}, \pi/4, 8)$. This is point P in Figure 1.4.4. For part (b), note that $2\pi/3 = \pi/2 + \pi/6$ and compute

$$x = r\cos\theta = 8\cos\frac{2\pi}{3} = -\frac{8}{2} = -4$$

and

$$y = r\sin\theta = 8\sin\frac{2\pi}{3} = 8\frac{\sqrt{3}}{2} = 4\sqrt{3}.$$

Thus, the Cartesian coordinates are $(-4, 4\sqrt{3}, -3)$. This is point Q in the figure. ▲

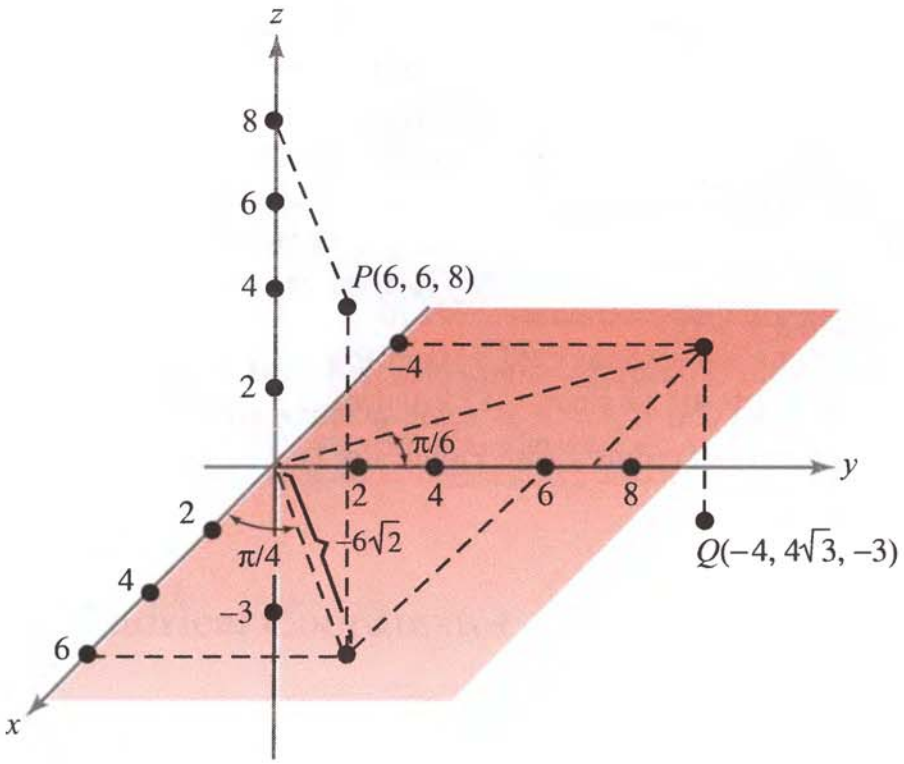


Figure 1.4.4 Some examples of the conversion between Cartesian and cylindrical coordinates.

Spherical Coordinates

Cylindrical coordinates are not the only possible generalization of polar coordinates to three dimensions. Recall that in two dimensions the magnitude of the vector $x\mathbf{i} + y\mathbf{j}$ (that is, $\sqrt{x^2 + y^2}$) is the r in the polar coordinate system. For cylindrical coordinates, the length of the vector $x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, namely,

$$\rho = \sqrt{x^2 + y^2 + z^2},$$

is not one of the coordinates of that system—instead, we used the magnitude $r = \sqrt{x^2 + y^2}$, the angle θ, and the "height" z.

We now modify this by introducing the *spherical coordinate* system, which *does use* ρ as a coordinate. Spherical coordinates are often useful for problems that possess spherical symmetry (symmetry about a point), whereas cylindrical coordinates can be applied when cylindrical symmetry (symmetry about a line) is involved.

Given a point $(x, y, z) \in \mathbb{R}^3$, let

$$\rho = \sqrt{x^2 + y^2 + z^2}$$

and represent x and y by polar coordinates in the xy plane:

$$x = r\cos\theta, \qquad y = r\sin\theta \tag{2}$$

where $r = \sqrt{x^2 + y^2}$ and θ is determined by formula (1) [see the expression for θ following formula (1)]. The coordinate z is given by

$$z = \rho\cos\phi,$$

where ϕ is the angle (chosen to lie between 0 and π, inclusive) that the radius vector $\mathbf{v} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ makes with the positive z axis, in the plane containing the vector $\mathbf{v}$ and the z axis (see Figure 1.4.5). Using the dot product, we can express ϕ as follows:

$$\cos\phi = \frac{\mathbf{v}\cdot\mathbf{k}}{\|\mathbf{v}\|}, \qquad \text{that is,} \qquad \phi = \cos^{-1}\left(\frac{\mathbf{v}\cdot\mathbf{k}}{\|\mathbf{v}\|}\right).$$

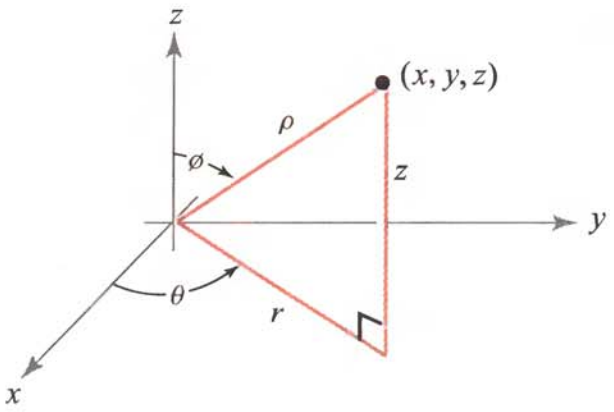


Figure 1.4.5 Spherical coordinates (ρ, θ, ϕ); the graph of points satisfying $\rho = a$ is a sphere.

We take as our coordinates the quantities ρ, θ, ϕ. Because

$$r = \rho\sin\phi,$$

we can use formula (2) to find x, y, z in terms of the spherical coordinates ρ, θ, ϕ.

DEFINITION The ***spherical coordinates*** of (x, y, z) is the triple (ρ, θ, ϕ), defined as follows:

$$x = \rho\sin\phi\cos\theta, \qquad y = \rho\sin\phi\sin\theta, \qquad z = \rho\cos\phi \tag{3}$$

where

$$\rho \geq 0, \qquad 0 \leq \theta < 2\pi, \qquad 0 \leq \phi \leq \pi.$$

Historical Note

In 1773, Joseph Louis Lagrange was working on Newton's gravitational theory as it applied to ellipsoids of revolution. In attempting to calculate the total gravitational attraction of such an ellipsoid, he encountered an integral that was difficult to evaluate. Motivated by this application, he introduced spherical coordinates, which allowed him to calculate the integral. We will be discussing the method of changing coordinates as it applies to multiple

integrals in Section 6.2, and applications to gravitation in Section 6.3, where we show how the inverse square law of gravity allowed Newton to consider spherical masses as point masses.

Spherical coordinates are also closely connected to navigation by latitude and longitude. To see the connection, first note that the sphere of radius a centered at the origin is described by a very simple equation in spherical coordinates, namely, $\rho = a$. Fixing the radius a, the spherical coordinates θ and ϕ are similar to the geographic coordinates of longitude and latitude if we take the earth's axis to be the z axis. There are differences, though: The geographical longitude is $|\theta|$ and is called east or west longitude, according to whether θ is a positive or negative measure from the Greenwich meridian; the geographical latitude is $|\pi/2 - \phi|$ and is called north or south latitude, according to whether $\pi/2 - \phi$ is positive or negative.

EXAMPLE 2

(a) Find the spherical coordinates of the Cartesian point $(1, -1, 1)$ and plot.

(b) Find the Cartesian coordinates of the spherical coordinate point $(3, \pi/6, \pi/4)$ and plot.

(c) Let a point have Cartesian coordinates $(2, -3, 6)$. Find its spherical coordinates and plot.

(d) Let a point have spherical coordinates $(1, -\pi/2, \pi/4)$. Find its Cartesian coordinates and plot.

SOLUTION

(a) $\rho = \sqrt{x^2 + y^2 + z^2} = \sqrt{1^2 + (-1)^2 + 1^2} = \sqrt{3},$

$$\theta = 2\pi + \tan^{-1}\left(\frac{y}{x}\right) = 2\pi + \tan^{-1}\left(\frac{-1}{1}\right) = 2\pi - \frac{\pi}{4} = \frac{7\pi}{4}$$

$$\phi = \cos^{-1}\left(\frac{z}{\rho}\right) = \cos^{-1}\left(\frac{1}{\sqrt{3}}\right) \approx 0.955 \approx 54.74°.$$

See Figure 1.4.6(a) and the formula for θ following formula (1).

(b) $$x = \rho \sin\phi \cos\theta = 3 \sin\left(\frac{\pi}{4}\right) \cos\left(\frac{\pi}{6}\right) = 3\left(\frac{1}{\sqrt{2}}\right)\frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{2\sqrt{2}},$$

$$y = \rho \sin\phi \sin\theta = 3 \sin\left(\frac{\pi}{4}\right) \sin\left(\frac{\pi}{6}\right) = 3\left(\frac{1}{\sqrt{2}}\right)\left(\frac{1}{2}\right) = \frac{3}{2\sqrt{2}},$$

$$z = \rho \cos\phi = 3 \cos\left(\frac{\pi}{4}\right) = \frac{3}{\sqrt{2}} = \frac{3\sqrt{2}}{2}.$$

See Figure 1.4.6(b).

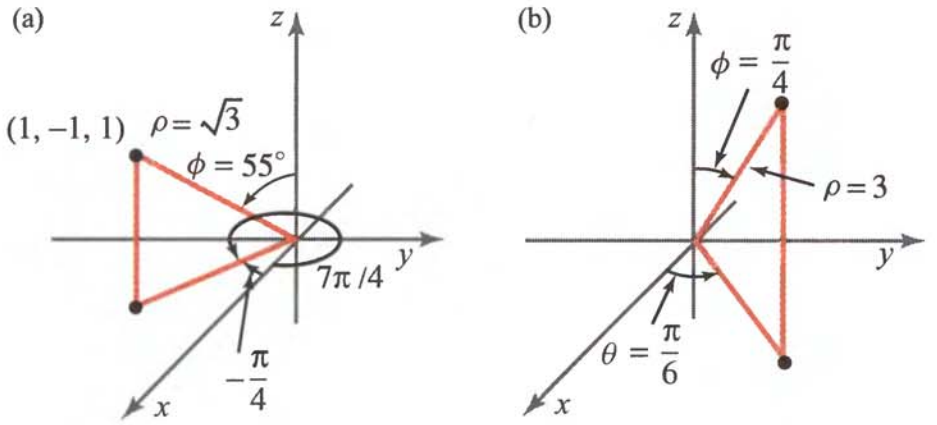


Figure 1.4.6 Finding (a) the spherical coordinates of the point (1, –1, 1), and (b) the Cartesian coordinates of $(3, \pi/6, \pi/4)$.

(c) $\rho = \sqrt{x^2 + y^2 + z^2} = \sqrt{2^2 + (-3)^2 + 6^2} = \sqrt{49} = 7,$

$$\theta = 2\pi + \tan^{-1}\left(\frac{y}{x}\right) = 2\pi + \tan^{-1}\left(\frac{-3}{2}\right) \approx 5.3004 \text{ radians} \approx 303.69°,$$

$$\phi = \cos^{-1}\left(\frac{z}{\rho}\right) = \cos^{-1}\left(\frac{6}{7}\right) \approx 0.541 \approx 31.0°.$$

See Figure 1.4.7(a).

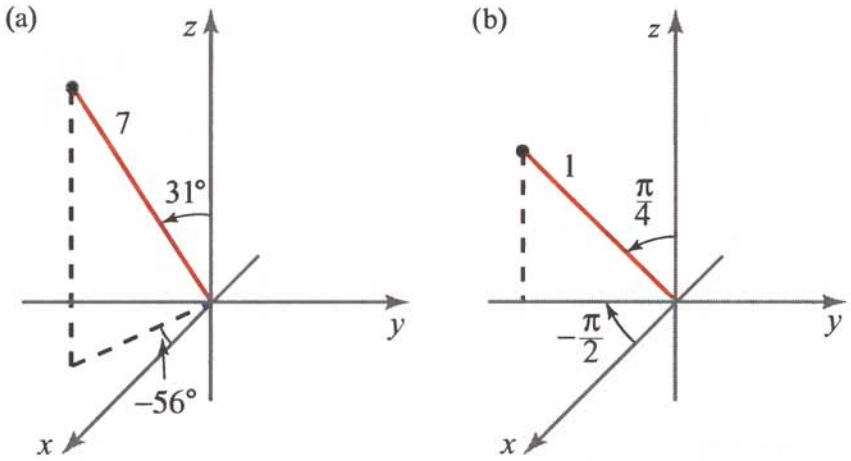


Figure 1.4.7 Finding (a) the spherical coordinates of the point (2, –3, 6), and (b) the Cartesian coordinates of $(1, -\pi/2, \pi/4)$.

(d) $x = \rho \sin\phi \cos\theta = 1 \sin\left(\frac{\pi}{4}\right) \cos\left(-\frac{\pi}{2}\right) = \left(\frac{\sqrt{2}}{2}\right) \cdot 0 = 0,$

$$y = \rho \sin\phi \sin\theta = 1 \sin\left(\frac{\pi}{4}\right) \sin\left(-\frac{\pi}{2}\right) = \left(\frac{\sqrt{2}}{2}\right)(-1) = -\frac{\sqrt{2}}{2},$$

$$z = \rho \cos\phi = 1 \cos\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}.$$

See Figure 1.4.7(b) ▲

EXAMPLE 3 Express (a) the surface $xz = 1$ and (b) the surface $x^2 + y^2 - z^2 = 1$ in spherical coordinates.

SOLUTION From formula (3), $x = \rho \sin\phi \cos\theta$, and $z = \rho \cos\phi$, and so the surface $xz = 1$ in (a) consists of all (ρ, θ, ϕ) such that

$$\rho^2 \sin\phi \cos\theta \cos\phi = 1, \qquad \text{that is,} \qquad \rho^2 \sin 2\phi \cos\theta = 2.$$

For part (b), we can write

$$x^2 + y^2 - z^2 = x^2 + y^2 + z^2 - 2z^2 = \rho^2 - 2\rho^2 \cos^2\phi,$$

so that the surface is $\rho^2(1 - 2\cos^2\phi) = 1$; that is, $-\rho^2 \cos(2\phi) = 1$. ▲

Associated with cylindrical and spherical coordinates are unit vectors that are the counterparts of **i**, **j**, and **k** for rectangular coordinates. They are shown in Figure 1.4.8. For example, $\mathbf{e}_r$ is the unit vector parallel to the xy plane and in the radial direction, so that $\mathbf{e}_r = (\cos\theta)\mathbf{i} + (\sin\theta)\mathbf{j}$. Similarly, in spherical coordinates, $\mathbf{e}_\phi$ is the unit vector tangent to the curve parametrized by the variable ϕ with the variables ρ and θ held fixed. We shall use these unit vectors later when we use cylindrical and spherical coordinates in vector calculations.

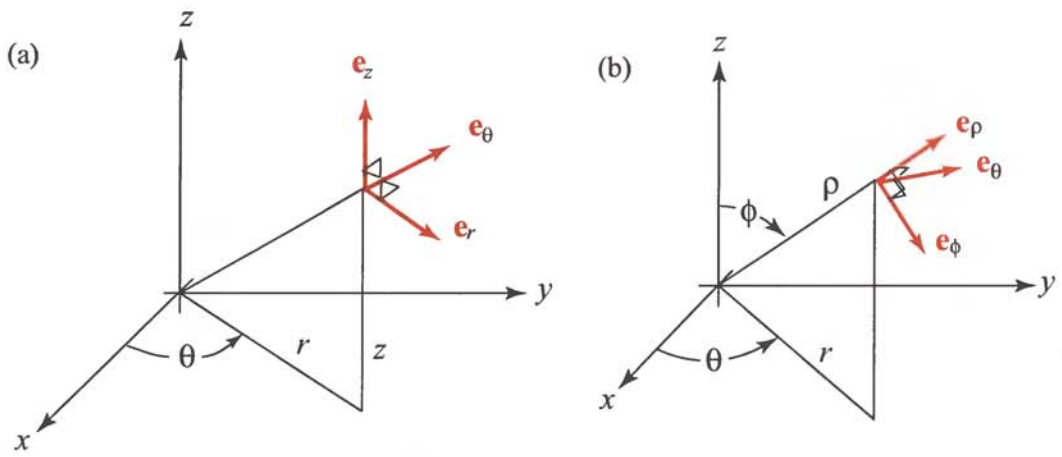


Figure 1.4.8 (a) Orthonormal vectors $\mathbf{e}_r$, $\mathbf{e}_\theta$, and $\mathbf{e}_z$ associated with cylindrical coordinates. The vector $\mathbf{e}_r$ is parallel to the line labeled r. (b) Orthonormal vectors $\mathbf{e}_\rho$, $\mathbf{e}_\theta$, and $\mathbf{e}_\phi$ associated with spherical coordinates.

EXERCISES

1. (a) The following points are given in cylindrical coordinates; express each in rectangular coordinates and spherical coordinates: $(1, 45°, 1)$, $(2, \pi/2, -4)$, $(0, 45°, 10)$, $(3, \pi/6, 4)$, $(1, \pi/6, 0)$, and $(2, 3\pi/4, -2)$. (Only the first point is solved in the Study Guide.)

(b) Change each of the following points from rectangular coordinates to spherical coordinates and to cylindrical coordinates: $(2, 1, -2)$, $(0, 3, 4)$, $(\sqrt{2}, 1, 1)$, $(-2\sqrt{3}, -2, 3)$. (Only the first point is solved in the Study Guide.)

2. Describe the geometric meaning of the following mappings in cylindrical coordinates:

(a) $(r, \theta, z) \mapsto (r, \theta, -z)$
(b) $(r, \theta, z) \mapsto (r, \theta + \pi, -z)$
(c) $(r, \theta, z) \mapsto (-r, \theta - \pi/4, z)$

3. Describe the geometric meaning of the following mappings in spherical coordinates:

(a) $(\rho, \theta, \phi) \mapsto (\rho, \theta + \pi, \phi)$
(b) $(\rho, \theta, \phi) \mapsto (\rho, \theta, \pi - \phi)$
(c) $(\rho, \theta, \phi) \mapsto (2\rho, \theta + \pi/2, \phi)$

4. (a) Describe the surfaces $r =$ constant, $\theta =$ constant, and $z =$ constant in the cylindrical coordinate system.

(b) Describe the surfaces $\rho =$ constant, $\theta =$ constant, and $\phi =$ constant in the spherical coordinate system.

5. Show that to represent each point in $\mathbb{R}^3$ by spherical coordinates it is necessary to take only values of θ between 0 and 2π, values of ϕ between 0 and π, and values of $\rho \geq 0$. Are coordinates unique if we allow $\rho \leq 0$?

6. Using cylindrical coordinates and the orthonormal (orthogonal normalized) vectors $\mathbf{e}_r$, $\mathbf{e}_\theta$, and $\mathbf{e}_z$ (see Figure 1.4.8),

(a) express each of $\mathbf{e}_r$, $\mathbf{e}_\theta$, and $\mathbf{e}_z$ in terms of $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$ and (x, y, z); and
(b) calculate $\mathbf{e}_\theta \times \mathbf{j}$ both analytically, using part (a), and geometrically.

7. Using spherical coordinates and the orthonormal (orthogonal normalized) vectors $\mathbf{e}_\rho$, $\mathbf{e}_\theta$, and $\mathbf{e}_\phi$ (see Figure 1.4.8(b)),

(a) express each of $\mathbf{e}_\rho$, $\mathbf{e}_\theta$, and $\mathbf{e}_\phi$ in terms of $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$ and (x, y, z); and
(b) calculate $\mathbf{e}_\theta \times \mathbf{j}$ and $\mathbf{e}_\phi \times \mathbf{j}$ both analytically and geometrically.

8. Express the plane $z = x$ in (a) cylindrical, and (b) spherical coordinates.

9. Show that in spherical coordinates:

(a) ρ is the length of $x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$.
(b) $\phi = \cos^{-1}(\mathbf{v} \cdot \mathbf{k}/\|\mathbf{v}\|)$, where $\mathbf{v} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$.
(c) $\theta = \cos^{-1}(\mathbf{u} \cdot \mathbf{i}/\|\mathbf{u}\|)$, where $\mathbf{u} = x\mathbf{i} + y\mathbf{j}$.

10. Two surfaces are described in spherical coordinates by the two equations $\rho = f(\theta, \phi)$ and $\rho = -2f(\theta, \phi)$, where $f(\theta, \phi)$ is a function of two variables. How is the second surface obtained geometrically from the first?

11. A circular membrane in space lies over the region $x^2 + y^2 \leq a^2$. The maximum z component of points in the membrane is b. Assume that (x, y, z) is a point on the membrane. Show that the corresponding point (r, θ, z) in cylindrical coordinates satisfies the conditions $0 \leq r \leq a, 0 \leq \theta \leq 2\pi, |z| \leq b$.

12. A tank in the shape of a right-circular cylinder of radius 10 ft and height 16 ft is half filled and lying on its side. Describe the air space inside the tank by suitably chosen cylindrical coordinates.

13. A vibrometer is to be designed that withstands the heating effects of its spherical enclosure of diameter d, which is buried to a depth $d/3$ in the earth, the upper portion being heated by the sun (assume the surface is flat). Heat conduction analysis requires a description of the buried portion of the enclosure in spherical coordinates. Find it.

14. An oil filter cartridge is a porous right-circular cylinder inside which oil diffuses from the axis to the outer curved surface. Describe the cartridge in cylindrical coordinates, if the diameter of the filter is 4.5 inches, the height is 5.6 inches, and the center of the cartridge is drilled (all the way through) from the top to admit a $\frac{5}{8}$-inch-diameter bolt.

15. Describe the surface given in spherical coordinates by $\rho = \cos 2\theta$.

1.5 n-Dimensional Euclidean Space

Vectors in n-space

In Sections 1.1 and 1.2 we studied the spaces $\mathbb{R} = \mathbb{R}^1$, $\mathbb{R}^2$, and $\mathbb{R}^3$ and gave geometric interpretations to them. For example, a point (x, y, z) in $\mathbb{R}^3$ can be thought of as a geometric object, namely, the directed line segment or vector emanating from the origin and ending at the point (x, y, z). We can therefore think of $\mathbb{R}^3$ in either of two ways:

(i) Algebraically, as a set of triples (x, y, z) where x, y, and z are real numbers

(ii) Geometrically, as a set of directed line segments

These two ways of looking at $\mathbb{R}^3$ are equivalent. For generalization it is easier to use definition (i). Specifically, we can define $\mathbb{R}^n$, where n is a positive integer (possibly greater than 3), to be the set of all ordered n-tuples $(x_1, x_2, \ldots, x_n)$, where the x_i are real numbers. For instance, $(1, \sqrt{5}, 2, \sqrt{3}) \in \mathbb{R}^4$.

The set $\mathbb{R}^n$ so defined is known as ***Euclidean n-space***, and its elements, which we write as $\mathbf{x} = (x_1, x_2, \ldots, x_n)$, are known as ***vectors*** or ***n-vectors***. By setting $n = 1, 2$, or 3, we recover the line, the plane, and three-dimensional space, respectively.

We launch our study of Euclidean n-space by introducing several algebraic operations. These are analogous to those introduced in Section 1.1 for $\mathbb{R}^2$ and $\mathbb{R}^3$. The first two, addition and scalar multiplication, are defined as follows:

(i) $(x_1, x_2, \ldots, x_n) + (y_1, y_2, \ldots, y_n) = (x_1 + y_1, x_2 + y_2, \ldots, x_n + y_n)$;

and

(ii) for any real number α,

$$\alpha(x_1, x_2, \ldots, x_n) = (\alpha x_1, \alpha x_2, \ldots, \alpha x_n).$$

The geometric significance of these operations for $\mathbb{R}^2$ and $\mathbb{R}^3$ was discussed in Section 1.1.

The n vectors

$$\mathbf{e}_1 = (1, 0, 0, \ldots, 0), \mathbf{e}_2 = (0, 1, 0, \ldots, 0), \ldots, \mathbf{e}_n = (0, 0, \ldots, 0, 1)$$

are called the ***standard basis vectors*** of $\mathbb{R}^n$, and they generalize the three mutually orthogonal unit vectors $\mathbf{i}, \mathbf{j}, \mathbf{k}$ of $\mathbb{R}^3$. The vector $\mathbf{x} = (x_1, x_2, \ldots, x_n)$ can then be written as $\mathbf{x} = x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + \cdots + x_n\mathbf{e}_n$.

For two vectors $\mathbf{x} = (x_1, x_2, x_3)$ and $\mathbf{y} = (y_1, y_2, y_3)$ in $\mathbb{R}^3$, we defined the *dot* or *inner product* $\mathbf{x} \cdot \mathbf{y}$ to be the real number $\mathbf{x} \cdot \mathbf{y} = x_1y_1 + x_2y_2 + x_3y_3$. This definition easily extends to $\mathbb{R}^n$; specifically, for $\mathbf{x} = (x_1, x_2, \ldots, x_n)$, $\mathbf{y} = (y_1, y_2, \ldots, y_n)$, we define the ***inner product*** of $\mathbf{x}$ and $\mathbf{y}$ to be $\mathbf{x} \cdot \mathbf{y} = x_1y_1 + x_2y_2 + \cdots + x_ny_n$. In $\mathbb{R}^n$, the notation $\langle \mathbf{x}, \mathbf{y} \rangle$ is often used in place of $\mathbf{x} \cdot \mathbf{y}$ for the inner product.

Continuing the analogy with $\mathbb{R}^3$, we are led to define the notion of the ***length*** or ***norm*** of a vector $\mathbf{x}$ by the formula

$$\text{Length of } \mathbf{x} = \|\mathbf{x}\| = \sqrt{\mathbf{x} \cdot \mathbf{x}} = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}.$$

If $\mathbf{x}$ and $\mathbf{y}$ are two vectors in the plane ($\mathbb{R}^2$) or in space ($\mathbb{R}^3$), then we know that the angle θ between them is given by the formula

$$\cos\theta = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|}.$$

The right side of this equation can be defined in $\mathbb{R}^n$ as well as in $\mathbb{R}^2$ or $\mathbb{R}^3$. It still represents the cosine of the angle between $\mathbf{x}$ and $\mathbf{y}$; this angle is geometrically well defined, because $\mathbf{x}$ and $\mathbf{y}$ lie in a two-dimensional subspace of $\mathbb{R}^n$ (the plane determined by $\mathbf{x}$ and $\mathbf{y}$) and our usual geometry ideas apply to such planes.

It will be useful to have available some algebraic properties of the inner product. These are summarized in the next theorem [compare with properties (i), (ii), (iii), and (iv) of Section 1.2].

THEOREM 3 For $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^n$ and α, β, real numbers, we have

(i) $(\alpha\mathbf{x} + \beta\mathbf{y}) \cdot \mathbf{z} = \alpha(\mathbf{x} \cdot \mathbf{z}) + \beta(\mathbf{y} \cdot \mathbf{z})$.

(ii) $\mathbf{x} \cdot \mathbf{y} = \mathbf{y} \cdot \mathbf{x}$.

(iii) $\mathbf{x} \cdot \mathbf{x} \geq 0$.

(iv) $\mathbf{x} \cdot \mathbf{x} = 0$ if and only if $\mathbf{x} = \mathbf{0}$.

PROOF Each of the four assertions can be proved by a simple computation. For example, to prove property (i) we write

$$\begin{aligned}(\alpha\mathbf{x} + \beta\mathbf{y}) \cdot \mathbf{z} &= (\alpha x_1 + \beta y_1, \alpha x_2 + \beta y_2, \ldots, \alpha x_n + \beta y_n) \cdot (z_1, z_2, \ldots, z_n) \\ &= (\alpha x_1 + \beta y_1)z_1 + (\alpha x_2 + \beta y_2)z_2 + \cdots + (\alpha x_n + \beta y_n)z_n \\ &= \alpha x_1 z_1 + \beta y_1 z_1 + \alpha x_2 z_2 + \beta y_2 z_2 + \cdots + \alpha x_n z_n + \beta y_n z_n \\ &= \alpha(\mathbf{x} \cdot \mathbf{z}) + \beta(\mathbf{y} \cdot \mathbf{z}).\end{aligned}$$

The other proofs are similar. ■

In Section 1.2, we proved an interesting property of dot products, called the Cauchy–Schwarz inequality.[4] For $\mathbb{R}^2$ our proof required the use of the law of cosines. For $\mathbb{R}^n$ we could also use this method, by confining our attention to a plane in $\mathbb{R}^n$. However, we can also give a direct, completely algebraic proof.

THEOREM 4: Cauchy–Schwarz Inequality in $\mathbb{R}^n$ Let $\mathbf{x}, \mathbf{y}$ be vectors in $\mathbb{R}^n$. Then

$$|\mathbf{x} \cdot \mathbf{y}| \leq \|\mathbf{x}\| \|\mathbf{y}\|.$$

PROOF Let $a = \mathbf{y} \cdot \mathbf{y}$ and $b = -\mathbf{x} \cdot \mathbf{y}$. If $a = 0$, the theorem is clearly valid, because then $\mathbf{y} = \mathbf{0}$ and both sides of the inequality reduce to 0. Thus, we may suppose $a \neq 0$. By Theorem 3 we have

$$\begin{aligned} 0 \leq (a\mathbf{x} + b\mathbf{y}) \cdot (a\mathbf{x} + b\mathbf{y}) &= a^2 \mathbf{x} \cdot \mathbf{x} + 2ab\mathbf{x} \cdot \mathbf{y} + b^2 \mathbf{y} \cdot \mathbf{y} \\ &= (\mathbf{y} \cdot \mathbf{y})^2 \mathbf{x} \cdot \mathbf{x} - (\mathbf{y} \cdot \mathbf{y})(\mathbf{x} \cdot \mathbf{y})^2. \end{aligned}$$

Dividing by $\mathbf{y} \cdot \mathbf{y}$ gives $0 \leq (\mathbf{y} \cdot \mathbf{y})(\mathbf{x} \cdot \mathbf{x}) - (\mathbf{x} \cdot \mathbf{y})^2$, that is, $(\mathbf{x} \cdot \mathbf{y})^2 \leq (\mathbf{x} \cdot \mathbf{x})(\mathbf{y} \cdot \mathbf{y}) = \|\mathbf{x}\|^2 \|\mathbf{y}\|^2$. Taking square roots on both sides of this inequality yields the desired result. ■

There is a useful consequence of the Cauchy–Schwarz inequality in terms of lengths. The triangle inequality is geometrically clear in $\mathbb{R}^3$ and was discussed in Section 1.2. The *analytic* proof of the triangle inequality that we gave in Section 1.2 works exactly the same in $\mathbb{R}^n$ and proves the following:

COROLLARY: Triangle Inequality in $\mathbb{R}^n$ Let $\mathbf{x}, \mathbf{y}$ be vectors in $\mathbb{R}^n$. Then

$$\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|.$$

If Theorem 4 and its corollary are written out algebraically, they become the following useful inequalities:

$$\left| \sum_{i=1}^{n} x_i y_i \right| \leq \left(\sum_{i=1}^{n} x_i^2 \right)^{1/2} \left(\sum_{i=1}^{n} y_i^2 \right)^{1/2};$$

$$\left(\sum_{i=1}^{n} (x_i + y_i)^2 \right)^{1/2} \leq \left(\sum_{i=1}^{n} x_i^2 \right)^{1/2} + \left(\sum_{i=1}^{n} y_i^2 \right)^{1/2}.$$

[4] Sometimes called the Cauchy–Bunyakovskii–Schwarz inequality, or simply the CBS inequality, because it was independently discovered in special cases by the French mathematician Cauchy, the Russian mathematician Bunyakovskii, and the German mathematician Schwarz.

EXAMPLE 1 Let $\mathbf{x} = (1, 2, 0, -1)$ and $\mathbf{y} = (-1, 1, 1, 0)$. Verify Theorem 4 and its corollary in this case.

SOLUTION

$$\|\mathbf{x}\| = \sqrt{1^2 + 2^2 + 0^2 + (-1)^2} = \sqrt{6}$$
$$\|\mathbf{y}\| = \sqrt{(-1)^2 + 1^2 + 1^2 + 0^2} = \sqrt{3}$$
$$\mathbf{x} \cdot \mathbf{y} = 1(-1) + 2 \cdot 1 + 0 \cdot 1 + (-1)0 = 1$$
$$\mathbf{x} + \mathbf{y} = (0, 3, 1, -1)$$
$$\|\mathbf{x} + \mathbf{y}\| = \sqrt{0^2 + 3^2 + 1^2 + (-1)^2} = \sqrt{11}.$$

We compute $\mathbf{x} \cdot \mathbf{y} = 1 \leq 4.24 \approx \sqrt{6}\sqrt{3} = \|\mathbf{x}\|\|\mathbf{y}\|$, which verifies Theorem 4. Similarly, we can check its corollary:

$$\|\mathbf{x} + \mathbf{y}\| = \sqrt{11} \approx 3.32$$
$$\leq 4.18 = 2.45 + 1.73 \approx \sqrt{6} + \sqrt{3} = \|\mathbf{x}\| + \|\mathbf{y}\|. \quad \blacktriangle$$

By analogy with $\mathbb{R}^3$, we can define the notion of distance in $\mathbb{R}^n$; namely, if $\mathbf{x}$ and $\mathbf{y}$ are points in $\mathbb{R}^n$, the ***distance between*** $\mathbf{x}$ ***and*** $\mathbf{y}$ is defined to be $\|\mathbf{x} - \mathbf{y}\|$, or the length of the vector $\mathbf{x} - \mathbf{y}$. We do not attempt to define the cross product on $\mathbb{R}^n$ except for $n = 3$.

General Matrices

Generalizing 2×2 and 3×3 matrices (see Section 1.3), we can consider $m \times n$ matrices, which are arrays of mn numbers:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}.$$

We shall also write A as $[a_{ij}]$. We define addition and multiplication by a scalar componentwise, just as we did for vectors. Given two $m \times n$ matrices A and B, we can add them to obtain a new $m \times n$ matrix $C = A + B$, whose ijth entry c_{ij} is the sum of a_{ij} and b_{ij}. It is clear that $A + B = B + A$.

EXAMPLE 2

(a) $$\begin{bmatrix} 2 & 1 & 0 \\ 3 & 4 & 1 \end{bmatrix} + \begin{bmatrix} -1 & 1 & 3 \\ 0 & 0 & 7 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 4 & 8 \end{bmatrix}.$$

(b) $[1 \quad 2] + [0 \quad -1] = [1 \quad 1]$.

(c) $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$. ▲

Given a scalar λ and an $m \times n$ matrix A, we can multiply A by λ to obtain a new $m \times n$ matrix $\lambda A = C$, whose ijth entry c_{ij} is the product λa_{ij}.

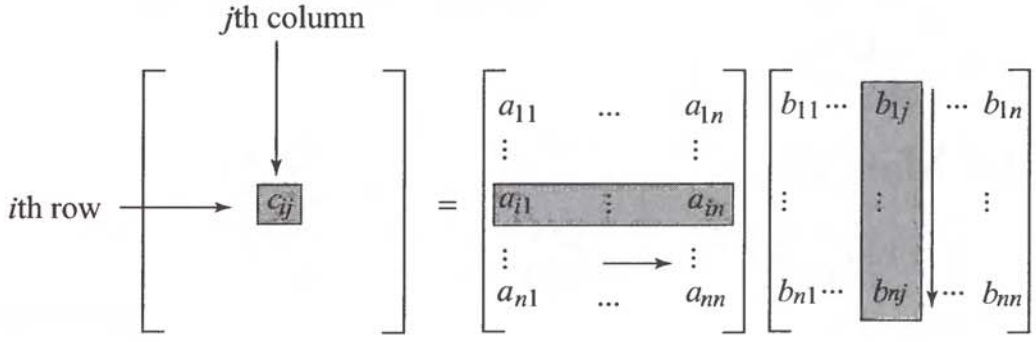
EXAMPLE 3

$$3\begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & 5 \\ 1 & 0 & 3 \end{bmatrix} = \begin{bmatrix} 3 & -3 & 6 \\ 0 & 3 & 15 \\ 3 & 0 & 9 \end{bmatrix}.$$ ▲

Next we turn to matrix multiplication. If $A = [a_{ij}]$, $B = [b_{ij}]$ are $n \times n$ matrices, then the product $AB = C$ has entries given by

$$c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj},$$

which is the dot product of the ith row of A and the jth column of B:



EXAMPLE 4 Let

$$A = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

Then

$$AB = \begin{bmatrix} 0 & 4 & 3 \\ 1 & 2 & 0 \\ 0 & 1 & 0 \end{bmatrix} \quad \text{and} \quad BA = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & 3 \\ 3 & 1 & 0 \end{bmatrix}.$$

Observe that $AB \neq BA$. ▲

Similarly, we can multiply an $m \times n$ matrix (m rows, n columns) by an $n \times p$ matrix (n rows, p columns) to obtain an $m \times p$ matrix (m rows, p columns) by the same rule. Note that for AB to be defined, the number of *columns* of A must equal the number of *rows* of B.

EXAMPLE 5 Let

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 1 & 1 & 2 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

Then

$$AB = \begin{bmatrix} 3 & 1 & 5 \\ 3 & 4 & 5 \end{bmatrix},$$

and BA is not defined. ▲

EXAMPLE 6 Let

$$A = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 3 \end{bmatrix} \quad \text{and} \quad B = [2 \quad 2 \quad 1 \quad 2].$$

Then

$$AB = \begin{bmatrix} 2 & 2 & 1 & 2 \\ 4 & 4 & 2 & 4 \\ 2 & 2 & 1 & 2 \\ 6 & 6 & 3 & 6 \end{bmatrix} \quad \text{and} \quad BA = [13]. \quad ▲$$

Any $m \times n$ matrix A determines a mapping of $\mathbb{R}^n$ to $\mathbb{R}^m$ defined as follows: Let $\mathbf{x} = (x_1, \ldots, x_n) \in \mathbb{R}^n$; consider the $n \times 1$ column matrix associated with $\mathbf{x}$, which we shall *temporarily* denote $\mathbf{x}^T$

$$\mathbf{x}^T = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

and multiply A by $\mathbf{x}^T$ (considered to be an $n \times 1$ matrix) to get a new $m \times 1$ matrix:

$$A\mathbf{x}^T = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix} = \mathbf{y}^T,$$

corresponding to the vector $\mathbf{y} = (y_1, \ldots, y_m)$.[5] Thus, although it may cause some confusion, we will write $\mathbf{x} = (x_1, \ldots, x_n)$ and $\mathbf{y} = (y_1, \ldots, y_m)$ as column matrices

$$\mathbf{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, \qquad \mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix}$$

when dealing with matrix multiplication; that is, we will *identify* these two forms of writing vectors. Thus, we will delete the T on $\mathbf{x}^T$ and view $\mathbf{x}^T$ and $\mathbf{x}$ as the same.

Thus, $A\mathbf{x} = \mathbf{y}$ will "really" mean the following: Write $\mathbf{x}$ as a column matrix, multiply it by A, and let $\mathbf{y}$ be the vector whose components are those of the resulting column matrix. The rule $\mathbf{x} \mapsto A\mathbf{x}$ therefore defines a mapping of $\mathbb{R}^n$ to $\mathbb{R}^m$. This mapping is linear; that is, it satisfies

$$A(\mathbf{x} + \mathbf{y}) = A\mathbf{x} + A\mathbf{y}$$
$$A(\alpha\mathbf{x}) = \alpha(A\mathbf{x}), \qquad \alpha \text{ a scalar},$$

as may be easily verified. One learns in a linear algebra course that, conversely, any linear transformation of $\mathbb{R}^n$ to $\mathbb{R}^m$ is representable in this way by an $m \times n$ matrix.

If $A = [a_{ij}]$ is an $m \times n$ matrix and $\mathbf{e}_j$ is the jth standard basis vector of $\mathbb{R}^n$, then $A\mathbf{e}_j$ is a vector in $\mathbb{R}^m$ with components the same as the jth column of A. That is, the ith component of $A\mathbf{e}_j$ is a_{ij}. In symbols, $(A\mathbf{e}_j)_i = a_{ij}$.

EXAMPLE 7 If

$$A = \begin{bmatrix} 1 & 0 & 3 \\ -1 & 0 & 1 \\ 2 & 1 & 2 \\ -1 & 2 & 1 \end{bmatrix},$$

then $\mathbf{x} \mapsto A\mathbf{x}$ of $\mathbb{R}^3$ to $\mathbb{R}^4$ is the mapping defined by

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \mapsto \begin{bmatrix} x_1 + 3x_3 \\ -x_1 + x_3 \\ 2x_1 + x_2 + 2x_3 \\ -x_1 + 2x_2 + x_3 \end{bmatrix}. \quad \blacktriangle$$

[5] To use a matrix A to get a mapping from vectors $\mathbf{x} = (x_1, \ldots, x_n)$ to vectors $\mathbf{y} = (y_1, \ldots, y_n)$ according to the equation $A\mathbf{x}^T = \mathbf{y}^T$, we write the vectors in the column form $\mathbf{x}^T$ instead of the row form $(x_1, \ldots, x_n)$. This sudden switch from writing $\mathbf{x}$ as a row to writing $\mathbf{x}$ as a column is necessitated by standard conventions on matrix multiplication.

EXAMPLE 8 The following illustrates what happens to a specific point when mapped by a 4×3 matrix:

$$A\mathbf{e}_2 = \begin{bmatrix} 4 & 2 & 9 \\ 3 & 5 & 4 \\ 1 & 2 & 3 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 2 \\ 1 \end{bmatrix} = \text{2nd column of } A. \quad \blacktriangle$$

Properties of Matrices

Matrix multiplication is not, in general, ***commutative***: If A and B are $n \times n$ matrices, then generally

$$AB \neq BA,$$

as Examples 4, 5, and 6 show.

An $n \times n$ matrix is said to be ***invertible*** if there is an $n \times n$ matrix B such that

$$AB = BA = I_n,$$

where

$$I_n = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix}$$

is the $n \times n$ identity matrix: I_n has the property that $I_n C = C I_n = C$ for any $n \times n$ matrix C. We denote B by A^{-1} and call A^{-1} the ***inverse*** of A. The inverse, when it exists, is unique.

EXAMPLE 9 If

$$A = \begin{bmatrix} 2 & 4 & 0 \\ 0 & 2 & 1 \\ 3 & 0 & 2 \end{bmatrix}, \quad \text{then} \quad A^{-1} = \frac{1}{20} \begin{bmatrix} 4 & -8 & 4 \\ 3 & 4 & -2 \\ -6 & 12 & 4 \end{bmatrix},$$

because $AA^{-1} = I_3 = A^{-1}A$, as may be checked by matrix multiplication. $\blacktriangle$

Methods of computing inverses are learned in linear algebra; we won't require these methods in this book. If A is invertible, the equation $A\mathbf{x} = \mathbf{y}$ can be solved for the vector $\mathbf{x}$ by multiplying both sides by A^{-1} to obtain[6] $\mathbf{x} = A^{-1}\mathbf{y}$.

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[6] In fact, Cramer's rule from Section 1.3 provides one way to invert matrices. Numerically more efficient methods based on elimination methods are learned in linear algebra or computer science.

In Section 1.3, we defined the determinant of a 3×3 matrix. This can be generalized by induction to $n \times n$ determinants. We illustrate here how to write the determinant of a 4×4 matrix in terms of the determinants of 3×3 matrices:

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{vmatrix} = a_{11} \begin{vmatrix} a_{22} & a_{23} & a_{24} \\ a_{32} & a_{33} & a_{34} \\ a_{42} & a_{43} & a_{44} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} & a_{24} \\ a_{31} & a_{33} & a_{34} \\ a_{41} & a_{43} & a_{44} \end{vmatrix}$$

$$+ a_{13} \begin{vmatrix} a_{21} & a_{22} & a_{24} \\ a_{31} & a_{32} & a_{34} \\ a_{41} & a_{42} & a_{44} \end{vmatrix} - a_{14} \begin{vmatrix} a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \\ a_{41} & a_{42} & a_{43} \end{vmatrix}$$

[see formula (2) of Section 1.3; the signs alternate $+, -, +, -$].

The basic properties of 3×3 determinants reviewed in Section 1.3 remain valid for $n \times n$ determinants. In particular, we note the fact that if A is an $n \times n$ matrix and B is the matrix formed by adding a scalar multiple of one row (or column) of A to another row (or, respectively, column) of A, then the determinant of A is equal to the determinant of B (see Example 10).

A basic theorem of linear algebra states that an $n \times n$ matrix A is invertible if and only if the determinant of A is not zero. Another basic property is that the determinant is multiplicative: $\det(AB) = (\det A)(\det B)$. In this text, we shall not make use of many details of linear algebra, and so we shall leave these assertions unproved.

EXAMPLE 10 Let

$$A = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ 2 & 1 & 0 & 1 \\ 1 & 1 & 0 & 2 \end{bmatrix}.$$

Find det A. Does A have an inverse?

SOLUTION Adding $(-1) \times$ first column to the third column, we get

$$\det A = \begin{vmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 \\ 2 & 1 & -2 & 1 \\ 1 & 1 & -1 & 2 \end{vmatrix} = 1 \begin{vmatrix} 1 & 0 & 1 \\ 1 & -2 & 1 \\ 1 & -1 & 2 \end{vmatrix}.$$

Adding $(-1) \times$ first column to the third column of this 3×3 determinant gives

$$\det A = \begin{vmatrix} 1 & 0 & 0 \\ 1 & -2 & 0 \\ 1 & -1 & 1 \end{vmatrix} = \begin{vmatrix} -2 & 0 \\ -1 & 1 \end{vmatrix} = -2.$$

Thus, $\det A = -2 \neq 0$, and so A has an inverse. ▲

If we have three matrices A, B, and C such that the products AB and BC are defined, then the products $(AB)C$ and $A(BC)$ are defined and are in fact equal (that is, matrix multiplication is *associative*). We call this the *triple product* of matrices and denote it by ABC.

EXAMPLE 11 Let

$$A = \begin{bmatrix} 3 \\ 5 \end{bmatrix}, \qquad B = [1 \quad 1], \qquad \text{and} \qquad C = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Then

$$ABC = A(BC) = \begin{bmatrix} 3 \\ 5 \end{bmatrix} [3] = \begin{bmatrix} 9 \\ 15 \end{bmatrix}. \quad \blacktriangle$$

EXAMPLE 12

$$\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 1 & 0 \end{bmatrix}. \quad \blacktriangle$$

Historical Note

The founder of modern (coordinate) geometry was René Descartes (see Figure 1.5.1), a great physicist, philosopher, and mathematician, as well as a founder of modern biology.

Born in Touraine, France, in 1596, Descartes had a fascinating life. After studying law, he settled in Paris, where he developed an interest in mathematics. In 1628, he moved to Holland, where he wrote his only mathematical work, *La Geometrie*, one of the origins of modern coordinate geometry.

Descartes had been highly critical of the geometry of the ancient Greeks, with all their undefined terms and with their proofs requiring ever newer and more ingenious approaches. For Descartes, this geometry was so tied to geometrical figures "that it can exercise the understanding only on condition of greatly fatiguing the imagination." He undertook to exploit, in geometry, the use of algebra, which had recently been developed. The result was *La Geometrie*, which made possible analytic or computational methods in geometry.

Remember that the Greeks were, like Descartes, philosophers as well as mathematicians and physicists. Their answer to the question of the meaning of space was "Euclidean geometry." Descartes had therefore succeeded in "algebrizing"the Greek model of space.

Gottfried Wilhelm Leibniz, cofounder (with Isaac Newton) of calculus, was also interested in "space analysis," but he did not think that Descartes'

Figure 1.5.1 René Descartes (1596–1650).

algebra went far enough. Leibniz called for a direct method of space analysis (*analysis situs*) that could be interpreted as a call for the development of vector analysis.

On September 8, 1679, Leibniz outlined his ideas in a letter to Christian Huygens:

> I am still not satisfied with algebra, because it does not give the shortest methods or the most beautiful constructions in geometry. This is why I believe that, so far as geometry is concerned, we need still another analysis which is distinctly geometrical or linear and which will express situation (situs) directly as algebra expresses magnitude directly. And I believe that I have found the way and that we can represent figures and even machines and movements by characters, as algebra represents numbers or magnitudes. I am sending you an essay which seems to me to be important.

In the essay, Leibniz described his ideas in greater detail:

> I have discovered certain elements of a new characteristic which is entirely different from algebra and which will have great advantages in representing to the mind, exactly and in a way faithful to its nature, even without figures, everything which depends on sense perception. Algebra is the characteristic for undetermined numbers of magnitudes only, but it does not express situation, angles, and

> motion directly. Hence it is often difficult to analyze the properties of a figure by calculation, and still more difficult to find very convenient geometrical demonstrations and constructions, even when the algebraic calculation is completed. But this new characteristic, which follows the visual figures, cannot fail to give the solution, the construction, and the geometric demonstration all at the same time, and in a natural way and in one analysis, that is, through determined procedure.

Leibniz's ideas influenced Hamilton and others. In the middle of the nineteenth century, Bolyai and Lobachevsky developed their "non-Euclidean" geometry, and Gauss studied and developed a theory of curved surfaces in three-dimensional space. Gauss developed two measures of curvature, the *mean curvature* and the *Gauss curvature*. For example, soap bubbles and soap films have constant mean curvature, whereas only soap bubbles have constant Gauss curvature. We discuss these ideas further in Section 7.7.

Bernhard Riemann (see Figure 1.5.2), possibly the greatest mathematical genius of all time, gave an inaugural address in 1854 before the faculty of Göttingen University, entitled "On the Hypotheses Which Lie at the Foundation of Geometry." It was this monumental work that would lay the foundation, 50 years later, of Einstein's general theory of relativity. Riemann, like Leibniz and the early Greeks, was interested in space, especially its metric (or distance) properties.

Figure 1.5.2 Bernhard Riemann (1826–1866).

Riemann called for the study of n-dimensional spaces and surfaces. He showed how to measure the curvature of three-, four-, ..., n-dimensional surfaces and (incredibly) showed that in order to be called "curved," a surface need not be "curving" inside anything else; curvature was simply a consequence of the intrinsic "metric properties of space." Once Riemann demonstrated that mathematical models permitted us to think of spaces of any dimension, the question naturally arose as to why our space is three-dimensional and not four-, five-, or more dimensional. Surprisingly, no one has yet put forth a convincing explanation why, at the moment of creation, space became three-dimensional.

Around 1910, Albert Einstein realized that gravity could be explained as a consequence of the curvature of a four-dimensional space–time (matter and energy curve space and time), and, thanks to Riemann, Einstein's space–time need not be enclosed in an ambient universe. Exactly how matter and energy curve space–time is the essence of Einstein's field equations in general relativity. In Section 7.7, we will discuss the ideas of curvature in greater depth and will indicate some of the ideas behind general relativity. The idea of n-dimensions also began to creep into mathematics from another, very different direction—from matrices.

The definition of a matrix, as an isolated abstract object, is due to the English mathematician Arthur Cayley. Cayley was born in 1821, and in 1863 was appointed Sedlesian Professor of Mathematics at Cambridge University. Around 1855, one year after Riemann's inaugural address, Cayley, in an effort to simplify notation for his study of linear equations (as we saw in Section 1.5), introduced the abstract idea of a matrix of m columns and n rows. Naturally, a $1 \times n$ matrix could also be viewed as a vector in an "n-dimensional space."

After this concept took hold, mathematicians realized that they lost little in working in general dimensions, and the subject of modern linear algebra was off and running. Again, physics was to be a major impetus. Modern, abstract, linear algebra, including abstract vector spaces, began to turn up in textbooks after the appearance in 1918 of Hermann Weyl's *Space–Time–Matter*.

EXERCISES

1. Calculate the dot product of $\mathbf{x} = (1, -1, 0, 2) \in \mathbb{R}^4$ and $\mathbf{y} = (1, 2, 3, 4) \in \mathbb{R}^4$.

2. In $\mathbb{R}^n$ show that

(a) $2\|\mathbf{x}\|^2 + 2\|\mathbf{y}\|^2 = \|\mathbf{x} + \mathbf{y}\|^2 + \|\mathbf{x} - \mathbf{y}\|^2$ (This is known as the ***parallelogram law***.)
(b) $\|\mathbf{x} - \mathbf{y}\|\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2$
(c) $4\langle \mathbf{x}, \mathbf{y} \rangle = \|\mathbf{x} + \mathbf{y}\|^2 - \|\mathbf{x} - \mathbf{y}\|^2$ (This is called the ***polarization identity***.)

Interpret these results geometrically in terms of the parallelogram formed by $\mathbf{x}$ and $\mathbf{y}$.

Verify the Cauchy–Schwarz inequality and the triangle inequality for the vectors in Exercises 3 to 6.

3. $\mathbf{x} = (2, 0, -1), \mathbf{y} = (4, 0, -2)$

4. $\mathbf{x} = (1, 0, 2, 6), \mathbf{y} = (3, 8, 4, 1)$

5. $\mathbf{x} = (1, -1, 1, -1, 1), \mathbf{y} = (3, 0, 0, 0, 2)$

6. $\mathbf{x} = (1, 0, 0, 1), \mathbf{y} = (-1, 0, 0, 1)$

7. Compute AB, $\det A$, $\det B$, $\det (AB)$, and $\det (A + B)$ for

$$A = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 3 & 2 \\ 3 & 1 & 1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} -2 & 0 & 2 \\ -1 & 1 & -1 \\ 1 & 4 & 3 \end{bmatrix}.$$

8. Compute AB, $\det A$, $\det B$, $\det (AB)$, and $\det (A + B)$ for

$$A = \begin{bmatrix} 3 & 0 & 1 \\ 1 & 2 & -1 \\ 1 & 0 & 1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}.$$

9. Use induction on k to prove that if $\mathbf{x}_1, \ldots, \mathbf{x}_k \in \mathbb{R}^n$, then

$$\|\mathbf{x}_1 + \cdots + \mathbf{x}_k\| \leq \|\mathbf{x}_1\| + \cdots + \|\mathbf{x}_k\|.$$

10. Prove using algebra, the ***identity of Lagrange***: For real numbers $x_1, \ldots, x_n$ and $y_1, \ldots, y_n$.

$$\left(\sum_{i=1}^{n} x_i y_i\right)^2 = \left(\sum_{i=1}^{n} x_i^2\right)\left(\sum_{i=1}^{n} y_i^2\right) - \sum_{i<j} (x_i y_j - x_j y_i)^2.$$

Use this to give another proof of the Cauchy–Schwarz inequality in $\mathbb{R}^n$,

11. Prove that if A is an $n \times n$ matrix, then

(a) $\det (\lambda A) = \lambda^n \det A$; and
(b) if B is a matrix obtained from A by multiplying any row or column by a scalar λ, then $\det B = \lambda \det A$.

In Exercises 12 to 14, A, B, and C denote $n \times n$ matrices.

12. Is $\det (A + B) = \det A + \det B$? Give a proof or counterexample.

13. Does $(A + B)(A - B) = A^2 - B^2$?

14. Assuming the law $\det (AB) = (\det A)(\det B)$, prove that $\det (ABC) = (\det A)(\det B)(\det C)$.

15. (This exercise assumes a knowledge of integration of continuous functions of one variable.) Note that the proof of the Cauchy–Schwarz inequality (Theorem 4) depends only on the properties of the inner product listed in Theorem 1. Use this observation to establish the following inequality for continuous functions $f, g: [0, 1] \to \mathbb{R}$:

$$\left| \int_0^1 f(x)g(x)\, dx \right| \leq \sqrt{\int_0^1 [f(x)]^2\, dx} \sqrt{\int_0^1 [g(x)]^2\, dx}.$$

Do this by

(a) verifying that the space of continuous functions from [0, 1] to $\mathbb{R}$ forms a vector space; that is, we may think of functions f, g abstractly as "vectors" that can be added to each other and multiplied by scalars.

(b) introducing the inner product of functions

$$f \cdot g = \int_0^1 f(x)g(x)\, dx$$

and verifying that it satisfies conditions (i) to (iv) of Theorem 3.

16. Define the transpose A^T of an $n \times n$ matrix A as follows: the ijth element of A^T is a_{ji} where a_{ij} is the ijth entry of A. Show that A^T is characterized by the following property: For all $\mathbf{x}, \mathbf{y}$ in $\mathbb{R}^n$,

$$(A^T\mathbf{x}) \cdot \mathbf{y} = \mathbf{x} \cdot (A\mathbf{y}).$$

17. Verify that the inverse of

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \text{is} \quad \frac{1}{ad - bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

18. Use your answer in Exercise 17 to show that the solution of the system

$$ax + by = e$$
$$cx + dy = f$$

is

$$\begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{ad - bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}\begin{bmatrix} e \\ f \end{bmatrix}.$$

19. Assuming the law $\det(AB) = (\det A)(\det B)$, verify that $(\det A)(\det A^{-1}) = 1$ and conclude that if A has an inverse, then $\det A \neq 0$.

REVIEW EXERCISES FOR CHAPTER 1

1. Let $\mathbf{v} = 3\mathbf{i} + 4\mathbf{j} + 5\mathbf{k}$ and $\mathbf{w} = \mathbf{i} - \mathbf{j} + \mathbf{k}$. Compute $\mathbf{v} + \mathbf{w}$, $3\mathbf{v}$, $6\mathbf{v} + 8\mathbf{w}$, $-2\mathbf{v}$, $\mathbf{v} \cdot \mathbf{w}$, $\mathbf{v} \times \mathbf{w}$. Interpret each operation geometrically by graphing the vectors.

2. Repeat Exercise 1 with $\mathbf{v} = 2\mathbf{j} + \mathbf{k}$ and $\mathbf{w} = -\mathbf{i} - \mathbf{k}$.

3. (a) Find the equation of the line through $(-1, 2, -1)$ in the direction of $\mathbf{j}$.
(b) Find the equation of the line passing through $(0, 2, -1)$ and $(-3, 1, 0)$.
(c) Find the equation for the plane perpendicular to the vector $(-2, 1, 2)$ and passing through the point $(-1, 1, 3)$.

4. (a) Find the equation of the line through $(0, 1, 0)$ in the direction of $3\mathbf{i} + \mathbf{k}$.
(b) Find the equation of the line passing through $(0, 1, 1)$ and $(0, 1, 0)$.
(c) Find an equation for the plane perpendicular to the vector $(-1, 1, -1)$ and passing through the point $(1, 1, 1)$.

5. Compute $\mathbf{v} \cdot \mathbf{w}$ for the following sets of vectors:

(a) $\mathbf{v} = -\mathbf{i} + \mathbf{j}; \mathbf{w} = \mathbf{k}$.
(b) $\mathbf{v} = \mathbf{i} + 2\mathbf{j} - \mathbf{k}; \mathbf{w} = 3\mathbf{i} + \mathbf{j}$.
(c) $\mathbf{v} = -2\mathbf{i} - \mathbf{j} + \mathbf{k}; \mathbf{w} = 3\mathbf{i} + 2\mathbf{j} - 2\mathbf{k}$.

6. Compute $\mathbf{v} \times \mathbf{w}$ for the vectors in Exercise 5. [Only part (b) is solved in the Study Guide.]

7. Find the cosine of the angle between the vectors in Exercise 5. [Only part (b) is solved in the Study Guide.]

8. Find the area of the parallelogram spanned by the vectors in Exercise 5. [Only part (b) is solved in the Study Guide.]

9. Use vector notation to describe the triangle in space whose vertices are the origin and the endpoints of vectors $\mathbf{a}$ and $\mathbf{b}$.

10. Show that three vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$ lie in the same plane through the origin if and only if there are three scalars α, β, γ, not all zero, such that $\alpha\mathbf{a} + \beta\mathbf{b} + \gamma\mathbf{c} = \mathbf{0}$.

11. For real numbers $a_1, a_2, a_3, b_1, b_2, b_3$, show that

$$(a_1b_1 + a_2b_2 + a_3b_3)^2 \leq (a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2).$$

12. Let $\mathbf{u}, \mathbf{v}, \mathbf{w}$ be unit vectors that are orthogonal to each other. If $\mathbf{a} = \alpha\mathbf{u} + \beta\mathbf{v} + \gamma\mathbf{w}$, show that

$$\alpha = \mathbf{a} \cdot \mathbf{u}, \qquad \beta = \mathbf{a} \cdot \mathbf{v}, \qquad \gamma = \mathbf{a} \cdot \mathbf{w}.$$

Interpret the results geometrically.

13. Let $\mathbf{a}, \mathbf{b}$ be two vectors in the plane, $\mathbf{a} = (a_1, a_2)$. $\mathbf{b} = (b_1, b_2)$, and let λ be a real number. Show that the area of the parallelogram determined by $\mathbf{a}$ and $\mathbf{b} + \lambda\mathbf{a}$ is the same as that determined by $\mathbf{a}$ and $\mathbf{b}$. Sketch. Relate this result to a known property of determinants.

14. Find the volume of the parallelepiped determined by the vertices $(0, 1, 0)$, $(1, 1, 1)$, $(0, 2, 0)$, $(3, 1, 2)$.

15. Given nonzero vectors $\mathbf{a}$ and $\mathbf{b}$ in $\mathbb{R}^3$, show that the vector $\mathbf{v} = \|\mathbf{a}\|\mathbf{b} + \|\mathbf{b}\|\mathbf{a}$ bisects the angle between $\mathbf{a}$ and $\mathbf{b}$.

16. Use vector methods to prove that the distance from the point (x_1, y_1) to the line $ax + by = c$ is

$$\frac{|ax_1 + by_1 - c|}{\sqrt{a^2 + b^2}}.$$

17. Verify that the direction of $\mathbf{b} \times \mathbf{c}$ is given by the right-hand rule, by choosing $\mathbf{b}$, $\mathbf{c}$ to be two of the vectors $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$.

18. (a) Suppose $\mathbf{a} \cdot \mathbf{b} = \mathbf{a}' \cdot \mathbf{b}$ for all $\mathbf{b}$. Show that $\mathbf{a} = \mathbf{a}'$.
(b) Suppose $\mathbf{a} \times \mathbf{b} = \mathbf{a}' \times \mathbf{b}$ for all $\mathbf{b}$. Is it true that $\mathbf{a} = \mathbf{a}'$?

19. (a) Using vector methods, show that the distance between two nonparallel lines l_1 and l_2 is given by

$$d = \frac{|(\mathbf{v}_2 - \mathbf{v}_1) \cdot (\mathbf{a}_1 \times \mathbf{a}_2)|}{\|\mathbf{a}_1 \times \mathbf{a}_2\|},$$

where $\mathbf{v}_1$, $\mathbf{v}_2$ are any two points on l_1 and l_2, respectively, and $\mathbf{a}_1$ and $\mathbf{a}_2$ are the directions of l_1 and l_2. [HINT: Consider the plane through l_2 that is parallel to l_1. Show that the vector $(\mathbf{a}_1 \times \mathbf{a}_2)/\|\mathbf{a}_1 \times \mathbf{a}_2\|$ is a unit normal for this plane; now project $\mathbf{v}_2 - \mathbf{v}_1$ onto this normal direction.]

(b) Find the distance between the line l_1 determined by the points $(-1, -1, 1)$ and $(0, 0, 0)$ and the line l_2 determined by the points $(0, -2, 0)$ and $(2, 0, 5)$.

20. Show that two planes given by the equations $Ax + By + Cz + D_1 = 0$ and $Ax + By + Cz + D_2 = 0$ are parallel, and that the distance between them is

$$\frac{|D_1 - D_2|}{\sqrt{A^2 + B^2 + C^2}}.$$

21. (a) Prove that the area of the triangle in the plane with vertices (x_1, y_1), (x_2, y_2), (x_3, y_3) is the absolute value of

$$\frac{1}{2}\begin{vmatrix} 1 & 1 & 1 \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{vmatrix}.$$

(b) Find the area of the triangle with vertices $(1, 2)$, $(0, 1)$, $(-1, 1)$.

22. Convert the following points from Cartesian to cylindrical and spherical coordinates and plot:

(a) $(0, 3, 4)$
(b) $(-\sqrt{2}, 1, 0)$
(c) $(0, 0, 0)$
(d) $(-1, 0, 1)$
(e) $(-2\sqrt{3}, -2, 3)$

23. Convert the following points from cylindrical to Cartesian and spherical coordinates and plot:

(a) $(1, \pi/4, 1)$
(b) $(3, \pi/6, -4)$

(c) $(0, \pi/4, 1)$
(d) $(2, -\pi/2, 1)$
(e) $(-2, -\pi/2, 1)$

24. Convert the following points from spherical to Cartesian and cylindrical coordinates and plot:

(a) $(1, \pi/2, \pi)$
(b) $(2, -\pi/2, \pi/6)$
(c) $(0, \pi/8, \pi/35)$
(d) $(2, -\pi/2, -\pi)$
(e) $(-1, \pi, \pi/6)$

25. Rewrite the equation $z = x^2 - y^2$ using cylindrical and spherical coordinates.

26. Using spherical coordinates, show that

$$\phi = \cos^{-1}\left(\frac{\mathbf{u} \cdot \mathbf{k}}{\|\mathbf{u}\|}\right)$$

where $\mathbf{u} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$. Interpret geometrically.

27. Verify the Cauchy–Schwarz and triangle inequalities for

$$\mathbf{x} = (3, 2, 1, 0) \quad \text{and} \quad \mathbf{y} = (1, 1, 1, 2).$$

28. Multiply the matrices

$$A = \begin{bmatrix} 3 & 0 & 1 \\ 2 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

Does $AB = BA$?

29. (a) Show that for two $n \times n$ matrices A and B, and $\mathbf{x} \in \mathbb{R}^n$,

$$(AB)\mathbf{x} = A(B\mathbf{x}).$$

(b) What does the equality in part (a) imply about the relationship between the composition of the mappings $\mathbf{x} \mapsto B\mathbf{x}$, $\mathbf{y} \mapsto A\mathbf{y}$, and matrix multiplication?

30. Find the volume of the parallelepiped spanned by the vectors

$$(1, 0, 1), \quad (1, 1, 1), \quad \text{and} \quad (-3, 2, 0).$$

31. (For students with some knowledge of linear algebra.) Verify that a linear mapping T of $\mathbb{R}^n$ to $\mathbb{R}^n$ is determined by an $n \times n$ matrix.

32. Find an equation for the plane that contains $(3, -1, 2)$ and the line with equation $\mathbf{v} = (2, -1, 0) + t(2, 3, 0)$.

33. The work W done in moving an object from $(0, 0)$ to $(7, 2)$ subject to a constant force $\mathbf{F}$ is $W = \mathbf{F} \cdot \mathbf{r}$, where $\mathbf{r}$ is the vector with its head at $(7, 2)$ and tail at $(0, 0)$. The units are feet and pounds.

(a) Suppose the force $\mathbf{F} = 10 \cos\theta\mathbf{i} + 10 \sin\theta\mathbf{j}$. Find W in terms of θ.

(b) Suppose the force $\mathbf{F}$ has magnitude of 6 lb and makes an angle of $\pi/6$ rad with the horizontal, pointing right. Find W in foot-pounds.

34. If a particle with mass m moves with velocity $\mathbf{v}$, its *momentum* is $\mathbf{p} = m\mathbf{v}$. In a game of marbles, a marble with mass 2 grams (g) is shot with velocity 2 meters per second (m/s), hits two marbles with mass 1 g each, and comes to a dead halt. One of the marbles flies off with a velocity of 3 m/s at an angle of 45° to the incident direction of the larger marble as in Figure 1.R.1. Assuming that the total momentum before and after the collision is the same (according to the law of conservation of momentum), at what angle and speed does the second marble move?

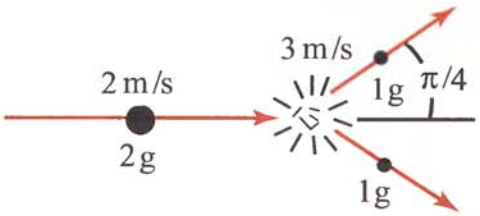


Figure 1.R.1 Momentum and marbles.

35. Show that for all x, y, z,

$$\begin{vmatrix} x+2 & y & z \\ z & y+1 & 10 \\ 5 & 5 & 2 \end{vmatrix} = -\begin{vmatrix} y & x+2 & z \\ 1 & z-x-2 & 10-z \\ 5 & 5 & 2 \end{vmatrix}.$$

36. Show that

$$\begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} \neq 0$$

if x, y, and z are all different.

37. Show that

$$\begin{vmatrix} 66 & 628 & 246 \\ 88 & 435 & 24 \\ 2 & -1 & 1 \end{vmatrix} = \begin{vmatrix} 68 & 627 & 247 \\ 86 & 436 & 23 \\ 2 & -1 & 1 \end{vmatrix}.$$

38. Show that

$$\begin{vmatrix} n & n+1 & n+2 \\ n+3 & n+4 & n+5 \\ n+6 & n+7 & n+8 \end{vmatrix}$$

has the same value no matter what n is. What is this value?

39. The volume of a tetrahedron with concurrent edges **a**, **b**, **c** is given by $V = \frac{1}{6}\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$.

(a) Express the volume as a determinant.
(b) Evaluate V when $\mathbf{a} = \mathbf{i} + \mathbf{j} + \mathbf{k}$, $\mathbf{b} = \mathbf{i} - \mathbf{j} + \mathbf{k}$, $\mathbf{c} = \mathbf{i} + \mathbf{j}$.

Use the following definition for Exercises 40 and 41: Let $\mathbf{r}_1, \ldots, \mathbf{r}_n$ *be vectors in* $\mathbb{R}^3$ *from* 0 *to the masses* $m_1, \ldots, m_n$. *The* ***center of mass*** *is the vector*

$$\mathbf{c} = \frac{\sum_{i=1}^{n} m_i \mathbf{r}_i}{\sum_{i=1}^{n} m_i}.$$

40. A tetrahedron sits in xyz coordinates with one vertex at (0, 0, 0), and the three edges concurrent at (0, 0, 0) are coincident with the vectors **a, b, c**.

(a) Draw a figure and label the heads of the vectors **a**, **b**, **c**.
(b) Find the center of mass of each of the four triangular faces of the tetrahedron if a unit mass is placed at each vertex.

41. Show that for any vector **r**, the center of mass of a system satisfies

$$\sum_{i=1}^{n} m_i \|\mathbf{r} - \mathbf{r}_i\|^2 = \sum_{i=1}^{n} m_i \|\mathbf{r}_i - \mathbf{c}\|^2 + m\|\mathbf{r} - \mathbf{c}\|^2,$$

where $m = \sum_{i=1}^{n} m_i$ is the total mass of the system.

In Exercises 42 to 47, find a unit vector that has the given property.

42. Parallel to the line $x = 3t + 1, y = 16t - 2, z = -(t + 2)$.

43. Orthogonal to the plane $x - 6y + z = 12$.

44. Parallel to both the planes $8x + y + z = 1$ and $x - y - z = 0$.

45. Orthogonal to $\mathbf{i} + 2\mathbf{j} - \mathbf{k}$ and to $\mathbf{k}$.

46. Orthogonal to the line $x = 2t - 1, y = -t - 1, z = t + 2$, and the vector $\mathbf{i} - \mathbf{j}$.

47. At an angle of 30° to $\mathbf{i}$ and making equal angles with $\mathbf{j}$ and $\mathbf{k}$.

2

Differentiation

I turn away with fright and horror from the lamentable evil of functions which do not have derivatives.

Charles Hermite,
in a letter to Thomas Jan Stieltjes

This chapter extends the principles of differential calculus for functions of one variable to functions of several variables. We begin in Section 2.1 with the geometry of real-valued functions and study the graphs of these functions as an aid in visualizing them. Section 2.2 gives some basic definitions relating to limits and continuity. This subject is treated briefly, because it requires time and mathematical maturity to develop fully and is therefore best left to a more advanced course. Fortunately, a complete understanding of all the subtleties of the limit concept is not necessary for our purposes; the student who has difficulty with this section should bear this in mind. However, we hasten to add that the notion of a limit is central to the definition of the derivative, but not to the computation of most derivatives in specific problems, as we already know from one-variable calculus. Sections 2.3 and 2.5 deal with the definition of the derivative, and establish some basic rules of calculus: namely, how to differentiate a sum, product, quotient, or composition. In Section 2.6, we study directional derivatives and tangent planes, relating these ideas to those in Section 2.1. Finally, the Internet supplement gives some of the technical proofs.

In generalizing calculus from one dimension to several, it is often convenient to use the language of matrix algebra. What we shall need has been summarized in Section 1.5.

2.1 The Geometry of Real-Valued Functions

We launch our investigation of real-valued functions by developing methods for visualizing them. In particular, we introduce the notions of a graph, a level curve, and a level surface of such functions.

Functions and Mappings

Let f be a function whose domain is a subset A of $\mathbb{R}^n$ and with a range contained in $\mathbb{R}^m$. By this we mean that to each $\mathbf{x} = (x_1, \ldots, x_n) \in A$, f assigns a value $f(\mathbf{x})$, an m-tuple in $\mathbb{R}^m$. Such functions f are called ***vector-valued functions***[1] if $m > 1$, and ***scalar-valued functions*** if $m = 1$. For example, the scalar-valued function $f(x, y, z) = (x^2 + y^2 + z^2)^{-3/2}$ maps the set A of $(x, y, z) \neq (0, 0, 0)$ in $\mathbb{R}^3$ ($n = 3$ in this case) to $\mathbb{R}$ ($m = 1$). To denote f we sometimes write

$$f\colon (x, y, z) \mapsto (x^2 + y^2 + z^2)^{-3/2}.$$

Note that in $\mathbb{R}^3$ we often use the notation (x, y, z) instead of (x_1, x_2, x_3). In general, the notation $\mathbf{x} \mapsto f(\mathbf{x})$ is useful for indicating the value to which a point $\mathbf{x} \in \mathbb{R}^n$ is sent. We write $f\colon A \subset \mathbb{R}^n \to \mathbb{R}^m$ to signify that A is the domain of f(a subset of $\mathbb{R}^n$) and the range is contained in $\mathbb{R}^m$. We also use the expression f maps A *into* $\mathbb{R}^m$. Such functions f are called ***functions of several variables*** if $A \subset \mathbb{R}^n$, $n > 1$.

As another example we can take the vector-valued function $g\colon \mathbb{R}^6 \to \mathbb{R}^2$ defined by the rule

$$g(\mathbf{x}) = g(x_1, x_2, x_3, x_4, x_5, x_6) = \left(x_1x_2x_3x_4x_5x_6, \sqrt{x_1^2 + x_6^2}\right).$$

The first coordinate of the value of g at $\mathbf{x}$ is the product of the coordinates of $\mathbf{x}$.

Functions from $\mathbb{R}^n$ to $\mathbb{R}^m$ are not just mathematical abstractions, they arise naturally in problems studied in all the sciences. For example, to specify the temperature T in a region A of space requires a function $T\colon A \subset \mathbb{R}^3 \to \mathbb{R}$ ($n = 3, m = 1$); thus, $T(x, y, z)$ is the temperature at the point (x, y, z). To specify the velocity of a fluid moving in space requires a map $\mathbf{V}\colon \mathbb{R}^4 \to \mathbb{R}^3$, where $\mathbf{V}(x, y, z, t)$ is the velocity vector of the fluid at the point (x, y, z) in space at time t (see Figure 2.1.1). To

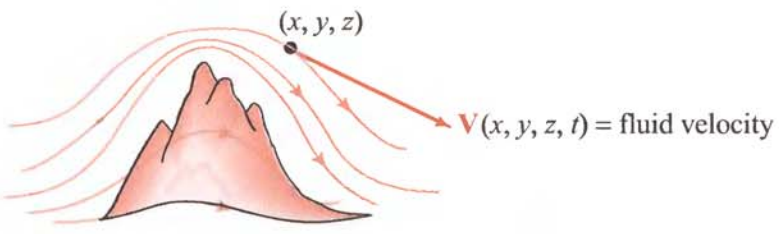


Figure 2.1.1 A fluid in motion defines a vector field $\mathbf{V}$ by specifying the velocity of the fluid particles at each point in space and time.

[1] Some mathematicians would write such an f in boldface, using the notation $\mathbf{f}(\mathbf{x})$, because the function is vector-valued. We did not do so, as a matter of personal taste. We use boldface primarily for mappings that are vector fields, introduced later. The notion of function was developed over many centuries, with the definition extended to cover more cases as they arose. For example, in 1667 James Gregory defined a function as "a quantity obtained from other quantities by a succession of algebraic operations or by any other operation imaginable." In 1755 Euler gave the following definition: "If some quantities depend on others in such a way as to undergo variation when the latter are varied then the former are called functions of the latter."

specify the reaction rate of a solution consisting of six reacting chemicals A, B, C, D, E, F in proportions x, y, z, w, u, v requires a map $\sigma\colon U \subset \mathbb{R}^6 \to \mathbb{R}$, where $\sigma(x, y, z, w, u, v)$ gives the rate when the chemicals are in the indicated proportions. To specify the cardiac vector (the vector giving the magnitude and direction of electric current flow in the heart) at time t requires a map $\mathbf{c}\colon \mathbb{R} \to \mathbb{R}^3$, $t \mapsto \mathbf{c}(t)$.

When $f\colon U \subset \mathbb{R}^n \to \mathbb{R}$, we say that f is a ***real-valued function of n variables with domain*** U. The reason we say "n variables" is simply that we regard the coordinates of a point $\mathbf{x} = (x_1, \ldots, x_n) \in U$ as n variables, and $f(\mathbf{x}) = f(x_1, \ldots, x_n)$ depends on these variables. We say "real-valued" because $f(x_1, \ldots, x_n)$ is a real number. A good deal of our work will be with real-valued functions, so we give them special attention.

Graphs of Functions

For $f\colon U \subset \mathbb{R} \to \mathbb{R}$ $(n = 1)$, the ***graph*** of f is the subset of $\mathbb{R}^2$ consisting of all points $(x, f(x))$ in the plane, for x in U. This subset can be thought of as a curve in $\mathbb{R}^2$. In symbols, we write this as

$$\text{graph} f = \{(x, f(x)) \in \mathbb{R}^2 \mid x \in U\},$$

where the curly braces mean "the set of all" and the vertical bar is read "such that." Drawing the graph of a function of one variable is a useful device to help visualize how the function actually behaves. (See Figure 2.1.2.) It will be helpful to generalize

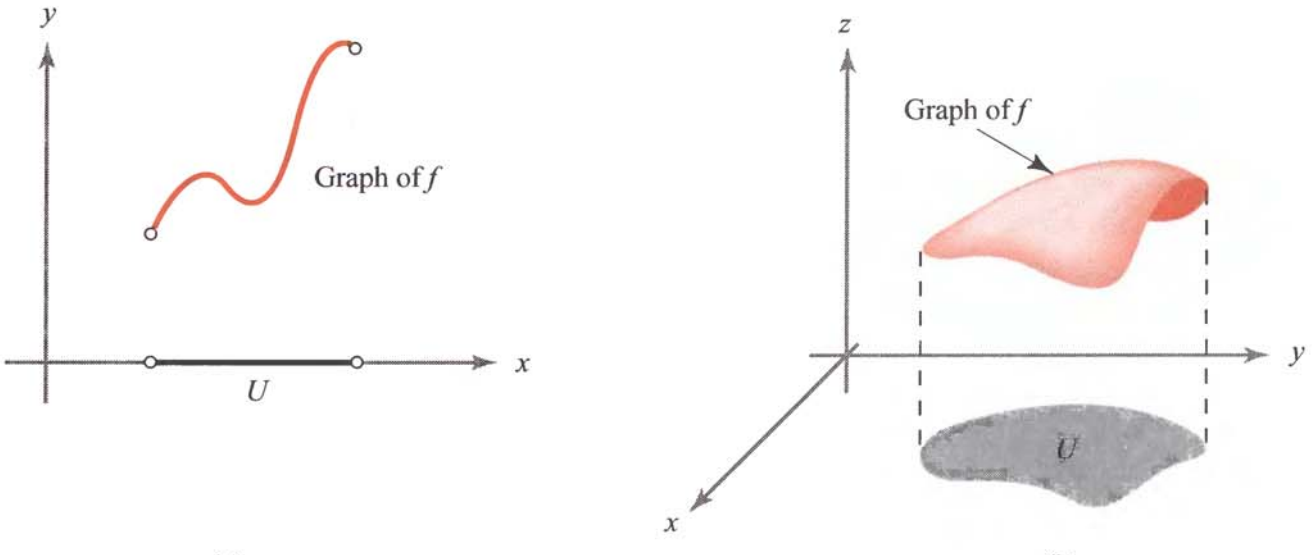


Figure 2.1.2 The graphs of (a) a function of one variable, and (b) a function of two variables.

the idea of a graph to functions of several variables. This leads to the following definition:

DEFINITION: Graph of a Function Let $f: U \subset \mathbb{R}^n \to \mathbb{R}$. Define the ***graph*** of f to be the subset of $\mathbb{R}^{n+1}$ consisting of all the points

$$(x_1, \ldots, x_n, f(x_1, \ldots, x_n))$$

in $\mathbb{R}^{n+1}$ for $(x_1, \ldots, x_n)$ in U. In symbols,

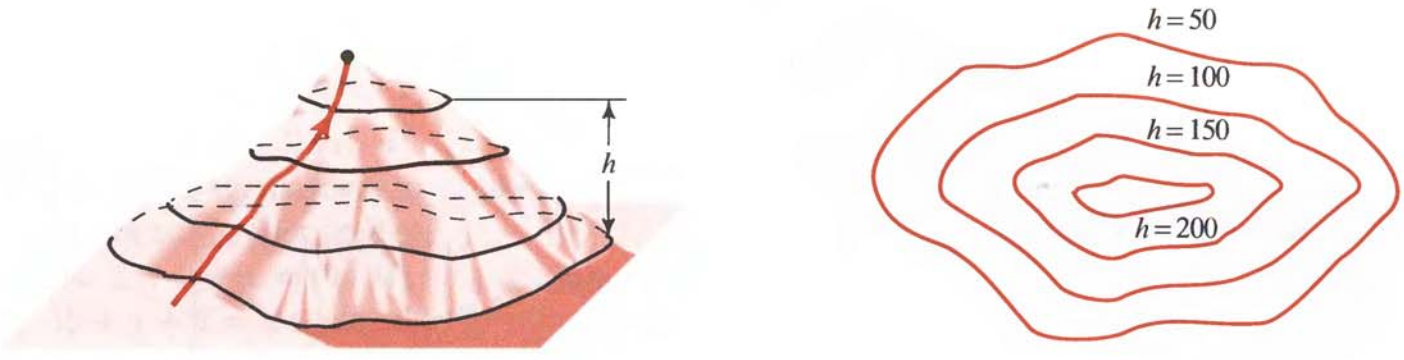
$$\text{graph } f = \{(x_1, \ldots, x_n, f(x_1, \ldots, x_n)) \in \mathbb{R}^{n+1} \mid (x_1, \ldots, x_n) \in U\}.$$

For the case $n = 1$ the graph is a curve in $\mathbb{R}^2$, while for $n = 2$ it is a surface in $\mathbb{R}^3$ (see Figure 2.1.2). For $n = 3$ it is difficult to visualize the graph, because, since we are humans living in a three-dimensional world, it is hard for us to envisage sets in $\mathbb{R}^4$. To help overcome this handicap, we introduce the idea of a level set.

Level Sets, Curves, and Surfaces

Suppose $f(x, y, z) = x^2 + y^2 + z^2$. A ***level set*** is a subset of $\mathbb{R}^3$ on which f is constant; for instance, the set where $x^2 + y^2 + z^2 = 1$ is a level set for f. This we can visualize: It is just a sphere of radius 1 in $\mathbb{R}^3$. Formally, a level set is the set of (x, y, z) such that $f(x, y, z) = c$, where c is a constant. The behavior or structure of a function is determined in part by the shape of its level sets; consequently, understanding these sets aids us in understanding the function in question. Level sets are also useful for understanding functions of two variables $f(x, y)$, in which case we speak of ***level curves*** or ***level contours***.

The idea is similar to that used to prepare contour maps, where one draws lines to represent constant altitudes; walking along such a line would mean walking on a level path. In the case of a hill rising from the xy plane, a graph of all the level curves gives us a good idea of the function $h(x, y)$, which represents the height of the hill at point (x, y) (see Figure 2.1.3).



(a) (b)

Figure 2.1.3 Level contours of a function are defined in the same manner as contour lines for a topographical map.

EXAMPLE 1 The constant function $f\colon \mathbb{R}^2 \to \mathbb{R}$, $(x, y) \mapsto 2$, that is, the function $f(x, y) = 2$, has as its graph the horizontal plane $z = 2$ in $\mathbb{R}^3$. The level curve of value c is empty if $c \neq 2$, and is the whole xy plane if $c = 2$. ▲

EXAMPLE 2 The function $f\colon \mathbb{R}^2 \to \mathbb{R}$, defined by $f(x, y) = x + y + 2$ has as its graph the inclined plane $z = x + y + 2$. This plane intersects the xy plane ($z = 0$) in the line $y = -x - 2$ and the z axis at the point $(0, 0, 2)$. For any value $c \in \mathbb{R}$, the level curve of value c is the straight line $y = -x + (c - 2)$; or in symbols, the set

$$L_c = \{(x, y) \mid y = -x + (c - 2)\} \subset \mathbb{R}^2.$$

We indicate a few of the level curves of the function in Figure 2.1.4. This is a contour map of the function f.

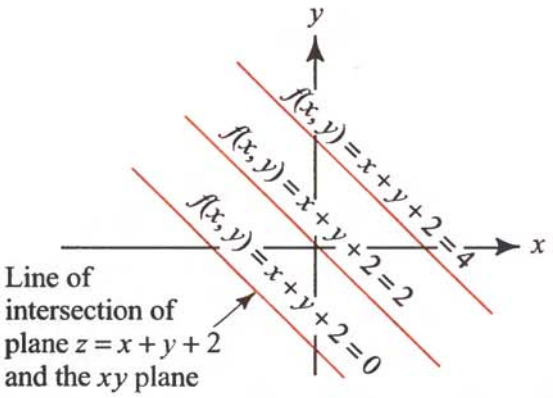


Figure 2.1.4 The level curves of $f(x, y) = x + y + 2$ show the sets on which f takes a given value.

From level curves labeled with the value or "height" of the function, the shape of the graph may be inferred by mentally elevating each level curve to the appropriate height, without stretching, tilting, or sliding it. If this procedure is visualized for all level curves, L_c, that is, for all values $c \in \mathbb{R}$, they will assemble to give the entire graph of f, as indicated by the shaded plane in Figure 2.1.5. If the graph is visualized

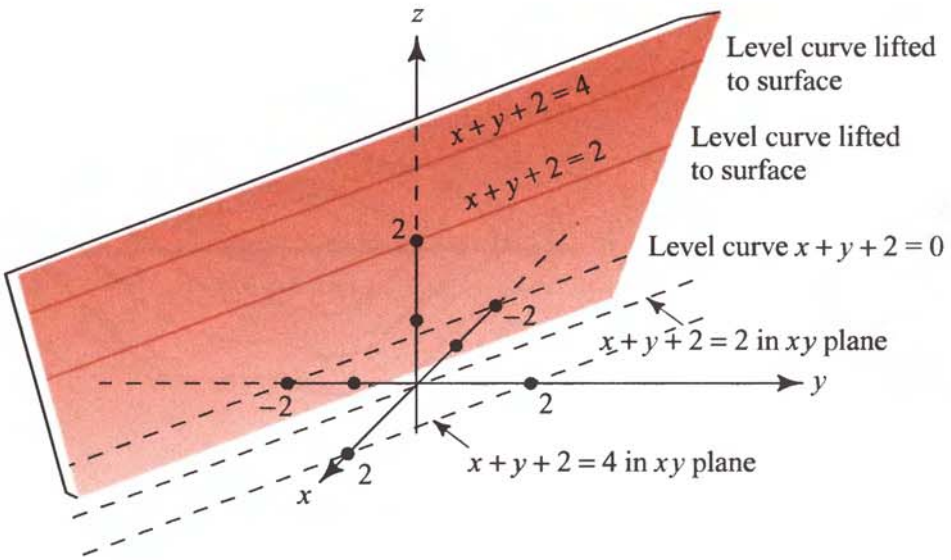


Figure 2.1.5 The relationship of level curves of Figure 2.1.4 to the graph of the function $f(x, y) = x + y + 2$, which is the plane $z = x + y + 2$.

using a finite number of level curves, a contour model is produced. If f is a smooth function, its graph will be a smooth surface, and so the contour model, mentally smoothed over, gives a good impression of the graph. ▲

DEFINITION: Level Curves and Surfaces Let $f: U \subset \mathbb{R}^n \to \mathbb{R}$ and let $c \in \mathbb{R}$. Then the ***level set of value*** c is defined to be the set of those points $\mathbf{x} \in U$ at which $f(\mathbf{x}) = c$. If $n = 2$, we speak of a ***level curve*** (of value c); and if $n = 3$, we speak of a ***level surface***. In symbols, the level set of value c is written

$$\{\mathbf{x} \in U \mid f(\mathbf{x}) = c\} \subset \mathbb{R}^n.$$

Note that the level set is always in the domain space.

EXAMPLE 3 Describe the graph of the quadratic function

$$f: \mathbb{R}^2 \to \mathbb{R}, (x, y) \mapsto x^2 + y^2.$$

SOLUTION The graph is the ***paraboloid of revolution*** $z = x^2 + y^2$, oriented upward from the origin, around the z axis. The level curve of value c is empty for $c < 0$; for $c > 0$ the level curve of value c is the set $\{(x, y) \mid x^2 + y^2 = c\}$, a circle of radius $\sqrt{c}$ centered at the origin. Thus, raised to height c above the xy plane, the level set is a circle of radius $\sqrt{c}$, indicating a parabolic shape (see Figures 2.1.6 and 2.1.7). ▲

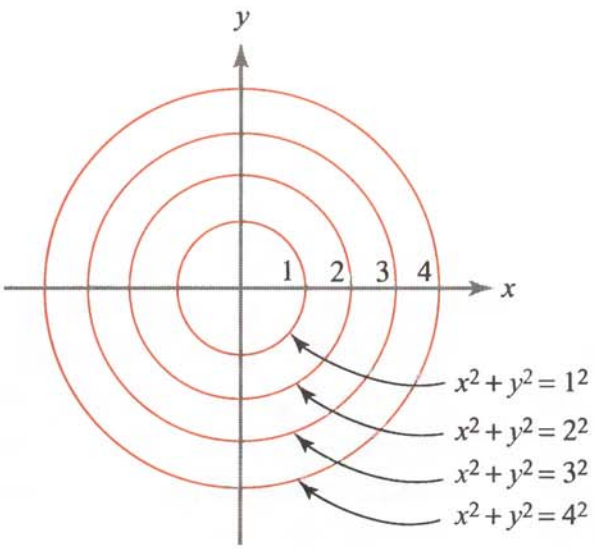


Figure 2.1.6 Some level curves for the function $f(x, y) = x^2 + y^2$.

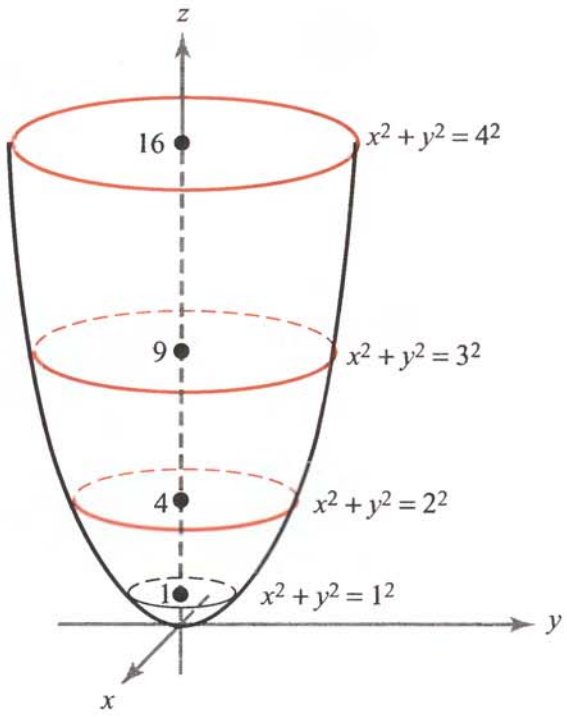


Figure 2.1.7 Level curves in Figure 2.1.6 raised to the graph.

The Method of Sections

By a ***section*** of the graph of f we mean the intersection of the graph and a (vertical) plane. For example, if P_1 is the xz plane in $\mathbb{R}^3$, defined by $y = 0$, then the section of f in Example 3 is the set

$$P_1 \cap \text{ graph } f = \{(x, y, z) \mid y = 0, z = x^2\},$$

which is a parabola in the xz plane. Similarly, if P_2 denotes the yz plane, defined by $x = 0$, then the section

$$P_2 \cap \text{ graph } f = \{(x, y, z) \mid x = 0, z = y^2\}$$

is a parabola in the yz plane (see Figure 2.1.8). It is usually helpful to compute at least one section to complement the information given by the level sets.

EXAMPLE 4 The graph of the quadratic function

$$f\colon \mathbb{R}^2 \to \mathbb{R}, (x, y) \mapsto x^2 - y^2$$

is called a ***hyperbolic paraboloid***, or ***saddle***, centered at the origin. Sketch the graph.

SOLUTION To visualize this surface, we first draw the level curves. To determine the level curves, we solve the equation $x^2 - y^2 = c$. Consider the values $c = 0, \pm 1, \pm 4$. For $c = 0$, we have $y^2 = x^2$, or $y = \pm x$, so that this level set consists of two straight lines through the origin. For $c = 1$, the level curve is $x^2 - y^2 = 1$, or $y = \pm\sqrt{x^2 - 1}$, which is a hyperbola that passes vertically through the x axis at the points $(\pm 1, 0)$ (see Figure 2.1.9). Similarly, for $c = 4$, the level curve is defined by $y = \pm\sqrt{x^2 - 4}$, the hyperbola passing vertically through the x axis at $(\pm 2, 0)$. For

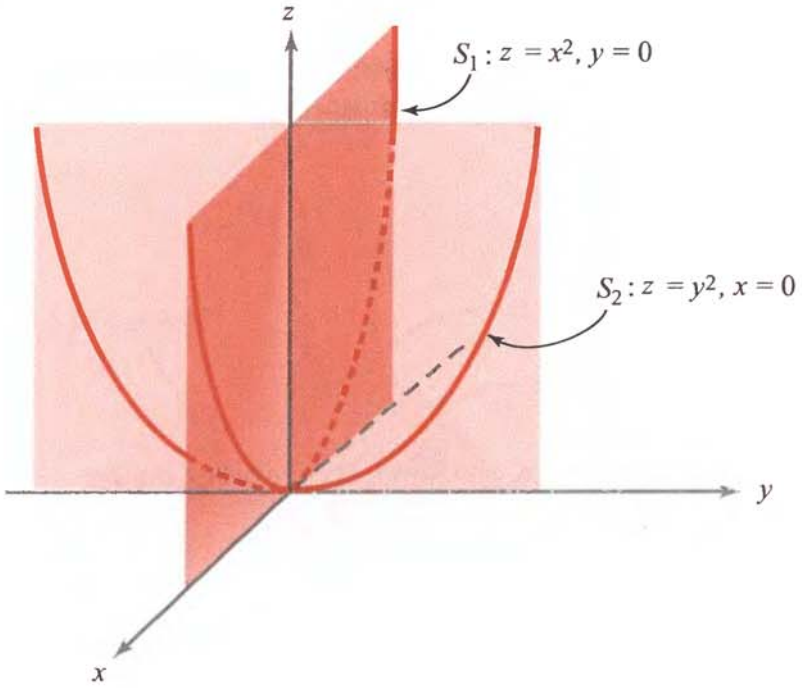


Figure 2.1.8 Two sections of the graph of $f(x, y) = x^2 + y^2$.

$c = -1$, we obtain the curve $x^2 - y^2 = -1$, that is, $x = \pm\sqrt{y^2 - 1}$, the hyperbola passing horizontally through the y axis at $(0, \pm 1)$. And for $c = -4$, the hyperbola through $(0, \pm 2)$ is obtained. These level curves are shown in Figure 2.1.9. Because it is not easy to visualize the graph of f from these data alone, we shall compute two sections, as in the previous example. For the section in the xz plane, we have

$$P_1 \cap \text{ graph of } f = \{(x, y, z) \mid y = 0, z = x^2\},$$

which is a parabola opening upward; and for the yz plane,

$$P_2 \cap \text{ graph } f = \{(x, y, z) \mid x = 0, z = -y^2\},$$

which is a parabola opening downward. The graph may now be visualized by lifting the level curves to the appropriate heights and smoothing out the resulting surface. Their

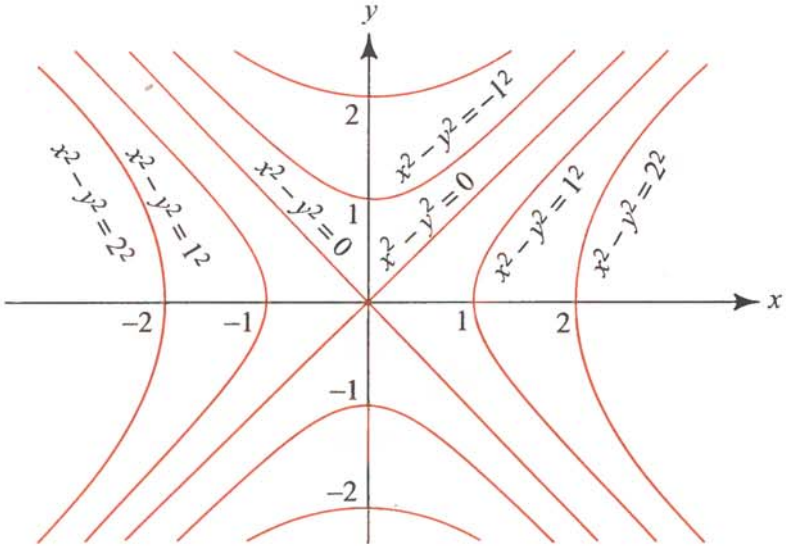


Figure 2.1.9 Level curves for the function $f(x, y) = x^2 - y^2$.

placement is aided by computing the parabolic sections. This procedure generates the hyperbolic saddle indicated in Figure 2.1.10. Compare this with the computer-generated graphs in Figure 2.1.11 (note that the orientation of the axes has been changed). ▲

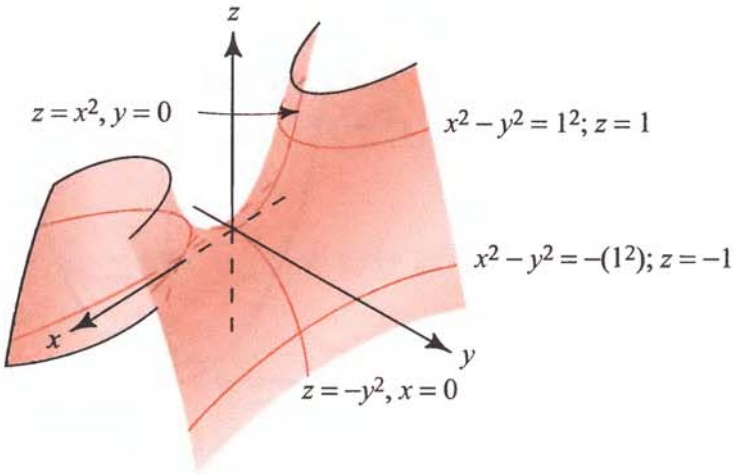


Figure 2.1.10 Some level curves on the graph of $f(x, y) = x^2 - y^2$.

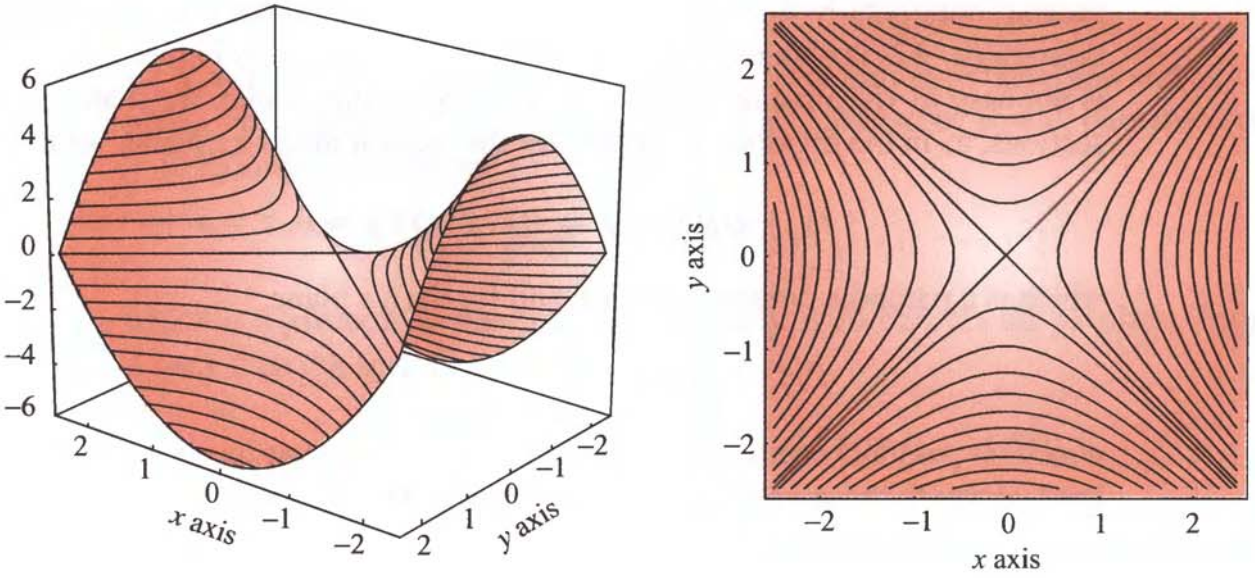


Figure 2.1.11 The graph of $z = x^2 - y^2$ and its level curves.

EXAMPLE 5 Describe the level sets of the function

$$f: \mathbb{R}^3 \to \mathbb{R}, (x, y, z) \mapsto x^2 + y^2 + z^2.$$

SOLUTION This is the three-dimensional analogue of Example 3. In this context, level sets are surfaces in the three-dimensional domain $\mathbb{R}^3$. The graph, in $\mathbb{R}^4$, cannot be visualized directly, but sections can nevertheless be computed.

The level set with value c is the set

$$L_c = \{(x, y, z) \mid x^2 + y^2 + z^2 = c\},$$

which is the sphere centered at the origin with radius $\sqrt{c}$ for $c > 0$, is a single point at the origin for $c = 0$, and is empty for $c < 0$. The level sets for $c = 0, 1, 4$, and 9 are indicated in Figure 2.1.12. ▲

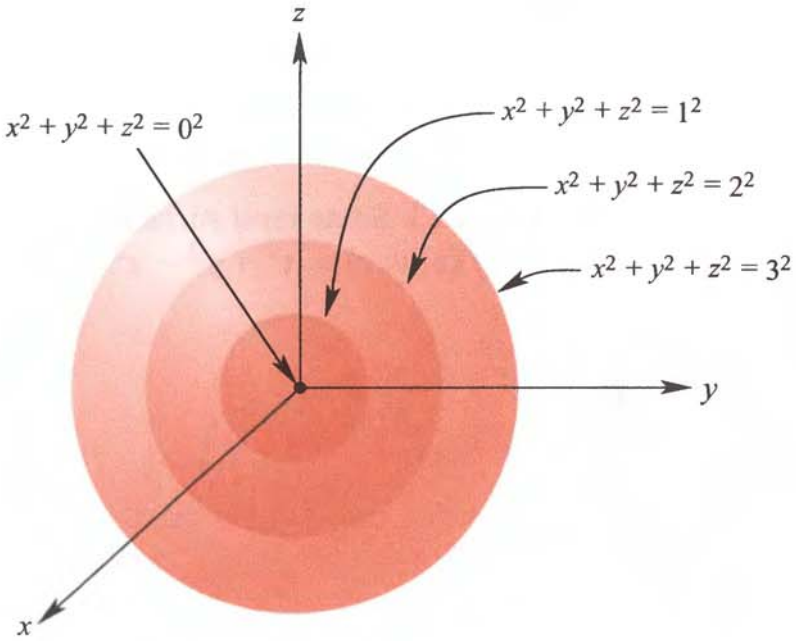


Figure 2.1.12 Some level surfaces for $f(x, y, z) = x^2 + y^2 + z^2$.

EXAMPLE 6 Describe the graph of the function $f: \mathbb{R}^3 \to \mathbb{R}$ defined by $f(x, y, z) = x^2 + y^2 - z^2$, which is the three-dimensional analogue of Example 4, and is also called a ***saddle***.

SOLUTION Formally, the graph of f is a subset of four-dimensional space. If we denote points in this space by (x, y, z, t), then the graph is given by

$$\{(x, y, z, t) \mid t = x^2 + y^2 - z^2\}.$$

The level surfaces of f are defined by

$$L_c = \{(x, y, z) \mid x^2 + y^2 - z^2 = c\}.$$

For $c = 0$, this is the cone $z = \pm\sqrt{x^2 + y^2}$ centered on the z axis. For c negative, say, $c = -a^2$, we obtain $z = \pm\sqrt{x^2 + y^2 + a^2}$, which is a hyperboloid of two sheets around the z axis, passing through the z axis at the points $(0, 0, \pm a)$. For c positive, say, $c = b^2$, the level surface is the ***single-sheeted hyperboloid of revolution*** around the z axis defined by $z = \pm\sqrt{x^2 + y^2 - b^2}$, which intersects the xy plane in the circle of radius $|b|$. These level surfaces are sketched in Figure 2.1.13.

Another view of the graph may be obtained from a section. For example, the subspace $S_{y=0} = \{(x, y, z, t) \mid y = 0\}$ intersects the graph in the section

$$S_{y=0} \cap \text{graph } f = \{(x, y, z, t) \mid y = 0, t = x^2 - z^2\},$$

that is, the set of points of the form $(x, 0, z, x^2 - z^2)$, which may be considered to be a surface in xzt space (see Figure 2.1.14). ▲

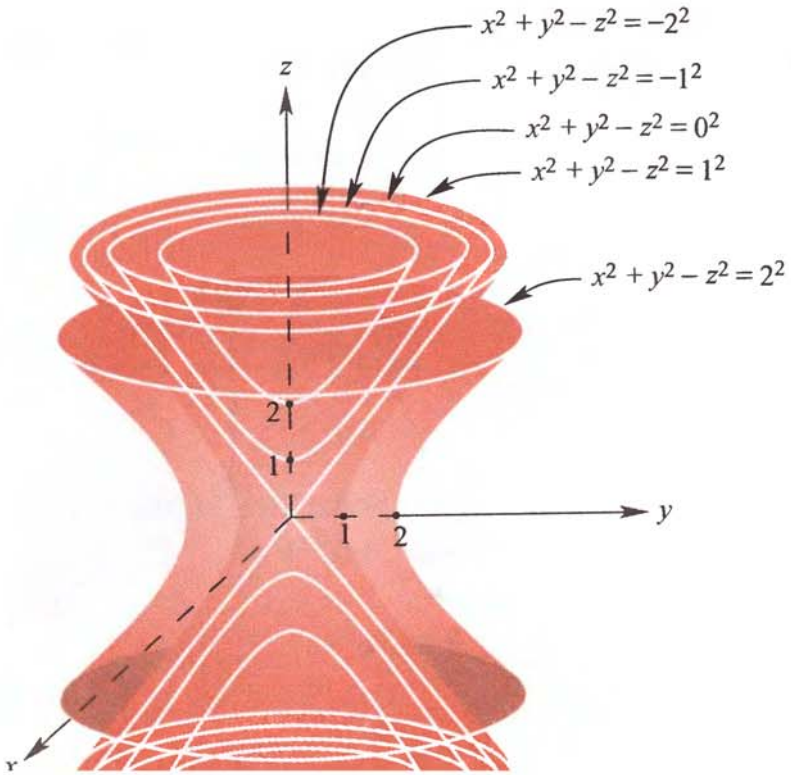
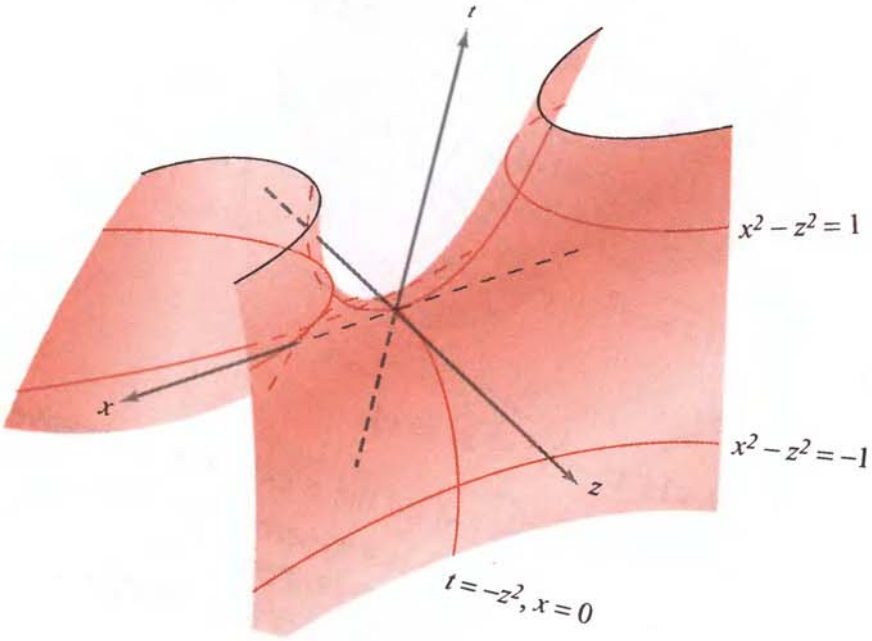


Figure 2.1.13 Some level surfaces of the function $f(x, y, z) = x^2 + y^2 - z^2$.



gure 2.1.14 The $y = 0$ section the graph of $f(x, y, z) =$ $+ y^2 - z^2$.

We have seen how the methods of sections and level sets can be used to understand the behavior of a function and its graph; these techniques can be quite useful to people who desire comprehensive visualization of complicated data. There are many computer programs available to do this, and we show the results of one such program in Figure 2.1.15.

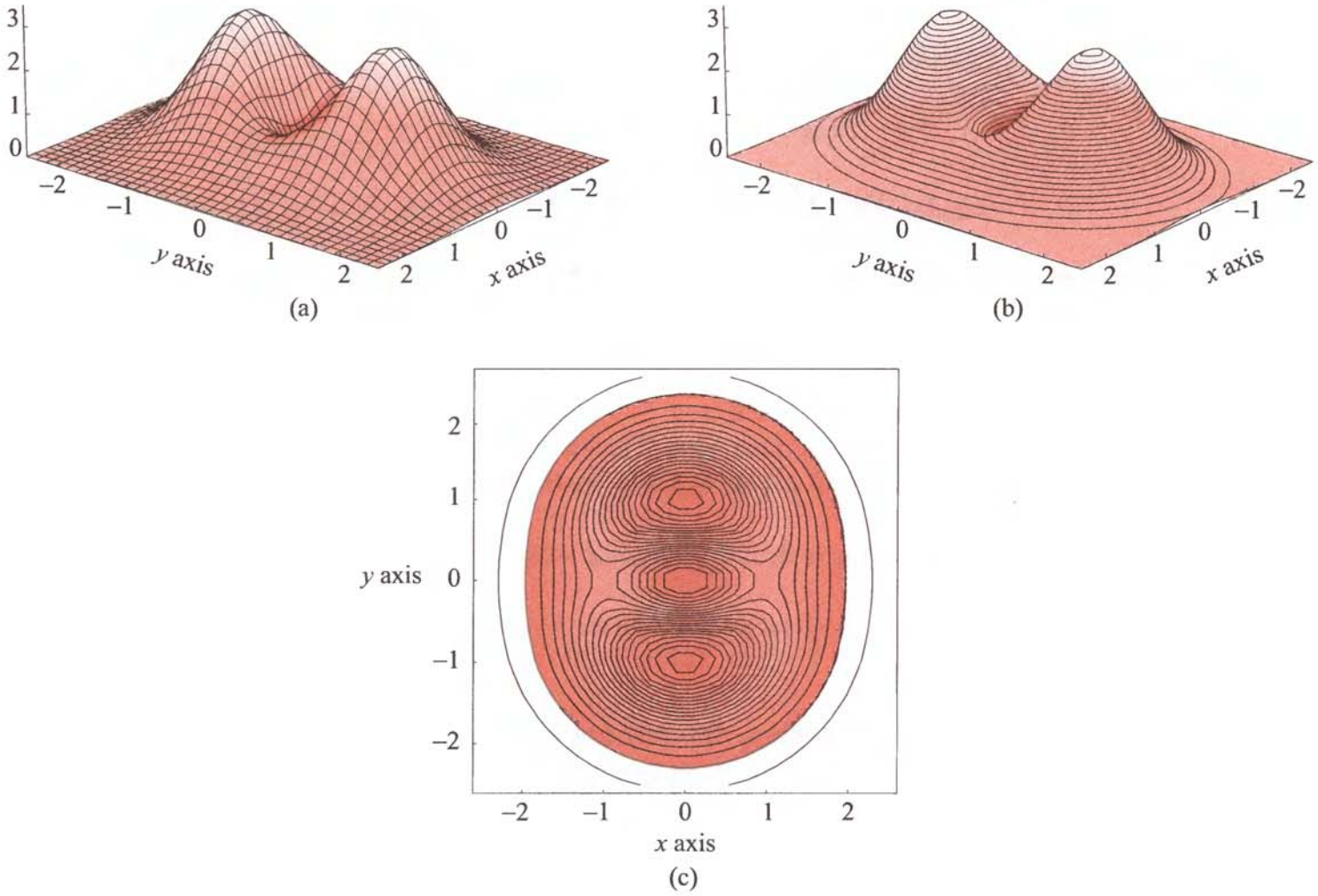


Figure 2.1.15 Computer-generated graph of $z = (x^2 + 3y^2)\exp(1 - x^2 - y^2)$ represented in three ways: (a) by sections, (b) by level curves on a graph, and (c) by level curves in the xy plane.

EXERCISES

1. Sketch the level curves and graphs of the following functions:

(a) $f\colon \mathbb{R}^2 \to \mathbb{R},\ (x, y) \mapsto x - y + 2$
(b) $f\colon \mathbb{R}^2 \to \mathbb{R},\ (x, y) \mapsto x^2 + 4y^2$
(c) $f\colon \mathbb{R}^2 \to \mathbb{R},\ (x, y) \mapsto -xy$

2. Describe the behavior, as c varies, of the level curve $f(x, y) = c$ for each of these functions:

(a) $f(x, y) = x^2 + y^2 + 1$ (b) $f(x, y) = 1 - x^2 - y^2$ (c) $f(x, y) = x^3 - x$

3. For the functions in Examples 2, 3, and 4, compute the section of the graph defined by the plane

$$S_\theta = \{(x, y, z) \mid y = x \tan\theta\}$$

for a given *constant* θ. Do this by expressing z as a function of r, where $x = r\cos\theta$, $y = r\sin\theta$. Determine which of these functions f have the property that the shape of the section $S_\theta \cap \text{graph } f$ is independent of θ. (The solution for Example 3 only is in the Study Guide.)

In Exercises 4 to 10, draw the level curves (in the xy plane) for the given function f and specified values of c. Sketch the graph of $z = f(x, y)$.

4. $f(x, y) = 4 - 3x + 2y, c = 0, 1, 2, 3, -1, -2, -3$

5. $f(x, y) = (100 - x^2 - y^2)^{1/2}, c = 0, 2, 4, 6, 8, 10$

6. $f(x, y) = (x^2 + y^2)^{1/2}, c = 0, 1, 2, 3, 4, 5$

7. $f(x, y) = x^2 + y^2, c = 0, 1, 2, 3, 4, 5$

8. $f(x, y) = 3x - 7y, c = 0, 1, 2, 3, -1, -2, -3$

9. $f(x, y) = x^2 + xy, c = 0, 1, 2, 3, -1, -2, -3$

10. $f(x, y) = x/y, c = 0, 1, 2, 3, -1, -2, -3$

In Exercises 11 to 13, sketch or describe the level surfaces and a section of the graph of each function.

11. $f: \mathbb{R}^3 \to \mathbb{R}, (x, y, z) \mapsto -x^2 - y^2 - z^2$

12. $f: \mathbb{R}^3 \to \mathbb{R}, (x, y, z) \mapsto 4x^2 + y^2 + 9z^2$

13. $f: \mathbb{R}^3 \to \mathbb{R}, (x, y, z) \mapsto x^2 + y^2$

In Exercises 14 to 18, describe the graph of each function by computing some level sets and sections.

14. $f: \mathbb{R}^3 \to \mathbb{R}, (x, y, z) \mapsto xy$

15. $f: \mathbb{R}^3 \to \mathbb{R}, (x, y, z) \mapsto xy + yz$

16. $f: \mathbb{R}^3 \to \mathbb{R}, (x, y, z) \mapsto xy + z^2$

17. $f: \mathbb{R}^2 \to \mathbb{R}, (x, y) \mapsto |y|$

18. $f: \mathbb{R}^2 \to \mathbb{R}, (x, y) \mapsto \max(|x|, |y|)$

Sketch or describe the surfaces in $\mathbb{R}^3$ of the equations presented in Exercises 19 to 31.

19. $4x^2 + y^2 = 16$

20. $x + 2z = 4$

21. $z^2 = y^2 + 4$

22. $x^2 + y^2 - 2x = 0$

23. $\dfrac{x}{4} = \dfrac{y^2}{4} + \dfrac{z^2}{9}$

24. $\dfrac{y^2}{9} + \dfrac{z^2}{4} = 1 + \dfrac{x^2}{16}$

25. $z = x^2$

26. $y^2 + z^2 = 4$

27. $z = \dfrac{y^2}{4} - \dfrac{x^2}{9}$

28. $y^2 = x^2 + z^2$

29. $4x^2 - 3y^2 + 2z^2 = 0$

30. $\dfrac{x^2}{9} + \dfrac{y^2}{12} + \dfrac{z^2}{9} = 1$

31. $x^2 + y^2 + z^2 + 4x - by + 9z - b = 0$, where b is a constant

32. Using polar coordinates, describe the level curves of the function defined by

$$f(x, y) = 2xy/(x^2 + y^2) \text{ if } (x, y) \neq (0, 0) \text{ and } f(0, 0) = 0.$$

33. Let $f\colon \mathbb{R}^2\backslash\{\mathbf{0}\} \to \mathbb{R}$ be given in polar coordinates by $f(r, \theta) = (\cos 2\theta)/r^2$. Sketch a few level curves in the xy plane. Here, $\mathbb{R}^2\backslash\{\mathbf{0}\} = \{\mathbf{x} \in \mathbb{R}^2 \mid \mathbf{x} \neq \mathbf{0}\}$.

34. Show that in Figure 2.1.15, the level "curve" $z = 3$ consists of two points.

2.2 Limits and Continuity

This section develops the concepts of open sets, limits, and continuity; open sets are needed to understand limits, and limits are in turn needed to understand continuity and differentiability.

As in elementary calculus, it is not necessary to completely master the limit concept in order to work problems in differentiation. For this reason, instructors may treat the following material with varying degrees of rigor. The student should consult with the instructor about the depth of understanding required.

Open Sets

We begin formulating the concept of an open set by defining an open disk. Let $\mathbf{x}_0 \in \mathbb{R}^n$ and let r be a positive real number. The ***open disk*** (or ***open ball***) of radius r and center $\mathbf{x}_0$ is defined to be the set of all points $\mathbf{x}$ such that $\|\mathbf{x} - \mathbf{x}_0\| < r$. This set is denoted $D_r(\mathbf{x}_0)$, and is the set of points $\mathbf{x}$ in $\mathbb{R}^n$ whose distance from $\mathbf{x}_0$ is less than r. Notice that we include only those $\mathbf{x}$ for which *strict* inequality holds. The disk $D_r(\mathbf{x}_0)$ is illustrated in Figure 2.2.1 for $n = 1, 2, 3$. For the case $n = 1$ and $x_0 \in \mathbb{R}$, the open disk $D_r(x_0)$ is the open interval $(x_0 - r, x_0 + r)$, which consists of all numbers $x \in \mathbb{R}$ *strictly* between $x_0 - r$ and $x_0 + r$. For the case $n = 2$, $\mathbf{x}_0 \in \mathbb{R}^2$, $D_r(\mathbf{x}_0)$ is the "inside" of the disk of radius r centered at $\mathbf{x}_0$. For the case $n = 3$, $\mathbf{x}_0 \in \mathbb{R}^3$, $D_r(\mathbf{x}_0)$ is the part strictly "inside" of the ball of radius r centered at $\mathbf{x}_0$.

DEFINITION: Open Sets Let $U \subset \mathbb{R}^n$ (that is, let U be a subset of $\mathbb{R}^n$). We call U an ***open set*** when for every point $\mathbf{x}_0$ in U there exists some $r > 0$ such that $D_r(\mathbf{x}_0)$ is contained within U; symbolically, we write $D_r(\mathbf{x}_0) \subset U$ (see Figure 2.2.2).

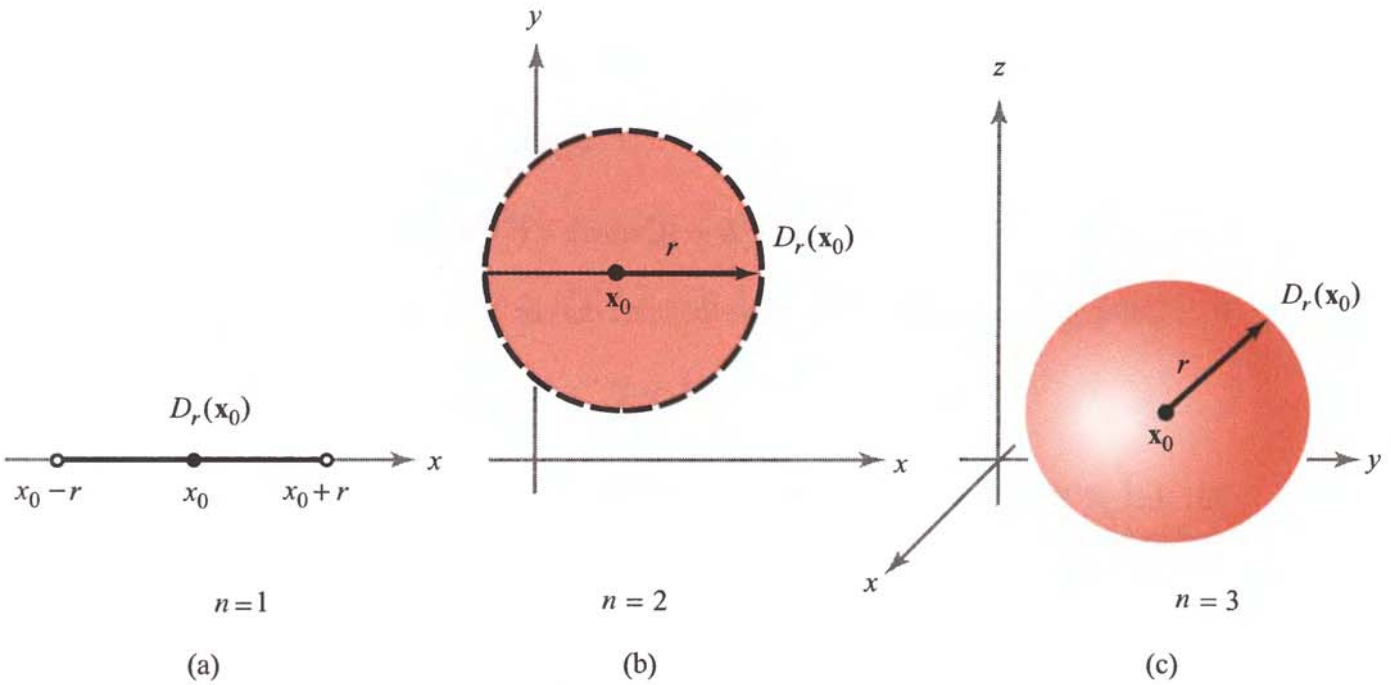


Figure 2.2.1 What disks $D_r(\mathbf{x}_0)$ look like in (a) one, (b) two, and (c) three dimensions.

The number $r > 0$ can depend on the point $\mathbf{x}_0$, and generally r will shrink as $\mathbf{x}_0$ gets closer to the "edge" of U. Intuitively speaking, a set U is open when the "boundary" points of U do not lie in U. In Figure 2.2.2, the dashed line is *not* included in U.

We establish the convention that the empty set $\emptyset$ (the set consisting of no elements) is open.

We have defined an open disk and an open set. From our choice of terms it would seem that an open disk should also be an open set. A little thought shows that this fact requires some proof. The following theorem does this.

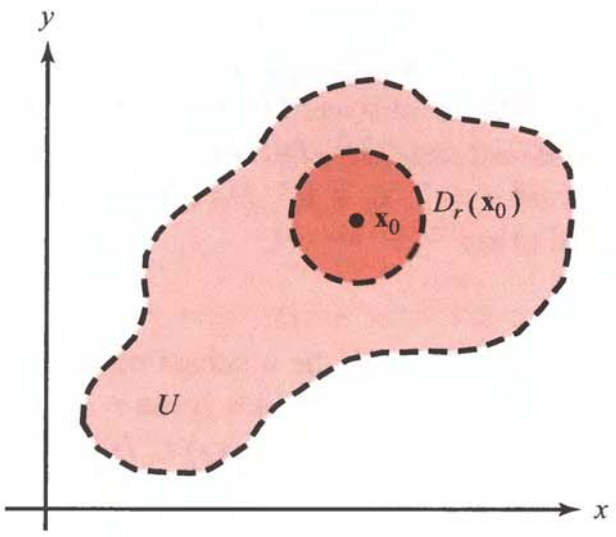
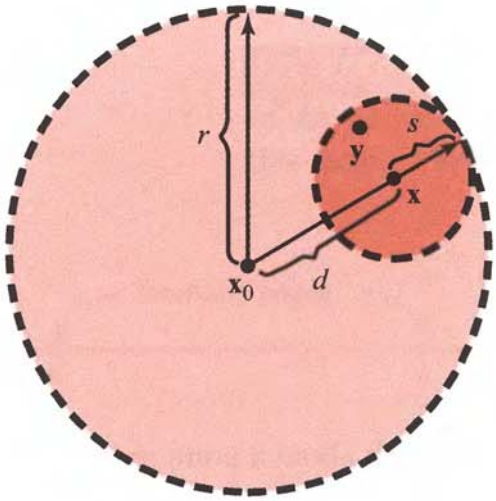


Figure 2.2.2 An open set U is one that completely encloses some disk $D_r(\mathbf{x}_0)$ about each of its points $\mathbf{x}_0$.

THEOREM 1 For each $\mathbf{x}_0 \in \mathbb{R}^n$ and $r > 0$, $D_r(\mathbf{x}_0)$ is an open set.

PROOF Let $\mathbf{x} \in D_r(\mathbf{x}_0)$, that is, let $\|\mathbf{x} - \mathbf{x}_0\| < r$. According to the definition of an open set, we must find an $s > 0$ such that $D_s(\mathbf{x}) \subset D_r(\mathbf{x}_0)$. Referring to Figure 2.2.3, we see that $s = r - \|\mathbf{x} - \mathbf{x}_0\|$ is a reasonable choice; note that $s > 0$, but that s becomes smaller if $\mathbf{x}$ is nearer the edge of $D_r(\mathbf{x}_0)$.



$d = \|\mathbf{x} - \mathbf{x}_0\|$
$s = r - \|\mathbf{x} - \mathbf{x}_0\|$

Figure 2.2.3 The geometry of the proof that an open disk is an open set.

To prove that $D_s(\mathbf{x}) \subset D_r(\mathbf{x}_0)$, let $\mathbf{y} \in D_s(\mathbf{x})$; that is, let $\|\mathbf{y} - \mathbf{x}\| < s$. We want to prove that $\mathbf{y} \in D_r(\mathbf{x}_0)$ as well. Proving this, in view of the definition of an r-disk, entails showing that $\|\mathbf{y} - \mathbf{x}_0\| < r$. This is done by using the triangle inequality for vectors in $\mathbb{R}^n$:

$$\|\mathbf{y} - \mathbf{x}_0\| = \|(\mathbf{y} - \mathbf{x}) + (\mathbf{x} - \mathbf{x}_0)\| \leq \|\mathbf{y} - \mathbf{x}\| + \|\mathbf{x} - \mathbf{x}_0\| < s + \|\mathbf{x} - \mathbf{x}_0\| = r.$$

Hence, $\|\mathbf{y} - \mathbf{x}_0\| < r$. ■

The following example illustrates some techniques that are useful in establishing the openness of sets.

EXAMPLE 1 Prove that $A = \{(x, y) \in \mathbb{R}^2 \mid x > 0\}$ is an open set.

SOLUTION The set is pictured in Figure 2.2.4.

Intuitively, this set is open, because no points on the "boundary," $x = 0$, are contained in the set. Such an argument will often suffice after one becomes accustomed to the concept of openness. At first, however, we should give details. To prove that A is open, we show that for every point $(x, y) \in A$ there exists an $r > 0$ such that $D_r(x, y) \subset A$. If $(x, y) \in A$, then $x > 0$. Choose $r = x$. If $(x_1, y_1) \in D_r(x, y)$, we have

$$|x_1 - x| = \sqrt{(x_1 - x)^2} \leq \sqrt{(x_1 - x)^2 + (y_1 - y)^2} < r = x,$$

and so $x_1 - x < x$ and $x - x_1 < x$. The latter inequality implies $x_1 > 0$, that is, $(x_1, y_1) \in A$. Hence $D_r(x, y) \subset A$, and therefore A is open (see Figure 2.2.5). ▲

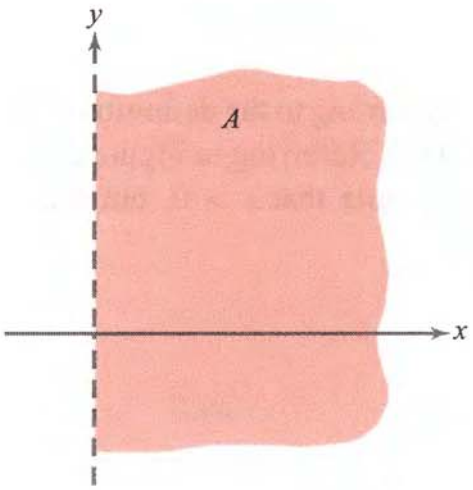


Figure 2.2.4 Show that A is an open set.

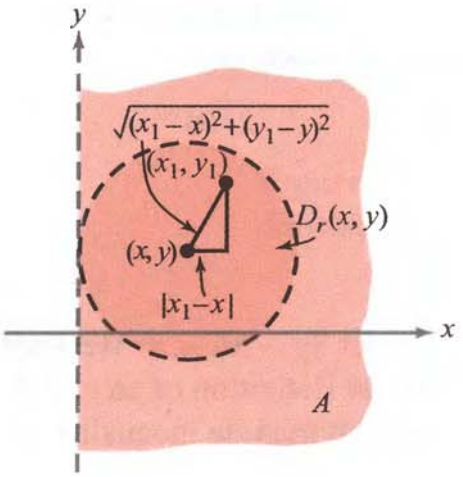


Figure 2.2.5 The construction of a disk about a point in A that is completely enclosed in A.

It is useful to have a special name for an open set containing a given point $\mathbf{x}$, because this idea arises often in the study of limits and continuity. Thus, by a ***neighborhood*** of $\mathbf{x} \in \mathbb{R}^n$ we merely mean an open set U containing the point $\mathbf{x}$. For example, $D_r(\mathbf{x}_0)$ is a neighborhood of $\mathbf{x}_0$ for any $r > 0$. The set A in Example 1 is a neighborhood of the point $\mathbf{x}_0 = (3, -10)$.

Boundary

Let us formally introduce the concept of a boundary point, which we alluded to in Example 1.

DEFINITION: Boundary Points Let $A \subset \mathbb{R}^n$. A point $\mathbf{x} \in \mathbb{R}^n$ is called a ***boundary point*** of A if every neighborhood of $\mathbf{x}$ contains at least one point in A and at least one point not in A.

In this definition, $\mathbf{x}$ itself may or may not be in A; if $\mathbf{x} \in A$, then $\mathbf{x}$ is a boundary point if every neighborhood of $\mathbf{x}$ contains at least one point *not* in A (it already contains a point of A, namely, $\mathbf{x}$). Similarly, if $\mathbf{x}$ is not in A, it is a boundary point if every neighborhood of $\mathbf{x}$ contains at least one point of A.

We shall be particularly interested in boundary points of open sets. By the definition of an open set, no point of an open set A can be a boundary point of A. Thus, *a point* $\mathbf{x}$ *is a boundary point of an open set* A *if and only if* $\mathbf{x}$ *is not in* A *and every neighborhood of* $\mathbf{x}$ *has a nonempty intersection with* A.

This expresses in precise terms the intuitive idea that a boundary point of A is a point just on the "edge" of A. In many examples it is perfectly clear what the boundary points are.

EXAMPLE 2 (a) Let $A = (a, b)$ in $\mathbb{R}$. Then the boundary points of A consist of the points a and b. A consideration of Figure 2.2.6 and the definition will make this clear. [The reader will be asked to prove this in Exercise 20(c).]

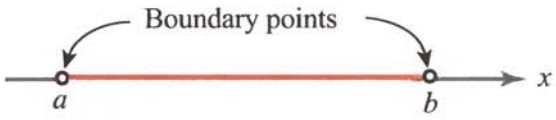


Figure 2.2.6 The boundary points of the interval (a, b).

(b) Let $A = D_r(x_0, y_0)$ be an r-disk about (x_0, y_0) in the plane. The boundary consists of points (x, y) with $(x - x_0)^2 + (y - y_0)^2 = r^2$ (Figure 2.2.7).

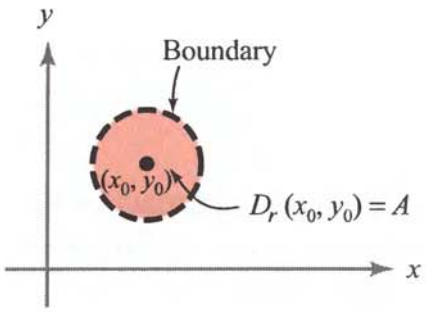


Figure 2.2.7 The boundary of A consists of points on the edge of A.

(c) Let $A = \{(x, y) \in \mathbb{R}^2 \mid x > 0\}$. Then the boundary of A consists of all points on the y axis (the student should draw a figure).

(d) Let A be $D_r(\mathbf{x}_0)$ minus the point $\mathbf{x}_0$ (a "punctured" disk about $\mathbf{x}_0$). Then $\mathbf{x}_0$ is a boundary point of A. ▲

Limits

We now turn our attention to the concept of a limit. *Throughout the following discussions the domain of definition of the function* f *will be an open set* A. We are interested in finding the limit of f as $\mathbf{x} \in A$ approaches either a point of A or a boundary point of A.

The reader should appreciate the fact that the limit concept is a basic and useful tool for the analysis of functions; it enables us to study derivatives, and hence

maxima and minima, asymptotes, improper integrals, and other important features of functions, as well as being useful for infinite series and sequences. We will present a theory of limits for functions of several variables that includes the theory for functions of one variable as a special case.

In one-variable calculus, the student has encountered the notion of $\lim_{x \to x_0} f(x) = l$ for a function $f\colon A \subset \mathbb{R} \to \mathbb{R}$ from a subset A of the real numbers to the real numbers. Intuitively, this means that as x gets closer and closer to x_0, the values $f(x)$ get closer and closer to (the limiting value) l. To put this intuitive idea on a firm, mathematical foundation, either the "epsilon (ε) and delta (δ) method" or the "neighborhood method" is usually introduced. The same is true for functions of several variables. In what follows we develop the neighborhood approach to limits. The epsilon-delta approach is left for optional study at the end of this section.

DEFINITION: Limit Let $f\colon A \subset \mathbb{R}^n \to \mathbb{R}^m$, where A is an open set. Let $\mathbf{x}_0$ be in A or be a boundary point of A, and let N be a neighborhood of $\mathbf{b} \in \mathbb{R}^m$. We say f is ***eventually in*** N ***as*** $\mathbf{x}$ ***approaches*** $\mathbf{x}_0$ if there exists a neighborhood U of $\mathbf{x}_0$ such that $\mathbf{x} \neq \mathbf{x}_0$, $\mathbf{x} \in U$, and $\mathbf{x} \in A$ imply $f(\mathbf{x}) \in N$. [The geometric meaning of this assertion is illustrated in Figure 2.2.8; note that $\mathbf{x}_0$ need not be in the set A, so that $f(\mathbf{x}_0)$ is not necessarily defined.] We say $f(\mathbf{x})$ ***approaches*** $\mathbf{b}$ as $\mathbf{x}$ approaches $\mathbf{x}_0$, or, in symbols,

$$\lim_{\mathbf{x} \to \mathbf{x}_0} f(\mathbf{x}) = \mathbf{b} \qquad \text{or} \qquad f(\mathbf{x}) \to \mathbf{b} \quad \text{as} \quad \mathbf{x} \to \mathbf{x}_0,$$

when, given *any* neighborhood N of $\mathbf{b}$, f is eventually in N as $\mathbf{x}$ approaches $\mathbf{x}_0$ [that is, "$f(\mathbf{x})$ is close to $\mathbf{b}$ if $\mathbf{x}$ is close to $\mathbf{x}_0$"]. It may be that as $\mathbf{x}$ approaches $\mathbf{x}_0$, the values $f(\mathbf{x})$ do not get close to any particular number. In this case, we say that $\lim_{\mathbf{x} \to \mathbf{x}_0} f(\mathbf{x})$ ***does not exist***.

Henceforth, whenever we consider the notion $\lim_{\mathbf{x} \to \mathbf{x}_0} f(\mathbf{x})$, we shall always assume that $\mathbf{x}_0$ either belongs to some open set on which f is defined or is on the boundary of such a set.

One reason we insist on $\mathbf{x} \neq \mathbf{x}_0$ in the definition of limit will become clear if we remember from one-variable calculus that we want to be able to define the derivative $f'(x_0)$ of a function f at a point x_0 by

$$f'(x_0) = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0},$$

and this expression is not defined at $x = x_0$.

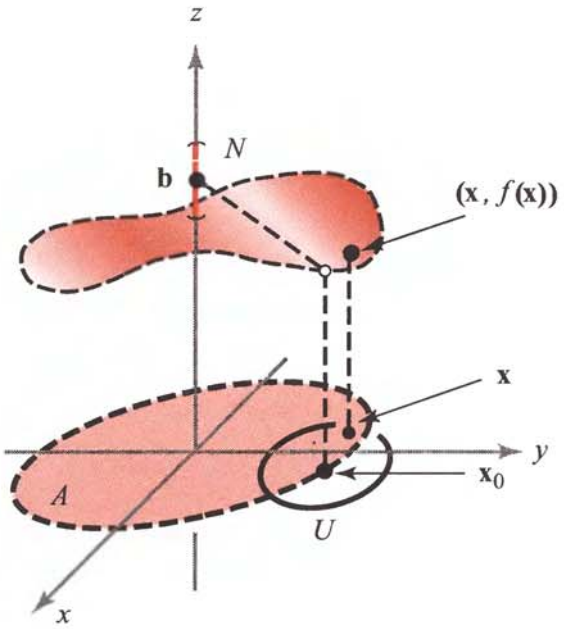


Figure 2.2.8 Limits in terms of neighborhoods; if $\mathbf{x}$ is in U, then $f(\mathbf{x})$ will be in N. (The little open circle denotes that the point does not lie on the graph.) In the figure, $f\colon A = \{(x, y) \mid x^2 + y^2 < 1\} \to \mathbb{R}$. (The dashed line is not in the graph of f.)

EXAMPLE 3 (a) This example illustrates a limit that does not exist. Consider the function $f\colon \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} 1 & \text{if } x > 0 \\ -1 & \text{if } x \leq 0. \end{cases}$$

The $\lim_{x \to 0} f(x)$ does not exist, since there are points x_1 arbitrarily close to 0 with $f(x_1) = 1$ and also points x_2 arbitrarily close to 0 with $f(x_2) = -1$; that is, there is no single number that f is close to when x is close to 0 (see Figure 2.2.9). If f is restricted to the domain $(0, 1)$ or $(-1, 0)$, then the limit does exist. Can you say why?

(b) This example illustrates a function whose limit does exist, but whose limiting value does not equal its value at the limiting point. Define $f\colon \mathbb{R} \to \mathbb{R}$ by

$$f(x) = \begin{cases} 0 & \text{if } x \neq 0 \\ 1 & \text{if } x = 0. \end{cases}$$

It is true that $\lim_{x \to 0} f(x) = 0$, since for any neighborhood U of 0, $x \in U$ and $x \neq 0$ implies that $f(x) = 0$. One sees from the graph in Figure 2.2.10 that f approaches 0 as $x \to 0$; we do not care that f happens to take on some other value at 0. ▲

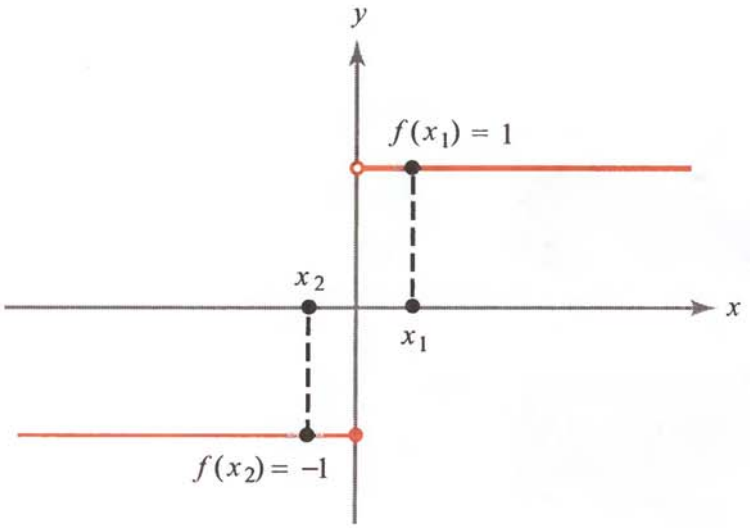


Figure 2.2.9 The limit of this function as $x \to 0$ does not exist.

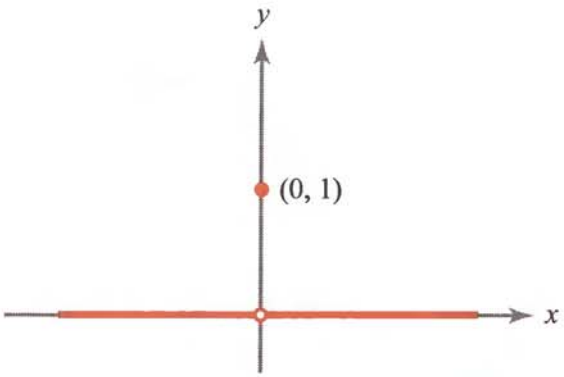


Figure 2.2.10 The limit of this function as $x \to 0$ is zero.

EXAMPLE 4 Use the definition to verify that the "obvious" $\lim_{\mathbf{x}\to\mathbf{x}_0} \mathbf{x} = \mathbf{x}_0$ holds, where $\mathbf{x}$ and $\mathbf{x}_0 \in \mathbb{R}^n$.

SOLUTION Let f be the function defined by $f(\mathbf{x}) = \mathbf{x}$, and let N be any neighborhood of $\mathbf{x}_0$. We must show that $f(\mathbf{x})$ is eventually in N as $\mathbf{x} \to \mathbf{x}_0$. According to the definition, we must find a neighborhood U of $\mathbf{x}_0$ with the property that if $\mathbf{x} \neq \mathbf{x}_0$ and $\mathbf{x} \in U$, then $f(\mathbf{x}) \in N$. Pick $U = N$. If $\mathbf{x} \in U$, then $\mathbf{x} \in N$; because $\mathbf{x} = f(\mathbf{x})$, it follows that $f(\mathbf{x}) \in N$. Thus, we have shown that $\lim_{\mathbf{x}\to\mathbf{x}_0} \mathbf{x} = \mathbf{x}_0$. In a similar way, we have

$$\lim_{(x,y)\to(x_0,y_0)} x = x_0, \qquad \text{etc.} \quad \blacktriangle$$

In what follows, the student may assume, without proof, the validity of limits from one-variable calculus. For example, $\lim_{x\to 1} \sqrt{x} = \sqrt{1} = 1$ and $\lim_{\theta\to 0} \sin\theta = \sin 0 = 0$ may be used.

EXAMPLE 5 (This example demonstrates another case in which the limit cannot simply be "read off" from the function.) Find $\lim_{x\to 1} g(x)$ where

$$g\colon x \mapsto \frac{x-1}{\sqrt{x}-1}.$$

SOLUTION This function is graphed in Figure 2.2.11(a).

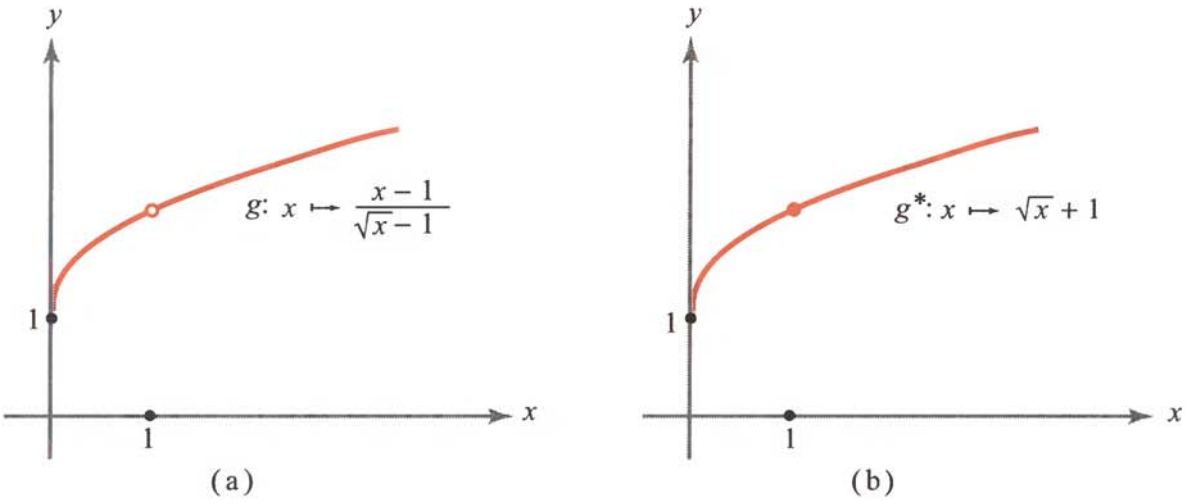


Figure 2.2.11 These graphs are the same except that in part (a), g is undefined at $x = 1$, whereas in part (b), g^* is defined for all $x \geq 0$.

We see that $g(1)$ is not defined, because division by zero is not defined. However, if we multiply the numerator and denominator of $g(x)$ by $\sqrt{x} + 1$, we find that for all x in the domain of g we have

$$g(x) = \frac{x - 1}{\sqrt{x} - 1} = \sqrt{x} + 1, \qquad x \neq 1.$$

The expression $g^*(x) = \sqrt{x} + 1$ is defined and takes the value 2 at $x = 1$; from one-variable calculus, $g^*(x) \to 2$ as $x \to 1$. But because $g^*(x) = g(x)$ for all $x \geq 0$, $x \neq 1$, we must have as well that $g(x) \to 2$ as $x \to 1$. ▲

We will consider other examples in two variables shortly.

Properties of Limits

To properly speak of *the* limit, we should establish that f can have *at most one* limit as $\mathbf{x} \to \mathbf{x}_0$. This is intuitively clear and we now state it formally. (See the Internet supplement for the proof.)

THEOREM 2: Uniqueness of Limits

If $\lim_{\mathbf{x} \to \mathbf{x}_0} f(\mathbf{x}) = \mathbf{b}_1$ and $\lim_{\mathbf{x} \to \mathbf{x}_0} f(\mathbf{x}) = \mathbf{b}_2$, then $\mathbf{b}_1 = \mathbf{b}_2$.

To carry out practical computations with limits, we require some rules for limits, for example, that the limit of a sum is the sum of the limits. These rules are summarized in the following theorem (see the Internet supplement for Chapter 2 for the proof).

THEOREM 3: Properties of Limits Let $f\colon A \subset \mathbb{R}^n \to \mathbb{R}^m$, $g\colon A \subset \mathbb{R}^n \to \mathbb{R}^m$, $\mathbf{x}_0$ be in A or be a boundary point of A, $\mathbf{b} \in \mathbb{R}^m$, and $c \in \mathbb{R}$; then

(i) If $\lim_{\mathbf{x}\to\mathbf{x}_0} f(\mathbf{x}) = \mathbf{b}$, then $\lim_{\mathbf{x}\to\mathbf{x}_0} cf(\mathbf{x}) = c\mathbf{b}$, where $cf\colon A \to \mathbb{R}^m$ is defined by $\mathbf{x} \mapsto c(f(\mathbf{x}))$.

(ii) If $\lim_{\mathbf{x}\to\mathbf{x}_0} f(\mathbf{x}) = \mathbf{b}_1$ and $\lim_{\mathbf{x}\to\mathbf{x}_0} g(\mathbf{x}) = \mathbf{b}_2$, then $\lim_{\mathbf{x}\to\mathbf{x}_0} (f+g)(\mathbf{x}) = \mathbf{b}_1 + \mathbf{b}_2$, where $(f+g)\colon A \to \mathbb{R}^m$ is defined by $\mathbf{x} \mapsto f(\mathbf{x}) + g(\mathbf{x})$.

(iii) If $m = 1$, $\lim_{\mathbf{x}\to\mathbf{x}_0} f(\mathbf{x}) = b_1$, and $\lim_{\mathbf{x}\to\mathbf{x}_0} g(\mathbf{x}) = b_2$, then $\lim_{\mathbf{x}\to\mathbf{x}_0} (fg)(\mathbf{x}) = b_1 b_2$, where $(fg)\colon A \to \mathbb{R}$ is defined by $\mathbf{x} \mapsto f(\mathbf{x})g(\mathbf{x})$.

(iv) If $m = 1$, $\lim_{\mathbf{x}\to\mathbf{x}_0} f(\mathbf{x}) = b \neq 0$, and $f(\mathbf{x}) \neq 0$ for all $\mathbf{x} \in A$, then $\lim_{\mathbf{x}\to\mathbf{x}_0} 1/f(\mathbf{x}) = 1/b$, where $1/f\colon A \to \mathbb{R}$ is defined by $\mathbf{x} \mapsto 1/f(\mathbf{x})$.

(v) If $f(\mathbf{x}) = (f_1(\mathbf{x}), \ldots, f_m(\mathbf{x}))$ where $f_i\colon A \to \mathbb{R}, i = 1, \ldots, m$, are the component functions of f, then $\lim_{\mathbf{x}\to\mathbf{x}_0} f(\mathbf{x}) = \mathbf{b} = (b_1, \ldots, b_m)$ if and only if $\lim_{\mathbf{x}\to\mathbf{x}_0} f_i(\mathbf{x}) = b_i$ for each $i = 1, \ldots, m$.

These results ought to be intuitively clear. For instance, rule (ii) says that if $f(\mathbf{x})$ is close to $\mathbf{b}_1$ and $g(\mathbf{x})$ is close to $\mathbf{b}_2$ when $\mathbf{x}$ is close to $\mathbf{x}_0$, then $f(\mathbf{x}) + g(\mathbf{x})$ is close to $\mathbf{b}_1 + \mathbf{b}_2$ when $\mathbf{x}$ is close to $\mathbf{x}_0$. The following example illustrates how this works.

EXAMPLE 6 Let $f\colon \mathbb{R}^2 \to \mathbb{R}$, $(x, y) \mapsto x^2 + y^2 + 2$. Compute the limit

$$\lim_{(x,y)\to(0,1)} f(x, y).$$

SOLUTION Here f is the sum of the three functions $(x, y) \mapsto x^2$, $(x, y) \mapsto y^2$, and $(x, y) \mapsto 2$. The limit of a sum is the sum of the limits, and the limit of a product is the product of the limits (Theorem 3). Hence, using the fact that $\lim_{(x,y)\to(x_0,y_0)} x = x_0$ (Example 4), we obtain

$$\lim_{(x,y)\to(x_0,y_0)} x^2 = \left(\lim_{(x,y)\to(x_0,y_0)} x\right)\left(\lim_{(x,y)\to(x_0,y_0)} x\right) = x_0^2$$

and, using the same reasoning, $\lim_{(x,y)\to(x_0,y_0)} y^2 = y_0^2$. Consequently,

$$\lim_{(x,y)\to(0,1)} f(x, y) = 0^2 + 1^2 + 2 = 3. \quad \blacktriangle$$

Continuous Functions

In single-variable calculus we learned that the idea of a continuous function is based on the intuitive notion of a function whose graph is an unbroken curve, that is, a curve that has no *jumps*, or the kind of curve that would be traced by a particle in motion or by a moving pencil point that is not lifted from the paper.

To perform a detailed analysis of functions, we need concepts more precise than this rather vague notion. An example may clarify these ideas. Consider the specific function $f\colon \mathbb{R} \to \mathbb{R}$ defined by $f(x) = -1$ if $x \leq 0$ and $f(x) = 1$ if $x > 0$. The graph of f is shown in Figure 2.2.12(a). [The little open circle denotes the fact that the point (0, 1) does *not* lie on the graph of f]. Clearly, the graph of f is broken at $x = 0$. Consider also the function $g\colon x \mapsto x^2$. This function is pictured in Figure 2.2.12(b). The graph of g is not broken at any point.

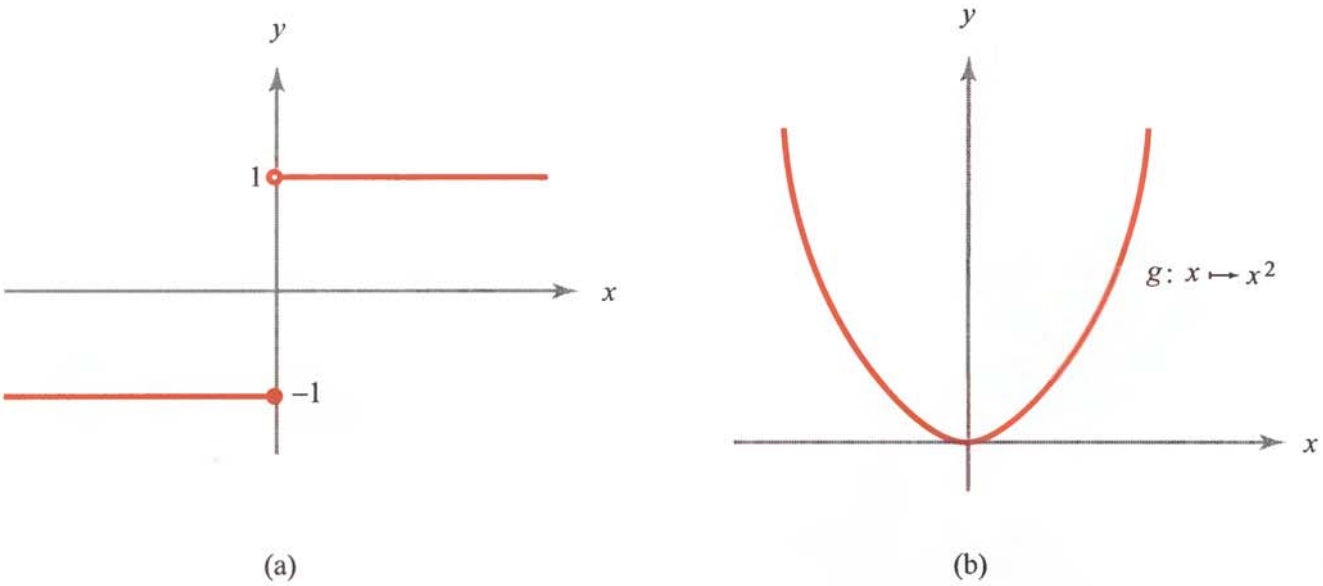


Figure 2.2.12 The function f in part (a) is not continuous, because its value jumps as x crosses 0, whereas the function g in part (b) is continuous.

If one examines examples of functions like f, whose graphs are broken at some point x_0, and functions like g, whose graphs are not broken, one sees that the principal difference between them is that for a function like g, the values of $g(x)$ get closer to $g(x_0)$ as x gets closer and closer to x_0. The same idea works for functions of several variables. But the notion of closer and closer does not suffice as a mathematical definition; thus, we shall formulate these concepts precisely in terms of limits.

Because the condition $\lim_{\mathbf{x}\to\mathbf{x}_0} f(\mathbf{x}) = f(\mathbf{x}_0)$ means that $f(\mathbf{x})$ is close to $f(\mathbf{x}_0)$ when $\mathbf{x}$ is close to $\mathbf{x}_0$, we see that this limit condition does indeed correspond to the requirement that the graph of f be unbroken (see Figure 2.2.13, where we illustrate the case $f\colon \mathbb{R} \to \mathbb{R}$). The case of several variables is easiest to visualize if we deal with real-valued functions, say $f\colon \mathbb{R}^2 \to \mathbb{R}$. In this case, we can visualize f by drawing its graph, which consists of all points (x, y, z) in $\mathbb{R}^3$ with $z = f(x, y)$. The continuity of f thus means that its graph has no "breaks" in it (see Figure 2.2.14).

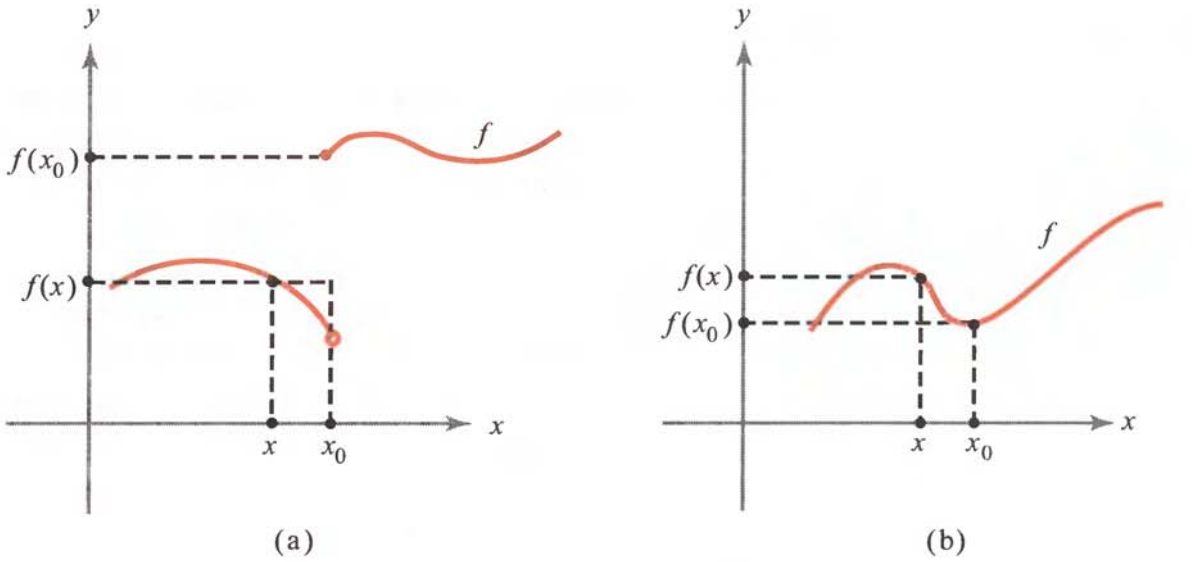


Figure 2.2.13 (a) Discontinuous function for which $\lim_{x \to x_0} f(x)$ does not exist. (b) Continuous function for which this limit exists and equals $f(x_0)$.

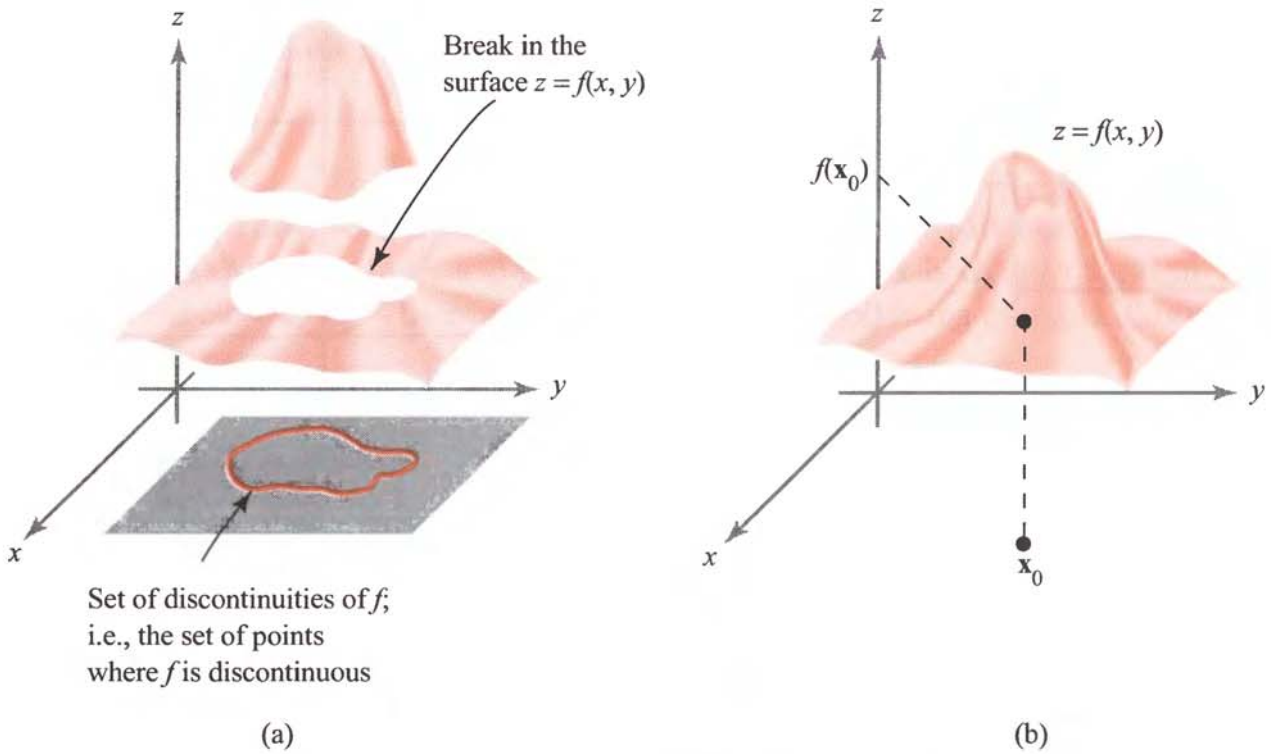


Figure 2.2.14 (a) A discontinuous function of two variables. (b) A continuous function.

DEFINITION: Continuity Let $f\colon A \subset \mathbb{R}^n \to \mathbb{R}^m$ be a given function with domain A. Let $\mathbf{x}_0 \in A$. We say f is ***continuous*** at $\mathbf{x}_0$ if and only if

$$\lim_{\mathbf{x} \to \mathbf{x}_0} f(\mathbf{x}) = f(\mathbf{x}_0).$$

If we just say that f is ***continuous***, we shall mean that f is continuous at each point $\mathbf{x}_0$ of A. If f is not continuous at $\mathbf{x}_0$, we say f is ***discontinuous*** at $\mathbf{x}_0$. If f is discontinuous at some point in its domain, we say f is ***discontinuous***.

EXAMPLE 7 Any polynomial $p(x) = a_0 + a_1x + \cdots + a_nx^n$ is continuous from $\mathbb{R}$ to $\mathbb{R}$. Indeed, from Theorem 3 and Example 4,

$$\begin{aligned}\lim_{x\to x_0}(a_0 + a_1x + \cdots + a_nx^n) &= \lim_{x\to x_0} a_0 + \lim_{x\to x_0} a_1x + \cdots + \lim_{x\to x_0} a_nx^n \\ &= a_0 + a_1x_0 + \cdots + a_nx_0^n,\end{aligned}$$

because the limit of a product is the product of the limits, which gives

$$\lim_{x\to x_0} x^n = \left(\lim_{x\to x_0} x\right)^n = x_0^n. \quad \blacktriangle$$

EXAMPLE 8 Let $f\colon \mathbb{R}^2 \to \mathbb{R}$, $f(x, y) = xy$. Then f is continuous, because, by the limit theorems and Example 4,

$$\lim_{(x,y)\to(x_0,y_0)} xy = \left(\lim_{(x,y)\to(x_0,y_0)} x\right)\left(\lim_{(x,y)\to(x_0,y_0)} y\right) = x_0y_0. \quad \blacktriangle$$

One can see by the same method that any polynomial $p(x, y)$ [for example, $p(x, y) = 3x^2 - 6xy^2 + y^3$] in x and y is continuous.

EXAMPLE 9 The function $f\colon \mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(x, y) = \begin{cases} 1 & \text{if } x \le 0 \text{ or } y \le 0 \\ 0 & \text{otherwise} \end{cases}$$

is not continuous at (0, 0) or at any point on the positive x axis or positive y axis. Indeed, if $(x_0, y_0) = \mathbf{u}$ is such a point (i.e., $x_0 = 0$ and $y_0 \ge 0$, or $y_0 = 0$ and $x_0 \ge 0$) and $\delta > 0$, there are points $(x, y) \in D_\delta(\mathbf{u})$, a neighborhood of $\mathbf{u}$, with $f(x, y) = 1$ and other points $(x, y) \in D_\delta(\mathbf{u})$ with $f(x, y) = 0$. Thus, it is *not* true that $f(x, y) \to f(x_0, y_0) = 1$ as $(x, y) \to (x_0, y_0)$. $\blacktriangle$

To prove that specific functions are continuous, we can avail ourselves of the limit theorems (see Theorem 3 and Example 7). If we transcribe those results in terms of continuity, we are led to the following:

THEOREM 4: Properties of Continuous Functions Suppose that $f\colon A \subset \mathbb{R}^n \to \mathbb{R}^m$, $g\colon A \subset \mathbb{R}^n \to \mathbb{R}^m$, and let c be a real number.

(i) If f is continuous at $\mathbf{x}_0$, so is cf, where $(cf)(\mathbf{x}) = c[f(\mathbf{x})]$.

(ii) If f and g are continuous at $\mathbf{x}_0$, so is $f + g$, where the sum of f and g is defined by $(f + g)(\mathbf{x}) = f(\mathbf{x}) + g(\mathbf{x})$.

(iii) If f and g are continuous at $\mathbf{x}_0$ and $m = 1$, then the product function fg defined by $(fg)(\mathbf{x}) = f(\mathbf{x})g(\mathbf{x})$ is continuous at $\mathbf{x}_0$.

(iv) If $f\colon A \subset \mathbb{R}^n \to \mathbb{R}$ is continuous at $\mathbf{x}_0$ and nowhere zero on A, then the quotient $1/f$ is continuous at $\mathbf{x}_0$, where $(1/f)(\mathbf{x}) = 1/f(\mathbf{x})$.

(v) If $f\colon A \subset \mathbb{R}^n \to \mathbb{R}^m$ and $f(\mathbf{x}) = (f_1(\mathbf{x}), \ldots, f_m(\mathbf{x}))$, then f is continuous at $\mathbf{x}_0$ if and only if each of the real-valued functions $f_1, \ldots, f_m$ is continuous at $\mathbf{x}_0$.

A variant of (iv) is often used: If $f(\mathbf{x}_0) \neq 0$ and f is continuous, then $f(\mathbf{x}) \neq 0$ in a neighborhood of $\mathbf{x}_0$ and so $1/f$ is defined in that neighborhood, and $1/f$ is continuous at $\mathbf{x}_0$.

EXAMPLE 10 Let $f\colon \mathbb{R}^2 \to \mathbb{R}^2, (x, y) \mapsto (x^2y, (y + x^3)/(1 + x^2))$. Show that f is continuous.

SOLUTION To see this, it is sufficient, by property (v) of Theorem 4, to show that each component is continuous. As we have mentioned, any polynomial in two variables is continuous; thus, the map $(x, y) \mapsto x^2y$ is continuous. Because $1 + x^2$ is continuous and nonzero, by property (iv), we know that $1/(1 + x^2)$ is continuous; hence, $(y + x^3)/(1 + x^2)$ is a product of continuous functions, and by (iii) is continuous. ▲

Similar reasoning applies to examples like the function $\mathbf{c}\colon \mathbb{R} \to \mathbb{R}^3$ given by $\mathbf{c}(t) = (t^2, 1, t^3/(1 + t^2))$ to show they are continuous as well.

Composition

Next we discuss *composition*, another basic operation that can be performed on functions. If g maps A to B and f maps B to C, the ***composition of*** g ***with*** f, or of f on g, denoted by $f \circ g$, maps A to C by sending $\mathbf{x} \mapsto f(g(\mathbf{x}))$ (see Figure 2.2.15). For example, $\sin(x^2)$ is the composition of $x \mapsto x^2$ with $y \mapsto \sin y$.

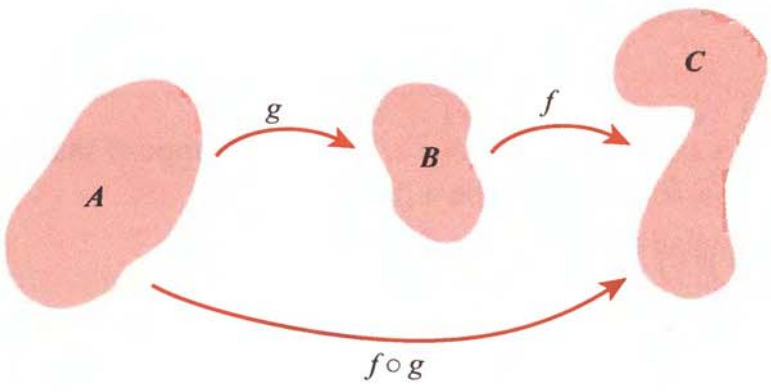


Figure 2.2.15 The composition of f on g.

THEOREM 5: Continuity of Compositions Let $g\colon A \subset \mathbb{R}^n \to \mathbb{R}^m$ and let $f\colon B \subset \mathbb{R}^m \to \mathbb{R}^p$. Suppose $g(A) \subset B$, so that $f \circ g$ is defined on A. If g is continuous at $\mathbf{x}_0 \in A$ and f is continuous at $\mathbf{y}_0 = g(\mathbf{x}_0)$, then $f \circ g$ is continuous at $\mathbf{x}_0$.

The intuition behind this is easy; the formal proof in the Internet supplement follows a similar pattern. Intuitively, we must show that as $\mathbf{x}$ gets close to $\mathbf{x}_0$, $f(g(\mathbf{x}))$ gets close to $f(g(\mathbf{x}_0))$. But as $\mathbf{x}$ gets close to $\mathbf{x}_0$, $g(\mathbf{x})$ gets close to $g(\mathbf{x}_0)$ (by continuity of g at $\mathbf{x}_0$); and as $g(\mathbf{x})$ gets close to $g(\mathbf{x}_0)$, $f(g(\mathbf{x}))$ gets close to $f(g(\mathbf{x}_0))$ [by continuity of f at $g(\mathbf{x}_0)$].

EXAMPLE 11 Let $f(x, y, z) = (x^2 + y^2 + z^2)^{30} + \sin z^3$. Show that f is continuous.

SOLUTION Here we can write f as a sum of the two functions $(x^2 + y^2 + z^2)^{30}$ and $\sin z^3$, so it suffices to show that each is continuous. The first is the composite of $(x, y, z) \mapsto (x^2 + y^2 + z^2)$ with $u \mapsto u^{30}$, and the second is the composite of $(x, y, z) \mapsto z^3$ with $u \mapsto \sin u$, and so we have continuity by Theorem 5. ▲

Limits in Terms of ε's and δ's

We now state a theorem (proved in the Internet supplement for Chapter 2) giving a useful formulation of the notion of limit in terms of epsilons and deltas that is often taken as the *definition* of limit. This is, in fact, another way of making precise the intuitive statement that "$f(\mathbf{x})$ is close to $\mathbf{b}$ when $\mathbf{x}$ is close to $\mathbf{x}_0$." To help understand this formulation, the reader should consider it with respect to each of the examples already presented.

THEOREM 6 Let $f\colon A \subset \mathbb{R}^n \to \mathbb{R}^m$ and let $\mathbf{x}_0$ be in A or be a boundary point of A. Then $\lim\limits_{\mathbf{x}\to\mathbf{x}_0} f(\mathbf{x}) = \mathbf{b}$ if and only if for every number $\varepsilon > 0$ there is a $\delta > 0$ such that for any $\mathbf{x} \in A$ satisfying $0 < \|\mathbf{x} - \mathbf{x}_0\| < \delta$, we have $\|f(\mathbf{x}) - \mathbf{b}\| < \varepsilon$ (see Figure 2.2.16).

To illustrate the methodology of the epsilon-delta technique in Theorem 6, we consider the following examples.

EXAMPLE 12 Show that $\lim\limits_{(x,y)\to(0,0)} x = 0$ using the ε-δ method.

SOLUTION Note that if $\delta > 0$, $\|(x, y) - (0, 0)\| = \sqrt{x^2 + y^2} < \delta$ implies $|x - 0| = |x| = \sqrt{x^2} \le \sqrt{x^2 + y^2} < \delta$. Thus, if $\|(x, y) - (0, 0)\| < \delta$, then $|x - 0|$ is also less than δ. Given $\varepsilon > 0$, we are required to find a $\delta > 0$ (generally depending on ε) with the property that $0 < \|(x, y) - (0, 0)\| < \delta$ implies $|x - 0| < \varepsilon$. What are we to pick as our δ? From the preceding calculation, we see that if we choose $\delta = \varepsilon$,

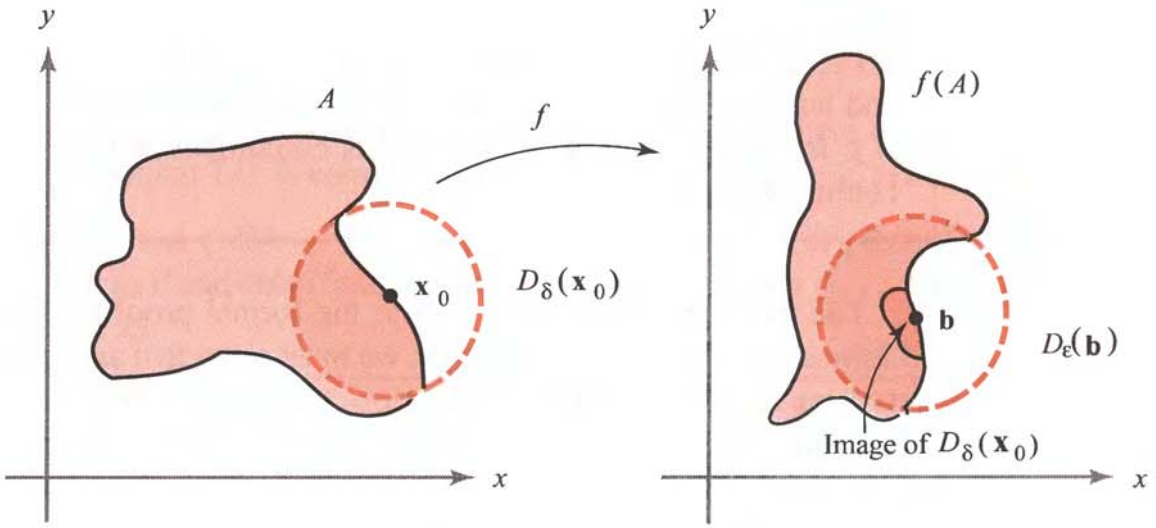


Figure 2.2.16 The geometry of the ε-δ definition of limit.

then $\|(x, y) - (0, 0)\| < \delta$ implies $|x - 0| < \varepsilon$. This shows that $\lim_{(x,y)\to(0,0)} x = 0$. Given $\varepsilon > 0$, we could have also chosen $\delta = \varepsilon/2$ or $\varepsilon/3$, but it suffices to find just one δ satisfying the requirements of the definition of a limit. ▲

EXAMPLE 13 Consider the function

$$f(x, y) = \frac{\sin(x^2 + y^2)}{x^2 + y^2}.$$

Even though f is not defined at $(0, 0)$, determine whether $f(x, y)$ approaches some number as (x, y) approaches $(0, 0)$.

SOLUTION From one-variable calculus or L'Hôpital's rule we know that

$$\lim_{\alpha\to 0} \frac{\sin \alpha}{\alpha} = 1.$$

Thus, it is reasonable to guess that

$$\lim_{\mathbf{v}\to(0,0)} f(\mathbf{v}) = \lim_{\mathbf{v}\to(0,0)} \frac{\sin \|\mathbf{v}\|^2}{\|\mathbf{v}\|^2} = 1.$$

Indeed, because $\lim_{\alpha\to 0}(\sin \alpha)/\alpha = 1$, given $\varepsilon > 0$ we are able to find a $\delta > 0$, with $0 < \delta < 1$, such that $0 < |\alpha| < \delta$ implies that $|(\sin \alpha)/\alpha - 1| < \varepsilon$. If $0 < \|\mathbf{v}\| < \delta$, then $0 < \|\mathbf{v}\|^2 < \delta^2 < \delta$, and therefore

$$|f(\mathbf{v}) - 1| = \left|\frac{\sin \|\mathbf{v}\|^2}{\|\mathbf{v}\|^2} - 1\right| < \varepsilon.$$

Thus, $\lim_{\mathbf{v}\to(0,0)} f(\mathbf{v}) = 1$. If we plot $[\sin(x^2 + y^2)]/(x^2 + y^2)$ on a computer, we get a graph that is indeed well behaved near $(0, 0)$ (Figure 2.2.17). ▲

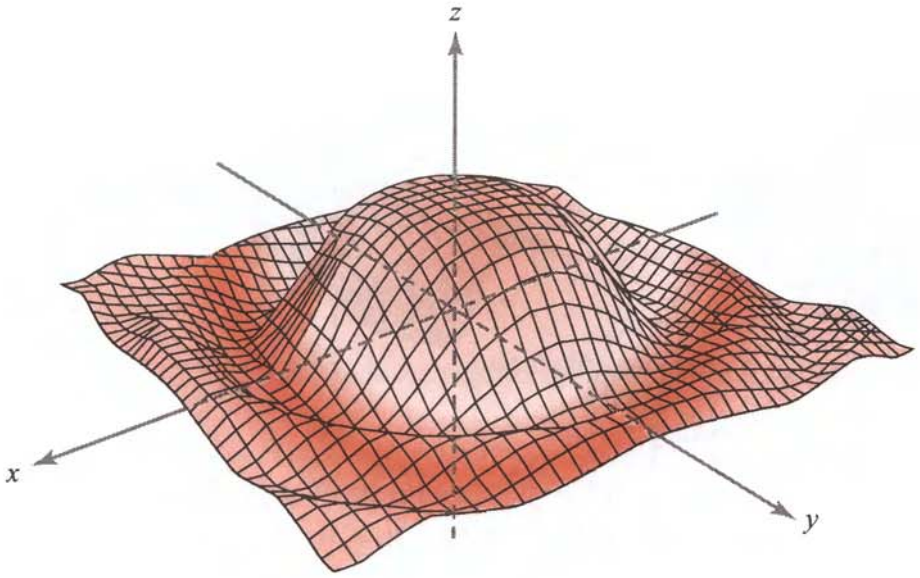


Figure 2.2.17 Graph of the function $f(x, y) = [\sin(x^2 + y^2)]/(x^2 + y^2)$.

EXAMPLE 14 Show that

$$\lim_{(x,y)\to(0,0)} \frac{x^2}{\sqrt{x^2+y^2}} = 0.$$

SOLUTION We must show that $x^2/\sqrt{x^2+y^2}$ is small when (x, y) is close to the origin. To do this, we use the following inequality:

$$0 \le \frac{x^2}{\sqrt{x^2+y^2}} \le \frac{x^2+y^2}{\sqrt{x^2+y^2}} \qquad \text{(because } y^2 \ge 0\text{)}$$
$$= \sqrt{x^2+y^2}.$$

Given $\varepsilon > 0$, choose $\delta = \varepsilon$. Then $\|(x, y) - (0, 0)\| = \|(x, y)\| = \sqrt{x^2+y^2}$, and so $\|(x, y) - (0, 0)\| < \delta$ implies that

$$\left|\frac{x^2}{\sqrt{x^2+y^2}} - 0\right| = \frac{x^2}{\sqrt{x^2+y^2}} \le \sqrt{x^2+y^2} = \|(x, y) - (0, 0)\| < \delta = \varepsilon.$$

Thus, the conditions of Theorem 6 have been fulfilled and the limit is verified. ▲

EXAMPLE 15 (a) Does

$$\lim_{(x,y)\to(0,0)} x^2/(x^2+y^2)$$

exist? [See Figure 2.2.18(a).]

(b) Prove that [see Figure 2.2.18(b)]

$$\lim_{(x,y)\to(0,0)} \frac{2x^2y}{x^2+y^2} = 0.$$

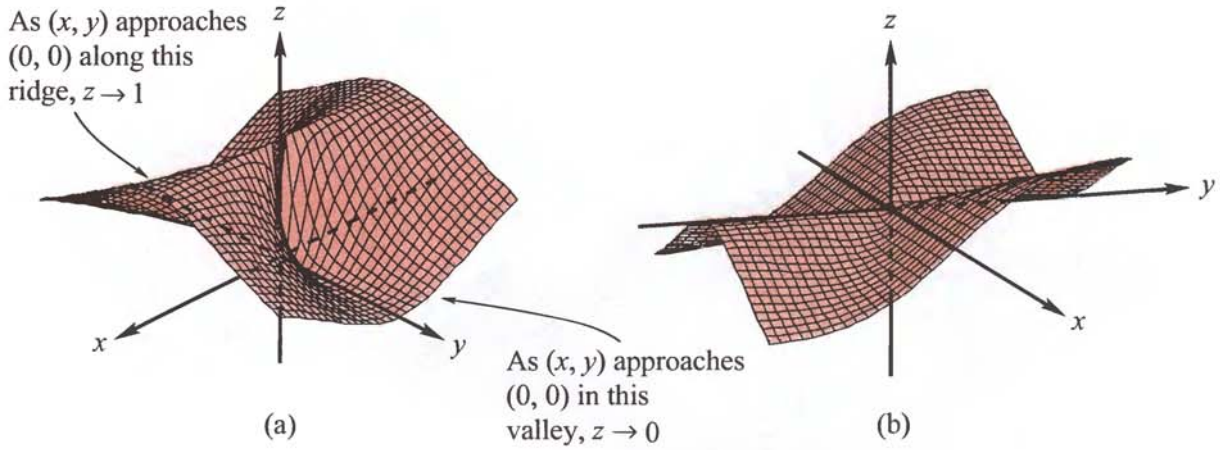


Figure 2.2.18 (a) The function $z = x^2/(x^2 + y^2)$ has no limit at (0, 0). (b) The function $z = (2x^2y)/(x^2 + y^2)$ has limit 0 at (0, 0).

SOLUTION (a) If the limit exists, $x^2/(x^2 + y^2)$ should approach a definite value, say a, as (x, y) gets near (0, 0). In particular, if (x, y) approaches zero along any given path, then $x^2/(x^2 + y^2)$ should approach the limiting value a. If (x, y) approaches (0, 0) along the line $y = 0$, the limiting value is clearly 1 (just set $y = 0$ in the preceding expression to get $x^2/x^2 = 1$). If (x, y) approaches (0, 0) along the line $x = 0$, the limiting value is

$$\lim_{y \to 0} \frac{0^2}{0^2 + y^2} = 0 \neq 1.$$

Hence, $\lim\limits_{(x,y)\to(0,0)} x^2/(x^2 + y^2)$ does not exist.

(b) Note that

$$\left| \frac{2x^2y}{x^2 + y^2} \right| \leq \left| \frac{2x^2y}{x^2} \right| = 2|y|.$$

Thus, given $\varepsilon > 0$, choose $\delta = \varepsilon/2$; then $0 < \|(x, y) - (0, 0)\| = \sqrt{x^2 + y^2} < \delta$ implies $|y| < \delta$, and thus

$$\left| \frac{2x^2y}{x^2 + y^2} - 0 \right| < 2\delta = \varepsilon. \quad \blacktriangle$$

Using the ε-δ notation, we are led to the following reformulation of the definition of continuity.

THEOREM 7 Let $f\colon A \subset \mathbb{R}^n \to \mathbb{R}^m$ be given. Then f is continuous at $\mathbf{x}_0 \in A$ if and only if for every number $\varepsilon > 0$ there is a number $\delta > 0$ such that

$$\mathbf{x} \in A \quad \text{and} \quad \|\mathbf{x} - \mathbf{x}_0\| < \delta \quad \text{implies} \quad \|f(\mathbf{x}) - f(\mathbf{x}_0)\| < \varepsilon.$$

The proof is almost immediate. Notice that in Theorem 6 we insisted that $0 < \|\mathbf{x} - \mathbf{x}_0\|$, that is, $\mathbf{x} \neq \mathbf{x}_0$. That is *not* imposed here; indeed, the conclusion of Theorem 7 is certainly valid when $\mathbf{x} = \mathbf{x}_0$, and so there is no need to exclude this

case. Here we do care about the value of f at $\mathbf{x}_0$; we want f at nearby points to be close to *this* value.

EXERCISES

In the following exercises the reader may assume that the exponential, sine, and cosine functions are continuous and may freely use techniques from one-variable calculus, such as L'Hôpital's rule.

Show that the subsets of the plane in Exercises 1–4 are open:

1. $A = \{(x, y) \mid -1 < x < 1, -1 < y < 1\}$

2. $B = \{(x, y) \mid y > 0\}$

3. $C = \{(x, y) \mid 2 < x^2 + y^2 < 4\}$

4. $D = \{(x, y) \mid x \neq 0 \text{ and } y \neq 0\}$

5. Compute the limits:

(a) $\lim\limits_{(x,y)\to(0,1)} x^3 y$

(b) $\lim\limits_{x\to 0} \dfrac{\cos x - 1}{x^2}$

(c) $\lim\limits_{h\to 0} \dfrac{e^h - 1}{h}$.

6. Compute the following limits:

(a) $\lim\limits_{(x,y)\to(0,1)} e^x y$

(b) $\lim\limits_{x\to 0} \dfrac{\sin^2 x}{x}$

(c) $\lim\limits_{x\to 0} \dfrac{\sin^2 x}{x^2}$

7. Compute the following limits:

(a) $\lim\limits_{x\to 3}(x^2 - 3x + 5)$

(b) $\lim\limits_{x\to 0} \sin x$

(c) $\lim\limits_{h\to 0} \dfrac{(x+h)^2 - x^2}{h}$

8. Compute the following limits if they exist:

(a) $\lim\limits_{(x,y)\to(0,0)} \dfrac{(x+y)^2 - (x-y)^2}{xy}$

(b) $\lim\limits_{(x,y)\to(0,0)} \dfrac{\sin xy}{y}$

(c) $\lim\limits_{(x,y)\to(0,0)} \dfrac{x^3 - y^3}{x^2 + y^2}$

9. Compute the following limits if they exist:

(a) $\lim\limits_{(x,y)\to(0,0)} \dfrac{e^{xy} - 1}{y}$

(b) $\lim\limits_{(x,y)\to(0,0)} \dfrac{\cos(xy) - 1}{x^2 y^2}$

(c) $\lim\limits_{(x,y)\to(0,0)} \dfrac{xy}{x^2 + y^2 + 2}$

10. Compute the following limits, if they exist:

(a) $\lim\limits_{(x,y)\to(0,0)} \dfrac{e^{xy}}{x+1}$

(b) $\lim\limits_{(x,y)\to(0,0)} \dfrac{\cos x - 1 - (x^2/2)}{x^4+y^4}$

(c) $\lim\limits_{(x,y)\to(0,0)} \dfrac{(x-y)^2}{x^2+y^2}$

11. Compute the following limits if they exist:

(a) $\lim\limits_{(x,y)\to(0,0)} \dfrac{\sin xy}{xy}$

(b) $\lim\limits_{(x,y,z)\to(0,0,0)} \dfrac{\sin(xyz)}{xyz}$

(c) $\lim\limits_{(x,y,z)\to(0,0,0)} f(x,y,z)$, where $f(x,y,z) = (x^2+3y^2)/(x+1)$.

12. Compute the following limits if they exist:

(a) $\lim\limits_{x\to 0} \dfrac{\sin 2x - 2x}{x^3}$

(b) $\lim\limits_{(x,y)\to(0,0)} \dfrac{\sin 2x - 2x + y}{x^3+y}$

(c) $\lim\limits_{(x,y,z)\to(0,0,0)} \dfrac{2x^2 y\cos z}{x^2+y^2}$

13. Compute $\lim\limits_{\mathbf{x}\to\mathbf{x}_0} f(\mathbf{x})$, if it exists, for the following cases:

(a) $f\colon \mathbb{R}\to\mathbb{R}, x\mapsto |x|, x_0 = 1$

(b) $f\colon \mathbb{R}^n\to\mathbb{R}, \mathbf{x}\mapsto \|\mathbf{x}\|$, arbitrary $\mathbf{x}_0$

(c) $f\colon \mathbb{R}\to\mathbb{R}^2, x\mapsto (x^2, e^x), x_0 = 1$

(d) $f\colon \mathbb{R}^2\backslash\{(0,0)\}\to\mathbb{R}^2, (x,y)\mapsto (\sin(x-y), e^{x(y+1)} - x - 1)/\|(x,y)\|, \mathbf{x}_0 = (0,0)$.

14. Let $A\subset\mathbb{R}^2$ be the open unit disk $D_1(0,0)$ with the point $\mathbf{x}_0 = (1,0)$ added, and let $f\colon A\to\mathbb{R}, \mathbf{x}\mapsto f(\mathbf{x})$ be the constant function $f(\mathbf{x}) = 1$. Show that $\lim\limits_{\mathbf{x}\to\mathbf{x}_0} f(\mathbf{x}) = 1$.

15. If $f\colon \mathbb{R}^n\to\mathbb{R}$ and $g\colon \mathbb{R}^n\to\mathbb{R}$ are continuous, show that the functions

$$f^2 g\colon \mathbb{R}^n\to\mathbb{R}, \mathbf{x}\mapsto [f(\mathbf{x})]^2 g(\mathbf{x})$$

and

$$f^2 + g\colon \mathbb{R}^n\to\mathbb{R}, x\mapsto [f(\mathbf{x})]^2 + g(\mathbf{x})$$

are continuous.

16. (a) Show that $f\colon \mathbb{R}\to\mathbb{R}, x\mapsto (1-x)^8 + \cos(1+x^3)$ is continuous.

(b) Show that the map $f\colon \mathbb{R}\to\mathbb{R}, x\mapsto x^2 e^x/(2-\sin x)$ is continuous.

17. (a) Can $[\sin(x+y)]/(x+y)$ be made continuous by suitably defining it at $(0,0)$?

(b) Can $xy/(x^2+y^2)$ be made continuous by suitably defining it at $(0,0)$?

(c) Prove that $f\colon \mathbb{R}^2\to\mathbb{R}, (x,y)\mapsto ye^x + \sin x + (xy)^4$ is continuous.

18. Using either ε's and δ's or spherical coordinates, show that

$$\lim_{(x,y,z)\to(0,0,0)} \frac{xyz}{x^2+y^2+z^2} = 0.$$

19. Use the ε-δ formulation of limits to prove that $x^2 \to 4$ as $x \to 2$. Give another proof using Theorem 3.

20. (a) Prove that for $\mathbf{x} \in \mathbb{R}^n$ and $s < t$, $D_s(\mathbf{x}) \subset D_t(\mathbf{x})$.
(b) Prove that if U and V are neighborhoods of $\mathbf{x} \in \mathbb{R}^n$, then so are $U \cap V$ and $U \cup V$.
(c) Prove that the boundary points of an open interval $(a, b) \subset \mathbb{R}$ are the points a and b.

21. Suppose $\mathbf{x}$ and $\mathbf{y}$ are in $\mathbb{R}^n$ and $\mathbf{x} \neq \mathbf{y}$. Show that there is a continuous function $f\colon \mathbb{R}^n \to \mathbb{R}$ with $f(\mathbf{x}) = 1$, $f(\mathbf{y}) = 0$, and $0 \leq f(\mathbf{z}) \leq 1$ for every $\mathbf{z}$ in $\mathbb{R}^n$.

22. Let $f\colon A \subset \mathbb{R}^n \to \mathbb{R}$ be given and let $\mathbf{x}_0$ be a boundary point of A. We say that $\lim\limits_{\mathbf{x}\to\mathbf{x}_0} f(\mathbf{x}) = \infty$ if for every $N > 0$ there is a $\delta > 0$ such that $0 < \|\mathbf{x} - \mathbf{x}_0\| < \delta$ and $\mathbf{x} \in A$ implies $f(\mathbf{x}) > N$.

(a) Prove that $\lim\limits_{x\to 1} (x-1)^{-2} = \infty$.
(b) Prove that $\lim\limits_{x\to 0} 1/|x| = \infty$. Is it true that $\lim\limits_{x\to 0} 1/x = \infty$?
(c) Prove that $\lim\limits_{(x,y)\to(0,0)} 1/(x^2+y^2) = \infty$.

23. Let $b \in \mathbb{R}$ and $f\colon \mathbb{R}\backslash[b] \to \mathbb{R}$ be a function. We write $\lim\limits_{x\to b-} f(x) = L$ and say that L is the ***left-hand limit*** of f at b, if for every $\varepsilon > 0$, there is a $\delta > 0$ such that $x < b$ and $0 < |x - b| < \delta$ implies $|f(x) - L| < \varepsilon$.

(a) Formulate a definition of ***right-hand limit***, or $\lim\limits_{x\to b+} f(x)$.
(b) Find $\lim\limits_{x\to 0-} 1/(1+e^{1/x})$ and $\lim\limits_{x\to 0+} 1/(1+e^{1/x})$.
(c) Sketch the graph of $1/(1+e^{1/x})$.

24. Show that f is continuous at $\mathbf{x}_0$ if and only if

$$\lim_{\mathbf{x}\to\mathbf{x}_0} \|f(\mathbf{x}) - f(\mathbf{x}_0)\| = 0.$$

25. Let $f\colon A \subset \mathbb{R}^n \to \mathbb{R}^m$ satisfy $\|f(\mathbf{x}) - f(\mathbf{y})\| \leq K\|\mathbf{x} - \mathbf{y}\|^\alpha$ for all $\mathbf{x}$ and $\mathbf{y}$ in A for positive constants K and α. Show that f is continuous. (Such functions are called ***Hölder-continuous*** or, if $\alpha = 1$, ***Lipschitz-continuous***.)

26. Show that $f\colon \mathbb{R}^n \to \mathbb{R}^m$ is continuous at all points if and only if the inverse image of every open set is open.

27. (a) Find a specific number $\delta > 0$ such that if $|a| < \delta$, then $|a^3 + 3a^2 + a| < 1/100$.
(b) Find a specific number $\delta > 0$ such that if $x^2 + y^2 < \delta^2$, then

$$|x^2 + y^2 + 3xy + 180xy^5| < 1/10{,}000.$$

2.3 Differentiation

In Section 2.1 we considered a few methods for graphing functions. By these methods alone it may be impossible to compute enough information to grasp even the general features of a complicated function. From elementary calculus we know that the idea

of the derivative can greatly aid us in this task; for example, it enables us to locate maxima and minima and to compute rates of change. The derivative also has many applications beyond this, as the student surely has discovered in elementary calculus.

Intuitively, we know from our work in Section 2.2 that a continuous function is one that has no "breaks" in its graph. A differentiable function from $\mathbb{R}^2$ to $\mathbb{R}$ ought to be such that not only are there no breaks in its graph, but there is a well-defined plane tangent to the graph at each point. Thus, there must not be any sharp folds, corners, or peaks in the graph (see Figure 2.3.1). In other words, the graph must be *smooth*.

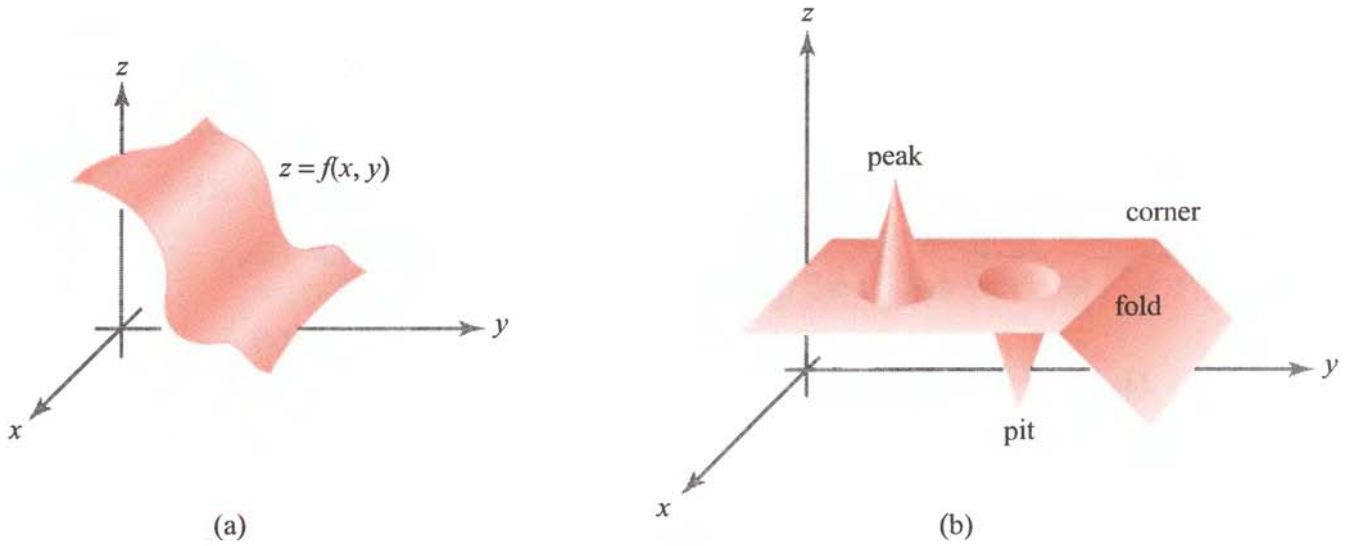


Figure 2.3.1 (a) A smooth graph and (b) a nonsmooth one.

Partial Derivatives

To make these ideas precise, we need a sound definition of what we mean by the phrase "$f(x_1, \ldots, x_n)$ is differentiable at $\mathbf{x} = (x_1, \ldots, x_n)$." Actually, this definition is not quite as simple as one might think. Toward this end, however, let us introduce the notion of the *partial derivative*. This notion relies only on our knowledge of one-variable calculus. (A quick review of the definition of the derivative in a one-variable calculus text might be advisable at this point.)

DEFINITION: Partial Derivatives Let $U \subset \mathbb{R}^n$ be an open set and suppose $f\colon U \subset \mathbb{R}^n \to \mathbb{R}$ is a real-valued function. Then $\partial f/\partial x_1, \ldots, \partial f/\partial x_n$, the ***partial derivatives*** of f with respect to the first, second, ... , nth variable, are the real-valued functions of n variables, which, at the point $(x_1, \ldots, x_n) = \mathbf{x}$, are defined by

$$\frac{\partial f}{\partial x_j}(x_1, \ldots, x_n) = \lim_{h \to 0} \frac{f(x_1, x_2, \ldots, x_j + h, \ldots, x_n) - f(x_1, \ldots, x_n)}{h}$$

$$= \lim_{h \to 0} \frac{f(\mathbf{x} + h\mathbf{e}_j) - f(\mathbf{x})}{h}$$

if the limits exist, where $1 \leq j \leq n$ and $\mathbf{e}_j$ is the jth standard basis vector defined by $\mathbf{e}_j = (0, \ldots, 1, \ldots, 0)$, with 1 in the jth slot (see Section 1.5). The domain of the function $\partial f/\partial x_j$ is the set of $\mathbf{x} \in \mathbb{R}^n$ for which the limit exists.

In other words, $\partial f/\partial x_j$ is just the derivative of f with respect to the variable x_j, with the other variables held fixed. If $f\colon \mathbb{R}^3 \to \mathbb{R}$, we shall often use the notation $\partial f/\partial x$, $\partial f/\partial y$, $\partial f/\partial z$ in place of $\partial f/\partial x_1$, $\partial f/\partial x_2$, $\partial f/\partial x_3$. If $f\colon U \subset \mathbb{R}^n \to \mathbb{R}^m$, then we can write

$$f(x_1, \ldots, x_n) = (f_1(x_1, \ldots, x_n), \ldots, f_m(x_1, \ldots, x_n)),$$

so that we can speak of the partial derivatives of each component; for example, $\partial f_m/\partial x_n$ is the partial derivative of the mth *component* with respect to x_n, the nth variable.

EXAMPLE 1 If $f(x, y) = x^2y + y^3$, find $\partial f/\partial x$ and $\partial f/\partial y$.

SOLUTION To find $\partial f/\partial x$ we hold y constant (think of it as some number, say 1) and differentiate only with respect to x; this yields

$$\frac{\partial f}{\partial x} = \frac{d(x^2y + y^3)}{dx} = 2xy.$$

Similarly, to find $\partial f/\partial y$ we hold x constant and differentiate only with respect to y:

$$\frac{\partial f}{\partial y} = \frac{d(x^2y + y^3)}{dy} = x^2 + 3y^2. \quad \blacktriangle$$

To indicate that a partial derivative is to be evaluated at a particular point, for example, at (x_0, y_0), we write

$$\frac{\partial f}{\partial x}(x_0, y_0) \qquad \text{or} \qquad \left.\frac{\partial f}{\partial x}\right|_{x=x_0, y=y_0} \qquad \text{or} \qquad \left.\frac{\partial f}{\partial x}\right|_{(x_0, y_0)}.$$

When we write $z = f(x, y)$ for the dependent variable, we sometimes write $\partial z/\partial x$ for $\partial f/\partial x$. Strictly speaking, this is an abuse of notation, but it is common practice to use these two notations interchangeably.

EXAMPLE 2 If $z = \cos xy + x \cos y = f(x, y)$, find the two partial derivatives $(\partial z/\partial x)\,(x_0, y_0)$ and $(\partial z/\partial y)(x_0, y_0)$.

SOLUTION First we fix y_0 and differentiate with respect to x, giving

$$\begin{aligned}\frac{\partial z}{\partial x}(x_0, y_0) &= \left.\frac{d(\cos xy_0 + x \cos y_0)}{dx}\right|_{x=x_0} \\ &= (-y_0 \sin xy_0 + \cos y_0)|_{x=x_0} \\ &= -y_0 \sin x_0y_0 + \cos y_0.\end{aligned}$$

Similarly, we fix x_0 and differentiate with respect to y to obtain

$$\begin{aligned}\frac{\partial z}{\partial y}(x_0, y_0) &= \left.\frac{d(\cos x_0 y + x_0 \cos y)}{dy}\right|_{y=y_0} \\ &= (-x_0 \sin x_0 y - x_0 \sin y)|_{y=y_0} \\ &= -x_0 \sin x_0 y_0 - x_0 \sin y_0. \quad \blacktriangle\end{aligned}$$

EXAMPLE 3 Find $\partial f/\partial x$ if $f(x, y) = xy/\sqrt{x^2 + y^2}$.

SOLUTION By the quotient rule,

$$\frac{\partial f}{\partial x} = \frac{y\sqrt{x^2+y^2} - xy(x/\sqrt{x^2+y^2})}{x^2+y^2} = \frac{y(x^2+y^2) - x^2 y}{(x^2+y^2)^{3/2}} = \frac{y^3}{(x^2+y^2)^{3/2}}. \quad \blacktriangle$$

A definition of differentiability that requires only the existence of partial derivatives turns out to be insufficient. Many standard results, such as the chain rule for functions of several variables would not follow, as Example 4 shows. Below, we shall see how to rectify this situation.

EXAMPLE 4 Let $f(x, y) = x^{1/3}y^{1/3}$. By definition,

$$\frac{\partial f}{\partial x}(0, 0) = \lim_{h\to 0} \frac{f(h, 0) - f(0, 0)}{h} = \lim_{h\to 0} \frac{0 - 0}{h} = 0,$$

and, similarly, $(\partial f/\partial y)(0, 0) = 0$ (these are not indeterminate forms!). It is necessary to use the original definition of partial derivatives, because the functions $x^{1/3}$ and $y^{1/3}$ are not themselves differentiable at 0. Suppose we restrict f to the line $y = x$ to get $f(x, x) = x^{2/3}$ (see Figure 2.3.2). We can view the substitution $y = x$ as the composition $f \circ g$ of the function g: $\mathbb{R} \to \mathbb{R}^2$, defined by $g(x) = (x, x)$, and f: $\mathbb{R}^2 \to \mathbb{R}$, defined by $f(x, y) = x^{1/3}y^{1/3}$.

Thus, the composite $f \circ g$ is given by $(f \circ g)(x) = x^{2/3}$. Each component of g is differentiable in x, and f has partial derivatives at $(0, 0)$, but $f \circ g$ *is not differentiable* at $x = 0$, in the sense of one-variable calculus. In other words, *the composition of f with g is not differentiable* in contrast to the calculus of functions of one variable, where the composition of differentiable functions *is* differentiable. Below, we shall give a definition of differentiability that has the pleasant consequence that the composition of differentiable functions *is* differentiable.

There is another reason for being dissatisfied with the mere existence of partial derivatives of $f(x, y) = x^{1/3}y^{1/3}$: There is no plane tangent, in any reasonable sense, to the graph at $(0, 0)$. The xy plane is tangent to the graph along the x and y axes because f has slope zero at $(0, 0)$ along these axes; that is, $\partial f/\partial x = 0$ and $\partial f/\partial y = 0$

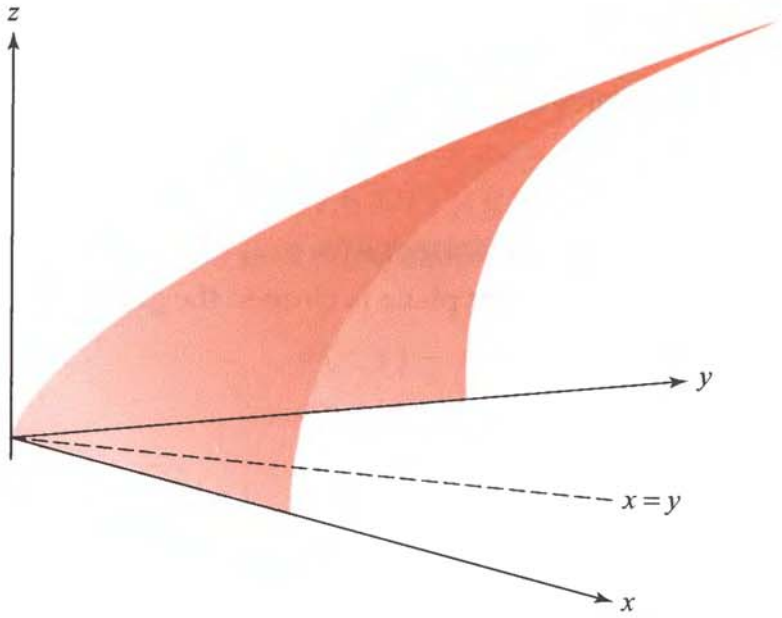


Figure 2.3.2 The portion of the graph of $x^{1/3}y^{1/3}$ in the first quadrant.

at (0, 0). Thus, if there is a tangent plane, it must be the xy plane. However, as is evident from Figure 2.3.2, the xy plane is not tangent to the graph in other directions, because the graph has a severe crinkle, and so the xy plane cannot be said to be tangent to the graph of f. ▲

The Linear Approximation

To "motivate" our definition of differentiability, let us compute what the equation of the plane tangent to the graph of $f\colon \mathbb{R}^2 \to \mathbb{R}$, $(x, y) \mapsto f(x, y)$ at (x_0, y_0) ought to be if f is smooth enough. In $\mathbb{R}^3$, a nonvertical plane has an equation of the form

$$z = ax + by + c.$$

If it is to be the plane tangent to the graph of f, the slopes along the x and y axes must be equal to $\partial f/\partial x$ and $\partial f/\partial y$, the rates of change of f with respect to x and y. Thus, $a = \partial f/\partial x$, $b = \partial f/\partial y$ [evaluated at (x_0, y_0)]. Finally, we may determine the constant c from the fact that $z = f(x_0, y_0)$ when $x = x_0$, $y = y_0$. Thus, we get the ***linear approximation***:

$$z = f(x_0, y_0) + \left[\frac{\partial f}{\partial x}(x_0, y_0)\right](x - x_0) + \left[\frac{\partial f}{\partial y}(x_0, y_0)\right](y - y_0), \tag{1}$$

which should be the equation of the plane tangent to the graph of f at (x_0, y_0), if f is "smooth enough" (see Figure 2.3.3).

Our definition of differentiability will mean in effect that the plane defined by the linear approximation (1) is a "good" approximation of f near (x_0, y_0). To get an idea of what one might mean by a good approximation, let us return for a moment to

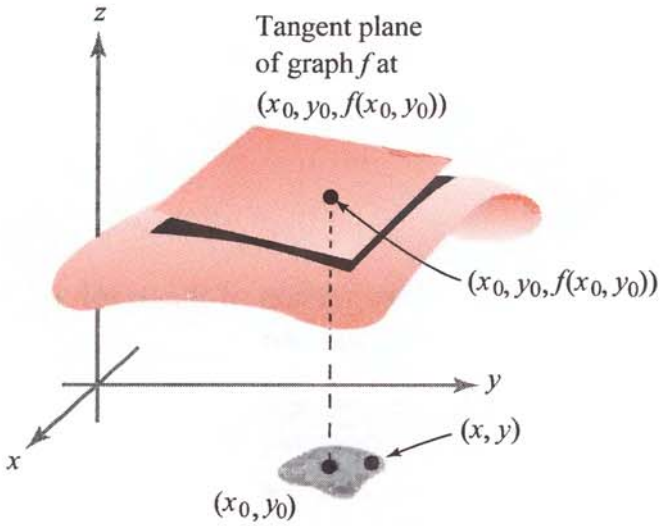


Figure 2.3.3 For points (x, y) near (x_0, y_0), the graph of the tangent plane is close to the graph of f.

one-variable calculus. If f is differentiable at a point x_0, then we know that

$$\lim_{\Delta x \to 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} = f'(x_0).$$

Let $x = x_0 + \Delta x$ and rewrite this as

$$\lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0} = f'(x_0).$$

Using the trivial limit $\lim_{x \to x_0} f'(x_0) = f'(x_0)$, we can rewrite the preceding equation as

$$\lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0} = \lim_{x \to x_0} f'(x_0);$$

that is,

$$\lim_{x \to x_0} \left[\frac{f(x) - f(x_0)}{x - x_0} - f'(x_0) \right] = 0;$$

that is,

$$\lim_{x \to x_0} \frac{f(x) - f(x_0) - f'(x_0)(x - x_0)}{x - x_0} = 0.$$

Thus, the tangent line l through $(x_0, f(x_0))$ with slope $f'(x_0)$ is close to f in the sense that the difference between $f(x)$ and $l(x) = f(x_0) + f'(x_0)(x - x_0)$, the equation of the tangent line goes to zero *even* when divided by $x - x_0$ as x goes to x_0. This is the notion of a "good approximation" that we will adapt to functions of several variables, with the tangent line replaced by the tangent plane [see equation (1), given earlier].

Differentiability for Functions of Two Variables

Using the linear approximation, we are ready to define the notion of differentiability.

DEFINITION: Differentiable: Two Variables Let $f: \mathbb{R}^2 \to \mathbb{R}$. We say f is ***differentiable*** at (x_0, y_0), if $\partial f/\partial x$ and $\partial f/\partial y$ exist at (x_0, y_0) and if

$$\frac{f(x, y) - f(x_0, y_0) - \left[\frac{\partial f}{\partial x}(x_0, y_0)\right](x - x_0) - \left[\frac{\partial f}{\partial y}(x_0, y_0)\right](y - y_0)}{\|(x, y) - (x_0, y_0)\|} \to 0 \tag{2}$$

as $(x, y) \to (x_0, y_0)$. This equation expresses what we mean by saying that

$$f(x_0, y_0) + \left[\frac{\partial f}{\partial x}(x_0, y_0)\right](x - x_0) + \left[\frac{\partial f}{\partial y}(x_0, y_0)\right](y - y_0)$$

is a ***good approximation*** to the function f.

It is not always easy to use this definition to see whether f is differentiable, but it will be easy to use another criterion, given shortly in Theorem 9.

Tangent Plane

We have used the informal notion of the plane tangent to the graph of a function to motivate our definition of differentiability. Now we are ready to adopt a formal definition of the tangent plane.

DEFINITION: Tangent Plane Let $f: \mathbb{R}^2 \to \mathbb{R}$ be differentiable at $\mathbf{x}_0 = (x_0, y_0)$. The plane in $\mathbb{R}^3$ defined by the equation

$$z = f(x_0, y_0) + \left[\frac{\partial f}{\partial x}(x_0, y_0)\right](x - x_0) + \left[\frac{\partial f}{\partial y}(x_0, y_0)\right](y - y_0),$$

is called the ***tangent plane*** of the graph of f at the point (x_0, y_0).

EXAMPLE 5 Compute the plane tangent to the graph of $z = x^2 + y^4 + e^{xy}$ at the point $(1, 0, 2)$.

SOLUTION Use formula (1), with $x_0 = 1$, $y_0 = 0$, and $z_0 = f(x_0, y_0) = 2$. The partial derivatives are

$$\frac{\partial z}{\partial x} = 2x + ye^{xy} \qquad \text{and} \qquad \frac{\partial z}{\partial y} = 4y^3 + xe^{xy}.$$

At (1, 0, 2), these partial derivatives are 2 and 1, respectively. Thus, by formula (1), the tangent plane is

$$z = 2(x-1) + 1(y-0) + 2, \qquad \text{that is,} \qquad z = 2x + y. \quad \blacktriangle$$

Let us write $\mathbf{D}f(x_0, y_0)$ for the row matrix

$$\left[\frac{\partial f}{\partial x}(x_0, y_0) \quad \frac{\partial f}{\partial y}(x_0, y_0)\right],$$

so that the definition of differentiability asserts that

$$f(x_0, y_0) + \mathbf{D}f(x_0, y_0)\begin{bmatrix} x - x_0 \\ y - y_0 \end{bmatrix}$$
$$= f(x_0, y_0) + \left[\frac{\partial f}{\partial x}(x_0, y_0)\right](x - x_0) + \left[\frac{\partial f}{\partial y}(x_0, y_0)\right](y - y_0) \qquad (3)$$

is our good approximation to f near (x_0, y_0). As earlier, "good" is taken in the sense that expression (3) differs from $f(x, y)$ by something small times $\sqrt{(x-x_0)^2 + (y-y_0)^2}$. We say that expression (3) is the ***best linear approximation*** to f near (x_0, y_0).

Differentiability: The General Case

Now we are ready to give a definition of differentiability for maps f of $\mathbb{R}^n$ to $\mathbb{R}^m$, using the preceding discussion as motivation. The derivative $\mathbf{D}f(\mathbf{x}_0)$ of $f = (f_1, \ldots, f_m)$ at a point $\mathbf{x}_0$ is a matrix $\mathbf{T}$ whose elements are $t_{ij} = \partial f_i / \partial x_j$ evaluated at $\mathbf{x}_0$.[2]

DEFINITION: Differentiable, n Variables, m Functions Let U be an open set in $\mathbb{R}^n$ and let $f\colon U \subset \mathbb{R}^n \to \mathbb{R}^m$ be a given function. We say that f is ***differentiable*** at $\mathbf{x}_0 \in U$ if the partial derivatives of f exist at $\mathbf{x}_0$ and if

$$\lim_{\mathbf{x} \to \mathbf{x}_0} \frac{\|f(\mathbf{x}) - f(\mathbf{x}_0) - \mathbf{T}(\mathbf{x} - \mathbf{x}_0)\|}{\|\mathbf{x} - \mathbf{x}_0\|} = 0, \qquad (4)$$

where $\mathbf{T} = \mathbf{D}f(\mathbf{x}_0)$ is the $m \times n$ matrix with matrix elements $\partial f_i / \partial x_j$ evaluated at $\mathbf{x}_0$ and $\mathbf{T}(\mathbf{x} - \mathbf{x}_0)$ means the product of $\mathbf{T}$ with $\mathbf{x} - \mathbf{x}_0$ (regarded as a column matrix). We call $\mathbf{T}$ the ***derivative*** of f at $\mathbf{x}_0$.

[2] It turns out that we need to postulate the existence of only *some* matrix giving the best linear approximation near $\mathbf{x}_0 \in \mathbb{R}^n$, because in fact this matrix is *necessarily* the matrix whose ijth entry is $\partial f_i / \partial x_j$ (see the Internet supplement for Chapter 2).

We shall always denote the derivative $\mathbf{T}$ of f at $\mathbf{x}_0$ by $\mathbf{D}f(\mathbf{x}_0)$, although in some books it is denoted $df(\mathbf{x}_0)$ and referred to as the ***differential*** of f. In the case where $m = 1$, the matrix $\mathbf{T}$ is just the row matrix

$$\left[\frac{\partial f}{\partial x_1}(\mathbf{x}_0) \quad \cdots \quad \frac{\partial f}{\partial x_n}(\mathbf{x}_0)\right].$$

(Sometimes, when there is danger of confusion, we separate the entries by commas.) Furthermore, setting $n = 2$ and putting the result back into equation (4), we see that conditions (2) and (4) do agree. Thus, if we let $\mathbf{h} = \mathbf{x} - \mathbf{x}_0$, a real-valued function f of n variables is differentiable at a point $\mathbf{x}_0$ if

$$\lim_{\mathbf{h}\to 0} \frac{1}{\|\mathbf{h}\|}\left|f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0) - \sum_{j=1}^{n} \frac{\partial f}{\partial x_j}(\mathbf{x}_0)h_j\right| = 0,$$

because

$$\mathbf{T}\mathbf{h} = \sum_{j=1}^{n} h_j \frac{\partial f}{\partial x_j}(\mathbf{x}_0).$$

For the general case of f mapping a subset of $\mathbb{R}^n$ to $\mathbb{R}^m$, the derivative is the $m \times n$ matrix given by

$$\mathbf{D}f(\mathbf{x}_0) = \begin{bmatrix} \dfrac{\partial f_1}{\partial x_1} & \cdots & \dfrac{\partial f_1}{\partial x_n} \\ \vdots & & \vdots \\ \dfrac{\partial f_m}{\partial x_1} & \cdots & \dfrac{\partial f_m}{\partial x_n} \end{bmatrix},$$

where $\partial f_i/\partial x_j$ is evaluated at $\mathbf{x}_0$. The matrix $\mathbf{D}f(\mathbf{x}_0)$ is, appropriately, called the ***matrix of partial derivatives of*** f ***at*** $\mathbf{x}_0$.

EXAMPLE 6 Calculate the matrices of partial derivatives for these functions.

(a) $f(x, y) = (e^{x+y} + y, y^2x)$

(b) $f(x, y) = (x^2 + \cos y, ye^x)$

(c) $f(x, y, z) = (ze^x, -ye^z)$

SOLUTION

(a) Here $f\colon \mathbb{R}^2 \to \mathbb{R}^2$ is defined by $f_1(x, y) = e^{x+y} + y$ and $f_2(x, y) = y^2x$. Hence, $\mathbf{D}f(x, y)$ is the 2×2 matrix

$$\mathbf{D}f(x, y) = \begin{bmatrix} e^{x+y} & e^{x+y} + 1 \\ y^2 & 2xy \end{bmatrix}.$$

(b) We have

$$\mathbf{D}f(x, y) = \begin{bmatrix} 2x & -\sin y \\ ye^x & e^x \end{bmatrix}.$$

(c) In this case,

$$\mathbf{D}f(x, y, z) = \begin{bmatrix} ze^x & 0 & e^x \\ 0 & -e^z & -ye^z \end{bmatrix}. \quad \blacktriangle$$

Gradients

For real-valued functions we use special terminology for the derivative.

DEFINITION: Gradient Consider the special case $f: U \subset \mathbb{R}^n \to \mathbb{R}$. Here $\mathbf{D}f(\mathbf{x})$ is a $1 \times n$ matrix:

$$\mathbf{D}f(\mathbf{x}) = \begin{bmatrix} \dfrac{\partial f}{\partial x_1} & \cdots & \dfrac{\partial f}{\partial x_n} \end{bmatrix}.$$

We can form the corresponding vector $(\partial f/\partial x_1, \ldots, \partial f/\partial x_n)$, called the ***gradient*** of f and denoted by ∇f or grad f.

From the definition, we see that for $f: \mathbb{R}^3 \to \mathbb{R}$,

$$\nabla f = \frac{\partial f}{\partial x}\mathbf{i} + \frac{\partial f}{\partial y}\mathbf{j} + \frac{\partial f}{\partial z}\mathbf{k},$$

while for $f: \mathbb{R}^2 \to \mathbb{R}$,

$$\nabla f = \frac{\partial f}{\partial x}\mathbf{i} + \frac{\partial f}{\partial y}\mathbf{j}.$$

The geometric significance of the gradient will be discussed in Section 2.6. In terms of inner products, we can write the derivative of f as

$$\mathbf{D}f(\mathbf{x})(\mathbf{h}) = \nabla f(\mathbf{x}) \cdot \mathbf{h}.$$

EXAMPLE 7 Let $f: \mathbb{R}^3 \to \mathbb{R}$, $f(x, y, z) = xe^y$. Then

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right) = (e^y, xe^y, 0). \quad \blacktriangle$$

EXAMPLE 8 If $f\colon \mathbb{R}^2 \to \mathbb{R}$ is given by $(x, y) \mapsto e^{xy} + \sin xy$, then

$$\begin{aligned}\nabla f(x, y) &= (ye^{xy} + y\cos xy)\mathbf{i} + (xe^{xy} + x\cos xy)\mathbf{j} \\ &= (e^{xy} + \cos xy)(y\mathbf{i} + x\mathbf{j}).\end{aligned}$$ ▲

In one-variable calculus it is shown that if f is differentiable, then f is continuous. We will state in Theorem 8 that this is also true for differentiable functions of several variables. As we know, there are plenty of functions of one variable that are continuous but not differentiable, such as $f(x) = |x|$. Before stating the result, let us give an example of a function of two variables whose *partial derivatives exist at a point, but which is not continuous at that point.*

EXAMPLE 9 Let $f\colon \mathbb{R}^2 \to \mathbb{R}$ be defined by

$$f(x, y) = \begin{cases} 1 & \text{if } x = 0 \text{ or if } y = 0 \\ 0 & \text{otherwise.} \end{cases}$$

Because f is constant on the x and y axes, where it equals 1,

$$\frac{\partial f}{\partial x}(0, 0) = 0 \qquad \text{and} \qquad \frac{\partial f}{\partial y}(0, 0) = 0.$$

But f is not continuous at $(0, 0)$, because $\lim\limits_{(x,y)\to(0,0)} f(x, y)$ does not exist. ▲

Some Basic Theorems

The first of these basic theorems relates differentiability and continuity.

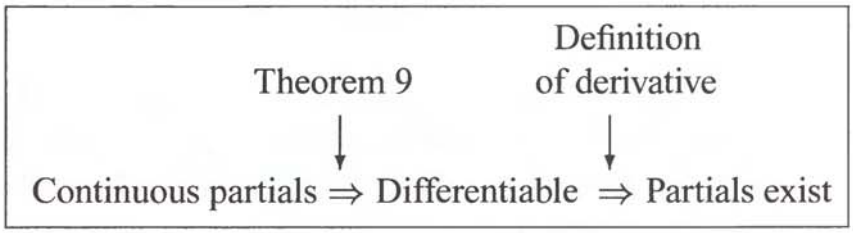
THEOREM 8 Let $f\colon U \subset \mathbb{R}^n \to \mathbb{R}^m$ be differentiable at $\mathbf{x}_0 \in U$. Then f is continuous at $\mathbf{x}_0$.

This result is very reasonable, because "differentiability" means that there is enough smoothness to have a tangent plane, which is stronger than just being continuous. Consult the Internet supplement for Chapter 2 for the formal proof.

As we have seen, it is usually easy to tell when the partial derivatives of a function exist using what we know from one-variable calculus. However, the definition of differentiability looks somewhat complicated, and the required approximation condition in equation (4) may seem, and sometimes is, difficult to verify. Fortunately, there is a simple criterion, given in the following theorem, that tells us when a function is differentiable.

THEOREM 9 Let $f\colon U \subset \mathbb{R}^n \to \mathbb{R}^m$. Suppose the partial derivatives $\partial f_i/\partial x_j$ of f all exist and are continuous in a neighborhood of a point $\mathbf{x} \in U$. Then f is differentiable at $\mathbf{x}$.

We give the proof in the Internet supplement for Chapter 2. Notice the following hierarchy:



Each converse statement, obtained by reversing an implication, is invalid. [For a counterexample to the converse of the first implication, use $f(x) = x^2 \sin(1/x)$, $f(0) = 0$; for the second, see Example 1 in the Internet supplement for Chapter 2 or use Example 4 in this section.]

A function whose partial derivatives exist and are continuous is said to be of ***class*** C^1. Thus, Theorem 9 says that *any* C^1 *function is differentiable.*

EXAMPLE 10 Let

$$f(x, y) = \frac{\cos x + e^{xy}}{x^2 + y^2}.$$

Show that f is differentiable at all points $(x, y) \neq (0, 0)$.

SOLUTION Observe that the partial derivatives

$$\frac{\partial f}{\partial x} = \frac{(x^2 + y^2)(ye^{xy} - \sin x) - 2x(\cos x + e^{xy})}{(x^2 + y^2)^2}$$

$$\frac{\partial f}{\partial y} = \frac{(x^2 + y^2)xe^{xy} - 2y(\cos x + e^{xy})}{(x^2 + y^2)^2}$$

are continuous except when $x = 0$ and $y = 0$ (by the results in Section 2.2). Thus, f is differentiable by Theorem 9. ▲

In the Internet supplement we show that $f(x, y) = xy/\sqrt{x^2 + y^2}$ [with $f(0, 0) = 0$] is continuous, has partial derivatives at (0, 0), yet is *not* differentiable there. See Figure 2.3.4. By Theorem 9, its partial derivatives cannot be continuous at (0, 0).

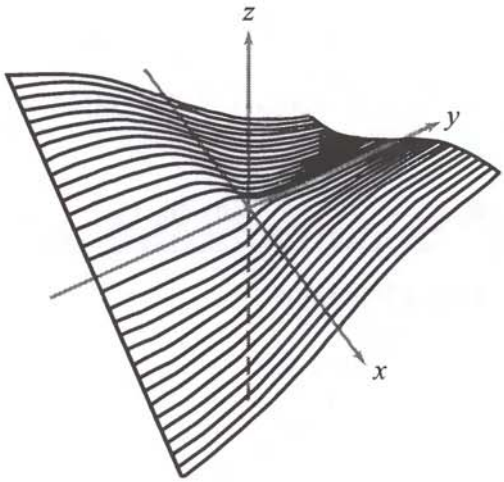


Figure 2.3.4 This function is not differentiable at (0, 0), because it is "crinkled."

EXERCISES

1. Find $\partial f/\partial x$, $\partial f/\partial y$ if

(a) $f(x, y) = xy$
(b) $f(x, y) = e^{xy}$
(c) $f(x, y) = x \cos x \cos y$
(d) $f(x, y) = (x^2 + y^2) \log (x^2 + y^2)$

2. Evaluate the partial derivatives $\partial z/\partial x$, $\partial z/\partial y$ for the given function at the indicated points.

(a) $z = \sqrt{a^2 - x^2 - y^2}$; (0, 0), $(a/2, a/2)$
(b) $z = \log \sqrt{1 + xy}$; (1, 2), (0, 0)
(c) $z = e^{ax} \cos (bx + y)$; $(2\pi/b, 0)$

3. In each case following, find the partial derivatives $\partial w/\partial x$, $\partial w/\partial y$.

(a) $w = xe^{x^2+y^2}$
(b) $w = \dfrac{x^2 + y^2}{x^2 - y^2}$
(c) $w = e^{xy} \log (x^2 + y^2)$
(d) $w = x/y$
(e) $w = \cos (ye^{xy}) \sin x$

4. Show that each of the following functions is differentiable at each point in its domain. Decide which of the functions are C^1.

(a) $f(x, y) = \dfrac{2xy}{(x^2 + y^2)^2}$
(b) $f(x, y) = \dfrac{x}{y} + \dfrac{y}{x}$
(c) $f(r, \theta) = \frac{1}{2} r \sin 2\theta, r > 0$
(d) $f(x, y) = \dfrac{xy}{\sqrt{x^2 + y^2}}$
(e) $f(x, y) = \dfrac{x^2 y}{x^4 + y^2}$

5. Find the equation of the plane tangent to the surface $z = x^2 + y^3$ at (3, 1, 10).

6. Using the respective functions in Exercise 1, compute the plane tangent to the graphs at the indicated points.

(a) (0, 0) (b) (0, 1) (c) $(0, \pi)$ (d) (0, 1)

7. Compute the matrix of partial derivatives of the following functions:

(a) $f: \mathbb{R}^2 \to \mathbb{R}^2$, $f(x, y) = (x, y)$
(b) $f: \mathbb{R}^2 \to \mathbb{R}^3$, $f(x, y) = (xe^y + \cos y, x, x + e^y)$
(c) $f: \mathbb{R}^3 \to \mathbb{R}^2$, $f(x, y, z) = (x + e^z + y, yx^2)$
(d) $f: \mathbb{R}^2 \to \mathbb{R}^3$, $f(x, y) = (xye^{xy}, x \sin y, 5xy^2)$

8. Compute the matrix of partial derivatives of

(a) $f(x, y) = (e^x, \sin xy)$
(b) $f(x, y, z) = (x - y, y + z)$
(c) $f(x, y) = (x + y, x - y, xy)$
(d) $f(x, y, z) = (x + z, y - 5z, x - y)$

9. Where does the plane tangent to $z = e^{x-y}$ at (1, 1, 1) meet the z axis?

10. Why should the graphs of $f(x, y) = x^2 + y^2$ and $g(x, y) = -x^2 - y^2 + xy^3$ be called "tangent" at (0, 0)?

11. Let $f(x, y) = e^{xy}$. Show that $x(\partial f/\partial x) = y(\partial f/\partial y)$.

12. Use the linear approximation to approximate a suitable function $f(x, y)$ and thereby estimate the following:

(a) $(0.99e^{0.02})^8$
(b) $(0.99)^3 + (2.01)^3 - 6(0.99)(2.01)$
(c) $\sqrt{(4.01)^2 + (3.98)^2 + (2.02)^2}$

13. Compute the gradients of the following functions:

(a) $f(x, y, z) = x \exp(-x^2 - y^2 - z^2)$ (Note that $\exp u = e^u$.)
(b) $f(x, y, z) = \dfrac{xyz}{x^2 + y^2 + z^2}$
(c) $f(x, y, z) = z^2 e^x \cos y$

14. Compute the tangent plane at (1, 0, 1) for each of the functions in Exercise 13. [The solution to part (c) only is in the Study Guide.]

15. Find the equation of the tangent plane to $z = x^2 + 2y^3$ at (1, 1, 3).

16. Calculate $\nabla h(1, 1, 1)$ if $h(x, y, z) = (x + z)e^{x-y}$.

17. Let $f(x, y, z) = x^2 + y^2 - z^2$. Calculate $\nabla f(0, 0, 1)$.

18. Evaluate the gradient of $f(x, y, z) = \log(x^2 + y^2 + z^2)$ at (1, 0, 1).

19. Describe all Hölder-continuous functions with $\alpha > 1$ (see Exercise 25, Section 2.2). (HINT: What is the derivative of such a function?)

20. Suppose $f\colon \mathbb{R}^n \to \mathbb{R}^m$ is a linear map. What is the derivative of f?

2.4 Introduction to Paths and Curves

In this section, we introduce some of the basic geometry and computational methods for paths in the plane and space. This will be an important ingredient for the chain rule treated in the next section. We will return to paths with additional topics in Chapter 4.

Paths and Curves

One often thinks of a curve as a line drawn on paper, such as a straight line, a circle, or a sine curve. It is useful to think of a curve C mathematically as the set of values of a function that maps an interval of real numbers into the plane or space. We shall call such a map a ***path***. We usually denote a path by $\mathbf{c}$. The image C of the path then corresponds to the curve we see on paper (see Figure 2.4.1). Often we write t for the independent variable and imagine it to be *time*, so that $\mathbf{c}(t)$ is the position at time t of a moving particle, which ***traces out*** a curve as t varies. We also say $\mathbf{c}$ ***parametrizes*** C. Strictly speaking, we should distinguish between $\mathbf{c}(t)$ as a *point* in space and as a *vector* based at the origin.

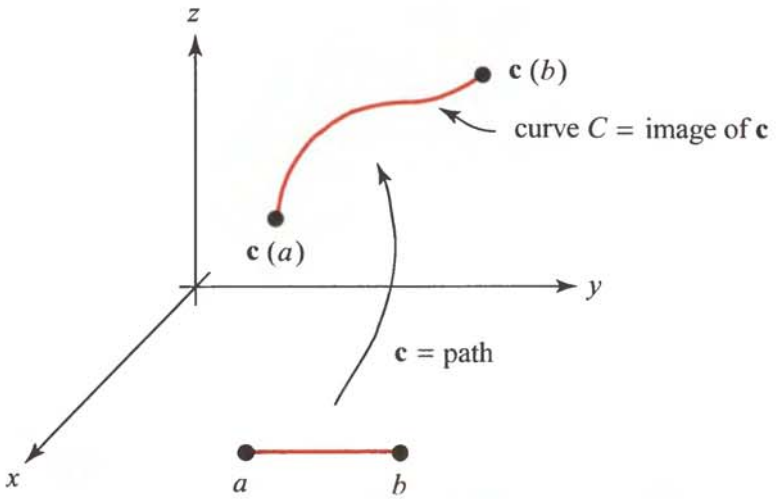


Figure 2.4.1 The map $\mathbf{c}$ is the path; its image C is the curve we "see."

EXAMPLE 1 The straight line L in $\mathbb{R}^3$ through the point (x_0, y_0, z_0) in the direction of vector $\mathbf{v}$ is the image of the path

$$\mathbf{c}(t) = (x_0, y_0, z_0) + t\mathbf{v}$$

for $t \in \mathbb{R}$ (see Figure 2.4.2). Thus, our notion of curve includes straight lines as special cases. ▲

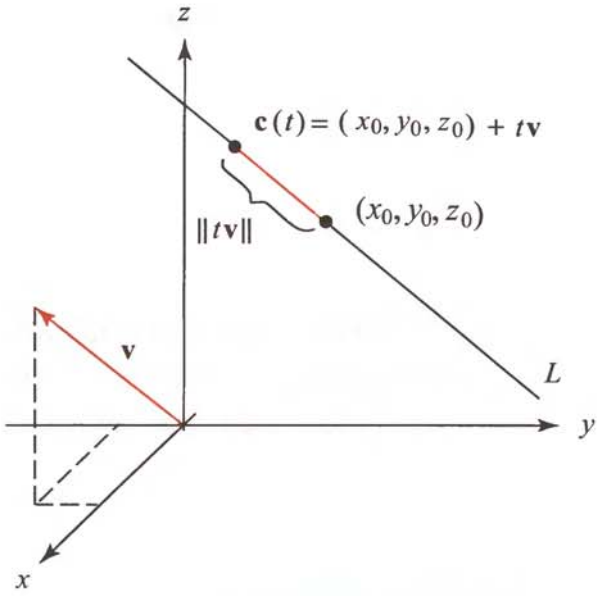


Figure 2.4.2 L is the straight line in space through (x_0, y_0, z_0) and in direction $\mathbf{v}$; its equation is $\mathbf{c}(t) = (x_0, y_0, z_0) + t\mathbf{v}$.

EXAMPLE 2 The unit circle C: $x^2 + y^2 = 1$ in the plane is the image of the path

$$\mathbf{c}: \mathbb{R} \to \mathbb{R}^2, \qquad \mathbf{c}(t) = (\cos t, \sin t), \qquad 0 \le t \le 2\pi,$$

(see Figure 2.4.3). The unit circle is also the image of the path $\tilde{\mathbf{c}}(t) = (\cos 2t, \sin 2t)$, $0 \le t \le \pi$. Thus, different paths may parametrize the same curve. ▲

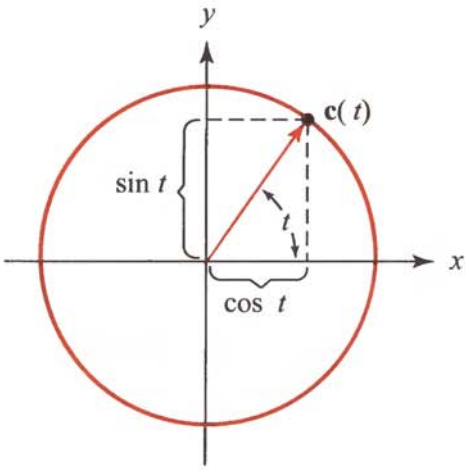


Figure 2.4.3 $\mathbf{c}(t) = (\cos t, \sin t)$ is a path whose image C is the unit circle.

Paths and Curves A ***path*** in $\mathbb{R}^n$ is a map $\mathbf{c}: [a, b] \to \mathbb{R}^n$; it is a ***path in the plane*** if $n = 2$ and a ***path in space*** if $n = 3$. The collection C of points $\mathbf{c}(t)$ as t varies in $[a, b]$ is called a ***curve***, and $\mathbf{c}(a)$ and $\mathbf{c}(b)$ are its ***endpoints***. The path $\mathbf{c}$ is said to ***parametrize*** the curve C. We also say $\mathbf{c}(t)$ ***traces out*** C as t varies.

If $\mathbf{c}$ is a path in $\mathbb{R}^3$, we can write $\mathbf{c}(t) = (x(t), y(t), z(t))$ and we call $x(t)$, $y(t)$, and $z(t)$ the ***component functions*** of $\mathbf{c}$. We form component functions similarly in $\mathbb{R}^2$ or, generally, in $\mathbb{R}^n$.

EXAMPLE 3 The path $\mathbf{c}(t) = (t, t^2)$ traces out a parabolic arc. This curve coincides with the graph $f(x) = x^2$ (see Figure 2.4.4). ▲

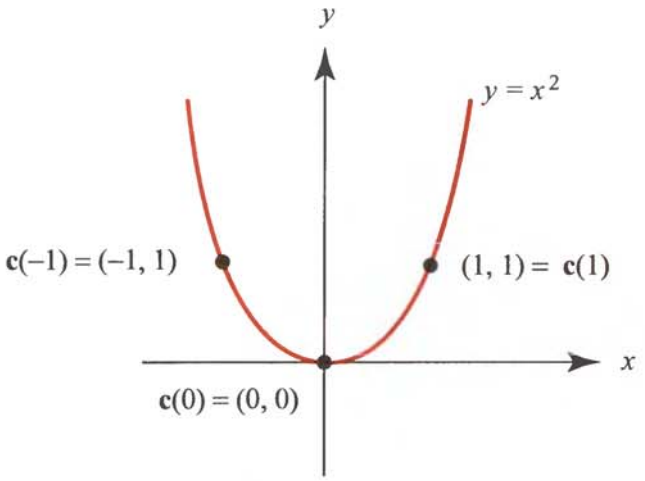


Figure 2.4.4 The image of $\mathbf{c}(t) = (t, t^2)$ is the parabola $y = x^2$.

EXAMPLE 4 A wheel of radius R rolls to the right along a straight line at speed v. Use vector methods to find the path $\mathbf{c}(t)$ of the point on the wheel that initially lies at a distance r below the center.

SOLUTION We place the wheel in the xy plane with its center initially at $(0, R)$, so that the position of the center at time t is given by the path $\mathbf{C}(t) = (vt, R)$. (Refer to Figure 2.4.5.)

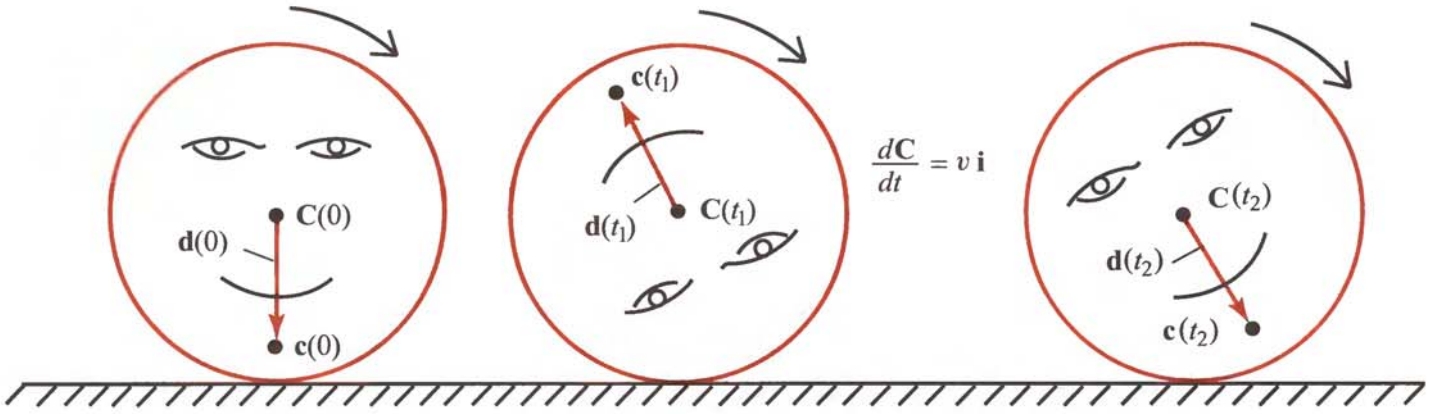


Figure 2.4.5 The vector $\mathbf{d}(t)$ points from the wheel's center, $\mathbf{C}(t)$, to the position $\mathbf{c}(t)$ of a point on the wheel and rotates in the clockwise direction while the wheel moves to the right.

The position of the point $\mathbf{c}(t)$ *relative to the center* is given by the vector $\mathbf{d}(t) = \mathbf{c}(t) - \mathbf{C}(t)$ that has the initial value $-r\mathbf{j}$ and rotates in the *clockwise* direction. The rate of rotation is such that the wheel makes a full rotation after the center has moved a distance $2\pi R$ (equal to the circumference of the wheel). This takes a time $2\pi R/v$, so the angular velocity $d\theta/dt$ of the wheel is v/R. Because the rotation is clockwise, the vector function $\mathbf{d}(t)$ is of the form

$$\mathbf{d}(t) = r\left(\cos\left[-\frac{v}{R}t + \theta\right]\mathbf{i} + \sin\left[-\frac{v}{R}t + \theta\right]\mathbf{j}\right)$$

for some initial angle θ. Because $\mathbf{d}(0) = -r\mathbf{j}$, we have $\cos\theta = 0$ and $\sin\theta = -1$, so $\theta = -\pi/2$, and hence

$$\mathbf{d}(t) = r\left(\cos\left[-\frac{v}{R}t - \frac{\pi}{2}\right]\mathbf{i} + \sin\left[-\frac{v}{R}t - \frac{\pi}{2}\right]\mathbf{j}\right).$$

Using $\cos(\varphi - \pi/2) = \sin\varphi$ and $\sin(\varphi - \pi/2) = -\cos\varphi$, along with $\cos(-\varphi) = \cos\varphi$ and $\sin(-\varphi) = -\sin\varphi$, we get

$$\mathbf{d}(t) = r\left(-\sin\frac{vt}{R}\,\mathbf{i} - \cos\frac{vt}{R}\,\mathbf{j}\right).$$

Finally, the path $\mathbf{c}(t)$ is given by adding the components of the vector function $\mathbf{d}(t)$ to the coordinates of the path $\mathbf{C}(t)$; the result is

$$\mathbf{c}(t) = \left(vt - r\sin\frac{vt}{R},\ R - r\cos\frac{vt}{R}\right).$$

In the special case $v = R = r = 1$, we get $\mathbf{c}(t) = (t - \sin t,\ 1 - \cos t)$. The image curve C of this path $\mathbf{c}$ is shown in Figure 2.4.6; it is called a ***cycloid***. ▲

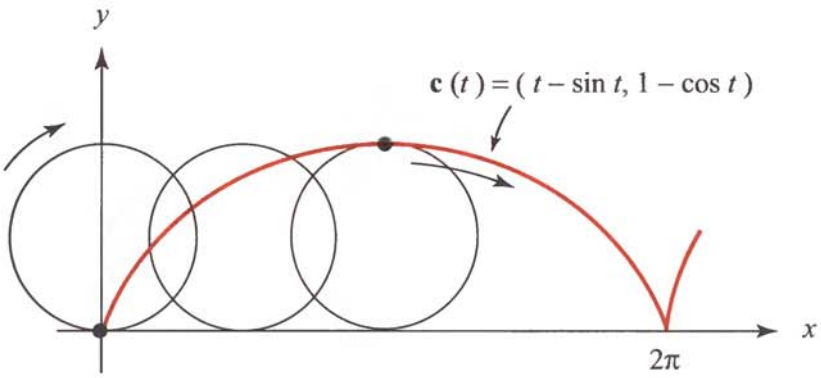


Figure 2.4.6 The curve traced by a point moving on the rim of a rolling circle is called a cycloid.

The preceding example considered the path of a point not necessarily on the rim of a wheel rolling along a straight line. When the wheel rolls on a circle, the resulting curve is called an ***epicycle***. These are the epicycles discussed in the Ptolemaic theory in the introduction. If the wheel is outside the circle and the point is on the rim, the curve is called an ***epicycloid***, and when the wheel is inside the circle it is a ***hypocycloid***. An example of the latter is shown in Figure 2.4.7.

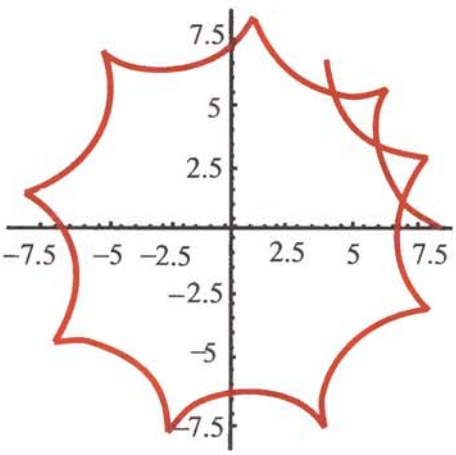


Figure 2.4.7 An example of a hypocycloid.

Historical Note

The French mathematician Blaise Pascal studied the cycloid in 1649 as a way of distracting himself at a time when he was suffering from a painful toothache. When the pain disappeared, he took it as a sign that God was not displeased with his thoughts. Pascal's results stimulated other mathematicians to investigate this curve, and subsequently numerous remarkable properties were found. One of these was discovered by the Dutchman Christian Huygens, who used it in the construction of a "perfect" pendulum clock.

Velocity and Tangents to Paths

If we think of $\mathbf{c}(t)$ as the curve traced out by a particle and t as time, it is reasonable to define the velocity vector as follows.

DEFINITION: Velocity Vector If $\mathbf{c}$ is a path and it is differentiable, we say $\mathbf{c}$ is a ***differentiable path***. The ***velocity*** of $\mathbf{c}$ at time t is defined by[3]

$$\mathbf{c}'(t) = \lim_{h \to 0} \frac{\mathbf{c}(t+h) - \mathbf{c}(t)}{h}.$$

We normally draw the vector $\mathbf{c}'(t)$ with its tail at the point $\mathbf{c}(t)$. The ***speed*** of the path $\mathbf{c}(t)$ is $s = \|\mathbf{c}'(t)\|$, the length of the velocity vector. If $\mathbf{c}(t) = (x(t), y(t))$ in $\mathbb{R}^2$, then

$$\mathbf{c}'(t) = (x'(t), y'(t)) = x'(t)\mathbf{i} + y'(t)\mathbf{j}$$

and if $\mathbf{c}(t) = (x(t), y(t), z(t))$ in $\mathbb{R}^3$, then

$$\mathbf{c}'(t) = (x'(t), y'(t), z'(t)) = x'(t)\mathbf{i} + y'(t)\mathbf{j} + z'(t)\mathbf{k}.$$

Here, $x'(t)$ is the one-variable derivative dx/dt. If we accept limits of vectors interpreted componentwise, the formulas for the velocity vector follow from the definition of the derivative. However, the limit can be interpreted in the sense of vectors as well. In Figure 2.4.8, we see that $[\mathbf{c}(t+h) - \mathbf{c}(t)]/h$ approaches the tangent to the path as $h \to 0$.

[3] If t lies at the endpoint of an interval, one should, as in one-variable calculus, take right- or left-handed limits.

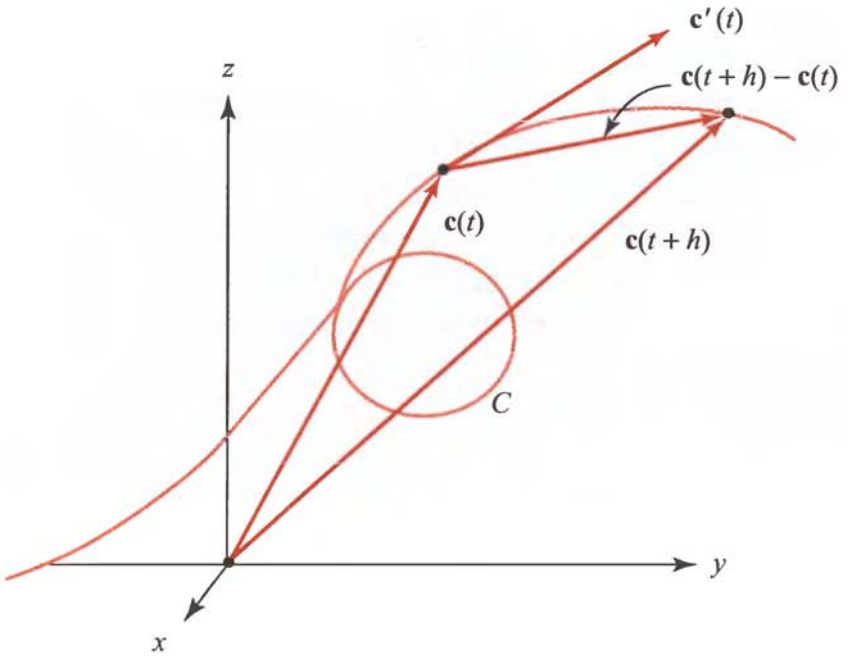


Figure 2.4.8 The vector $\mathbf{c}'(t)$ is tangent to the path $\mathbf{c}(t)$.

Tangent Vector The velocity $\mathbf{c}'(t)$ is a vector ***tangent*** to the path $\mathbf{c}(t)$ at time t. If C is a curve traced out by $\mathbf{c}$ and if $\mathbf{c}'(t)$ is not equal to $\mathbf{0}$, then $\mathbf{c}'(t)$ is a vector tangent to the curve C at the point $\mathbf{c}(t)$.

If we think of the derivative $\mathbf{D}\mathbf{c}(t)$ as a matrix, it will be a column vector with the entries $x'(t)$, $y'(t)$, and $z'(t)$. Thus, the derivative here is consistent with our earlier notion.

EXAMPLE 5 Compute the tangent vector to the path $\mathbf{c}(t) = (t, t^2, e^t)$ at $t = 0$.

SOLUTION Here $\mathbf{c}'(t) = (1, 2t, e^t)$, and so at $t = 0$ we obtain the tangent vector $(1, 0, 1)$. ▲

EXAMPLE 6 Describe the path $\mathbf{c}(t) = (\cos t, \sin t, t)$. Find the velocity vector at the point on the image curve where $t = \pi/2$.

SOLUTION For a given t, the point $(\cos t, \sin t, 0)$ lies on the circle $x^2 + y^2 = 1$ in the xy plane. Therefore, the point $(\cos t, \sin t, t)$ lies t units above the point $(\cos t, \sin t, 0)$ if t is positive and $-t$ units below $(\cos t, \sin t, 0)$ if t is negative. As t increases, $(\cos t, \sin t, t)$ wraps around the cylinder $x^2 + y^2 = 1$ with the z coordinate increasing. The curve this traces out is called a ***helix***, which is depicted in Figure 2.4.9. At $t = \pi/2$, $\mathbf{c}'(\pi/2) = (-\sin \pi/2, \cos \pi/2, 1) = (-1, 0, 1) = -\mathbf{i} + \mathbf{k}$. ▲

EXAMPLE 7 The cycloidal path of a particle on the edge of a wheel of radius R with velocity v is given by $\mathbf{c}(t) = (vt - R\sin(vt/R), R - R\cos(vt/R))$. (See Example 4.) Find the velocity $\mathbf{c}'(t)$ of the particle as a function of t. When is the velocity zero? Is the velocity vector ever vertical?

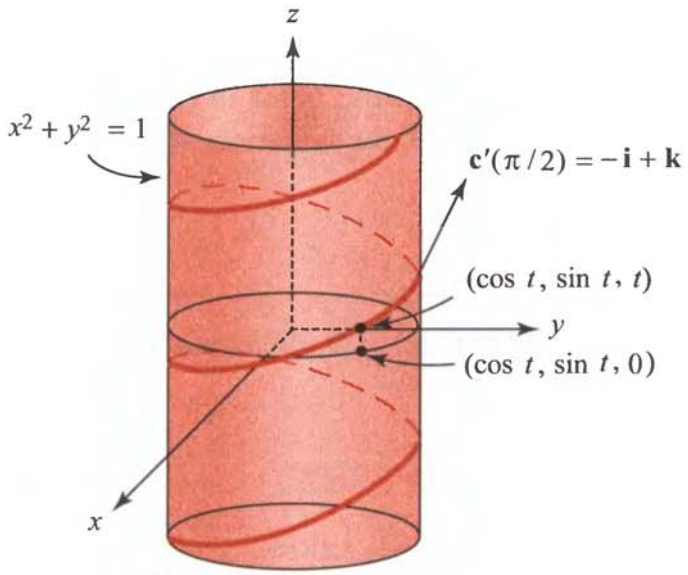


Figure 2.4.9 The helix $\mathbf{c}(t) = (\cos t, \sin t, t)$ wraps around the cylinder $x^2 + y^2 = 1$.

SOLUTION To find the velocity, we differentiate:

$$\mathbf{c}'(t) = \left(\frac{d}{dt}\left(vt - R\sin\frac{vt}{R}\right), \frac{d}{dt}\left(R - R\cos\frac{vt}{R}\right)\right)$$
$$= \left(v - v\cos\frac{vt}{R}, v\sin\frac{vt}{R}\right).$$

In vector notation, $\mathbf{c}'(t) = (v - v\cos(vt/R))\mathbf{i} + (v\sin(vt/R))\mathbf{j}$. The component in the direction of $\mathbf{i}$ is $v(1 - \cos(vt/R))$, which is zero whenever vt/R is an integer multiple of 2π. For such values of t, $\sin(vt/R)$ is zero as well, so the only times at which the velocity is zero are when $t = 2\pi nR/v$ for some integer n. At such times, $\mathbf{c}(t) = (2\pi nR, 0)$, so the moving point is touching the ground. These moments occur at time intervals of $2\pi R/v$ (more frequently for small wheels, as well as for rapidly rolling ones).

The velocity vector is never vertical, because the horizontal component vanishes only when the vertical one does as well. ▲

Figure 2.4.10 shows some velocity vectors superimposed on the cycloidal path of Figure 2.4.6.

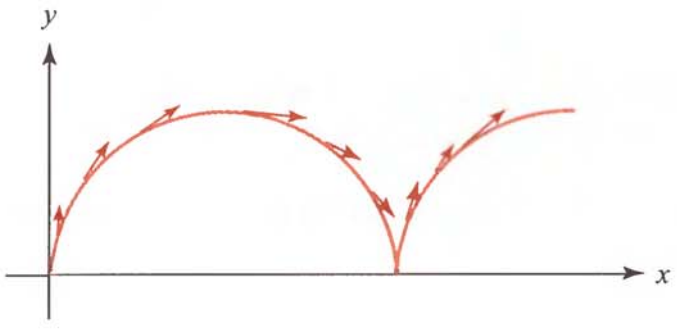


Figure 2.4.10 Velocity vectors for the curve traced out by a point on the rim of a rolling wheel.

Tangent Line

The tangent line to a path at a point is the line through the point in the direction of the tangent vector. Using the point-direction form of the equation of a line, we obtain the parametric equation for the tangent line.

Tangent Line to a Path If $\mathbf{c}(t)$ is a path, and if $\mathbf{c}'(t_0) \neq \mathbf{0}$, the equation of its ***tangent line*** at the point $\mathbf{c}(t_0)$ is

$$\boldsymbol{l}(t) = \mathbf{c}(t_0) + (t - t_0)\mathbf{c}'(t_0).$$

If C is the curve traced out by $\mathbf{c}$, then the line traced out by $\boldsymbol{l}$ is the tangent line to the curve C at $\mathbf{c}(t_0)$.

Notice that we have written the equation in such a way that $\boldsymbol{l}$ goes through the point $\mathbf{c}(t_0)$ at $t = t_0$ (rather than $t = 0$). See Figure 2.4.11.

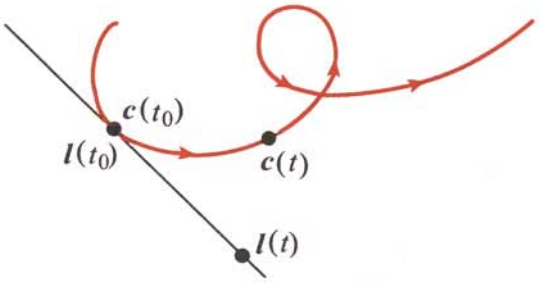


Figure 2.4.11 The tangent line to a path.

EXAMPLE 8 A path in $\mathbb{R}^3$ goes through the point $(3, 6, 5)$ at $t = 0$ with tangent vector $\mathbf{i} - \mathbf{j}$. Find the equation of the tangent line.

SOLUTION The equation of the tangent line is

$$\boldsymbol{l}(t) = (3, 6, 5) + t(\mathbf{i} - \mathbf{j}) = (3, 6, 5) + t(1, -1, 0) = (3 + t, 6 - t, 5).$$

In (x, y, z) coordinates, the tangent line is $x = 3 + t$, $y = 6 - t$, $z = 5$. ▲

Physically, we can interpret motion along the tangent line as the path that a particle on a curve would follow if it were set free at a certain moment.

EXAMPLE 9 Suppose that a particle follows the path $\mathbf{c}(t) = (e^t, e^{-t}, \cos t)$ until it flies off on a tangent at $t = 1$. Where is it at $t = 3$?

SOLUTION The velocity vector is $(e^t, -e^{-t}, -\sin t)$, which at $t = 1$ is the vector $(e, -1/e, -\sin 1)$. The particle is at $(e, 1/e, \cos 1)$ at $t = 1$. The equation of the tangent line is $\boldsymbol{l}(t) = (e, 1/e, \cos 1) + (t - 1)(e, -1/e, -\sin 1)$. At $t = 3$, the position on this line is

$$\boldsymbol{l}(3) = \left(e, \frac{1}{e}, \cos 1\right) + 2\left(e, -\frac{1}{e}, -\sin 1\right) = \left(3e, -\frac{1}{e}, \cos 1 - 2\sin 1\right)$$
$$\cong (8.155, -0.368, -1.143). \quad \blacktriangle$$

EXERCISES

Sketch the curves that are the images of the paths in Exercises 1 to 4.

1. $x = \sin t$, $y = 4\cos t$, where $0 \leq t \leq 2\pi$

2. $x = 2\sin t$, $y = 4\cos t$, where $0 \leq t \leq 2\pi$

3. $\mathbf{c}(t) = (2t - 1, t + 2, t)$

4. $\mathbf{c}(t) = (-t, 2t, 1/t)$, where $1 \leq t \leq 3$

In Exercises 5 to 8, determine the velocity vector of the given path.

5. $\mathbf{c}(t) = 6t\mathbf{i} + 3t^2\mathbf{j} + t^3\mathbf{k}$

6. $\mathbf{c}(t) = (\sin 3t)\mathbf{i} + (\cos 3t)\mathbf{j} + 2t^{3/2}\mathbf{k}$

7. $\mathbf{r}(t) = (\cos^2 t, 3t - t^3, t)$

8. $\mathbf{r}(t) = (4e^t, 6t^4, \cos t)$

In Exercises 9 to 12, compute the tangent vector to the given path.

9. $\mathbf{c}(t) = (e^t, \cos t)$

10. $\mathbf{c}(t) = (3t^2, t^3)$

11. $\mathbf{c}(t) = (t\sin t, 4t)$

12. $\mathbf{c}(t) = (t^2, e^2)$

13. When is the velocity vector of a point on the rim of a rolling wheel *horizontal*? What is the speed at this point?

14. If the position of a particle in space is $(6t, 3t^2, t^3)$ at time t, what is its velocity vector at $t = 0$?

In Exercises 15 and 16, determine the equation of the tangent line to the given path at the specified value of t.

15. $(\sin 3t, \cos 3t, 2t^{5/2}); t = 1$

16. $(\cos^2 t, 3t - t^3, t); t = 0$

In Exercises 17 to 20, suppose that a particle following the given path $\mathbf{c}(t)$ *flies off on a tangent at* $t = t_0$. *Compute the position of the particle at the given time* t_1.

17. $\mathbf{c}(t) = (t^2, t^3 - 4t, 0)$, where $t_0 = 2, t_1 = 3$

18. $\mathbf{c}(t) = (e^t, e^{-t}, \cos t)$, where $t_0 = 1, t_1 = 2$

19. $\mathbf{c}(t) = (4e^t, 6t^4, \cos t)$, where $t_0 = 0, t_1 = 1$

20. $\mathbf{c}(t) = (\sin e^t, t, 4 - t^3)$, where $t_0 = 1, t_1 = 2$

2.5 Properties of the Derivative

In elementary calculus, we learn how to differentiate sums, products, quotients, and composite functions. We now generalize these ideas to functions of several variables, paying particular attention to the differentiation of composite functions. The rule for differentiating composites, called the *chain rule*, takes on a more profound form for functions of several variables than for those of one variable.

If f is a real-valued function of one variable, written as $z = f(y)$, and y is a function of x, written $y = g(x)$, then z becomes a function of x through substitution, namely, $z = f(g(x))$, and we have the familiar chain rule:

$$\frac{dz}{dx} = \frac{dz}{dy}\frac{dy}{dx} = f'(g(x))g'(x).$$

If f is a real-valued function of three variables u, v, and w, written in the form $z = f(u, v, w)$, and the variables u, v, w are each functions of x, $u = g(x)$, $v = h(x)$, and $w = k(x)$, then by substituting $g(x)$, $h(x)$, and $k(x)$ for u, v, and w, we obtain z as a function of x: $z = f(g(x), h(x), k(x))$. The chain rule in this case reads:

$$\frac{dz}{dx} = \frac{\partial z}{\partial u}\frac{du}{dx} + \frac{\partial z}{\partial v}\frac{dv}{dx} + \frac{\partial z}{\partial w}\frac{dw}{dx}.$$

One of the goals of this section is to explain such formulas in detail.

Sums, Products, Quotients

These rules work just as they do in one-variable calculus.

THEOREM 10: Sums, Products, Quotients

(i) **Constant Multiple Rule.** Let $f: U \subset \mathbb{R}^n \to \mathbb{R}^m$ be differentiable at $\mathbf{x}_0$ and let c be a real number. Then $h(\mathbf{x}) = cf(\mathbf{x})$ is differentiable at $\mathbf{x}_0$ and

$$\mathbf{D}h(\mathbf{x}_0) = c\mathbf{D}f(\mathbf{x}_0) \qquad \text{(equality of matrices).}$$

(ii) **Sum Rule.** Let $f: U \subset \mathbb{R}^n \to \mathbb{R}^m$ and $g: U \subset \mathbb{R}^n \to \mathbb{R}^m$ be differentiable at $\mathbf{x}_0$. Then $h(\mathbf{x}) = f(\mathbf{x}) + g(\mathbf{x})$ is differentiable at $\mathbf{x}_0$ and

$$\mathbf{D}h(\mathbf{x}_0) = \mathbf{D}f(\mathbf{x}_0) + \mathbf{D}g(\mathbf{x}_0) \qquad \text{(sum of matrices).}$$

(iii) **Product Rule.** Let $f: U \subset \mathbb{R}^n \to \mathbb{R}$ and $g: U \subset \mathbb{R}^n \to \mathbb{R}$ be differentiable at $\mathbf{x}_0$ and let $h(\mathbf{x}) = g(\mathbf{x})f(\mathbf{x})$. Then $h: U \subset \mathbb{R}^n \to \mathbb{R}$ is differentiable at $\mathbf{x}_0$ and

$$\mathbf{D}h(\mathbf{x}_0) = g(\mathbf{x}_0)\mathbf{D}f(\mathbf{x}_0) + f(\mathbf{x}_0)\mathbf{D}g(\mathbf{x}_0).$$

(Note that each side of this equation is a $1 \times n$ matrix; a more general product rule is presented in Exercise 29 at the end of this section.)

(iv) **Quotient Rule.** With the same hypotheses as in rule (iii), let $h(\mathbf{x}) = f(\mathbf{x})/g(\mathbf{x})$ and suppose g is never zero on U. Then h is differentiable at $\mathbf{x}_0$ and

$$\mathbf{D}h(\mathbf{x}_0) = \frac{g(\mathbf{x}_0)\mathbf{D}f(\mathbf{x}_0) - f(\mathbf{x}_0)\mathbf{D}g(\mathbf{x}_0)}{[g(\mathbf{x}_0)]^2}.$$

PROOF The proofs of rules (i) through (iv) proceed almost exactly as in the one-variable case with a slight difference in notation. We shall prove rules (i) and (ii), leaving the proofs of rules (iii) and (iv) as Exercise 25.

(i) To show that $\mathbf{D}h(\mathbf{x}_0) = c\mathbf{D}f(\mathbf{x}_0)$, we must show that

$$\lim_{\mathbf{x}\to\mathbf{x}_0} \frac{\|h(\mathbf{x}) - h(\mathbf{x}_0) - c\mathbf{D}f(\mathbf{x}_0)(\mathbf{x} - \mathbf{x}_0)\|}{\|\mathbf{x} - \mathbf{x}_0\|} = 0,$$

that is, that

$$\lim_{\mathbf{x}\to\mathbf{x}_0} \frac{\|cf(\mathbf{x}) - cf(\mathbf{x}_0) - c\mathbf{D}f(\mathbf{x}_0)(\mathbf{x} - \mathbf{x}_0)\|}{\|\mathbf{x} - \mathbf{x}_0\|} = 0,$$

[see equation (4) of Section 2.3]. This is certainly true, since f is differentiable and the constant c can be factored out [see Theorem 3(i), Section 2.2].

(ii) By the triangle inequality, we may write

$$\frac{\|h(\mathbf{x}) - h(\mathbf{x}_0) - [\mathbf{D}f(\mathbf{x}_0) + \mathbf{D}g(\mathbf{x}_0)](\mathbf{x} - \mathbf{x}_0)\|}{\|\mathbf{x} - \mathbf{x}_0\|}$$

$$= \frac{\|f(\mathbf{x}) - f(\mathbf{x}_0) - [\mathbf{D}f(\mathbf{x}_0)](\mathbf{x} - \mathbf{x}_0) + g(\mathbf{x}) - g(\mathbf{x}_0) - [\mathbf{D}g(\mathbf{x}_0)](\mathbf{x} - \mathbf{x}_0)\|}{\|\mathbf{x} - \mathbf{x}_0\|}$$

$$\leq \frac{\|f(\mathbf{x}) - f(\mathbf{x}_0) - [\mathbf{D}f(\mathbf{x}_0)](\mathbf{x} - \mathbf{x}_0)\|}{\|\mathbf{x} - \mathbf{x}_0\|} + \frac{\|g(\mathbf{x}) - g(\mathbf{x}_0) - [\mathbf{D}g(\mathbf{x}_0)](\mathbf{x} - \mathbf{x}_0)\|}{\|\mathbf{x} - \mathbf{x}_0\|},$$

and each term approaches 0 as $\mathbf{x} \to \mathbf{x}_0$. Hence, rule (ii) holds. ■

EXAMPLE 1 Verify the formula for $\mathbf{D}h$ in rule (iv) of Theorem 10 with

$$f(x, y, z) = x^2 + y^2 + z^2 \text{ and } g(x, y, z) = x^2 + 1.$$

SOLUTION Here

$$h(x, y, z) = \frac{x^2 + y^2 + z^2}{x^2 + 1},$$

so that by direct differentiation

$$\mathbf{D}h(x, y, z) = \left[\frac{\partial h}{\partial x}, \frac{\partial h}{\partial y}, \frac{\partial h}{\partial z}\right] = \left[\frac{(x^2 + 1)2x - (x^2 + y^2 + z^2)2x}{(x^2 + 1)^2}, \frac{2y}{x^2 + 1}, \frac{2z}{x^2 + 1}\right]$$

$$= \left[\frac{2x(1 - y^2 - z^2)}{(x^2 + 1)^2}, \frac{2y}{x^2 + 1}, \frac{2z}{x^2 + 1}\right].$$

By rule (iv), we get

$$\mathbf{D}h = \frac{g\mathbf{D}f - f\mathbf{D}g}{g^2} = \frac{(x^2 + 1)[2x, 2y, 2z] - (x^2 + y^2 + z^2)[2x, 0, 0]}{(x^2 + 1)^2},$$

which is the same as what we obtained directly. ▲

Chain Rule

As we mentioned earlier, it is in the differentiation of composite functions that we meet apparently substantial alterations of the formula from one-variable calculus. However, if we use the **D** notation, that is, matrix notation for derivatives, the chain rule for functions of several variables looks similar to the one-variable rule.

THEOREM 11: Chain Rule Let $U \subset \mathbb{R}^n$ and $V \subset \mathbb{R}^m$ be open sets. Let $g\colon U \subset \mathbb{R}^n \to \mathbb{R}^m$ and $f\colon V \subset \mathbb{R}^m \to \mathbb{R}^p$ be given functions such that g maps U into V, so that $f \circ g$ is defined. Suppose g is differentiable at $\mathbf{x}_0$ and f is differentiable at $\mathbf{y}_0 = g(\mathbf{x}_0)$. Then $f \circ g$ is differentiable at $\mathbf{x}_0$ and

$$\mathbf{D}(f \circ g)(\mathbf{x}_0) = \mathbf{D}f(\mathbf{y}_0)\mathbf{D}g(\mathbf{x}_0). \tag{1}$$

The right-hand side is the matrix product of $\mathbf{D}f(\mathbf{y}_0)$ with $\mathbf{D}g(\mathbf{x}_0)$.

We shall now give a proof of the chain rule *under the additional assumption that the partial derivatives of f are continuous*, building up to the general case by developing two special cases that are themselves important. (The complete proof of Theorem 11 without the additional assumption of continuity is given in the Internet supplement for Chapter 2.)

First Special Case of the Chain Rule

Suppose $\mathbf{c}\colon \mathbb{R} \to \mathbb{R}^3$ is a differentiable path and $f\colon \mathbb{R}^3 \to \mathbb{R}$. Let $h(t) = f(\mathbf{c}(t)) = f(x(t), y(t), z(t))$, where $\mathbf{c}(t) = (x(t), y(t), z(t))$. Then

$$\frac{dh}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt} + \frac{\partial f}{\partial z}\frac{dz}{dt}. \tag{2}$$

That is,

$$\frac{dh}{dt} = \nabla f(\mathbf{c}(t)) \cdot \mathbf{c}'(t),$$

where $\mathbf{c}'(t) = (x'(t), y'(t), z'(t))$.

This is the special case of Theorem 11 in which we take $\mathbf{c} = g$ and f to be real-valued, and $m = 3$. Notice that

$$\nabla f(\mathbf{c}(t)) \cdot \mathbf{c}'(t) = \mathbf{D}f(\mathbf{c}(t))\mathbf{D}\mathbf{c}(t),$$

where the product on the left-hand side is the dot product of vectors, while the product on the right-hand side is matrix multiplication, and where we regard $\mathbf{D}f(\mathbf{c}(t))$ as a *row* matrix and $\mathbf{D}\mathbf{c}(t)$ as *a column* matrix. The vectors $\nabla f(\mathbf{c}(t))$ and $\mathbf{c}'(t)$ have the same components as their matrix equivalents; the notational change indicates the switch from matrices to vectors.

PROOF OF EQUATION (2). By definition,

$$\frac{dh}{dt}(t_0) = \lim_{t \to t_0} \frac{h(t) - h(t_0)}{t - t_0}.$$

Adding and subtracting two terms, we write

$$\begin{aligned}\frac{h(t)-h(t_0)}{t-t_0} &= \frac{f(x(t),y(t),z(t))-f(x(t_0),y(t_0),z(t_0))}{t-t_0}\\ &= \frac{f(x(t),y(t),z(t))-f(x(t_0),y(t),z(t))}{t-t_0}\\ &\quad+\frac{f(x(t_0),y(t),z(t))-f(x(t_0),y(t_0),z(t))}{t-t_0}\\ &\quad+\frac{f(x(t_0),y(t_0),z(t))-f(x(t_0),y(t_0),z(t_0))}{t-t_0}.\end{aligned}$$

Now we invoke the *mean-value theorem* from one-variable calculus, which states: *If* $g\colon [a,b]\to\mathbb{R}$ *is continuous and is differentiable on the open interval* (a,b), *then there is a point* c *in* (a,b) *such that* $g(b)-g(a)=g'(c)(b-a)$. Applying this to f as a function of x, we can assert that for some c between x and x_0,

$$f(x,y,z)-f(x_0,y,z)=\left[\frac{\partial f}{\partial x}(c,y,z)\right](x-x_0).$$

In this way, we find that

$$\begin{aligned}\frac{h(t)-h(t_0)}{t-t_0} &= \left[\frac{\partial f}{\partial x}(c,y(t),z(t))\right]\frac{x(t)-x(t_0)}{t-t_0}+\left[\frac{\partial f}{\partial y}(x(t_0),d,z(t))\right]\frac{y(t)-y(t_0)}{t-t_0}\\ &\quad+\left[\frac{\partial f}{\partial z}(x(t_0),y(t_0),e)\right]\frac{z(t)-z(t_0)}{t-t_0},\end{aligned}$$

where c, d, and e lie between $x(t)$ and $x(t_0)$, between $y(t)$ and $y(t_0)$, and between $z(t)$ and $z(t_0)$, respectively. Taking the limit $t\to t_0$, using the continuity of the partials $\partial f/\partial x$, $\partial f/\partial y$, $\partial f/\partial z$, and the fact that c, d, and e converge to $x(t_0)$, $y(t_0)$, and $z(t_0)$, respectively, we obtain formula (2). ■

Second Special Case of the Chain Rule

Let $f\colon \mathbb{R}^3\to\mathbb{R}$ and let $g\colon \mathbb{R}^3\to\mathbb{R}^3$. Write

$$g(x,y,z)=(u(x,y,z),v(x,y,z),w(x,y,z))$$

and define $h\colon \mathbb{R}^3\to\mathbb{R}$ by setting

$$h(x,y,z)=f(u(x,y,z),v(x,y,z),w(x,y,z)).$$

In this case, the chain rule states that

$$\begin{bmatrix}\dfrac{\partial h}{\partial x} & \dfrac{\partial h}{\partial y} & \dfrac{\partial h}{\partial z}\end{bmatrix}=\begin{bmatrix}\dfrac{\partial f}{\partial u} & \dfrac{\partial f}{\partial v} & \dfrac{\partial f}{\partial w}\end{bmatrix}\begin{bmatrix}\dfrac{\partial u}{\partial x} & \dfrac{\partial u}{\partial y} & \dfrac{\partial u}{\partial z}\\ \dfrac{\partial v}{\partial x} & \dfrac{\partial v}{\partial y} & \dfrac{\partial v}{\partial z}\\ \dfrac{\partial w}{\partial x} & \dfrac{\partial w}{\partial y} & \dfrac{\partial w}{\partial z}\end{bmatrix}. \tag{3}$$

In this special case, we have taken $n = m = 3$ and $p = 1$ for concreteness, and $U = \mathbb{R}^3$ and $V = \mathbb{R}^3$ for simplicity, and have written out the matrix product $[\mathbf{D}f(\mathbf{y}_0)][\mathbf{D}g(\mathbf{x}_0)]$ explicitly (with the arguments $\mathbf{x}_0$ and $\mathbf{y}_0$ suppressed in the matrices).

PROOF OF THE SECOND SPECIAL CASE OF THE CHAIN RULE. By definition, $\partial h/\partial x$ is obtained by differentiating h with respect to x, holding y and z fixed. But then $(u(x, y, z), v(x, y, z), w(x, y, z))$ may be regarded as a vector function of the single variable x. The first special case applies to this situation and, after the variables are renamed, gives

$$\frac{\partial h}{\partial x} = \frac{\partial f}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial f}{\partial v}\frac{\partial v}{\partial x} + \frac{\partial f}{\partial w}\frac{\partial w}{\partial x}. \tag{3'}$$

Similarly,

$$\frac{\partial h}{\partial y} = \frac{\partial f}{\partial u}\frac{\partial u}{\partial y} + \frac{\partial f}{\partial v}\frac{\partial v}{\partial y} + \frac{\partial f}{\partial w}\frac{\partial w}{\partial y} \tag{3''}$$

and

$$\frac{\partial h}{\partial z} = \frac{\partial f}{\partial u}\frac{\partial u}{\partial z} + \frac{\partial f}{\partial v}\frac{\partial v}{\partial z} + \frac{\partial f}{\partial w}\frac{\partial w}{\partial z}. \tag{3'''}$$

These equations are exactly what would be obtained by multiplying out the matrices in equation (3). ■

PROOF OF THEOREM 11. The general case in equation (1) may be proved in two steps. First, equation (2) is generalized to m variables; that is, for $f(x_1, \ldots, x_m)$ and $\mathbf{c}(t) = (x_1(t), \ldots, x_m(t))$, one has

$$\frac{dh}{dt} = \sum_{i=1}^{m} \frac{\partial f}{\partial x_i}\frac{dx_i}{dt},$$

where $h(t) = f(x_1(t), \ldots, x_m(t))$. Second, the result obtained in the first step is used to obtain the formula

$$\frac{\partial h_j}{\partial x_i} = \sum_{k=1}^{m} \frac{\partial f_j}{\partial y_k}\frac{\partial y_k}{\partial x_i},$$

where $f = (f_1, \ldots, f_p)$ is a vector function of arguments $y_1, \ldots, y_m$; $g(x_1, \ldots, x_n) = (y_1(x_1, \ldots, x_n), \ldots, y_m(x_1, \ldots, x_n))$; and $h_j(x_1, \ldots, x_n) = f_j(y_1(x_1, \ldots, x_n), \ldots, y_m(x_1, \ldots, x_n))$. (Using the letter y for both functions and arguments is an abuse of notation, but it can help one remember the formula.) This formula is equivalent to formula (1) after the matrices are multiplied out. ■

The pattern of the chain rule will become clear once the student has worked some additional examples. For instance,

$$\frac{\partial}{\partial x} f(u(x, y), v(x, y), w(x, y), z(x, y)) = \frac{\partial f}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial f}{\partial v}\frac{\partial v}{\partial x} + \frac{\partial f}{\partial w}\frac{\partial w}{\partial x} + \frac{\partial f}{\partial z}\frac{\partial z}{\partial x},$$

with a similar formula for $\partial f/\partial y$.

The chain rule can help us understand the relationship between the geometry of a mapping $f\colon \mathbb{R}^2 \to \mathbb{R}^2$ and the geometry of curves in $\mathbb{R}^2$. (Similar statements may be made about $\mathbb{R}^3$ or, generally, $\mathbb{R}^n$.) If $\mathbf{c}(t)$ is a path in the plane, then as we saw in Section 2.4, $\mathbf{c}'(t)$ represents the tangent (or velocity) vector of the path $\mathbf{c}(t)$, and this tangent (or velocity) vector is thought of as beginning at $\mathbf{c}(t)$. Now let $\mathbf{p}(t) = f(\mathbf{c}(t))$, where $f\colon \mathbb{R}^2 \to \mathbb{R}^2$. The path $\mathbf{p}$ represents the image of the path $\mathbf{c}(t)$ under the mapping f. The tangent vector to $\mathbf{p}$ is given by the chain rule:

$$\mathbf{p}'(t) = \underbrace{\mathbf{D}f(\mathbf{c}(t))}_{\text{matrix}} \underbrace{\mathbf{c}'(t)}_{\text{column vector}}.$$

matrix multiplication

In other words, *the derivative matrix of f maps the tangent (or velocity) vector of a path $\mathbf{c}$ to the tangent (or velocity) vector of the corresponding image path $\mathbf{p}$* (see Figure 2.5.1). Thus, points are mapped by f, while tangent vectors to curves are mapped by the derivative of f, evaluated at the base point of the tangent vector in the domain.

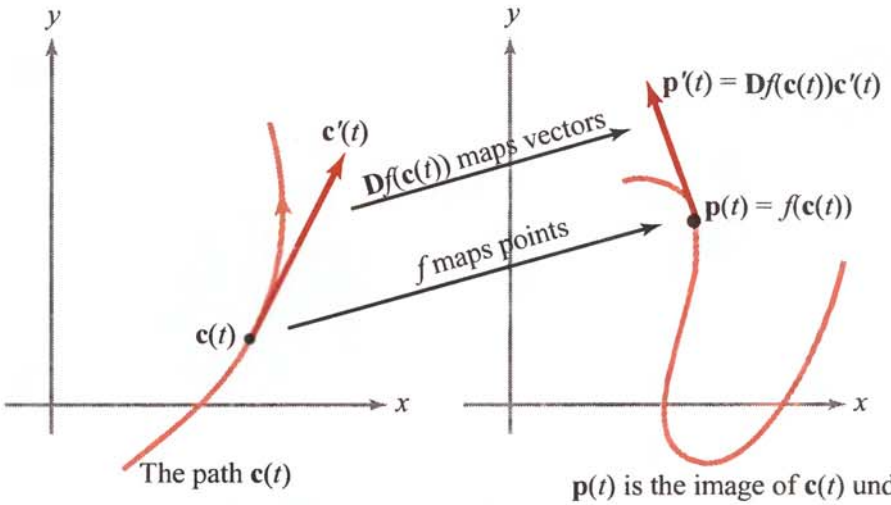


Figure 2.5.1 Tangent vectors are mapped by the derivative matrix.

EXAMPLE 2 Verify the chain rule in the form of formula (3′) for

$$f(u, v, w) = u^2 + v^2 - w,$$

where

$$u(x, y, z) = x^2 y, \qquad v(x, y, z) = y^2, \qquad w(x, y, z) = e^{-xz}.$$

SOLUTION Here

$$h(x, y, z) = f(u(x, y, z), v(x, y, z), w(x, y, z))$$
$$= (x^2y)^2 + y^4 - e^{-xz} = x^4y^2 + y^4 - e^{-xz}.$$

Thus, differentiating directly,

$$\frac{\partial h}{\partial x} = 4x^3y^2 + ze^{-xz}.$$

On the other hand, using the chain rule,

$$\frac{\partial h}{\partial x} = \frac{\partial f}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial f}{\partial v}\frac{\partial v}{\partial x} + \frac{\partial f}{\partial w}\frac{\partial w}{\partial x} = 2u(2xy) + 2v \cdot 0 + (-1)(-ze^{-xz})$$
$$= (2x^2y)(2xy) + ze^{-xz},$$

which is the same as the preceding equation. ▲

EXAMPLE 3 Given $g(x, y) = (x^2 + 1, y^2)$ and $f(u, v) = (u + v, u, v^2)$, compute the derivative of $f \circ g$ at the point $(x, y) = (1, 1)$ using the chain rule.

SOLUTION The matrices of partial derivatives are

$$\mathbf{D}f(u, v) = \begin{bmatrix} \frac{\partial f_1}{\partial u} & \frac{\partial f_1}{\partial v} \\ \frac{\partial f_2}{\partial u} & \frac{\partial f_2}{\partial v} \\ \frac{\partial f_3}{\partial u} & \frac{\partial f_3}{\partial v} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 2v \end{bmatrix} \quad \text{and} \quad \mathbf{D}g(x, y) = \begin{bmatrix} 2x & 0 \\ 0 & 2y \end{bmatrix}.$$

When $(x, y) = (1, 1)$, note that $g(x, y) = (u, v) = (2, 1)$. Hence,

$$\mathbf{D}(f \circ g)(1, 1) = \mathbf{D}f(2, 1)\mathbf{D}g(1, 1) = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 2 \end{bmatrix}\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 2 & 0 \\ 0 & 4 \end{bmatrix}$$

is the required derivative. ▲

EXAMPLE 4 Let $f(x, y)$ be given and make the substitution $x = r\cos\theta$, $y = r\sin\theta$ (polar coordinates). Write a formula for $\partial f/\partial\theta$.

SOLUTION By the chain rule,

$$\frac{\partial f}{\partial \theta} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial \theta}$$

that is,

$$\frac{\partial f}{\partial \theta} = -r \sin\theta \frac{\partial f}{\partial x} + r \cos\theta \frac{\partial f}{\partial y}. \quad \blacktriangle$$

EXAMPLE 5 Let $f(x, y) = (\cos y + x^2, e^{x+y})$ and $g(u, v) = (e^{u^2}, u - \sin v)$. (a) Write a formula for $f \circ g$. (b) Calculate $\mathbf{D}(f \circ g)(0, 0)$ using the chain rule.

SOLUTION (a) We have

$$\begin{aligned}(f \circ g)(u, v) &= f(e^{u^2}, u - \sin v) \\ &= \left(\cos(u - \sin v) + e^{2u^2}, e^{e^{u^2} + u - \sin v}\right).\end{aligned}$$

(b) By the chain rule,

$$\mathbf{D}(f \circ g)(0, 0) = [\mathbf{D}f(g(0, 0))][\mathbf{D}g(0, 0)] = [\mathbf{D}f(1, 0)][\mathbf{D}g(0, 0)].$$

Now

$$\mathbf{D}g(0, 0) = \begin{bmatrix} 2ue^{u^2} & 0 \\ 1 & -\cos v \end{bmatrix}_{(u,v)=(0,0)} = \begin{bmatrix} 0 & 0 \\ 1 & -1 \end{bmatrix}$$

and

$$\mathbf{D}f(1, 0) = \begin{bmatrix} 2x & -\sin y \\ e^{x+y} & e^{x+y} \end{bmatrix}_{(x,y)=(1,0)} = \begin{bmatrix} 2 & 0 \\ e & e \end{bmatrix}.$$

[Remember that $\mathbf{D}f$ is evaluated at $g(0, 0)$, not at $(0, 0)$!] Thus,

$$\mathbf{D}(f \circ g)(0, 0) = \begin{bmatrix} 2 & 0 \\ e & e \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ e & -e \end{bmatrix}. \quad \blacktriangle$$

EXAMPLE 6 Let $f\colon U \subset \mathbb{R}^n \to \mathbb{R}^m$ be differentiable, with $f = (f_1, \ldots, f_m)$, and let $g(\mathbf{x}) = \sin[f(\mathbf{x}) \cdot f(\mathbf{x})]$. Compute $\mathbf{D}g(\mathbf{x})$.

SOLUTION By the chain rule, $\mathbf{D}g(\mathbf{x}) = \cos[f(\mathbf{x}) \cdot f(\mathbf{x})]\mathbf{D}h(\mathbf{x})$, where $h(\mathbf{x}) = [f(\mathbf{x}) \cdot f(\mathbf{x})] = f_1^2(\mathbf{x}) + \cdots + f_m^2(\mathbf{x})$. Then

$$\begin{aligned}\mathbf{D}h(x) &= \begin{bmatrix} \dfrac{\partial h}{\partial x_1} & \cdots & \dfrac{\partial h}{\partial x_n} \end{bmatrix} \\ &= \begin{bmatrix} 2f_1 \dfrac{\partial f_1}{\partial x_1} + \cdots + 2f_m \dfrac{\partial f_m}{\partial x_1} & \cdots & 2f_1 \dfrac{\partial f_1}{\partial x_n} + \cdots + 2f_m \dfrac{\partial f_m}{\partial x_n} \end{bmatrix},\end{aligned}$$

which can be written $2f(\mathbf{x})\mathbf{D}f(\mathbf{x})$, where we regard f as a row matrix,

$$f = [f_1 \quad \cdots \quad f_m] \qquad \text{and} \qquad \mathbf{D}f = \begin{bmatrix} \dfrac{\partial f_1}{\partial x_1} & \cdots & \dfrac{\partial f_1}{\partial x_n} \\ \vdots & & \vdots \\ \dfrac{\partial f_m}{\partial x_1} & \cdots & \dfrac{\partial f_m}{\partial x_n} \end{bmatrix}.$$

Thus, $\mathbf{D}g(\mathbf{x}) = 2[\cos(f(\mathbf{x}) \cdot f(\mathbf{x}))]f(\mathbf{x})\mathbf{D}f(\mathbf{x})$. ▲

EXERCISES

1. If $f\colon U \subset \mathbb{R}^n \to \mathbb{R}$ is differentiable, prove that $\mathbf{x} \mapsto f^2(\mathbf{x}) + 2f(\mathbf{x})$ is differentiable as well, and compute its derivative in terms of $\mathbf{D}f(\mathbf{x})$.

2. Prove that the following functions are differentiable, and find their derivatives at an arbitrary point:

(a) $f\colon \mathbb{R}^2 \to \mathbb{R}, (x, y) \mapsto 2$
(b) $f\colon \mathbb{R}^2 \to \mathbb{R}, (x, y) \mapsto x + y$
(c) $f\colon \mathbb{R}^2 \to \mathbb{R}, (x, y) \mapsto 2 + x + y$
(d) $f\colon \mathbb{R}^2 \to \mathbb{R}, (x, y) \mapsto x^2 + y^2$
(e) $f\colon \mathbb{R}^2 \to \mathbb{R}, (x, y) \mapsto e^{xy}$
(f) $f\colon U \to \mathbb{R}, (x, y) \mapsto \sqrt{1 - x^2 - y^2}$, where $U = \{(x, y) \mid x^2 + y^2 < 1\}$
(g) $f\colon \mathbb{R}^2 \to \mathbb{R}, (x, y) \mapsto x^4 - y^4$

3. Write out the chain rule for each of the following functions and justify your answer in each case using Theorem 11.

(a) $\partial h/\partial x$ where $h(x, y) = f(x, u(x, y))$
(b) dh/dx where $h(x) = f(x, u(x), v(x))$
(c) $\partial h/\partial x$ where $h(x, y, z) = f(u(x, y, z), v(x, y), w(x))$

4. Verify the chain rule for $\partial h/\partial x$, where $h(x, y) = f(u(x, y), v(x, y))$ and

$$f(u, v) = \frac{u^2 + v^2}{u^2 - v^2}, \qquad u(x, y) = e^{-x-y}, \qquad v(x, y) = e^{xy}.$$

5. Verify the first special case of the chain rule for the composition $f \circ \mathbf{c}$ in each of the cases:

(a) $f(x, y) = xy$, $\mathbf{c}(t) = (e^t, \cos t)$
(b) $f(x, y) = e^{xy}$, $\mathbf{c}(t) = (3t^2, t^3)$
(c) $f(x, y) = (x^2 + y^2)\log\sqrt{x^2 + y^2}$, $\mathbf{c}(t) = (e^t, e^{-t})$
(d) $f(x, y) = x\exp(x^2 + y^2)$, $\mathbf{c}(t) = (t, -t)$

6. What is the velocity vector for each path $\mathbf{c}(t)$ in Exercise 5? [The solution to part (b) only is in the Study Guide to this text.]

7. Let $f\colon \mathbb{R}^3 \to \mathbb{R}$ and $g\colon \mathbb{R}^3 \to \mathbb{R}$ be differentiable. Prove that

$$\nabla(fg) = f\nabla g + g\nabla f.$$

8. Let $f\colon \mathbb{R}^3 \to \mathbb{R}$ be differentiable. Making the substitution

$$x = \rho\cos\theta\sin\phi, \qquad y = \rho\sin\theta\sin\phi, \qquad z = \rho\cos\phi$$

(spherical coordinates) into $f(x, y, z)$, compute $\partial f/\partial\rho$, $\partial f/\partial\theta$, and $\partial f/\partial\phi$ in terms of $\partial f/\partial x$, $\partial f/\partial y$, and $\partial f/\partial z$.

9. Let $f(u, v) = (\tan(u-1) - e^v, u^2 - v^2)$ and $g(x, y) = (e^{x-y}, x - y)$. Calculate $f \circ g$ and $\mathbf{D}(f \circ g)(1, 1)$.

10. Let $f(u, v, w) = (e^{u-w}, \cos(v + u) + \sin(u + v + w))$ and $g(x, y) = (e^x, \cos(y - x), e^{-y})$. Calculate $f \circ g$ and $\mathbf{D}(f \circ g)(0, 0)$.

11. Find $(\partial/\partial s)(f \circ T)(1, 0)$, where $f(u, v) = \cos u \sin v$ and $T\colon \mathbb{R}^2 \to \mathbb{R}^2$ is defined by $T(s, t) = (\cos(t^2 s), \log\sqrt{1 + s^2})$.

12. Suppose that the temperature at the point (x, y, z) in space is $T(x, y, z) = x^2 + y^2 + z^2$. Let a particle follow the right-circular helix $\sigma(t) = (\cos t, \sin t, t)$ and let $T(t)$ be its temperature at time t.

(a) What is $T'(t)$?
(b) Find an approximate value for the temperature at $t = (\pi/2) + 0.01$.

13. Suppose that a duck is swimming in the circle $x = \cos t$, $y = \sin t$ and that the water temperature is given by the formula $T = x^2 e^y - xy^3$. Find dT/dt, the rate of change in temperature the duck might feel: (a) by the chain rule; (b) by expressing T in terms of t and differentiating.

14. Let $f\colon \mathbb{R}^n \to \mathbb{R}^m$ be a linear mapping so that (by Exercise 20, Section 2.3) $\mathbf{D}f(\mathbf{x})$ is the matrix of f. Check the validity of the chain rule directly for linear mappings.

15. Let $f\colon \mathbb{R}^2 \to \mathbb{R}^2; (x, y) \mapsto (e^{x+y}, e^{x-y})$. Let $\mathbf{c}(t)$ be a path with $\mathbf{c}(0) = (0, 0)$ and $\mathbf{c}'(0) = (1, 1)$. What is the tangent vector to the image of $\mathbf{c}(t)$ under f at $t = 0$?

16. Let $f(x, y) = 1/\sqrt{x^2 + y^2}$. Compute $\nabla f(x, y)$.

17. (a) Let $y(x)$ be defined implicitly by $G(x, y(x)) = 0$, where G is a given function of two variables. Prove that if $y(x)$ and G are differentiable, then

$$\frac{dy}{dx} = -\frac{\partial G/\partial x}{\partial G/\partial y} \qquad \text{if} \qquad \frac{\partial G}{\partial y} \neq 0.$$

(b) Obtain a formula analogous to that in part (a) if y_1, y_2 are defined implicitly by

$$G_1(x, y_1(x), y_2(x)) = 0,$$

$$G_2(x, y_1(x), y_2(x)) = 0.$$

(c) Let y be defined implicitly by

$$x^2 + y^3 + e^y = 0.$$

Compute dy/dx in terms of x and y.

18. Thermodynamics texts[4] use the relationship

$$\left(\frac{\partial y}{\partial x}\right)\left(\frac{\partial z}{\partial y}\right)\left(\frac{\partial x}{\partial z}\right) = -1.$$

Explain the meaning of this equation and prove that it is true. [HINT: Start with a relationship $F(x, y, z) = 0$ that defines $x = f(y, z)$, $y = g(x, z)$, and $z = h(x, y)$ and differentiate implicitly.]

19. Dieterici's equation of state for a gas is

$$P(V - b)e^{a/RVT} = RT,$$

where a, b, and R are constants. Regard volume V as a function of temperature T and pressure P and prove that

$$\frac{\partial V}{\partial T} = \left(R + \frac{a}{TV}\right) \Big/ \left(\frac{RT}{V - b} - \frac{a}{V^2}\right).$$

20. This exercise gives another example of the fact that the chain rule is not applicable if f is not differentiable. Consider the function

$$f(x, y) = \begin{cases} \dfrac{xy^2}{x^2 + y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0). \end{cases}$$

Show that

(a) $\partial f/\partial x$ and $\partial f/\partial y$ exist at $(0, 0)$.

(b) If $\mathbf{g}(t) = (at, bt)$ for constants a and b, then $f \circ \mathbf{g}$ is differentiable and $(f \circ \mathbf{g})'(0) = ab^2/(a^2 + b^2)$, but $\nabla f(0, 0) \cdot \mathbf{g}'(0) = 0$.

21. Prove that if $f\colon U \subset \mathbb{R}^n \to \mathbb{R}$ is differentiable at $\mathbf{x}_0 \in U$, there is a neighborhood V of $\mathbf{0} \in \mathbb{R}^n$ and a function $R_1\colon V \to \mathbb{R}$ such that for all $\mathbf{h} \in V$, we have $\mathbf{x}_0 + \mathbf{h} \in U$,

$$f(\mathbf{x}_0 + \mathbf{h}) = f(\mathbf{x}_0) + [\mathbf{D}f(\mathbf{x}_0)]\mathbf{h} + R_1(\mathbf{h})$$

---

[4]See S. M. Binder, "Mathematical Methods in Elementary Thermodynamics," *J. Chem. Educ.* 43 (1966): 85–92. A proper understanding of partial differentiation can be of significant use in applications; for example, see M. Feinberg, "Constitutive Equation for Ideal Gas Mixtures and Ideal Solutions as Consequences of Simple Postulates," *Chem. Eng. Sci.* 32 (1977): 75–78.

and

$$\frac{R_1(\mathbf{h})}{\|\mathbf{h}\|} \to 0 \quad \text{as} \quad \mathbf{h} \to \mathbf{0}.$$

22. Suppose $\mathbf{x}_0 \in \mathbb{R}^n$ and $0 \leq r_1 < r_2$. Show that there is a C^1 function $f\colon \mathbb{R}^n \to \mathbb{R}$ such that $f(\mathbf{x}) = 0$ for $\|\mathbf{x} - \mathbf{x}_0\| \geq r_2$; $0 < f(\mathbf{x}) < 1$ for $r_1 < \|\mathbf{x} - \mathbf{x}_0\| < r_2$; and $f(\mathbf{x}) = 1$ for $\|\mathbf{x} - \mathbf{x}_0\| \leq r_1$. [HINT: Apply a cubic polynomial with $g(r_1^2) = 1$ and $g(r_2^2) = g'(r_2^2) = g'(r_1^2) = 0$ to $\|\mathbf{x} - \mathbf{x}_0\|^2$ when $r_1 < \|\mathbf{x} - \mathbf{x}_0\| < r_2$.]

23. Find a C^1 mapping $f\colon \mathbb{R}^3 \to \mathbb{R}^3$ that takes the vector $\mathbf{i} + \mathbf{j} + \mathbf{k}$ emanating from the origin to $\mathbf{i} - \mathbf{j}$ emanating from $(1, 1, 0)$ and takes $\mathbf{k}$ emanating from $(1, 1, 0)$ to $\mathbf{k} - \mathbf{i}$ emanating from the origin.

24. What is wrong with the following argument? Suppose $w = f(x, y, z)$ and $z = g(x, y)$. By the chain rule,

$$\frac{\partial w}{\partial x} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial x} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial x} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial x} = \frac{\partial w}{\partial x} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial x}.$$

Hence, $0 = (\partial w/\partial z)(\partial z/\partial x)$, and so $\partial w/\partial z = 0$ or $\partial z/\partial x = 0$, which is, in general, absurd.

25. Prove rules (iii) and (iv) of Theorem 10. (HINT: Use the same addition and subtraction tricks as in the one-variable case and Theorem 8.)

26. Show that $h\colon \mathbb{R}^n \to \mathbb{R}^m$ is differentiable if and only if each of the m components $h_i\colon \mathbb{R}^n \to \mathbb{R}$ is differentiable. (HINT: Use the coordinate projection function and the chain rule for one implication and consider

$$\left[\frac{\|h(\mathbf{x}) - h(\mathbf{x}_0) - \mathbf{D}h(\mathbf{x}_0)(\mathbf{x} - \mathbf{x}_0)\|}{\|\mathbf{x} - \mathbf{x}_0\|}\right]^2 = \frac{\sum_{i=1}^m [h_i(\mathbf{x}) - h_i(\mathbf{x}_0)\mathbf{D}h_i(\mathbf{x}_0)(\mathbf{x} - \mathbf{x}_0)]^2}{\|\mathbf{x} - \mathbf{x}_0\|^2}$$

to obtain the other.)

27. Use the chain rule and differentiation under the integral sign, namely,

$$\frac{d}{dx}\int_a^b f(x, y)\,dy = \int_a^b \frac{\partial f}{\partial x}(x, y)\,dy,$$

to show that

$$\frac{d}{dx}\int_0^x f(x, y)\,dy = f(x, x) + \int_0^x \frac{\partial f}{\partial x}(x, y)\,dy.$$

28. For what integers $p > 0$ is

$$f(x) = \begin{cases} x^p \sin(1/x) & x \neq 0 \\ 0 & x = 0 \end{cases}$$

differentiable? For what p is the derivative continuous?

29. Suppose $f\colon \mathbb{R}^n \to \mathbb{R}$ and $g\colon \mathbb{R}^n \to \mathbb{R}^m$ are differentiable. Show that the product function $h(\mathbf{x}) = f(\mathbf{x})g(\mathbf{x})$ from $\mathbb{R}^n$ to $\mathbb{R}^m$ is differentiable and that if $\mathbf{x}_0$ and $\mathbf{y}$ are in $\mathbb{R}^n$, then $[\mathbf{D}h(\mathbf{x}_0)]\mathbf{y} = f(\mathbf{x}_0)\{[\mathbf{D}g(\mathbf{x}_0)]\mathbf{y}\} + \{[\mathbf{D}f(\mathbf{x}_0)]\mathbf{y}\}g(\mathbf{x}_0)$.

2.6 Gradients and Directional Derivatives

In Section 2.1 we studied the graphs of real-valued functions. Now we take up this study again, using the methods of calculus. Specifically, gradients will be used to obtain a formula for the plane tangent to a level surface.

Gradients in $\mathbb{R}^3$

Let us recall the definition.

DEFINITION: The Gradient If $f\colon U \subset \mathbb{R}^3 \to \mathbb{R}$ is differentiable, the ***gradient*** of f at (x, y, z) is the vector in space given by

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right).$$

This vector is also denoted $\nabla f(x, y, z)$. Thus, ∇f is just the matrix of the derivative $\mathbf{D}f$, written as a vector.

EXAMPLE 1 Let $f(x, y, z) = \sqrt{x^2 + y^2 + z^2} = r$, the distance from $\mathbf{0}$ to (x, y, z). Then

$$\begin{aligned}\nabla f(x, y, z) &= \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right) \\ &= \left(\frac{x}{\sqrt{x^2 + y^2 + z^2}}, \frac{y}{\sqrt{x^2 + y^2 + z^2}}, \frac{z}{\sqrt{x^2 + y^2 + z^2}}\right) = \frac{\mathbf{r}}{r},\end{aligned}$$

where $\mathbf{r}$ is the point (x, y, z). Thus, ∇f is the unit vector in the direction of (x, y, z). ▲

EXAMPLE 2 If $f(x, y, z) = xy + z$, then

$$\nabla f(x, y, z) = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right) = (y, x, 1). \quad ▲$$

Suppose $f\colon \mathbb{R}^3 \to \mathbb{R}$ is a real-valued function. Let $\mathbf{v}$ and $\mathbf{x} \in \mathbb{R}^3$ be fixed vectors and consider the function from $\mathbb{R}$ to $\mathbb{R}$ defined by $t \mapsto f(\mathbf{x} + t\mathbf{v})$. The set of points of the form $\mathbf{x} + t\mathbf{v}$, $t \in \mathbb{R}$, is the line L through the point $\mathbf{x}$ parallel to the vector $\mathbf{v}$ (see Figure 2.6.1).

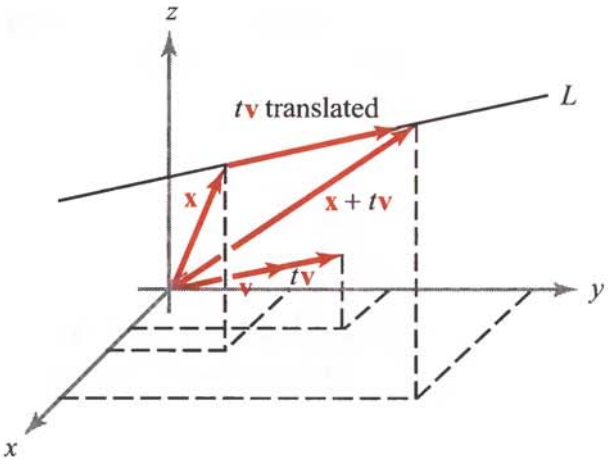


Figure 2.6.1 The equation of L is $\mathbf{l}(t) = \mathbf{x} + t\mathbf{v}$.

Directional Derivatives

The function $t \mapsto f(\mathbf{x} + t\mathbf{v})$ represents the function f restricted to the line L. For example, if a bird flies along this line with velocity $\mathbf{v}$ so that $\mathbf{x} + t\mathbf{v}$ is its position at time t, and if f represents the temperature as a function of position, then $f(\mathbf{x} + t\mathbf{v})$ is the temperature at time t. We may ask: How fast are the values of f changing along the line L at the point $\mathbf{x}$? Because the rate of change of a function is given by a derivative, we could say that the answer to this question is the value of the derivative of this function of t at $t = 0$ (when $t = 0$, $\mathbf{x} + t\mathbf{v}$ reduces to $\mathbf{x}$). This would be the derivative of f at the point $\mathbf{x}$ in the direction of L, that is, of $\mathbf{v}$. We can formalize this concept as follows.

DEFINITION: Directional Derivatives If $f\colon \mathbb{R}^3 \to \mathbb{R}$, the ***directional derivative*** of f at $\mathbf{x}$ along the vector $\mathbf{v}$ is given by

$$\frac{d}{dt} f(\mathbf{x} + t\mathbf{v})\bigg|_{t=0}$$

if this exists.

In the definition of a directional derivative, we normally choose $\mathbf{v}$ to be a *unit* vector. In this case we are moving in the direction $\mathbf{v}$ with unit speed and we refer to $\frac{d}{dt} f(\mathbf{x} + t\mathbf{v})|_{t=0}$ as the ***directional derivative of f in the direction $\mathbf{v}$***.

We now elaborate on why a *unit vector* is chosen in the definition of the directional derivative. Suppose that f measures the temperature in degrees and that we are interested in how fast the temperature changes as we move in a particular direction. If we are measuring distance in meters, then the rate of change of temperature will be measured in degrees per meter. Suppose, for simplicity, that the temperature is changing at a constant rate—say, two degrees per meter—as we move in a given

direction $\mathbf{v}$ starting at $\mathbf{x}$. Thus, when we go one meter ahead, the temperature changes by two degrees. That is,

$$f(\mathbf{x}+\mathbf{v}) - f(\mathbf{x}) = 2$$

Such a relation is going to hold only when $\mathbf{v}$ is a unit vector, reflecting the fact that we are going ahead by *one* meter. More generally, the definition of the directional derivative is going to truly measure only the rate of change of f with respect to distance along a line in a given direction if $\mathbf{v}$ is a unit vector.

From the definition, we can see that the directional derivative can also be defined by the formula

$$\lim_{h\to 0} \frac{f(\mathbf{x}+h\mathbf{v}) - f(\mathbf{x})}{h}.$$

THEOREM 12 If $f\colon \mathbb{R}^3 \to \mathbb{R}$ is differentiable, then all directional derivatives exist. The directional derivative at $\mathbf{x}$ in the direction $\mathbf{v}$ is given by

$$\mathbf{D}f(\mathbf{x})\mathbf{v} = \operatorname{grad} f(\mathbf{x})\cdot\mathbf{v} = \nabla f(\mathbf{x})\cdot\mathbf{v} = \left[\frac{\partial f}{\partial x}(\mathbf{x})\right]v_1 + \left[\frac{\partial f}{\partial y}(\mathbf{x})\right]v_2 + \left[\frac{\partial f}{\partial z}(\mathbf{x})\right]v_3,$$

where $\mathbf{v} = (v_1, v_2, v_3)$.

PROOF Let $\mathbf{c}(t) = \mathbf{x} + t\mathbf{v}$, so that $f(\mathbf{x}+t\mathbf{v}) = f(\mathbf{c}(t))$. By the first special case of the chain rule, $(d/dt)f(\mathbf{c}(t)) = \nabla f(\mathbf{c}(t))\cdot\mathbf{c}'(t)$. However, $\mathbf{c}(0) = \mathbf{x}$ and $\mathbf{c}'(0) = \mathbf{v}$, and so

$$\frac{d}{dt} f(\mathbf{x}+t\mathbf{v})\bigg|_{t=0} = \nabla f(\mathbf{x})\cdot\mathbf{v},$$

as we were required to prove. ■

Notice that one does not have to use straight lines when computing the rate of change of f in a specific direction $\mathbf{v}$. Indeed, for a general path $\mathbf{c}(t)$ with $\mathbf{c}(0) = \mathbf{x}$ and $\mathbf{c}'(0) = \mathbf{v}$, we have from the chain rule,

$$\frac{d}{dt} f(\mathbf{c}(t))\bigg|_{t=0} = \nabla f(\mathbf{c}(t))\cdot\mathbf{c}'(t)\bigg|_{t=0} = \nabla f(\mathbf{x})\cdot\mathbf{v}.$$

EXAMPLE 3 Let $f(x, y, z) = x^2 e^{-yz}$. Compute the rate of change of f in the direction of the unit vector

$$\mathbf{v} = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right) \quad \text{at the point} \quad (1, 0, 0).$$

SOLUTION The required rate of change is, using Theorem 12,

$$\nabla f \cdot \mathbf{v} = (2xe^{-yz}, -x^2ze^{-yz}, -x^2ye^{-yz}) \cdot \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right),$$

which, at the point (1, 0, 0), becomes

$$(2, 0, 0) \cdot \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right) = \frac{2}{\sqrt{3}}. \quad \blacktriangle$$

Directions of Fastest Increase

From Theorem 12 we can also obtain the geometrical significance of the gradient:

THEOREM 13 Assume $\nabla f(\mathbf{x}) \neq \mathbf{0}$. Then $\nabla f(\mathbf{x})$ points in the direction along which f is increasing the fastest.

PROOF If $\mathbf{n}$ is a unit vector, the rate of change of f in direction $\mathbf{n}$ is given by $\nabla f(\mathbf{x}) \cdot \mathbf{n} = \|\nabla f(\mathbf{x})\| \cos\theta$, where θ is the angle between $\mathbf{n}$ and $\nabla f(\mathbf{x})$. This is maximum when $\theta = 0$; that is, when $\mathbf{n}$ and ∇f are parallel. [If $\nabla f(\mathbf{x}) = \mathbf{0}$ this rate of change is 0 for any $\mathbf{n}$.] ■

In other words, if one wishes to move in a direction in which f will increase most quickly, one should proceed in the direction $\nabla f(\mathbf{x})$. Analogously, if one wishes to move in a direction in which f *decreases* the fastest, one should proceed in the direction $-\nabla f(\mathbf{x})$.

EXAMPLE 4 In what direction from (0, 1) does $f(x, y) = x^2 - y^2$ increase the fastest?

SOLUTION The gradient is

$$\nabla f = 2x\mathbf{i} - 2y\mathbf{j},$$

and so at (0, 1) this is

$$\nabla f|_{(0,1)} = -2\mathbf{j}.$$

By Theorem 13, f increases fastest in the direction $-\mathbf{j}$. (Can you see why this answer is consistent with Figure 2.1.9?) ▲

Gradients and Tangent Planes to Level Sets

Now we find the relationship between the gradient of a function f and its level surfaces. The gradient points in the direction in which the values of f change most rapidly, whereas a level surface lies in the directions in which they do not change

at all. If f is reasonably well behaved, the gradient and the level surface will be perpendicular.

THEOREM 14: The Gradient is Normal to Level Surfaces Let $f: \mathbb{R}^3 \to \mathbb{R}$ be a C^1 map and let (x_0, y_0, z_0) lie on the level surface S defined by $f(x, y, z) = k$, for k a constant. Then $\nabla f(x_0, y_0, z_0)$ is normal to the level surface in the following sense: If $\mathbf{v}$ is the tangent vector at $t = 0$ of a path $\mathbf{c}(t)$ in S with $\mathbf{c}(0) = (x_0, y_0, z_0)$, then $\nabla f(x_0, y_0, z_0) \cdot \mathbf{v} = 0$ (see Figure 2.6.2).

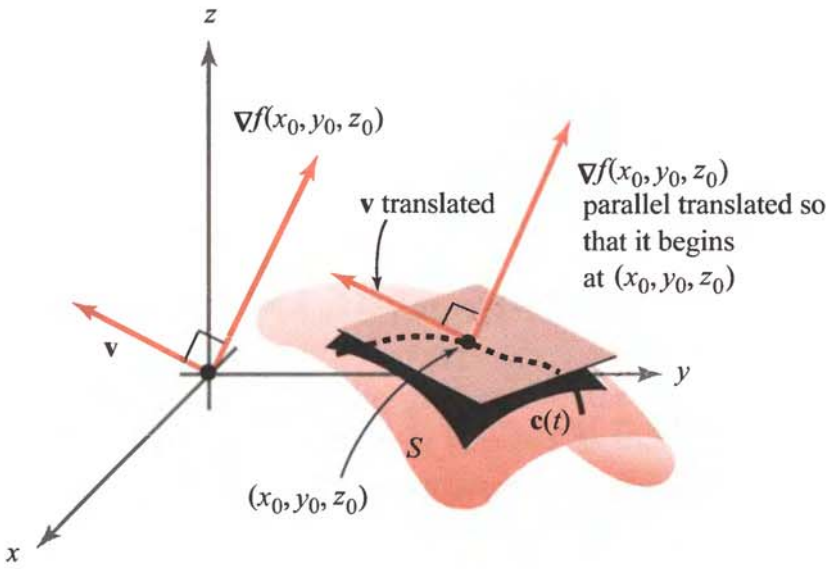


Figure 2.6.2 Geometric significance of the gradient: ∇f is orthogonal to the surface S on which f is constant.

PROOF Let $\mathbf{c}(t)$ lie in S; then $f(\mathbf{c}(t)) = k$. Let $\mathbf{v}$ be as in the hypothesis; then $\mathbf{v} = \mathbf{c}'(0)$. Hence, the fact that $f(\mathbf{c}(t))$ is constant in t, and the chain rule give

$$0 = \frac{d}{dt} f(\mathbf{c}(t))\bigg|_{t=0} = \nabla f(\mathbf{c}(0)) \cdot \mathbf{v}. \quad \blacksquare$$

If we study the conclusion of Theorem 14, we see that it is reasonable to *define* the plane tangent to S as the orthogonal plane to the gradient.

DEFINITION: Tangent Planes to Level Surfaces Let S be the surface consisting of those (x, y, z) such that $f(x, y, z) = k$, for k a constant. The ***tangent plane*** of S at a point (x_0, y_0, z_0) of S is defined by the equation

$$\nabla f(x_0, y_0, z_0) \cdot (x - x_0, y - y_0, z - z_0) = 0 \qquad (1)$$

if $\nabla f(x_0, y_0, z_0) \neq \mathbf{0}$. That is, the tangent plane is the set of points (x, y, z) that satisfy equation (1).

This extends the definition we gave earlier for the tangent plane of the graph of a function (see Exercise 11 at the end of this section).

EXAMPLE 5 Compute the equation of the plane tangent to the surface defined by $3xy + z^2 = 4$ at $(1, 1, 1)$.

SOLUTION Here $f(x, y, z) = 3xy + z^2$ and $\nabla f = (3y, 3x, 2z)$, which at $(1, 1, 1)$ is the vector $(3, 3, 2)$. Thus, the tangent plane is

$$(3, 3, 2) \cdot (x - 1, y - 1, z - 1) = 0;$$

that is,

$$3x + 3y + 2z = 8. \quad \blacktriangle$$

In Theorem 14 and the definition following it, we could just as well have worked in two dimensions as in three. Thus, if we have $f\colon \mathbb{R}^2 \to \mathbb{R}$ and consider a level *curve*

$$C = \{(x, y) \mid f(x, y) = k\},$$

then $\nabla f(x_0, y_0)$ is perpendicular to C for any point (x_0, y_0) on C. Likewise, the tangent line to C at (x_0, y_0) has the equation

$$\nabla f(x_0, y_0) \cdot (x - x_0, y - y_0) = 0 \tag{2}$$

if $\nabla f(x_0, y_0) \neq \mathbf{0}$; that is, the tangent line is the set of points (x, y) that satisfy equation (2) (see Figure 2.6.3).

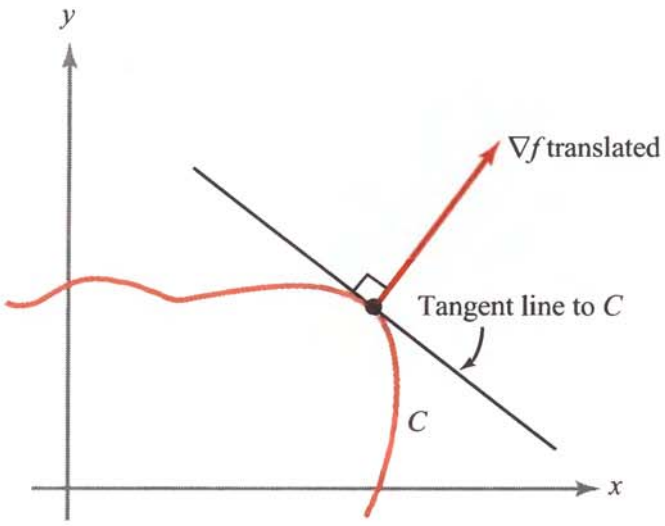


Figure 2.6.3 In the plane, the gradient ∇f is orthogonal to the curve f = constant.

The Gradient Vector Field

We often speak of ∇f as a ***gradient vector field***. The word "field" means that ∇f assigns a vector to each point in the domain of f. In Figure 2.6.4 we describe the gradient ∇f not by drawing its graph, which, if $f\colon \mathbb{R}^3 \to \mathbb{R}$, would be a subset of

$\mathbb{R}^6$, that is, the set of tuples $(\mathbf{x}, \nabla f(\mathbf{x}))$, but by representing $\nabla f(\mathrm{P})$, for each point P, as a vector emanating from the point P rather than from the origin. Like a graph, this pictorial method of depicting ∇f contains the point P and the value $\nabla f(\mathrm{P})$ in the same picture.

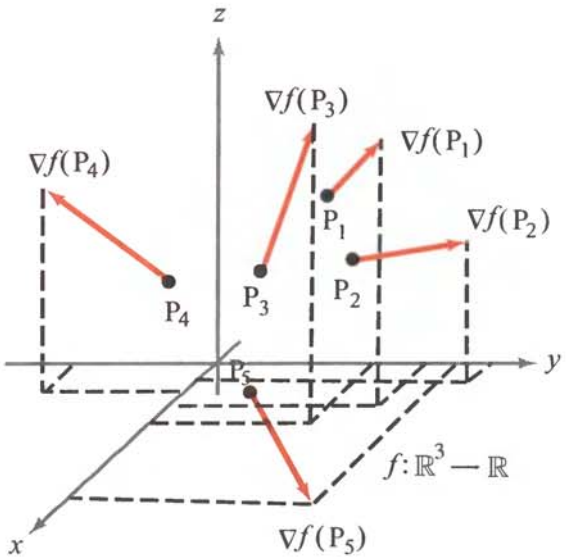


Figure 2.6.4 The gradient ∇f of a function $f\colon \mathbb{R}^3 \to \mathbb{R}$ is a vector field on $\mathbb{R}^3$; at each point P_i, $\nabla f(\mathrm{P}_i)$ is a vector emanating from P_i.

The gradient vector field has important geometric significance. It shows the direction in which f is increasing the fastest and the direction that is orthogonal to the level surfaces (or curves in the plane) of f. That it does both of these at once is quite plausible. To see this, imagine a hill as shown in Figure 2.6.5(a). Let h be the height function, a function of two variables. If we draw level curves of h, these are just level contours of the hill. We could imagine them as level paths on the hill [see Figure 2.6.5(b)]. One thing should be obvious to anyone who has gone for a hike: To get to the top of the hill the fastest, we should walk perpendicular to level contours.[5] This is consistent with Theorems 13 and 14, which state that the direction of fastest increase (the gradient) is orthogonal to the level curves.

EXAMPLE 6 The gravitational force on a unit mass m at (x, y, z) produced by a mass M at the origin in $\mathbb{R}^3$ is, according to Newton's law of gravitation, given by

$$\mathbf{F} = -\frac{GmM}{r^2}\mathbf{n},$$

where G is a constant; $r = \|\mathbf{r}\| = \sqrt{x^2 + y^2 + z^2}$, which is the distance of (x, y, z) from the origin; and $\mathbf{n} = \mathbf{r}/r$, the unit vector in the direction of $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, which is the position vector from the origin to (x, y, z).

Note that $\mathbf{F} = \nabla(GmM/r) = -\nabla V$, that is, $\mathbf{F}$ is the negative of the gradient of the gravitational potential $V = -GmM/r$. This can be verified as in Example 1.

[5] This discussion assumes that one walks at the same speed in all directions. Of course, hikers know that this is not necessarily realistic.

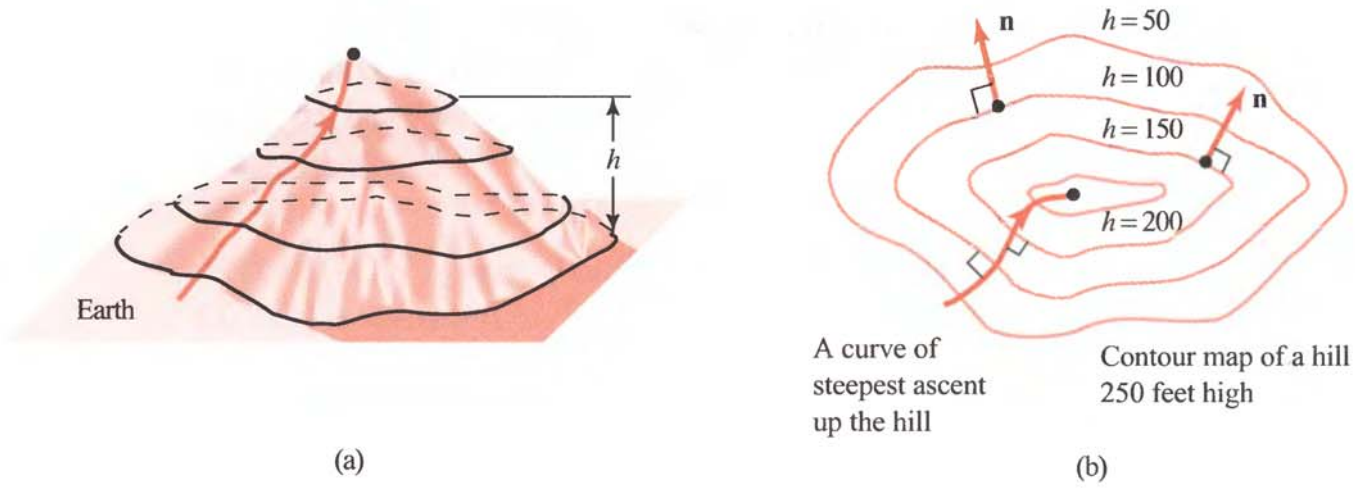


Figure 2.6.5 A physical illustration of the two facts (a) ∇f is the direction of fastest increase of f, and (b) ∇f is orthogonal to the level curves.

Notice that $\mathbf{F}$ is directed inward toward the origin. Also, the level surfaces of V are spheres. The gradient vector field $\mathbf{F}$ is normal to these spheres, which confirms the result of Theorem 14. ▲

EXAMPLE 7 Find a unit vector normal to the surface S given by $z = x^2y^2 + y + 1$ at the point $(0, 0, 1)$.

SOLUTION Let $f(x, y, z) = x^2y^2 + y + 1 - z$, and consider the level surface defined by $f(x, y, z) = 0$. Because this is the set of points (x, y, z) with $z = x^2y^2 + y + 1$, we see that this level set coincides with the surface S. The gradient is given by

$$\nabla f(x, y, z) = \frac{\partial f}{\partial x}\mathbf{i} + \frac{\partial f}{\partial y}\mathbf{j} + \frac{\partial f}{\partial z}\mathbf{k} = 2xy^2\mathbf{i} + (2x^2y + 1)\mathbf{j} - \mathbf{k},$$

and so

$$\nabla f(0, 0, 1) = \mathbf{j} - \mathbf{k}.$$

This vector is perpendicular to S at $(0, 0, 1)$, and so to find a unit normal $\mathbf{n}$ we divide this vector by its length to obtain

$$\mathbf{n} = \frac{\nabla f(0, 0, 1)}{\|\nabla f(0, 0, 1)\|} = \frac{1}{\sqrt{2}}(\mathbf{j} - \mathbf{k}). \quad ▲$$

EXAMPLE 8 Consider two conductors, one charged positively and the other negatively. Between them, an electric potential is set up. This potential is a function $\phi\colon \mathbb{R}^3 \to \mathbb{R}$ (an example of a *scalar* field). The electric field is given by $\mathbf{E} = -\nabla\phi$. From Theorem 14 we know that $\mathbf{E}$ is perpendicular to level surfaces of ϕ. These level surfaces are called ***equipotential surfaces***, because the potential is constant on them (see Figure 2.6.6). ▲

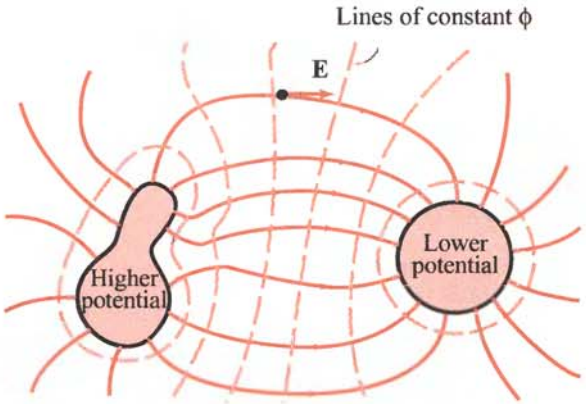


Figure 2.6.6 Equipotential surfaces (the dotted lines) are orthogonal to the electric force field **E**.

EXERCISES

1. Show that the directional derivative of $f(x, y, z) = z^2x + y^3$ at $(1, 1, 2)$ in the direction $(1/\sqrt{5})\mathbf{i} + (2/\sqrt{5})\mathbf{j}$ is $2\sqrt{5}$.

2. Compute the directional derivatives of the following functions at the indicated points in the given directions:

(a) $f(x, y) = x + 2xy - 3y^2$, $(x_0, y_0) = (1, 2)$, $\mathbf{v} = \frac{3}{5}\mathbf{i} + \frac{4}{5}\mathbf{j}$
(b) $f(x, y) = \log \sqrt{x^2 + y^2}$, $(x_0, y_0) = (1, 0)$, $\mathbf{v} = (1/\sqrt{5})(2\mathbf{i} + \mathbf{j})$
(c) $f(x, y) = e^x \cos(\pi y)$, $(x_0, y_0) = (0, -1)$, $\mathbf{v} = -(1/\sqrt{5})\mathbf{i} + (2/\sqrt{5})\mathbf{j}$
(d) $f(x, y) = xy^2 + x^3y$, $(x_0, y_0) = (4, -2)$, $\mathbf{v} = (1/\sqrt{10})\mathbf{i} + (3/\sqrt{10})\mathbf{j}$

3. Compute the directional derivatives of the following functions along unit vectors at the indicated points in directions *parallel* to the given vector:

(a) $f(x, y) = x^y$, $(x_0, y_0) = (e, e)$, $\mathbf{d} = 5\mathbf{i} + 12\mathbf{j}$
(b) $f(x, y, z) = e^x + yz$, $(x_0, y_0, z_0) = (1, 1, 1)$, $\mathbf{d} = (1, -1, 1)$
(c) $f(x, y, z) = xyz$, $(x_0, y_0, z_0) = (1, 0, 1)$, $\mathbf{d} = (1, 0, -1)$

4. Find the planes tangent to the following surfaces at the indicated points:

(a) $x^2 + 2y^2 + 3xz = 10$, at the point $(1, 2, \frac{1}{3})$
(b) $y^2 - x^2 = 3$, at the point $(1, 2, 8)$
(c) $xyz = 1$, at the point $(1, 1, 1)$

5. Find the equation for the plane tangent to each surface $z = f(x, y)$ at the indicated point:

(a) $z = x^3 + y^3 - 6xy$, at the point $(1, 2, -3)$
(b) $z = (\cos x)(\cos y)$, at the point $(0, \pi/2, 0)$
(c) $z = (\cos x)(\sin y)$, at the point $(0, \pi/2, 1)$

6. Compute the gradient ∇f for each of the following functions:

(a) $f(x, y, z) = 1/\sqrt{x^2 + y^2 + z^2}$
(b) $f(x, y, z) = xy + yz + xz$
(c) $f(x, y, z) = \dfrac{1}{x^2 + y^2 + z^2}$

7. For the functions in Exercise 6, what is the direction of fastest increase at (1, 1, 1)? [The solution to part (c) only is in the Study Guide to this text.]

8. Show that a unit normal to the surface $x^3y^3 + y - z + 2 = 0$ at (0, 0, 2) is given by $\mathbf{n} = (1/\sqrt{2})(\mathbf{j} - \mathbf{k})$.

9. Find a unit normal to the surface $\cos(xy) = e^z - 2$ at $(1, \pi, 0)$.

10. Verify Theorems 13 and 14 for $f(x, y, z) = x^2 + y^2 + z^2$.

11. Show that the definition following Theorem 14 yields, as a special case, the formula for the plane tangent to the graph of $f(x, y)$ by regarding the graph as a level surface of $F(x, y, z) = f(x, y) - z$ (see Section 2.3).

12. Let $f(x, y) = -(1 - x^2 - y^2)^{1/2}$ for (x, y) such that $x^2 + y^2 < 1$. Show that the plane tangent to the graph of f at $(x_0, y_0, f(x_0, y_0))$ is orthogonal to the vector with components $(x_0, y_0, f(x_0, y_0))$. Interpret this geometrically.

13. For the following functions $f\colon \mathbb{R}^3 \to \mathbb{R}$ and $\mathbf{g}\colon \mathbb{R} \to \mathbb{R}^3$, find ∇f and $\mathbf{g}'$ and evaluate $(f \circ \mathbf{g})'(1)$.

(a) $f(x, y, z) = xz + yz + xy$, $\mathbf{g}(t) = (e^t, \cos t, \sin t)$
(b) $f(x, y, z) = e^{xyz}$, $\mathbf{g}(t) = (6t, 3t^2, t^3)$
(c) $f(x, y, z) = (x^2 + y^2 + z^2) \log \sqrt{x^2 + y^2 + z^2}$, $\mathbf{g}(t) = (e^t, e^{-t}, t)$

14. Compute the directional derivative of f in the given directions $\mathbf{v}$ at the given points P.

(a) $f(x, y, z) = xy^2 + y^2z^3 + z^3x$, $\mathrm{P} = (4, -2, -1)$, $\mathbf{v} = 1/\sqrt{14}(\mathbf{i} + 3\mathbf{j} + 2\mathbf{k})$
(b) $f(x, y, z) = x^{yz}$, $\mathrm{P} = (e, e, 0)$, $\mathbf{v} = \frac{12}{13}\mathbf{i} + \frac{3}{13}\mathbf{j} + \frac{4}{13}\mathbf{k}$

15. Let $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ and $r = \|\mathbf{r}\|$. Prove that

$$\nabla\left(\frac{1}{r}\right) = -\frac{\mathbf{r}}{r^3}.$$

16. Captain Ralph is in trouble near the sunny side of Mercury. The temperature of the ship's hull when he is at location (x, y, z) will be given by $T(x, y, z) = e^{-x^2-2y^2-3z^2}$, where x, y, and z are measured in meters. He is currently at (1, 1, 1).

(a) In what direction should he proceed in order to decrease the temperature most rapidly?

(b) If the ship travels at e^8 meters per second, how fast will be the temperature decrease if he proceeds in that direction?

(c) Unfortunately, the metal of the hull will crack if cooled at a rate greater than $\sqrt{14}e^2$ degrees per second. Describe the set of possible directions in which he may proceed to bring the temperature down at no more than that rate.

17. A function $f\colon \mathbb{R}^2 \to \mathbb{R}$ is said to be *independent of the second variable* if there is a function $g\colon \mathbb{R} \to \mathbb{R}$ such that $f(x, y) = g(x)$ for all x in $\mathbb{R}$. In this case, calculate ∇f in terms of g'.

18. Let f and g be functions from $\mathbb{R}^3$ to $\mathbb{R}$. Suppose f is differentiable and $\nabla f(\mathbf{x}) = g(\mathbf{x})\mathbf{x}$. Show that spheres centered at the origin are contained in the level sets for f; that is, f is constant on such spheres.

19. A function $f\colon \mathbb{R}^n \to \mathbb{R}$ is called an *even* function if $f(\mathbf{x}) = f(-\mathbf{x})$ for every $\mathbf{x}$ in $\mathbb{R}^n$. If f is differentiable and even, find $\mathbf{D}f$ at the origin.

20. Suppose that a mountain has the shape of an elliptic paraboloid $z = c - ax^2 - by^2$, where a, b, and c are positive constants, x and y are the east-west and north-south map coordinates, and z is the altitude above sea level (x, y, z are all measured in meters). At the point (1, 1), in what direction is the altitude increasing most rapidly? If a marble were released at (1, 1), in what direction would it begin to roll?

21. An engineer wishes to build a railroad up the mountain of Exercise 20. Straight up the mountain is much too steep for the power of the engines. At the point (1, 1), in what directions may the track be laid so that it will be climbing with a 3% grade—that is, an angle whose tangent is 0.03? (There are two possibilities.) Make a sketch of the situation indicating the two possible directions for a 3% grade at (1, 1).

22. In electrostatics, the force $\mathbf{P}$ of attraction between two particles of opposite charge is given by $\mathbf{P} = k(\mathbf{r}/\|\mathbf{r}\|^3)$ (*Coulomb's law*), where k is a constant and $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$. Show that $\mathbf{P}$ is the gradient of $f = -k/\|\mathbf{r}\|$.

23. The electrostatic potential V due to two infinite parallel filaments with linear charge densities λ and $-\lambda$ is $V = (\lambda/2\pi\varepsilon_0)\ln(r_2/r_1)$, where $r_1^2 = (x - x_0)^2 + y^2$ and $r_2^2 = (x + x_0)^2 + y^2$. We think of the filaments as being in the z direction, passing through the xy plane at $(-x_0, 0)$ and $(x_0, 0)$. Find $\nabla V(x, y)$.

24. For each of the following, find the maximum and minimum values attained by the function f along the path $\mathbf{c}(t)$:

(a) $f(x, y) = xy$; $\mathbf{c}(t) = (\cos t, \sin t)$; $0 \le t \le 2\pi$.
(b) $f(x, y) = x^2 + y^2$; $\mathbf{c}(t) = (\cos t, 2\sin t)$; $0 \le t \le 2\pi$.

25. Suppose that a particle is ejected from the surface $x^2 + y^2 - z^2 = -1$ at the point $(1, 1, \sqrt{3})$ along the normal directed toward the xy plane to the surface at time $t = 0$ with a speed of 10 units per second. When and where does it cross the xy plane?

26. Let $f\colon \mathbb{R}^3 \to \mathbb{R}$ and regard $\mathbf{D}f(x, y, z)$ as a linear map of $\mathbb{R}^3$ to $\mathbb{R}$. Show that the kernel (that is, the set of vectors mapped to zero) of $\mathbf{D}f$ is the plane in $\mathbb{R}^3$ orthogonal to ∇f.

REVIEW EXERCISES FOR CHAPTER 2

1. Describe the graphs of:

(a) $f(x, y) = 3x^2 + y^2$ (b) $f(x, y) = xy + 3x$

2. Describe some appropriate level surfaces and sections of the graphs of:

(a) $f(x, y, z) = 2x^2 + y^2 + z^2$

(b) $f(x, y, z) = x^2$
(c) $f(x, y, z) = xyz$

3. Compute the derivative $\mathbf{D}f(\mathbf{x})$ of each of the following functions:

(a) $f(x, y) = (x^2 y, e^{-xy})$
(b) $f(x) = (x, x)$
(c) $f(x, y, z) = e^x + e^y + e^z$
(d) $f(x, y, z) = (x, y, z)$

4. Suppose $f(x, y) = f(y, x)$ for all (x, y). Prove that

$$(\partial f/\partial x)(a, b) = (\partial f/\partial y)(b, a).$$

5. Let $f(x, y) = (1 - x^2 - y^2)^{1/2}$. Show that the plane tangent to the graph of f at $(x_0, y_0, f(x_0, y_0))$ is orthogonal to the vector $(x_0, y_0, f(x_0, y_0))$. Interpret geometrically.

6. Let $F(u, v)$ and $u = h(x, y, z)$, $v = k(x, y, z)$ be given (differentiable) real-valued functions and let $f(x, y, z)$ be defined by $f(x, y, z) = F(h(x, y, z), k(x, y, z))$. Write a formula for the gradient of f in terms of the partial derivatives of F, h, and k.

7. Find an equation for the tangent plane of the graph of f at the point $(x_0, y_0, f(x_0, y_0))$ for:

(a) $f: \mathbb{R}^2 \to \mathbb{R}, (x, y) \mapsto x - y + 2$, $(x_0, y_0) = (1, 1)$
(b) $f: \mathbb{R}^2 \to \mathbb{R}, (x, y) \mapsto x^2 + 4y^2$, $(x_0, y_0) = (2, -1)$
(c) $f: \mathbb{R}^2 \to \mathbb{R}, (x, y) \mapsto xy$, $(x_0, y_0) = (-1, -1)$
(d) $f(x, y) = \log(x + y) + x \cos y + \arctan(x + y)$, $(x_0, y_0) = (1, 0)$
(e) $f(x, y) = \sqrt{x^2 + y^2}$, $(x_0, y_0) = (1, 1)$
(f) $f(x, y) = xy$, $(x_0, y_0) = (2, 1)$

8. Compute an equation for the tangent planes of the following surfaces at the indicated points.

(a) $x^2 + y^2 + z^2 = 3$, $(1, 1, 1)$
(b) $x^3 - 2y^3 + z^3 = 0$, $(1, 1, 1)$
(c) $(\cos x)(\cos y)e^z = 0$, $(\pi/2, 1, 0)$
(d) $e^{xyz} = 1$, $(1, 1, 0)$

9. Draw some level curves for the following functions:

(a) $f(x, y) = 1/xy$
(b) $f(x, y) = x^2 - xy - y^2$

10. Consider a temperature function $T(x, y) = x \sin y$. Plot a few level curves. Compute ∇T and explain its meaning.

11. Find the following limits if they exist:

(a) $\lim_{(x,y)\to(0,0)} \dfrac{\cos xy - 1}{x}$

(b) $\lim_{(x,y)\to(0,0)} \sqrt{|(x + y)/(x - y)|}, x \neq y$

12. Compute the first partial derivatives and gradients of the following functions:

(a) $f(x, y, z) = xe^z + y\cos x$
(b) $f(x, y, z) = (x + y + z)^{10}$
(c) $f(x, y, z) = (x^2 + y)/z$

13. Compute $\dfrac{\partial}{\partial x}[x \exp(1 + x^2 + y^2)]$

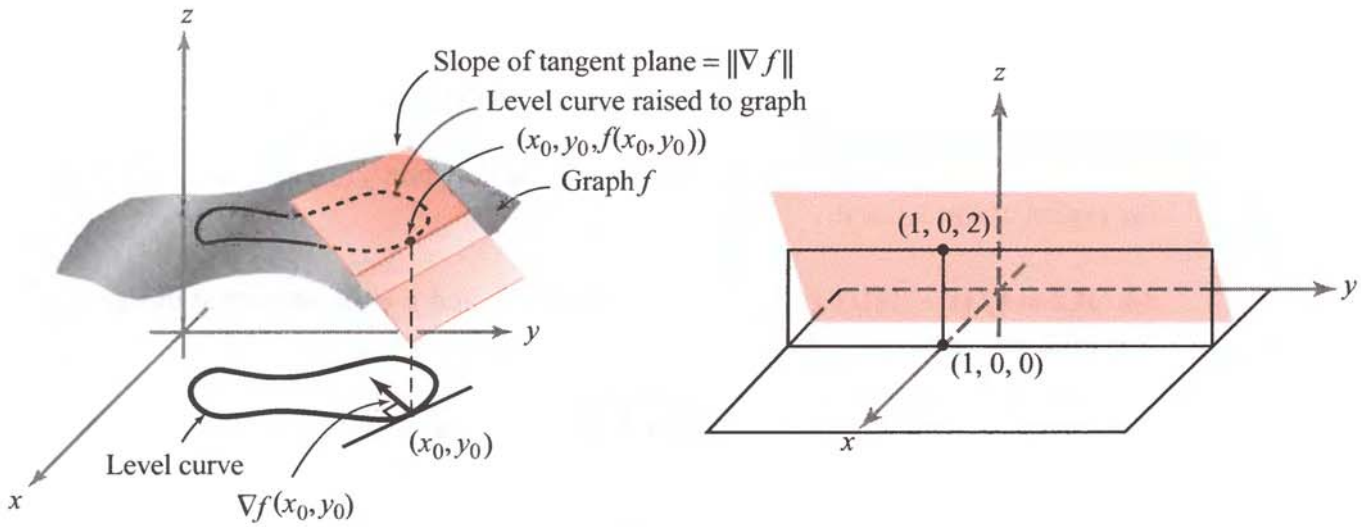
14. Let $y(x)$ be a differentiable function defined implicitly by $F(x, y(x)) = 0$. From Exercise 17(a), Section 2.5, we know that

$$\frac{dy}{dx} = -\frac{\partial F/\partial x}{\partial F/\partial y}.$$

Consider the surface $z = F(x, y)$, and suppose F is increasing as a function of x and as a function of y; that is, $\partial F/\partial x > 0$ and $\partial F/\partial y > 0$. By considering the graph and the plane $z = 0$, show that for z fixed at $z = 0$, y should *decrease* as x increases and x should *decrease* as y increases. Does this agree with the minus sign in the formula for dy/dx?

15. (a) Consider the graph of a function $f(x, y)$ [Figure 2.R.1(a)]. Let (x_0, y_0) lie on a level curve C, so $\nabla f(x_0, y_0)$ is perpendicular to this curve. Show that the tangent plane of the graph is the plane that (i) contains the line perpendicular to $\nabla f(x_0, y_0)$ and lying in the horizontal plane $z = f(x_0, y_0)$, and (ii) has slope $\|\nabla f(x_0, y_0)\|$ relative to the xy plane. (By the *slope* of a plane P relative to the xy plane we mean the tangent of the angle θ, $0 \le \theta \le \pi$, between the upward-pointing normal $\mathbf{p}$ to P and the unit vector $\mathbf{k}$.)

(b) Use this method to show that the tangent plane of the graph of $f(x, y) = (x + \cos y)x^2$ at $(1, 0, 2)$ is as sketched in Figure 2.R.1(b).



(a) (b)

Figure 2.R.1 (a) The relationship between the gradient of a function and the tangent plane to the graph [Exercise 15(a)]. A specific instance of the plane in (b) for Exercise 15(b).

16. Find the plane tangent to the surface $z = x^2 + y^2$ at the point $(1, -2, 5)$. Explain the geometric significance, for this surface, of the gradient of $f(x, y) = x^2 + y^2$ (see Exercise 15).

17. In which direction is the directional derivative of $f(x, y) = (x^2 - y^2)/(x^2 + y^2)$ at $(1, 1)$ equal to zero?

18. Find the directional derivative of the given function at the given point and in the direction of the given vector.

(a) $f(x, y, z) = e^x \cos(yz)$, $p_0 = (0, 0, 0)$, $\mathbf{v} = (2, 1, -2)$
(b) $f(x, y, z) = xy + yz + zx$, $p_0 = (1, 1, 2)$, $\mathbf{v} = (10, -1, 2)$

19. Find the tangent plane and normal to the hyperboloid $x^2 + y^2 - z^2 = 18$ at $(3, 5, -4)$.

20. Let $(x(t), y(t))$ be a path in the plane, $0 \leq t \leq 1$, and let $f(x, y)$ be a C^1 function of two variables. Assume that $(dx/dt)f_x + (dy/dt)f_y \leq 0$. Show that $f(x(1), y(1)) \leq f(x(0), y(0))$.

21. A bug finds itself in a toxic environment. The toxicity level is given by $T(x, y) = 2x^2 - 4y^2$. The bug is at $(-1, 2)$. In what direction should it move to lower the toxicity the fastest?

22. Find the direction in which the function $w = x^2 + xy$ increases most rapidly at the point $(-1, 1)$. What is the magnitude of ∇w at this point? Interpret this magnitude geometrically.

23. Let f be defined on an open set S in $\mathbb{R}^n$. We say that f is ***homogeneous of degree*** p over S if $f(\lambda \mathbf{x}) = \lambda^p f(\mathbf{x})$ for every real λ and for every $\mathbf{x}$ in S for which $\lambda \mathbf{x} \in S$.

(a) If such a function is differentiable at $\mathbf{x}$, show that $\mathbf{x} \cdot \nabla f(\mathbf{x}) = pf(\mathbf{x})$. This is known as ***Euler's theorem*** for homogeneous functions. [HINT: For fixed $\mathbf{x}$, define $g(\lambda) = f(\lambda \mathbf{x})$ and compute $g'(1)$.]

(b) Find p and check Euler's theorem for the function $f(x, y, z) = x - 2y - \sqrt{xz}$, on the region where $xz > 0$.

24. If $z = [f(x - y)]/y$ (where f is differentiable and $y \neq 0$), show that the identity $z + y(\partial z/\partial x) + y(\partial z/\partial y) = 0$ holds.

25. Given $z = f((x + y)/(x - y))$ for f a C^1 function, show that

$$x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = 0.$$

26. Let f have partial derivatives $\partial f(\mathbf{x})/\partial x_i$, where $i = 1, 2, \ldots, n$, at each point $\mathbf{x}$ of an open set U in $\mathbb{R}^n$. If f has a local maximum or a local minimum at the point $\mathbf{x}_0$ in U, show that $\partial f(\mathbf{x}_0)/\partial x_i = 0$ for each i.

27. Consider the functions defined in $\mathbb{R}^2$ by the following formulas:

(i) $f(x, y) = xy/(x^2 + y^2)$ if $(x, y) \neq (0, 0)$, $f(0, 0) = 0$
(ii) $f(x, y) = x^2y^2/(x^2 + y^4)$ if $(x, y) \neq (0, 0)$, $f(0, 0) = 0$

(a) In each case, show that the partial derivatives $\partial f(x, y)/\partial x$ and $\partial f(x, y)/\partial y$ exist for every (x, y) in $\mathbb{R}^2$, and evaluate these derivatives explicitly in terms of x and y.

(b) Explain why the functions described in (i) and (ii) are or are not differentiable at $(0, 0)$.

28. Compute the gradient vector $\nabla f(x, y)$ at all points (x, y) in $\mathbb{R}^2$ for each of the following functions:

(a) $f(x, y) = x^2y^2 \log(x^2 + y^2)$ if $(x, y) \neq (0, 0)$, $f(0, 0) = 0$
(b) $f(x, y) = xy \sin[1/(x^2 + y^2)]$ if $(x, y) \neq (0, 0)$, $f(0, 0) = 0$

29. Find the directional derivatives of the following functions at the point $(1, 1)$ in the direction $(\mathbf{i} + \mathbf{j})/\sqrt{2}$:

(a) $f(x, y) = x \tan^{-1}(x/y)$
(b) $f(x, y) = \cos(\sqrt{x^2 + y^2})$
(c) $f(x, y) = \exp(-x^2 - y^2)$

30. (a) Let $\mathbf{u} = \mathbf{i} - 2\mathbf{j} + 2\mathbf{k}$ and $\mathbf{v} = 2\mathbf{i} + \mathbf{j} - 3\mathbf{k}$. Find: $\|\mathbf{u}\|$, $\mathbf{u} \cdot \mathbf{v}$, $\mathbf{u} \times \mathbf{v}$, and a vector in the same direction as $\mathbf{u}$, but of unit length.

(b) Find the rate of change of $e^{xy} \sin(xyz)$ in the direction $\mathbf{u}$ at $(0, 1, 1)$.

31. Let $h(x, y) = 2e^{-x^2} + e^{-3y^2}$ denote the height on a mountain at position (x, y). In what direction from $(1, 0)$ should one begin walking in order to climb the fastest?

32. Compute an equation for the plane tangent to the graph of

$$f(x, y) = \frac{e^x}{x^2 + y^2}$$

at $x = 1, y = 2$.

33. (a) Give a careful statement of the general form of the chain rule.

(b) Let $f(x, y) = x^2 + y$ and $\mathbf{h}(u) = (\sin 3u, \cos 8u)$. Let $g(u) = f(\mathbf{h}(u))$. Compute dg/du at $u = 0$ both directly *and* by using the chain rule.

34. (a) Sketch the level curves of $f(x, y) = -x^2 - 9y^2$ for $c = 0, -1, -10$.

(b) On your sketch, draw in ∇f at $(1, 1)$. Discuss.

35. At time $t = 0$, a particle is ejected from the surface $x^2 + 2y^2 + 3z^2 = 6$ at the point $(1, 1, 1)$ in a direction normal to the surface at a speed of 10 units per second. At what time does it cross the sphere $x^2 + y^2 + z^2 = 103$?

36. At what point(s) on the surface in Exercise 35 is the normal vector parallel to the line $x = y = z$?

37. Compute $\partial z/\partial x$ and $\partial z/\partial y$ if

$$z = \frac{u^2 + v^2}{u^2 - v^2}, \qquad u = e^{-x-y}, \qquad v = e^{xy}$$

(a) by substitution and direct calculation, and (b) by the chain rule.

38. Compute the partial derivatives as in Exercise 37 if $z = uv$, $u = x + y$, and $v = x - y$.

39. What is wrong with the following argument? Suppose that $w = f(x, y)$ and $y = x^2$. By the chain rule,

$$\frac{\partial w}{\partial x} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial x} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial x} = \frac{\partial w}{\partial x} + 2x\frac{\partial w}{\partial y}.$$

Hence, $0 = 2x(\partial w/\partial y)$, and so $\partial w/\partial y = 0$. Choose an explicit example to really see that this is incorrect.

40. A boat is sailing northeast at 20 km/h. Assuming that the temperature drops at a rate of 0.2°C/km in the northerly direction and 0.3°C/km in the easterly direction, what is the time rate of change of temperature as observed on the boat?

41. Use the chain rule to find a formula for $(d/dt)\exp[f(t)g(t)]$.

42. Use the chain rule to find a formula for $(d/dt)(f(t)^{g(t)})$.

43. Verify the chain rule for the function $f(x, y, z) = [\ln(1 + x^2 + 2z^2)]/(1 + y^2)$ and the path $\mathbf{c}(t) = (t, 1 - t^2, \cos t)$.

44. Verify the chain rule for the function $f(x, y) = x^2/(2 + \cos y)$ and the path $x = e^t$, $y = e^{-t}$.

45. Suppose that $u(x, t)$ satisfies the differential equation $u_t + uu_x = 0$ and that x, as a function $x = f(t)$ of t, satisfies $dx/dt = u(x, t)$. Prove that $u(f(t), t)$ is constant in t.

46. The displacement at time t and horizontal position on a line x of a certain violin string is given by $u = \sin(x - 6t) + \sin(x + 6t)$. Calculate the velocity of the string at $x = 1$ when $t = \frac{1}{3}$.

47. The ***ideal gas law*** $PV = nRT$ involves a constant R, the number n of moles of the gas, the volume V, the Kelvin temperature T, and the pressure P.

(a) Show that each of n, P, T, V is a function of the remaining variables, and determine explicitly the defining equations.

(b) Calculate $\partial V/\partial T$, $\partial T/\partial P$, $\partial P/\partial V$ and show that their product equals -1.

48. The ***potential temperature*** θ is defined in terms of temperature T and pressure p by

$$\theta = T\left(\frac{1000}{p}\right)^{0.286}.$$

The temperature and pressure may be thought of as functions of position (x, y, z) in the atmosphere and also of time t.

(a) Find formulas for $\partial\theta/\partial x$, $\partial\theta/\partial y$, $\partial\theta/\partial z$, $\partial\theta/\partial t$ in terms of partial derivatives of T and p.

(b) The condition $\partial\theta/\partial z < 0$ is regarded as an unstable atmosphere, for it leads to large vertical excursions of air parcels from a single upward or downward impetus. Meteorologists use the formula

$$\frac{\partial\theta}{\partial z} = \frac{\theta}{T}\left(\frac{\partial T}{\partial z} + \frac{g}{C_p}\right)$$

where $g = 32.2$ and C_p is a positive constant. How does the temperature change in the upward direction for an unstable atmosphere?

49. The specific volume V, pressure P, and temperature T of a van der Waals gas are related by $P = RT/(V - \beta) - \alpha/V^2$, where α, β, and R are constants.

(a) Explain why any two of V, P, and T can be considered independent variables that determine the third variable.

(b) Find $\partial T/\partial P$, $\partial P/\partial V$, $\partial V/\partial T$. Identify which variables are constant, and interpret each partial derivative physically.

(c) Verify that $(\partial T/\partial P)(\partial P/\partial V)(\partial V/\partial T) = -1$(not $+1$!).

50. The height h of the Hawaiian volcano Mauna Loa is (roughly) described by the function $h(x, y) = 2.59 - 0.00024y^2 - 0.00065x^2$, where h is the height above sea level in miles and x and y measure east-west and north-south distances in miles from the top of the mountain. At $(x, y) = (-2, -4)$:

(a) How fast is the height increasing in the direction (1, 1) (that is, northeastward)? Express your answer in miles of height per mile of horizontal distance traveled.

(b) In what direction is the steepest upward path?

51. (a) In what direction is the directional derivative of $f(x, y) = (x^2 - y^2)/(x^2 + y^2)$ at (1, 1) equal to zero?

(b) How about at an arbitrary point (x_0, y_0) in the first quadrant?

(c) Describe the level curves of f. In particular, discuss them in terms of the result of part (b).

52. (a) Show that the curve $x^2 - y^2 = c$, for *any* value of c, satisfies the differential equation $dy/dx = x/y$.

(b) Draw in a few of the curves $x^2 - y^2 = c$, say for $c = \pm 1$. At several points (x, y) along each of these curves, draw a short segment of slope x/y; check that these segments appear to be tangent to the curve. What happens when $y = 0$? What happens when $c = 0$?

53. Suppose that f is a differentiable function of one variable and that a function $u = g(x, y)$ is defined by

$$u = g(x, y) = xyf\left(\frac{x + y}{xy}\right).$$

Show that u satisfies a (partial) differential equation of the form

$$x^2\frac{\partial u}{\partial x} - y^2\frac{\partial u}{\partial y} = G(x, y)u$$

and find the function $G(x, y)$.

54. (a) Let F be a function of one variable and f a function of two variables. Show that the gradient vector of $g(x, y) = F(f(x, y))$ is parallel to the gradient vector of $f(x, y)$.

(b) Let $f(x, y)$ and $g(x, y)$ be functions such that $\nabla f = \lambda\nabla g$ for some function $\lambda(x, y)$. What is the relation between the level curves of f and g? Explain why there might be a function F such that $g(x, y) = F(f(x, y))$.

3

Higher-Order Derivatives: Maxima and Minima

Leonhard Euler (by Emanuel Handman) (1707–1783).

All that is superfluous displeases God and Nature.
All that displeases God and Nature is evil.

Dante Alighieri, circa 1300

... namely, because the shape of the whole universe is most perfect, and, in fact, designed by the wisest creator, nothing in all of the world will occur in which no maximum or minimum rule is somehow shining forth.

Leonhard Euler

In one-variable calculus, to test a function $f(x)$ for a local maximum or minimum, one often uses the second derivative. We look for critical points x_0, that is, points x_0 for which $f'(x_0) = 0$, and at each such point we check the sign of the second derivative $f''(x_0)$. If $f''(x_0) < 0$, $f(x_0)$ is a local maximum of f; if $f''(x_0) > 0$, $f(x_0)$ is a local minimum of f; if $f''(x_0) = 0$, the test fails.

This chapter extends these methods to real-valued functions of several variables. We begin in Section 3.1 with a discussion of iterated and higher-order partial derivatives, and in Section 3.2 we discuss the multivariable form of Taylor's theorem; this is then used in Section 3.3 to derive tests for maxima, minima, and saddle points. As with functions of one variable, such methods help one to visualize the shape of a graph.

In Section 3.4, we study the problem of maximizing a real-valued function subject to supplementary conditions, also referred to as constraints. For example, we might

wish to maximize $f(x, y, z)$ among those (x, y, z) constrained to lie on the unit sphere, $x^2 + y^2 + z^2 = 1$. Section 3.5 discusses a technical theorem (the implicit function theorem) useful for studying constraints. It will also be useful later in our study of surfaces.

3.1 Iterated Partial Derivatives

The preceding chapter developed considerable information concerning the derivative of a map and investigated the geometry associated with the derivative of real-valued functions by making use of the gradient. In this section, we proceed to study higher-order derivatives, with the goal of proving the equality of the "mixed second partial derivatives" of a function. We begin by defining the necessary terms.

Let $f: \mathbb{R}^3 \to \mathbb{R}$ be of class C^1. Recall that this means that $\partial f/\partial x$, $\partial f/\partial y$, and $\partial f/\partial z$ exist and are continuous. If these derivatives, in turn, have continuous partial derivatives, we say that f is of ***class*** C^2, or is ***twice continuously differentiable***. Likewise, if we say f is of class C^3, we mean f has continuous iterated partial derivatives of third order, and so on. Here are a few examples of how second-order derivatives are written:

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right), \qquad \frac{\partial^2 f}{\partial x\, \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right), \qquad \frac{\partial^2 f}{\partial z\, \partial y} = \frac{\partial}{\partial z}\left(\frac{\partial f}{\partial y}\right), \qquad \text{etc.}$$

The process can, of course, be repeated for third-order derivatives, and so on. If f is a function of only x and y and $\partial f/\partial x$, $\partial f/\partial y$ are continuously differentiable, then by taking second partial derivatives we get the four functions

$$\frac{\partial^2 f}{\partial x^2}, \qquad \frac{\partial^2 f}{\partial y^2}, \qquad \frac{\partial^2 f}{\partial x\, \partial y}, \qquad \text{and} \qquad \frac{\partial^2 f}{\partial y\, \partial x}.$$

All of these are called ***iterated partial derivatives***, while $\partial^2 f/\partial x\, \partial y$ and $\partial^2 f/\partial y\, \partial x$ are called ***mixed partial derivatives***.

EXAMPLE 1 Find all second partial derivatives of $f(x, y) = xy + (x + 2y)^2$.

SOLUTION The first partials are

$$\frac{\partial f}{\partial x} = y + 2(x + 2y), \qquad \frac{\partial f}{\partial y} = x + 4(x + 2y).$$

Now differentiate each of these expressions with respect to x and y:

$$\frac{\partial^2 f}{\partial x^2} = 2, \qquad \frac{\partial^2 f}{\partial y^2} = 8$$

$$\frac{\partial^2 f}{\partial x\, \partial y} = 5, \qquad \frac{\partial^2 f}{\partial y\, \partial x} = 5. \quad \blacktriangle$$

EXAMPLE 2 Find all second partial derivatives of $f(x, y) = \sin x \sin^2 y$.

SOLUTION We proceed just as in Example 1:

$$\frac{\partial f}{\partial x} = \cos x \sin^2 y, \qquad \frac{\partial f}{\partial y} = 2 \sin x \sin y \cos y = \sin x \sin 2y;$$

$$\frac{\partial^2 f}{\partial x^2} - = -\sin x \sin^2 y, \qquad \frac{\partial^2 f}{\partial y^2} = 2 \sin x \cos 2y;$$

$$\frac{\partial^2 f}{\partial x\, \partial y} = \cos x \sin 2y, \qquad \frac{\partial^2 f}{\partial y\, \partial x} = 2 \cos x \sin y \cos y = \cos x \sin 2y. \quad \blacktriangle$$

EXAMPLE 3 Let $f(x, y, z) = e^{xy} + z \cos x$. Then

$$\frac{\partial f}{\partial x} = ye^{xy} - z \sin x, \qquad \frac{\partial f}{\partial y} = xe^{xy}, \qquad \frac{\partial f}{\partial z} = \cos x,$$

$$\frac{\partial^2 f}{\partial z\, \partial x} = -\sin x, \qquad \frac{\partial^2 f}{\partial x\, \partial z} = -\sin x, \qquad \text{etc.} \quad \blacktriangle$$

The Mixed Partials are Equal

In all these examples note that the pairs of mixed partial derivatives, such as $\partial^2 f/\partial x\, \partial y$ and $\partial^2 f/\partial y\, \partial x$, or $\partial^2 f/\partial z\, \partial x$ and $\partial^2 f/\partial x\, \partial z$, are equal. It is a basic and perhaps surprising fact that *this is always the case for C^2 functions*. We shall prove this in the next theorem for functions $f(x, y)$ of two variables, but the proof can be readily extended to functions of n variables.

THEOREM 1: Equality of Mixed Partials If $f(x, y)$ is of class C^2 (is twice continuously differentiable), then the mixed partial derivatives are equal; that is,

$$\frac{\partial^2 f}{\partial x\, \partial y} = \frac{\partial^2 f}{\partial y\, \partial x}.$$

PROOF Consider the following expression (see Figure 3.1.1):

$$S(\Delta x, \Delta y) = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0 + \Delta x, y_0) - f(x_0, y_0 + \Delta y) + f(x_0, y_0).$$

Holding y_0 and Δy fixed, define

$$g(x) = f(x, y_0 + \Delta y) - f(x, y_0),$$

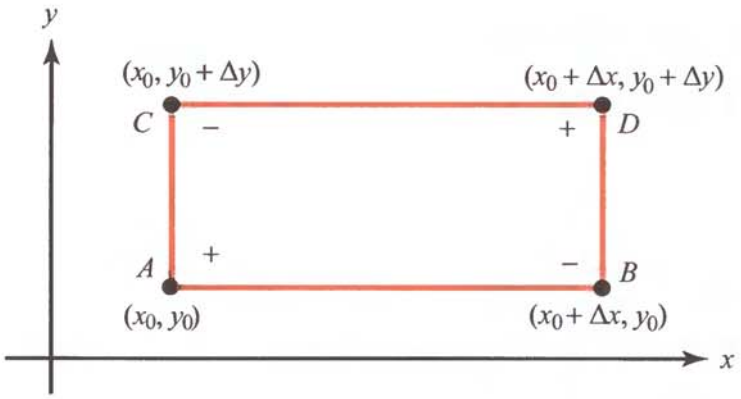


Figure 3.1.1 The algebra behind the equality of mixed partials: writing the difference of differences in two ways.

so that $S(\Delta x, \Delta y) = g(x_0 + \Delta x) - g(x_0)$, *which expresses S as a difference of differences*. By the mean-value theorem for functions of one variable, $g(x_0 + \Delta x) - g(x_0)$ equals $g'(\bar{x})\Delta x$ for some $\bar{x}$ between x_0 and $x_0 + \Delta x$. Hence,

$$S(\Delta x, \Delta y) = \left[\frac{\partial f}{\partial x}(\bar{x}, y_0 + \Delta y) - \frac{\partial f}{\partial x}(\bar{x}, y_0)\right]\Delta x.$$

Applying the mean-value theorem again, there is a $\bar{y}$ between y_0 and $y_0 + \Delta y$ such that

$$S(\Delta x, \Delta y) = \frac{\partial^2 f}{\partial y\, \partial x}(\bar{x}, \bar{y})\, \Delta x\, \Delta y.$$

Because $\partial^2 f/\partial y\, \partial x$ is continuous, it follows that

$$\frac{\partial^2 f}{\partial y\, \partial x}(x_0, y_0) = \lim_{(\Delta x, \Delta y) \to (0,0)} \frac{1}{\Delta x\, \Delta y}[S(\Delta x, \Delta y)].$$

Noting that S is symmetric in Δx and Δy, one shows in a similar way that $\partial^2 f/\partial x\, \partial y$ is given by the *same limit formula*, which proves the result. ■

Historical Note

The equality of mixed partial derivatives is one of the most important results of multivariable calculus. It will reappear on several occasions later in the book, when we study vector identities.

In the next historical note, we will discuss the role of partial derivatives in the formulation of many of the basic equations governing physical phenomena. One of the giants in this era was Leonhard Euler (1707–1783), who developed the equations of fluid mechanics that bear his name—the Euler equations. It was in connection with the needs of this development that he discovered, around 1734, the equality of mixed partial derivatives. Euler was about 27 years old at the time.

In Exercise 11 we ask the reader to deduce from Theorem 1 that for a C^3 function of x, y, and z,

$$\frac{\partial^3 f}{\partial x\,\partial y\,\partial z} = \frac{\partial^3 f}{\partial z\,\partial y\,\partial x} = \frac{\partial^3 f}{\partial y\,\partial z\,\partial x}, \qquad \text{etc.}$$

In other words, we can compute iterated partial derivatives in any order we please.

EXAMPLE 4 Verify the equality of the mixed second partial derivatives for the function

$$f(x, y) = xe^y + yx^2.$$

SOLUTION Here

$$\frac{\partial f}{\partial x} = e^y + 2xy, \qquad \frac{\partial f}{\partial y} = xe^y + x^2,$$

$$\frac{\partial^2 f}{\partial y\,\partial x} = e^y + 2x, \qquad \frac{\partial^2 f}{\partial x\,\partial y} = e^y + 2x,$$

and so we have

$$\frac{\partial^2 f}{\partial y\,\partial x} = \frac{\partial^2 f}{\partial x\,\partial y}. \quad \blacktriangle$$

Sometimes the notation f_x, f_y, f_z is used for the partial derivatives: $f_x = \partial f/\partial x$, and so on. With this notation, we write $f_{xy} = (f_x)_y$, and so equality of the mixed partials is denoted by $f_{xy} = f_{yx}$. Notice that $f_{xy} = \partial^2 f/\partial y\,\partial x$, so the order of x and y is reversed in the two notations; fortunately, the equality of mixed partials makes this potential ambiguity irrelevant. The following example illustrates this subscript notation.

EXAMPLE 5 Let

$$z = f(x, y) = e^x \sin xy$$

and write $x = g(s, t)$, $y = h(s, t)$ for certain functions g and h. Let

$$k(s, t) = f(g(s, t), h(s, t)).$$

Calculate k_{st}.

SOLUTION By the chain rule,

$$k_s = f_x g_s + f_y h_s = (e^x \sin xy + ye^x \cos xy)g_s + (xe^x \cos xy)h_s.$$

Differentiating in t using the product rules gives

$$k_{st} = (f_x)_t g_s + f_x (g_s)_t + (f_y)_t h_s + f_y (h_s)_t.$$

Applying the chain rule again to $(f_x)_t$ and $(f_y)_t$ gives

$$(f_x)_t = f_{xx} g_t + f_{xy} h_t \qquad \text{and} \qquad (f_y)_t = f_{yx} g_t + f_{yy} h_t,$$

and so k_{st} becomes

$$\begin{aligned} k_{st} &= (f_{xx} g_t + f_{xy} h_t) g_s + f_x g_{st} + (f_{yx} g_t + f_{yy} h_t) h_s + f_y h_{st} \\ &= f_{xx} g_t g_s + f_{xy} (h_t g_s + h_s g_t) + f_{yy} h_t h_s + f_x g_{st} + f_y h_{st}. \end{aligned}$$

Notice that this last formula is symmetric in (s, t), verifying the equality $k_{st} = k_{ts}$. Computing f_{xx}, f_{xy}, and f_{yy}, we get

$$\begin{aligned} k_{st} = {} & (e^x \sin xy + 2ye^x \cos xy - y^2 e^x \sin xy) g_t g_s \\ & + (xe^x \cos xy + e^x \cos xy - xye^x \sin xy)(h_t g_s + h_s g_t) \\ & - (x^2 e^x \sin xy) h_t h_s + (e^x \sin xy + ye^x \cos xy) g_{st} + (xe^x \cos xy) h_{st}, \end{aligned}$$

in which it is understood that $x = g(s, t)$ and $y = h(s, t)$. ▲

Historical Note

Some Partial Differential Equations

> Philosophy [nature] is written in that great book which ever is before our eyes—I mean the universe—but we cannot understand it if we do not first learn the language and grasp the symbols in which it is written. The book is written in mathematical language, and the symbols are triangles, circles and other geometrical figures, without whose help it is impossible to comprehend a single word of it; without which one wanders in vain through a dark labyrinth.
>
> GALILEO

This quotation illustrates the Greek belief, again popular in the time of Galileo, that much of nature could be described using mathematics. In the latter part of the seventeenth century this thinking was dramatically reinforced when Newton used his law of gravitation to derive Kepler's three laws of celestial motion (see Section 4.1) to explain the tides, and to show that

the earth was flattened at the poles. The impact of this philosophy on mathematics was substantial, and many mathematicians sought to "mathematize" nature. The extent to which mathematics pervades the physical sciences today (and, to an increasing amount, economics and the social and life sciences) is testament to the success of these endeavors. Correspondingly, the attempts to mathematize nature have often led to new mathematical discoveries.

Many of the laws of nature were described in terms of either ordinary differential equations (ODEs, equations involving the derivatives of functions of one variable alone, such as the laws of planetary motion) or partial differential equations (PDEs), that is, equations involving partial derivatives of functions. To give the reader some historical perspective and offer motivation for studying partial derivatives, we present a brief description of three of the most famous partial differential equations: the heat equation, the potential equation (or Laplace's equation), and the wave equation. (Further information on some PDEs is given in Section 8.5.)

THE HEAT EQUATION. In the early part of the nineteenth century the French mathematician Joseph Fourier (1768–1830) took up the study of heat. Heat flow had obvious applications to both industrial and scientific problems: A better understanding of it would, for example, make possible more efficient smelting of metals and would enable scientists to determine the temperature of a body given the temperature at its boundary, and to approximate the temperature of the earth's interior.

Let a homogeneous body $\mathcal{B} \subset \mathbb{R}^3$ (Figure 3.1.2) be represented by some region in 3-space. Let $T(x, y, z, t)$ denote the temperature of the body at the

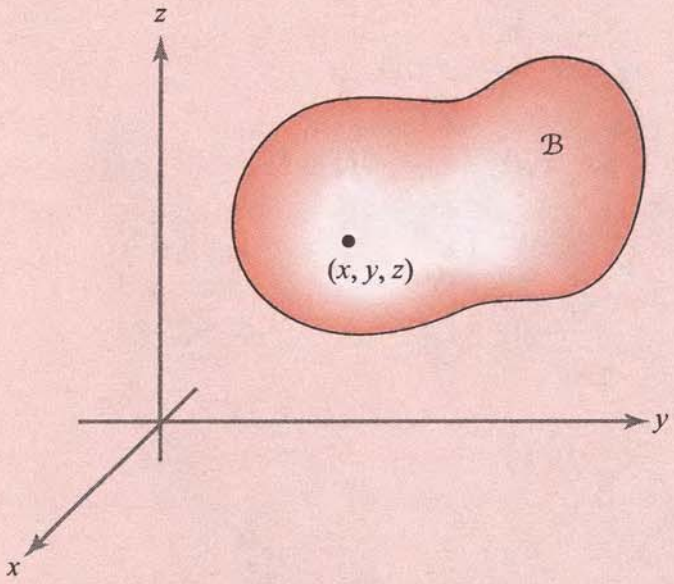


Figure 3.1.2 A homogeneous body in space.

point (x, y, z) at time t. Fourier proved, on the basis of physical principles (described in Section 8.5), that T must satisfy the partial differential equation called the ***heat equation***,

$$k\left(\frac{\partial^2 T}{\partial x^2}+\frac{\partial^2 T}{\partial y^2}+\frac{\partial^2 T}{\partial z^2}\right)=\frac{\partial T}{\partial t}, \tag{1}$$

where k is a constant whose value depends on the conductivity of the material comprising the body.

Fourier used this equation to solve problems in heat conduction. In fact, his investigations into the solutions of equation (1) led him to the discovery of *Fourier series*.

THE POTENTIAL EQUATION. Consider the gravitational potential V (often called Newton's potential) of a mass m at a point (x, y, z) caused by a point mass M situated at the origin. This potential is given by $V=-GmM/r$, where $r=\sqrt{x^2+y^2+z^2}$. The potential V satisfies the equation

$$\frac{\partial^2 V}{\partial x^2}+\frac{\partial^2 V}{\partial y^2}+\frac{\partial^2 V}{\partial z^2}=0 \tag{2}$$

everywhere except at the origin, as we will check in the next chapter (see also Exercise 23). This equation is known as ***Laplace's equation***. Pierre-Simon de Laplace (1749–1827) had worked on the gravitational attraction of nonpoint masses and was the first to consider equation (2) with regard to gravitational attraction. He gave arguments (later shown to be incorrect) that equation (2) held for any body and any point whether inside or outside that body. However, Laplace was not the first person to write down equation (2). The potential equation appeared for the first time in one of Euler's major papers in 1752, "Principles of the Motions of Fluids," in which he derived the potential equation with regard to the motion of (incompressible) fluids. Euler remarked that he had no idea how to solve equation (2). Poisson later showed that if (x, y, z) lies inside an attracting body, then V satisfies the equation

$$\frac{\partial^2 V}{\partial x^2}+\frac{\partial^2 V}{\partial y^2}+\frac{\partial^2 V}{\partial z^2}=-4\pi\rho, \tag{3}$$

where ρ is the mass density of the attracting body. Equation (3) is now called ***Poisson's equation***. Poisson was also the first to point out the importance of this equation for problems involving electric fields. Notice that if the temperature T is constant in time, then the heat equation (1) reduces to Laplace's equation (2).

Laplace's and Poisson's equations are fundamental to many fields besides fluid mechanics, gravitational fields, and electrostatic fields. For example, they are useful for studying soap films and liquid crystals (see *The Parsimonious Universe: Shape and Form in the Natural World* by S. Hildebrandt and A. Tromba, Springer-Verlag, New York/Berlin, 1995).

THE WAVE EQUATION. The linear wave equation in space has the form

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = c^2 \frac{\partial^2 f}{\partial t^2}. \tag{4}$$

The one-dimensional wave equation

$$\frac{\partial^2 f}{\partial x^2} = c^2 \frac{\partial^2 f}{\partial t^2} \tag{4'}$$

was derived in about 1727 by Johann II Bernoulli and several years later by Jean Le Rond d'Alembert in the study of how to determine the motion of a vibrating string (such as a violin string). Equation (4) became useful in the study of both vibrating bodies and elasticity. As we shall see when we consider Maxwell's equations for electromagnetism in Section 8.5, this equation also arises in the study of the propagation of electromagnetic radiation and sound waves.

EXAMPLE 6 The partial differential equation $u_t + u_{xxx} + uu_x = 0$, called the ***Korteweg–de Vries equation*** (or KdV equation, for short), describes the motion of water waves in a shallow channel.

(a) Show that for any positive constant c, the function

$$u(x, t) = 3c \operatorname{sech}^2\left[\tfrac{1}{2}(x - ct)\sqrt{c}\right]$$

is a solution of the Korteweg–de Vries equation. (This solution represents a traveling "hump" of water in the channel and is called a ***soliton***.)[1]

(b) How do the shape and speed of the soliton depend on c?

SOLUTION (a) We compute u_t, u_x, u_{xx}, and u_{xxx} using the chain rule and the differentiation formula $(d/dx)\operatorname{sech} x = -\operatorname{sech} x \tanh x$ from one-variable calculus.

[1] Solitons were first observed by J. Scott Russell around 1840 in barge canals near Edinburgh. He reported his results in *Trans. R. Soc. Edinburgh* **14** (1840): 47–109.

Letting $\alpha = (x - ct)\sqrt{c}/2$,

$$u_t = 6c \operatorname{sech} \alpha \frac{\partial}{\partial t} \operatorname{sech} \alpha = -6c \operatorname{sech}^2 \alpha \tanh \alpha \frac{\partial \alpha}{\partial t}$$

$$= 3c^{5/2} \operatorname{sech}^2 \alpha \tanh \alpha = c^{3/2} u \tanh \alpha.$$

Also,

$$u_x = -6c \operatorname{sech}^2 \alpha \tanh \alpha \frac{\partial \alpha}{\partial x}$$

$$= -3c^{3/2} \operatorname{sech}^2 \alpha \tanh \alpha = -\sqrt{c}\, u \tanh \alpha,$$

and so $u_t + cu_x = 0$ and

$$u_{xx} = -\sqrt{c}\left[u_x \tanh \alpha + u(\operatorname{sech}^2 \alpha)\frac{\sqrt{c}}{2}\right] = -\sqrt{c}\,(\tanh \alpha)u_x - \frac{u^2}{6}$$

$$= c(\tanh^2 \alpha)u - \frac{u^2}{6} = c(1 - \operatorname{sech}^2 \alpha)u - \frac{u^2}{6}$$

$$= cu - \frac{u^2}{3} - \frac{u^2}{6} = cu - \frac{u^2}{2}.$$

Thus,

$$u_{xxx} = cu_x - uu_x; \qquad \text{that is,} \qquad u_{xxx} + uu_x = cu_x.$$

Hence,

$$u_t + u_{xxx} + uu_x = u_t + cu_x = 0.$$

(b) The speed of the soliton is c, because $u(x + ct, t) = u(x, 0)$. The soliton is higher and thinner when c is larger. Its shape at time $t = 10$ is shown in Figure 3.1.3. ▲

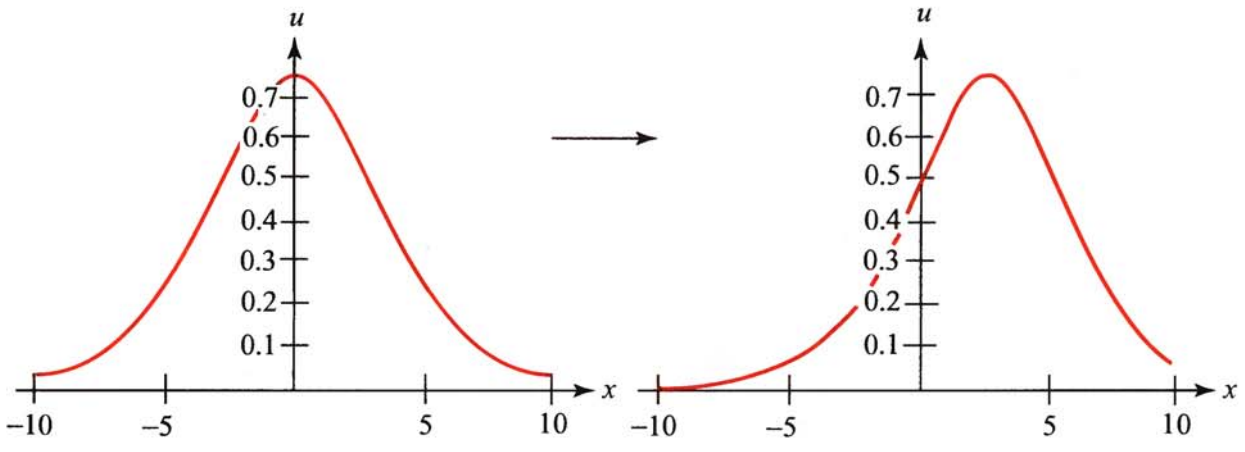


Figure 3.1.3 The graph of $u(x, t) = 3 \operatorname{sech}^2(\sqrt{c}(x - ct)/2)$ for $c = \frac{1}{4}$ at times (a) $t = 0$ and (b) $t = 10$.

EXERCISES

In Exercises 1 to 6, compute the second partial derivatives $\partial^2 f/\partial x^2$, $\partial^2 f/\partial x\,\partial y$, $\partial^2 f/\partial y\,\partial x$, $\partial^2 f/\partial y^2$ *for each of the following functions. Verify Theorem 1 in each case.*

1. $f(x, y) = 2xy/(x^2 + y^2)^2$, on the region where $(x, y) \neq (0, 0)$

2. $f(x, y, z) = e^z + (1/x) + xe^{-y}$, on the region where $x \neq 0$

3. $f(x, y) = \cos(xy^2)$

4. $f(x, y) = e^{-xy^2} + y^3x^4$

5. $f(x, y) = 1/(\cos^2 x + e^{-y})$

6. $f(x, y) = \log(x - y)$

7. Find $\partial^2 z/\partial x^2$, $\partial^2 z/\partial x\,\partial y$, $\partial^2 z/\partial y\,\partial x$, and $\partial^2 z/\partial y^2$ for

(a) $z = 3x^2 + 2y^2$
(b) $z = (2x^2 + 7x^2y)/3xy$, on the region where $x \neq 0$ and $y \neq 0$

8. Find all the second partial derivatives of

(a) $z = \sin(x^2 - 3xy)$
(b) $z = x^2y^2e^{2xy}$

9. Find f_{xy}, f_{yz}, f_{zx}, and f_{xyz} for

$$f(x, y, z) = x^2y + xy^2 + yz^2.$$

10. Let $z = x^4y^3 - x^8 + y^4$.

(a) Compute $\partial^3 z/\partial y\,\partial x\,\partial x$, $\partial^3 z/\partial x\,\partial y\,\partial x$, and $\partial^3 z/\partial x\,\partial x\,\partial y$ (also denoted $\partial^3 z/\partial x^2\partial y$).
(b) Compute $\partial^3 z/\partial x\,\partial y\,\partial y$, $\partial^3 z/\partial y\,\partial x\,\partial y$, and $\partial^3 z/\partial y\,\partial y\,\partial x$ (also denoted $\partial^3 z/\partial y^2\partial x$).

11. Use Theorem 1 to show that if $f(x, y, z)$ is of class C^3, then

$$\frac{\partial^3 f}{\partial x\,\partial y\,\partial z} = \frac{\partial^3 f}{\partial y\,\partial z\,\partial x}.$$

12. Verify that

$$\frac{\partial^3 f}{\partial x\,\partial y\,\partial z} = \frac{\partial^3 f}{\partial z\,\partial y\,\partial x}$$

for $f(x, y, z) = ze^{xy} + yz^3x^2$.

13. Verify that $f_{xzw} = f_{zwx}$ for $f(x, y, z, w) = e^{xyz}\sin(xw)$.

14. If $f(x, y, z, w)$ is of class C^3, show that $f_{xzw} = f_{zwx}$.

15. Evaluate all first and second partial derivatives of the following functions:

(a) $f(x, y) = x \arctan(x/y)$
(b) $f(x, y) = \cos\sqrt{x^2 + y^2}$
(c) $f(x, y) = \exp(-x^2 - y^2)$

16. Let $w = f(x, y)$ be a function of two variables and let $x = u + v$, $y = u - v$. Show that

$$\frac{\partial^2 w}{\partial u\, \partial v} = \frac{\partial^2 w}{\partial x^2} - \frac{\partial^2 w}{\partial y^2}.$$

17. Let $f\colon \mathbb{R}^2 \to \mathbb{R}$ be a C^2 function and let $\mathbf{c}(t)$ be a C^2 curve in $\mathbb{R}^2$. Write a formula for the second derivative $(d^2/dt^2)((f \circ \mathbf{c})(t))$ using the chain rule twice.

18. Let $f(x, y, z) = e^{xz} \tan(yz)$ and let $x = g(s, t)$, $y = h(s, t)$, $z = k(s, t)$, and define the function $m(s, t) = f(g(s, t), h(s, t), k(s, t))$. Find a formula for m_{st} using the chain rule and verify that your answer is symmetric in s and t.

19. A function $u = f(x, y)$ with continuous second partial derivatives satisfying Laplace's equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

is called a ***harmonic function***. Show that the function $u(x, y) = x^3 - 3xy^2$ is harmonic.

20. Which of the following functions are harmonic? (See Exercise 19.)

(a) $f(x, y) = x^2 - y^2$
(b) $f(x, y) = x^2 + y^2$
(c) $f(x, y) = xy$
(d) $f(x, y) = y^3 + 3x^2y$
(e) $f(x, y) = \sin x \cosh y$
(f) $f(x, y) = e^x \sin y$

21. Let f and g be C^2 functions of one variable. Set $\phi = f(x - t) + g(x + t)$.

(a) Prove that ϕ satisfies the wave equation: $\partial^2\phi/\partial t^2 = \partial^2\phi/\partial x^2$.
(b) Sketch the graph of ϕ against t and x if $f(x) = x^2$ and $g(x) = 0$.

22. (a) Show that function $g(x, t) = 2 + e^{-t} \sin x$ satisfies the heat equation: $g_t = g_{xx}$. [Here $g(x, t)$ represents the temperature in a metal rod at position x and time t.]

(b) Sketch the graph of g for $t \geq 0$. (HINT: Look at sections by the planes $t = 0$, $t = 1$, and $t = 2$.)

(c) What happens to $g(x, t)$ as $t \to \infty$? Interpret this limit in terms of the behavior of heat in the rod.

23. Show that Newton's potential $V = -GmM/r$ satisfies Laplace's equation

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0 \qquad \text{for} \qquad (x, y, z) \neq (0, 0, 0).$$

24. Let

$$f(x, y) = \begin{cases} xy(x^2 - y^2)/(x^2 + y^2), & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

(see Figure 3.1.4).

(a) If $(x, y) \neq (0, 0)$, calculate $\partial f/\partial x$ and $\partial f/\partial y$.
(b) Show that $(\partial f/\partial x)(0, 0) = 0 = (\partial f/\partial y)(0, 0)$.
(c) Show that $(\partial^2 f/\partial x\, \partial y)(0, 0) = 1$, $(\partial^2 f/\partial y\, \partial x)(0, 0) = -1$.
(d) What went wrong? Why are the mixed partials not equal?

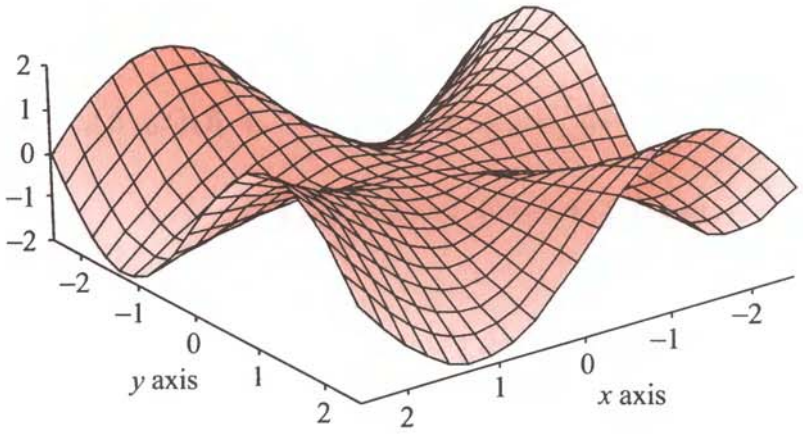


Figure 3.1.4 The graph of the function in Exercise 24.

3.2 Taylor's Theorem

When we introduced the derivative in Chapter 2, we saw that the *linear approximation* of a function played an essential role for a geometric reason—finding the equation of a tangent plane—as well as an analytic reason—finding approximate values of functions. Taylor's theorem deals with the important issue of finding *quadratic and higher-order approximations*.

Taylor's theorem is a central tool for finding accurate numerical approximations of functions, and as such plays an important role in many areas of applied and computational mathematics. We shall use it in the next section to develop the second derivative test for maxima and minima of functions of several variables.

The strategy used to prove Taylor's theorem is to reduce it to the one-variable case by probing a function of many variables along lines of the form $\mathbf{l}(t) = \mathbf{x}_0 + t\mathbf{h}$

emanating from a point $\mathbf{x}_0$ and heading in the direction $\mathbf{h}$. Thus, it will be useful for us to begin by reviewing Taylor's theorem from one-variable calculus.

Single-Variable Taylor Theorem

When recalling a theorem from an earlier course, it is helpful to ask these basic questions: What is the main point of the theorem? What are the key ideas in the proof? Can I understand the result better the second time around?

The main point of the single-variable Taylor theorem is to find approximations of a function near a given point that are accurate to a higher order than the linear approximation. The key idea in the proof is to use the *fundamental theorem of calculus*, followed by *integration by parts*. In fact, just by recalling these basic ideas, one can reconstruct the entire proof. Thinking in this way will help organize all the pieces that need to come together to develop a mastery of Taylor approximations of functions of one and several variables.

For a smooth function $f\colon \mathbb{R} \to \mathbb{R}$ of one variable, Taylor's theorem asserts that:

$$f(x_0+h) = f(x_0) + f'(x_0)\cdot h + \frac{f''(x_0)}{2}h^2 + \cdots + \frac{f^{(k)}(x_0)}{k!}h^k + R_k(x_0, h), \quad (1)$$

where

$$R_k(x_0, h) = \int_{x_0}^{x_0+h} \frac{(x_0+h-\tau)^k}{k!} f^{k+1}(\tau)\, d\tau$$

is the remainder. For small h, this remainder is small to order k in the sense that

$$\lim_{h\to 0} \frac{R_k(x_0, h)}{h^k} = 0. \quad (2)$$

In other words, $R_k(x_0, h)$ is small compared to the already small quantity h^k.

The preceding is the formal statement of Taylor's theorem. What about the proof? As promised, we begin with the fundamental theorem of calculus, written in the form:

$$f(x_0+h) = f(x_0) + \int_{x_0}^{x_0+h} f'(\tau)\, d\tau.$$

Next, we write $d\tau = -d(x_0 + h - \tau)$ and integrate parts[2] to give:

$$f(x_0 + h) = f(x_0) + f'(x_0)h + \int_{x_0}^{x_0+h} f''(\tau)(x_0 + h - \tau)\, d\tau,$$

which is the first-order Taylor formula. Integrating by parts again:

$$\begin{aligned}
&\int_{x_0}^{x_0+h} f''(\tau)(x_0 + h - \tau)\, d\tau \\
&\quad = -\frac{1}{2}\int_{x_0}^{x_0+h} f''(\tau)\, d(x_0 + h - \tau)^2 \\
&\quad = \frac{1}{2}f''(x_0)h^2 + \frac{1}{2}\int_{x_0}^{x_0+h} f'''(\tau)(x_0 + h - \tau)^2\, d\tau,
\end{aligned}$$

which, when substituted into the preceding formula, gives the ***second-order Taylor formula***:

$$f(x_0 + h) = f(x_0) + f'(x_0)h + \frac{1}{2}f''(x_0)h^2 + \frac{1}{2}\int_{x_0}^{x_0+h} f'''(\tau)(x_0 + h - \tau)^2\, d\tau.$$

This is Taylor's theorem for $k = 2$.

Taylor's theorem for general k proceeds by repeated integration by parts. The statement (2) that $R_k(x_0, h)/h^k \to 0$ as $h \to 0$ is seen as follows. For τ in the interval $[x_0, x_0 + h]$, we have $|x_0 + h - \tau| \le |h|$ and $f^{k+1}(\tau)$, being continuous, is bounded; say, $|f^{k+1}(\tau)| \le M$. Then:

$$|R_k(x_0, h)| = \left| \int_{x_0}^{x_0+h} \frac{(x_0 + h - \tau)^k}{k!} f^{k+1}(\tau)\, d\tau \right| \le \frac{|h|^{k+1}}{k!} M$$

and, in particular, $|R_k(x_0, h)/h^k| \le |h|\, M/k! \to 0$ as $h \to 0$.

---

[2]Recall that integration by parts (the product rule for the derivative read backwards) reads as:

$$\int_a^b u\, dv = uv\Big|_a^b - \int_a^b v\, du.$$

Here we choose $u = f'(\tau)$ and $v = x_0 + h - \tau$.

Taylor's Theorem for Many Variables

Our next goal in this section is to prove an analogous theorem that is valid for functions of several variables. We already know a first-order version, that is, when $k = 1$. Indeed, if $f\colon \mathbb{R}^n \to \mathbb{R}$ is differentiable at $\mathbf{x}_0$ and we define

$$R_1(\mathbf{x}_0, \mathbf{h}) = f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0) - [\mathbf{D}f(\mathbf{x}_0)](\mathbf{h}),$$

so that

$$f(\mathbf{x}_0 + \mathbf{h}) = f(\mathbf{x}_0) + [\mathbf{D}f(\mathbf{x}_0)](\mathbf{h}) + R_1(\mathbf{x}_0, \mathbf{h}),$$

then by the definition of differentiability,

$$\frac{|R_1(\mathbf{x}_0, \mathbf{h})|}{\|\mathbf{h}\|} \to 0 \qquad \text{as} \qquad \mathbf{h} \to 0,$$

that is, $R_1(\mathbf{x}_0, \mathbf{h})$ vanishes to first order at $\mathbf{x}_0$. In summary, we have:

THEOREM 2: First-Order Taylor Formula Let $f\colon U \subset \mathbb{R}^n \to \mathbb{R}$ be differentiable at $\mathbf{x}_0 \in U$. Then

$$f(\mathbf{x}_0 + \mathbf{h}) = f(\mathbf{x}_0) + \sum_{i=1}^{n} h_i \frac{\partial f}{\partial x_i}(\mathbf{x}_0) + R_1(\mathbf{x}_0, \mathbf{h}),$$

where $R_1(\mathbf{x}_0, \mathbf{h})/\|\mathbf{h}\| \to 0$ as $\mathbf{h} \to \mathbf{0}$ in $\mathbb{R}^n$.

The second-order version is as follows:

THEOREM 3: Second-Order Taylor Formula Let $f\colon U \subset \mathbb{R}^n \to \mathbb{R}$ have continuous partial derivatives of third order.[3] Then we may write

$$f(\mathbf{x}_0 + \mathbf{h}) = f(\mathbf{x}_0) + \sum_{i=1}^{n} h_i \frac{\partial f}{\partial x_i}(\mathbf{x}_0) + \frac{1}{2} \sum_{i,j=1}^{n} h_i h_j \frac{\partial^2 f}{\partial x_i \, \partial x_j}(\mathbf{x}_0) + R_2(\mathbf{x}_0, \mathbf{h}),$$

where $R_2(\mathbf{x}_0, \mathbf{h})/\|\mathbf{h}\|^2 \to 0$ as $\mathbf{h} \to \mathbf{0}$ and the second sum is over all i's and j's between 1 and n (so there are n^2 terms).

[3] For the statement of the theorem as given here, f actually needs only to be of class C^2, but for a convenient form of the remainder we assume f is of class C^3.

Notice that this result can be written in matrix form as

$$f(\mathbf{x}_0+\mathbf{h}) = f(\mathbf{x}_0) + \left(\frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_n}\right)\begin{pmatrix} h_1 \\ \vdots \\ h_n \end{pmatrix}$$

$$+\frac{1}{2}(h_1, \ldots, h_n)\begin{pmatrix} \dfrac{\partial^2 f}{\partial x_1\,\partial x_1} & \dfrac{\partial^2 f}{\partial x_1\,\partial x_2} & \cdots & \dfrac{\partial^2 f}{\partial x_1\,\partial x_n} \\ \dfrac{\partial^2 f}{\partial x_2\,\partial x_1} & \dfrac{\partial^2 f}{\partial x_2\,\partial x_2} & \cdots & \dfrac{\partial^2 f}{\partial x_2\,\partial x_n} \\ \vdots & & & \\ \dfrac{\partial^2 f}{\partial x_n\,\partial x_1} & \dfrac{\partial^2 f}{\partial x_n\,\partial x_2} & \cdots & \dfrac{\partial^2 f}{\partial x_n\,\partial x_n} \end{pmatrix}\begin{pmatrix} h_1 \\ h_2 \\ \vdots \\ h_n \end{pmatrix},$$

$$+ R_2(\mathbf{x}_0, \mathbf{h}),$$

where the derivatives of f are evaluated at $\mathbf{x}_0$.

In the course of the proof of the Theorem 3, we shall obtain a useful explicit formula for the remainder, as in the single-variable theorem.

PROOF OF THEOREM 3 Let $g(t) = f(\mathbf{x}_0 + t\mathbf{h})$ with $\mathbf{x}_0$ and $\mathbf{h}$ fixed, which is a C^3 function of t. Now apply the single-variable Taylor theorem (1) to g, with $k = 2$ to obtain

$$\left.\begin{aligned} g(1) &= g(0) + g'(0) + \frac{g''(0)}{2!} + R_2, \\ \text{where} \qquad R_2 &= \int_0^1 \frac{(t-1)^2}{2!} g'''(t)\,dt. \end{aligned}\right\} \tag{3}$$

By the chain rule,

$$g'(t) = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(\mathbf{x}_0 + t\mathbf{h})\mathbf{h}_i; \qquad g''(t) = \sum_{i,j=1}^{n} \frac{\partial^2 f}{\partial x_i\,\partial x_j}(\mathbf{x}_0 + t\mathbf{h})h_i h_j,$$

and

$$g'''(t) = \sum_{i,j,k=1}^{n} \frac{\partial^3 f}{\partial x_i\,\partial x_j\,\partial x_k}(\mathbf{x}_0 + t\mathbf{h})h_i h_j h_k.$$

Writing $R_2 = R_2(\mathbf{x}_0, \mathbf{h})$ we have thus proved:

$$\left.\begin{aligned} f(\mathbf{x}_0+\mathbf{h}) &= f(\mathbf{x}_0)+\sum_{i=1}^{n} h_i \frac{\partial f}{\partial x_i}(\mathbf{x}_0)+\frac{1}{2}\sum_{i,j=1}^{n} h_i h_j \frac{\partial^2 f}{\partial x_i\,\partial x_j}(\mathbf{x}_0)+R_2(\mathbf{x}_0,\mathbf{h}), \\ &\text{where} \\ R_2(\mathbf{x}_0,\mathbf{h}) &= \sum_{i,j,k=1}^{n}\int_0^1 \frac{(t-1)^2}{2}\frac{\partial^3 f}{\partial x_i\,\partial x_j\,\partial x_k}(\mathbf{x}_0+t\mathbf{h})h_i h_j h_k\,dt. \end{aligned}\right\} \tag{4}$$

The integrand is a continuous function of t and is therefore bounded by a positive constant C on a small neighborhood of $\mathbf{x}_0$ (because it has to be close to its value at $\mathbf{x}_0$). Also note that $|h_i| \le \|\mathbf{h}\|$, for $\|\mathbf{h}\|$ small, and so

$$|R_2(\mathbf{x}_0,\mathbf{h})| \le \|\mathbf{h}\|^3 C.$$

In particular,

$$\frac{|R_2(\mathbf{x}_0,\mathbf{h})|}{\|\mathbf{h}\|^2} \le \|\mathbf{h}\|C \to 0 \qquad \text{as} \qquad \mathbf{h}\to\mathbf{0},$$

as required by the theorem.

The proof of Theorem 2 follows analogously from the Taylor formula (1) with $k = 1$. A similar argument for R_1 shows that $|R_1(\mathbf{x}_0,\mathbf{h})|/\|\mathbf{h}\| \to 0$ as $\mathbf{h}\to\mathbf{0}$, although this also follows directly from the definition of differentiability. ■

Forms of the Remainder In Theorem 2,

$$R_1(\mathbf{x}_0,\mathbf{h}) = \sum_{i,j=1}^{n}\int_0^1 (1-t)\frac{\partial^2 f}{\partial x_i\partial x_j}(\mathbf{x}_0+t\mathbf{h})h_i h_j\,dt = \sum_{i,j=1}^{n}\frac{1}{2}\frac{\partial^2 f}{\partial x_i\partial x_j}(\mathbf{c}_{ij})h_i h_j, \tag{5}$$

where $\mathbf{c}_{ij}$ lies somewhere on the line joining $\mathbf{x}_0$ to $\mathbf{x}_0+\mathbf{h}$.

In Theorem 3,

$$\begin{aligned} R_2(\mathbf{x}_0,\mathbf{h}) &= \sum_{i,j,k=1}^{n}\int_0^1 \frac{(t-1)^2}{2}\frac{\partial^3 f}{\partial x_i\,\partial x_j\,\partial x_k}(\mathbf{x}_0+t\mathbf{h})h_i h_j h_k\,dt \\ &= \sum_{i,j,k=1}^{n}\frac{1}{3!}\frac{\partial^3 f}{\partial x_i\,\partial x_j\,\partial x_k}(\mathbf{c}_{ijk})h_i h_j h_k, \end{aligned} \tag{5'}$$

where $\mathbf{c}_{ijk}$ lies somewhere on the line joining $\mathbf{x}_0$ to $\mathbf{x}_0+\mathbf{h}$.

The formulas involving $\mathbf{c}_{ij}$ and $\mathbf{c}_{ijk}$ (called Lagrange's form of the remainder) are obtained by making use of the *second mean-value theorem for integrals*. This states that

$$\int_a^b h(t)g(t)\,dt = h(c)\int_a^b g(t)\,dt,$$

provided h and g are continuous and $g \geq 0$ on $[a, b]$; here c is some number between a and b.[4] This is applied in formula (5) for the explicit form of the remainder with $h(t) = (\partial^2 f/\partial x_i \partial x_j)(\mathbf{x}_0 + t\mathbf{h})$ and $g(t) = 1 - t$.

The third-order Taylor formula is

$$f(\mathbf{x}_0 + \mathbf{h}) = f(\mathbf{x}_0) + \sum_{i=1}^{n} h_i \frac{\partial f}{\partial x_i}(\mathbf{x}_0) + \frac{1}{2}\sum_{i,j=1}^{n} h_i h_j \frac{\partial^2 f}{\partial x_i\,\partial x_j}(\mathbf{x}_0)$$
$$+ \frac{1}{3!}\sum_{i,j,k=1}^{n} h_i h_j h_k \frac{\partial^3 f}{\partial x_i\,\partial x_j\,\partial x_k}(\mathbf{x}_0) + R_3(\mathbf{x}_0, \mathbf{h}),$$

where $R_3(\mathbf{x}_0, \mathbf{h})/\|\mathbf{h}\|^3 \to 0$ as $\mathbf{h} \to \mathbf{0}$, and so on. The general formula can be proved by induction, using the method of proof already given.

EXAMPLE 1 Compute the second-order Taylor formula for the function $f(x, y) = \sin(x + 2y)$, about the point $\mathbf{x}_0 = (0, 0)$.

SOLUTION Notice that

$$f(0, 0) = 0,$$
$$\frac{\partial f}{\partial x}(0, 0) = \cos(0 + 2 \cdot 0) = 1, \qquad \frac{\partial f}{\partial y}(0, 0) = 2\cos(0 + 2 \cdot 0) = 2,$$
$$\frac{\partial^2 f}{\partial x^2}(0, 0) = 0, \qquad \frac{\partial^2 f}{\partial y^2}(0, 0) = 0, \qquad \frac{\partial^2 f}{\partial x\,\partial y}(0, 0) = 0.$$

Thus,

$$f(\mathbf{h}) = f(h_1, h_2) = h_1 + 2h_2 + R_2(\mathbf{0}, \mathbf{h}),$$

---

[4] *Proof* If $g = 0$, the result is clear, so we can suppose $g \neq 0$; thus, we can assume $\int_a^b g(t)\,dt > 0$. Let M and m be the maximum and minimum values of h, achieved at t_M and t_m, respectively. Because $g(t) \geq 0$,

$$m\int_a^b g(t)\,dt \leq \int_a^b h(t)g(t)\,dt \leq M\int_a^b g(t)\,dt.$$

Thus, $\left(\int_a^b h(t)g(t)\,dt\right)\Big/\left(\int_a^b g(t)\,dt\right)$ lies between $m = h(t_m)$ and $M = h(t_M)$ and therefore, by the intermediate-value theorem, equals $h(c)$ for some intermediate c. ■

where

$$\frac{R_2(\mathbf{0}, \mathbf{h})}{\|\mathbf{h}\|^2} \to 0 \qquad \text{as} \qquad \mathbf{h} \to \mathbf{0}. \quad \blacktriangle$$

EXAMPLE 2 Compute the second-order Taylor formula for $f(x, y) = e^x \cos y$ about the point $x_0 = 0$, $y_0 = 0$.

SOLUTION Here

$$f(0, 0) = 1, \qquad \frac{\partial f}{\partial x}(0, 0) = 1, \qquad \frac{\partial f}{\partial y}(0, 0) = 0,$$

$$\frac{\partial^2 f}{\partial x^2}(0, 0) = 1, \qquad \frac{\partial^2 f}{\partial y^2}(0, 0) = -1, \qquad \frac{\partial^2 f}{\partial x\, \partial y}(0, 0) = 0,$$

and so

$$f(\mathbf{h}) = f(h_1, h_2) = 1 + h_1 + \tfrac{1}{2}h_1^2 - \tfrac{1}{2}h_2^2 + R_2(\mathbf{0}, \mathbf{h}),$$

where

$$\frac{R_2(\mathbf{0}, \mathbf{h})}{\|\mathbf{h}\|^2} \to 0 \qquad \text{as} \qquad \mathbf{h} \to \mathbf{0}. \quad \blacktriangle$$

In the case of functions of one variable, one can expand $f(x)$ in an infinite power series, called the ***Taylor series***:

$$f(x_0 + h) = f(x_0) + f'(x_0)h + \frac{f''(x_0)h^2}{2} + \cdots + \frac{f^{(k)}(x_0)h^k}{k!} + \cdots,$$

provided one can show that $R_k(x_0, h) \to 0$ as $k \to \infty$. Similarly, for functions of several variables the preceding terms are replaced by the corresponding ones involving partial derivatives, as we have seen in Theorem 3. Again, one can represent such a function by its Taylor series provided one can show that $R_k \to 0$ as $k \to \infty$. This point is examined further in Exercise 7.

EXAMPLE 3 Find the first- and second-order Taylor approximations to $f(x, y) = \sin(xy)$ at the point $(x_0, y_0) = (1, \pi/2)$.

SOLUTION Here

$$f(x_0, y_0) = \sin(x_0 y_0) = \sin(\pi/2) = 1$$

$$f_x(x_0, y_0) = y_0 \cos(x_0 y_0) = \frac{\pi}{2} \cos(\pi/2) = 0$$

$$f_y(x_0, y_0) = x_0 \cos(x_0 y_0) = \cos(\pi/2) = 0$$

$$f_{xx}(x_0, y_0) = -y_0^2 \sin(x_0 y_0) = -\frac{\pi^2}{4} \sin(\pi/2) = -\pi^2/4$$

$$f_{xy}(x_0, y_0) = \cos(x_0 y_0) - x_0 y_0 \sin(x_0 y_0) = -\frac{\pi}{2} \sin(\pi/2) = -\pi/2$$

$$f_{yy}(x_0, y_0) = -x_0^2 \sin(x_0 y_0) = -\sin(\pi/2) = -1.$$

Thus, the linear (first-order) approximation is

$$\begin{aligned} l(x, y) &= f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) \\ &= 1 + 0 + 0 = 1, \end{aligned}$$

and the second-order (or quadratic) approximation is

$$\begin{aligned} g(x, y) &= 1 + 0 + 0 + \frac{1}{2}\left(-\frac{\pi^2}{4}\right)(x-1)^2 + \left(-\frac{\pi}{2}\right)(x-1)\left(y - \frac{\pi}{2}\right) \\ &\quad + \frac{1}{2}(-1)\left(y - \frac{\pi}{2}\right)^2 \\ &= 1 - \frac{\pi^2}{8}(x-1)^2 - \frac{\pi}{2}(x-1)\left(y - \frac{\pi}{2}\right) - \frac{1}{2}\left(y - \frac{\pi}{2}\right)^2. \end{aligned}$$

See Figure 3.2.1. ▲

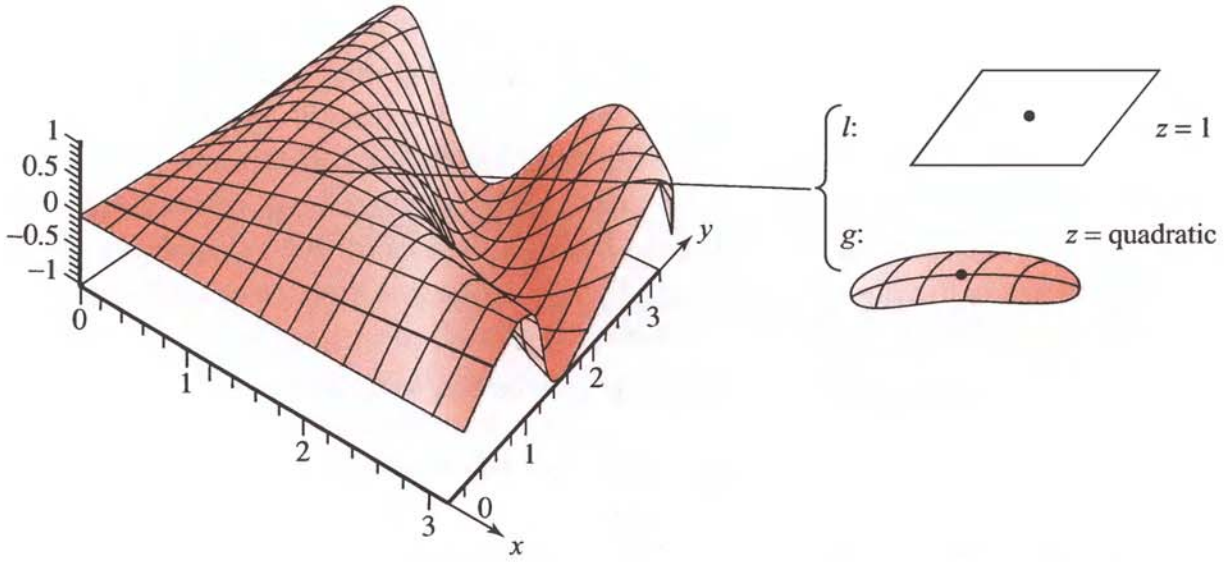


Figure 3.2.1 The linear and quadratic approximations to $z = \sin(xy)$ near $(1, \pi/2)$.

EXAMPLE 4 Find linear and quadratic approximations to the expression $(3.98 - 1)^2/(5.97 - 3)^2$. Compare with the exact value.

SOLUTION Let $f(x, y) = (x-1)^2/(y-3)^2$. The desired expression is close to $f(4, 6) = 1$. To find the approximations, we differentiate:

$$f_x = \frac{2(x-1)}{(y-3)^2}, \qquad f_y = \frac{-2(x-1)^2}{(y-3)^3}$$

$$f_{xy} = f_{yx} = \frac{-4(x-1)}{(y-3)^3}, \qquad f_{xx} = \frac{2}{(y-3)^2}, \qquad f_{yy} = \frac{6(x-1)^2}{(y-3)^4}.$$

At the point of approximation, we have

$$f_x(4, 6) = \frac{2}{3}, \qquad f_y = -\frac{2}{3}, \qquad f_{xy} = f_{yx} = -\frac{4}{9}, \qquad f_{xx} = \frac{2}{9}, \qquad f_{yy} = \frac{2}{3}.$$

The linear approximation is then

$$1 + \frac{2}{3}(-0.02) - \frac{2}{3}(-0.03) = 1.00666.$$

The quadratic approximation is

$$1 + \frac{2}{3}(-0.02) - \frac{2}{3}(-0.03) + \frac{2}{9}\frac{(-0.02)^2}{2} - \frac{4}{9}(-0.02)(-0.03) + \frac{2}{3}\frac{(-0.03)^2}{2}$$
$$= 1.00674.$$

The "exact" value using a calculator is 1.00675. ▲

EXERCISES

In each of Exercises 1 to 6, determine the second-order Taylor formula for the given function about the given point (x_0, y_0).

1. $f(x, y) = (x + y)^2$, where $x_0 = 0, y_0 = 0$

2. $f(x, y) = 1/(x^2 + y^2 + 1)$, where $x_0 = 0, y_0 = 0$

3. $f(x, y) = e^{x+y}$, where $x_0 = 0, y_0 = 0$

4. $f(x, y) = e^{-x^2-y^2}\cos(xy)$, where $x_0 = 0, y_0 = 0$

5. $f(x, y) = \sin(xy) + \cos(xy)$, where $x_0 = 0, y_0 = 0$

6. $f(x, y) = e^{(x-1)^2}\cos y$, where $x_0 = 1, y_0 = 0$

7. (Challenging) A function $f\colon \mathbb{R} \to \mathbb{R}$ is called an ***analytic*** function provided

$$f(x + h) = f(x) + f'(x)h + \cdots + \frac{f^{(k)}(x)}{k!}h^k + \cdots$$

[i.e., the series on the right-hand side converges and equals $f(x + h)$].

(a) Suppose f satisfies the following condition: On any closed interval $[a, b]$, there is a constant M such that for all $k = 1, 2, 3, \ldots$, $|f^{(k)}(x)| \leq M^k$ for all $x \in [a, b]$. Prove that f is analytic.

(b) Let $f(x) = \begin{cases} e^{-1/x} & x > 0 \\ 0 & x \leq 0. \end{cases}$

Show that f is a C^∞ function, but f is not analytic.

(c) Give a definition of analytic functions from $\mathbb{R}^n$ to $\mathbb{R}$. Generalize the proof of part (a) to this class of functions.

(d) Develop $f(x, y) = e^{x+y}$ in a power series about $x_0 = 0$, $y_0 = 0$.

3.3 Extrema of Real-Valued Functions

Historical Note

As we saw in the book's Historical introduction, the early Greeks sought to mathematize nature and to find, as in the geometric Ptolemaic model of planetary motion, mathematical laws governing the universe. With the revival of Greek learning during the Renaissance, this point of view again took hold and the search for these laws recommenced. In particular, the question was raised as to whether there was *one* law, one mathematical principle that governed and superseded all others, a principle that the Creator used in His Grand Design of the Universe.

MAUPERTUIS'S PRINCIPLE. In 1744, the French scientist Pierre-Louis de Maupertuis (see Figure 3.3.1) put forth his grand scheme of the world. The "metaphysical principle" of Maupertuis is the assumption that nature always operates with the greatest possible economy. In short, physical laws are a consequence of a principle of "economy of means"; nature always acts in such a way as to minimize some quantity that Maupertuis called the *action*. Action was nothing more than the expenditure of energy over time, or energy × time. In applications, the type of energy changes with each case. For example, physical systems often try to "rearrange themselves" to have a minimum energy—such as a ball rolling from a mountain peak to a valley, or the primordial irregular Earth assuming a more nearly spherical shape. As another example, the spherical shape of soap bubbles is connected with the fact that spheres are the surfaces of least area containing a fixed volume.

We state Maupertuis's principle formally as: *Nature always minimizes action.* Maupertuis saw in this principle an expression of the wisdom of the Supreme Being, of God, according to which everything in nature is

Figure 3.3.1 Pierre-Louis de Maupertuis (1698–1759).

performed in the most economical way. He wrote:

> What satisfaction for the human spirit that, in contemplating these laws which contain the principle of motion and of rest for all bodies in the universe, he finds the proof of existence of Him who governs the world.

Maupertuis indeed believed that he had discovered God's fundamental law, the very secret of Creation itself, but he was actually not the first person to pose this principle.

In 1707, Leibniz wrote down the principle of least action in a letter to Johann Bernoulli, which became lost until 1913, when it was discovered in Germany's Gotha library. For Leibniz, this principle was a natural outgrowth of his great philosophical treatise *The Theodicy*, in which he argues that God may indeed think of all possible worlds, but would want to

create only the best among them; and hence our world is necessarily the *best of all possible worlds*.

Action, as defined by Leibniz, was motivated by the following reasoning, used in his letter. Think of a hiker walking along a road, and consider how to describe his action. If he travels 2 kilometers in 1 hour, you would say that he has carried out twice as much action as he would if he traveled 2 kilometers in 2 hours. However, you would also say that he carries out twice as much action in traveling 2 kilometers in 2 hours as he would in traveling 1 kilometer in 1 hour. Altogether then, our hiker, by walking 2 kilometers in 1 hour, carries out 4 times as much action as he would in traveling 1 kilometer in 1 hour.

Using this intuitive idea, Maupertuis defined action as the product of distance, velocity, and mass:

$$\text{Action} = \text{Mass} \times \text{Distance} \times \text{Velocity}.$$

Mass is included in this definition to account for the hiker's backpack. Moreover, according to Leibniz, the kinetic energy E is given by the formula:

$$E = \frac{1}{2} \times \text{Mass} \times (\text{Velocity})^2.$$

So action has the same physical dimension as

$$\text{Energy} \times \text{Time},$$

because velocity is distance divided by time.

Principle of Least Action. In the 250 years after Maupertuis formulated his principle, this *principle of least action* has been found to be a "theoretical basis" for Newton's law of gravity, Maxwell's equations for electromagnetism, Schrödinger's equation of quantum mechanics, and Einstein's field equation in general relativity.

Max Planck (see Figure 3.3.2), one of the greatest scientists of the modern era and the discoverer of the "quantization" of nature, was also a profound believer in the mathematical design of the universe. On June 29, 1922, on "Leibniz Day" in Berlin, Germany, just a few years after World War I, with all its terrible carnage, Planck delivered an address honoring this great scholar.

What follows are some excerpts from Planck's remarks:

> *The Theodicy* culminates with the statement that whatever occurs in our world, in the large as in the small, in nature as in spiritual life, is once and for all regulated by divine reason, and in such a way that our world is the best among possible worlds.
>
> Would Leibniz reaffirm this statement even today, in view of the misery of the present time, in view of the bitter failure of many efforts not immediately aimed at material gain, in view of the

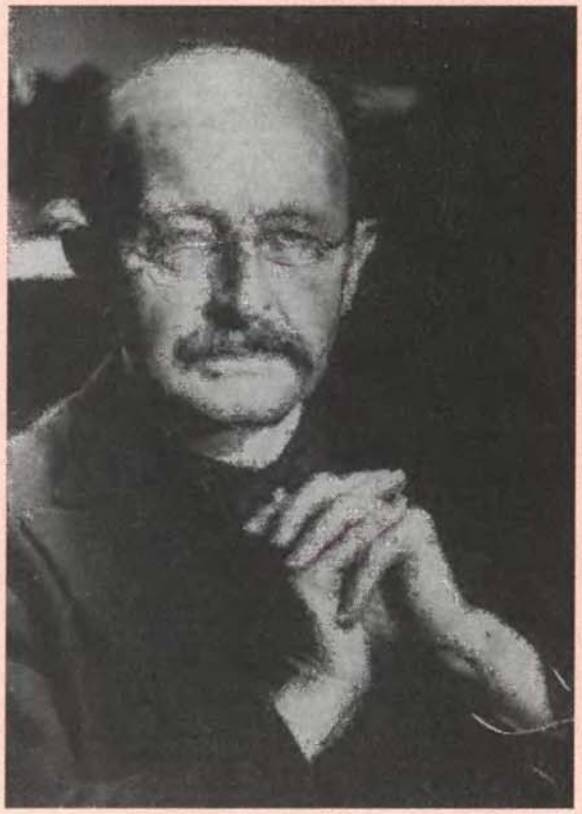


Figure 3.3.2 Max Planck (1858–1947).

undeniable fact that the imagined general harmony of people today seems to be further away from its realization than ever? No doubt, we should have to answer this question in the affirmative, even if we did not know that Leibniz never ceased to earnestly occupy himself until his last years despite an adverse fate and disappointments of all kinds, and we shall hardly err in assuming that it was exactly the *Theodicy* that gave him support and comfort in the most sorrowful days of his life. This once again is a touching example of the old truth that our most profound and most sacred principles are firmly rooted in our innermost soul, independent of experiences in the outer world.

Modern science, in particular under the influence of the development of the notion of causality, has moved far away from Leibniz's teleological point of view. Science has abandoned the assumption of a special, anticipating reason, and it considers each event in the natural and spiritual world, at least in principle, as reducible to prior states. But still we notice a fact, particularly in the most exact science, which, at least in this context, is most surprising. Present-day physics, as far as it is theoretically organized, is completely governed by a system of space–time differential equations which state that each process in nature is totally determined by the events which occur in its immediate temporal and spatial neighborhood. This entire rich system of differential equations, though they differ in detail, since they refer to mechanical, electric, magnetic, and thermal processes, is now

completely contained in a single dictum—*the principle of least action*. This, in short, states that, of all possible processes, the only ones that actually occur are those that involve minimum expenditure of action. As we can see, only a short step is required to recognize in the preference for the smallest quantity of action the ruling of divine reason, and thus to discover a part of Leibniz's teleological ordering of the universe.[5]

In present-day physics the principle of least action plays a relatively minor role. It does not quite fit into the framework of present theories. Of course, admittedly it is a correct statement; yet usually it serves not as the foundation of the theory, but as a true but dispensable appendix, because present theoretical physics is entirely tailored to the principle of infinitesimal local effects, and sees extensions to larger spaces and times as an unnecessary and uneconomical complication of the method of treatment. Hence, physics is inclined to view the principle of least action more as a formal and accidental curiosity than as a pillar of physical knowledge.

There is much more to the story of the least action principle, which we will revisit in Section 4.1.

[5]For more information and history, consult S. Hildebrandt and A. J. Tromba, *The Parsimonious Universe: Shape and Form in the Natural World*, Springer-Verlag, New York/Berlin, 1995.

Maxima and Minima for Functions of n-Variables

As the previous remarks show, for Leibniz, Euler, and Maupertuis, and for much of modern science as well, all in nature is a consequence of some maximum or minimum principle. To make such grand schemes—as well as some that are more down to earth—effective, one must first learn the techniques of how to find maxima and minima of functions of n variables.

Extreme Points

Among the most basic geometric features of the graph of a function are its extreme points, at which the function attains its greatest and least values. In this section, we derive a method for determining these points. In fact, the method locates local extrema as well. These are points at which the function attains a maximum or minimum value relative only to nearby points. Let us begin by defining our terms.

DEFINITION If $f: U \subset \mathbb{R}^n \to \mathbb{R}$ is a given scalar function, a point $\mathbf{x}_0 \in U$ is called a ***local minimum*** of f if there is a neighborhood V of $\mathbf{x}_0$ such that for all points $\mathbf{x}$ in V, $f(\mathbf{x}) \geq f(\mathbf{x}_0)$. (See Figure 3.3.3.) Similarly, $\mathbf{x}_0 \in U$ is a ***local maximum*** if there is a neighborhood V of $\mathbf{x}_0$ such that $f(\mathbf{x}) \leq f(\mathbf{x}_0)$ for all $\mathbf{x} \in V$. The point

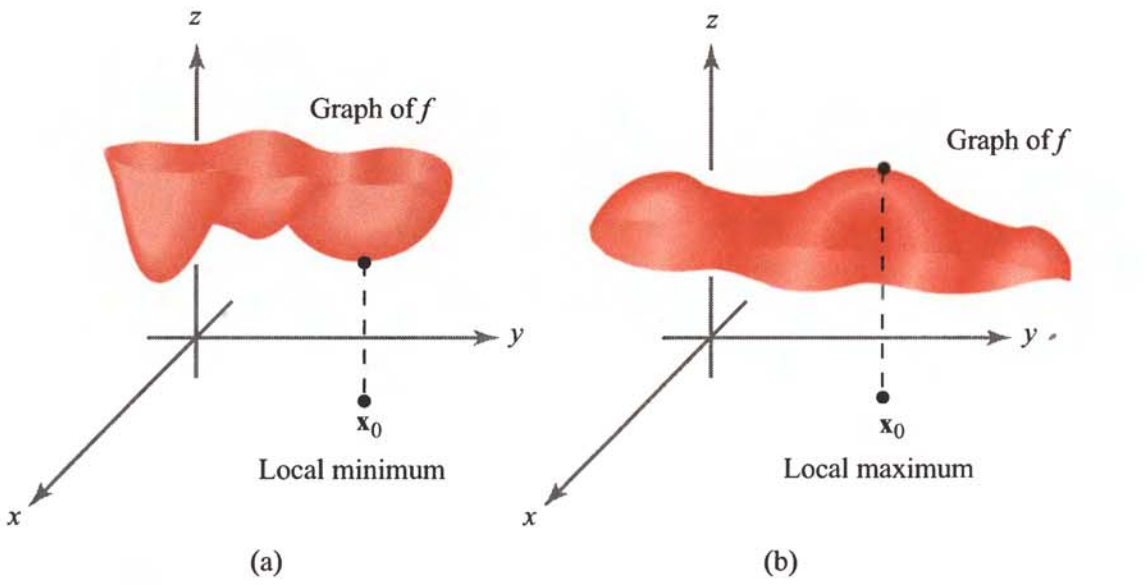


Figure 3.3.3 (a) Local minimum and (b) local maximum points for a function of two variables.

$\mathbf{x}_0 \in U$ is said to be a ***local***, or ***relative***, ***extremum*** if it is either a local minimum or a local maximum. A point $\mathbf{x}_0$ is a ***critical point*** of f if either f is *not differentiable at* $\mathbf{x}_0$, or if it is, $\mathbf{D}f(\mathbf{x}_0) = \mathbf{0}$. A critical point that is not a local extremum is called a ***saddle point***.[6]

First Derivative Test for Local Extrema

The location of extrema is based on the following fact, which should be familiar from one-variable calculus (the case $n = 1$): *Every extremum is a critical point.*

THEOREM 4: First Derivative Test for Local Extrema If $U \subset \mathbb{R}^n$ is open, the function $f: U \subset \mathbb{R}^n \to \mathbb{R}$ is differentiable, and $\mathbf{x}_0 \in U$ is a local extremum, then $\mathbf{D}f(\mathbf{x}_0) = \mathbf{0}$; that is, $\mathbf{x}_0$ is a critical point of f.

PROOF Suppose that f achieves a local maximum at $\mathbf{x}_0$. Then for any $\mathbf{h} \in \mathbb{R}^n$, the function $g(t) = f(\mathbf{x}_0 + t\mathbf{h})$ has a local maximum at $t = 0$. Thus, from one-variable calculus $g'(0) = 0$.[7] On the other hand, by the chain rule,

$$g'(0) = [\mathbf{D}f(\mathbf{x}_0)]\mathbf{h}.$$

Thus, $[\mathbf{D}f(\mathbf{x}_0)]\mathbf{h} = 0$ for every $\mathbf{h}$, and so $\mathbf{D}f(\mathbf{x}_0) = \mathbf{0}$. The case in which f achieves a local minimum at $\mathbf{x}_0$ is entirely analogous. ■

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[6]The term "saddle point" is sometimes not used this generally; we shall discuss saddle points further in the subsequent development.

[7]Recall the proof from one-variable calculus: Because $g(0)$ is a local maximum, $g(t) \le g(0)$ for small $t > 0$, so $g(t) - g(0) \le 0$, and hence $g'(0) = \lim_{t \to 0^+} (g(t) - g(0))/t \le 0$, where $\lim_{t \to 0^+}$ means the limit as $t \to 0$, $t > 0$. For small $t < 0$, we similarly have $g'(0) = \lim_{t \to 0^-} (g(t) - g(0))/t \ge 0$. Therefore, $g'(0) = 0$.

If we remember that $\mathbf{D}f(\mathbf{x}_0) = \mathbf{0}$ means that all the components of $\mathbf{D}f(\mathbf{x}_0)$ are zero, we can rephrase the result of Theorem 4: If $\mathbf{x}_0$ is a local extremum, then

$$\frac{\partial f}{\partial x_i}(\mathbf{x}_0) = 0, \qquad i = 1, \ldots, n;$$

that is, each partial derivative is zero at $\mathbf{x}_0$. In other words, $\nabla f(\mathbf{x}_0) = \mathbf{0}$, where ∇f is the gradient of f.

If we seek to find the extrema or local extrema of a function, then Theorem 4 states that we should look among the critical points. Sometimes these can be tested by inspection, but usually we use tests (to be developed below) analogous to the second-derivative test in one-variable calculus.

EXAMPLE 1 Find the maxima and minima of the function $f\colon \mathbb{R}^2 \to \mathbb{R}$, defined by $f(x, y) = x^2 + y^2$. (Ignore the fact that this example can be done by inspection.)

SOLUTION We first identify the critical points of f by solving the two equations $\partial f/\partial x = 0$ and $\partial f/\partial y = 0$, for x and y. But

$$\frac{\partial f}{\partial x} = 2x \qquad \text{and} \qquad \frac{\partial f}{\partial y} = 2y,$$

so the only critical point is the origin (0, 0), where the value of the function is zero. Because $f(x, y) \geq 0$, this point is a relative minimum—in fact, an absolute, or global, minimum—of f. Because (0, 0) is the only critical point, there are no maxima. ▲

EXAMPLE 2 Consider the function $f\colon \mathbb{R}^2 \to \mathbb{R}$, $(x, y) \mapsto x^2 - y^2$. Ignoring for the moment that this function has a saddle and no extrema, apply the method of Theorem 4 for the location of extrema.

SOLUTION As in Example 1, we find that f has only one critical point, at the origin, and the value of f there is zero. Examining values of f directly for points near the origin, we see that $f(x, 0) \geq f(0, 0)$ and $f(0, y) \leq f(0, 0)$, with strict inequalities when $x \neq 0$ and $y \neq 0$. Because x or y can be taken arbitrarily small, the origin cannot be either a relative minimum or a relative maximum (so it is a saddle point). Therefore, this function can have no relative extrema (see Figure 3.3.4). ▲

EXAMPLE 3 Find all the critical points of $z = x^2y + y^2x$.

SOLUTION Differentiating, we obtain

$$\frac{\partial z}{\partial x} = 2xy + y^2, \qquad \frac{\partial z}{\partial y} = 2xy + x^2.$$

Equating the partial derivatives to zero yields

$$2xy + y^2 = 0, \qquad 2xy + x^2 = 0.$$

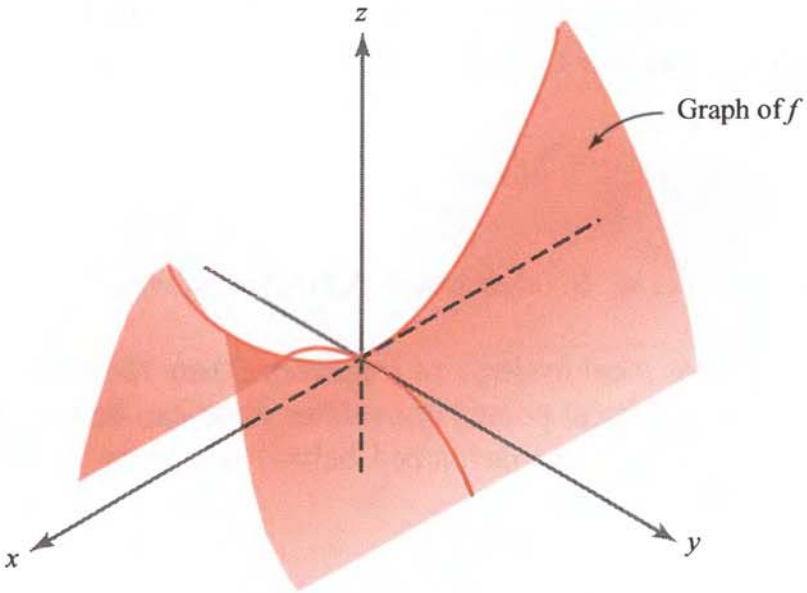

Figure 3.3.4 A function of two variables with a saddle point.

Subtracting, we obtain $x^2 = y^2$. Thus, $x = \pm y$. Substituting $x = +y$ in the first of the two preceding equations, we find that

$$2y^2 + y^2 = 3y^2 = 0,$$

so that $y = 0$ and thus $x = 0$. If $x = -y$, then

$$-2y^2 + y^2 = -y^2 = 0,$$

so $y = 0$ and therefore $x = 0$. Hence, the only critical point is $(0, 0)$. For $x = y$, $z = 2x^3$, which is both positive and negative for x near zero. Thus, $(0, 0)$ is not a relative extremum. ▲

EXAMPLE 4 Refer to Figure 3.3.5, a computer-drawn graph of the function $z = 2(x^2 + y^2)\, e^{-x^2-y^2}$. Where are the critical points?

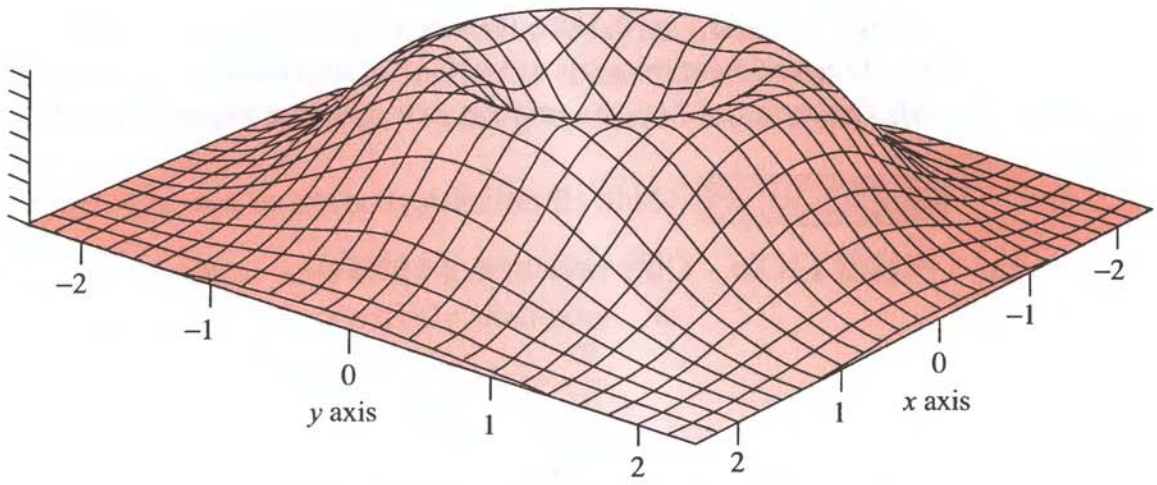

Figure 3.3.5 The volcano: $z = 2(x^2 + y^2)\exp(-x^2 - y^2)$.

SOLUTION Because $z = 2(x^2 + y^2)e^{-x^2-y^2}$, we have

$$\begin{aligned}\frac{\partial z}{\partial x} &= 4x(e^{-x^2-y^2}) + 2(x^2 + y^2)e^{-x^2-y^2}(-2x) \\ &= e^{-x^2-y^2}[4x - 4x(x^2 + y^2)] \\ &= 4x(e^{-x^2-y^2})(1 - x^2 - y^2)\end{aligned}$$

and

$$\frac{\partial z}{\partial y} = 4y(e^{-x^2-y^2})(1 - x^2 - y^2).$$

These both vanish when $x = y = 0$ or when $x^2 + y^2 = 1$. This is consistent with the figure: Points on the crater's rim are maxima and the origin is a minimum. ▲

Second Derivative Test for Local Extrema

The remainder of this section is devoted to deriving a criterion, depending on the second derivative, for a critical point to be a relative extremum. In the special case $n = 1$, our criterion will reduce to the familiar condition from one-variable calculus: $f''(x_0) > 0$ for a minimum and $f''(x_0) < 0$ for a maximum. But in the general context, the second derivative is a fairly complicated mathematical object. To state our criterion, we will introduce a version of the second derivative called the Hessian, which in turn is related to quadratic functions. ***Quadratic functions*** are functions $g\colon \mathbb{R}^n \to \mathbb{R}$ that have the form

$$g(h_1, \ldots, h_n) = \sum_{i,j=1}^{n} a_{ij} h_i h_j$$

for a $n \times n$ matrix $[a_{ij}]$. In terms of matrix multiplication, we can write

$$g(h_1, \ldots, h_n) = [h_1 \cdots h_n] \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} h_1 \\ \vdots \\ h_n \end{bmatrix}.$$

For example, if $n = 3$,

$$g(h_1, h_2, h_3) = h_1^2 - 2h_1h_2 + h_3^2 = [h_1\, h_2\, h_3] \begin{bmatrix} 1 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} h_1 \\ h_2 \\ h_3 \end{bmatrix}$$

is a quadratic function.

We can, if we wish, assume that $[a_{ij}]$ is symmetric; in fact, g is unchanged if we replace $[a_{ij}]$ by the symmetric matrix $[b_{ij}]$, where $b_{ij} = \frac{1}{2}(a_{ij} + a_{ji})$, because

$h_i h_j = h_j h_i$ and the sum is over all i and j. The *quadratic* nature of g is reflected in the identity

$$g(\lambda h_1, \ldots, \lambda h_n) = \lambda^2 g(h_1, \ldots, h_n),$$

which follows from the definition.

Now we are ready to define Hessian functions (named after Ludwig Otto Hesse, who introduced them in 1844).

DEFINITION Suppose that $f\colon U \subset \mathbb{R}^n \to \mathbb{R}$ has second-order continuous derivatives $(\partial^2 f/\partial x_i\, \partial x_j)(\mathbf{x}_0)$, for $i, j = 1, \ldots, n$, at a point $\mathbf{x}_0 \in U$. The ***Hessian of f at*** $\mathbf{x}_0$ is the quadratic function defined by

$$Hf(\mathbf{x}_0)(\mathbf{h}) = \frac{1}{2}\sum_{i,j=1}^{n} \frac{\partial^2 f}{\partial x_i\, \partial x_j}(\mathbf{x}_0) h_i h_j$$

$$= \frac{1}{2}(h_1, \ldots, h_n) \begin{pmatrix} \dfrac{\partial^2 f}{\partial x_1\, \partial x_1} & \cdots & \dfrac{\partial^2 f}{\partial x_1\, \partial x_n} \\ \vdots & & \\ \dfrac{\partial^2 f}{\partial x_n\, \partial x_1} & \cdots & \dfrac{\partial^2 f}{\partial x_n\, \partial x_n} \end{pmatrix} \begin{pmatrix} h_1 \\ \vdots \\ h_n \end{pmatrix}.$$

Notice that, by equality of mixed partials, the second derivative matrix is symmetric.

This function is usually used at critical points $\mathbf{x}_0 \in U$. In this case, $\mathbf{D}f(\mathbf{x}_0) = \mathbf{0}$, so the Taylor formula (see Theorem 2, Section 3.2) may be written in the form

$$f(\mathbf{x}_0 + \mathbf{h}) = f(\mathbf{x}_0) + Hf(\mathbf{x}_0)(\mathbf{h}) + R_2(\mathbf{x}_0, \mathbf{h}).$$

Thus, *at a critical point the Hessian equals the first nonconstant term in the Taylor series of f.*

A quadratic function $g\colon \mathbb{R}^n \to \mathbb{R}$ is called ***positive-definite*** if $g(\mathbf{h}) \geq 0$ for all $\mathbf{h} \in \mathbb{R}^n$ and $g(\mathbf{h}) = 0$ only for $\mathbf{h} = \mathbf{0}$. Similarly, g is ***negative-definite*** if $g(\mathbf{h}) \leq 0$ and $g(\mathbf{h}) = 0$ for $\mathbf{h} = \mathbf{0}$ only. Note that if $n = 1$, $Hf(x_0)(h) = \frac{1}{2} f''(x_0) h^2$, which is positive-definite if and only if $f''(x_0) > 0$.

THEOREM 5: Second Derivative Test for Local Extrema If $f\colon U \subset \mathbb{R}^n \to \mathbb{R}$ is of class C^3, $\mathbf{x}_0 \in U$ is a critical point of f, and the Hessian $Hf(\mathbf{x}_0)$ is positive-definite, then $\mathbf{x}_0$ is a relative minimum of f. Similarly, if $Hf(\mathbf{x}_0)$ is negative-definite, then $\mathbf{x}_0$ is a relative maximum.

Actually, we shall prove that the extrema given by this criterion are *strict*. A relative maximum $\mathbf{x}_0$ is said to be ***strict*** if $f(\mathbf{x}) < f(\mathbf{x}_0)$ for nearby $\mathbf{x} \neq \mathbf{x}_0$. A strict relative minimum is defined similarly. Also, the theorem is valid even if f is only C^2 but we have assumed C^3 for simplicity.

The proof of Theorem 5 requires Taylor's theorem and the following result from linear algebra.

LEMMA 1 If $B = [b_{ij}]$ is an $n \times n$ real matrix, and if the associated quadratic function

$$H\colon \mathbb{R}^n \to \mathbb{R}, (h_1, \ldots, h_n) \mapsto \frac{1}{2} \sum_{i,j=1}^{n} b_{ij} h_i h_j$$

is positive-definite, then there is a constant $M > 0$ such that for all $\mathbf{h} \in \mathbb{R}^n$,

$$H(\mathbf{h}) \geq M\|\mathbf{h}\|^2.$$

PROOF For $\|\mathbf{h}\| = 1$, set $g(\mathbf{h}) = H(\mathbf{h})$. Then g is a continuous function of $\mathbf{h}$ for $\|\mathbf{h}\| = 1$ and so achieves a minimum value, say M.[8] Because H is quadratic, we have

$$H(\mathbf{h}) = H\left(\frac{\mathbf{h}}{\|\mathbf{h}\|}\|\mathbf{h}\|\right) = H\left(\frac{\mathbf{h}}{\|\mathbf{h}\|}\right)\|\mathbf{h}\|^2 = g\left(\frac{\mathbf{h}}{\|\mathbf{h}\|}\right)\|\mathbf{h}\|^2 \geq M\|\mathbf{h}\|^2$$

for any $\mathbf{h} \neq \mathbf{0}$. (The result is obviously valid if $\mathbf{h} = \mathbf{0}$). ■

Note that the quadratic function associated with the symmetric matrix $\frac{1}{2}(\partial^2 f/\partial x_i\, \partial x_j)$ is exactly the Hessian.

PROOF OF THEOREM 5 Recall that if $f\colon U \subset \mathbb{R}^n \to \mathbb{R}$ is of class C^3 and $\mathbf{x}_0 \in U$ is a critical point, Taylor's theorem may be expressed in the form

$$f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0) = Hf(\mathbf{x}_0)(\mathbf{h}) + R_2(\mathbf{x}_0, \mathbf{h}),$$

where $(R_2(\mathbf{x}_0, \mathbf{h}))/\|\mathbf{h}\|^2 \to 0$ as $\mathbf{h} \to \mathbf{0}$.

Because $Hf(\mathbf{x}_0)$ is positive-definite, lemma 1 assures us of a constant $M > 0$ such that for all $\mathbf{h} \in \mathbb{R}^n$

$$Hf(\mathbf{x}_0)(\mathbf{h}) \geq M\|\mathbf{h}\|^2.$$

[8] Here we are using, without proof, a theorem analogous to a theorem in calculus that states that every continuous function on an interval $[a, b]$ achieves a maximum and a minimum; see Theorem 7.

Because $R_2(\mathbf{x}_0, \mathbf{h})/\|\mathbf{h}\|^2 \to 0$ as $\mathbf{h} \to \mathbf{0}$, there is a $\delta > 0$ such that for $0 < \|\mathbf{h}\| < \delta$

$$|R_2(\mathbf{x}_0, \mathbf{h})| < M\|\mathbf{h}\|^2.$$

Thus, $0 < Hf(\mathbf{x}_0)(\mathbf{h}) + \boldsymbol{R}_2(\mathbf{x}_0, \mathbf{h}) = f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0)$ for $0 < \|\mathbf{h}\| < \delta$, so that $\mathbf{x}_0$ is a relative minimum, in fact, a strict relative minimum.

The proof in the negative-definite case is similar, or else follows by applying the preceding to $-f$, and is left as an exercise. ■

EXAMPLE 5 Consider again the function $f\colon \mathbb{R}^2 \to \mathbb{R}$, $(x, y) \mapsto x^2 + y^2$. Then $(0, 0)$ is a critical point, and f is already in the form of Taylor's theorem:

$$f((0, 0) + (h_1, h_2)) = f(0, 0) + (h_1^2 + h_2^2) + 0.$$

We can see directly that the Hessian at $(0, 0)$ is

$$Hf(\mathbf{0})(\mathbf{h}) = h_1^2 + h_2^2,$$

which is clearly positive-definite. Thus, $(0, 0)$ is a relative minimum. This simple case can, of course, be done without calculus. Indeed, it is clear that $f(x, y) > 0$ for all $(x, y) \neq (0, 0)$. ▲

For functions of two variables $f(x, y)$, the Hessian may be written as follows:

$$Hf(x, y)(\mathbf{h}) = \tfrac{1}{2}[h_1, h_2] \begin{bmatrix} \dfrac{\partial^2 f}{\partial x^2} & \dfrac{\partial^2 f}{\partial y\, \partial x} \\ \dfrac{\partial^2 f}{\partial x\, \partial y} & \dfrac{\partial^2 f}{\partial y^2} \end{bmatrix} \begin{bmatrix} h_1 \\ h_2 \end{bmatrix}.$$

Now we shall give a useful criterion for when a quadratic function defined by such a 2×2 matrix is positive-definite. This will be useful in conjunction with Theorem 5.

LEMMA 2 Let

$$B = \begin{bmatrix} a & b \\ b & c \end{bmatrix} \quad \text{and} \quad H(\mathbf{h}) = \tfrac{1}{2}[h_1, h_2]B \begin{bmatrix} h_1 \\ h_2 \end{bmatrix}.$$

Then $H(\mathbf{h})$ is positive-definite if and only if $a > 0$ and $\det B = ac - b^2 > 0$.

PROOF We have

$$H(\mathbf{h}) = \tfrac{1}{2}[h_1, h_2] \begin{bmatrix} ah_1 + bh_2 \\ bh_1 + ch_2 \end{bmatrix} = \tfrac{1}{2}(ah_1^2 + 2bh_1h_2 + ch_2^2).$$

Let us complete the square, writing

$$H(\mathbf{h}) = \tfrac{1}{2}a\left(h_1 + \frac{b}{a}h_2\right)^2 + \frac{1}{2}\left(c - \frac{b^2}{a}\right)h_2^2.$$

Suppose H is positive-definite. Setting $h_2 = 0$, we see that $a > 0$. Setting $h_1 = -(b/a)h_2$, we get $c - b^2/a > 0$ or $ac - b^2 > 0$. Conversely, if $a > 0$ and $c - b^2/a > 0$, $H(\mathbf{h})$ is a sum of squares, so that $H(\mathbf{h}) \geq 0$. If $H(\mathbf{h}) = 0$, then each square must be zero. This implies that both h_1 and h_2 must be zero, so that $H(\mathbf{h})$ is positive-definite. ■

Similarly, one can see that $H(\mathbf{h})$ is negative-definite if and only if $a < 0$ and $ac - b^2 > 0$. We note that an alternative formulation is that $H(\mathbf{h})$ is positive- (respectively, negative-) definite if $a + c = \text{trace } B > 0$ (respectively, < 0) and $\det B > 0$.

Determinant Test for Positive Definiteness

There are similar criteria for an $n \times n$ symmetric matrix B. Consider the n square submatrices along the diagonal (see Figure 3.3.6). B is positive-definite (that is, the quadratic function associated with B is positive-definite) if and only if the determinants of these diagonal submatrices are all greater than zero. For negative-definite B, the signs should be alternately < 0 and > 0. We shall not prove this general case here.[9] In case the determinants of the diagonal submatrices are all nonzero, but the matrix is not positive- or negative-definite, the critical point is of ***saddle type***; in this case, one can show that the point is neither a maximum nor a minimum in the manner of Example 2.

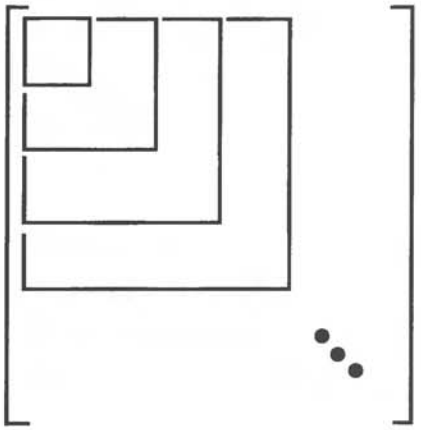

Figure 3.3.6 "Diagonal" submatrices are used in the criterion for positive definiteness; they must all have determinant > 0.

[9]This is proved in, for example, K. Hoffman and R. Kunze, *Linear Algebra*, Prentice Hall, Englewood Cliffs, N.J., 1961, pp. 249–251. For students with sufficient background in linear algebra, it should be noted that B is positive-definite when all of its eigenvalues (which are necessarily real, because B is symmetric) are positive.

Second Derivative Test

Lemma 2 and Theorem 5 imply the following result:

THEOREM 6: Second Derivative Maximum-Minimum Test for Functions of Two Variables Let $f(x, y)$ be of class C^3 on an open set U in $\mathbb{R}^2$. A point (x_0, y_0) is a (strict) local minimum of f provided the following three conditions hold:

(i) $\dfrac{\partial f}{\partial x}(x_0, y_0) = \dfrac{\partial f}{\partial y}(x_0, y_0) = 0$

(ii) $\dfrac{\partial^2 f}{\partial x^2}(x_0, y_0) > 0$

(iii) $D = \left(\dfrac{\partial^2 f}{\partial x^2}\right)\left(\dfrac{\partial^2 f}{\partial y^2}\right) - \left(\dfrac{\partial^2 f}{\partial x\, \partial y}\right)^2 > 0$ at (x_0, y_0)

(D is called the ***discriminant*** of the Hessian.) If in (ii) we have <0 instead of >0 and condition (iii) is unchanged, then we have a (strict) local maximum.

EXAMPLE 6 Classify the critical points of the function $f\colon \mathbb{R}^2 \to \mathbb{R}$, defined by $(x, y) \mapsto x^2 - 2xy + 2y^2$.

SOLUTION As in Example 5, we find that $f(0, 0) = 0$, the origin is the only critical point, and the Hessian is

$$Hf(\mathbf{0})(\mathbf{h}) = h_1^2 - 2h_1h_2 + 2h_2^2 = (h_1 - h_2)^2 + h_2^2,$$

which is clearly positive-definite. Thus, f has a relative minimum at $(0, 0)$. Alternatively, we can apply Theorem 6. At $(0, 0)$, $\partial^2 f/\partial x^2 = 2$, $\partial^2 f/\partial y^2 = 4$, and $\partial^2 f/\partial x\, \partial y = -2$. Conditions (i), (ii), and (iii) hold, so f has a relative minimum at $(0, 0)$ ▲

If $D < 0$ in Theorem 6, then we have a saddle point. In fact, one can prove that $f(x, y)$ is larger than $f(x_0, y_0)$ as we move away from (x_0, y_0) in some direction and smaller in the orthogonal direction (see Exercise 26). The general appearance is thus similar to that shown in Figure 3.3.4. The appearance of the graph near (x_0, y_0) in the case $D = 0$ must be determined by further analysis.

We summarize the procedure for dealing with functions of two variables: After all critical points have been found and their associated Hessians computed, some of these Hessians may be positive-definite, indicating relative minima; some may be negative-definite, indicating relative maxima; and some may be neither positive- nor negative-definite, indicating saddle points. The shape of the graph at a saddle point

where $D < 0$ is like that in Figure 3.3.4. Critical points for which $D \neq 0$ are called ***nondegenerate critical points***. Such points are maxima, minima, or saddle points. The remaining critical points, where $D = 0$, may be tested directly, with level sets and sections or by some other method. Such critical points are said to be ***degenerate***; the methods developed in this chapter fail to provide a picture of the behavior of a function near such points, so we examine them case by case.

EXAMPLE 7 Locate the relative maxima, minima, and saddle points of the function

$$f(x, y) = \log(x^2 + y^2 + 1).$$

SOLUTION We must first locate the critical points of this function; therefore, according to Theorem 3, we calculate

$$\nabla f(x, y) = \frac{2x}{x^2 + y^2 + 1}\,\mathbf{i} + \frac{2y}{x^2 + y^2 + 1}\,\mathbf{j}.$$

Thus, $\nabla f(x, y) = \mathbf{0}$ if and only if $(x, y) = (0, 0)$, and so the only critical point of f is $(0, 0)$. Now we must determine whether this is a maximum, a minimum, or a saddle point. The second partial derivatives are

$$\frac{\partial^2 f}{\partial x^2} = \frac{2(x^2 + y^2 + 1) - (2x)(2x)}{(x^2 + y^2 + 1)^2},$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{2(x^2 + y^2 + 1) - (2y)(2y)}{(x^2 + y^2 + 1)^2},$$

and

$$\frac{\partial^2 f}{\partial x\,\partial y} = \frac{-2x(2y)}{(x^2 + y^2 + 1)^2}.$$

Therefore,

$$\frac{\partial^2 f}{\partial x^2}(0, 0) = 2 = \frac{\partial^2 f}{\partial y^2}(0, 0) \qquad \text{and} \qquad \frac{\partial^2 f}{\partial x\,\partial y}(0, 0) = 0,$$

which yields

$$D = 2 \cdot 2 = 4 > 0.$$

Because $(\partial^2 f/\partial x^2)(0, 0) > 0$, we conclude by Theorem 6 that $(0, 0)$ is a local minimum. (Can you show this just from the fact that $\log t$ is an increasing function of $t > 0$?) ▲

EXAMPLE 8 The graph of the function $g(x, y) = 1/xy$ is a surface S in $\mathbb{R}^3$. Find the points on S that are closest to the origin (0, 0, 0). (See Figure 3.3.7.)

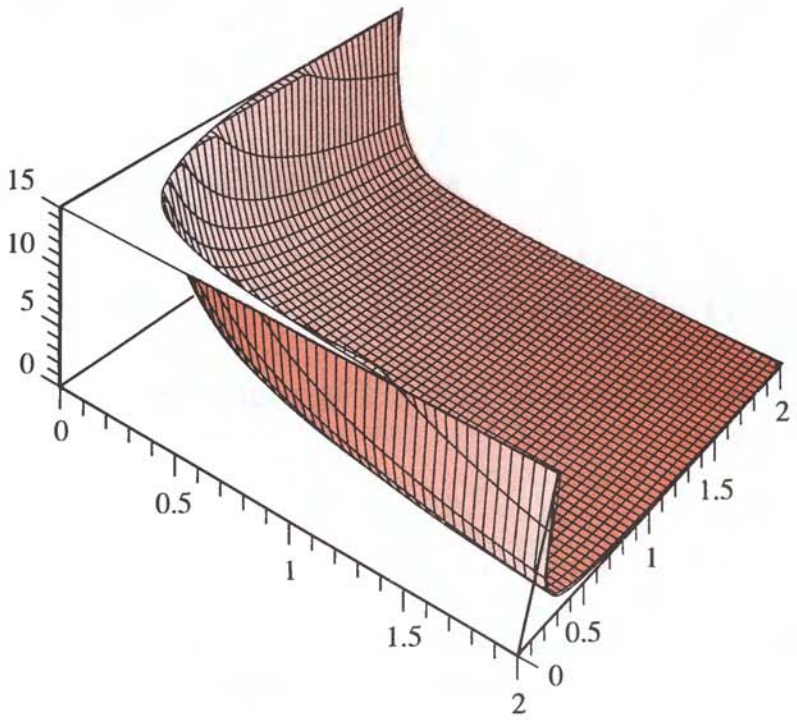

Figure 3.3.7 The surface $z = 1/xy$ in the first quadrant. (There are similar figures in the other quadrants, but notice that $z < 0$ in the second and fourth quadrants.)

SOLUTION Each point on S is of the form $(x, y, 1/xy)$. The distance from this point to the origin is

$$d(x, y) = \sqrt{x^2 + y^2 + \frac{1}{x^2y^2}}.$$

It is easier to work with the square of d, so let $f(x, y) = x^2 + y^2 + (1/x^2y^2)$, which will have the same minimum point. Notice that $f(x, y)$ becomes very large as x and y get larger and larger; $f(x, y)$ also becomes very large as (x, y) approaches the x or y axis where f is not defined, so f must attain a minimum at some critical point. The critical points are determined by:

$$\frac{\partial f}{\partial x} = 2x - \frac{2}{x^3y^2} = 0,$$

$$\frac{\partial f}{\partial y} = 2y - \frac{2}{y^3x^2} = 0,$$

that is, $x^4y^2 - 1 = 0$, and $x^2y^4 - 1 = 0$. From the first equation we get $y^2 = 1/x^4$, and, substituting this into the second equation, we obtain

$$\frac{x^2}{x^8} = 1 = \frac{1}{x^6}.$$

Thus, $x = \pm 1$ and $y = \pm 1$, and it therefore follows that f has four critical points, namely, $(1, 1)$, $(1, -1)$, $(-1, 1)$, and $(-1, -1)$. Note that f has the value 3 for all these points, so they are all minima. Therefore, the points on the surface closest to the point $(0, 0, 0)$ are $(1, 1, 1)$, $(1, -1, -1)$, $(-1, 1, -1)$, and $(-1, -1, 1)$ and the minimum distance is $\sqrt{3}$. Is this consistent with the graph in Figure 3.3.7? ▲

EXAMPLE 9 Analyze the behavior of $z = x^5y + xy^5 + xy$ at its critical points.

SOLUTION The first partial derivatives are

$$\frac{\partial z}{\partial x} = 5x^4y + y^5 + y = y(5x^4 + y^4 + 1)$$

and

$$\frac{\partial z}{\partial y} = x(5y^4 + x^4 + 1).$$

The terms $5x^4 + y^4 + 1$ and $5y^4 + x^4 + 1$ are always greater than or equal to 1, and so it follows that the only critical point is $(0, 0)$.

The second partial derivatives are

$$\frac{\partial^2 z}{\partial x^2} = 20x^3y, \qquad \frac{\partial^2 z}{\partial y^2} = 20xy^3,$$

and

$$\frac{\partial^2 z}{\partial x\, \partial y} = 5x^4 + 5y^4 + 1.$$

Thus, at $(0, 0)$, $D = -1$, and so $(0, 0)$ is a nondegenerate saddle point and the graph of z near $(0, 0)$ looks like the graph in Figure 3.3.4. ▲

Global Maxima and Minima

We end this section with a discussion of the theory of *absolute*, or *global*, maxima and minima of functions of several variables. Unfortunately, the location of absolute maxima and minima for functions on $\mathbb{R}^n$ is, in general, a more difficult problem than for functions of one variable.

DEFINITION Suppose $f\colon A \to \mathbb{R}$ is a function defined on a set A in $\mathbb{R}^2$ or $\mathbb{R}^3$. A point $\mathbf{x}_0 \in A$ is said to be an ***absolute maximum*** (or ***absolute minimum***) point of f if $f(\mathbf{x}) \leq f(\mathbf{x}_0)$ [or $f(\mathbf{x}) \geq f(\mathbf{x}_0)$] for all $\mathbf{x} \in A$.

In one-variable calculus, one learns—but often does not prove—that every continuous function on a closed interval I assumes its absolute maximum (or minimum) value at some point $\mathbf{x}_0$ in I. A generalization of this theoretical fact also holds in $\mathbb{R}^n$.

Such theorems guarantee that the maxima or minima one is seeking actually exist; therefore, the search for them is not in vain.

DEFINITION A set $D \in \mathbb{R}^n$ is said to be ***bounded*** if there is a number $M > 0$ such that $\|\mathbf{x}\| < M$ for all $\mathbf{x} \in D$. A set is ***closed*** if it contains all its boundary points.

Thus, a set is bounded if it can be strictly contained in some (large) ball. The appropriate generalization of the one-variable theorem on maxima and minima is the following result, stated without proof.

THEOREM 7: Global Existence Theorem for Maxima and Minima
Let D be closed and bounded in $\mathbb{R}^n$ and let $f\colon D \to \mathbb{R}$ be continuous. Then f assumes its absolute maximum and minimum values at some points $\mathbf{x}_0$ and $\mathbf{x}_1$ of D.

Simply stated, $\mathbf{x}_0$ and $\mathbf{x}_1$ are points where f assumes its largest and smallest values. As in one-variable calculus, these points need not be uniquely determined.

Suppose now that $D = U \cup \partial U$, where U is open and ∂U is its boundary. If $D \subset \mathbb{R}^2$, we suppose that ∂U is a piecewise smooth curve; that is, D is a region bounded by a collection of smooth curves—for example, a square or the sets depicted in Figure 3.3.8.

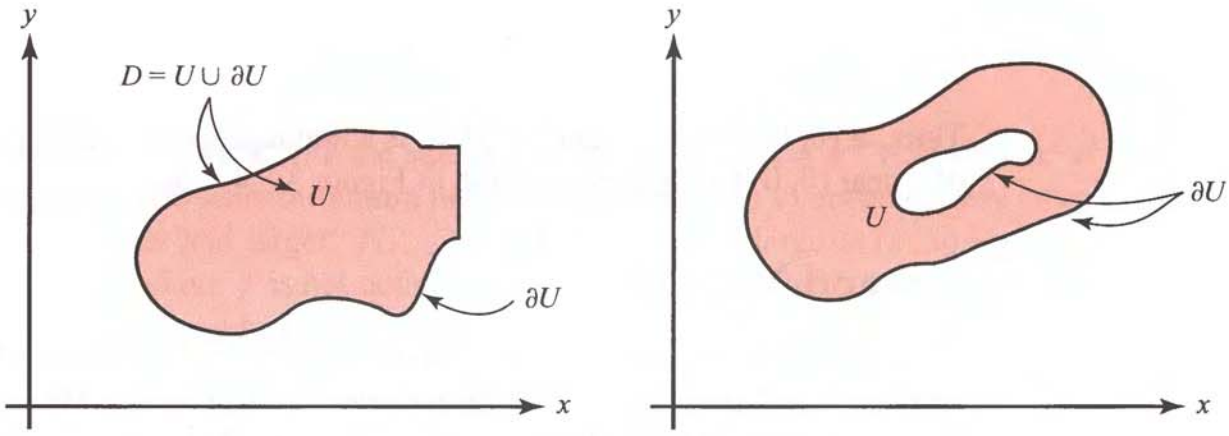

Figure 3.3.8 $D = U \cup \partial U$: Two examples of regions whose boundary is a piecewise smooth curve.

If $\mathbf{x}_0$ and $\mathbf{x}_1$ are in U, we know from Theorem 4 that they are critical points of f. If they are in ∂U, and ∂U is a smooth curve (i.e., the image of a smooth path $\mathbf{c}$ with $\mathbf{c}' \neq 0$), then they are maximum or minimum points of f viewed as a function on ∂U. These observations provide a method of finding the absolute maximum and minimum values of f on a region D.

Strategy for Finding the Absolute Maxima and Minima on a Region with Boundary Let f be a continuous function of two variables defined on a closed and bounded region D in $\mathbb{R}^2$, which is bounded by a smooth closed curve. To find the absolute maximum and minimum of f on D:

(i) Locate all critical points for f in U.

(ii) Find the maximum and minimum of f viewed as a function only on ∂U.

(iii) Compute the value of f at all of these critical points.

(iv) Compare all these values and select the largest and the smallest.

If D is a region bounded by a collection of smooth curves (such as a square), then one follows a similar procedure, but including in step (iii) the points where the curves meet (such as the corners of the square).

All the steps except step (ii) should now be familiar to the student. To carry out step (ii) in the plane, one way is to find a smooth parametrization of ∂U; that is, we find a path $\mathbf{c}\colon I \to \partial U$, where I is some interval, which is onto ∂U. Second, we consider the function of one variable $t \mapsto f(\mathbf{c}(t))$, where $t \in I$, and locate the maximum and minimum points $t_0, t_1 \in I$ (remember to check the endpoints!). Then $\mathbf{c}(t_0)$, $\mathbf{c}(t_1)$ will be maximum and minimum *points* for f as a function on ∂U. Another method for dealing with step (ii) is the Lagrange multiplier method, to be presented in the next section.

EXAMPLE 10 Find the maximum and minimum values of the function $f(x, y) = x^2 + y^2 - x - y + 1$ in the disk D defined by $x^2 + y^2 \leq 1$.

SOLUTION (i) To find the critical points we set $\partial f/\partial x = \partial f/\partial y = 0$. Thus, $2x - 1 = 0$, $2y - 1 = 0$, and hence $(x, y) = (\frac{1}{2}, \frac{1}{2})$ is the only critical point in the open disk $U = \{(x, y) \mid x^2 + y^2 < 1\}$.

(ii) The boundary ∂U can be parametrized by $\mathbf{c}(t) = (\sin t, \cos t)$, $0 \leq t \leq 2\pi$. Thus,

$$f(\mathbf{c}(t)) = \sin^2 t + \cos^2 t - \sin t - \cos t + 1 = 2 - \sin t - \cos t = g(t).$$

To find the maximum and minimum of f on ∂U, it suffices to locate the maximum and minimum of g. Now $g'(t) = 0$ only when

$$\sin t = \cos t, \qquad \text{that is, when} \qquad t = \frac{\pi}{4}, \frac{5\pi}{4}.$$

Thus, the candidates for the maximum and minimum for f on ∂U are the points $\mathbf{c}(\pi/4)$, $\mathbf{c}(5\pi/4)$ and the endpoints $\mathbf{c}(0) = \mathbf{c}(2\pi)$.

(iii) The values of f at the critical points are: $f(\frac{1}{2}, \frac{1}{2}) = \frac{1}{2}$ from step (i) and, from step (ii),

$$f\left(\mathbf{c}\left(\frac{\pi}{4}\right)\right) = f\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right) = 2 - \sqrt{2},$$

$$f\left(\mathbf{c}\left(\frac{5\pi}{4}\right)\right) = f\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right) = 2 + \sqrt{2},$$

and

$$f(\mathbf{c}(0)) = f(\mathbf{c}(2\pi)) = f(0, 1) = 1.$$

(iv) Comparing all the values $\frac{1}{2}, 2 - \sqrt{2}, 2 + \sqrt{2}, 1$, it is clear that the absolute minimum occurs at $(1/2, 1/2)$ and the absolute maximum occurs at $(-\sqrt{2}/2, -\sqrt{2}/2)$. ▲

In Section 3.4, we shall consider a generalization of the strategy for finding the absolute maximum and minimum to regions D in $\mathbb{R}^n$.

EXERCISES

In Exercises 1 to 16, find the critical points of the given function and then determine whether they are local maxima, local minima, or saddle points.

1. $f(x, y) = x^2 - y^2 + xy$

2. $f(x, y) = x^2 + y^2 - xy$

3. $f(x, y) = x^2 + y^2 + 2xy$

4. $f(x, y) = x^2 + y^2 + 3xy$

5. $f(x, y) = e^{1+x^2-y^2}$

6. $f(x, y) = x^2 - 3xy + 5x - 2y + 6y^2 + 8$

7. $f(x, y) = 3x^2 + 2xy + 2x + y^2 + y + 4$

8. $f(x, y) = \sin(x^2 + y^2)$ [consider only the critical point $(0, 0)$]

9. $f(x, y) = \cos(x^2 + y^2)$ [consider only the three critical points $(0, 0)$, $(\sqrt{\pi/2}, \sqrt{\pi/2})$, and $(0, \sqrt{\pi})$]

10. $f(x, y) = y + x \sin y$

11. $f(x, y) = e^x \cos y$

12. $f(x, y) = (x - y)(xy - 1)$

13. $f(x, y) = xy + \dfrac{1}{x} + \dfrac{1}{y}$

14. $f(x, y) = \log(2 + \sin xy)$

15. $f(x, y) = x \sin y$

16. $f(x, y) = (x + y)(xy + 1)$

17. Find the local maxima and minima for $z = (x^2 + 3y^2)\, e^{1-x^2-y^2}$. (See Figure 2.1.15.)

18. Let $f(x, y) = x^2 + y^2 + kxy$. If you imagine the graph changing as k increases, at what values of k does the shape of the graph change qualitatively?

19. An examination of the function $f\colon \mathbb{R}^2 \to \mathbb{R}$, $(x, y) \mapsto (y - 3x^2)(y - x^2)$ will give an idea of the difficulty of finding conditions that guarantee that a critical point is a relative extremum when Theorem 6 fails.[10] Show that

(a) the origin is a critical point of f;
(b) f has a relative minimum at $(0, 0)$ on every straight line through $(0, 0)$; that is, if $g(t) = (at, bt)$, then $f \circ g\colon \mathbb{R} \to \mathbb{R}$ has a relative minimum at 0, for every choice of a and b;
(c) the origin is not a relative minimum of f.

20. Let $f(x, y) = Ax^2 + E$ where A and E are constants. What are the critical points of f? Are they local maxima or local minima?

21. Let $f(x, y) = x^2 - 2xy + y^2$. Here $D = 0$. Can you say whether the critical points are local minima, local maxima, or saddle points?

22. Find the point on the plane $2x - y + 2z = 20$ nearest the origin.

23. Show that a rectangular box of given volume has minimum surface area when the box is a cube.

24. Show that the rectangular parallelepiped with fixed surface area and maximum volume is a cube.

25. Write the number 120 as a sum of three numbers so that the sum of the products taken two at a time is a maximum.

26. Show that if (x_0, y_0) is a critical point of a quadratic function $f(x, y)$ and $D < 0$, then there are points (x, y) near (x_0, y_0) at which $f(x, y) > f(x_0, y_0)$ and, similarly, points for which $f(x, y) < f(x_0, y_0)$.

27. Determine the nature of the critical points of the function

$$f(x, y, z) = x^2 + y^2 + z^2 + xy.$$

28. Let n be an integer greater than 2 and set $f(x, y) = ax^n + cy^n$, where $ac \neq 0$. Determine the nature of the critical points of f.

---

[10] This interesting phenomenon was first pointed out by the famous mathematician Giuseppe Peano (1858–1932). Another curious "pathology" is given in Exercise 41.

29. Determine the nature of the critical points of $f(x, y) = x^3 + y^2 - 6xy + 6x + 3y$.

30. Find the absolute maximum and minimum values of the function $f(x, y) = (x^2 + y^2)^4$ on the disk $x^2 + y^2 \leq 1$. (You do not *have* to use calculus.)

31. Repeat Exercise 30 for the function $f(x, y) = x^2 + xy + y^2$.

32. A curve C in space is defined *implicitly* on the cylinder $x^2 + y^2 = 1$ by the additional equation $x^2 - xy + y^2 - z^2 = 1$. Find the point or points on C closest to the origin.

33. Find the absolute maximum and minimum values for $f(x, y) = \sin x + \cos y$ on the rectangle R defined by $0 \leq x \leq 2\pi$, $0 \leq y \leq 2\pi$.

34. Find the absolute maximum and minimum values for the function $f(x, y) = xy$ on the rectangle R defined by $-1 \leq x \leq 1$, $-1 \leq y \leq 1$.

35. Determine the nature of the critical points of $f(x, y) = xy + 1/x + 8/y$.

In Exercises 36 through 40, D denotes the unit disk.

36. Let u be a C^2 function on D which is "strictly subharmonic"; that is, the following inequality holds: $\nabla^2 u = (\partial^2 u/\partial x^2) + (\partial^2 u/\partial y^2) > 0$. Show that u cannot have a maximum point in $D \backslash \partial D$ (the set of points in D, but not in ∂D).

37. Let u be a harmonic function on D; that is, $\nabla^2 u = 0$ on $D \backslash \partial D$ and be continuous on D. Show that if u achieves its maximum value in $D \backslash \partial D$, it also achieves it on ∂D. This is sometimes called the "weak maximum principle" for harmonic functions. [HINT: Consider $\nabla^2(u + \varepsilon e^x)$, $\varepsilon > 0$. You can use the following fact, which is proved in more advanced texts: Given a sequence $\{\mathbf{p}_n\}$, $n = 1, 2, \ldots$, of points in a closed bounded set A in $\mathbb{R}^2$ or $\mathbb{R}^3$, there exists a point $\mathbf{q}$ such that every neighborhood of $\mathbf{q}$ contains at least one member of $\{\mathbf{p}_n\}$.]

38. Define the notion of a strict superharmonic function u on D by mimicking Exercise 36. Show that u cannot have a minimum in $D \backslash \partial D$.

39. Let u be harmonic in D as in Exercise 37. Show that if u achieves its minimum value in $D \backslash \partial D$, it also achieves it on ∂D. This is sometimes called the "weak minimum principle" for harmonic functions.

40. Let ϕ: $\partial D \to \mathbb{R}$ be continuous and let T be a solution on D to $\nabla^2 T = 0$, continuous on D and $T = \phi$ on ∂D.

(a) Use Exercises 36 to 39 to show that such a solution, if it exists, must be unique.

(b) Suppose that $T(x, y)$ represents a temperature function that is independent of time, with ϕ representing the temperature of a circular plate at its boundary. Can you give a physical interpretation of the principle stated in part (a)?

41. (a) Let f be a C^1 function on the real line $\mathbb{R}$. Suppose that f has exactly one critical point x_0 that is a strict local minimum for f. Show that x_0 is also an absolute minimum for f, that is, that $f(x) \geq f(x_0)$ for all x.

(b) The next example shows that the conclusion of part (a) does not hold for functions of more than one variable. Let $f\colon \mathbb{R}^2 \to \mathbb{R}$ be defined by

$$f(x, y) = -y^4 - e^{-x^2} + 2y^2\sqrt{e^x + e^{-x^2}}.$$

(i) Show that (0, 0) is the only critical point for f and that it is a local minimum.
(ii) Argue informally that f has no absolute minimum.

42. Suppose that a pentagon is composed of a rectangle topped by an isosceles triangle (see Figure 3.3.9). If the length of the perimeter is fixed, find the maximum possible area.

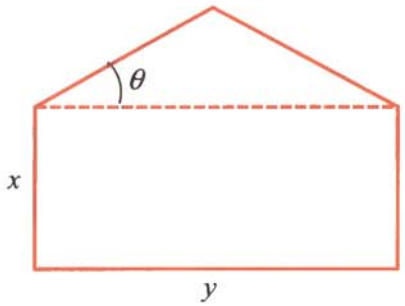

Figure 3.3.9 Maximize the area for fixed perimeter.

3.4 Constrained Extrema and Lagrange Multipliers

Often one is required to maximize or minimize a function subject to certain *constraints* or *side conditions*. For example, we might need to maximize $f(x, y)$ subject to the condition that $x^2 + y^2 = 1$; that is, that (x, y) lie on the unit circle. More generally, we might need to maximize or minimize $f(x, y)$ subject to the side condition that (x, y) also satisfies an equation $g(x, y) = c$ where g is some function and c equals a constant [in the preceding example, $g(x, y) = x^2 + y^2$, and $c = 1$]. The set of such (x, y) is a level curve for g.

The purpose of this section is to develop some methods for handling this sort of problem. In Figure 3.4.1 we picture a graph of a function $f(x, y)$. In this picture,

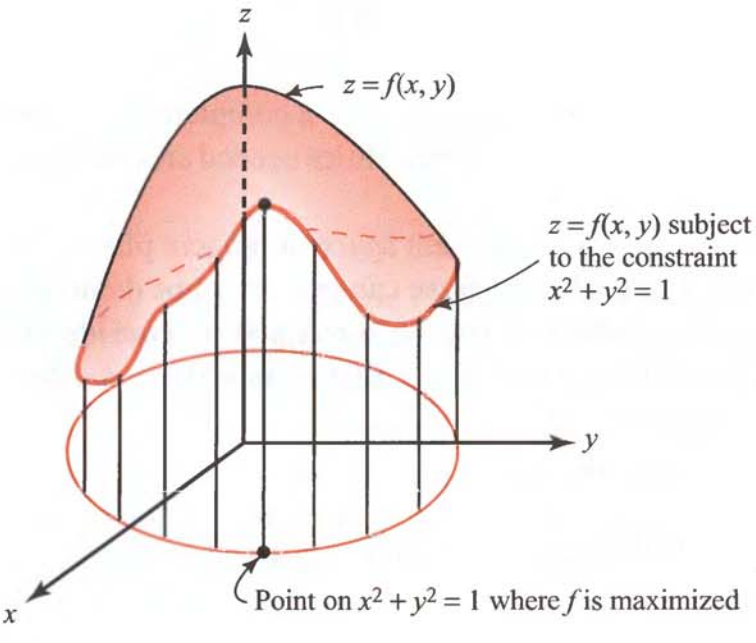

Figure 3.4.1 The geometric meaning of maximizing f subject to the constraint $x^2 + y^2 = 1$.

the maximum of f might be at (0, 0). However, suppose we are not interested in this maximum but only the maximum of $f(x, y)$ when (x, y) belongs to the unit circle; that is, when $x^2 + y^2 = 1$. The cylinder over $x^2 + y^2 = 1$ intersects the graph of $z = f(x, y)$ in a curve that lies on this graph. The problem of maximizing or minimizing $f(x, y)$ subject to the constraint $x^2 + y^2 = 1$ amounts to finding the point on this curve where z is the greatest or the least.

The Lagrange Multiplier Method

In general, let $f: U \subset \mathbb{R}^n \to \mathbb{R}$ and $g: U \subset \mathbb{R}^n \to \mathbb{R}$ be given C^1 functions, and let S be the level set for g with value c [recall that this is the set of points $\mathbf{x} \in \mathbb{R}^n$ with $g(\mathbf{x}) = c$].

When f is restricted to S we again have the notion of local maxima or local minima of f (local extrema), and an absolute maximum (largest value) or absolute minimum (smallest value) must be a local extremum. The following method provides a necessary condition for a constrained extremum:

THEOREM 8: The Method of Lagrange Multipliers Suppose that $f: U \subset \mathbb{R}^n \to \mathbb{R}$ and $g: U \subset \mathbb{R}^n \to \mathbb{R}$ are given C^1 real-valued functions. Let $\mathbf{x}_0 \in U$ and $g(\mathbf{x}_0) = c$, and let S be the level set for g with value c (recall that this is the set of points $\mathbf{x} \in \mathbb{R}^n$ satisfying $g(\mathbf{x}) = c$). Assume $\nabla g(\mathbf{x}_0) \neq \mathbf{0}$.

If $f|S$, which denotes "f restricted to S," has a local maximum or minimum on S at $\mathbf{x}_0$, then there is a real number λ such that

$$\nabla f(\mathbf{x}_0) = \lambda \nabla g(\mathbf{x}_0). \tag{1}$$

In general, a point $\mathbf{x}_0$ where equation (1) holds is said to be a ***critical point*** of $f|S$.

PROOF We have not developed enough techniques to give a complete proof, but we can provide the essential points. (The additional technicalities needed are discussed in Section 3.5 and in the Internet supplement.)

In Section 2.6 we learned that for $n = 3$ the tangent space or tangent plane of S at $\mathbf{x}_0$ is the space orthogonal to $\nabla g(\mathbf{x}_0)$. For arbitrary n we can give the same definition for the tangent space of S at $\mathbf{x}_0$. This definition can be motivated by considering tangents to paths $\mathbf{c}(t)$ that lie in S, as follows: If $\mathbf{c}(t)$ is a path in S and $\mathbf{c}(0) = \mathbf{x}_0$, then $\mathbf{c}'(0)$ is a tangent vector to S at $\mathbf{x}_0$, but

$$\frac{d}{dt} g(\mathbf{c}(t)) = \frac{d}{dt} c = 0,$$

and on the other hand, by the chain rule,

$$\frac{d}{dt}g(\mathbf{c}(t))\Big|_{t=0} = \nabla g(\mathbf{x}_0)\cdot\mathbf{c}'(0),$$

so that $\nabla g(\mathbf{x}_0)\cdot\mathbf{c}'(0) = 0$; that is, $\mathbf{c}'(0)$ is orthogonal to $\nabla g(\mathbf{x}_0)$.

If $f|S$ has a maximum at $\mathbf{x}_0$, then $f(\mathbf{c}(t))$ has a maximum at $t = 0$. By one-variable calculus, $df(\mathbf{c}(t))/dt|_{t=0} = 0$. Hence, by the chain rule,

$$0 = \frac{d}{dt}f(\mathbf{c}(t))\Big|_{t=0} = \nabla f(\mathbf{x}_0)\cdot\mathbf{c}'(0).$$

Thus, $\nabla f(\mathbf{x}_0)$ is perpendicular to the tangent of every curve in S and so is perpendicular to the whole tangent space to S at $\mathbf{x}_0$. Because the space perpendicular to this tangent space is a line, $\nabla f(\mathbf{x}_0)$ and $\nabla g(\mathbf{x}_0)$ are parallel. Because $\nabla g(\mathbf{x}_0) \neq \mathbf{0}$, it follows that $\nabla f(\mathbf{x}_0)$ is a multiple of $\nabla g(\mathbf{x}_0)$, which is the conclusion of the theorem. ■

Let us extract some geometry from this proof.

THEOREM 9 If f, when constrained to a surface S, has a maximum or minimum at $\mathbf{x}_0$, then $\nabla f(\mathbf{x}_0)$ is perpendicular to S at $\mathbf{x}_0$ (see Figure 3.4.2).

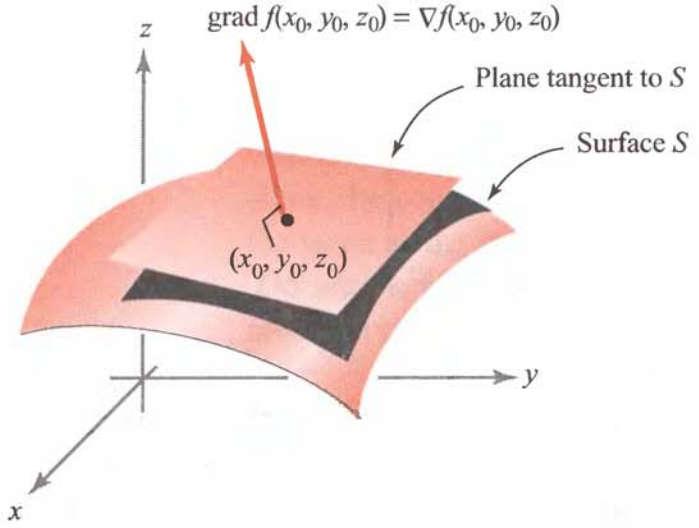

Figure 3.4.2 The geometry of constrained extrema.

These results tell us that in order to find the constrained extrema of f, we must look among those points $\mathbf{x}_0$ satisfying the conclusions of these two theorems. We shall give several illustrations of how to use each.

When the method of Theorem 8 is used, we look for a point $\mathbf{x}_0$ and a constant λ, called a ***Lagrange multiplier***, such that $\nabla f(\mathbf{x}_0) = \lambda\nabla g(\mathbf{x}_0)$. This method is more analytic in nature than the geometric method of Theorem 9. Surprisingly, Euler introduced these multipliers in 1744, some 40 years before Lagrange!

Equation (1) says that the partial derivatives of f are proportional to those of g. Finding such points $\mathbf{x}_0$ at which this occurs means solving the simultaneous equations

$$\left.\begin{aligned} \frac{\partial f}{\partial x_1}(x_1, \ldots, x_n) &= \lambda \frac{\partial g}{\partial x_1}(x_1, \ldots, x_n) \\ \frac{\partial f}{\partial x_2}(x_1, \ldots, x_n) &= \lambda \frac{\partial g}{\partial x_2}(x_1, \ldots, x_n) \\ &\vdots \\ \frac{\partial f}{\partial x_n}(x_1, \ldots, x_n) &= \lambda \frac{\partial g}{\partial x_n}(x_1, \ldots, x_n) \\ g(x_1, \ldots, x_n) &= c \end{aligned}\right\} \tag{2}$$

for $x_1, \ldots, x_n$ and λ.

Another way of looking at these equations is as follows: Think of λ as an additional variable and form the auxiliary function

$$h(x_1, \ldots, x_n, \lambda) = f(x_1, \ldots, x_n) - \lambda[g(x_1, \ldots, x_n) - c].$$

The Lagrange multiplier theorem says that to find the extreme points of $f|S$, we should examine the critical points of h. These are found by solving the equations

$$\left.\begin{aligned} 0 = \frac{\partial h}{\partial x_1} &= \frac{\partial f}{\partial x_1} - \lambda \frac{\partial g}{\partial x_1} \\ &\vdots \\ 0 = \frac{\partial h}{\partial x_n} &= \frac{\partial f}{\partial x_n} - \lambda \frac{\partial g}{\partial x_n} \\ 0 = \frac{\partial h}{\partial \lambda} &= g(x_1, \ldots, x_n) - c \end{aligned}\right\}, \tag{3}$$

which are the same as equations (2) above.

Second derivative tests for maxima and minima analogous to those in Section 3.3 will be given in Theorem 10 later in this section. However, in many problems it is possible to distinguish between maxima and minima by direct observation or by geometric means. Because this is often simpler, we consider examples of the latter type first.

EXAMPLE 1 Let $S \subset \mathbb{R}^2$ be the line through $(-1, 0)$ inclined at 45°, and let $f\colon \mathbb{R}^2 \to \mathbb{R}, (x, y) \mapsto x^2 + y^2$. Find the extrema of $f|S$.

SOLUTION Here $S = \{(x, y) \mid y - x - 1 = 0\}$, and therefore we set $g(x, y) = y - x - 1$ and $c = 0$. We have $\nabla g(x, y) = -\mathbf{i} + \mathbf{j} \neq \mathbf{0}$. The relative extrema of $f|S$ must be found among the points at which ∇f is orthogonal to S, that is, inclined at $-45°$. But $\nabla f(x, y) = (2x, 2y)$, which has the desired slope only when $x = -y$, or when (x, y) lies on the line L through the origin inclined at $-45°$. This can occur

in the set S only for the single point at which L and S intersect (see Figure 3.4.3). Reference to the level curves of f indicates that this point, $(-1/2, 1/2)$, is a relative minimum of $f|S$ (but not of f).

Notice that in this problem, f on S has a minimum but no maximum. ▲

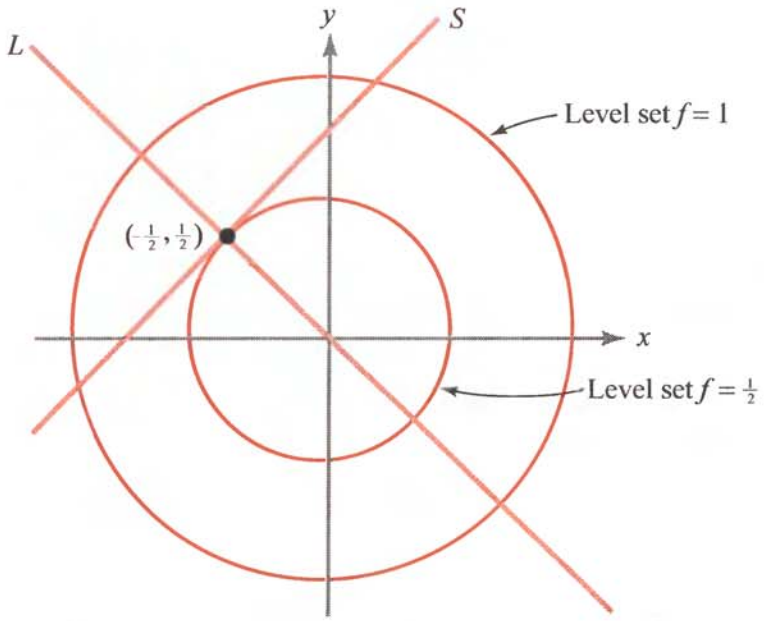

Figure 3.4.3 The geometry associated with finding the extrema of $f(x, y) = x^2 + y^2$ restricted to $S = \{(x, y) \mid y - x - 1 = 0\}$.

EXAMPLE 2 Let $f: \mathbb{R}^2 \to \mathbb{R}, (x, y) \mapsto x^2 - y^2$, and let S be the circle of radius 1 around the origin. Find the extrema of $f|S$.

SOLUTION The set S is the level curve for g with value 1, where $g: \mathbb{R}^2 \to \mathbb{R}$, $(x, y) \mapsto x^2 + y^2$. Because both of these functions have been studied in previous examples, we know their level curves; these are shown in Figure 3.4.4. In two dimensions, the condition that $\nabla f = \lambda \nabla g$ at $\mathbf{x}_0$, that is, that ∇f and ∇g are parallel at $\mathbf{x}_0$ is the same as the level curves being tangent at $\mathbf{x}_0$ (why?). Thus, the extreme points of $f|S$ are $(0, \pm 1)$ and $(\pm 1, 0)$. Evaluating f, we find $(0, \pm 1)$ are minima and $(\pm 1, 0)$ are maxima.

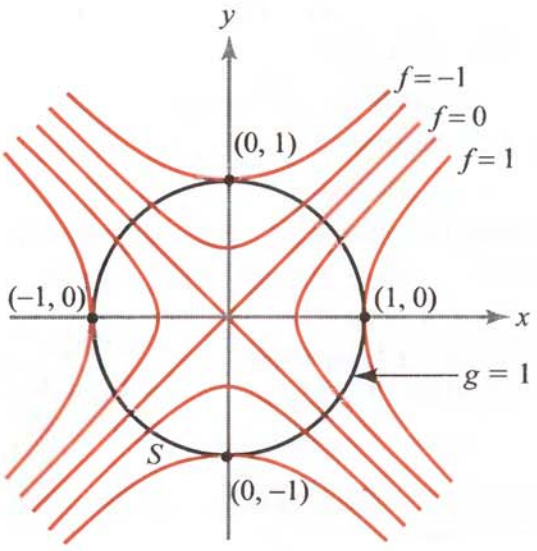

Figure 3.4.4 The geometry associated with the problem of finding the extrema of $x^2 - y^2$ on $S = \{(x, y) \mid x^2 + y^2 = 1\}$.

Let us also do this problem analytically by the method of Lagrange multipliers. Clearly,

$$\nabla f(x, y) = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right) = (2x, -2y) \qquad \text{and} \qquad \nabla g(x, y) = (2x, 2y).$$

Note that $\nabla g(x, y) \neq \mathbf{0}$ if $x^2 + y^2 = 1$. Thus, according to the Lagrange multiplier theorem, we must find a λ such that

$$(2x, -2y) = \lambda(2x, 2y) \qquad \text{and} \qquad (x, y) \in S, \qquad \text{i.e., } x^2 + y^2 = 1.$$

These conditions yield three equations, which can be solved for the three unknowns x, y, and λ. From $2x = \lambda 2x$ we conclude that either $x = 0$ or $\lambda = 1$. If $x = 0$, then $y = \pm 1$ and $-2y = \lambda 2y$ implies $\lambda = -1$. If $\lambda = 1$, then $y = 0$ and $x = \pm 1$. Thus, we get the points $(0, \pm 1)$ and $(\pm 1, 0)$, as before. As we have mentioned, this method only locates potential extrema; whether they are maxima, minima, or neither must be determined by other means, such as geometric arguments or the second derivative test given below.[11] ▲

EXAMPLE 3 Maximize the function $f(x, y, z) = x + z$ subject to the constraint $x^2 + y^2 + z^2 = 1$.

SOLUTION By Theorem 7 we know that the function f restricted to the unit sphere $x^2 + y^2 + z^2 = 1$ has a maximum (and also a minimum). To find the maximum, we again use the Lagrange multiplier theorem. We seek λ and (x, y, z) such that

$$1 = 2x\lambda, \qquad 0 = 2y\lambda, \qquad \text{and} \qquad 1 = 2z\lambda,$$

and

$$x^2 + y^2 + z^2 = 1.$$

From the first or the third equation, we see that $\lambda \neq 0$. Thus, from the second equation, we get $y = 0$. From the first and third equations, $x = z$, and so from the fourth, $x = \pm 1/\sqrt{2} = z$. Hence, our points are $(1/\sqrt{2}, 0, 1/\sqrt{2})$ and $(-1/\sqrt{2}, 0, -1/\sqrt{2})$. Comparing the values of f at these points, we can see that the first point yields the maximum of f (restricted to the constraint) and the second the minimum. ▲

EXAMPLE 4 Assume that among all rectangular boxes with fixed surface area of 10 square meters there is a box of largest possible volume. Find its dimensions.

SOLUTION If x, y, and z are the lengths of the sides, $x \geq 0$, $y \geq 0$, $z \geq 0$, respectively, and the volume is $f(x, y, z) = xyz$. The constraint is $2(xy + xz + yz) = 10$;

[11] In these examples, $\nabla g(\mathbf{x}_0) \neq \mathbf{0}$ on the surface S, as required by the Lagrange multiplier theorem. If $\nabla g(\mathbf{x}_0)$ were zero for some $\mathbf{x}_0$ on S, then it would have to be included among the possible extrema.

that is, $xy + xz + yz = 5$. Thus, the Lagrange multiplier conditions are

$$\begin{aligned} yz &= \lambda(y + z) \\ xz &= \lambda(x + z) \\ xy &= \lambda(y + x) \\ xy + xz + yz &= 5. \end{aligned}$$

First of all, $x \neq 0$, because $x = 0$ implies $yz = 5$ and $0 = \lambda z$, so that $\lambda = 0$ and we get the contradictory equation $yz = 0$. Similarly, $y \neq 0, z \neq 0, x + y \neq 0$. Elimination of λ from the first two equations gives $yz/(y + z) = xz/(x + z)$, which gives $x = y$; similarly, $y = z$. Substituting these values into the last equation, we obtain $3x^2 = 5$, or $x = \sqrt{5/3}$. Thus, we get the solution $x = y = z = \sqrt{5/3}$, and $xyz = (5/3)^{3/2}$. This (cubical) shape must therefore maximize the volume, assuming there is a box of maximum volume. ▲

Existence of Solutions

We should note that the solution to Example 4 does *not* demonstrate that the cube is the rectangular box of largest volume with a given fixed surface area; it proves that the cube is the only possible candidate for a maximum. We shall sketch a proof that it really is the maximum later. The distinction between showing that there is *only one possible solution* to a problem and that, in fact, *a solution exists* is a subtle one that many (even great) mathematicians have overlooked.

Queen Dido (ca. 900 B.C.) realized that among all planar regions with fixed circumference, the disc is the region of maximum area. It is not difficult to prove this fact under the assumption that there is a region of maximum area; however, proving that such a region of maximum area exists is quite another (difficult) matter. A complete proof was not given until the second half of the nineteenth century by the German mathematician Weierstrass.

Let us consider a nonmathematical parallel to this situation. Put yourself in the place of Lord Peter Wimsey, Dorothy Sayers' famous detective:

> "Undoubtedly," said Wimsey, "but if you think that this identification is going to make life one grand, sweet song for you, you are mistaken. . . . Since we have devoted a great deal of time and thought to the case on the assumption that it was murder, it's a convenience to know that the assumption is correct."

Wimsey has found the body of a dead man, and after some time has located ten suspects. He is sure that no one else other than one of the suspects could be the murderer. By collecting all the evidence and checking alibis, he then reduces the number of suspects one by one, until, finally, only the butler remains; hence he is the murderer! But wait, Peter is a very cautious man. By checking everything once again, he discovers that the man died by suicide; so there is no murder. You see the

point: It does not suffice to find a clear and uniquely determined suspect in a criminal case where murder is suspected; you must prove that a murder actually took place.

The same goes for our cube; the fact that it is the only possible candidate for a maximum does not prove that it is maximum. (For more information see *The Parsimonious Universe: Shape and Form in the Natural World*, by S. Hildebrandt and A. Tromba, Springer-Verlag, New York/Berlin, 1995.)

The key to showing that $f(x, y, z) = xyz$ really has a maximum lies in the fact that f is a continuous function that is defined on the unbounded surface $S: xy + xz + yz = 5$, and not on a bounded set, which includes its boundary, where Theorem 7 of Section 3.3 would apply. We have already seen problems of this sort for functions of one and two variables.

The way to show that $f(x, y, z) = xyz \geq 0$ does indeed have a maximum on $xy + yz + xz = 5$ is to show that if either x, y, or z tend to ∞, then $f(x, y, z) \to 0$. We may then conclude that the maximum of f on S must exist by appealing to Theorem 7 (the student should supply the details). So, suppose (x, y, z) lies in S and $x \to \infty$, then $y \to 0$ and $z \to 0$ (why?). Multiplying the equation defining S by z we obtain the equation $xyz + xz^2 + yz^2 = 5z \to 0$ as $x \to \infty$. Because $x, y, z \geq 0$, $xyz = f(x, y, z) \to 0$. Similarly, $f(x, y, z) \to 0$ if either y or z tend to ∞. Thus, a box of maximum volume must exist.

Some general guidelines may be useful for maximum and minimum problems with constraints. First of all, if the surface S is bounded (as an ellipsoid is, for example), then f must have a maximum and a minimum on S. (See Theorem 7 in the preceding section.) In particular, if f has only two points satisfying the conditions of the Lagrange multiplier theorems or Theorem 9, then one must be a maximum and one must be a minimum. Evaluating f at each point will tell the maximum from the minimum. However, if there are more than two such points, some can be saddle points. Also, if S is not bounded (for example, if it is a hyperboloid), then f need not have any maxima or minima.

Several Constraints

If a surface S is defined by a number of constraints, namely,

$$\left.\begin{aligned} g_1(x_1, \ldots, x_n) &= c_1 \\ g_2(x_1, \ldots, x_n) &= c_2 \\ &\vdots \\ g_k(x_1, \ldots, x_n) &= c_k \end{aligned}\right\}, \tag{4}$$

then the Lagrange multiplier theorem may be generalized as follows: *If f has a maximum or a minimum at $\mathbf{x}_0$ on S, there must exist constants $\lambda_1, \ldots, \lambda_k$ such that*[12]

$$\nabla f(\mathbf{x}_0) = \lambda_1 \nabla g_1(\mathbf{x}_0) + \cdots + \lambda_k \nabla g_k(\mathbf{x}_0). \tag{5}$$

[12] As with the hypothesis $\nabla g(\mathbf{x}_0) \neq \mathbf{0}$ in the Lagrange multiplier theorem, here one must assume that the vectors $\nabla g_1(\mathbf{x}_0), \ldots, \nabla g_k(\mathbf{x}_0)$ are linearly independent; that is, each $\nabla g_i(\mathbf{x}_0)$ is not a linear combination of the other $\nabla g_j(\mathbf{x}_0)$, $j \neq i$.

This case may be proved by generalizing the method used to prove the Lagrange multiplier theorem. Let us give an example of how this more general formulation is used.

EXAMPLE 5 Find the extreme points of $f(x, y, z) = x + y + z$ subject to the two conditions $x^2 + y^2 = 2$ and $x + z = 1$.

SOLUTION Here there are two constraints:

$$g_1(x, y, z) = x^2 + y^2 - 2 = 0 \qquad \text{and} \qquad g_2(x, y, z) = x + z - 1 = 0.$$

Thus, we must find x, y, z, λ_1, and λ_2 such that

$$\nabla f(x, y, z) = \lambda_1 \nabla g_1(x, y, z) + \lambda_2 \nabla g_2(x, y, z),$$
$$g_1(x, y, z) = 0, \qquad \text{and} \qquad g_2(x, y, z) = 0.$$

Computing the gradients and equating components, we get

$$1 = \lambda_1 \cdot 2x + \lambda_2 \cdot 1,$$
$$1 = \lambda_1 \cdot 2y + \lambda_2 \cdot 0,$$
$$1 = \lambda_1 \cdot 0 + \lambda_2 \cdot 1,$$
$$x^2 + y^2 = 2, \qquad \text{and} \qquad x + z = 1.$$

These are five equations for x, y, z, λ_1, and λ_2. From the third equation, $\lambda_2 = 1$, and so $2x\lambda_1 = 0$, $2y\lambda_1 = 1$. Because the second implies $\lambda_1 \neq 0$, we have $x = 0$. Thus, $y = \pm\sqrt{2}$ and $z = 1$. Hence, the possible extrema are $(0, \pm\sqrt{2}, 1)$. By inspection, $(0, \sqrt{2}, 1)$ gives a relative maximum, and $(0, -\sqrt{2}, 1)$ a relative minimum.

The condition $x^2 + y^2 = 2$ implies that x and y must be bounded. The condition $x + z = 1$ implies that z is also bounded. If follows that the constraint set S is closed and bounded. By Theorem 7 it follows that f has a maximum and minimum on S that must therefore occur at $(0, \sqrt{2}, 1)$ and $(0, -\sqrt{2}, 1)$, respectively. ▲

The method of Lagrange multipliers provides us with another tool to locate the absolute maxima and minima of differentiable functions on bounded regions in $\mathbb{R}^2$ (see the strategy for finding absolute maximum and minimum in Section 3.3).

EXAMPLE 6 Find the absolute maximum of $f(x, y) = xy$ on the unit disk D, where D is the set of points (x, y) with $x^2 + y^2 \leq 1$.

SOLUTION By Theorem 7 of Section 3.3, we know the absolute maximum exists. First, we find all the critical points of f in U, the set of points (x, y) with

$x^2 + y^2 < 1$. Because

$$\frac{\partial f}{\partial x} = y \qquad \text{and} \qquad \frac{\partial f}{\partial y} = x,$$

$(0, 0)$ is the only critical point of f in U. Now consider f on the unit circle, the level curve $g(x, y) = 1$, where $g(x, y) = x^2 + y^2$. To locate the maximum and minimum of f on C, we write down the Lagrange multiplier equations: $\nabla f(x, y) = (y, x) = \lambda \nabla g(x, y) = \lambda(2x, 2y)$ and $x^2 + y^2 = 1$. Rewriting these in component form, we get

$$\begin{aligned} y &= 2\lambda x, \\ x &= 2\lambda y, \\ x^2 + y^2 &= 1. \end{aligned}$$

Thus,

$$y = 4\lambda^2 y,$$

or $\lambda = \pm 1/2$ and $y = \pm x$, which means that $x^2 + x^2 = 2x^2 = 1$ or $x = \pm 1/\sqrt{2}$, $y = \pm 1/\sqrt{2}$. On C we compute four candidates for the absolute maximum and minimum, namely,

$$\left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right), \quad \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right), \quad \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right), \quad \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right).$$

The value of f at both $(-1/\sqrt{2}, -1/\sqrt{2})$ and $(1/\sqrt{2}, 1/\sqrt{2})$ is $1/2$. The value of f at $(-1/\sqrt{2}, 1/\sqrt{2})$ and $(1/\sqrt{2}, -1/\sqrt{2})$ is $-1/2$, and the value of f at $(0, 0)$ is 0. Therefore, the absolute maximum of f is $1/2$ and the absolute minimum is $-1/2$, both occurring on C. At $(0, 0)$, $\partial^2 f/\partial x^2 = 0$, $\partial^2 f/\partial y^2 = 0$ and $\partial^2 f/\partial x\, \partial y = 1$, so the discriminant is -1 and thus $(0, 0)$ is a saddle point. ▲

EXAMPLE 7 Find the absolute maximum and minimum of $f(x, y) = \frac{1}{2}x^2 + \frac{1}{2}y^2$ in the elliptical region D defined by $\frac{1}{2}x^2 + y^2 \leq 1$.

SOLUTION Again by Theorem 7, Section 3.3, the absolute maximum exists. We first locate the critical points of f in U, the set of points (x, y) with $\frac{1}{2}x^2 + y^2 < 1$. Because

$$\frac{\partial f}{\partial x} = x, \qquad \frac{\partial f}{\partial y} = y,$$

the only critical point is the origin $(0, 0)$.

We now find the maximum and minimum of f on C, the boundary of U, which is the level curve $g(x, y) = 1$, where $g(x, y) = \frac{1}{2}x^2 + y^2$. The Lagrange multiplier

equations are

$$\nabla f(x, y) = (x, y) = \lambda \nabla g(x, y) = \lambda(x, 2y)$$

and $(x^2/2) + y^2 = 1$. In other words,

$$\begin{aligned} x &= \lambda x \\ y &= 2\lambda y \\ \frac{x^2}{2} + y^2 &= 1. \end{aligned}$$

If $x = 0$, then $y = \pm 1$ and $\lambda = \frac{1}{2}$. If $y = 0$, then $x = \pm\sqrt{2}$ and $\lambda = 1$. If $x \neq 0$ and $y \neq 0$, we get both $\lambda = 1$ and $1/2$, which is impossible. Thus, the candidates for the maxima and minima of f on C are $(0, \pm 1)$, $(\pm\sqrt{2}, 0)$ and for f inside D, the candidate is $(0, 0)$. The value of f at $(0, \pm 1)$ is $1/2$, at $(\pm\sqrt{2}, 0)$ it is 1, and at $(0, 0)$ it is 0. Thus, the absolute minimum of f occurs at $(0, 0)$ and is 0. The absolute maximum of f on D is thus 1 and occurs at the points $(\pm\sqrt{2}, 0)$. ▲

Global Maxima and Minima

The method of Lagrange multipliers enhances our techniques for finding global maxima and minima. In this respect, the following is useful.

DEFINITION Let U be an open region in $\mathbb{R}^n$ with boundary ∂U. We say that ∂U is ***smooth*** if ∂U is the level set of a smooth function g whose gradient ∇g never vanishes (i.e., $\nabla g \neq \mathbf{0}$). Then we have the following strategy.

Lagrange Multiplier Strategy for Finding Absolute Maxima and Minima on Regions with Boundary Let f be a differentiable function on a closed and bounded region $D = U \cup \partial U$, U open in $\mathbb{R}^n$, with smooth boundary ∂U.

To find the absolute maximum and minimum of f on D:

(i) Locate all critical points of f in U.

(ii) Use the method of Lagrange multiplier to locate all the critical points of $f|\partial U$.

(iii) Compute the values of f at all these critical points.

(iv) Select the largest and the smallest.

EXAMPLE 8 Find the absolute maximum and minimum of the function $f(x, y, z) = x + y + z$ on the set $D = \{(x, y, z) \mid x^2 + y^2 + z^2 \leq 1\}$.

SOLUTION As in the previous examples, we know the absolute maximum and minimum exists. Now $D = U \cup \partial U$, where

$$U = \{(x, y, z) \mid x^2 + y^2 + z^2 < 1\}$$

and

$$\partial U = \{(x, y, z) \mid x^2 + y^2 + z^2 = 1\}.$$

The gradient of f is $\nabla f = (1, 1, 1)$, and so f has no critical points in U. Therefore, the maximum and minimum values of f must occur on ∂U.

Let $g(x, y, z) = x^2 + y^2 + z^2$. Then ∂U is the level set $g(x, y, z) = 1$. By the method of Lagrange multipliers, the maximum and minimum must occur at a critical point of $f|\partial U$, that is, at a point $\mathbf{x}_0$ where $\nabla f(\mathbf{x}_0) = \lambda \nabla g(\mathbf{x}_0)$ for some scalar λ.

Thus,

$$(1, 1, 1) = \lambda(2x, 2y, 2z); \qquad \text{that is,} \qquad x = \frac{1}{2\lambda}, \qquad y = \frac{1}{2\lambda}, \qquad z = \frac{1}{2\lambda}.$$

Because $x^2 + y^2 + z^2 = 1$, we obtain $\lambda = \pm\sqrt{3}/2$ and so $\mathbf{x}_0 = \pm(1/\sqrt{3}, 1/\sqrt{3}, 1/\sqrt{3})$. Clearly, $-(1/\sqrt{3}, 1/\sqrt{3}, 1/\sqrt{3})$ is the point where f assumes its absolute minimum (namely, $-\sqrt{3}$) and $(1/\sqrt{3}, 1/\sqrt{3}, 1\sqrt{3})$, the point where f assumes its maximum value $\sqrt{3}$. ▲

Two Additional Applications

We now present two further applications of the mathematical techniques developed in this section to geometry and to economics. We shall begin wth a geometric example.

EXAMPLE 9 Suppose we have a curve defined by the equation

$$\phi(x, y) = Ax^2 + 2Bxy + Cy^2 - 1 = 0.$$

Find the maximum and minimum distance of the curve to the origin. (These are the lengths of the ***semimajor*** and the ***semiminor*** axis of this quadric.)

SOLUTION The problem is equivalent to finding the extreme values of $f(x, y) = x^2 + y^2$ subject to the constraining condition $\phi(x, y) = 0$. Using the Lagrange multiplier method, we have the following equations:

$$2x + \lambda(2Ax + 2By) = 0 \tag{6}$$

$$2y + \lambda(2Bx + 2Cy) = 0 \tag{7}$$

$$Ax^2 + 2Bxy + Cy^2 = 1. \tag{8}$$

Adding x times equation (6) to y times equation (7), we obtain

$$2(x^2 + y^2) + 2\lambda(Ax^2 + 2Bxy + Cy^2) = 0.$$

By equation (8), it follows that $x^2 + y^2 + \lambda = 0$. Let $t = -1/\lambda = 1/(x^2 + y^2)$ [the case $\lambda = 0$ is impossible, because $(0, 0)$ is not on the curve $\phi(x, y) = 0$]. Then equations (6) and (7) can be written as follows:

$$\begin{aligned} 2(A - t)x + 2By &= 0 \\ 2Bx + 2(C - t)y &= 0. \end{aligned} \tag{9}$$

If these two equations are to have a nontrivial solution [remember that $(x, y) = (0, 0)$ is not on our curve and so is not a solution], it follows from a theorem of linear algebra that their determinant vanishes:[13]

$$\begin{vmatrix} A - t & B \\ B & C - t \end{vmatrix} = 0.$$

Because this equation is quadratic in t, there are two solutions, which we shall call t_1 and t_2. Because $-\lambda = x^2 + y^2$, we have $\sqrt{x^2 + y^2} = \sqrt{-\lambda}$. Now $\sqrt{x^2 + y^2}$ is the distance from the point (x, y) to the origin. Therefore, if (x_1, y_1) and (x_2, y_2) denote the nontrivial solutions to equation (9) corresponding to t_1 and t_2, and if t_1 and t_2 are positive, we get $\sqrt{x_2^2 + y_2^2} = 1/\sqrt{t_2}$ and $\sqrt{x_1^2 + y_1^2} = 1/\sqrt{t_1}$. Consequently, if $t_1 > t_2$, the lengths of the semiminor and semimajor axes are $1/\sqrt{t_1}$ and $1/\sqrt{t_2}$, respectively. If the curve is an ellipse, both t_1 and t_2 are, in fact, real and positive. What happens with a hyperbola or a parabola? ▲

Finally, we discuss an application to economics.

EXAMPLE 10 Suppose that the output of a manufacturing firm is a quantity Q of a certain product, where Q is a function $f(K, L)$, where K is the amount of capital equipment (or investment) and L is the amount of labor used. If the price of labor is p, the price of capital is q, and the firm can spend no more than B dollars, how can we find the amount of capital and labor to maximize the output Q?

SOLUTION We would expect that if the amount of capital or labor is increased, then the output Q should also increase; that is,

$$\frac{\partial Q}{\partial K} \geq 0 \qquad \text{and} \qquad \frac{\partial Q}{\partial L} \geq 0.$$

[13] The matrix of coefficients of the equations cannot have an inverse, because this would imply that the solution is zero. Recall that a matrix that does not have an inverse has determinant zero.

We also expect that as more labor is added to a given amount of capital equipment, we get less additional output for our effort; that is,

$$\frac{\partial^2 Q}{\partial L^2} < 0.$$

Similarly,

$$\frac{\partial^2 Q}{\partial K^2} < 0.$$

With these assumptions on Q, it is reasonable to expect the level curves of output (called ***isoquants***) $Q(K, L) = c$ to look something like the curves sketched in Figure 3.4.5, with $c_1 < c_2 < c_3$.

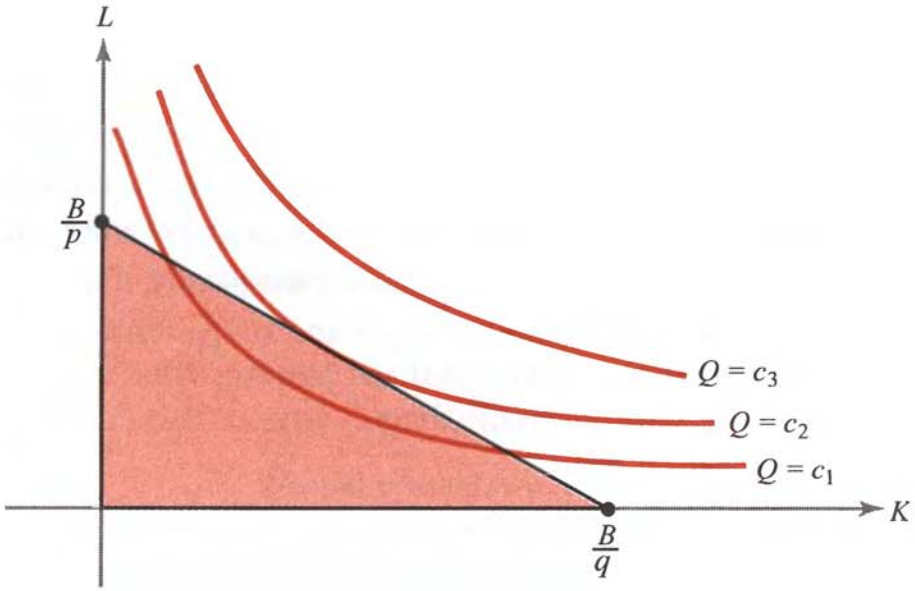

Figure 3.4.5 What is the largest value of Q in the shaded triangle?

We can interpret the convexity of the isoquants as follows: As one moves to the right along a given isoquant, it takes more and more capital to replace a unit of labor and still produce the same output. The budget constraint means that we must stay inside the triangle bounded by the axes and the line $pL + qK = B$. Geometrically, it is clear that we produce the most by spending all our money in such a way as to pick the isoquant that just touches, but does not cross, the budget line.

Because the maximum point lies on the boundary of our domain, we apply the method of Lagrange multipliers to find the maximum. To maximize $Q = f(K, L)$ subject to the constraint $pL + qK = B$, we look for critical points of the auxiliary function,

$$h(K, L, \lambda) = f(K, L) - \lambda(pL + qK - B).$$

Thus, we want

$$\frac{\partial Q}{\partial K} = \lambda q, \qquad \frac{\partial Q}{\partial L} = \lambda p, \qquad \text{and} \qquad pL + qK = B.$$

These are the conditions we must meet in order to maximize output. (The reader is asked to work out a specific case in Exercise 31.) ▲

In the preceding example, λ represents something interesting. Let $k = qK$ and $l = pL$, so that k is the dollar value of the capital used and l is the dollar value of the labor used. Then the first two equations become

$$\frac{\partial Q}{\partial k} = \frac{1}{q}\frac{\partial Q}{\partial K} = \lambda = \frac{1}{p}\frac{\partial Q}{\partial L} = \frac{\partial Q}{\partial l}.$$

Thus, at the optimum production point the marginal change in output per dollar's worth of additional capital investment is equal to the marginal change of output per dollar's worth of additional labor, and λ is this common value. At the optimum point, the exchange of a dollar's worth of capital for a dollar's worth of labor does not change the output. Away from the optimum point the marginal outputs are different, and one exchange or the other will increase the output.

A Second Derivative Test for Constrained Extrema

In Section 3.3, we developed a second derivative test for extrema of functions of several variables by looking at the second-degree term in the Taylor series of f. If the Hessian matrix of second partial derivatives is either positive-definite or negative-definite at a critical point of f, this point is a relative minimum or maximum, respectively.

The question naturally arises as to whether there is a second derivative test for maximum and minimum problems *in the presence of constraints*. The answer is yes and the test involves a matrix called a *bordered Hessian*. We will first discuss the test and how to apply it for the case of a function $f(x, y)$ of two variables subject to the constraint $g(x, y) = c$.

THEOREM 10 Let $f\colon U \subset \mathbb{R}^2 \to \mathbb{R}$ and $g\colon U \subset \mathbb{R}^2 \to \mathbb{R}$ be smooth (at least C^2) functions. Let $\mathbf{v}_0 \in U$, $g(\mathbf{v}_0) = c$, and S be the level curve for g with value c. Assume that $\nabla g(\mathbf{v}_0) \neq \mathbf{0}$ and that there is a real number λ such that $\nabla f(\mathbf{v}_0) = \lambda \nabla g(\mathbf{v}_0)$. Form the auxiliary function $h = f - \lambda g$ and the ***bordered Hessian*** determinant

$$|\bar{H}| = \begin{vmatrix} 0 & -\dfrac{\partial g}{\partial x} & -\dfrac{\partial g}{\partial y} \\ -\dfrac{\partial g}{\partial x} & \dfrac{\partial^2 h}{\partial x^2} & \dfrac{\partial^2 h}{\partial x\,\partial y} \\ -\dfrac{\partial g}{\partial y} & \dfrac{\partial^2 h}{\partial x\,\partial y} & \dfrac{\partial^2 h}{\partial y^2} \end{vmatrix} \text{ evaluated at } \mathbf{v}_0.$$

(i) If $|\bar{H}| > 0$, then $\mathbf{v}_0$ is a local maximum point for $f|S$.

(ii) If $|\bar{H}| < 0$, then $\mathbf{v}_0$ is a local minimum point for $f|S$.

(iii) If $|\bar{H}| = 0$, the test is inconclusive and $\mathbf{v}_0$ may be a minimum, a maximum, or neither.

This theorem is proved in the Internet supplement for this section.

EXAMPLE 11 Find extreme points of $f(x, y) = (x - y)^n$ subject to the constraint $x^2 + y^2 = 1$, where $n \geq 1$.

SOLUTION We set the first derivatives of the auxiliary function h defined by $h(x, y, \lambda) = (x - y)^n - \lambda(x^2 + y^2 - 1)$ equal to 0:

$$\begin{aligned} n(x - y)^{n-1} - 2\lambda x &= 0 \\ -n(x - y)^{n-1} - 2\lambda y &= 0 \\ -(x^2 + y^2 - 1) &= 0. \end{aligned}$$

From the first two equations we see that $\lambda(x + y) = 0$. If $\lambda = 0$, then $x = y = \pm\sqrt{2}/2$. If $\lambda \neq 0$, then $x = -y$. The four critical points are represented in Figure 3.4.6 and the corresponding values of $f(x, y)$ are listed below:

(A)	$x = \sqrt{2}/2$	$y = \sqrt{2}/2$	$\lambda = 0$	$f(x, y) = 0$
(B)	$x = \sqrt{2}/2$	$y = -\sqrt{2}/2$	$\lambda = n(\sqrt{2})^{n-2}$	$f(x, y) = (\sqrt{2})^n$
(C)	$x = -\sqrt{2}/2$	$y = -\sqrt{2}/2$	$\lambda = 0$	$f(x, y) = 0$
(D)	$x = -\sqrt{2}/2$	$y = \sqrt{2}/2$	$\lambda = (-1)^{n-2} n(\sqrt{2})^{n-2}$	$f(x, y) = (-\sqrt{2})^n$.

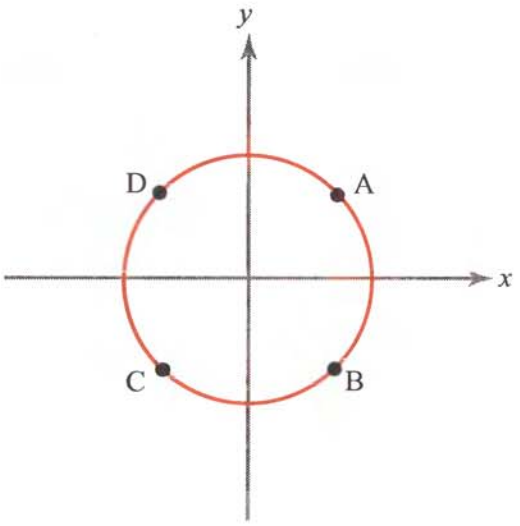

Figure 3.4.6 The four critical points in Example 11.

By inspection, we see that if n is even, then A and C are minimum points and B and D are maxima. If n is odd, then B is a maximum point, D is a minimum, and A and C are neither. Let us see whether Theorem 10 is consistent with these observations.

The bordered Hessian determinant is

$$|\bar{H}| = \begin{vmatrix} 0 & -2x & -2y \\ -2x & n(n-1)(x-y)^{n-2} - 2\lambda & -n(n-1)(x-y)^{n-2} \\ -2y & -n(n-1)(x-y)^{n-2} & n(n-1)(x-y)^{n-2} - 2\lambda \end{vmatrix}$$

$$= -4n(n-1)(x-y)^{n-2}(x+y)^2 + 8\lambda(x^2 - y^2).$$

If $n = 1$ or if $n \geq 3$, $|\bar{H}| = 0$ at A, B, C, and D. If $n = 2$, then $|\bar{H}| = 0$ at B and D and -16 at A and C. Thus, the second-derivative test picks up the minima at A and C, but is inconclusive in testing the maxima at B and D for $n = 2$. It is also inconclusive for all other values of n. ▲

Just as in the unconstrained case, there is also a second-derivative test for functions of more than two variables. If we are to find extreme points for $f(x_1, \ldots, x_n)$ subject to a single constraint $g(x_1, \ldots, x_n) = c$, we first form the bordered Hessian for the auxiliary function $h(x_1, \ldots, x_n) = f(x_1, \ldots, x_n) - \lambda(g(x_1, \ldots, x_n) - c)$ as follows:

$$\begin{vmatrix} 0 & \frac{-\partial g}{\partial x_1} & \frac{-\partial g}{\partial x_2} & \cdots & \frac{-\partial g}{\partial x_n} \\ \frac{-\partial g}{\partial x_1} & \frac{\partial^2 h}{\partial x_1^2} & \frac{\partial^2 h}{\partial x_1\, \partial x_2} & \cdots & \frac{\partial^2 h}{\partial x_1\, \partial x_n} \\ \frac{-\partial g}{\partial x_2} & \frac{\partial^2 h}{\partial x_1\, \partial x_2} & \frac{\partial^2 h}{\partial x_2^2} & \cdots & \frac{\partial^2 h}{\partial x_2\, \partial x_n} \\ \vdots & \vdots & \vdots & & \vdots \\ \frac{-\partial g}{\partial x_n} & \frac{\partial^2 h}{\partial x_1\, \partial x_n} & \frac{\partial^2 h}{\partial x_2\, \partial x_n} & \cdots & \frac{\partial^2 h}{\partial x_n^2} \end{vmatrix}.$$

Second, we examine the determinants of the diagonal submatrices of order ≥ 3 at the critical points of h. If they are all negative, that is, if

$$\begin{vmatrix} 0 & -\frac{\partial g}{\partial x_1} & -\frac{\partial g}{\partial x_2} \\ -\frac{\partial g}{\partial x_1} & \frac{\partial^2 h}{\partial x_1^2} & \frac{\partial^2 h}{\partial x_1\, \partial x_2} \\ -\frac{\partial g}{\partial x_2} & \frac{\partial^2 h}{\partial x_1\, \partial x_2} & \frac{\partial^2 h}{\partial x_2^2} \end{vmatrix} < 0, \quad \begin{vmatrix} 0 & -\frac{\partial g}{\partial x_1} & -\frac{\partial g}{\partial x_2} & -\frac{\partial g}{\partial x_3} \\ -\frac{\partial g}{\partial x_1} & \frac{\partial^2 h}{\partial x_1^2} & \frac{\partial^2 h}{\partial x_1\, \partial x_2} & \frac{\partial^2 h}{\partial x_1\, \partial x_3} \\ -\frac{\partial g}{\partial x_2} & \frac{\partial^2 h}{\partial x_1\, \partial x_2} & \frac{\partial^2 h}{\partial x_2^2} & \frac{\partial^2 h}{\partial x_2\, \partial x_3} \\ -\frac{\partial g}{\partial x_3} & \frac{\partial^2 h}{\partial x_1\, \partial x_3} & \frac{\partial^2 h}{\partial x_2\, \partial x_3} & \frac{\partial^2 h}{\partial x_3^2} \end{vmatrix} < 0, \ldots,$$

then we are at a local minimum of $f|S$. If they start out with a positive 3×3 subdeterminant and alternate in sign (that is, >0, <0, >0, <0, ...), then we are at a local

maximum. If they are all nonzero and do not fit one of these patterns, then the point is neither a maximum nor a minimum (it is said to be of the saddle type).[14]

EXAMPLE 12 Study the local extreme points of $f(x, y, z) = xyz$ on the surface of the unit sphere $x^2 + y^2 + z^2 = 1$ using the second-derivative test.

SOLUTION Setting the partial derivatives of the auxiliary function $h(x, y, z, \lambda) = xyz - \lambda(x^2 + y^2 + z^2 - 1)$ equal to zero gives

$$\begin{aligned} yz &= 2\lambda x \\ xz &= 2\lambda y \\ xy &= 2\lambda z \\ x^2 + y^2 + z^2 &= 1. \end{aligned}$$

Thus, $3xyz = 2\lambda(x^2 + y^2 + z^2) = 2\lambda$. If $\lambda = 0$, the solutions are $(x, y, z, \lambda) = (\pm 1, 0, 0, 0)$, $(0, \pm 1, 0, 0)$, and $(0, 0, \pm 1, 0)$. If $\lambda \neq 0$, then we have $2\lambda = 3xyz = 6\lambda z^2$ and so $z^2 = \frac{1}{3}$. Similarly, $x^2 = y^2 = \frac{1}{3}$. Thus, the solutions are given by $\lambda = \frac{3}{2}xyz = \pm\sqrt{3}/6$. The critical points of h and the corresponding values of f are given in Table 3.1. From it, we see that points E, F, G, and K are minima. Points D, H, I, and J are maxima. To see whether this is in accord with the second-derivative

Table 3.1 The critical points A, B, . . . , J, K of h and corresponding values of f

	x	y	z	λ	$f(x, y, z)$
±A	±1	0	0	0	0
±B	0	±1	0	0	0
±C	0	0	±1	0	0
D	$\sqrt{3}/3$	$\sqrt{3}/3$	$\sqrt{3}/3$	$\sqrt{3}/6$	$\sqrt{3}/9$
E	$-\sqrt{3}/3$	$\sqrt{3}/3$	$\sqrt{3}/3$	$-\sqrt{3}/6$	$-\sqrt{3}/9$
F	$\sqrt{3}/3$	$-\sqrt{3}/3$	$\sqrt{3}/3$	$-\sqrt{3}/6$	$-\sqrt{3}/9$
G	$\sqrt{3}/3$	$\sqrt{3}/3$	$-\sqrt{3}/3$	$-\sqrt{3}/6$	$-\sqrt{3}/9$
H	$\sqrt{3}/3$	$-\sqrt{3}/3$	$-\sqrt{3}/3$	$\sqrt{3}/6$	$\sqrt{3}/9$
I	$-\sqrt{3}/3$	$\sqrt{3}/3$	$-\sqrt{3}/3$	$\sqrt{3}/6$	$\sqrt{3}/9$
J	$-\sqrt{3}/3$	$-\sqrt{3}/3$	$\sqrt{3}/3$	$\sqrt{3}/6$	$\sqrt{3}/9$
K	$-\sqrt{3}/3$	$-\sqrt{3}/3$	$-\sqrt{3}/3$	$-\sqrt{3}/6$	$-\sqrt{3}/9$

[14] For a detailed discussion, see C. Caratheodory, *Calculus of Variations and Partial Differential Equations*, Holden-Day, San Francisco, 1965; Y. Murata, *Mathematics for Stability and Optimization of Economic Systems*, Academic Press, New York, 1977, pp. 263–271; or D. Spring, *Am. Math. Mon.* 92 (1985): 631–643.

test, we need to consider two determinants. First, we look at the following:

$$|\bar{H}_2| = \begin{vmatrix} 0 & -\partial g/\partial x & -\partial g/\partial y \\ -\partial g/\partial x & \partial^2 h/\partial x^2 & \partial^2/\partial x\,\partial y \\ -\partial g/\partial y & \partial^2 h/\partial x\,\partial y & \partial^2 h/\partial y^2 \end{vmatrix} = \begin{vmatrix} 0 & -2x & -2y \\ -2x & -2\lambda & z \\ -2y & z & -2\lambda \end{vmatrix}$$
$$= 8\lambda x^2 + 8\lambda y^2 + 8xyz = 8\lambda(x^2 + y^2 + 2z^2).$$

Observe that sign $(|\bar{H}_2|) =$ sign $\lambda =$ sign (xyz), where the sign of a number is 1 if that number is positive, or is -1 if that number is negative. Second, we consider

$$|\bar{H}_3| = \begin{vmatrix} 0 & -\partial g/\partial x & -\partial g/\partial y & -\partial g/\partial z \\ -\partial g/\partial x & \partial^2 h/\partial x^2 & \partial^2 h/\partial x\,\partial y & \partial^2 h/\partial x\,\partial z \\ -\partial g/\partial y & \partial^2 h/\partial x\,\partial y & \partial^2 h/\partial y^2 & \partial^2 h/\partial y\,\partial z \\ -\partial g/\partial z & \partial^2 h/\partial x\,\partial z & \partial^2 h/\partial y\,\partial z & \partial^2 h/\partial z^2 \end{vmatrix}$$
$$= \begin{vmatrix} 0 & -2x & -2y & -2z \\ -2x & -2\lambda & z & y \\ -2y & z & -2\lambda & x \\ -2z & y & x & -2\lambda \end{vmatrix},$$

which works out to be $+4$ at points $\pm$A, $\pm$B, and $\pm$C and $-\frac{16}{3}$ at the other eight points. At E, F, G, and K, we have $|\bar{H}_2| < 0$ *and* $|\bar{H}_3| < 0$, and so the test indicates these are local minima. At D, H, I, and J we have $|\bar{H}_2| > 0$ and $|\bar{H}_3| < 0$, and so the test says these are local maxima. Finally, the second-derivative test shows that $\pm$A, $\pm$B, and $\pm$C are saddle points. ▲

EXERCISES

In Exercises 1 to 5 find the extrema of f subject to the stated constraints.

1. $f(x, y, z) = x - y + z$, subject to $x^2 + y^2 + z^2 = 2$

2. $f(x, y) = x - y$, subject to $x^2 - y^2 = 2$

3. $f(x, y) = x$, subject to $x^2 + 2y^2 = 3$

4. $f(x, y, z) = x + y + z$, subject to $x^2 - y^2 = 1, 2x + z = 1$

5. $f(x, y) = 3x + 2y$, subject to $2x^2 + 3y^2 = 3$

Find the relative extrema of $f|S$ in Exercises 6 to 9.

6. $f\colon \mathbb{R}^2 \to \mathbb{R}, (x, y) \mapsto x^2 + y^2, S = \{(x, 2) \mid x \in \mathbb{R}\}$

7. $f\colon \mathbb{R}^2 \to \mathbb{R}, (x, y) \mapsto x^2 + y^2, S = \{(x, y) \mid y \geq 2\}$

8. $f\colon \mathbb{R}^2 \to \mathbb{R}, (x, y) \mapsto x^2 - y^2, S = \{(x, \cos x) \mid x \in \mathbb{R}\}$

9. $f\colon \mathbb{R}^3 \to \mathbb{R}, (x, y, z) \mapsto x^2 + y^2 + z^2, S = \{(x, y, z) \mid z \geq 2 + x^2 + y^2\}$

10. Use the method of Lagrange multipliers to find the absolute maximum and minimum values of $f(x, y) = x^2 + y^2 - x - y + 1$ on the unit disk (see Example 10 of Section 3.3).

11. Consider the function $f(x, y) = x^2 + xy + y^2$ defined on the unit disk, namely, $D = \{(x, y) \mid x^2 + y^2 \leq 1\}$. Use the method of Lagrange multipliers to locate the maximum and minimum points for f on the unit circle. Use this to determine the absolute maximum and minimum values for f on D.

12. A rectangular box with no top is to have a surface area of 16 m^2. Find the dimensions that maximize its volume.

13. Design a cylindrical can (with a lid) to contain 1 liter ($= 1000$ cm^3) of water, using the minimum amount of metal.

14. Show that solutions of equations (4) and (5) are in one-to-one correspondence with the critical points of

$$h(x_1, \ldots, x_n, \lambda_1, \ldots, \lambda_k) = f(x_1, \ldots, x_n) - \lambda_1[g_1(x_1, \ldots, x_n) - c_1] - \cdots - \lambda_k[g_k(x_1, \ldots, x_n) - c_k].$$

15. Find the absolute maximum and minimum for the function $f(x, y, z) = x + y - z$ on the ball $B = \{(x, y, z) \mid x^2 + y^2 + z^2 \leq 1\}$.

16. Repeat Exercise 15 for $f(x, y, z) = x + yz$.

17. A rectangular mirror with area A square feet is to have trim along the edges. If the trim along the horizontal edges costs p cents per foot and that for the vertical edges costs q cents per foot, find the dimensions that will minimize the total cost.

18. An irrigation canal in Arizona has concrete sides and bottom with trapezoidal cross section of area $A = y(x + y \tan\theta)$ and wetted perimeter $P = x + 2y/\cos\theta$, where $x =$ bottom width, $y =$ water depth, $\theta =$ side inclination, measured from vertical. The best design for a fixed inclination θ is found by solving $P =$ minimum subject to the condition $A =$ constant. Show that $y^2 = (A\cos\theta)/(2 - \sin\theta)$.

19. Apply the second-derivative test to study the nature of the extrema in Exercises 1 and 5.

20. A light ray travels from point A to point B crossing a boundary between two media (see Figure 3.4.7). In the first medium its speed is v_1, and in the second it is v_2. Show that the trip is made in minimum time when *Snell's law* holds:

$$\frac{\sin\theta_1}{\sin\theta_2} = \frac{v_1}{v_2}.$$

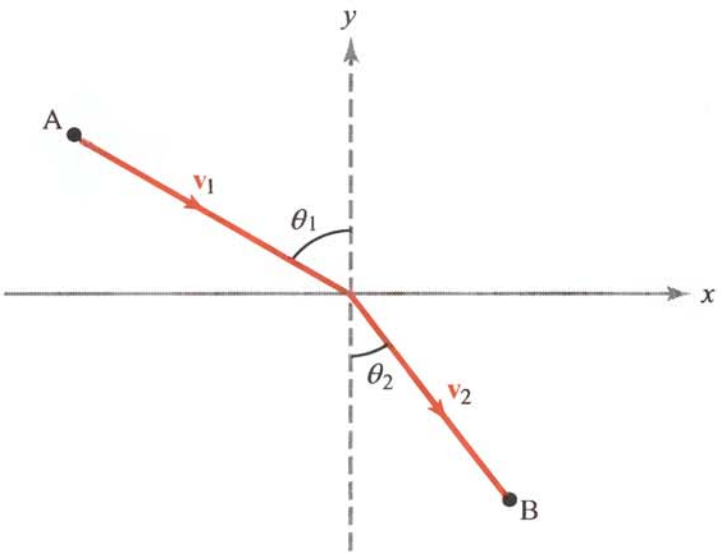

Figure 3.4.7 Snell's law of refraction.

21. A parcel delivery service requires that the dimensions of a rectangular box be such that the length plus twice the width plus twice the height be no more than 108 inches ($l + 2w + 2h \leq 108$). What is the volume of the largest-volume box the company will deliver?

22. Let P be a point on a surface S in $\mathbb{R}^3$ defined by the equation $f(x, y, z) = 1$, where f is of class C^1. Suppose that P is a point where the distance from the origin to S is maximized. Show that the vector emanating from the origin and ending at P is perpendicular to S.

23. Let A be a nonzero symmetric 3×3 matrix. Thus, its entries satisfy $a_{ij} = a_{ji}$. Consider the function $f(\mathbf{x}) = \frac{1}{2}(A\mathbf{x}) \cdot \mathbf{x}$.

(a) What is ∇f?

(b) Consider the restriction of f to the unit sphere $S = \{(x, y, z) \mid x^2 + y^2 + z^2 = 1\}$ in $\mathbb{R}^3$. By Theorem 7 we know that f must have a maximum and a minimum on S. Show that there must be an $\mathbf{x} \in S$ and a $\lambda \neq 0$ such that $A\mathbf{x} = \lambda\mathbf{x}$. (The vector $\mathbf{x}$ is called an ***eigenvector***, while the scalar λ is called an ***eigenvalue***.)

(c) What are the maxima and minima for f on $B = \{(x, y, z) \mid x^2 + y^2 + z^2 \leq 1\}$?

24. Suppose that A in the function f defined in Exercise 23 is not necessarily symmetric.

(a) What is ∇f?

(b) Can one conclude the existence of an eigenvector and eigenvalues as in Exercise 23?

25. (a) Find the critical points of $x + y^2$ subject to the constraint $2x^2 + y^2 = 1$.

(b) Use the bordered Hessian to classify the critical points.

26. Answer the question posed in the last line of Example 9.

27. Try to find the extrema of $xy + yz$ among points satisfying $xz = 1$.

28. A company's production function is $Q(x, y) = xy$. The cost of production is $C(x, y) = 2x + 3y$. If this company can spend $C(x, y) = 10$, what is the maximum quantity that can be produced?

29. Find the point on the curve $(\cos t, \sin t, \sin(t/2))$ that is farthest from the origin.

30. A firm uses wool and cotton fiber to produce cloth. The amount of cloth produced is given by $Q(x, y) = xy - x - y + 1$, where x is the number of pounds of wool, y the number of pounds of cotton, $x > 1$, and $y > 1$. If wool costs p dollars per pound, and cotton q dollars per pound and the firm can spend B dollars on material, what should the ratio of cotton and wool be to produce the most cloth?

31. Carry out the analysis of Example 10 for the production function $Q(K, L) = AK^{\alpha}L^{1-\alpha}$, where A and α are positive constants and $0 < \alpha < 1$. This is called a ***Cobb–Douglas production function*** and is sometimes used as a simple model for the national economy. Q is then the aggregate output of the economy for a given input of capital and labor.

3.5 The Implicit Function Theorem

In this section, we state two versions of the *implicit function theorem*, arguably the most important theorem in all of mathematical analysis. The entire theoretical basis of the idea of a surface as well as the method of Lagrange multipliers depends on it. Moreover, it is a cornerstone of several fields of mathematics, such as differential topology and geometry.

The One-Variable Implicit Function Theorem

In one-variable calculus, we learn the importance of the inversion process. For example, $x = \ln y$ is the inverse of $y = e^x$, and $x = \sin^{-1} y$ is the inverse of $y = \sin x$. The inversion process is also important for functions of several variables; for example, the switch between Cartesian and polar coordinates in the plane involves inverting two functions of two variables.

Recall from one-variable calculus that if $y = f(x)$ is a C^1 function and $f'(x_0) \neq 0$, then locally near x_0 we can solve for x to give the inverse function: $x = f^{-1}(y)$. We learn that $(f^{-1})'(y) = 1/f'(x)$; that is, $dx/dy = 1/(dy/dx)$. That $y = f(x)$ can be inverted is plausible because $f'(x_0) \neq 0$ means that the slope of $y = f(x)$ is nonzero, so that the graph is rising or falling near x_0. Thus, if we reflect the graph across the line $y = x$, it is still a graph *near* (x_0, y_0) where $y_0 = f(x_0)$. For example, in Figure 3.5.1, we can invert $y = f(x)$ in the shaded box, so in this range, $x = f^{-1}(y)$ is defined.

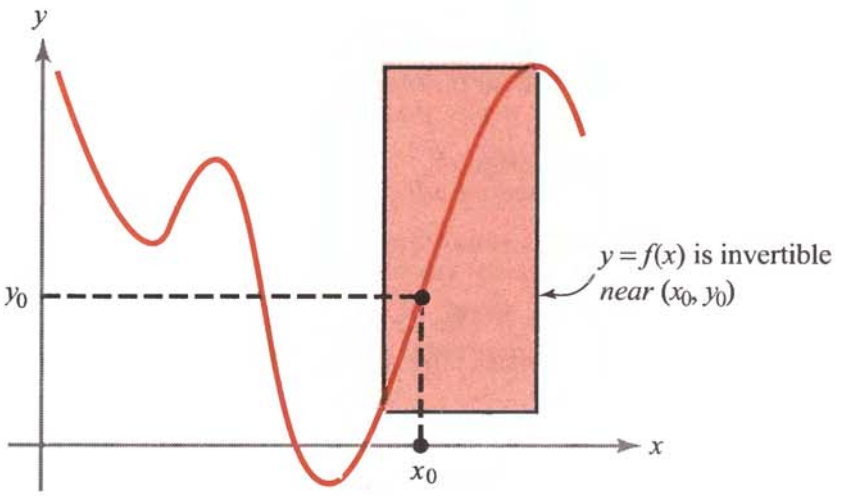

Figure 3.5.1 If $f'(x_0) \neq 0$, then $y = f(x)$ is locally invertible.

A Special Result

We next turn to the situation for real-valued functions of variables $x_1, \ldots, x_n$ and z.

THEOREM 11: Special Implicit Function Theorem Suppose that $F\colon \mathbb{R}^{n+1} \to \mathbb{R}$ has continuous partial derivatives. Denoting points in $\mathbb{R}^{n+1}$ by $(\mathbf{x}, z)$, where $\mathbf{x} \in \mathbb{R}^n$ and $z \in \mathbb{R}$, assume that $(\mathbf{x}_0, z_0)$ satisfies

$$F(\mathbf{x}_0, z_0) = 0 \qquad \text{and} \qquad \frac{\partial F}{\partial z}(\mathbf{x}_0, z_0) \neq 0.$$

Then there is a ball U containing $\mathbf{x}_0$ in $\mathbb{R}^n$ and a neighborhood V of z_0 in $\mathbb{R}$ such that there is a unique function $z = g(\mathbf{x})$ defined for $\mathbf{x}$ in U and z in V that satisfies

$$F(\mathbf{x}, g(\mathbf{x})) = 0.$$

Moreover, if $\mathbf{x}$ in U and z in V satisfy $F(\mathbf{x}, z) = 0$, then $z = g(\mathbf{x})$. Finally, $z = g(\mathbf{x})$ is continuously differentiable, with the derivative given by

$$\mathbf{D}g(\mathbf{x}) = -\frac{1}{\dfrac{\partial F}{\partial z}(\mathbf{x}, z)}\, \mathbf{D}_{\mathbf{x}} F(\mathbf{x}, z)\Big|_{z=g(\mathbf{x})},$$

where $\mathbf{D}_{\mathbf{x}} F$ denotes the (partial) derivative of F with respect to the variable $\mathbf{x}$, that is, we have $\mathbf{D}_{\mathbf{x}} F = [\partial F/\partial x_1, \ldots, \partial F/\partial x_n]$; in other words,

$$\frac{\partial g}{\partial x_i} = -\frac{\partial F/\partial x_i}{\partial F/\partial z}, \qquad i = 1, \ldots, n. \tag{1}$$

A proof of this theorem is given in the Internet supplement.

Once it is known that $z = g(\mathbf{x})$ exists and is differentiable, formula (1) may be checked by implicit differentiation; to see this, note that the chain rule applied to $F(\mathbf{x}, g(\mathbf{x})) = 0$ gives

$$\mathbf{D}_{\mathbf{x}} F(\mathbf{x}, g(\mathbf{x})) + \left[\frac{\partial F}{\partial z}(\mathbf{x}, g(\mathbf{x}))\right][\mathbf{D}g(\mathbf{x})] = 0,$$

which is equivalent to formula (1).

EXAMPLE 1 In the special implicit function theorem, it is important to recognize the necessity of taking sufficiently small neighborhoods U and V. For example, consider the equation

$$x^2 + z^2 - 1 = 0,$$

that is, $F(x, z) = x^2 + z^2 - 1$, with $n = 1$. Here $(\partial F/\partial z)(x, z) = 2z$, and so the special implicit function theorem applies to a point (x_0, z_0) satisfying $x_0^2 + z_0^2 - 1 = 0$ and $z_0 \neq 0$. Thus, near such points, z is a unique function of x. This function is

$z = \sqrt{1 - x^2}$ if $z_0 > 0$ and $z = -\sqrt{1 - x^2}$ if $z_0 < 0$. Note that z is defined for $|x| < 1$ only (U must not be too big) and z is unique only if it is near z_0 (V must not be too big). These facts and the nonexistence of $\partial z/\partial x$ at $z_0 = 0$ are, of course, clear from the fact that $x^2 + z^2 = 1$ defines a circle in the xz plane (Figure 3.5.2). ▲

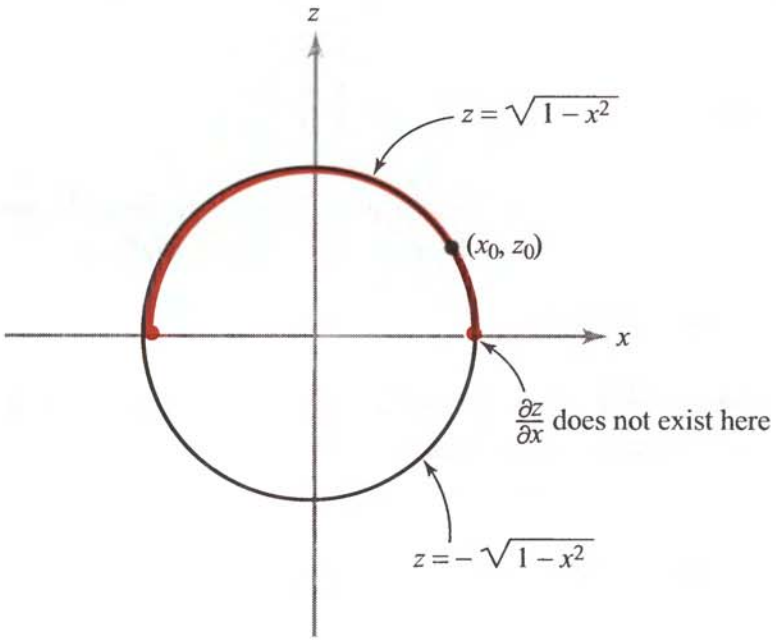

Figure 3.5.2 It is necessary to take small neighborhoods in the implicit function theorem.

The Implicit Function Theorem and Surfaces

Let us apply Theorem 11 to the study of surfaces. We are concerned with the level set of a function g: $U \subset \mathbb{R}^n \to \mathbb{R}$, that is, with the surface S consisting of the set of $\mathbf{x}$ satisfying $g(\mathbf{x}) = c_0$, where $c_0 = g(\mathbf{x}_0)$ and where $\mathbf{x}_0$ is given. Let us take $n = 3$ for concreteness. Thus, we are dealing with the level surface of a function $g(x, y, z)$ through a given point (x_0, y_0, z_0). As in the Lagrange multiplier theorem, assume that $\nabla g(x_0, y_0, z_0) \neq \mathbf{0}$. This means that at least one of the partial derivatives of g is nonzero. For definiteness, suppose that $(\partial g/\partial z)(x_0, y_0, z_0) \neq 0$. By applying Theorem 11 to the function $(x, y, z) \mapsto g(x, y, z) - c_0$, we know there is a unique function $z = k(x, y)$ satisfying $g(x, y, k(x, y)) = c_0$ for (x, y) near (x_0, y_0) and z near z_0. Thus, near z_0 the surface S is the graph of the function k. Because k is continuously differentiable, this surface has a tangent plane at (x_0, y_0, z_0) given by

$$z = z_0 + \left[\frac{\partial k}{\partial x}(x_0, y_0)\right](x - x_0) + \left[\frac{\partial k}{\partial y}(x_0, y_0)\right](y - y_0). \tag{2}$$

But by formula (1),

$$\frac{\partial k}{\partial x}(x_0, y_0) = -\frac{\dfrac{\partial g}{\partial x}(x_0, y_0, z_0)}{\dfrac{\partial g}{\partial z}(x_0, y_0, z_0)} \quad \text{and} \quad \frac{\partial k}{\partial y}(x_0, y_0) = -\frac{\dfrac{\partial g}{\partial y}(x_0, y_0, z_0)}{\dfrac{\partial g}{\partial z}(x_0, y_0, z_0)}.$$

Substituting these two equations into the equation for the tangent plane gives this equivalent description:

$$0 = (z - z_0)\frac{\partial g}{\partial z}(x_0, y_0, z_0) + (x - x_0)\frac{\partial g}{\partial x}(x_0, y_0, z_0) + (y - y_0)\frac{\partial g}{\partial y}(x_0, y_0, z_0);$$

that is,

$$(x - x_0, y - y_0, z - z_0) \cdot \nabla g(x_0, y_0, z_0) = 0.$$

Thus, the tangent plane to the level surface of g is the orthogonal complement to $\nabla g(x_0, y_0, z_0)$ through the point (x_0, y_0, z_0). This agrees with our characterization of tangent planes to level sets from Chapter 2.

We are now ready to complete the proof of the Lagrange multiplier theorem. To do this, we must show that every vector tangent to S at (x_0, y_0, z_0) is tangent to a curve in S. By Theorem 11, we need only show this for a graph of the form $z = k(x, y)$. However, if $\mathbf{v} = (x - x_0, y - y_0, z - z_0)$ is tangent to the graph [that is, if it satisfies equation (2)], then $\mathbf{v}$ is tangent to the path in S given by

$$\mathbf{c}(t) = (x_0 + t(x - x_0), y_0 + t(y - y_0), k(x_0 + t(x - x_0), y_0 + t(y - y_0)))$$

at $t = 0$. This can be checked by using the chain rule. (See Figure 3.5.3.)

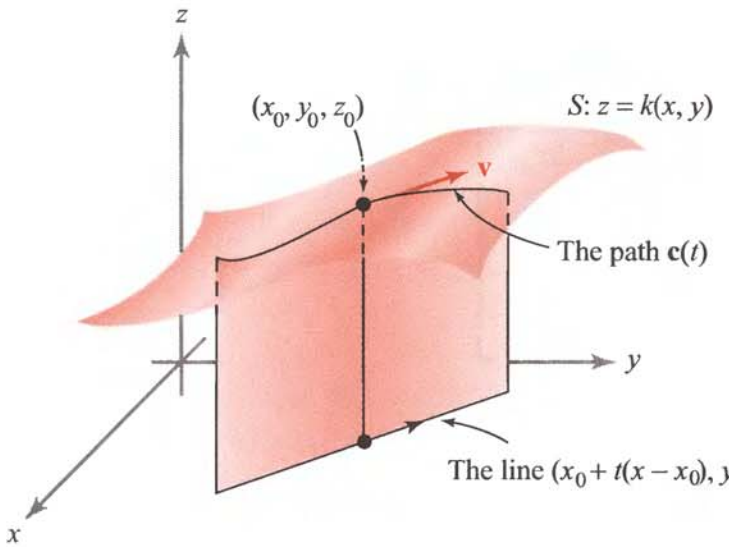

Figure 3.5.3 The construction of a path $\mathbf{c}(t)$ in the surface S whose tangent vector is $\mathbf{v}$.

EXAMPLE 2 Near what points may the surface

$$x^3 + 3y^2 + 8xz^2 - 3z^3y = 1$$

be represented as a graph of a differentiable function $z = k(x, y)$?

SOLUTION Here we take $F(x, y, z) = x^3 + 3y^2 + 8xz^2 - 3z^3y - 1$ and attempt to solve $F(x, y, z) = 0$ for z as a function of (x, y). By Theorem 11, this may be done near a point (x_0, y_0, z_0) if $(\partial F/\partial z)(x_0, y_0, z_0) \neq 0$, that is, if

$$z_0(16x_0 - 9z_0y_0) \neq 0,$$

which means, in turn,

$$z_0 \neq 0 \qquad \text{and} \qquad 16x_0 \neq 9z_0y_0. \quad \blacktriangle$$

General Implicit Function Theorem

Next we shall state, without proof, the *general implicit function theorem*.[15] Instead of attempting to solve one equation for one variable, we attempt to solve m equations for m variables $z_1, \ldots, z_m$:

$$\begin{aligned} F_1(x_1, \ldots, x_n, z_1, \ldots, z_m) &= 0 \\ F_2(x_1, \ldots, x_n, z_1, \ldots, z_m) &= 0 \\ \vdots \qquad\qquad & \vdots \qquad\qquad \vdots \\ F_m(x_1, \ldots, x_n, z_1, \ldots, z_m) &= 0. \end{aligned} \tag{3}$$

In Theorem 11 we had the condition $\partial F/\partial z \neq 0$. The condition appropriate to the general implicit function theorem is that $\Delta \neq 0$,[16] where Δ is the determinant of the $m \times m$ matrix

$$\begin{bmatrix} \dfrac{\partial F_1}{\partial z_1} & \cdots & \dfrac{\partial F_1}{\partial z_m} \\ \vdots & & \vdots \\ \dfrac{\partial F_m}{\partial z_1} & \cdots & \dfrac{\partial F_m}{\partial z_m} \end{bmatrix}$$

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[15] For three different proofs of the general case, consult:

(a) E. Goursat, *A Course in Mathematical Analysis*, I, Dover, New York, 1959, p. 45. (This proof derives the general theorem by successive application of Theorem 11.)

(b) T. M. Apostol, *Mathematical Analysis*, 2d ed., Addison-Wesley, Reading, Mass., 1974.

(c) J. E. Marsden and M. Hoffman, *Elementary Classical Analysis*, 2d ed., Freeman, New York, 1993.

Of these sources, the last two use more sophisticated ideas that are usually not covered until a junior-level course in analysis. The first, however, is easily understood by the reader who has some knowledge of linear algebra.

[16] For students who have had linear algebra: The condition $\Delta \neq 0$ has a simple interpretation in the case that F *is linear*; namely, $\Delta \neq 0$ is equivalent to the rank of F being equal to m, which in turn is equivalent to the fact that the solution space of $F = 0$ is m-dimensional.

evaluated at the point $(\mathbf{x}_0, \mathbf{z}_0)$; in the neighborhood of such a point, we can uniquely solve for $\mathbf{z}$ in terms of $\mathbf{x}$.

THEOREM 12: General Implicit Function Theorem If $\Delta \neq 0$, then near the point $(\mathbf{x}_0, \mathbf{z}_0)$, equation (3) defines unique (smooth) functions

$$z_i = k_i(x_1, \ldots, x_n) \qquad (i = 1, \ldots, m).$$

Their derivatives may be computed by implicit differentiation.

EXAMPLE 3 Show that near the point $(x, y, u, v) = (1, 1, 1, 1)$, we can solve

$$xu + yvu^2 = 2$$
$$xu^3 + y^2v^4 = 2$$

uniquely for u and v as functions of x and y. Compute $\partial u/\partial x$ at the point $(1, 1)$.

SOLUTION To check solvability, we form the equations

$$F_1(x, y, u, v) = xu + yvu^2 - 2$$
$$F_2(x, y, u, v) = xu^3 + y^2v^4 - 2$$

and the determinant

$$\Delta = \begin{vmatrix} \dfrac{\partial F_1}{\partial u} & \dfrac{\partial F_1}{\partial v} \\ \dfrac{\partial F_2}{\partial u} & \dfrac{\partial F_2}{\partial v} \end{vmatrix} \qquad \text{at} \qquad (1, 1, 1, 1)$$

$$= \begin{vmatrix} x + 2yuv & yu^2 \\ 3u^2x & 4y^2v^3 \end{vmatrix} \qquad \text{at} \qquad (1, 1, 1, 1)$$

$$= \begin{vmatrix} 3 & 1 \\ 3 & 4 \end{vmatrix} = 9.$$

Because $\Delta \neq 0$, solvability is assured by the general implicit function theorem. To find $\partial u/\partial x$, we implicitly differentiate the given equations in x using the chain rule:

$$x\frac{\partial u}{\partial x} + u + y\frac{\partial v}{\partial x}u^2 + 2yvu\frac{\partial u}{\partial x} = 0$$

$$3xu^2\frac{\partial u}{\partial x} + u^3 + 4y^2v^3\frac{\partial v}{\partial x} = 0.$$

Setting $(x, y, u, v) = (1, 1, 1, 1)$ gives

$$3\frac{\partial u}{\partial x} + \frac{\partial v}{\partial x} = -1$$

$$3\frac{\partial u}{\partial x} + 4\frac{\partial v}{\partial x} = -1.$$

Solving for $\partial u/\partial x$ by multiplying the first equation by 4 and subtracting gives $\partial u/\partial x = -\frac{1}{3}$. ▲

Inverse Function Theorem

A special case of the general implicit function theorem is the *inverse function theorem*. Here we attempt to solve the n equations

$$\left.\begin{aligned} f_1(x_1, \ldots, x_n) &= y_1 \\ &\cdots \\ f_n(x_1, \ldots, x_n) &= y_n \end{aligned}\right\} \tag{4}$$

for $x_1, \ldots, x_n$ as functions of $y_1, \ldots, y_n$; that is, we are trying to invert the equations of system (4). This is analogous to forming the inverses of functions like $\sin x = y$ and $e^x = y$, with which the reader should be familiar from elementary calculus. Now, however, we are concerned with functions of several variables. The question of solvability is answered by the general implicit function theorem applied to the functions $y_i - f_i(x_1, \ldots, x_n)$ with the unknowns $x_1, \ldots, x_n$ (called $z_1, \ldots, z_n$ earlier). The condition for solvability in a neighborhood of a point $\mathbf{x}_0$ is $\Delta \neq 0$, where Δ is the determinant of the matrix $\mathbf{D}f(\mathbf{x}_0)$, and $f = (f_1, \ldots, f_n)$. The quantity Δ is denoted by $\partial(f_1, \ldots, f_n)/\partial(x_1, \ldots, x_n)$, or $\partial(y_1, \ldots, y_n)/\partial(x_1, \ldots, x_n)$ or $J(f)(\mathbf{x}_0)$ and is called the ***Jacobian determinant*** of f. Explicitly,

$$\left.\frac{\partial(f_1, \ldots, f_n)}{\partial(x_1, \ldots, x_n)}\right|_{\mathbf{x}=\mathbf{x}_0} = J(f)(\mathbf{x}_0) = \begin{vmatrix} \dfrac{\partial f_1}{\partial x_1}(\mathbf{x}_0) & \cdots & \dfrac{\partial f_1}{\partial x_n}(\mathbf{x}_0) \\ \vdots & & \vdots \\ \dfrac{\partial f_n}{\partial x_1}(\mathbf{x}_0) & \cdots & \dfrac{\partial f_n}{\partial x_n}(\mathbf{x}_0) \end{vmatrix}. \tag{5}$$

The reader should note that in the case when f is linear, for example $f(x) = Ax$, where A is an $n \times n$ matrix, the condition $\Delta \neq 0$ is equivalent to the fact that the determinant of A, $\det A \neq 0$, and from Section 1.5 we know that A, and therefore f, has an inverse.

The Jacobian determinant will play an important role in our work on integration (see Chapter 5). The following theorem summarizes this discussion:

THEOREM 13: Inverse Function Theorem Let $U \subset \mathbb{R}^n$ be open and let $f_1: U \to \mathbb{R}, \ldots, f_n: U \to \mathbb{R}$ have continuous partial derivatives. Consider the equations (4) near a given solution $\mathbf{x}_0, \mathbf{y}_0$. If $J(f)(\mathbf{x}_0)$ [defined by equation (5)] is nonzero, then equation (4) can be solved uniquely as $\mathbf{x} = g(\mathbf{y})$ for $\mathbf{x}$ near $\mathbf{x}_0$ and $\mathbf{y}$ near $\mathbf{y}_0$. Moreover, the function g has continuous partial derivatives.

EXAMPLE 4 Consider the equations

$$\frac{x^4 + y^4}{x} = u, \qquad \sin x + \cos y = v.$$

Near which points (x, y) can we solve for x, y in terms of u, v?

SOLUTION Here the functions are $u = f_1(x, y) = (x^4 + y^4)/x$ and $v = f_2(x, y) = \sin x + \cos y$. We want to know the points near which we can solve for x, y as functions of u and v. According to the inverse function theorem, we must first compute the Jacobian determinant $\partial(f_1, f_2)/\partial(x, y)$. We take the domain of $f = (f_1, f_2)$ to be $U = \{(x, y) \in \mathbb{R}^2 \mid x \neq 0\}$. Now

$$\frac{\partial(f_1, f_2)}{\partial(x, y)} = \begin{vmatrix} \dfrac{\partial f_1}{\partial x} & \dfrac{\partial f_1}{\partial y} \\ \dfrac{\partial f_2}{\partial x} & \dfrac{\partial f_2}{\partial y} \end{vmatrix} = \begin{vmatrix} \dfrac{3x^4 - y^4}{x^2} & \dfrac{4y^3}{x} \\ \cos x & -\sin y \end{vmatrix} = \frac{\sin y}{x^2}(y^4 - 3x^4) - \frac{4y^3}{x}\cos x.$$

Therefore, at points where this does not vanish we can solve for x, y in terms of u and v. In other words, we can solve for x, y near those x, y for which $x \neq 0$ and $(\sin y)(y^4 - 3x^4) \neq 4xy^3 \cos x$. Such conditions generally cannot be solved explicitly. For example, if $x_0 = \pi/2$, $y_0 = \pi/2$, we can solve for x, y near (x_0, y_0) because there, $\partial(f_1, f_2)/\partial(x, y) \neq 0$. ▲

EXERCISES

1. Let $F(x, y) = 0$ define a curve in the xy plane through the point (x_0, y_0), where F is C^1. Assume that $(\partial F/\partial y)(x_0, y_0) \neq 0$. Show that this curve can be locally represented by the graph of a function $y = g(x)$. Show that (i) the line orthogonal to $\nabla F(x_0, y_0)$ agrees with (ii) the tangent line to the graph of $y = g(x)$.

2. Show that $xy + z + 3xz^5 = 4$ is solvable for z as a function of (x, y) near $(1, 0, 1)$. Compute $\partial z/\partial x$ and $\partial z/\partial y$ at $(1, 0)$.

3. (a) Check directly (i.e., without using Theorem 11) where we can solve the equation $F(x, y) = y^2 + y + 3x + 1 = 0$ for y in terms of x.

(b) Check that your answer in part (a) agrees with the answer you expect from the implicit function theorem. Compute dy/dx.

4. Repeat Exercise 3 with $F(x, y) = xy^2 - 2y + x^2 + 2 = 0$.

5. Show that $x^3z^2 - z^3yx = 0$ is solvable for z as a function of (x, y) near $(1, 1, 1)$, but not near the origin. Compute $\partial z/\partial x$ and $\partial z/\partial y$ at $(1, 1)$.

6. Discuss the solvability in the system

$$\begin{aligned} 3x + 2y + z^2 + u + v^2 &= 0 \\ 4x + 3y + z + u^2 + v + w + 2 &= 0 \\ x + z + w + u^2 + 2 &= 0 \end{aligned}$$

for u, v, w in terms of x, y, z near $x = y = z = 0$, $u = v = 0$, and $w = -2$.

7. Discuss the solvability of

$$\begin{aligned} y + x + uv &= 0 \\ uxy + v &= 0 \end{aligned}$$

for u, v in terms of x, y near $x = y = u = v = 0$ and check directly.

8. Investigate whether or not the system

$$\begin{aligned} u(x, y, z) &= x + xyz \\ v(x, y, z) &= y + xy \\ w(x, y, z) &= z + 2x + 3z^2 \end{aligned}$$

can be solved for x, y, z in terms of u, v, w near $(x, y, z) = (0, 0, 0)$.

9. Consider $f(x, y) = ((x^2 - y^2)/(x^2 + y^2), xy/(x^2 + y^2))$. Does this map of $\mathbb{R}^2\backslash(0, 0)$ to $\mathbb{R}^2$ have a local inverse near $(x, y) = (0, 1)$?

10. (a) Define $x\colon \mathbb{R}^2 \to \mathbb{R}$ by $x(r, \theta) = r\cos\theta$ and define $y\colon \mathbb{R}^2 \to \mathbb{R}$ by $y(r, \theta) = r\sin\theta$. Show that

$$\left.\frac{\partial(x, y)}{\partial(r, \theta)}\right|_{(r_0,\theta_0)} = r_0.$$

(b) When can we form a smooth inverse function $(r(x, y), \theta(x, y))$? Check directly and with the inverse function theorem.

(c) Consider the following transformations for spherical coordinates (see Section 1.4):

$$\begin{aligned} x(\rho, \phi, \theta) &= \rho\sin\phi\cos\theta \\ x(\rho, \phi, \theta) &= \rho\sin\phi\sin\theta \\ z(\rho, \phi, \theta) &= \rho\cos\phi. \end{aligned}$$

Show that the Jacobian determinant is given by

$$\frac{\partial(x, y, z)}{\partial(\rho, \phi, \theta)} = \rho^2 \sin\phi.$$

(d) When can we solve for (ρ, ϕ, θ) in terms of (x, y, z)?

11. Let (x_0, y_0, z_0) be a point of the locus defined by $z^2 + xy - a = 0, z^2 + x^2 - y^2 - b = 0$, where a and b are constants.

(a) Under what conditions may the part of the locus near (x_0, y_0, z_0) be represented in the form $x = f(z), y = g(z)$?

(b) Compute $f'(z)$ and $g'(z)$.

12. Is it possible to solve the system of equations

$$xy^2 + xzu + yv^2 = 3$$
$$u^3yz + 2xv - u^2v^2 = 2$$

for $u(x, y, z), v(x, y, z)$ near $(x, y, z) = (1, 1, 1), (u, v) = (1, 1)$? Compute $\partial v/\partial y$ at $(x, y, z) = (1, 1, 1)$.

13. The problem of factoring a polynomial $x^n + a_{n-1}x^{n-1} + \cdots + a_0$ into linear factors is, in a sense, an "inverse function" problem. The coefficients a_i may be thought of as functions of the n roots r_j. We would like to find the roots as functions of the coefficients in some region. With $n = 3$, apply the inverse function theorem to this problem and state what it tells you about the possibility of doing this.

REVIEW EXERCISES FOR CHAPTER 3

1. Analyze the behavior of the following functions at the indicated points. [Your answer in part (b) may depend on the constant C.]

(a) $z = x^2 - y^2 + 3xy, \quad (x, y) = (0, 0)$
(b) $z = x^2 - y^2 + Cxy, \quad (x, y) = (0, 0)$

2. Find and classify the extreme values (if any) of the functions on $\mathbb{R}^2$ defined by the following expressions:

(a) $y^2 - x^3$ (b) $(x - 1)^2 + (x - y)^2$ (c) $x^2 + xy^2 + y^4$

3. (a) Find the minimum distance from the origin in $\mathbb{R}^3$ to the surface $z = \sqrt{x^2 - 1}$.
(b) Repeat part (a) for the surface $z = 6xy + 7$.

4. Find the first few terms in the Taylor expansion of $f(x, y) = e^{xy}\cos x$ about $x = 0$, $y = 0$.

5. Prove that

$$z = \frac{3x^4 - 4x^3 - 12x^2 + 18}{12(1 + 4y^2)}$$

has one local maximum, one local minimum, and one saddle point. (The graph is shown in Figure 3.R.1.)

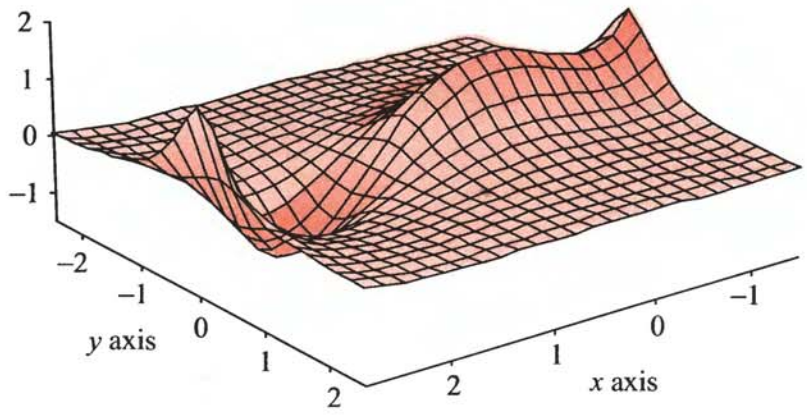

Figure 3.R.1 Graph of $z = (3x^4 - 4x^3 - 12x^2 + 18)/12(1 + 4y^2)$.

6. Find the maxima, minima, and saddles of the function $z = (2 + \cos \pi x)(\sin \pi y)$, which is graphed in Figure 3.R.2.

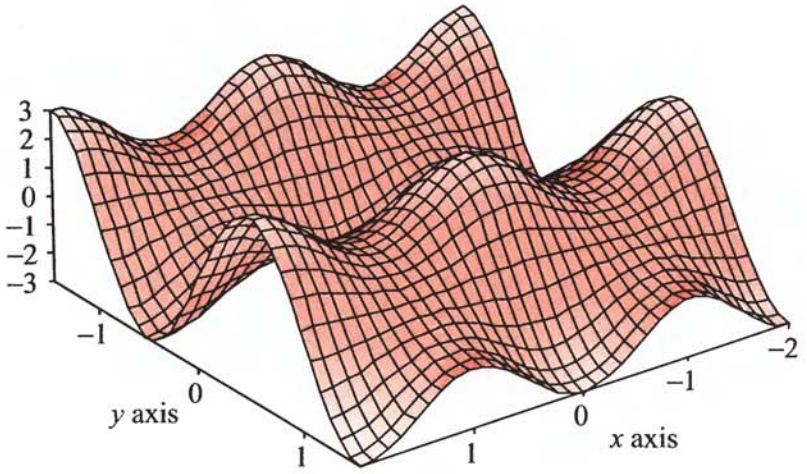

Figure 3.R.2 Graph of $z = (2 + \cos \pi x)\,(\sin \pi y)$.

7. Find and describe the critical points of $f(x, y) = y \sin(\pi x)$. (See Figure 3.R.3.)

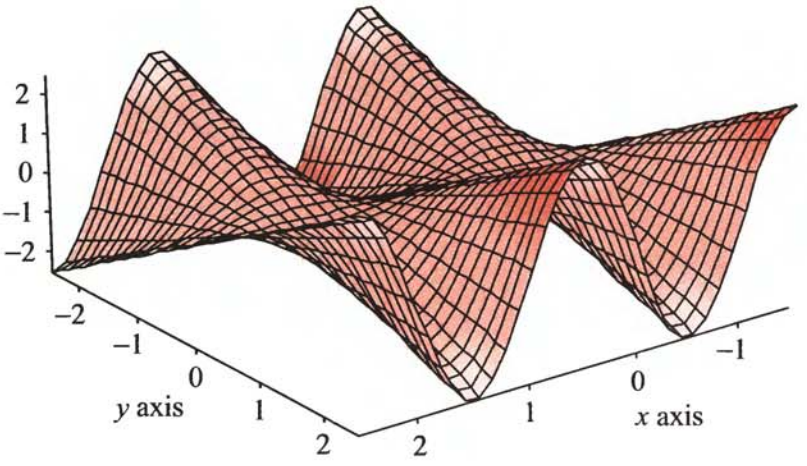

Figure 3.R.3 Graph of $z = y \sin(\pi x)$.

8. A graph of the function $z = \sin(\pi x)/(1 + y^2)$ is shown in Figure 3.R.4. Verify that this function has alternating maxima and minima on the x axis, with no other critical points.

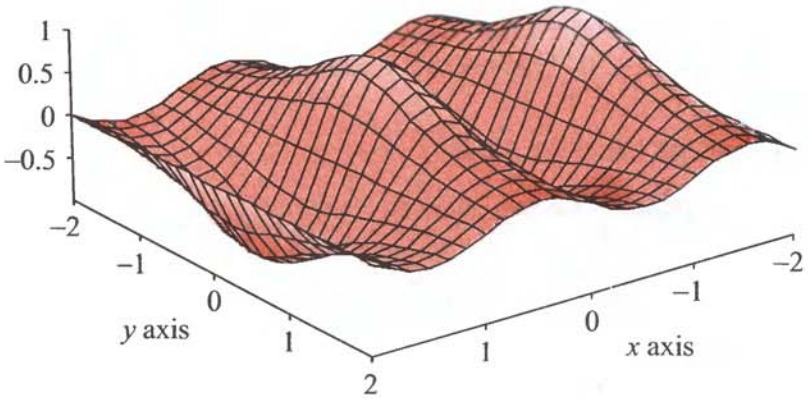

Figure 3.R.4 Graph of $z = \sin(\pi x)/(1 + y^2)$.

In Exercises 9 to 14 find the extrema of the given functions subject to the given constraints.

9. $f(x, y) = x^2 - 2xy + 2y^2$, subject to $x^2 + y^2 = 1$

10. $f(x, y) = xy - y^2$, subject to $x^2 + y^2 = 1$

11. $f(x, y) = \cos(x^2 - y^2)$, subject to $x^2 + y^2 = 1$

12. $f(x, y) = \dfrac{x^2 - y^2}{x^2 + y^2}$, subject to $x + y = 1$

13. $z = xy$, subject to the condition $x + y = 1$.

14. $z = \cos^2 x + \cos^2 y$, subject to the condition $x + y = \pi/4$.

15. Find the points on the surface $z^2 - xy = 1$ nearest to the origin.

16. Use the implicit function theorem to compute dy/dx for

(a) $x/y = 10$ (b) $x^3 - \sin y + y^4 = 4$ (c) $e^{x+y^2} + y^3 = 0$

17. Find the shortest distance from the point $(0, b)$ to the parabola $x^2 - 4y = 0$. Solve this problem using the Lagrange multiplier method and also without using Lagrange's method.

18. Solve the following geometric problems by Lagrange's method.

(a) Find the shortest distance from the point (a_1, a_2, a_3) in $\mathbb{R}^3$ to the plane whose equation is given by $b_1x_1 + b_2x_2 + b_3x_3 + b_0 = 0$, where $(b_1, b_2, b_3) \neq (0, 0, 0)$.

(b) Find the point on the line of intersection of the two planes $a_1x_1 + a_2x_2 + a_3x_3 = 0$ and $b_1x_1 + b_2x_2 + b_3x_3 + b_0 = 0$ that is nearest to the origin.

(c) Show that the volume of the largest rectangular parallelepiped that can be inscribed in the ellipsoid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

is $8abc/3\sqrt{3}$.

19. A particle moves in a potential $V(x, y) = x^3 - y^2 + x^2 + 3xy$. Determine whether $(0, 0)$ is a stable equilibrium point—that is, whether or not $(0, 0)$ is a strict local minimum of V.

20. Study the nature of the function $f(x, y) = x^3 - 3xy^2$ near $(0, 0)$. Show that the point $(0, 0)$ is a degenerate critical point, that is, $D = 0$. This surface is called a *monkey saddle*.

21. Find the maximum of $f(x, y) = xy$ on the curve $(x + 1)^2 + y^2 = 1$.

22. Find the maximum and minimum of $f(x, y) = xy - y + x - 1$ on the set $x^2 + y^2 \leq 2$.

23. The Baraboo, Wisconsin, plant of International Widget Co., Inc., uses aluminium, iron, and magnesium to produce high-quality widgets. The quantity of widgets that may be produced using x tons of aluminum, y tons of iron, and z tons of magnesium is $Q(x, y, z) = xyz$. The cost of raw materials is aluminum, \$6 per ton; iron, \$4 per ton; and magnesium, \$8 per ton. How many tons each of aluminum, iron, and magnesium should be used to manufacture 1000 widgets at the lowest possible cost? (HINT: Find an extreme value for what function subject to what constraint?)

24. Let $f\colon \mathbb{R} \to \mathbb{R}$ be of class C^1 and let

$$\begin{aligned} u &= f(x) \\ v &= -y + xf(x). \end{aligned}$$

If $f'(x_0) \neq 0$, show that this transformation of $\mathbb{R}^2$ to $\mathbb{R}^2$ is invertible near (x_0, y_0) and its inverse is given by

$$\begin{aligned} x &= f^{-1}(u) \\ y &= -v + uf^{-1}(u). \end{aligned}$$

25. Show that the pair of equations

$$\begin{aligned} x^2 - y^2 - u^3 + v^2 + 4 &= 0 \\ 2xy + y^2 - 2u^2 + 3v^4 + 8 &= 0 \end{aligned}$$

determine functions $u(x, y)$ and $v(x, y)$ defined for (x, y) near $x = 2$ and $y = -1$ such that $u(2, -1) = 2$ and $v(2, -1) = 1$. Compute $\partial u/\partial x$ at $(2, -1)$.

26. Show that there are positive numbers p and q and unique functions u and v from the interval $(-1 - p, -1 + p)$ into the interval $(1 - q, 1 + q)$ satisfying

$$xe^{u(x)} + u(x)e^{v(x)} = 0 = xe^{v(x)} + v(x)e^{u(x)}$$

for all x in the interval $(-1 - p, -1 + p)$ with $u(-1) = 1 = v(-1)$.

27. To work this exercise, the reader should be familiar with the technique of diagonalizing a 2×2 matrix. Let $a(x)$, $b(x)$, and $c(x)$ be three continuous functions defined on $U \cup \partial U$,

where U is an open set and ∂U denotes its set of boundary points (see Section 2.2). Use the notation of Lemma 2 in Section 3.3, and assume that for each $x \in U \cup \partial U$ the quadratic form defined by the matrix

$$\begin{bmatrix} a & b \\ b & c \end{bmatrix}$$

is positive-definite. For a C^2 function v on $U \cup \partial U$, we define a differential operator L by $Lv = a(\partial^2 v/\partial x^2) + 2b(\partial^2 v/\partial x \partial y) + c(\partial^2 v/\partial y^2)$. With this positive-definite condition, such an operator is said to be ***elliptic***. A function v is said to be ***strictly subharmonic relative*** to L if $Lv > 0$. Show that a strictly subharmonic function cannot have a maximum point in U.

28. A function v is said to be in the ***kernel*** of the operator L described in Exercise 27 if $Lv = 0$ on $U \cup \partial U$. Arguing as in Exercise 37 of Section 3.3, show that if v achieves its maximum on U it also achieves it on ∂U. This is called the weak maximum principle for elliptic operators.

29. Let L be an elliptic differential operator as in Exercises 27 and 28.

(a) Define the notion of a strict superharmonic function.
(b) Show that such functions cannot achieve a minimum on U.
(c) If v is as in Exercise 28, show that if v achieves its minimum on U it also achieves it on ∂U.

*The following **method of least squares** should be applied to Exercises 30 to 35.*

It sometimes happens that the theory behind an experiment indicates that the experimental data should lie approximately along a straight line of the form $y = mx + b$. The actual results, of course, never match the theory exactly. We are then faced with the problem of finding the straight line that *best fits* some set of experimental data $(x_1, y_1), \ldots, (x_n, y_n)$ as in Figure 3.R.5. If we guess at a straight line $y = mx + b$ to fit the data, each point will deviate vertically from the line by an amount $d_i = y_i - (mx_i + b)$.

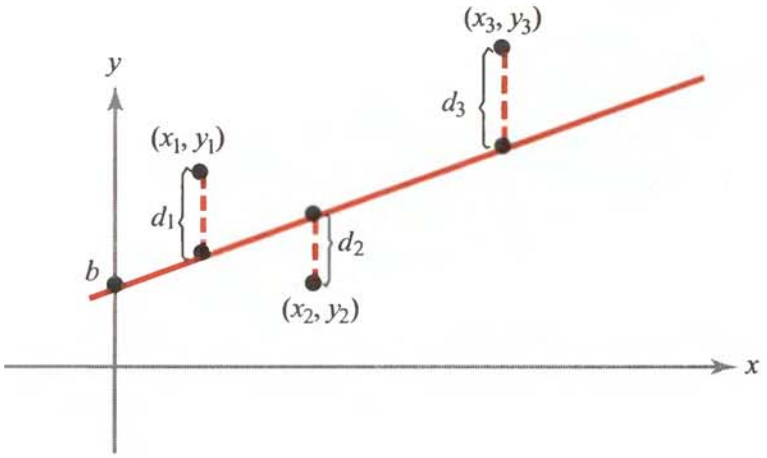

Figure 3.R.5 The method of least squares tries to find a straight line that best approximates a set of data.

We would like to choose m and b in such a way as to make the total effect of these deviations as small as possible. However, because some are negative and some positive, we

could get a lot of cancellations and still have a pretty bad fit. This leads us to suspect that a better measure of the total error might be the sum of the *squares* of these deviations. Thus, we are led to the problem of finding the m and b that minimize the function

$$s = f(m, b) = d_1^2 + d_2^2 + \cdots + d_n^2 = \sum_{i=1}^{n} (y_i - mx_i - b)^2,$$

where $x_1, \ldots, x_n$ and $y_1, \ldots, y_n$ are the given data.

30. For each set of three data points, plot the points, write down the function $f(m, b)$ from the preceding equation, find m and b to give the best straight-line fit according to the method of least squares, and plot the straight line.

(a) $(x_1, y_1) = (1, 1)$
$(x_2, y_2) = (2, 3)$
$(x_3, y_3) = (4, 3)$

(b) $(x_1, y_1) = (0, 0)$
$(x_2, y_2) = (1, 2)$
$(x_3, y_3) = (2, 3)$

31. Show that if only two data points (x_1, y_1) and (x_2, y_2) are given, this method produces the line through (x_1, y_1) and (x_2, y_2).

32. Show that the equations for a critical point, $\partial s/\partial b = 0$ and $\partial s/\partial m = 0$, are equivalent to

$$m\left(\sum x_i\right) + nb = \left(\sum y_i\right) \quad \text{and} \quad m\left(\sum x_i^2\right) + b\left(\sum x_i\right) = \left(\sum x_i y_i\right),$$

where all the sums run from $i = 1$ to $i = n$.

33. If $y = mx + b$ is the best-fitting straight line to the data points $(x_1, y_1), \ldots, (x_n, y_n)$ according to the least-square method, show that

$$\sum_{i=1}^{n} (y_i - mx_i - b) = 0;$$

that is, the positive and negative deviations cancel (see Exercise 32).

34. Use the second derivative test to show that the critical point of f is a minimum.

35. Use the method of least squares to find the straight line that best fits the points (0, 1), (1, 3), (2, 2), (3, 4), and (4, 5). Plot the points and line.[17]

---

[17]The method of least squares may be varied and generalized in a number of ways. The basic idea can be applied to equations of more complicated curves than the straight line. For example, this might be done to find the parabola that best fits a given set of data points. These ideas also formed part of the basis for the development of the science of cybernetics by Norbert Wiener. Another version of the data is the following problem of least-square approximation: Given a function f defined and integrable on an interval $[a, b]$, find a polynomial P of degree $\leq n$ such that the mean square error

$$\int_a^b |f(x) - P(x)|^2 \, dx$$

is as small as possible.

4

Vector-Valued Functions

. . . who by vigor of mind almost divine, the motions and figures of the planets, the paths of comets, and the tides of the seas first demonstrated.

Newton's Epitaph

Chapters 2 and 3 focused on *real*-valued functions. This chapter is largely concerned with *vector*-valued functions. We begin in the first section with a continuation of our study of paths, adding applications of Newton's second law. Then we study arc length of paths. Following this, we introduce the divergence and curl of a vector field which, in addition to the gradient, are basic operations in vector *differential* calculus. The basic geometry and calculus of the divergence and curl are studied. The associated *integral* calculus will be given in Chapter 8.

4.1 Acceleration and Newton's Second Law

In Section 2.4, we studied the basic geometry of paths, learning how to sketch curves (the images of paths) and compute tangent lines. We also learned to think of, as the name suggests, a path as the trajectory of a particle and to regard the derivative of the path as its velocity vector. In this section, we continue our study of paths, including additional topics, especially acceleration and Newton's second law.

Differentiation of Paths

Recall that a path in $\mathbb{R}^n$ is a map $\mathbf{c}$ of $\mathbb{R}$ or an interval in $\mathbb{R}$ to $\mathbb{R}^n$. If the path is differentiable, its derivative at each time t is an $n \times 1$ matrix. Specifically, if $x_1(t), \ldots, x_n(t)$ are the component functions of $\mathbf{c}$, the derivative matrix is

$$\mathbf{c}'(t) = \begin{bmatrix} dx_1/dt \\ dx_2/dt \\ \vdots \\ dx_n/dt \end{bmatrix},$$

which can also be written in vector form as

$$(dx_1/dt, \ldots, dx_n/dt) \qquad \text{or as} \qquad (x_1'(t), \ldots, x_n'(t)).$$

Recall from Section 2.4 that $\mathbf{c}'(t)$ is the *tangent vector* to the path at the point $\mathbf{c}(t)$. Also recall that if $\mathbf{c}$ represents the path of a moving particle, then its *velocity vector* is

$$\mathbf{v} = \mathbf{c}'(t),$$

and its *speed* is $s = \|\mathbf{v}\|$.

The differentiation of paths is facilitated by the following rules.

Differentiation Rules Let $\mathbf{b}(t)$ and $\mathbf{c}(t)$ be differentiable paths in $\mathbb{R}^3$ and $p(t)$ and $q(t)$ be differentiable scalar functions:

Sum Rule:
$$\frac{d}{dt}[\mathbf{b}(t) + \mathbf{c}(t)] = \mathbf{b}'(t) + \mathbf{c}'(t)$$

Scalar Multiplication Rule:
$$\frac{d}{dt}[p(t)\mathbf{c}(t)] = p'(t)\mathbf{c}(t) + p(t)\mathbf{c}'(t)$$

Dot Product Rule:
$$\frac{d}{dt}[\mathbf{b}(t) \cdot \mathbf{c}(t)] = \mathbf{b}'(t) \cdot \mathbf{c}(t) + \mathbf{b}(t) \cdot \mathbf{c}'(t)$$

Cross Product Rule:
$$\frac{d}{dt}[\mathbf{b}(t) \times \mathbf{c}(t)] = \mathbf{b}'(t) \times \mathbf{c}(t) + \mathbf{b}(t) \times \mathbf{c}'(t)$$

Chain Rule:
$$\frac{d}{dt}[\mathbf{c}(q(t))] = q'(t)\mathbf{c}'(q(t)).$$

These rules follow by applying the usual differentiation rules to the components.

EXAMPLE 1 Show that if $\mathbf{c}(t)$ is a vector function such that $\|\mathbf{c}(t)\|$ is constant, then $\mathbf{c}'(t)$ is perpendicular to $\mathbf{c}(t)$ for all t.

SOLUTION Because $\|\mathbf{c}(t)\|$ is constant, so is its square $\|\mathbf{c}(t)\|^2 = \mathbf{c}(t) \cdot \mathbf{c}(t)$. The derivative of this constant is zero, so by the dot product rule,

$$0 = \frac{d}{dt}[\mathbf{c}(t) \cdot \mathbf{c}(t)] = \mathbf{c}'(t) \cdot \mathbf{c}(t) + \mathbf{c}(t) \cdot \mathbf{c}'(t) = 2\mathbf{c}(t) \cdot \mathbf{c}'(t);$$

thus, $\mathbf{c}(t) \cdot \mathbf{c}'(t) = 0$; that is, $\mathbf{c}'(t)$ is perpendicular to $\mathbf{c}(t)$. ▲

For a path describing uniform rectilinear motion, the velocity vector is constant. In general, the velocity vector is a vector function $\mathbf{v} = \mathbf{c}'(t)$ that depends on t. The derivative $\mathbf{a} = d\mathbf{v}/dt = \mathbf{c}''(t)$ is called the ***acceleration*** of the curve. If the curve is

$(x(t), y(t), z(t))$, then the acceleration at time t is given by

$$\mathbf{a}(t) = x''(t)\mathbf{i} + y''(t)\mathbf{j} + z''(t)\mathbf{k}.$$

EXAMPLE 2 A particle moves in such a way that its acceleration is constantly equal to $-\mathbf{k}$. If the position when $t = 0$ is $(0, 0, 1)$ and the velocity at $t = 0$ is $\mathbf{i} + \mathbf{j}$, when and where does the particle fall below the plane $z = 0$? Describe the path traveled by the particle (assume $t \geq 0$).

SOLUTION Let $(x(t), y(t), z(t))$ be the path traced out by the particle, so that the velocity vector is $\mathbf{c}'(t) = x'(t)\mathbf{i} + y'(t)\mathbf{j} + z'(t)\mathbf{k}$. The acceleration $\mathbf{c}''(t)$ is $-\mathbf{k}$, so $x''(t) = 0$, $y''(t) = 0$, and $z''(t) = -1$. It follows that $x'(t)$ and $y'(t)$ are constant functions, and $z'(t)$ is a linear function with slope -1. Because $\mathbf{c}'(0) = \mathbf{i} + \mathbf{j}$, we get $\mathbf{c}'(t) = \mathbf{i} + \mathbf{j} - t\mathbf{k}$. Integrating again and using the initial position $(0, 0, 1)$, we find that $(x(t), y(t), z(t)) = (t, t, 1 - \frac{1}{2}t^2)$. The particle drops below the plane $z = 0$ when $1 - \frac{1}{2}t^2 = 0$; that is, $t = \sqrt{2}$ (because $t \geq 0$). At that instant, the position is $(\sqrt{2}, \sqrt{2}, 0)$. The path traveled by the particle is a parabola in the plane $y = x$ (see Figure 4.1.1), because in this plane the equation is described by $z = 1 - \frac{1}{2}x^2$. ▲

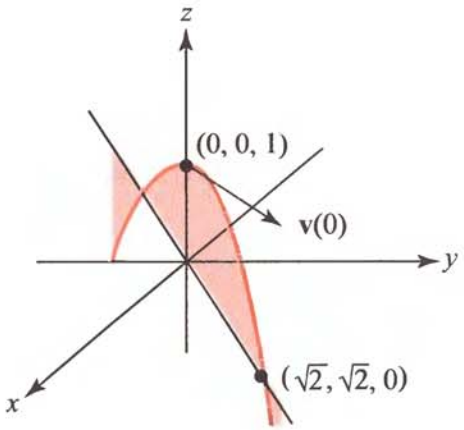

Figure 4.1.1 The path of the particle with initial position $(0, 0, 1)$, initial velocity $\mathbf{i} + \mathbf{j}$, and constant acceleration $-\mathbf{k}$ is a parabola in the plane $y = x$.

The image of a C^1 path is not necessarily "very smooth"; indeed, it may have sharp bends or changes of direction. For instance, the cycloid $\mathbf{c}(t) = (t - \sin t, 1 - \cos t)$ shown in Figure 2.4.6 has cusps at all points where $\mathbf{c}$ touches the x axis (that is, when $1 - \cos t = 0$, which happens when $t = 2\pi n$, $n = 0, \pm 1, \ldots$). Another example is the ***hypocycloid of four cusps***, $\mathbf{c}: [0, 2\pi] \to \mathbb{R}^2$, $t \mapsto (\cos^3 t, \sin^3 t)$, which has cusps at four points (Figure 4.1.2). At all such points, however, $\mathbf{c}'(t) = \mathbf{0}$, and the tangent line is not well defined. Evidently, the direction of $\mathbf{c}'(t)$ may change abruptly at points where it slows to rest.

A differentiable path $\mathbf{c}$ is said to be ***regular*** at $t = t_0$ if $\mathbf{c}'(t_0) \neq \mathbf{0}$. If $\mathbf{c}'(t) \neq \mathbf{0}$ for all t, we say that c is a regular path. In this case, the image curve looks smooth.

EXAMPLE 3 A particle moves along a hypocycloid according to the equations

$$x = \cos^3 t, \qquad y = \sin^3 t, \qquad a \leq t \leq b.$$

What are the velocity and speed of the particle?

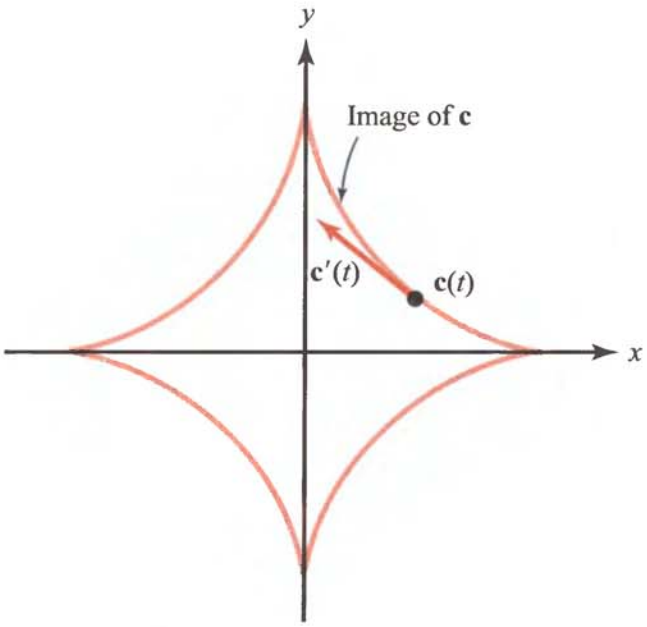

Figure 4.1.2 The image of the smooth path $\mathbf{c}(t) = (\cos^3 t, \sin^3 t)$, a hypocycloid, does not "look smooth."

SOLUTION The velocity vector of the particle is

$$\mathbf{v} = \frac{dx}{dt}\mathbf{i} + \frac{dy}{dt}\mathbf{j} = -(3 \sin t \cos^2 t)\mathbf{i} + (3 \cos t \sin^2 t)\mathbf{j},$$

and its speed is

$$s = \|\mathbf{v}\| = (9 \sin^2 t \cos^4 t + 9 \cos^2 t \sin^4 t)^{1/2} = 3\,|\sin t|\,|\cos t|. \quad \blacktriangle$$

Newton's Second Law

If a particle of mass m moves in $\mathbb{R}^3$, the force $\mathbf{F}$ acting on it at the point $\mathbf{c}(t)$ is related to the acceleration $\mathbf{a}(t)$ by ***Newton's second law***:[1]

$$\mathbf{F}(\mathbf{c}(t)) = m\mathbf{a}(t).$$

In particular, if no forces act on a particle, then $\mathbf{a}(t) = \mathbf{0}$, so $\mathbf{c}'(t)$ is constant and the particle follows a straight line.

Acceleration and Newton's Second Law The ***acceleration*** of a path $\mathbf{c}(t)$ is

$$\mathbf{a}(t) = \mathbf{c}''(t).$$

If $\mathbf{F}$ is the force acting and m is the mass of the particle, then

$$\mathbf{F} = m\mathbf{a}.$$

[1] Most scientists acknowledge that $\mathbf{F} = m\mathbf{a}$ is the single most important equation in all of science and engineering.

In the problem of determining the path $\mathbf{c}(t)$ of a particle under the influence of a given force field, $\mathbf{F}$, Newton's law becomes a differential equation (i.e., an equation involving derivatives) for $\mathbf{c}(t)$.

For example, the motion of a planet moving along a path $\mathbf{r}(t)$ around the sun (considered to be located at the origin in $\mathbb{R}^3$) obeys the law

$$m\mathbf{r}'' = -\frac{GmM}{r^3}\mathbf{r},$$

where M is the mass of the sun, m that of the planet, $r = \|\mathbf{r}\|$, and G is the gravitational constant. The relation used in determining the force, $\mathbf{F} = -GmM\mathbf{r}/r^3$, is called ***Newton's law of gravitation*** (see Figure 4.1.3). We shall not make a general study of such equations in this book, but content ourselves with the special case of circular orbits. (More general orbits—the conic sections—are discussed in the Internet supplement.)

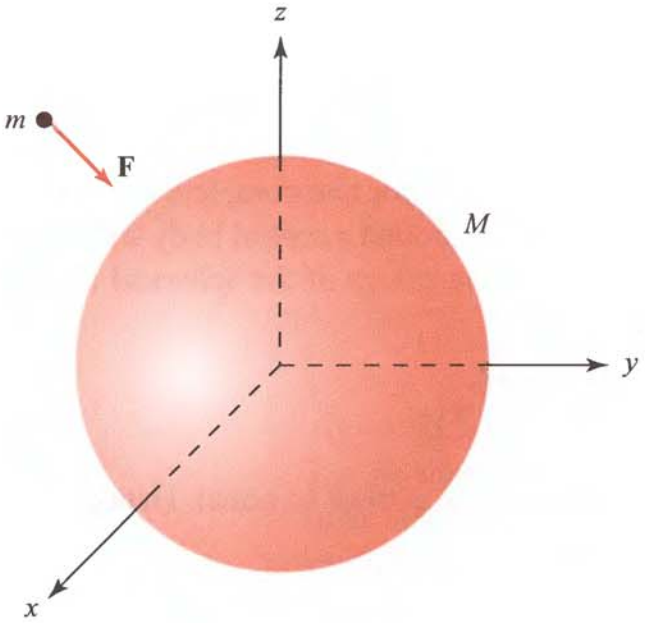

Figure 4.1.3 A mass M attracts a mass m with a force $\mathbf{F}$ given by Newton's law of gravitation: $\mathbf{F} = -GmM\mathbf{r}/r^3$.

Circular Orbits

Consider a particle of mass m moving at constant speed s in a circular path of radius r_0. Supposing that it moves in the xy plane, we can suppress the third component and write its location as

$$\mathbf{r}(t) = \left(r_0 \cos\frac{st}{r_0}, r_0 \sin\frac{st}{r_0}\right).$$

Note that this is a circle of radius r_0 and that its speed is given by $\|\mathbf{r}'(t)\| = s$. The quantity s/r_0 is called the ***frequency*** and is denoted ω. Thus,

$$\mathbf{r}(t) = (r_0 \cos \omega t,\ r_0 \sin \omega t).$$

The acceleration is given by

$$\mathbf{a}(t) = \mathbf{r}''(t) = \left(-\frac{s^2}{r_0}\cos\frac{st}{r_0}, -\frac{s^2}{r_0}\sin\frac{st}{r_0}\right) = -\frac{s^2}{r_0^2}\mathbf{r}(t) = -\omega^2\mathbf{r}(t).$$

Thus, the acceleration is in a direction opposite to $\mathbf{r}(t)$; that is, it is directed toward the center of the circle (see Figure 4.1.4). This acceleration multiplied by the mass of the particle is called the ***centripetal force***. Even though the speed is constant, the direction of the velocity is continuously changing and therefore the acceleration, which is a rate of change in either speed or direction or both, is nonzero.

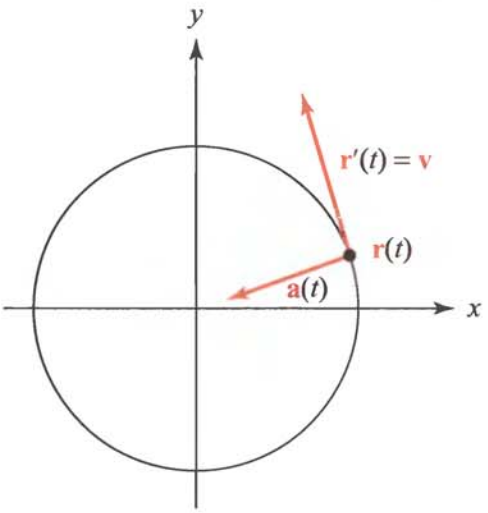

Figure 4.1.4 The position, velocity, and acceleration of a particle in circular motion.

Newton's law helps us discover a relationship between the radius of the orbit of a revolving body and the period, that is, the time it takes for one complete revolution. Consider a satellite of mass m moving with a speed s around a central body with mass M in a *circular* orbit of radius r_0 (distance from the *center* of the spherical central body). By Newton's second law $\mathbf{F} = m\mathbf{a}$, we get

$$-\frac{s^2 m}{r_0^2}\mathbf{r}(t) = -\frac{GmM}{r_0^3}\mathbf{r}(t).$$

The lengths of the vectors on both sides of this equation must be equal. Hence,

$$s^2 = \frac{GM}{r_0}.$$

If T denotes the period, then $s = 2\pi r_0 / T$; substituting this value for s in the preceding equation and solving for T, we obtain the following:

Kepler's Law

$$T^2 = r_0^3 \frac{(2\pi)^2}{GM}.$$

Thus, the *square of the period is proportional to the cube of the radius.*

We have defined two basic concepts associated with a path; its velocity and its acceleration. Both involve *differential* calculus. The basic concept of the length of a path, which involves *integral* calculus, will be taken up in the next section.

EXAMPLE 4 Suppose that a satellite is to be in a circular orbit about the earth such that it stays fixed in the sky over one point on the equator. What is the radius of

such a *geosynchronous* orbit? (The mass of the earth is 5.98×10^{24} kilograms and $G = 6.67 \times 10^{-11}$ in the meter-kilogram-second system of units.)

SOLUTION The period of the satellite should be 1 day, so $T = 60 \times 60 \times 24 = 86{,}400$ seconds. From the formula $T^2 = r_0^3(2\pi)^2/GM$, we get $r_0^3 = T^2GM/(2\pi)^2$, and so

$$r_0^3 = \frac{T^2GM}{(2\pi)^2} = \frac{(86{,}400)^2 \times (6.67 \times 10^{-11}) \times (5.98 \times 10^{24})}{(2\pi)^2} \approx 7.54 \times 10^{22}\,\text{m}^3.$$

Thus, $r_0 = 4.23 \times 10^7$ m $= 42{,}300$ km $\approx 26{,}200$ mi. ▲

Supplement to Section 4.1: Planetary Orbits, Hamilton's Principle, and Spacecraft Trajectories

In this section, we have been studying paths in space and Newton's second law. Hopefully, the student realizes that these ideas apply to the real world—the motion of our earth around the sun, for example, is governed by these laws. But there is more to the story, and we will try to convey some of it here.

Historical Note

Kepler, Newton, and Hamilton

As we discussed in the historical introduction, the law of planetary motion stating that the square of the period is proportional to the cube of the radius of an orbit is one of the three that Kepler observed before Newton formulated his laws of motion, known more generally as Newton's mechanics. These mechanics enable one to compute the period of a satellite about the earth or a planet about the sun (when the radius of its orbit is given), and, as we will indicate shortly, trajectories of space missions.

Kepler discovered and used results like this not only for circular orbits but more generally for elliptical orbits. Newton was able to derive Kepler's three celestial laws from his own law of gravitation. The neat mathematical order of the universe that these laws provided had a great impact on eighteenth-century thought.

Newton never wrote down his laws of mechanics as differential equations. This was first done by Euler around 1730. Newton made most of his deductions (at least those in published form) by geometric methods. Euler also showed how Newton's equations followed from Maupertuis's action principle. The clearest version of the action principle in mechanics, now known as ***Hamilton's principle***, is due to William Rowan Hamilton around 1830, who, as we all should now know, happens to also be the father of vector calculus. Hamilton's version of Maupertuis's principle was elegantly presented by Richard Feynman, as we discuss next.

Feynman and Hamilton's Principle

In his legendary Caltech *Lectures on Physics*, Nobel Prize–winning physicist Richard Phillips Feynman (see Figure 4.1.5) included what he called a "Special Lecture" on a topic clearly very close to his heart—one that he first heard about from his New York high school teacher, Mr. Bader. Mr. Bader told his (apparently bored) student Feynman how principles of maxima and minima apply to the trajectories of moving objects and in particular how the action principle of Maupertuis, Leibniz, and Hamilton (discussed in Section 3.3) applies to Newton's mechanics, governed by $\mathbf{F} = m\mathbf{a}$.

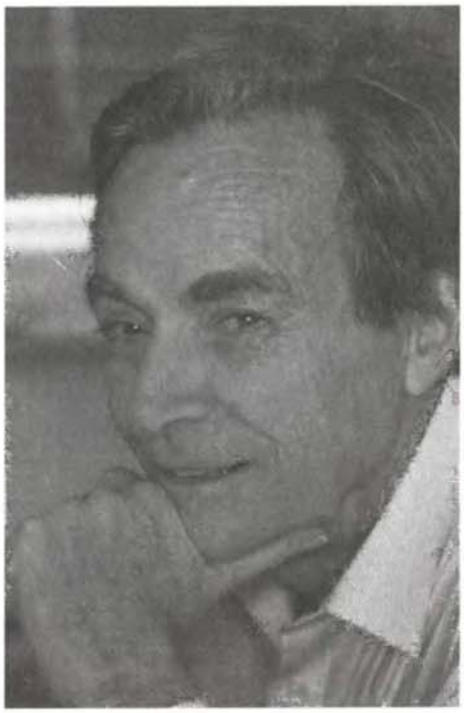

Figure 4.1.5 Richard P. Feynman (1918–1988).

Professor Feynman, at the end of his lecture, notes that "a physicist, a student of Mr. Bader, in 1942 showed how this action principle applied to quantum mechanics." That student was Feynman himself, who received the Nobel Prize for his insights, which also included the discovery of *Feynman integrals*. The moral here is *pay attention to your teachers—especially the best ones*!

We include the first part of Feynman's lecture here and more of it in the Internet supplement; see Volume II, Lecture 19, of the *Feynman Lectures on Physics* for the entire lecture.

The Principle of Least Action, by Richard Feynman

When I was in high school, my physics teacher—whose name was Mr. Bader—called me down one day after physics class and said, "You look bored; I want to tell you something interesting." Then he told me

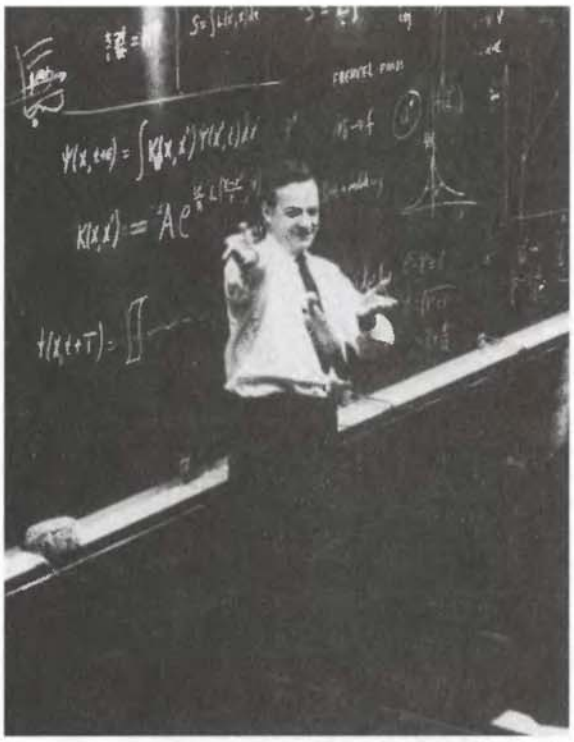

Figure 4.1.6 Feynman lecturing at Caltech.

something which I found absolutely fascinating, and have, since then, always found fascinating. Every time the subject comes up, I work on it. In fact, when I began to prepare this lecture I found myself making more analyses on the thing. Instead of worrying about the lecture, I got involved in a new problem. The subject is this—the principle of least action.

Mr. Bader told me the following: Suppose you have a particle (in a gravitational field, for instance) which starts somewhere and moves to some other point by free motion—you throw it, and it goes up and comes down [see Figure 4.1.7].

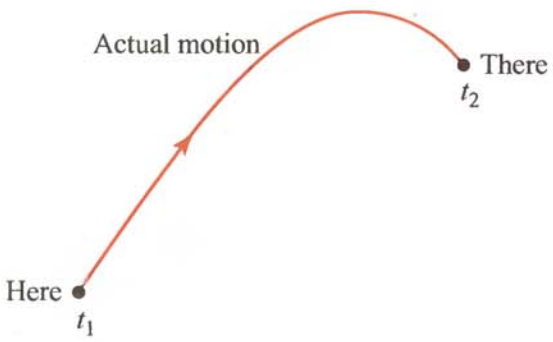

Figure 4.1.7

It goes from the original place to the final place in a certain amount of time. Now, you try a different motion. Suppose that to get from here to there, it went like this [see Figure 4.1.8], but got there in just the same amount of time.

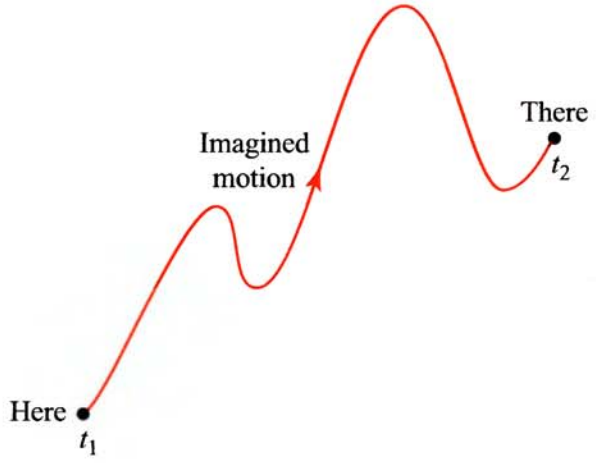

Figure 4.1.8

Then he said this: "If you calculate the kinetic energy at every moment on the path, take away the potential energy, and integrate it over the time during the whole path, you'll find that the number you'll get is bigger than that for the actual motion."

In other words, the laws of Newton could be stated not in the form $F = ma$ but in the form: The average kinetic energy less the average potential energy is as little as possible for the path of an object going from one point to another.

Let me illustrate a little better what this means. If you take the case of the gravitational field, then if the particle has the path $x(t)$ (let's just take one dimension for a moment; we take a trajectory that goes up and down and not sideways), where x is the height above the ground, the kinetic energy is $\frac{1}{2}m(dx/dt)^2$, and the potential energy at any time is mgx. Now I take the kinetic energy minus the potential energy at every moment along the path and integrate that with respect to time from the initial time to the final time. Let's suppose that at the original time t_1 we started at some height and at the end of the time t_2 we are definitely ending at some other place [see Figure 4.1.9].

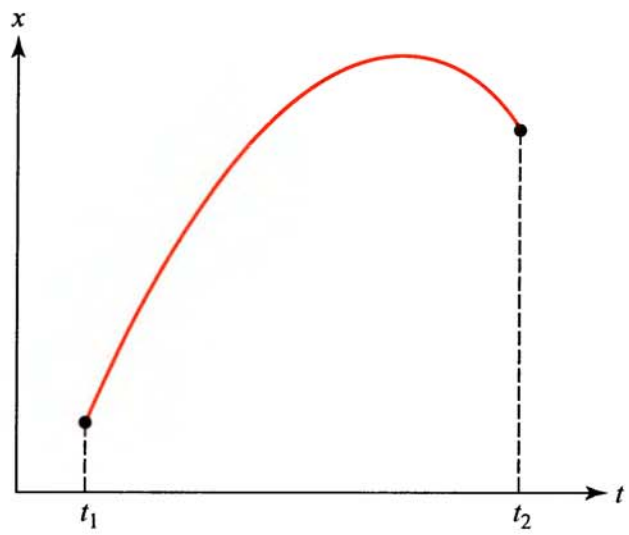

Figure 4.1.9

Then the integral is

$$\int_{t_1}^{t_2} \left[\frac{1}{2} m \left(\frac{dx}{dt} \right)^2 - mgx \right] dt.$$

The actual motion is some kind of curve—it's a parabola if we plot against the time—and gives a certain value for the integral. But we could *imagine* some other motion that went very high and came up and down in some peculiar way [see Figure 4.1.10].

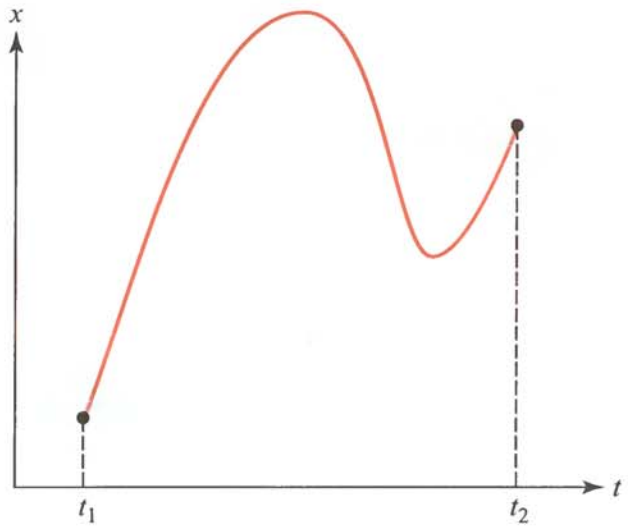

Figure 4.1.10

We can calculate the kinetic energy minus the potential energy and integrate for such a path . . . or for any other path we want. The miracle is that the true path is the one for which that *integral is least*.

Real-Life Trajectories

Interesting paths in $\mathbb{R}^3$ that obey Newton's second law occur in our own solar system and are used by NASA to plan space missions. One such mission, the *Genesis* Discovery Mission, launched from earth August 8, 2001 (and is due to return to earth in September 2004), has a particularly interesting trajectory, as shown in Figure 4.1.11. More information about this trajectory and the mission objectives can be found at http://genesismission.jpl.nasa.gov/.

The points denoted L_1 and L_2 in this figure denote places of balance (discovered by Euler) between the earth and the sun. A motionless spacecraft positioned there will remain there. There are periodic orbits about these points that we have (loosely)

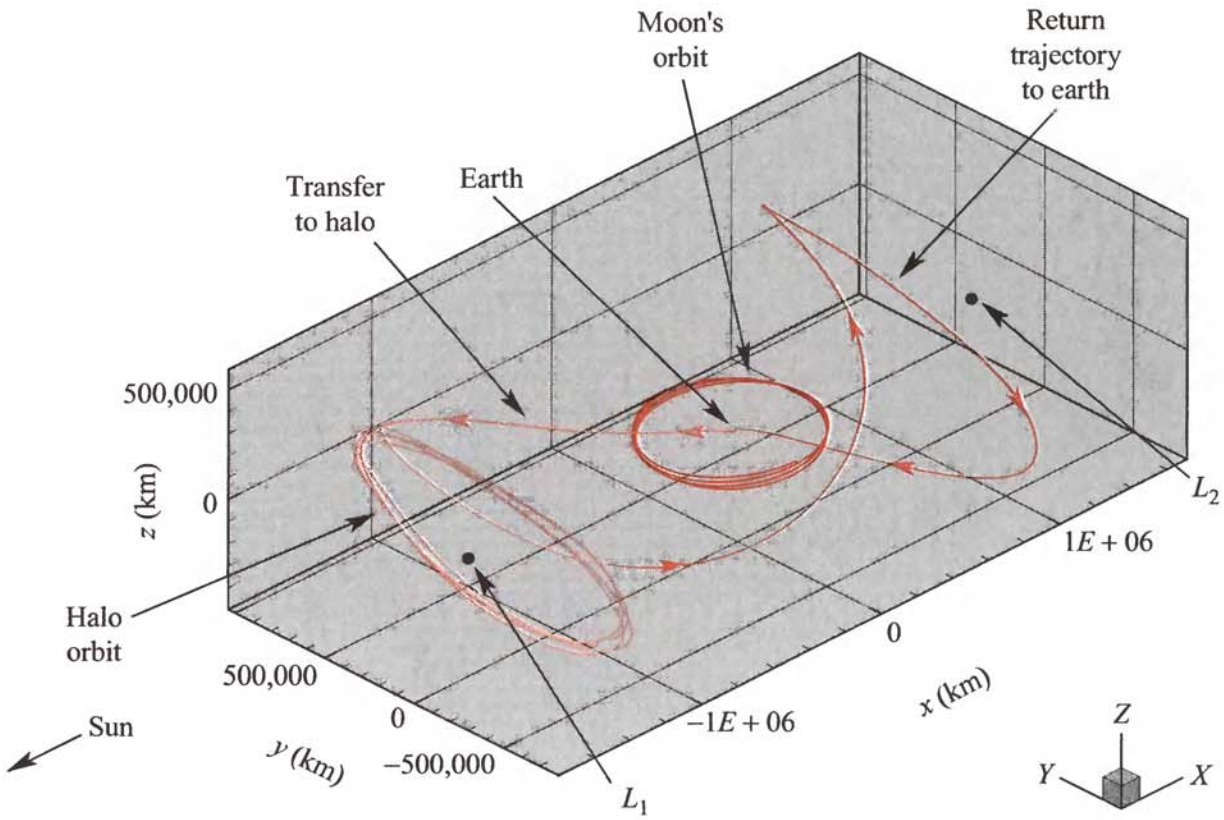

Figure 4.1.11 Trajectory of the *Genesis* spacecraft from the earth to a periodic orbit about a million and a half kilometers from earth and the interesting return trajectory to earth.

called *halo orbits*. The main dynamics of the spacecraft is governed by the pull of both the earth and the sun (and to a very small extent the moon) on the spacecraft. This is thus part of the famous *three-body problem* studied and made famous by Poincaré around 1890.[2]

Emmy Noether and Hamilton's Principle

Emmy Noether (1882–1935) (see Figure 4.1.12) is perhaps best known for her work in algebra, but she made a significant contribution to Hamilton's principle as well.[3] For planetary motion, the angular momentum vector $\mathbf{J} = \mathbf{r}(t) \times m\dot{\mathbf{r}}(t)$ is time-independent (so is a *conserved quantity*), as one can readily see by computing the time derivative of $\mathbf{J}$ and using $\mathbf{F} = m\mathbf{a}$ (see Exercise 20). What Noether discovered was a deep connection between such conserved quantities and symmetries in Hamilton's principle—in the case of angular momentum, this is rotational symmetry. Noether's discoveries have had a profound influence on the study of mechanical systems, from classical to quantum, ever since.

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[2]For more information about Poincaré, see F. Diacu and P. Holmes, *Celestial Encounters. The Origins of Chaos and Stability*, Princeton University Press: Princeton, NJ, 1996.

[3]"Invariante Variationsprobleme," *Göttingen Math. Phys.* **2** (1918): 235–257.

Figure 4.1.12 Emmy Noether (1882–1935).

EXERCISES

In Exercises 1 to 4, find the velocity and acceleration vectors and the equation of the tangent line for each of the following curves, at the given value of t.

1. $\mathbf{r}(t) = (\cos t)\mathbf{i} + (\sin 2t)\mathbf{j}$, at $t = 0$
2. $\mathbf{c}(t) = (t\,\sin t, t\,\cos t, \sqrt{3}t)$, at $t = 0$
3. $\mathbf{r}(t) = \sqrt{2}t\mathbf{i} + e^t\mathbf{j} + e^{-t}\mathbf{k}$, at $t = 0$
4. $\mathbf{c}(t) = t\mathbf{i} + t\mathbf{j} + \frac{2}{3}t^{3/2}\mathbf{k}$, at $t = 9$

In Exercises 5 to 8, let $\mathbf{c}_1(t) = e^t\mathbf{i} + (\sin t)\mathbf{j} + t^3\mathbf{k}$ and $\mathbf{c}_2(t) = e^{-t}\mathbf{i} + (\cos t)\mathbf{j} - 2t^3\mathbf{k}$. Find each of the stated derivatives in two different ways to verify the rules in the box preceding Example 1.

5. $\dfrac{d}{dt}[\mathbf{c}_1(t) + \mathbf{c}_2(t)]$
6. $\dfrac{d}{dt}[\mathbf{c}_1(t) \cdot \mathbf{c}_2(t)]$
7. $\dfrac{d}{dt}[\mathbf{c}_1(t) \times \mathbf{c}_2(t)]$
8. $\dfrac{d}{dt}\{\mathbf{c}_1(t) \cdot [2\mathbf{c}_2(t) + \mathbf{c}_1(t)]\}$
9. If $\mathbf{r}(t) = 6t\mathbf{i} + 3t^2\mathbf{j} + t^3\mathbf{k}$, what force acts on a particle of mass m moving along $\mathbf{r}$ at $t = 0$?

10. Let a particle of mass 1 gram (g) follow the path in Exercise 1, with units in seconds and centimeters. What force acts on it at $t = 0$? (Give the units in your answer.)

11. A body of mass 2 kilograms moves on a circle of radius 3 meters, making one revolution every 5 seconds. Find the centripetal force acting on the body.

12. Find the centripetal force acting on a body of mass 4 kilograms, moving on a circle of radius 10 meters with a frequency of 2 revolutions per second.

13. Show that if the acceleration of an object is always perpendicular to the velocity, then the speed of the object is constant. (HINT: See Example 1.)

14. Show that, at a local maximum or minimum of $\|\mathbf{r}(t)\|$, the vector $\mathbf{r}'(t)$ is perpendicular to $\mathbf{r}(t)$.

15. A satellite is in a circular orbit 500 miles above the surface of the earth. What is the period of the orbit? (You may take the radius of the earth to be 4000 miles, or 6.436×10^6 meters).

16. What is the acceleration of the satellite in Exercise 15? The centripetal force?

17. Find the path $\mathbf{c}$ such that $\mathbf{c}(0) = (0, -5, 1)$ and $\mathbf{c}'(t) = (t, e^t, t^2)$.

18. Let $\mathbf{c}$ be a path in $\mathbb{R}^3$ with zero acceleration. Prove that $\mathbf{c}$ is a straight line or a point.

19. Find paths $\mathbf{c}(t)$ that represent the following curves or trajectories.

(a) $\{(x, y) \mid y = e^x\}$
(b) $\{(x, y) \mid 4x^2 + y^2 = 1\}$
(c) A straight line in $\mathbb{R}^3$ passing through the origin and the point (a, b, c)
(d) $\{(x, y) \mid 9x^2 + 16y^2 = 4\}$

20. Let $\mathbf{c}(t)$ be a path, $\mathbf{v}(t)$ its velocity, and $\mathbf{a}(t)$ the acceleration. Suppose $\mathbf{F}$ is a C^1 mapping of $\mathbb{R}^3$ to $\mathbb{R}^3$, $m > 0$, and $\mathbf{F}(\mathbf{c}(t)) = m\mathbf{a}(t)$ (Newton's second law). Prove that

$$\frac{d}{dt}[m\mathbf{c}(t) \times \mathbf{v}(t)] = \mathbf{c}(t) \times \mathbf{F}(\mathbf{c}(t))$$

(i.e., "rate of change of angular momentum = torque"). What can you conclude if $\mathbf{F}(\mathbf{c}(t))$ is parallel to $\mathbf{c}(t)$? Is this the case in planetary motion?

21. Continue the investigations in Exercise 20 to prove Kepler's law that a planet moving under the influence of gravity about the sun does so in a fixed plane.

4.2 Arc Length

Definition of Arc Length

What is the length of a path $\mathbf{c}(t)$? Because the speed $\|\mathbf{c}'(t)\|$ is the rate of change of distance traveled with respect to time, the distance traveled by a point moving along the curve should be the integral of speed with respect to the time over the interval

$[t_0, t_1]$ of travel time; that is, the length of the path, also called its ***arc length***, is

$$L(\mathbf{c}) = \int_{t_0}^{t_1} \|\mathbf{c}'(t)\| dt.$$

There is the question as to whether or not this formula actually corresponds to the true arc length. For example, suppose we take a curve in space and glue a string tightly to it, cutting the string so it exactly fits the curve. If we then remove the string, straighten it out and measure it with a straight edge, we surely should obtain the length of the curve. That our formula for arc length agrees with such a process is justified in the supplement at the end of this section.

EXAMPLE 1 The arc length of the path $\mathbf{c}(t) = (r\cos t, r\sin t)$, for t lying in the interval $[0, 2\pi]$; that is, for $0 \le t \le 2\pi$, is

$$L(\mathbf{c}) = \int_0^{2\pi} \sqrt{(-r\sin t)^2 + (r\cos t)^2}\, dt = 2\pi r,$$

which is the circumference of a circle of radius r. If we had allowed $0 \le t \le 4\pi$, we would have obtained $4\pi r$, because the path traverses the same circle *twice* (Figure 4.2.1). ▲

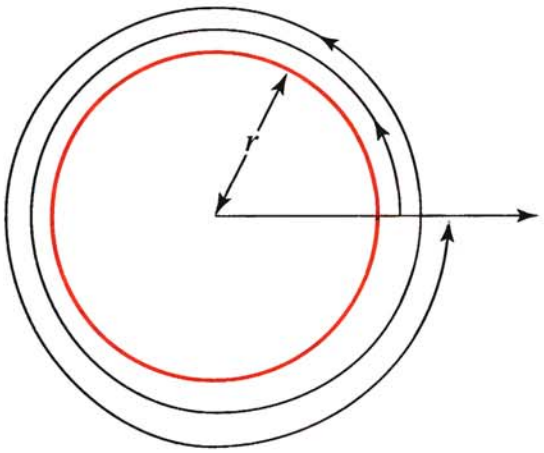


Figure 4.2.1 The arc length of a circle traversed twice is $4\pi r$.

Arc Length The length of the path $\mathbf{c}(t) = (x(t), y(t), z(t))$ for $t_0 \le t \le t_1$, is

$$L(\mathbf{c}) = \int_{t_0}^{t_1} \sqrt{[x'(t)]^2 + [y'(t)]^2 + [z'(t)]^2}\, dt.$$

For planar curves, one omits the $z'(t)$ term, as in Example 1. Here is an example in $\mathbb{R}^3$.

EXAMPLE 2 Find the arc length of $(\cos t, \sin t, t^2)$, $0 \le t \le \pi$.

SOLUTION The path $\mathbf{c}(t) = (\cos t, \sin t, t^2)$ has the velocity vector given by $\mathbf{v} = (-\sin t, \cos t, 2t)$. Because

$$\|\mathbf{v}\| = \sqrt{\sin^2 t + \cos^2 t + 4t^2} = \sqrt{1 + 4t^2} = 2\sqrt{t^2 + \left(\frac{1}{2}\right)^2},$$

the arc length is

$$L(\mathbf{c}) = \int_0^{\pi} 2\sqrt{t^2 + \left(\frac{1}{2}\right)^2}\, dt.$$

This integral may be evaluated using the following formula from the table of integrals:

$$\int \sqrt{x^2 + a^2}\, dx = \frac{1}{2}\left[x\sqrt{x^2 + a^2} + a^2 \log\left(x + \sqrt{x^2 + a^2}\right)\right] + C.$$

Thus,

$$L(\mathbf{c}) = 2 \cdot \frac{1}{2}\left[t\sqrt{t^2 + \left(\frac{1}{2}\right)^2} + \left(\frac{1}{2}\right)^2 \log\left(t + \sqrt{t^2 + \left(\frac{1}{2}\right)^2}\right)\right]\Bigg|_{t=0}^{\pi}$$

$$= \pi\sqrt{\pi^2 + \frac{1}{4}} + \frac{1}{4}\log\left(\pi + \sqrt{\pi^2 + \frac{1}{4}}\right) - \frac{1}{4}\log\left(\sqrt{\frac{1}{4}}\right)$$

$$= \frac{\pi}{2}\sqrt{1 + 4\pi^2} + \frac{1}{4}\log\left(2\pi + \sqrt{1 + 4\pi^2}\right) \approx 10.63.$$

As a check on our answer, we may note that the path $\mathbf{c}$ connects the points $(1, 0, 0)$ and $(-1, 0, \pi^2)$. The distance between these points is $\sqrt{4 + \pi^2} \approx 3.72$, which is less than 10.63, as it should be. ▲

If a curve is made up of a finite number of pieces each of which is C^1 (with bounded derivative), we compute the arc length by adding the lengths of the component pieces. Such curves are called ***piecewise*** C^1. Sometimes we just say "piecewise smooth."

EXAMPLE 3 A billiard ball on a pool table follows the path $\mathbf{c}$: $[-1, 1] \to \mathbb{R}^3$ defined by $\mathbf{c}(t) = (x(t), y(t), z(t)) = (|t|, |t - \frac{1}{2}|, 0)$. Find the distance traveled by the ball.

SOLUTION This path is not smooth, because $x(t) = |t|$ is not differentiable at 0, nor is $y(t) = |t - \frac{1}{2}|$ differentiable at $\frac{1}{2}$. However, if we divide the interval $[-1, 1]$ into the pieces $[-1, 0]$, $[0, \frac{1}{2}]$, and $[\frac{1}{2}, 1]$, we see that $x(t)$ and $y(t)$ have continuous derivatives on each of the intervals $[-1, 0]$, $[0, \frac{1}{2}]$, and $[\frac{1}{2}, 1]$. (See Figure 4.2.2.)

On $[-1, 0]$, $x(t) = -t$, $y(t) = -t + \frac{1}{2}$, and $z(t) = 0$, so $\|\mathbf{c}'(t)\| = \sqrt{2}$. Hence, the arc length of $\mathbf{c}$ between -1 and 0 is $\int_{-1}^{0} \sqrt{2}\, dt = \sqrt{2}$. Similarly, on $[0, \frac{1}{2}]$, $x(t) = t$, $y(t) = -t + \frac{1}{2}$, $z(t) = 0$, and again $\|\mathbf{c}'(t)\| = \sqrt{2}$, so that the arc length of $\mathbf{c}$ between 0

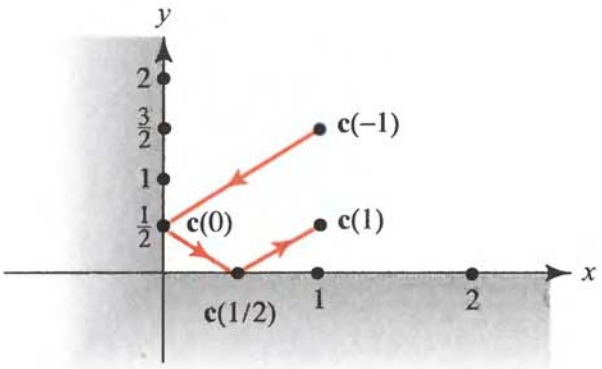


Figure 4.2.2 A piecewise smooth path.

and $\frac{1}{2}$ is $\frac{1}{2}\sqrt{2}$. Finally, on $[\frac{1}{2}, 1]$ we have $x(t) = t$, $y(t) = t - \frac{1}{2}$, $z(t) = 0$, and the arc length of $\mathbf{c}$ between $\frac{1}{2}$ and 1 is $\frac{1}{2}\sqrt{2}$. Thus, the total arc length of $\mathbf{c}$ is $2\sqrt{2}$. Of course, one can also compute the answer as the sum of the distances from $\mathbf{c}(-1)$ to $\mathbf{c}(0)$ to $\mathbf{c}(\frac{1}{2})$ to $\mathbf{c}(1)$. ▲

EXAMPLE 4 Consider the point with position function

$$\mathbf{c}(t) = (t - \sin t, 1 - \cos t),$$

which traces out the cycloid discussed in Section 2.4 (see Figure 2.4.6). Find the velocity, the speed, and the length of one arch.

SOLUTION The velocity vector is $\mathbf{c}'(t) = (1 - \cos t, \sin t)$, so the speed of the point $\mathbf{c}(t)$ is

$$\|\mathbf{c}'(t)\| = \sqrt{(1 - \cos t)^2 + \sin^2 t} = \sqrt{2 - 2\cos t}.$$

Hence, $\mathbf{c}(t)$ moves at variable speed although the circle rolls at constant speed. Furthermore, the speed of $\mathbf{c}(t)$ is zero when t is an integral multiple of 2π. At these values of t, the y coordinate of the point $\mathbf{c}(t)$ is zero and so the point lies on the x axis. The arc length of one cycle is

$$L(\mathbf{c}) = \int_0^{2\pi} \sqrt{2 - 2\cos t}\, dt = 2\int_0^{2\pi} \sqrt{\frac{1 - \cos t}{2}}\, dt$$

$$= 2\int_0^{2\pi} \sin\frac{t}{2}\, dt \left(\text{because } 1 - \cos t = 2\sin^2\frac{t}{2} \text{ and } \sin\frac{t}{2} \geq 0 \text{ on } [0, 2\pi]\right)$$

$$= 4\left(-\cos\frac{t}{2}\right)\Big|_0^{2\pi} = 8. \quad ▲$$

The Differential of Arc Length

The arc-length formula suggests that one introduce the following notation, which will be useful in Chapter 7 in our discussion of line integrals.

Arc-Length Differential An ***infinitesimal displacement*** of a particle following a path $\mathbf{c}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$ is

$$d\mathbf{s} = dx\mathbf{i} + dy\mathbf{j} + dz\mathbf{k} = \left(\frac{dx}{dt}\mathbf{i} + \frac{dy}{dt}\mathbf{j} + \frac{dz}{dt}\mathbf{k}\right) dt,$$

and its length

$$ds = \sqrt{dx^2 + dy^2 + dz^2} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\, dt$$

is the ***differential of arc length***. See Figure 4.2.3.

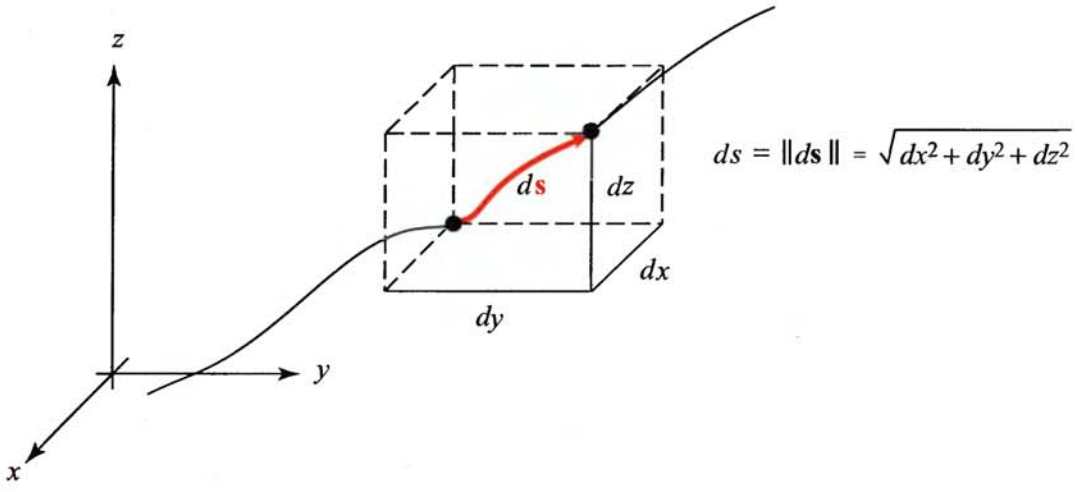


Figure 4.2.3 Differential of arc length.

These formulas help us remember the arc-length formula as

$$\text{arc length} = \int_{t_0}^{t_1} ds.$$

As we have done before with such geometric concepts as length and angle, we can extend the notion of arc length to paths in n-dimensional space.

Arc Length in $\mathbb{R}^n$ Let $\mathbf{c}\colon [t_0, t_1] \to \mathbb{R}^n$ be a piecewise C^1 path. Its ***length*** is defined to be

$$L(\mathbf{c}) = \int_{t_0}^{t_1} \|\mathbf{c}'(t)\|\, dt.$$

The integrand is the square root of the sum of the squares of the coordinate functions of $\mathbf{c}'(t)$: If

$$\mathbf{c}(t) = (x_1(t), x_2(t), \ldots, x_n(t)),$$

then

$$L(\mathbf{c}) = \int_{t_0}^{t_1} \sqrt{(x_1'(t))^2 + (x_2'(t))^2 + \cdots + (x_n'(t))^2}\, dt.$$

EXAMPLE 5 Find the length of the path $\mathbf{c}(t) = (\cos t, \sin t, \cos 2t, \sin 2t)$ in $\mathbb{R}^4$, defined on the interval from 0 to π.

SOLUTION We have $\mathbf{c}'(t) = (-\sin t, \cos t, -2 \sin 2t, 2 \cos 2t)$, and so

$$\|\mathbf{c}'(t)\| = \sqrt{\sin^2 t + \cos^2 t + 4 \sin^2 2t + 4 \cos^2 2t} = \sqrt{1+4} = \sqrt{5},$$

a constant, so the length of the path is

$$\int_0^{\pi} \sqrt{5}\, dt = \sqrt{5}\pi. \quad \blacktriangle$$

It is common to introduce the ***arc-length function*** $s(t)$ associated to a path $\mathbf{c}(t)$ given by

$$s(t) = \int_a^t \|\mathbf{c}'(u)\|\, du,$$

so that (by the fundamental theorem of calculus)

$$s'(t) = \|\mathbf{c}'(t)\|$$

and

$$\int_a^b s'(t)\, dt = s(b) - s(a) = s(b).$$

EXAMPLE 6 Consider the graph of a function of one variable $y = f(x)$ for x in the interval $[a, b]$. We can consider it to be a curve parametrized by $t = x$, namely, $\mathbf{c}(x) = (x, f(x))$ for x ranging from a to b. The arc-length formula gives

$$L(\mathbf{c}) = \int_a^b \sqrt{1 + [f'(x)]^2}\, dx,$$

which agrees with the formula for the length of a graph from one-variable calculus. $\blacktriangle$

Justification for the Arc-Length Formula

The following discussion assumes an acquaintance with the definite integral defined in terms of Riemann sums. If your background in this topic needs reinforcement, the material may be postponed until after Chapter 5.

In $\mathbb{R}^3$ there is another way to justify the arc-length formula based on polygonal approximations. We partition the interval $[a, b]$ into N subintervals of equal length:

$$a = t_0 < t_1 < \cdots < t_N = b;$$

$$t_{i+1} - t_i = \frac{b-a}{N} \qquad \text{for} \qquad 0 \le i \le N-1.$$

We then consider the polygonal line obtained by joining the successive pairs of points $\mathbf{c}(t_i)$, $\mathbf{c}(t_{i+1})$ for $0 \le i \le N-1$. This yields a polygonal approximation to $\mathbf{c}$ as in Figure 4.2.4. By the formula for distance in $\mathbb{R}^3$, it follows that the line segment from $\mathbf{c}(t_i)$ to $\mathbf{c}(t_{i+1})$ has length

$$\|\mathbf{c}(t_{i+1}) - \mathbf{c}(t_i)\| = \sqrt{[x(t_{i+1}) - x(t_i)]^2 + [y(t_{i+1}) - y(t_i)]^2 + [z(t_{i+1}) - z(t_i)]^2},$$

where $\mathbf{c}(t) = (x(t), y(t), z(t))$. Applying the mean-value theorem to $x(t)$, $y(t)$, and $z(t)$ on $[t_i, t_{i+1}]$, we obtain three points t_i^*, t_i^{**}, and t_i^{***} such that

$$x(t_{i+1}) - x(t_i) = x'(t_i^*)(t_{i+1} - t_i),$$
$$y(t_{i+1}) - y(t_i) = y'(t_i^{**})(t_{i+1} - t_i),$$

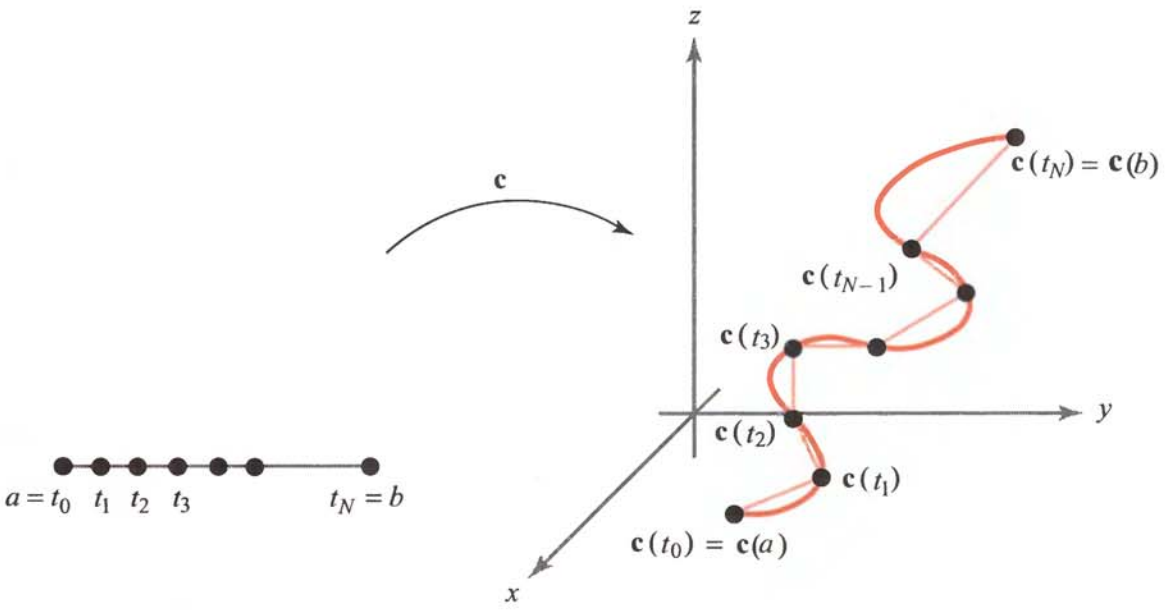


Figure 4.2.4 A path $\mathbf{c}$ may be approximated by a polygonal path obtained by joining each $\mathbf{c}(t_i)$ to $\mathbf{c}(t_{i+1})$ by a straight line.

and

$$z(t_{i+1}) - z(t_i) = z'(t_i^{***})(t_{i+1} - t_i).$$

Thus, the line segment from $\mathbf{c}(t_i)$ to $\mathbf{c}(t_{i+1})$ has length

$$\sqrt{[x'(t_i^*)]^2 + [y'(t_i^{**})]^2 + [z'(t_i^{***})]^2}(t_{i+1} - t_i).$$

Therefore, the length of our approximating polygonal line is

$$S_N = \sum_{i=0}^{N-1} \sqrt{[x'(t_i^*)]^2 + [y'(t_i^{**})]^2 + [z'(t_i^{***})]^2}(t_{i+1} - t_i).$$

As $N \to \infty$, this polygonal line approximates the image of $\mathbf{c}$ more closely. Therefore, we define the arc length of $\mathbf{c}$ as the limit, if it exists, of the sequence S_N as $N \to \infty$. Because the derivatives x', y', and z' are all assumed to be continuous on $[a, b]$, we can conclude that, in fact, the limit does exist and is given by

$$\lim_{N\to\infty} S_N = \int_a^b \sqrt{[x'(t)]^2 + [y'(t)]^2 + [z'(t)]^2}\, dt.$$

(The theory of integration relates the integral to sums by the formula

$$\int_a^b f(t)\, dt = \lim_{N\to\infty} \sum_{i=0}^{N-1} f(t_i^*)(t_{i+1} - t_i),$$

where $t_0, \ldots, t_N$ is a partition of $[a, b]$, $t_i^* \in [t_i, t_{i+1}]$ is arbitrary, and f is a continuous function. Here we have *possibly different points* t_i^*, t_i^{**}, and t_i^{***}, and so this formula must be extended slightly.)

EXERCISES

Find the arc length of the given curve on the specified interval in Exercises 1 to 6.[4]

1. $(2\cos t, 2\sin t, t)$, for $0 \le t \le 2\pi$

2. $(1, 3t^2, t^3)$, for $0 \le t \le 1$

3. $(\sin 3t, \cos 3t, 2t^{3/2})$, for $0 \le t \le 1$

[4]Several of these problems make use of the formula

$$\int \sqrt{x^2 + a^2}\, dx = \tfrac{1}{2}\left[x\sqrt{x^2 + a^2} + a^2 \log(x + \sqrt{x^2 + a^2})\right] + C$$

from the table of integrals in the back of the book.

4. $\left(t+1, \frac{2\sqrt{2}}{3}t^{3/2}+7, \frac{1}{2}t^2\right)$, for $1 \le t \le 2$ **5.** (t, t, t^2), for $1 \le t \le 2$

6. $(t, t\sin t, t\cos t)$, for $0 \le t \le \pi$

7. Find the length of the path $\mathbf{c}(t)$, defined by $\mathbf{c}(t) = (2\cos t, 2\sin t, t)$, if $0 \le t \le 2\pi$ and $\mathbf{c}(t) = (2, t-2\pi, t)$, if $2\pi \le t \le 4\pi$.

8. Let $\mathbf{c}$ be the path $\mathbf{c}(t) = (t, t\sin t, t\cos t)$. Find the arc length of $\mathbf{c}$ between the two points $(0, 0, 0)$ and $(\pi, 0, -\pi)$.

9. Let $\mathbf{c}$ be the path $\mathbf{c}(t) = (2t, t^2, \log t)$, defined for $t > 0$. Find the arc length of $\mathbf{c}$ between the points $(2, 1, 0)$ and $(4, 4, \log 2)$.

10. The arc-length function $s(t)$ for a given path $\mathbf{c}(t)$, defined by $s(t) = \int_a^t \|\mathbf{c}'(\tau)\|\, d\tau$, represents the distance a particle traversing the trajectory of $\mathbf{c}$ will have traveled by time t if it starts out at time a; that is, it gives the length of $\mathbf{c}$ between $\mathbf{c}(a)$ and $\mathbf{c}(t)$. Find the arc-length functions for the curves $\alpha(t) = (\cosh t,\ \sinh t, t)$ and $\beta(t) = (\cos t,\ \sin t, t)$, with $a = 0$.

11. Let $\mathbf{c}(t)$ be a given path, $a \le t \le b$. Let $s = \alpha(t)$ be a new variable, where α is a strictly increasing C^1 function given on $[a, b]$. For each s in $[\alpha(a), \alpha(b)]$ there is a unique t with $\alpha(t) = s$. Define the function $\mathbf{d}\colon [\alpha(a), \alpha(b)] \to \mathbb{R}^3$ by $\mathbf{d}(s) = \mathbf{c}(t)$.

(a) Argue that the image curves of $\mathbf{c}$ and $\mathbf{d}$ are the same.
(b) Show that $\mathbf{c}$ and $\mathbf{d}$ have the same arc length.
(c) Let $s = \alpha(t) = \int_a^t \|\mathbf{c}'(\tau)\|\, d\tau$. Define $\mathbf{d}$ as above by $\mathbf{d}(s) = \mathbf{c}(t)$. Show that

$$\left\| \frac{d}{ds}\mathbf{d}(s) \right\| = 1.$$

The path $s \mapsto \mathbf{d}(s)$ is said to be an ***arc length reparametrization*** of $\mathbf{c}$ (see also Exercise 13).

Exercises 12 to 17 develop some of the classic differential geometry of curves.

12. Let $\mathbf{c}\colon [a, b] \to \mathbb{R}^3$ be an infinitely differentiable path (derivatives of all orders exist). Assume $\mathbf{c}'(t) \neq \mathbf{0}$ for any t. The vector $\mathbf{c}'(t)/\|\mathbf{c}'(t)\| = \mathbf{T}(t)$ is tangent to $\mathbf{c}$ at $\mathbf{c}(t)$, and, because $\|\mathbf{T}(t)\| = 1$, $\mathbf{T}$ is called the ***unit tangent*** to $\mathbf{c}$.

(a) Show that $\mathbf{T}'(t) \cdot \mathbf{T}(t) = 0$. [HINT: Differentiate $\mathbf{T}(t) \cdot \mathbf{T}(t) = 1$.]
(b) Write down a formula for $\mathbf{T}'(t)$ in terms of $\mathbf{c}$.

13. (a) A path $\mathbf{c}(s)$ is said to be ***parametrized by arc length*** or, what is the same thing, to have ***unit speed*** if $\|\mathbf{c}'(s)\| = 1$. For a path parametrized by arc length on $[a, b]$, show that $l(\mathbf{c}) = b - a$.

(b) The ***curvature*** at a point $\mathbf{c}(s)$ on a path is defined by $k = \|\mathbf{T}'(s)\|$ when the path is parametrized by arc length. Show that $k = \|\mathbf{c}''(s)\|$.

(c) If $\mathbf{c}$ is given in terms of some other parameter t and $\mathbf{c}'(t)$ is never $\mathbf{0}$, show that $k = \|\mathbf{c}'(t) \times \mathbf{c}''(t)\|/\|\mathbf{c}'(t)\|^3$.

(d) Calculate the curvature of the helix $\mathbf{c}(t) = (1/\sqrt{2})(\cos t,\ \sin t, t)$. (This $\mathbf{c}$ is a scalar multiple of the right-circular helix.)

14. If $\mathbf{T}'(t) \neq \mathbf{0}$, it follows from Exercise 12 that $\mathbf{N}(t) = \mathbf{T}'(t)/\|\mathbf{T}'(t)\|$ is normal (i.e., perpendicular) to $\mathbf{T}(t)$; $\mathbf{N}$ is called the ***principal normal vector***. Let a third unit vector that is perpendicular to both $\mathbf{T}$ and $\mathbf{N}$ be defined by $\mathbf{B} = \mathbf{T} \times \mathbf{N}$; $\mathbf{B}$ is called the ***binormal vector***. Together, $\mathbf{T}$, $\mathbf{N}$, and $\mathbf{B}$ form a right-handed system of mutually orthogonal vectors that may be thought of as moving along the path (Figure 4.2.5). Show that

(a) $\dfrac{d\mathbf{B}}{dt} \cdot \mathbf{B} = 0.$

(b) $\dfrac{d\mathbf{B}}{dt} \cdot \mathbf{T} = 0.$

(c) $d\mathbf{B}/dt$ is a scalar multiple of $\mathbf{N}$.

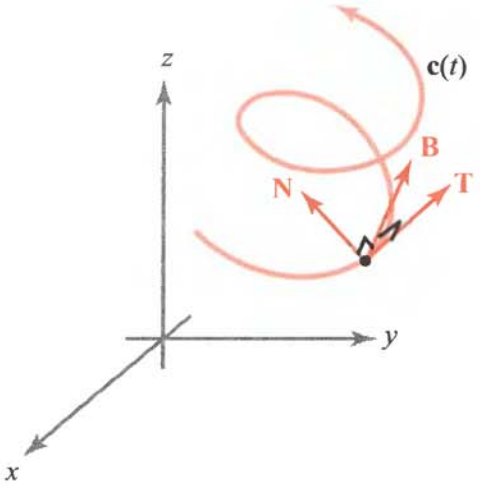


Figure 4.2.5 The tangent $\mathbf{T}$, principal normal $\mathbf{N}$, and binormal $\mathbf{B}$.

15. If $\mathbf{c}(s)$ is parametrized by arc length, we use the result of Exercise 14(c) to define a scalar-valued function τ, called the ***torsion***, by

$$\frac{d\mathbf{B}}{ds} = -\tau\mathbf{N}.$$

(a) Show that $\tau = [\mathbf{c}'(s) \times \mathbf{c}''(s)] \cdot \mathbf{c}'''(s)/\|\mathbf{c}''(s)\|^2$.

(b) Show that if $\mathbf{c}$ is given in terms of some other parameter t,

$$\tau = \frac{[\mathbf{c}'(t) \times \mathbf{c}''(t)] \cdot \mathbf{c}'''(t)}{\|\mathbf{c}'(t) \times \mathbf{c}''(t)\|^2}.$$

Compare with Exercise 13(c).

(c) Compute the torsion of the helix $\mathbf{c}(t) = (1/\sqrt{2})(\cos t,\ \sin t, t)$.

16. Show that if a path lies in a plane, then the torsion is zero. Do this by demonstrating that $\mathbf{B}$ is constant and is a normal vector to the plane in which $\mathbf{c}$ lies. (If the torsion is not zero, it gives a measure of how fast the curve is twisting out of the plane of $\mathbf{T}$ and $\mathbf{N}$.)

17. (a) Use the results of Exercises 13, 14, and 15 to prove the following ***Frenet formulas*** for a unit-speed curve:

$$\frac{d\mathbf{T}}{ds} = k\mathbf{N}; \qquad \frac{d\mathbf{N}}{ds} = -k\mathbf{T} + \tau\mathbf{B}; \qquad \frac{d\mathbf{B}}{ds} = -\tau\mathbf{N}.$$

(b) Reexpress the results of part (a) as

$$\frac{d}{ds}\begin{pmatrix}\mathbf{T}\\ \mathbf{N}\\ \mathbf{B}\end{pmatrix} = \omega \times \begin{pmatrix}\mathbf{T}\\ \mathbf{N}\\ \mathbf{B}\end{pmatrix}$$

for a suitable vector ω.

18. In special relativity, the ***proper time*** of a path $\mathbf{c}: [a, b] \to \mathbb{R}^4$ with components given by $\mathbf{c}(\lambda) = (x(\lambda), y(\lambda), z(\lambda), t(\lambda))$ is defined to be the quantity

$$\int_a^b \sqrt{-[x'(\lambda)]^2 - [y'(\lambda)]^2 - [z'(\lambda)]^2 + c^2[t'(\lambda)]^2}\, d\lambda,$$

where c is the velocity of light, a constant. In Figure 4.2.6, show that, using self-explanatory notation, the "twin paradox inequality" holds:

$$\text{proper time (AB)} + \text{proper time (BC)} < \text{proper time (AC)}.$$

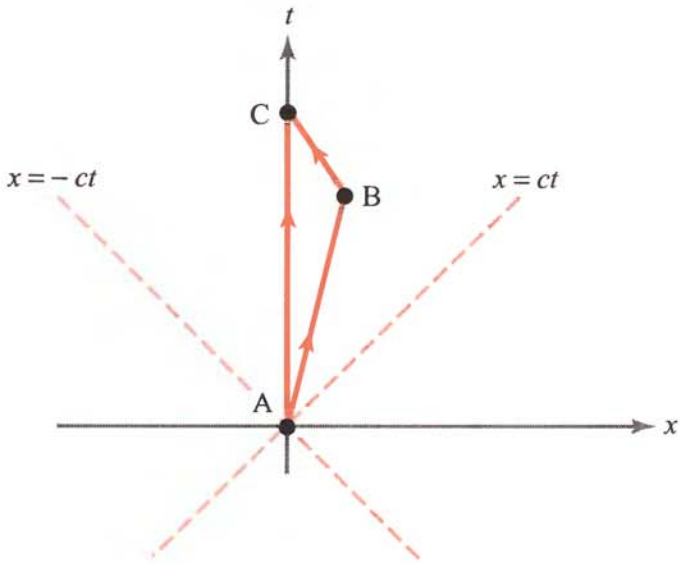


Figure 4.2.6 The relativistic triangle inequality.

19. The early Greeks knew that a straight line was the shortest possible path between two points. Euclid, in his book *Optics*, stated the "principle of the reflection of light"—that is, light traveling in a plane travels in a straight line, and when it is reflected across a mirror, the angle of incidence equals the angle of reflection.

The Greeks could not have had a proof that straight lines provided the shortest path between two points because they, in the first place, had no definition of the length of a path. They saw this property of straight lines as more or less "obvious."

Using the justification of arc length in this section and the triangle inequality of Section 1.5, argue that if $\mathbf{c}_0$ is the straight-line path $c_0(t) = t\mathrm{P} + (1 - t)\mathrm{Q}$ between P and Q in $\mathbb{R}^3$, then

$$L(\mathbf{c}_0) \leq L(\mathbf{c})$$

for any other path $\mathbf{c}$ joining P and Q.

4.3 Vector Fields

The Concept of a Vector Field

In Chapter 2, we introduced a particular kind of vector field, the gradient. In this section we study *general* vector fields, discussing their geometric and physical significance.

Vector Fields A ***vector field*** in $\mathbb{R}^n$ is a map $\mathbf{F}: A \subset \mathbb{R}^n \to \mathbb{R}^n$ that assigns to each point $\mathbf{x}$ in its domain A a vector $\mathbf{F}(\mathbf{x})$. If $n = 2$, $\mathbf{F}$ is called a ***vector field in the plane***, and if $n = 3$, $\mathbf{F}$ is a ***vector field in space***.

Picture $\mathbf{F}$ as attaching an *arrow* to each point (Figure 4.3.1). By contrast, a map $f: A \subset \mathbb{R}^n \to \mathbb{R}$ that assigns a *number* to each point is a ***scalar field***. A vector field $\mathbf{F}(x, y, z)$ on $\mathbb{R}^3$ has three ***component scalar fields*** F_1, F_2, and F_3, so that

$$\mathbf{F}(x, y, z) = (F_1(x, y, z), F_2(x, y, z), F_3(x, y, z)).$$

Similarly, a vector field on $\mathbb{R}^n$ has n components $F_1, \ldots, F_n$. If each component is a C^k function, we say the vector field $\mathbf{F}$ is of ***class*** C^k. Vector fields will be assumed to be at least of class C^1 unless otherwise noted.

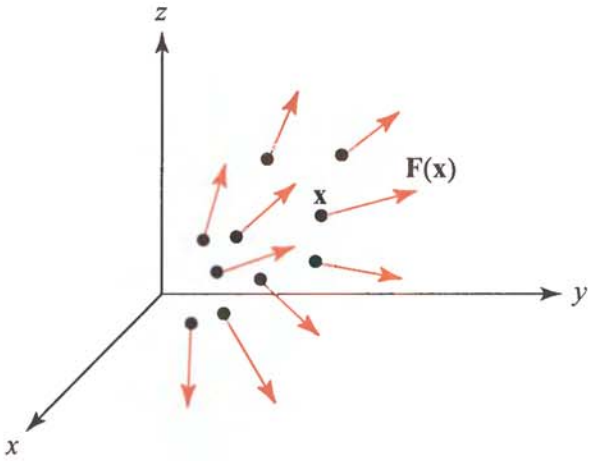


Figure 4.3.1 A vector field $\mathbf{F}$ assigns a vector $\mathbf{F}(\mathbf{x})$ to each point $\mathbf{x}$ of its domain.

In many applications, $\mathbf{F}(\mathbf{x})$ represents a physical vector quantity (force, velocity, etc.) associated with the position $\mathbf{x}$, as in the following examples.

EXAMPLE 1 The flow of water through a pipe is said to be ***steady*** if, at each point inside the pipe, the velocity of the fluid passing through that point does not change with time. (Note that this is quite different from saying that the water in the pipe is not moving.) Attaching to each point the fluid velocity at that point, we obtain the ***velocity field*** $\mathbf{V}$ of the fluid (see Figure 4.3.2). Notice that the length of the arrows (the speed), as well as the direction of flow, may change from point to point. ▲

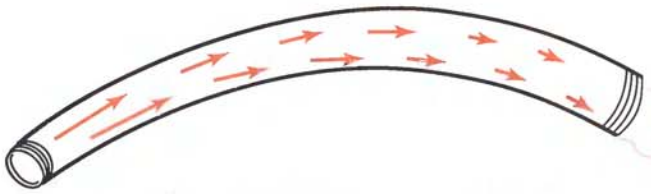


Figure 4.3.2 A vector field describing the velocity of flow in a pipe.

EXAMPLE 2 Some forms of rotary motion (such as the motion of particles on a turntable) can be described by the vector field

$$\mathbf{V}(x, y) = -y\mathbf{i} + x\mathbf{j}.$$

See Figure 4.3.3, in which we have shown instead of $\mathbf{V}$ the shorter vector field $\frac{1}{4}\mathbf{V}$ so that the arrows do not overlap. This is a common convention in drawing pictures of vector fields. ▲

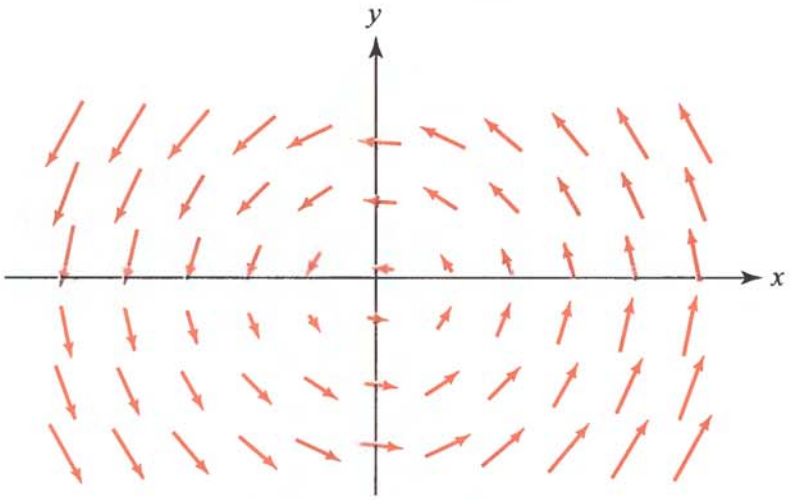


Figure 4.3.3 A rotary vector field.

EXAMPLE 3 In the plane, $\mathbb{R}^2$, let the vector field x be defined by

$$\mathbf{V}(x, y) = \frac{y\mathbf{i}}{x^2 + y^2} - \frac{x\mathbf{j}}{x^2 + y^2} = \left(\frac{y}{x^2 + y^2}, -\frac{x}{x^2 + y^2}\right)$$

(except at the origin, where $\mathbf{V}$ is not defined). This vector field is a good approximation to the planar part of the velocity of water flowing toward a hole in the bottom of a tub (Figure 4.3.4). Notice that the velocity becomes *larger* as you approach the hole. ▲

Gradient Vector Fields

In Section 2.6 we introduced the gradient of a function by

$$\nabla f(x, y, z) = \frac{\partial f}{\partial x}(x, y, z)\mathbf{i} + \frac{\partial f}{\partial y}(x, y, z)\mathbf{j} + \frac{\partial f}{\partial z}(x, y, z)\mathbf{k}.$$

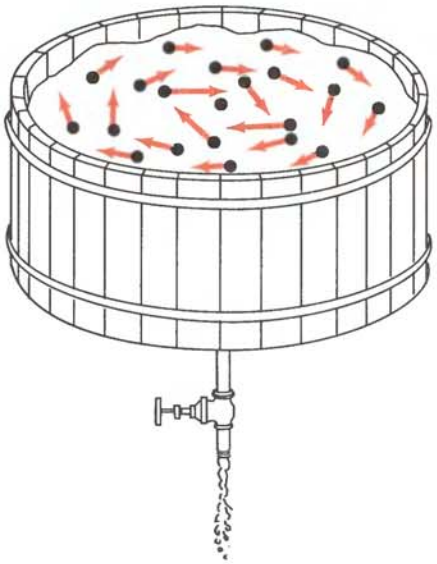


Figure 4.3.4 The vector field describing circular flow in a tub.

Now we want to think of this as an example of a vector field—it assigns a vector to each point (x, y, z). As such, we refer to ∇f as a ***gradient vector field***. Gradient fields come up in a variety of situations, as the next two examples show.

EXAMPLE 4 A piece of material is heated on one side and cooled on another. The temperature at each point within the body is described at a given moment by a scalar field $T(x, y, z)$. The flow of heat may be marked by a field of arrows indicating the direction and magnitude of the flow (Figure 4.3.5). This ***energy*** or ***heat flux vector field*** is given by $\mathbf{J} = -k\nabla T$, where $k > 0$ is a constant called the ***conductivity*** and ∇T is the gradient of the real-valued function T. Level sets of T are called ***isotherms***. Note that the heat flows from hot regions toward cold ones, since $-\nabla T$ points in the direction of decreasing T. ▲

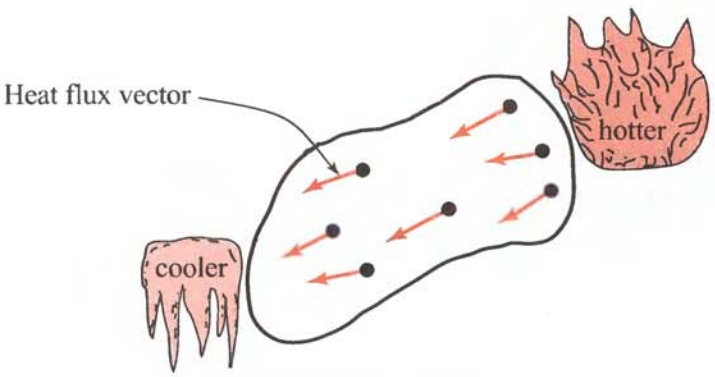


Figure 4.3.5 A vector field describing the direction and magnitude of heat flow.

EXAMPLE 5 The force of attraction of the earth on a mass m can be described by a vector field called the ***gravitational force field***. Place the origin of a coordinate system at the center of the earth (assumed spherical). According to Newton's law of gravity, this field is given by

$$\mathbf{F} = -\frac{mMG}{r^3}\mathbf{r},$$

where $\mathbf{r}(x, y, z) = (x, y, z)$, and $r = \|\mathbf{r}\|$ (see Figure 4.3.6). The domain of this vector field consists of those $\mathbf{r}$ for which $\|\mathbf{r}\|$ is greater than the radius of the earth. As we saw in Example 6, Section 2.6, $\mathbf{F}$ is a gradient field, $\mathbf{F} = -\nabla V$, where

$$V = -\frac{mMG}{r}$$

is the ***gravitational potential***. Note again that $\mathbf{F}$ points in the direction of *decreasing* V. Writing $\mathbf{F}$ in terms of components, we see that

$$\mathbf{F}(x, y, z) = \left(\frac{-mMG}{r^3}x, \frac{-mMG}{r^3}y, \frac{-mMG}{r^3}z\right). \quad \blacktriangle$$

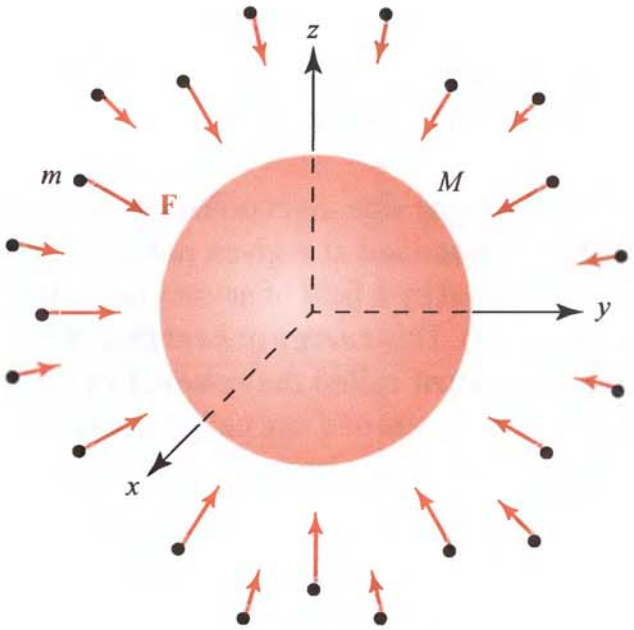


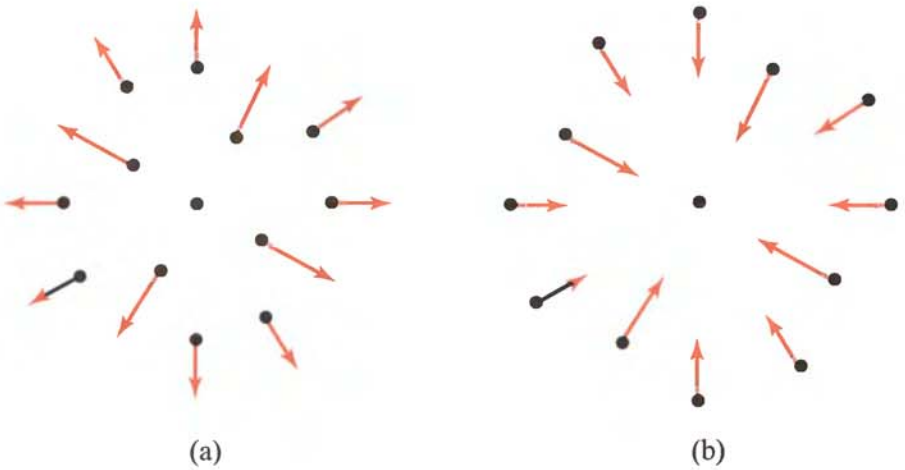
Figure 4.3.6 The vector field $\mathbf{F}$ given by Newton's law of gravitation.

EXAMPLE 6 According to ***Coulomb's law***, the force acting on a charge e at position $\mathbf{r}$ due to a charge Q at the origin is

$$\mathbf{F} = \frac{\varepsilon Qe}{r^3}\mathbf{r} = -\nabla V,$$

where $V = \varepsilon Qe/r$ and ε is a constant that depends on the units used. For $Qe > 0$ (like charges) the force is repulsive [Figure 4.3.7(a)], and for $Qe < 0$ (unlike charges) the force is attractive [Figure 4.3.7(b)]. Because the potential V is constant on the level surfaces of V, they are called ***equipotential surfaces***. Note that the force field is orthogonal to the equipotential surfaces (the force field is radial and the equipotential surfaces are concentric spheres). $\blacktriangle$

The next example shows that not every vector field is a gradient.



(a) (b)

Figure 4.3.7 The vector fields associated with (a) like charges ($Qe > 0$), and (b) unlike charges ($Qe < 0$).

EXAMPLE 7 Show that the vector field $\mathbf{V}$ on $\mathbb{R}^2$ defined by $\mathbf{V}(x, y) = y\mathbf{i} - x\mathbf{j}$ is not a gradient vector field; that is, there is no C^1 function f such that

$$\mathbf{V}(x, y) = \nabla f(x, y) = \frac{\partial f}{\partial x}\mathbf{i} + \frac{\partial f}{\partial y}\mathbf{j}.$$

SOLUTION Suppose that such an f exists. Then $\partial f/\partial x = y$ and $\partial f/\partial y = -x$. Because these are C^1 functions, f itself must have continuous first- *and* second-order partial derivatives. But, $\partial^2 f/\partial x\,\partial y = -1$, and $\partial^2 f/\partial y\,\partial x = 1$, which violates the equality of mixed partials. Thus, $\mathbf{V}$ cannot be a gradient vector field. ▲

Conservation of Energy and Escaping the Earth's Gravitational Field

Consider a particle of mass m moving in a force field $\mathbf{F}$ that is a potential field. That is, assume $\mathbf{F} = -\nabla V$ for a real-valued function V, and that the particle moves according to $\mathbf{F} = m\mathbf{a}$. Thus, if the path is $\mathbf{r}(t)$, then

$$m\ddot{\mathbf{r}}(t) = -\nabla V(\mathbf{r}(t)). \tag{1}$$

A basic fact about such motion is the *conservation of energy*. The energy E of the particle is defined to be the sum of the kinetic and potential energies, defined as

$$E = \frac{1}{2}m\|\dot{\mathbf{r}}(t)\|^2 + V(\mathbf{r}(t)). \tag{2}$$

The principle of *conservation of energy* states that if Newton's second law holds, then E is independent of time; that is, $dE/dt = 0$. The proof of this fact is a simple calculation; we use equation (2), the chain rule, and equation (1):

$$\begin{aligned}\frac{dE}{dt} &= m\dot{\mathbf{r}}\cdot\ddot{\mathbf{r}} + (\nabla V)\cdot\dot{\mathbf{r}} \\ &= \dot{\mathbf{r}}\cdot(-\nabla V + \nabla V) = 0.\end{aligned}$$

Escape Velocity

As an application of conservation of energy, we compute the velocity required for a rocket to escape the earth's gravitational influence. Assume the rocket has mass m and is at a distance R_0 from the center of the earth (or another planet) when its escape velocity v_e has been reached, and that it will coast thereafter. The energy at this time is

$$E_0 = \frac{1}{2}mv_e^2 - \frac{mMG}{R_0}. \tag{3}$$

By conservation of energy, E_0 will equal the energy at a later time, which we write as

$$E_0 = E = \frac{1}{2}mv^2 - \frac{mMG}{R}, \tag{4}$$

where v is the velocity and R is the distance from the center of the earth (or the other planet). What we mean by the term *escape velocity* is that v_e is chosen such that the rocket gets to great distances, at which time it is barely moving. That is, v is close to zero and R is very large. Thus, from equation (4), we see that $E = 0$ and hence $E_0 = 0$; solving $E_0 = 0$ for v_e using equation (3) gives:

$$v_e = \sqrt{\frac{2MG}{R_0}}.$$

Now GM/R_0^2 is exactly g, the acceleration due to gravity at the distance R_0 from the center of the planet. Thus, we can write:

$$v_e = \sqrt{2gR_0}.$$

For the earth, if the escape velocity were to be achieved at the surface of the earth (of course, this is a bit unrealistic), this would give

$$v_e = \sqrt{2 \cdot 9.8 \text{ m/s}^2 \cdot 6{,}371{,}000 \text{ m}} = 11{,}127 \text{ m/s}.$$

However, this is a good approximation to the velocity that a satellite in low earth orbit needs in order to escape the earth's gravitational field.

Flow Lines

An important concept related to general (not necessarily gradient) vector fields is that of a flow line, defined as follows.

Flow Lines If $\mathbf{F}$ is a vector field, a ***flow line*** for $\mathbf{F}$ is a path $\mathbf{c}(t)$ such that

$$\mathbf{c}'(t) = \mathbf{F}(\mathbf{c}(t)).$$

That is, $\mathbf{F}$ yields the velocity field of the path $\mathbf{c}(t)$.

In the context of Example 1, a flow line is the path followed by a small particle suspended in the fluid (Figure 4.3.8). Flow lines are also appropriately called ***streamlines*** or ***integral curves***.

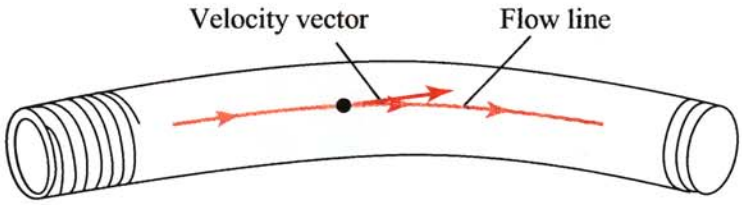


Figure 4.3.8 The velocity vector of a fluid is tangent to a flow line.

Geometrically, a flow line for a given vector filed $\mathbf{F}$ is a curve that threads its way through the domain of the vector field in such a way that the tangent vector of the curve coincides with the vector field, as in Figure 4.3.9.

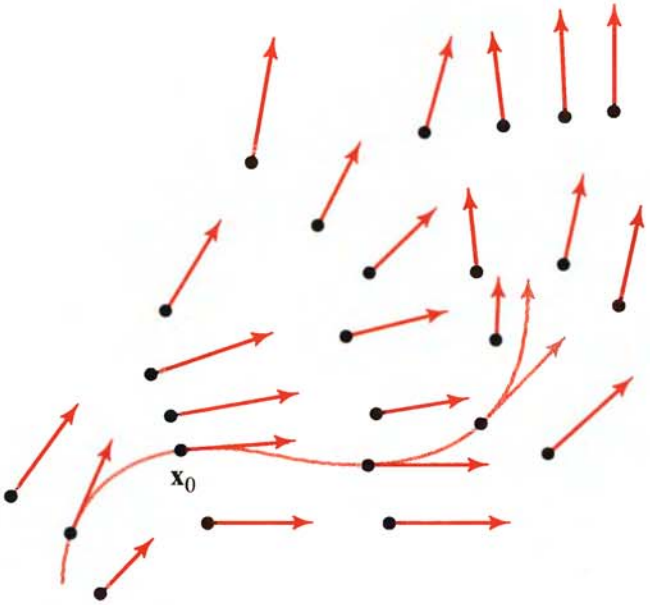


Figure 4.3.9 A flow line threading its way through a vector field in the plane.

A flow line may be viewed as a solution of a system of differential equations. Indeed, we can write the definition $\mathbf{c}'(t) = \mathbf{F}(\mathbf{c}(t))$ as

$$x'(t) = P(x(t), y(t), z(t)),$$
$$y'(t) = Q(x(t), y(t), z(t)),$$
$$z'(t) = R(x(t), y(t), z(t)),$$

where $\mathbf{c}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$, and where

$$\mathbf{F} = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}.$$

One learns about such systems in courses on differential equations, but we are not presuming such a course has been taken.

EXAMPLE 8 Show that the path $\mathbf{c}(t) = (\cos t,\ \sin t)$ is a flow line of the vector field $\mathbf{F}(x, y) = -y\mathbf{i} + x\mathbf{j}$. Can you find others?

SOLUTION We must verify that $\mathbf{c}'(t) = \mathbf{F}(\mathbf{c}(t))$. The left side is $(-\sin t)\mathbf{i} + (\cos t)\mathbf{j}$, while the right side is $\mathbf{F}(\cos t,\ \sin t) = (-\sin t)\mathbf{i} + (\cos t)\mathbf{j}$, so we have a flow line. As suggested by Figure 4.3.3, the other flow lines are also circles. They have the form

$$\mathbf{c}(t) = (r\ \cos(t - t_0), r\ \sin(t - t_0))$$

for constants r and t_0. ▲

In many cases, explicit formulas for flow lines are not available, so one must resort to numerical methods. Figure 4.3.10 shows some output from a program that computes flow lines numerically and plots them on the screen.

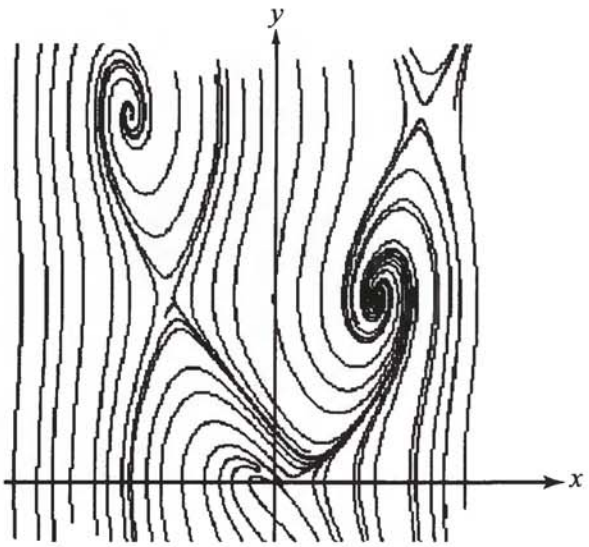


Figure 4.3.10 Computer-generated integral curves of the vector field $\mathbf{F}(x, y) = (\sin y)\mathbf{i} + (x^2 - y)\mathbf{j}$. This figure was created using *3D-XplorMath*, available from Richard Palais' Web site at rsp.math.brandeis.edu/3D-XplorMath.

Historical Note

The Field Concept

The concept of a "field," such as a vector field, has had an enormous impact on the development of conceptual frameworks for physics and engineering. It is truly one of the great breakthrough ideas in the history of human thought. It is the notion that allows one to describe, in a systematic way, influences on objects and between objects that are spatially separated.

The idea of a field began with Newton's concept of the gravitational field. In this case, the gravitational field describes the attractive influence of one body or group of bodies on one another. Similarly, the electric field produced by a charged object or group of objects creates, according to

Coulomb's law, a force on another charged object. Using vector fields to describe these sorts of forces has led to a deeper understanding of attractive and repulsive forces in nature.

However, it was the monumental discovery of the Maxwell field equations, which describe the propagation of electromagnetic energy, that cemented the concept of "field" in scientific thought. This example is particularly interesting because these fields can *propagate*. The contrast between the electromagnetic field that can propagate and the gravitational field that involves instantaneous *action at a distance* has caused great interest among philosophers of science.

Einstein's idea is that gravitation can be described in terms of the metric properties of space–time and that in this theory the associated field can also propagate, just like the electromagnetic field, thus providing profound philosophical evidence that Einstein's version of gravity may be correct. These ideas have also led to modern efforts to detect gravitational waves. For a further discussion of Einstein's work, see Section 7.7.

The idea of a field is also used in engineering to describe elastic systems and interesting microstructural properties of materials. In modern theoretical physics, the field concept is used to describe elementary particles and is central to attempts by modern theoretical physicists to unify gravity with the quantum mechanical physics of elementary particles. It is impossible to imagine a modern theoretical framework that does not incorporate some sort of field concept as a central ingredient.

EXERCISES

In Exercises 1 to 8, sketch the given vector field or a small multiple of it.

1. $\mathbf{F}(x, y) = (2, 2)$

2. $\mathbf{F}(x, y) = (4, 0)$

3. $\mathbf{F}(x, y) = (x, y)$

4. $\mathbf{F}(x, y) = (-x, y)$

5. $\mathbf{F}(x, y) = (2y, x)$

6. $\mathbf{F}(x, y) = (y, -2x)$

7. $\mathbf{F}(x, y) = \left(\dfrac{x}{\sqrt{x^2 + y^2}}, \dfrac{y}{\sqrt{x^2 + y^2}}\right)$

8. $\mathbf{F}(x, y) = \left(\dfrac{y}{\sqrt{x^2 + y^2}}, \dfrac{x}{\sqrt{x^2 + y^2}}\right)$

In Exercises 9 to 12, sketch a few flow lines of the given vector field.

9. $\mathbf{F}(x, y) = (y, -x)$

10. $\mathbf{F}(x, y) = (x, -y)$

11. $\mathbf{F}(x, y) = (x, x^2)$

12. $\mathbf{F}(x, y, z) = (y, -x, 0)$

In Exercises 13 to 16, show that the given curve $\mathbf{c}(t)$ *is a flow line of the given velocity vector field* $\mathbf{F}(x, y, z)$.

13. $\mathbf{c}(t) = (e^{2t}, \log|t|, 1/t), t \neq 0; \mathbf{F}(x, y, z) = (2x, z, -z^2)$

14. $\mathbf{c}(t) = (t^2, 2t - 1, \sqrt{t}), t > 0; \mathbf{F}(x, y, z) = (y + 1, 2, 1/2z)$

15. $\mathbf{c}(t) = (\sin t, \cos t, e^t); \mathbf{F}(x, y, z) = (y, -x, z)$

16. $\mathbf{c}(t) = \left(\frac{1}{t^3}, e^t, \frac{1}{t}\right); \mathbf{F}(x, y, z) = (-3z^4, y, -z^2)$

17. Show that it takes half as much energy to launch a satellite into an orbit just above the earth as it does to escape the earth. (Ignore the rotation of the earth.)

18. Let $\mathbf{c}(t)$ be a flow line of a gradient field $\mathbf{F} = -\nabla V$. Prove that $V(\mathbf{c}(t))$ is a decreasing function of t.

19. Suppose that the isotherms in a region are all concentric spheres centered at the origin. Prove that the energy flux vector field points either toward or away from the origin.

20. Sketch the gradient field $-\nabla V$ for $V(x, y) = (x + y)/(x^2 + y^2)$ and the equipotential surface $V = 1$.

4.4 Divergence and Curl

For each of the divergence and curl operations, we will make use of the ***del operator***, defined by

$$\nabla = \mathbf{i}\frac{\partial}{\partial x} + \mathbf{j}\frac{\partial}{\partial y} + \mathbf{k}\frac{\partial}{\partial z}.$$

For functions of one variable, taking a derivative can be thought of as an operation or process; that is, given a function $y = f(x)$, its derivative is the result of *operating* on y by the derivative *operator* d/dx. Similarly, we can write the gradient as

$$\nabla f = \left(\mathbf{i}\frac{\partial}{\partial x} + \mathbf{j}\frac{\partial}{\partial y}\right) f = \mathbf{i}\frac{\partial f}{\partial x} + \mathbf{j}\frac{\partial f}{\partial y}$$

for functions of two variables, and

$$\nabla f = \left(\mathbf{i}\frac{\partial}{\partial x} + \mathbf{j}\frac{\partial}{\partial y} + \mathbf{k}\frac{\partial}{\partial z}\right) f = \mathbf{i}\frac{\partial f}{\partial x} + \mathbf{j}\frac{\partial f}{\partial y} + \mathbf{k}\frac{\partial f}{\partial z}$$

for three variables. In operational terms, the gradient of f is obtained by taking the ∇ operator and applying it to f.

Definition of Divergence

We define the divergence of a vector field $\mathbf{F}$ by taking the *dot product* of ∇ with $\mathbf{F}$.

Divergence If $\mathbf{F} = F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k}$, the ***divergence*** of $\mathbf{F}$ is the scalar field

$$\operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}.$$

Similarly, if $\mathbf{F} = (F_1, \ldots, F_n)$ is a vector field on $\mathbb{R}^n$, its divergence is

$$\operatorname{div} \mathbf{F} = \sum_{i=1}^{n} \frac{\partial F_i}{\partial x_i} = \frac{\partial F_1}{\partial x_1} + \frac{\partial F_2}{\partial x_2} + \cdots + \frac{\partial F_n}{\partial x_n}.$$

EXAMPLE 1 Compute the divergence of

$$\mathbf{F} = x^2 y\mathbf{i} + z\mathbf{j} + xyz\mathbf{k}.$$

SOLUTION

$$\operatorname{div} \mathbf{F} = \frac{\partial}{\partial x}(x^2 y) + \frac{\partial}{\partial y}(z) + \frac{\partial}{\partial z}(xyz) = 2xy + 0 + xy = 3xy. \quad \blacktriangle$$

Interpretation

The divergence has an important physical interpretation. If we imagine $\mathbf{F}$ to be the velocity field of a gas (or a fluid), then div $\mathbf{F}$ *represents the rate of expansion per unit volume under the flow of the gas (or fluid)*. If div $\mathbf{F} < 0$, the gas (or fluid) is *compressing*. For a vector field $\mathbf{F}(x, y) = F_1\mathbf{i} + F_2\mathbf{j}$ on the plane, the *divergence*

$$\nabla \cdot \mathbf{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y}$$

measures the rate of expansion of area.

This interpretation is explained graphically, as follows. Choose a small region W about a point $\mathbf{x}_0$. For each point $\mathbf{x}$ in W, let $\mathbf{x}(t)$ be the flow line emanating from $\mathbf{x}$. The set of points $\mathbf{x}(t)$ describe how the set W flows after time t (see Figure 4.4.1).

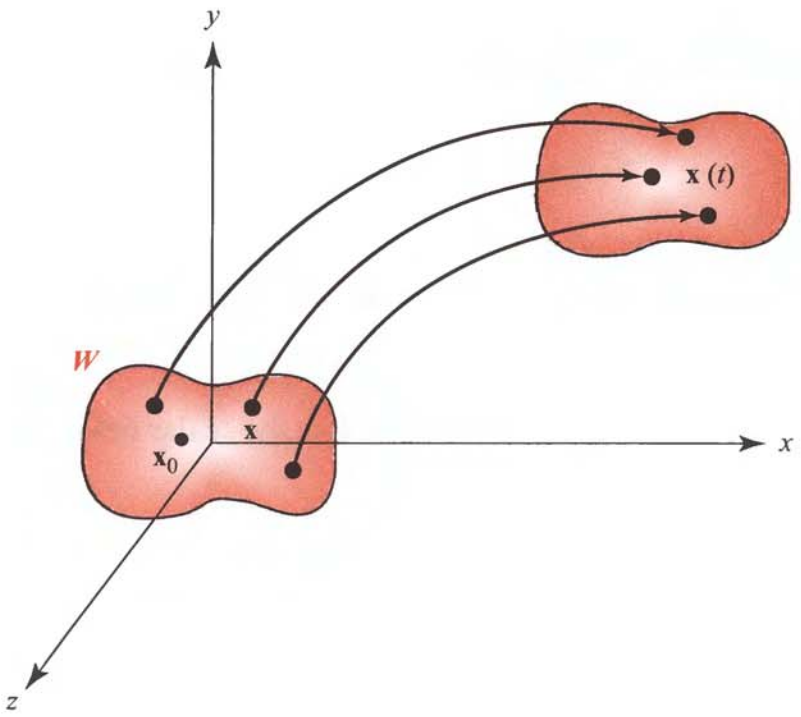


Figure 4.4.1 Flowing a region W along the flow lines of a vector field.

Call the region that results after time t has elapsed $W(t)$, and let $\mathcal{V}(t)$ be its volume (or area in two dimensions). Then the relative rate of change of volume is the divergence; more precisely,

$$\frac{1}{\mathcal{V}(0)}\frac{d}{dt}\mathcal{V}(t)\bigg|_{t=0} \approx \operatorname{div} \mathbf{F}(\mathbf{x}_0),$$

with the approximation being more exact as W shrinks to $\mathbf{x}_0$. A direct proof of this is given in the Internet supplement, but a more natural argument is given in Chapter 8, in the context of the integral theorems of vector calculus.

EXAMPLE 2 Consider the vector field in the plane given by $\mathbf{V}(x, y) = x\mathbf{i}$. Relate the sign of the divergence of $\mathbf{V}$ with the rate of change of areas under the flow.

SOLUTION We think of $\mathbf{V}$ as the velocity field of a fluid in the plane. The vector field $\mathbf{V}$ points to the right for $x > 0$ and to the left if $x < 0$, as we see in Figure 4.4.2. The length of $\mathbf{V}$ gets shorter toward the origin. As the fluid moves, it expands (the area of the shaded rectangle increases), so we expect div $\mathbf{V} > 0$. Indeed, div $\mathbf{V} = 1$. ▲

EXAMPLE 3 The flow lines of the vector field $\mathbf{F} = x\mathbf{i} + y\mathbf{j}$ are straight lines directed away from the origin (Figure 4.4.3).

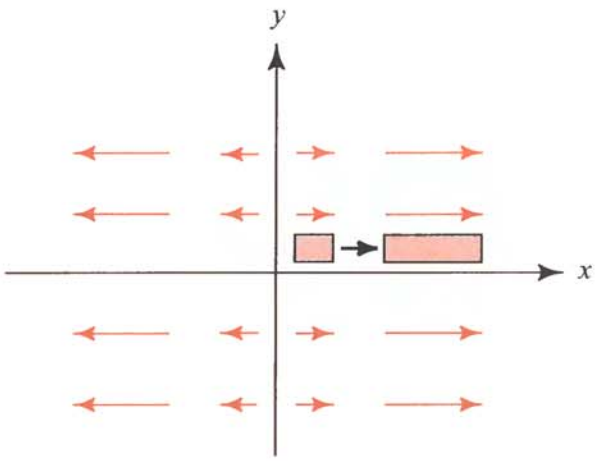


Figure 4.4.2 This fluid is expanding.

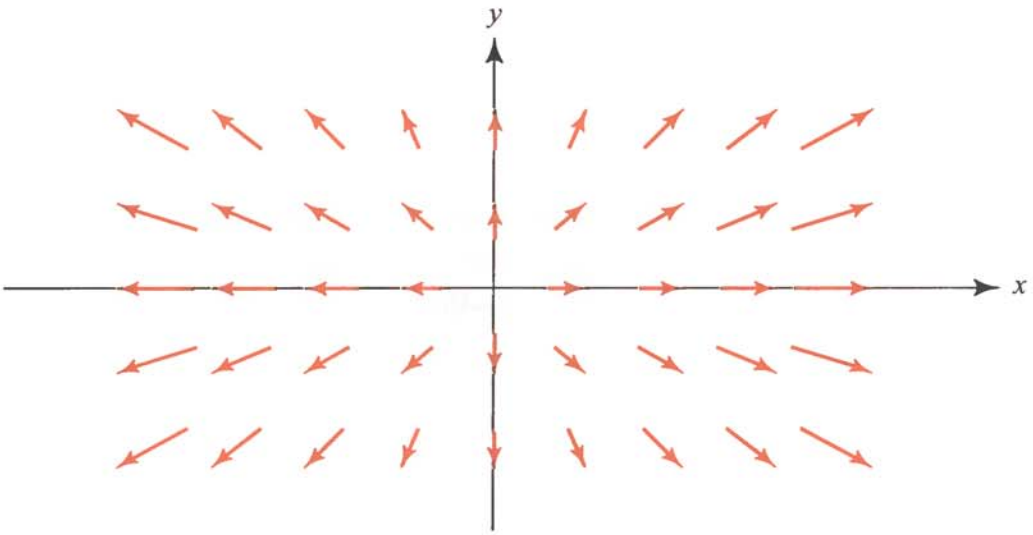


Figure 4.4.3 The vector field $\mathbf{F}(x, y) = x\mathbf{i} + y\mathbf{j}$.

If these flow lines are those of a fluid, the fluid is expanding as it moves out from the origin, so div $\mathbf{F}$ should be positive. In fact,

$$\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}x + \frac{\partial}{\partial y}y = 2 > 0. \quad \blacktriangle$$

EXAMPLE 4 Consider the vector field $\mathbf{F} = -x\mathbf{i} - y\mathbf{j}$. Here the flow lines point toward the origin instead of away from it (see Figure 4.4.4). Therefore, the fluid is compressing, so we expect (div $\mathbf{F}$) < 0. Calculating, we see that

$$\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(-x) + \frac{\partial}{\partial y}(-y) = -1 - 1 = -2 < 0. \quad \blacktriangle$$

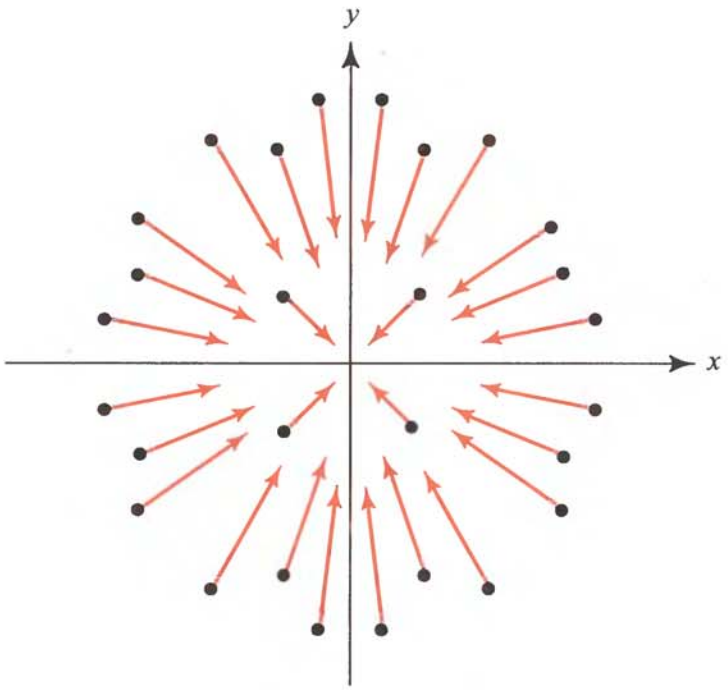


Figure 4.4.4 The vector field $\mathbf{F}(x, y) = -x\mathbf{i} - y\mathbf{j}$.

EXAMPLE 5 As we saw in the last section, the flow lines of $\mathbf{F} = -y\mathbf{i} + x\mathbf{j}$ are concentric circles about the origin, moving counterclockwise (see Figure 4.4.5). From this figure, it appears that the fluid is neither compressing nor expanding. This is confirmed by calculating

$$\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(-y) + \frac{\partial}{\partial y}(x) = 0 + 0 = 0. \quad \blacktriangle$$

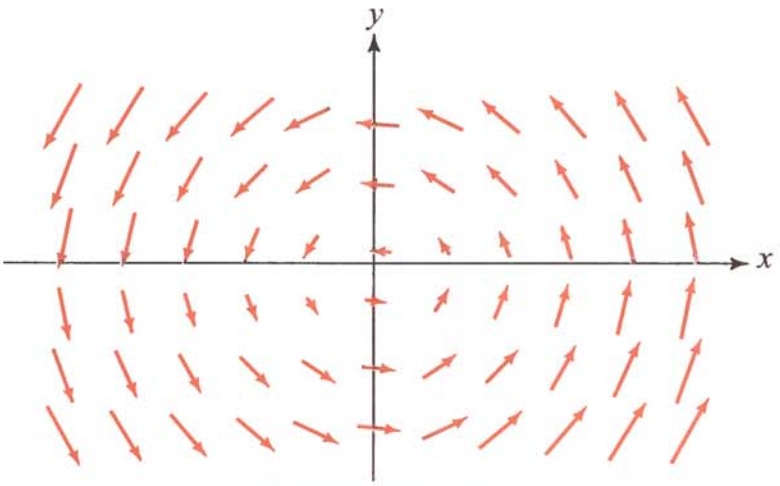


Figure 4.4.5 The vector field $\mathbf{F}(x, y) = -y\mathbf{i} + x\mathbf{j}$ has zero divergence.

EXAMPLE 6 Some flow lines of $\mathbf{F} = x\mathbf{i} - y\mathbf{j}$ are shown in Figure 4.4.6. Here our intuition about expansion or compression is less clear. However, it is true that the shaded regions shown have the same area, and we calculate that

$$\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}x + \frac{\partial}{\partial y}(-y) = 1 + (-1) = 0. \quad \blacktriangle$$

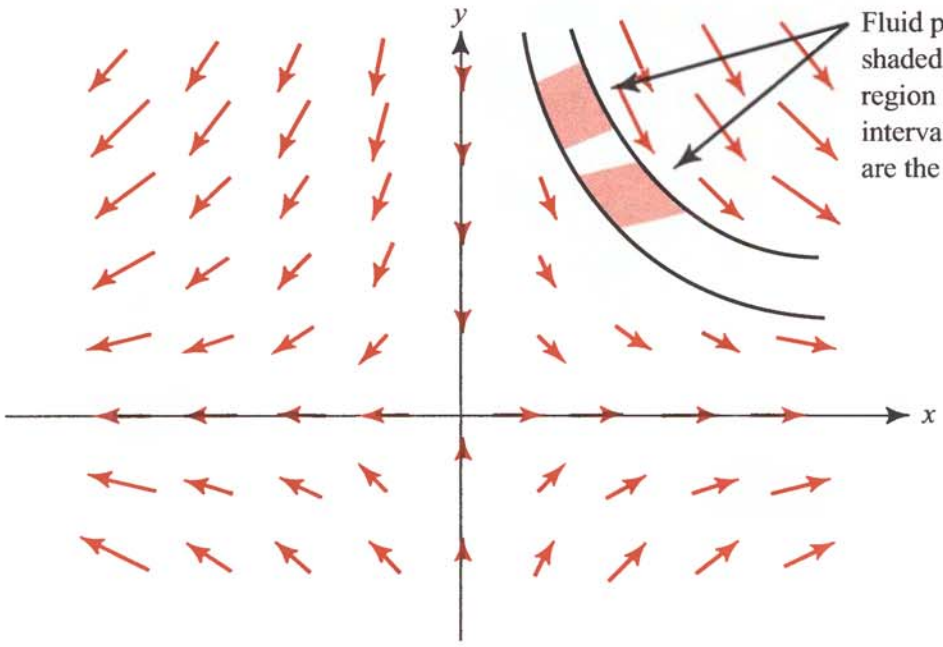


Figure 4.4.6 The vector field $\mathbf{F}(x, y) = x\mathbf{i} - y\mathbf{j}$.

Curl

To calculate the curl, the second basic operation performed on vector fields, we take the *cross product* of ∇ with $\mathbf{F}$.

Curl of a Vector Field If $\mathbf{F} = F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k}$, the ***curl*** of $\mathbf{F}$ is the *vector field*

$$\operatorname{curl} \mathbf{F} = \nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix}$$

$$= \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}\right)\mathbf{i} + \left(\frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x}\right)\mathbf{j} + \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right)\mathbf{k}.$$

If we write $\mathbf{F} = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}$, which is alternative notation, the same formula for the curl reads

$$\operatorname{curl} \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}$$

$$= \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}\right)\mathbf{i} - \left(\frac{\partial R}{\partial x} - \frac{\partial P}{\partial z}\right)\mathbf{j} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\mathbf{k}.$$

EXAMPLE 7 Let $\mathbf{F}(x, y, z) = x\mathbf{i} + xy\mathbf{j} + \mathbf{k}$. Find $\nabla \times \mathbf{F}$.

SOLUTION We use the preceding formula:

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ x & xy & 1 \end{vmatrix} = (0-0)\mathbf{i} - (0-0)\mathbf{j} + (y-0)\mathbf{k}.$$

Thus, $\nabla \times \mathbf{F} = y\mathbf{k}$. ▲

EXAMPLE 8 Find the curl of $xy\mathbf{i} - \sin z\mathbf{j} + \mathbf{k}$.

SOLUTION Letting $\mathbf{F} = xy\mathbf{i} - \sin z\mathbf{j} + \mathbf{k}$,

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ xy & -\sin z & 1 \end{vmatrix}$$

$$= \begin{vmatrix} \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ -\sin z & 1 \end{vmatrix} \mathbf{i} - \begin{vmatrix} \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial z} \\ xy & 1 \end{vmatrix} \mathbf{j} + \begin{vmatrix} \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} \\ xy & -\sin z \end{vmatrix} \mathbf{k}$$

$$= \cos z\mathbf{i} - x\mathbf{k}. \quad ▲$$

Unlike the divergence, which can be defined in $\mathbb{R}^n$ for any n, we define the curl only in three-dimensional space (or for planar vector fields, regarding their third component as zero).

The Curl and Rotations

The physical significance of the curl will be discussed in Chapter 8, when we study Stokes' theorem. However, we can now consider a specific situation, in which the curl is associated with rotations.

EXAMPLE 9 Consider a solid rigid body B rotating about an axis L. The rotational motion of the body can be described by a vector ω along the axis of rotation, the direction being chosen so that the body rotates about ω, as in Figure 4.4.7. We call ω the ***angular velocity vector***. The length $\omega = \|\omega\|$ is taken to be the angular speed of the body B, that is, the speed of any point in B divided by its distance from the axis L of rotation. The motion of points in the rotating body is described by the vector field $\mathbf{v}$ whose value at each point is the velocity at that point. To find $\mathbf{v}$, let Q be any point in B and let α be the distance from Q to L.

Figure 4.4.7 shows that $\alpha = \|\mathbf{r}\| \sin\theta$, where $\mathbf{r}$ is the vector whose initial point is the origin and whose terminal point is Q and θ is the angle between $\mathbf{r}$ and the axis L of rotation. The tangential velocity $\mathbf{v}$ of Q is directed counterclockwise along the

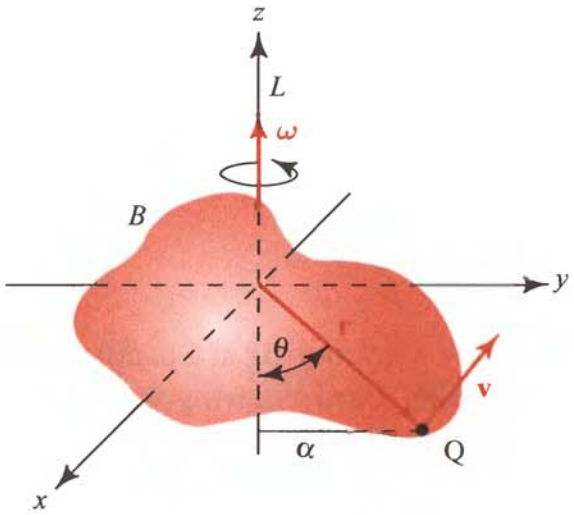


Figure 4.4.7 The velocity $\mathbf{v}$ and angular velocity $\boldsymbol{\omega}$ of a rotating body are related by $\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$.

tangent to a circle parallel to the xy plane with radius α and has magnitude

$$\|\mathbf{v}\| = \omega\alpha = \omega\|\mathbf{r}\| \sin\theta = \|\boldsymbol{\omega}\|\|\mathbf{r}\| \sin\theta.$$

The direction and magnitude of $\mathbf{v}$ imply that $\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$. Selecting a coordinate system in which L is the z axis, we can write $\boldsymbol{\omega} = \omega\mathbf{k}$ and $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$. Thus,

$$\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r} = -\omega y\mathbf{i} + \omega x\mathbf{j},$$

and so

$$\operatorname{curl} \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ -\omega y & \omega x & 0 \end{vmatrix} = 2\omega\mathbf{k} = 2\boldsymbol{\omega}.$$

Hence, for the rotation of a rigid body, the curl of the velocity vector field is a vector field whose value is the same at each point. It is directed along the axis of rotation with magnitude *twice* the angular speed. ▲

The Curl and Rotational Flow

If a vector field represents the flow of a *fluid*, then the value of $\nabla \times \mathbf{F}$ at a point is twice the angular velocity vector of a *rigid* body that rotates as the fluid does near that point. In particular, $\nabla \times \mathbf{F} = \mathbf{0}$ at a point P means that the fluid is free from rigid rotations at P; that is, it has no whirlpools. Another justification of this idea depends on Stokes' theorem from Chapter 8. However, we can say informally that curl $\mathbf{F} = \mathbf{0}$ means that if a *small* rigid paddle wheel is placed in the fluid, it will move with the fluid but will not rotate around its own axis. Such a vector field is called ***irrotational***. For example, it has been determined from experiments that fluid draining from a tub is usually irrotational except right at the center, even though the fluid is "rotating"

around the drain (see Figure 4.4.8). In Example 10, *the flow lines of the vector field* $\mathbf{V}$ *are circles about the origin, yet we show that the flow is irrotational.* Thus, the reader should be warned of the possible confusion the word "irrotational" can cause.

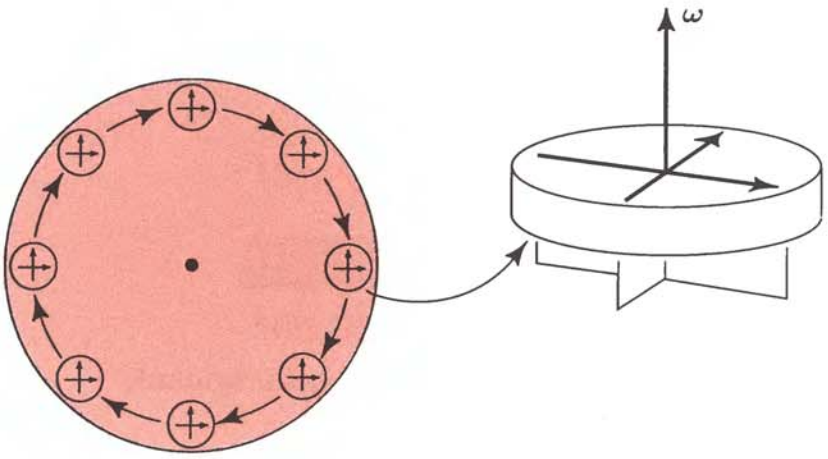


Figure 4.4.8 Looking at a paddle wheel from above a moving fluid. The velocity field $\mathbf{V}(x, y, z) = (y\mathbf{i} - x\mathbf{j})/(x^2 + y^2)$ is irrotational; the paddle wheel does not rotate around its axis ω.

EXAMPLE 10 Verify that the vector field

$$\mathbf{V}(x, y, z) = \frac{y\mathbf{i} - x\mathbf{j}}{x^2 + y^2}$$

is irrotational when $(x, y) \neq (0, 0)$ (i.e., except where $\mathbf{V}$ is not defined).

SOLUTION The curl is

$$\nabla \times \mathbf{V} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ \dfrac{y}{x^2 + y^2} & \dfrac{-x}{x^2 + y^2} & 0 \end{vmatrix}$$

$$= 0\mathbf{i} + 0\mathbf{j} + \left[\frac{\partial}{\partial x}\left(\frac{-x}{x^2 + y^2}\right) - \frac{\partial}{\partial y}\left(\frac{y}{x^2 + y^2}\right)\right]\mathbf{k}$$

$$= \left[\frac{-(x^2 + y^2) + 2x^2}{(x^2 + y^2)^2} + \frac{-(x^2 + y^2) + 2y^2}{(x^2 + y^2)^2}\right]\mathbf{k} = \mathbf{0}. \quad \blacktriangle$$

Gradients are Curl Free

The following identity is a basic relation between the gradient and curl, which should be compared with the fact that for any vector $\mathbf{v}$, we have $\mathbf{v} \times \mathbf{v} = \mathbf{0}$.

THEOREM 1: Curl of a Gradient For any C^2 function f,

$$\nabla \times (\nabla f) = \mathbf{0}.$$

That is, the curl of any gradient is the zero vector.

PROOF Because $\nabla f = (\partial f/\partial x, \partial f/\partial y, \partial f/\partial z)$ we have, by definition,

$$\nabla \times \nabla f = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ \dfrac{\partial f}{\partial x} & \dfrac{\partial f}{\partial y} & \dfrac{\partial f}{\partial z} \end{vmatrix}$$

$$= \left(\frac{\partial^2 f}{\partial y\,\partial z} - \frac{\partial^2 f}{\partial z\,\partial y}\right)\mathbf{i} + \left(\frac{\partial^2 f}{\partial z\,\partial x} - \frac{\partial^2 f}{\partial x\,\partial z}\right)\mathbf{j} + \left(\frac{\partial^2 f}{\partial x\,\partial y} - \frac{\partial^2 f}{\partial y\,\partial x}\right)\mathbf{k}.$$

Each component is zero because of the equality of mixed partial derivatives. ■

The converse to this theorem (a vector field with zero curl is a gradient, under suitable hypotheses) will be discussed in Chapter 8.

EXAMPLE 11 Let $\mathbf{V}(x, y, z) = y\mathbf{i} - x\mathbf{j}$. Show that $\mathbf{V}$ is not a gradient field.

SOLUTION If $\mathbf{V}$ were a gradient field, then it would satisfy curl $\mathbf{V} = \mathbf{0}$ by Theorem 1. But

$$\operatorname{curl} \mathbf{V} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ y & -x & 0 \end{vmatrix} = -2\mathbf{k} \neq \mathbf{0},$$

so $\mathbf{V}$ cannot be a gradient. ▲

Scalar Curl

There is an operation on vector fields in the plane that is closely related to the curl. If $\mathbf{F} = P(x, y)\mathbf{i} + Q(x, y)\mathbf{j}$ is a vector field in the plane, it can also be regarded as a vector field in space for which the $\mathbf{k}$ component is zero and the other two components are independent of z. The curl of $\mathbf{F}$ then reduces to

$$\nabla \times \mathbf{F} = \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\mathbf{k}$$

and always points in the **k** direction. The function

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$$

of x and y is called the ***scalar curl*** of **F**.

EXAMPLE 12 Find the scalar curl of $\mathbf{V}(x, y) = -y^2\mathbf{i} + x\mathbf{j}$.

SOLUTION The curl is

$$\nabla \times \mathbf{V} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ -y^2 & x & 0 \end{vmatrix} = (1 + 2y)\,\mathbf{k},$$

so the scalar curl, which is the coefficient of **k**, is $1 + 2y$. ▲

Curls are Divergence Free

A basic relation between the divergence and curl operations is given next.

THEOREM 2: Divergence of a Curl For any C^2 vector field **F**,

$$\text{div curl } \mathbf{F} = \nabla \cdot (\nabla \times \mathbf{F}) = 0.$$

That is, the divergence of any curl is zero.

As with the curl of a gradient, the proof rests on the equality of the mixed partial derivatives. The student should write out the details. A converse will be discussed in Chapter 8.

EXAMPLE 13 Show that the vector field $\mathbf{V}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ cannot be the curl of some vector field **F**; that is, there is no **F** with $\mathbf{V} = \text{curl } \mathbf{F}$.

SOLUTION If this were so, then div **V** would be zero by Theorem 2. But

$$\text{div } \mathbf{V} = \frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z} = 3 \neq 0,$$

so **V** cannot be curl **F** for any **F**. ▲

Laplacian

The ***Laplace operator*** ∇^2, which operates on functions f, is defined to be the divergence of the gradient:

$$\nabla^2 f = \nabla \cdot (\nabla f) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}.$$

This operator plays an important role in many physical laws, as we have mentioned in Section 3.1.

EXAMPLE 14 Show that $\nabla^2 f = 0$ for

$$f(x, y, z) = \frac{1}{\sqrt{x^2 + y^2 + z^2}} = \frac{1}{r} \quad \text{and} \quad (x, y, z) \neq (0, 0, 0),$$

where $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ and $r = \|\mathbf{r}\|$.

SOLUTION The first derivatives are

$$\frac{\partial f}{\partial x} = \frac{-x}{(x^2 + y^2 + z^2)^{3/2}}, \quad \frac{\partial f}{\partial y} = \frac{-y}{(x^2 + y^2 + z^2)^{3/2}}, \quad \frac{\partial f}{\partial z} = \frac{-z}{(x^2 + y^2 + z^2)^{3/2}}.$$

Computing the second derivatives, we find that

$$\begin{aligned}
\frac{\partial^2 f}{\partial x^2} &= \frac{3x^2}{(x^2 + y^2 + z^2)^{5/2}} - \frac{1}{(x^2 + y^2 + z^2)^{3/2}}, \\
\frac{\partial^2 f}{\partial y^2} &= \frac{3y^2}{(x^2 + y^2 + z^2)^{5/2}} - \frac{1}{(x^2 + y^2 + z^2)^{3/2}}, \\
\frac{\partial^2 f}{\partial z^2} &= \frac{3z^2}{(x^2 + y^2 + z^2)^{5/2}} - \frac{1}{(x^2 + y^2 + z^2)^{3/2}}.
\end{aligned}$$

Thus,

$$\begin{aligned}
\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} &= \frac{3(x^2 + y^2 + z^2)}{(x^2 + y^2 + z^2)^{5/2}} - \frac{3}{(x^2 + y^2 + z^2)^{3/2}} \\
&= \frac{3}{(x^2 + y^2 + z^2)^{3/2}} - \frac{3}{(x^2 + y^2 + z^2)^{3/2}} = 0. \quad \blacktriangle
\end{aligned}$$

Vector Identities

We now have these basic operations on hand: gradient, divergence, curl, and the Laplace operator. The following box contains some basic general formulas that are useful when computing with vector fields.

Basic Identities of Vector Analysis

1. $\nabla(f+g) = \nabla f + \nabla g$
2. $\nabla(cf) = c\nabla f$, for a constant c
3. $\nabla(fg) = f\nabla g + g\nabla f$
4. $\nabla(f/g) = (g\nabla f - f\nabla g)/g^2$, at points $\mathbf{x}$ where $g(\mathbf{x}) \neq 0$
5. $\text{div}\,(\mathbf{F}+\mathbf{G}) = \text{div}\,\mathbf{F} + \text{div}\,\mathbf{G}$
6. $\text{curl}\,(\mathbf{F}+\mathbf{G}) = \text{curl}\,\mathbf{F} + \text{curl}\,\mathbf{G}$
7. $\text{div}\,(f\mathbf{F}) = f\,\text{div}\,\mathbf{F} + \mathbf{F}\cdot\nabla f$
8. $\text{div}\,(\mathbf{F}\times\mathbf{G}) = \mathbf{G}\cdot\text{curl}\,\mathbf{F} - \mathbf{F}\cdot\text{curl}\,\mathbf{G}$
9. $\text{div curl}\,\mathbf{F} = 0$
10. $\text{curl}\,(f\mathbf{F}) = f\,\text{curl}\,\mathbf{F} + \nabla f\times\mathbf{F}$
11. $\text{curl}\,\nabla f = \mathbf{0}$
12. $\nabla^2(fg) = f\nabla^2 g + g\nabla^2 f + 2(\nabla f\cdot\nabla g)$
13. $\text{div}\,(\nabla f\times\nabla g) = 0$
14. $\text{div}\,(f\nabla g - g\nabla f) = f\nabla^2 g - g\nabla^2 f$

EXAMPLE 15 Prove identity 7 in the preceding box.

SOLUTION The vector field $f\,\mathbf{F}$ has components fF_i, for $i = 1, 2, 3$, and so

$$\text{div}\,(f\,\mathbf{F}) = \frac{\partial}{\partial x}(fF_1) + \frac{\partial}{\partial y}(fF_2) + \frac{\partial}{\partial z}(fF_3).$$

However, $(\partial/\partial x)(fF_1) = f\,\partial F_1/\partial x + F_1\,\partial f/\partial x$ by the product rule, with similar expressions for the other terms. Therefore,

$$\begin{aligned}\text{div}\,(f\,\mathbf{F}) &= f\left(\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}\right) + F_1\frac{\partial f}{\partial x} + F_2\frac{\partial f}{\partial y} + F_3\frac{\partial f}{\partial z}\\ &= f(\nabla\cdot\mathbf{F}) + \mathbf{F}\cdot\nabla f. \quad \blacktriangle\end{aligned}$$

Let us use these identities to redo Example 14.

EXAMPLE 16 Show that for $\mathbf{r} \neq \mathbf{0}$, $\nabla^2(1/r) = 0$.

SOLUTION As in the case of the gravitational potential, $\nabla(1/r) = -\mathbf{r}/r^3$. In general, $\nabla(r^n) = nr^{n-2}\mathbf{r}$ (see Exercise 30). By the identity $\nabla \cdot (f\,\mathbf{F}) = f\nabla \cdot \mathbf{F} + \nabla f \cdot \mathbf{F}$, we get

$$\nabla \cdot \left(\frac{\mathbf{r}}{r^3}\right) = \frac{1}{r^3}\nabla \cdot \mathbf{r} + \mathbf{r} \cdot \nabla\left(\frac{1}{r^3}\right)$$

$$= \frac{3}{r^3} + \mathbf{r} \cdot \left(\frac{-3\mathbf{r}}{r^5}\right) = \frac{3}{r^3} - \frac{3}{r^3} = 0. \quad \blacktriangle$$

Historical Note

Divergence and Curl

William Rowan Hamilton, in his investigation of quaternions (discussed in Section 1.3) introduced the *del operator*, defined formally as

$$\nabla = \frac{\partial}{\partial x}\mathbf{i} + \frac{\partial}{\partial y}\mathbf{j} + \frac{\partial}{\partial z}\mathbf{k}.$$

Hamilton firmly believed in the significance of this operator. If $f(x, y, z)$ is a scalar function on $\mathbb{R}^3$, then "multiplication" by ∇ gives the gradient of f:

$$\nabla f = \frac{\partial f}{\partial x}\mathbf{i} + \frac{\partial f}{\partial y}\mathbf{j} + \frac{\partial f}{\partial z}\mathbf{k},$$

which, of course, gives the direction of steepest ascent (see Section 2.6). If

$$\mathbf{V}(x,y,z) = V_1(x,y,z)\mathbf{i} + V_2(x,y,z)\mathbf{j} + V_3(x,y,z)\mathbf{k}$$

is a vector field, then the "quaternionic multiplication" of ∇ with $\mathbf{V}$ yields

$$\nabla\mathbf{V} = -\text{div}\,\mathbf{V} + \text{curl}\,\mathbf{V}.$$

Thus, what we now call the divergence of $\mathbf{V}$ is the negative of the scalar part of this product, and curl $\mathbf{V}$ is the vector part (c.f. the quaternion discussion in Section 1.3).

As far as we are aware, Hamilton never gave a physical interpretation of divergence and curl, but he surely believed that, as a consequence of his faith in them, they must have an important physical interpretation. His faith in his mathematical formalism was justified, but a physical explanation of divergence and curl had to wait for James Clerk Maxwell's *Treatise on Electricity and Magnetism*. Here, Maxwell used both the divergence and the

curl in his equations for the interaction of electric and magnetic fields (the Maxwell equations are discussed in Chapter 8).

Curiously, Maxwell referred to divergence as *convergence* and to curl as *rotation*, a term still used in the literature. It was Josiah Gibbs (Figure 4.4.9) who renamed convergence and rotation as the more familiar terms we use today—divergence and curl.

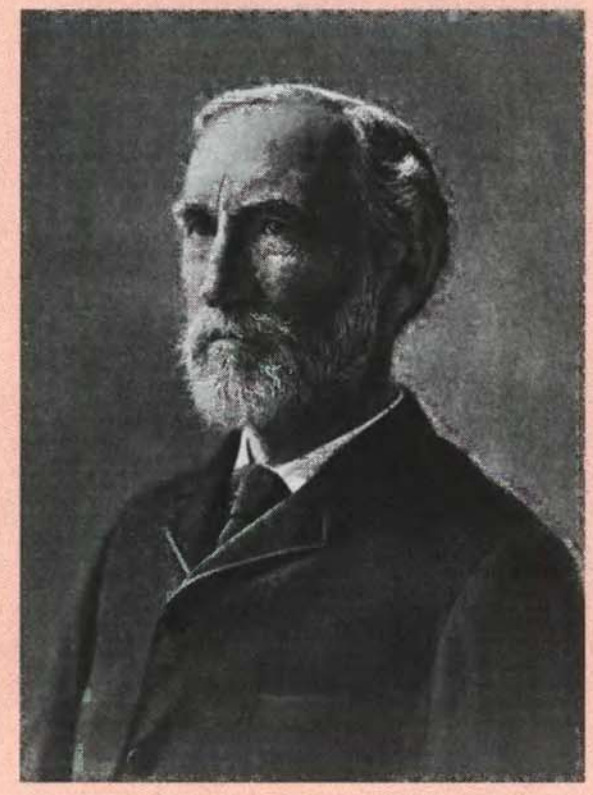


Figure 4.4.9 Josiah Willard Gibbs (1839–1903).

Maxwell gave a physical interpretation of the divergence using the Gauss divergence theorem, as we do in Section 8.4. His physical interpretation of the curl as a rotation was rather brief. Gibbs provided a more elementary interpretation of divergence, as we do in this section. In the spirit of Leibniz (who believed in infinitesimal quantities dx,dy,dz), Gibbs imagined placing a small cube of dimensions dx by dy by dz in a fluid. The faces of this cube have areas $dx\,dy$, $dy\,dz$, and $dx\,dz$.

At this point, students may be interested to hear Gibbs through the words of his student E. B. Wilson:

> Consider the amount of fluid which passes through those faces of the cube which are parallel to the YZ plane, i.e., perpendicular to the X axis [see Figure 4.4.10].
>
> The normal to the face whose x coordinate is the lesser, that is, the normal to the left-hand face of the cube is $-\mathbf{i}$. The flux of substance through this face is
>
> $$-\mathbf{i} \cdot \mathbf{V}(x, y, z)\, dy\, dz.$$

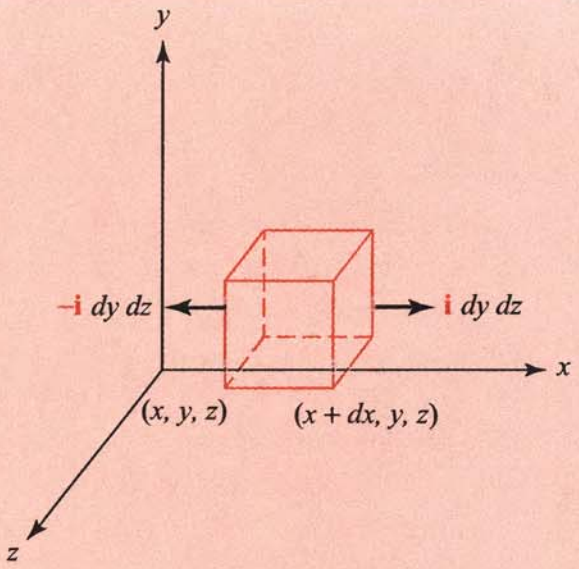


Figure 4.4.10 Cube with faces parallel to the *YZ* plane.

The normal to the opposite face, the face whose x coordinate is greater by the amount dx, is $+\mathbf{i}$, and the flux through it is therefore

$$\mathbf{i}\cdot\mathbf{V}(x+dx,y,z)\,dy\,dz=\mathbf{i}\cdot\left[\mathbf{V}(x,y,z)+\frac{\partial\mathbf{V}}{\partial x}\,dx\right]dy\,dz$$

$$=\mathbf{i}\cdot\mathbf{V}(x,y,z)\,dy\,dz+\mathbf{i}\cdot\frac{\partial\mathbf{V}}{\partial x}\,dx\,dy\,dz.$$

The total flux outward from the cube through these two faces is therefore the algebraic sum of these quantities. This is simply

$$\mathbf{i}\cdot\frac{\partial\mathbf{V}}{\partial x}\,dx\,dy\,dz=\frac{\partial V_1}{\partial x}\,dx\,dy\,dz.$$

In like manner the fluxes through the other pairs of faces of the cube are

$$\mathbf{j}\cdot\frac{\partial\mathbf{V}}{\partial y}\,dx\,dy\,dz \qquad \text{and} \qquad \mathbf{k}\cdot\frac{\partial\mathbf{V}}{\partial z}\,dx\,dy\,dz.$$

The total flux out from the cube is therefore

$$\left(\mathbf{i}\cdot\frac{\partial\mathbf{V}}{\partial x}+\mathbf{j}\cdot\frac{\partial\mathbf{V}}{\partial y}+\mathbf{k}\cdot\frac{\partial\mathbf{V}}{\partial z}\right)dx\,dy\,dz.$$

This is the net quantity of fluid that leaves the cube per unit time. The quotient of this by the volume $dx\,dy\,dz$ of the cube gives the

rate of diminution of density. This is

$$\nabla \cdot \mathbf{V} = \mathbf{i} \cdot \frac{\partial \mathbf{V}}{\partial x} + \mathbf{j} \cdot \frac{\partial \mathbf{V}}{\partial y} + \mathbf{k} \cdot \frac{\partial \mathbf{V}}{\partial z} = \frac{\partial V_1}{\partial x} + \frac{\partial V_2}{\partial y} + \frac{\partial V_3}{\partial z}.$$

Because $\nabla \cdot \mathbf{V}$ thus represents the diminution of density or the rate at which matter is leaving a point per unit volume per unit time, it is called the *divergence*. Maxwell employed the term *convergence* to denote the rate at which fluid approaches a point per unit volume per unit time. This is the negative of the divergence. In the case that the fluid is *incompressible*, as much matter must leave the cube as enters it. The total change of contents must therefore be zero. For this reason, the characteristic differential equation that any incompressible fluid must satisfy is

$$\nabla \cdot \mathbf{V} = 0,$$

where $\mathbf{V}$ is the velocity of the fluid. This equation is often known as the *hydrodynamic equation*. It is satisfied by any flow of water, since water is practically incompressible. The great importance of the equation for work in electricity is due to the fact that according to Maxwell's hypothesis, electric displacement obeys the same laws as an incompressible fluid. If, then, $\mathbf{D}$ is the electric displacement,

$$\operatorname{div} \mathbf{D} = \nabla \cdot \mathbf{D} = 0.$$

Gibbs' interpretation of curl was much like the one we gave in Example 9 for the rotation of a rigid body. Wilson remarks that an analysis of the meaning of curl for fluid motion was "rather difficult." It remains a bit elusive, even today, as can be seen from our discussion following Example 9. We provide another interpretation in Chapter 8.

EXERCISES

Find the divergence of the vector fields in Exercises 1 to 4.

1. $\mathbf{V}(x, y, z) = e^{xy}\mathbf{i} - e^{xy}\mathbf{j} + e^{yz}\mathbf{k}$

2. $\mathbf{V}(x, y, z) = yz\mathbf{i} + xz\mathbf{j} + xy\mathbf{k}$

3. $\mathbf{V}(x, y, z) = x\mathbf{i} + (y + \cos x)\mathbf{j} + (z + e^{xy})\mathbf{k}$

4. $\mathbf{V}(x, y, z) = x^2\mathbf{i} + (x + y)^2\mathbf{j} + (x + y + z)^2\mathbf{k}$

5. Figure 4.4.11 shows some flow lines and moving regions for a fluid moving in the plane field velocity field $\mathbf{V}$. Where is div $\mathbf{V} > 0$, and also where is div $\mathbf{V} < 0$?

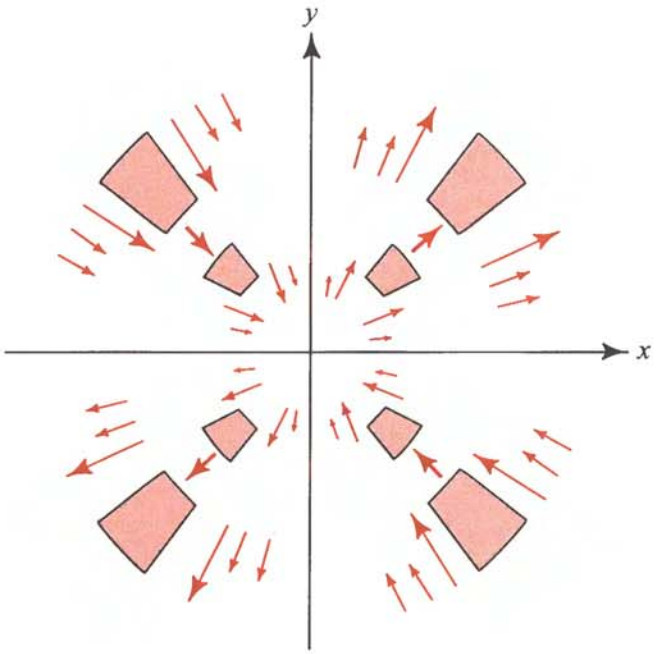


Figure 4.4.11 The flow lines of a fluid moving in the plane.

6. Let $V(x, y, z) = x\mathbf{i}$ be the velocity field of a fluid in space. Relate the sign of the divergence with the rate of change of volume under the flow.

7. Sketch a few flow lines for $\mathbf{F}(x, y) = y\mathbf{i}$. Calculate $\nabla \cdot \mathbf{F}$ and explain why your answer is consistent with your sketch.

8. Sketch a few flow lines for $\mathbf{F}(x, y) = -3x\mathbf{i} - y\mathbf{j}$. Calculate $\nabla \cdot \mathbf{F}$ and explain why your answer is consistent with your sketch.

Calculate the divergence of the vector fields in Exercises 9 to 12.

9. $\mathbf{F}(x, y) = x^3\mathbf{i} - x \sin(xy)\mathbf{j}$

10. $\mathbf{F}(x, y) = y\mathbf{i} - x\mathbf{j}$

11. $\mathbf{F}(x, y) = \sin(xy)\mathbf{i} - \cos(x^2y)\mathbf{j}$

12. $\mathbf{F}(x, y) = xe^y\mathbf{i} - [y/(x + y)]\mathbf{j}$

Compute the curl, $\nabla \times \mathbf{F}$, of the vector fields in Exercises 13 to 16.

13. $\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$

14. $\mathbf{F}(x, y, z) = yz\mathbf{i} + xz\mathbf{j} + xy\mathbf{k}$

15. $\mathbf{F}(x, y, z) = (x^2 + y^2 + z^2)(3\mathbf{i} + 4\mathbf{j} + 5\mathbf{k})$

16. $\mathbf{F}(x, y, z) = \dfrac{yz\mathbf{i} - xz\mathbf{j} + xy\mathbf{k}}{x^2 + y^2 + z^2}$

Calculate the scalar curl of each of the vector fields in Exercises 17 to 20.

17. $\mathbf{F}(x, y) = \sin x\mathbf{i} + \cos x\mathbf{j}$

18. $\mathbf{F}(x, y) = y\mathbf{i} - x\mathbf{j}$

19. $\mathbf{F}(x, y) = xy\mathbf{i} + (x^2 - y^2)\mathbf{j}$

20. $\mathbf{F}(x, y) = x\mathbf{i} + y\mathbf{j}$

Verify that $\nabla \times (\nabla f) = \mathbf{0}$ *for the functions in Exercises 21 to 24.*

21. $f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$

22. $f(x, y, z) = xy + yz + xz$

23. $f(x, y, z) = 1/(x^2 + y^2 + z^2)$

24. $f(x, y, z) = x^2y^2 + y^2z^2$

25. Show that $\mathbf{F} = y(\cos x)\mathbf{i} + x(\sin y)\mathbf{j}$ is *not* a gradient vector field.

26. Show that $\mathbf{F} = (x^2 + y^2)\mathbf{i} - 2xy\mathbf{j}$ is *not* a gradient field.

27. Prove identity 10 in the list of vector identities.

28. Suppose that $\nabla \cdot \mathbf{F} = 0$ and $\nabla \cdot \mathbf{G} = 0$. Which of the following necessarily have zero divergence?

(a) $\mathbf{F} + \mathbf{G}$ (b) $\mathbf{F} \times \mathbf{G}$

29. Let $\mathbf{F} = 2xz^2\mathbf{i} + \mathbf{j} + y^3zx\mathbf{k}$ and $f = x^2y$. Compute the following quantities:

(a) ∇f (b) $\nabla \times \mathbf{F}$ (c) $\mathbf{F} \times \nabla f$ (d) $\mathbf{F} \cdot (\nabla f)$

30. Let $\mathbf{r}(x, y, z) = (x, y, z)$ and $r = \sqrt{x^2 + y^2 + z^2} = \|\mathbf{r}\|$. Prove the following identities.

(a) $\nabla(1/r) = -\mathbf{r}/r^3, r \neq 0$; and, in general, $\nabla(r^n) = nr^{n-2}\mathbf{r}$ and $\nabla(\log r) = \mathbf{r}/r^2$.
(b) $\nabla^2(1/r) = 0, r \neq 0$; and, in general, $\nabla^2 r^n = n(n + 1)r^{n-2}$.
(c) $\nabla \cdot (\mathbf{r}/r^3) = 0$; and, in general, $\nabla \cdot (r^n\mathbf{r}) = (n + 3)r^n$.
(d) $\nabla \times \mathbf{r} = \mathbf{0}$; and, in general, $\nabla \times (r^n\mathbf{r}) = \mathbf{0}$.

31. Does $\nabla \times \mathbf{F}$ have to be perpendicular to $\mathbf{F}$?

32. Let $\mathbf{F}(x, y, z) = 3x^2y\mathbf{i} + (x^3 + y^3)\mathbf{j}$.

(a) Verify that curl $\mathbf{F} = \mathbf{0}$.
(b) Find a function f such that $\mathbf{F} = \nabla f$. (Techniques for constructing f in general are given in Chapter 8. The one in this problem should be sought by trial and error.)

33. Show that the real and imaginary parts of each of the following complex functions form the components of an irrotational and incompressible vector field in the plane; here $i = \sqrt{-1}$.

(a) $(x - iy)^2$ (b) $(x - iy)^3$ (c) $e^{x-iy} = e^x(\cos y - i \sin y)$

REVIEW EXERCISES FOR CHAPTER 4

For Exercises 1 to 4, at the indicated point, compute the velocity vector, the acceleration vector, the speed, and the equation of the tangent line.

1. $\mathbf{c}(t) = (t^3 + 1, e^{-t}, \cos(\pi t/2))$, at $t = 1$

2. $\mathbf{c}(t) = (t^2 - 1, \cos(t^2), t^4)$, at $t = \sqrt{\pi}$

3. $\mathbf{c}(t) = (e^t, \sin t, \cos t)$, at $t = 0$

4. $\mathbf{c}(t) = \dfrac{t^2}{1 + t^2}\mathbf{i} + t\mathbf{j} + \mathbf{k}$, at $t = 2$

5. Calculate the tangent and acceleration vectors for the helix $\mathbf{c}(t) = (\cos t, \sin t, t)$ at $t = \pi/4$.

6. Calculate the tangent and acceleration vector for the cycloid $\mathbf{c}(t) = (t - \sin t, 1 - \cos t)$ at $t = \pi/4$ and sketch.

7. Let a particle of mass m move on the path $\mathbf{c}(t) = (t^2, \sin t, \cos t)$. Compute the force acting on the particle at $t = 0$.

8. (a) Let $\mathbf{c}(t)$ be a path with $\|\mathbf{c}(t)\| =$ constant; that is, the curve lies on a sphere. Show that $\mathbf{c}'(t)$ is orthogonal to $\mathbf{c}(t)$.

(b) Let $\mathbf{c}$ be a path whose speed is never zero. Show that $\mathbf{c}$ has constant speed if and only if the acceleration vector $\mathbf{c}''$ is always perpendicular to the velocity vector $\mathbf{c}'$.

9. Express the arc length of the curve $x^2 = y^3 = z^5$ between $x = 1$ and $x = 4$ as an integral, using a suitable parametrization.

10. Find the arc length of $\mathbf{c}(t) = t\mathbf{i} + (\log t)\mathbf{j} + 2\sqrt{2t}\mathbf{k}$ for $1 \le t \le 2$.

11. A particle is constrained to move around the unit circle in the xy plane according to the formula $(x, y, z) = (\cos(t^2), \sin(t^2), 0)$, $t \ge 0$.

(a) What are the velocity vector and speed of the particle as functions of t?

(b) At what point on the circle should the particle be released to hit a target at (2, 0, 0)? (Be careful about which direction the particle is moving around the circle.)

(c) At what time t should the release take place? (Use the smallest $t > 0$ that will work.)

(d) What are the velocity and speed at the time of release?

(e) At what time is the target hit?

12. A particle of mass m moves under the influence of a force $\mathbf{F} = -k\mathbf{r}$, where k is a constant and $\mathbf{r}(t)$ is the position of the particle at time t.

(a) Write down differential equations for the components of $\mathbf{r}(t)$.
(b) Solve the equations in part (a) subject to the initial conditions $\mathbf{r}(0) = \mathbf{0}$, $\mathbf{r}'(0) = 2\mathbf{j} + \mathbf{k}$.

13. Write the curve described by the equations $x - 1 = 2y + 1 = 3z + 2$ in parametric form.

14. Write the curve $x = y^3 = z^2 + 1$ in parametric form.

15. Show that $\mathbf{c}(t) = (1/(1-t), 0, e^t/(1-t))$ is a flow line of the vector field defined by $\mathbf{F}(x, y, z) = (x^2, 0, z(1+x))$.

16. Let $\mathbf{F}(x, y) = f(x^2 + y^2)[-y\mathbf{i} + x\mathbf{j}]$ for a function f of one variable. What equation must $g(t)$ satisfy for

$$\mathbf{c}(t) = [\cos g(t)]\mathbf{i} + [\sin g(t)]\mathbf{j}$$

to be a flow line for $\mathbf{F}$?

Compute $\nabla \cdot \mathbf{F}$ and $\nabla \times \mathbf{F}$ for the vector fields in Exercises 17 to 20.

17. $\mathbf{F} = 2x\mathbf{i} + 3y\mathbf{j} + 4z\mathbf{k}$

18. $\mathbf{F} = x^2\mathbf{i} + y^2\mathbf{j} + z^2\mathbf{k}$

19. $\mathbf{F} = (x + y)\mathbf{i} + (y + z)\mathbf{j} + (z + x)\mathbf{k}$

20. $\mathbf{F} = x\mathbf{i} + 3xy\mathbf{j} + z\mathbf{k}$

Compute the divergence and curl of the vector fields in Exercises 21 and 22 at the points indicated.

21. $\mathbf{F}(x, y, z) = y\mathbf{i} + z\mathbf{j} + x\mathbf{k}$, at the point (1, 1, 1)

22. $\mathbf{F}(x, y, z) = (x + y)^3\mathbf{i} + (\sin xy)\mathbf{j} + (\cos xyz)\mathbf{k}$, at the point (2, 0, 1)

Calculate the gradients of the functions in Exercises 23 to 26, and verify that $\nabla \times \nabla f = \mathbf{0}$.

23. $f(x, y) = e^{xy} + \cos(xy)$

24. $f(x, y) = \dfrac{x^2 - y^2}{x^2 + y^2}$

25. $f(x, y) = e^{x^2} - \cos(xy^2)$

26. $f(x, y) = \tan^{-1}(x^2 + y^2)$

27. (a) Let $f(x, y, z) = xyz^2$; compute ∇f.

(b) Let $\mathbf{F}(x, y, z) = xy\mathbf{i} + yz\mathbf{j} + zy\mathbf{k}$; compute $\nabla \times \mathbf{F}$.

(c) Compute $\nabla \times (f\mathbf{F})$ using identity 10 of the list of vector identities. Compare with a direct computation.

28. (a) Let $\mathbf{F} = 2xye^z\mathbf{i} + e^z x^2\mathbf{j} + (x^2 ye^z + z^2)\mathbf{k}$. Compute $\nabla \cdot \mathbf{F}$ and $\nabla \times \mathbf{F}$.

(b) Find a function $f(x, y, z)$ such that $\mathbf{F} = \nabla f$.

29. Let $\mathbf{F}(x, y) = f(x^2 + y^2)[-y\mathbf{i} + x\mathbf{j}]$, as in Exercise 16. Calculate div $\mathbf{F}$ and curl $\mathbf{F}$ and discuss your answers in view of the results of Exercise 16.

30. Let a particle of mass m move along the elliptical helix $\mathbf{c}(t) = (4\cos t, \sin t, t)$.

(a) Find the equation of the tangent line to the helix at $t = \pi/4$.

(b) Find the force acting on the particle at time $t = \pi/4$.

(c) Write an expression (in terms of an integral) for the arc length of the curve $\mathbf{c}(t)$ between $t = 0$ and $t = \pi/4$.

31. (a) Let $g(x, y, z) = x^3 + 5yz + z^2$ and let $h(u)$ be a function of one variable such that $h'(1) = 1/2$. Let $f = h \circ g$. Starting at (1, 0, 0), in what directions is f changing at 50% of its maximum rate?

(b) For $g(x, y, z) = x^3 + 5yz + z^2$, calculate $\mathbf{F} = \nabla g$, the gradient of g, and verify directly that $\nabla \times \mathbf{F} = 0$ at each point (x, y, z).

32. (a) Write in parametric form the curve that is the intersection of the surfaces $x^2 + y^2 + z^2 = 3$ and $y = 1$.

(b) Find the equation of the line tangent to this curve at (1, 1, 1).

(c) Write an integral expression for the arc length of this curve. What is the value of this integral?

33. In meteorology, the ***negative pressure gradient*** **G** is a vector quantity that points from regions of high pressure to regions of low pressure, normal to the lines of constant pressure (***isobars***).

(a) In an xy coordinate system,

$$\mathbf{G} = -\frac{\partial P}{\partial x}\mathbf{i} - \frac{\partial P}{\partial y}\mathbf{j}.$$

Write a formula for the magnitude of the negative pressure gradient.

(b) If the horizontal pressure gradient provided the only horizontal force acting on the air, the wind would blow directly across the isobars in the direction of **G**, and for a given air mass, with acceleration proportional to the magnitude of **G**. Explain, using Newton's second law.

(c) Because of the rotation of the earth, the wind does not blow in the direction that part (b) would suggest. Instead, it obeys ***Buys–Ballot's law***, which states: "If in the Northern Hemisphere, you stand with your back to the wind, the high pressure is on your right and the low pressure is on your left." Draw a figure and introduce xy coordinates so that **G** points in the proper direction.

(d) State and graphically illustrate Buys–Ballot's law for the Southern Hemisphere, in which the orientation of high and low pressure is reversed.

34. A sphere of mass m, radius a, and uniform density has potential u and gravitational force $\mathbf{F}$, at a distance r from the center (0, 0, 0), given by

$$u = \frac{3m}{2a} - \frac{mr^2}{2a^3}, \qquad \mathbf{F} = -\frac{m}{a^3}\mathbf{r} \qquad (r \le a);$$

$$u = \frac{m}{r}, \qquad \mathbf{F} = -\frac{m}{r^3}\mathbf{r} \qquad (r > a).$$

Here, $r = \|\mathbf{r}\|$, $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$.

(a) Verify that $\mathbf{F} = \nabla u$ on the inside and outside of the sphere.

(b) Check that u satisfies Poisson's equation: $\partial^2 u/\partial x^2 + \partial^2 u/\partial y^2 + \partial^2 u/\partial z^2 =$ constant inside the sphere.

(c) Show that u satisfies Laplace's equation: $\partial^2 u/\partial x^2 + \partial^2 u/\partial y^2 + \partial^2 u/\partial z^2 = 0$ outside the sphere.

35. A circular helix that lies on the cylinder $x^2 + y^2 = R^2$ with pitch ρ may be described parametrically by

$$x = R\cos\theta, \qquad y = R\sin\theta, \qquad z = \rho\theta, \qquad \theta \ge 0.$$

A particle slides under the action of gravity (which acts parallel to the z axis) without friction along the helix. If the particle starts out at the height $z_0 > 0$, then when it reaches the height z along the helix, its speed is given by

$$\frac{ds}{dt} = \sqrt{(z_0 - z)2g},$$

where s is arc length along the helix, g is the constant of gravity, t is time, and $0 \le z \le z_0$.

(a) Find the length of the part of the helix between the planes $z = z_0$ and $z = z_1$, $0 \le z_1 < z_0$.

(b) Compute the time T_0 it takes the particle to reach the plane $z = 0$.

36. A sphere of radius 10 centimeters (cm) with center at (0, 0, 0) rotates about the z axis with angular velocity 4 in such a direction that the rotation looks counterclockwise from the positive z axis.

(a) Find the rotation vector ω (see Example 9, in Section 4.4).

(b) Find the velocity $\mathbf{v} = \omega \times \mathbf{r}$ when $\mathbf{r} = 5\sqrt{2}(\mathbf{i} - \mathbf{j})$ is on the "equator."

(c) Find the velocity of the point $(0, 5\sqrt{3}, 5)$ on the sphere.

37. Find the speed of the students in a classroom located at a latitude 49°N due to the rotation of the earth. (Ignore the motion of the earth about the sun, the sun in the galaxy, etc.; the radius of the earth is 3960 miles.)

5

Double and Triple Integrals

It is to Archimedes himself (c. 225 B.C.) that we owe the nearest approach to actual integration to be found among the Greeks. His first noteworthy advance in this direction was concerned with his proof that the area of a parabolic segment is four thirds of the triangle with the same base and vertex, or two thirds of the circumscribed parallelogram.

D. E. Smith, History of Mathematics

In this chapter and the next we study the integration of real-valued functions of several variables; this chapter treats integrals of functions of two and three variables, or *double* and *triple integrals*. The double integral has a basic geometric interpretation as volume, and can be defined rigorously as a limit of approximating sums. We shall present several techniques for evaluating double and triple integrals and consider some applications.

5.1 Introduction

This section discusses some geometric aspects of the double integral, deferring a more rigorous discussion in terms of Riemann sums until Section 5.2.

Double Integrals as Volumes

Consider a continuous function of two variables $f: R \subset \mathbb{R}^2 \to \mathbb{R}$ whose domain R is a rectangle with sides parallel to the coordinate axes. The rectangle R can be described in terms of the two closed intervals $[a, b]$ and $[c, d]$, representing the sides of R along the x and y axes, respectively, as in Figure 5.1.1. In this case, we say that R is the ***Cartesian product*** of $[a, b]$ and $[c, d]$ and write $R = [a, b] \times [c, d]$.

Assume that $f(x, y) \geq 0$ on R, so that the graph of $z = f(x, y)$ is a surface lying above the rectangle R. This surface, the rectangle R, and the four planes $x = a$, $x = b$, $y = c$, and $y = d$ form the boundary of a region V in space (see Figure 5.1.1).

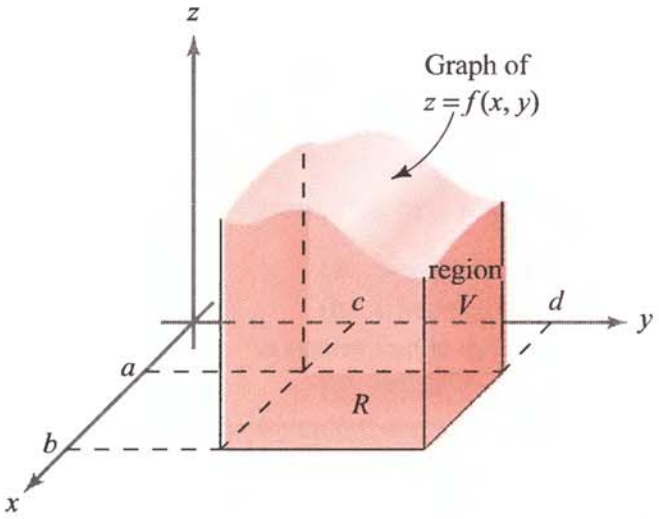


Figure 5.1.1 The region V in space is bounded by the graph of f, the rectangle R, and the four vertical sides indicated.

The problem of how to rigorously define the volume of V has to be faced, and we shall solve it in Section 5.2 by the classic method of exhaustion, or rather, in more modern terms, the method of Riemann sums. To gain an intuitive grasp of the double integral, we provisionally assume that the volume of a region has been defined.

Double Integrals The volume of the region above R and under the graph of a nonnegative function f is called the ***(double) integral*** of f over R and is denoted by

$$\iint_R f(x, y)\, dA, \qquad \text{or} \qquad \iint_R f(x, y)\, dx\, dy.$$

EXAMPLE 1 (a) If f is defined by $f(x, y) = k$, where k is a positive constant, then

$$\iint_R f(x, y)\, dA = k(b - a)(d - c),$$

because the integral is equal to the volume of a rectangular box with base R and height k.

(b) If $f(x, y) = 1 - x$ and $R = [0, 1] \times [0, 1]$, then

$$\iint_R f(x, y)\, dA = \frac{1}{2},$$

because the integral is equal to the volume of the triangular solid shown in Figure 5.1.2. ▲

EXAMPLE 2 Suppose $z = f(x, y) = x^2 + y^2$ and $R = [-1, 1] \times [0, 1]$. Then the integral $\iint_R (x^2 + y^2)\, dx\, dy$ is equal to the volume of the solid sketched in Figure 5.1.3. We shall compute this integral in Example 3. ▲

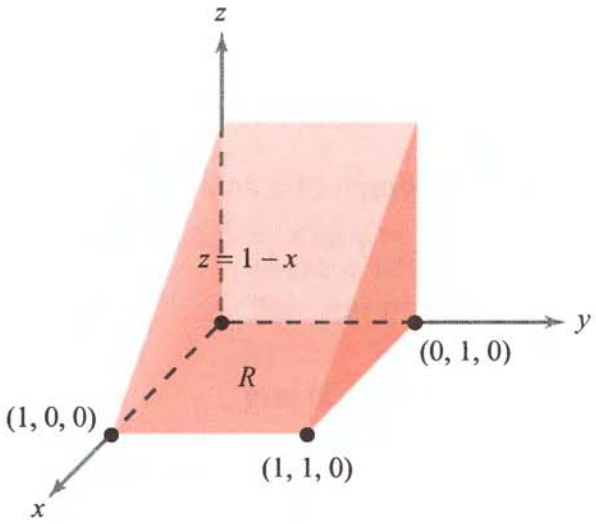


Figure 5.1.2 Volume under the graph $z = 1 - x$ and over $R = [0, 1] \times [0, 1]$.

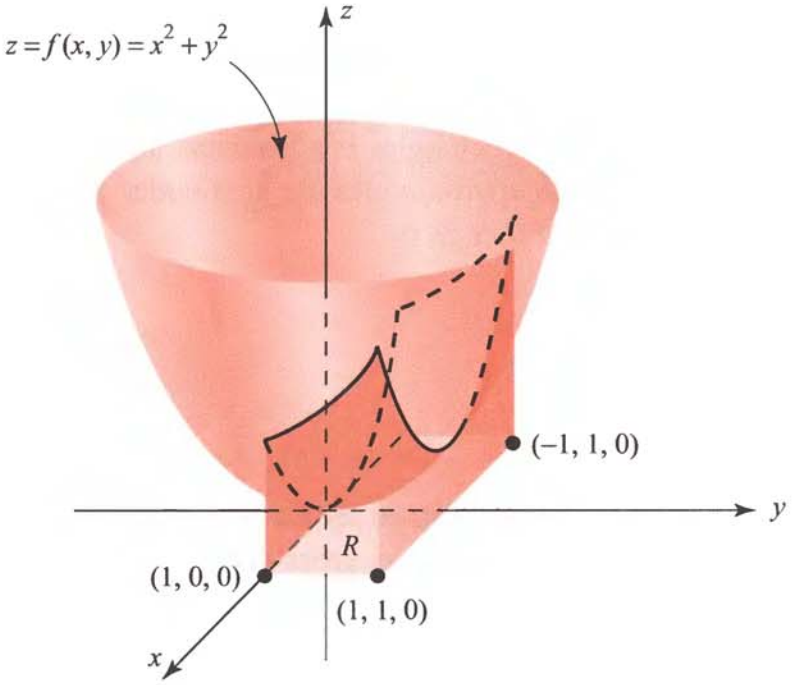


Figure 5.1.3 Volume under $z = x^2 + y^2$ and over $R = [-1, 1] \times [0, 1]$.

These ideas are similar to those for a single integral $\int_a^b f(x)\,dx$, which represents the area under the graph of f if $f \geq 0$; see Figure 5.1.4.[1]

Single integrals $\int_a^b f(x)\,dx$ can be rigorously defined, without recourse to the area concept, as a limit of Riemann sums. The idea is to approximate $\int_a^b f(x)\,dx$ by choosing a partition $a = x_0 < x_1 < \cdots < x_n = b$ of $[a, b]$, selecting points $c_i \in [x_i, x_{i+1}]$, and forming the Riemann sum

$$\sum_{i=0}^{n-1} f(c_i)(x_{i+1} - x_i) \approx \int_a^b f(x)\,dx$$

[1] Readers not already familiar with this idea should review the appropriate sections of their introductory calculus text.

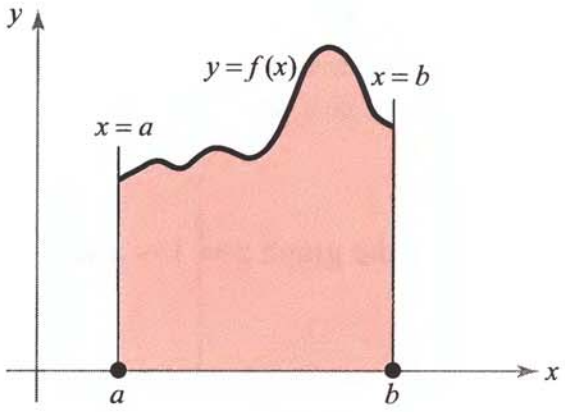


Figure 5.1.4 Area under the graph of a nonnegative continuous function f from $x = a$ to $x = b$ is $\int_a^b f(x)\,dx$.

(see Figure 5.1.5). We examine the analogous process for double integrals in the next section.

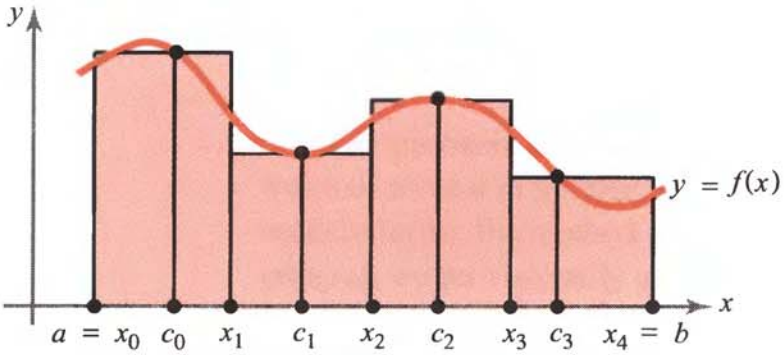


Figure 5.1.5 The sum of the areas of the shaded rectangles is a Riemann sum, which approximates the area under f from $x = a$ to $x = b$.

Cavalieri's Principle

There is a useful method for computing volumes, known as *Cavalieri's principle.* Suppose we have a solid body and we let $A(x)$ denote its cross-sectional area in a plane P_x measured at a distance x from a reference plane (Figure 5.1.6).

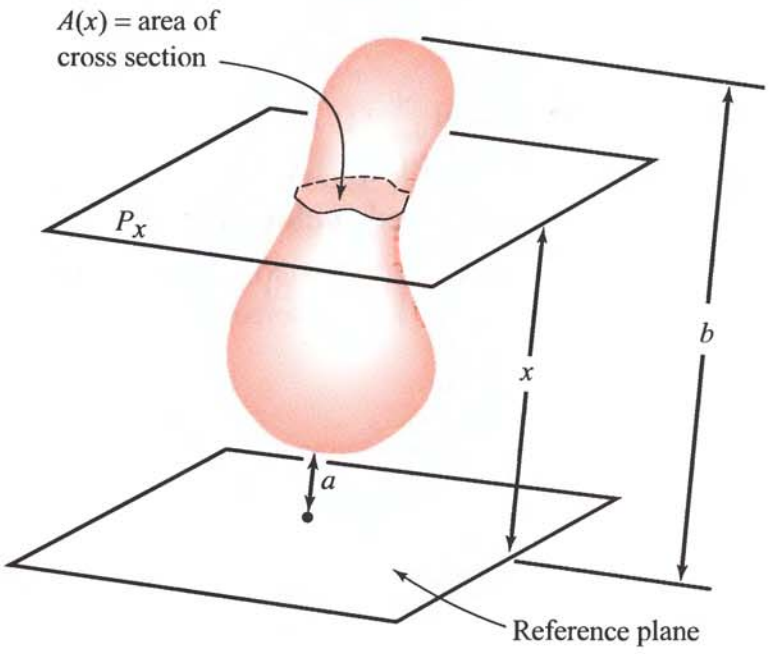


Figure 5.1.6 A solid body with cross-sectional area $A(x)$ at distance x from a reference plane.

According to Cavalieri's principle, the volume of the body is given by

$$\text{volume} = \int_a^b A(x)\,dx,$$

where a and b are the minimum and maximum distances from the reference plane. This can be made intuitively clear as follows. If we partition $[a, b]$ into $a = x_0 < x_1 < \cdots < x_n = b$, then an approximating Riemann sum for the preceding integral is

$$\sum_{i=0}^{n-1} A(c_i)(x_{i+1} - x_i).$$

But this sum also approximates the volume of the body, because $A(x)\,\Delta x$ is the volume of a slab with cross-sectional area $A(x)$ and thickness Δx (Figure 5.1.7). Therefore, it is reasonable to accept the preceding formula for the volume. A more careful justification of this method is given in the Internet supplement for Chapter 5.

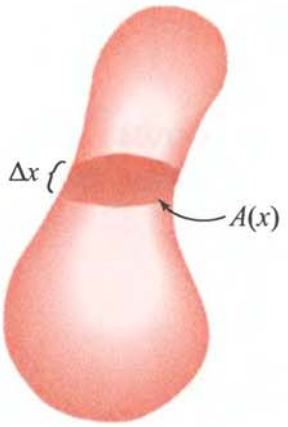


Figure 5.1.7 Volume of a slab with cross-sectional area $A(x)$ and thickness Δx equals $A(x)\Delta x$. The total volume of the body is $\int_a^b A(x)\,dx$.

The Slice Method — Cavalieri's Principle Let S be a solid and, for x satisfying $a \le x \le b$, let P_x be a family of parallel planes such that:

1. S lies between P_a and P_b;
2. The area of the slice of S cut by P_x is $A(x)$.

Then the volume of S is equal to

$$\int_a^b A(x)\,dx.$$

Historical Note

Bonaventura Cavalieri (1598–1647) was a pupil of Galileo and a professor in Bologna. His investigations into area and volume were important building blocks of the foundations of calculus. Although his methods were criticized by his contemporaries, similar ideas had been used by Archimedes in antiquity, and were later taken up by the "fathers" of calculus, Newton and Leibniz.

Reduction to Iterated Integrals

We now use Cavalieri's principle to evaluate double integrals. Consider the solid region under a graph $z = f(x, y)$ defined on the region $[a, b] \times [c, d]$, where f is continuous and greater than zero. There are two natural cross-sectional area functions: one obtained by using cutting planes perpendicular to the x axis, and the other obtained by using cutting planes perpendicular to the y axis. The cross section determined by a cutting plane $x = x_0$, of the first sort, is the plane region under the graph of $z = f(x_0, y)$ from $y = c$ to $y = d$ (Figure 5.1.8).

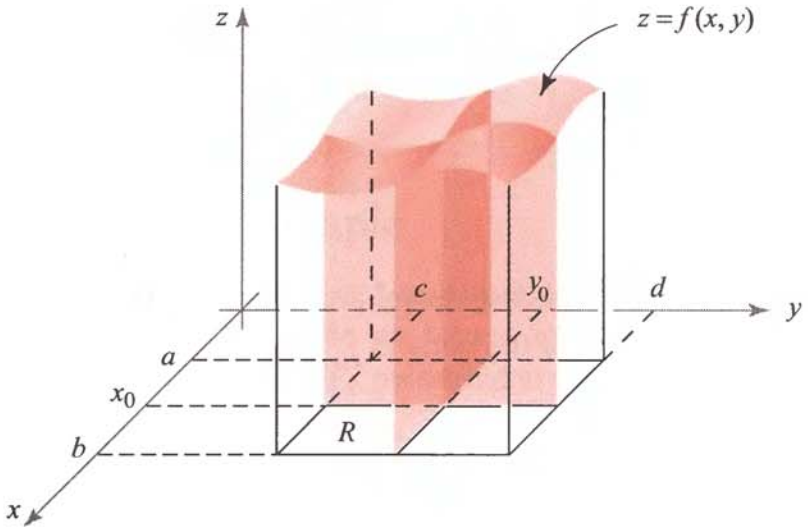


Figure 5.1.8 Two different cross sections sweeping out the volume under $z = f(x, y)$.

When we fix $x = x_0$, we obtain the function $y \mapsto f(x_0, y)$, which is continuous on $[c, d]$. The cross-sectional area $A(x_0)$ is, therefore, equal to the integral $\int_c^d f(x_0, y)\,dy$. Thus, the cross-sectional area function A has domain $[a, b]$, and is given by the rule $A: x \mapsto \int_c^d f(x, y)\,dy$. By Cavalieri's principle, the volume V of the region under $z = f(x, y)$ must be equal to

$$V = \int_a^b A(x)\,dx = \int_a^b \left[\int_c^d f(x, y)\,dy \right] dx.$$

The integral $\int_a^b \left[\int_c^d f(x, y)\,dy \right] dx$ is known as an ***iterated integral*** because it is obtained by integrating with respect to y and then integrating the result with respect to x. Because $\iint_R f(x, y)\,dA$ is equal to the volume V, we get the following result.

Double and Iterated Integrals

$$\iint_R f(x, y)\, dA = \int_a^b \left[\int_c^d f(x, y)\, dy \right] dx. \tag{1}$$

If we use cutting planes perpendicular to the y axis, we obtain

$$\iint_R f(x, y)\, dA = \int_c^d \left[\int_a^b f(x, y)\, dx \right] dy. \tag{2}$$

The expression on the right of formula (2) is the iterated integral obtained by integrating with respect to x and then integrating the result with respect to y.

Thus, if our intuition about volumes is correct, formulas (1) and (2) ought to be valid. This is in fact true when the concepts we are discussing are defined rigorously, and is known as *Fubini's theorem*. We give a proof of this theorem in the next section.

As the following examples illustrate, the notion of the iterated integral and equations (1) and (2) provide a powerful method for *computing* the double integral of a function of two variables.

EXAMPLE 3 Evaluate the integral

$$\iint_R (x^2 + y^2)\, dx\, dy,$$

where $R = [-1, 1] \times [0, 1]$.

SOLUTION By equation (2),

$$\iint_R (x^2 + y^2)\, dx\, dy = \int_0^1 \left[\int_{-1}^1 (x^2 + y^2)\, dx \right] dy.$$

To find $\int_{-1}^1 (x^2 + y^2)\, dx$, we treat y as a constant and integrate with respect to x. Because $x^3/3 + y^2 x$ is an antiderivative of $x^2 + y^2$ with respect to x, we can integrate, using the fundamental theorem of calculus, to obtain

$$\int_{-1}^1 (x^2 + y^2)\, dx = \left[\frac{x^3}{3} + y^2 x \right]_{x=-1}^1 = \frac{2}{3} + 2y^2.$$

Next, we integrate $\frac{2}{3} + 2y^2$ with respect to y from 0 to 1, to obtain

$$\int_0^1 \left(\frac{2}{3} + 2y^2 \right) dy = \left[\frac{2}{3} y + \frac{2}{3} y^3 \right]_{y=0}^1 = \frac{4}{3}.$$

Hence, the volume of the solid we saw in Figure 5.1.3 is 4/3.

For completeness, let us evaluate $\iint_R (x^2 + y^2)\,dx\,dy$ using equation (1)—that is, integrating with respect to y first and then with respect to x. We have

$$\iint_R (x^2 + y^2)\,dx\,dy = \int_{-1}^{1} \left[\int_0^1 (x^2 + y^2)\,dy \right] dx.$$

Treating x as a constant in the y integration, we obtain

$$\int_0^1 (x^2 + y^2)\,dy = \left[x^2 y + \frac{y^3}{3} \right]_{y=0}^{1} = x^2 + \frac{1}{3}.$$

Next, we evaluate $\int_{-1}^{1} \left(x^2 + \frac{1}{3}\right) dx$ to obtain

$$\int_{-1}^{1} \left(x^2 + \frac{1}{3} \right) dx = \left[\frac{x^3}{3} + \frac{x}{3} \right]_{x=-1}^{1} = \frac{4}{3},$$

which agrees with our previous answer. ▲

EXAMPLE 4 Compute the double integral $\iint_S \cos x \sin y\,dx\,dy$, where S is the square $[0, \pi/2] \times [0, \pi/2]$ (see Figure 5.1.9).

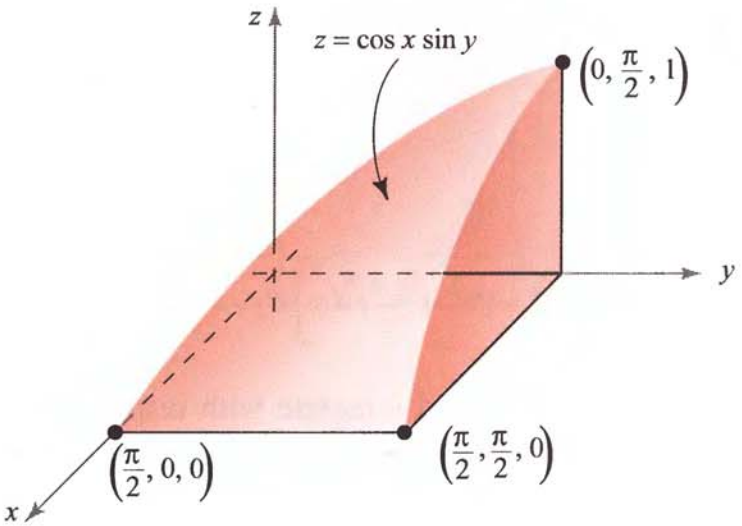


Figure 5.1.9 Volume under $z = \cos x \sin y$ and over the rectangle $[0, \pi/2] \times [0, \pi/2]$.

SOLUTION By equation (2),

$$\iint_S \cos x \sin y\,dx\,dy = \int_0^{\pi/2} \left[\int_0^{\pi/2} \cos x \sin y\,dx \right] dy$$

$$= \int_0^{\pi/2} \sin y \left[\int_0^{\pi/2} \cos x\,dx \right] dy = \int_0^{\pi/2} \sin y\,dy = 1. \quad ▲$$

In the next section, we shall use Riemann sums to rigorously define the double integral for a large class of functions of two variables without recourse to the notion of volume. Although we shall drop the requirement that $f(x, y) \geq 0$, equations (1) and (2) will remain valid. Therefore, the iterated integral will again provide the key to computing the double integral. In Section 5.3, we treat double integrals over regions more general than rectangles.

Finally, we remark that it is common to delete the brackets in iterated integrals such as equations (1) and (2) and write

$$\int_a^b \int_c^d f(x, y)\, dy\, dx \qquad \text{in place of} \qquad \int_a^b \left[\int_c^d f(x, y)\, dy \right] dx$$

and

$$\int_c^d \int_a^b f(x, y)\, dx\, dy \qquad \text{in place of} \qquad \int_c^d \left[\int_a^b f(x, y)\, dx \right] dy.$$

EXERCISES

1. Evaluate the following iterated integrals:

(a) $\int_{-1}^{1} \int_{0}^{1} (x^4 y + y^2)\, dy\, dx$

(b) $\int_{0}^{\pi/2} \int_{0}^{1} (y \cos x + 2)\, dy\, dx$

(c) $\int_{0}^{1} \int_{0}^{1} (xye^{x+y})\, dy\, dx$

(d) $\int_{-1}^{0} \int_{1}^{2} (-x \log y)\, dy\, dx$

2. Evaluate the integrals in Exercise 1 by integrating with respect to x and then with respect to y. [The solution to part (b) only is in the Study Guide to this text.]

3. Use Cavalieri's principle to show that the volumes of two cylinders with the same base and height are equal (see Figure 5.1.10).

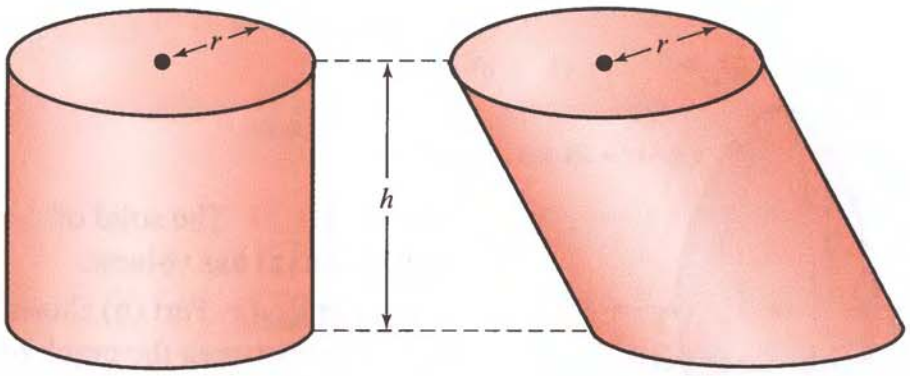


Figure 5.1.10 Two cylinders with the same base and height have the same volume.

4. Using Cavalieri's principle, compute the volume of the structure shown in Figure 5.1.11; each cross section is a rectangle of length 5 and width 3.